

MICROCOMPUTER MN101E

MN101EF56K/57G/76K
LSI User's Manual Vol.2

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About This Manual

■Objective

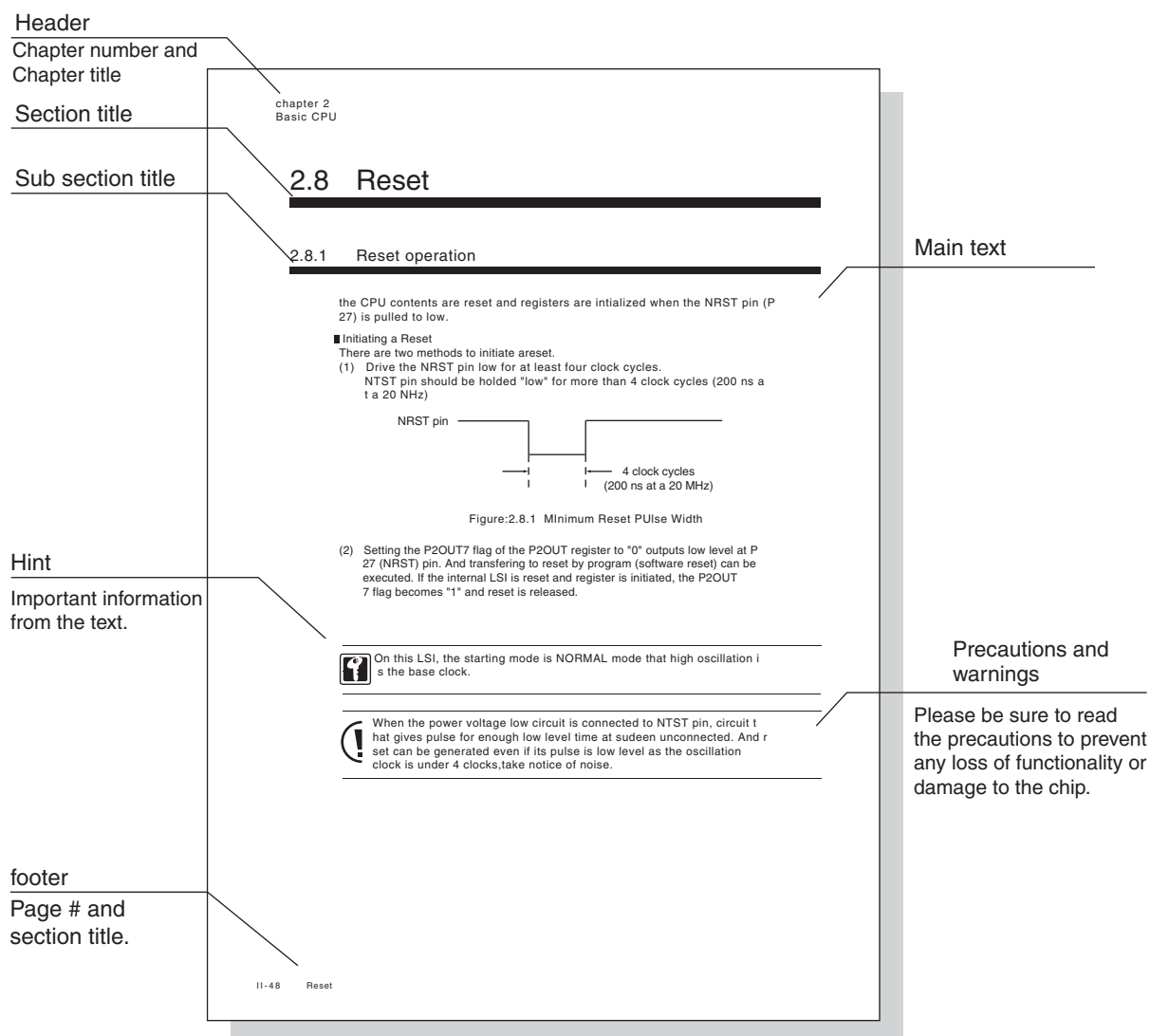
The primary objective of this LSI manual is to describe the features of this product including an overview, CPU basic functions, interrupt, port, timer, serial interface, and other peripheral hardware functions.

Each section consists of brief functional information, block diagrams and the details of control registers including operation methods and setting examples.

■Structure of This Manual

Each section of this manual consists of a title, summary, main text, hint, precautions and warnings, and references.

The layout and definition of each section are shown below.



This page serves as an example to the explanations above. It may be different on an actual page.

■Finding Desired Information

This manual provides three methods for finding the desired information quickly and easily.

- 1.Refer to the index at the front of the manual to locate the beginning of each section.
- 2.Refer to the table of contents at the front of the manual to locate the desired titles.
- 3.The chapter number and chapter title are located at the top corner of each page, and the section titles are located at the bottom corner of each page.

■Related Manuals

Note that the following documents related to MN101E series are available.

- "MN101E Series Instruction Manual"
<Describes the instruction set.>
- "MN101C/MN101E Series C Compiler User's Manual Usage Guide"
<Describes the installation, commands and options of the C Compiler.>
- "MN101C/MN101E Series C Compiler User's Manual Language Description"
<Describes the syntax of the C Compiler.>
- "MN101C/MN101E Series C Compiler User's Manual Library Reference"
<Describes the standard library of the C Compiler.>
- "MN101C/MN101E Series Cross-assembler User's Manual"
<Describes the assembler syntax and notation.>
- "PanaX EX Installation Manual"
<Describes the steps to install the Integrated Development Environment (DebugFactory Builder), C compiler and the real-time OS. It also describes the procedure to setup the on-board environment.>
- "In-Circuit Emulator Installation Manual"
<Describes the steps to install the Integrated Development Environment (DebugFactory Builder), C compiler and the real-time OS. It also describes the procedure to setup the in-circuit emulator.>
- "PanaX EX Commander User's Manual"
< Describes the usage and notes on designing target board.>

■Caution

The related documents listed above are subject to change without notice.
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■Contact Information

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9.1 Overview

This LSI consists of five 8-bit timers/baud rate timers (Timer 0, 1, 2, 3 and 4). Timer 0 and 1 or Timer 2 and 3 can be cascaded to form 16-bit timers. Under cascade connection, Timer 0, 2 and 4 form the lower 8 bits of the 16-bit timer while Timer 1 and 3 form the upper 8 bits of the 16-bit timer. In addition, cascading Timer 0 to 2 form a 24-bit counter, and cascading Timer 0 to 3 form a 32-bit counter.

8-bit timer consists of two prescalers which can be used simultaneously. Each prescaler counts $f_{pll-div}$ and f_s as the base clock. Configurations of the hardware are shown below.

Prescaler 0 (based on $f_{pll-div}$) 7-bit prescaler
Prescaler 1 (based on f_s) 3-bit prescaler

Prescaler 0 outputs $f_{pll-div}/4$, $f_{pll-div}/16$, $f_{pll-div}/32$, $f_{pll-div}/64$ and $f_{pll-div}/128$.

Prescaler 1 outputs $f_s/2$, $f_s/4$ and $f_s/8$.

Divided clock of $f_{pll-div}$ or f_s can be selected as the clock source for each timer by using the prescaler output.

Pins to be used can be switched to TMnIOA/TMnIOB/TMnIOC.

TM0IOA (PA0)
TM0IOB (P04)
TM1IOA (PA1)
TM1IOB (P62)
TM2IOA (PA2)
TM2IOB (P04)
TM3IOA (PA3)
TM3IOB (P63)
TM4IOA (PA4)
TM4IOB (P64)

Table:9.1.1 8-bit Timer Pin Functions

Table remarks √: With function -: Without function

Functions	Pin Name	MN101EF76K	MN101EF57G	MN101EF56K
TM0IOA	PA0	√	√	√
TM0IOB	P04	√	√	√
TM0IOC	P10	√	-	√
TM1IOA	PA1	√	√	√
TM1IOB	P62	√	√	√
TM1IOC	P12	√	-	√
TM2IOA	PA2	√	√	√
TM2IOB	P04	√	√	√
TM2IOC	P11	√	-	√
TM3IOA	PA3	√	√	√
TM3IOB	P63	√	√	√
TM3IOC	P13	√	-	√



In this manual, if there is not much difference in the function between Pin A, B and C, “A”, “B”, and “C” of the pin names are omitted.



When changing fpll-div frequency by bp7 to bp4 of OSCCNT register, it should be executed after the 8-bit timer is stopped.

9.1.1 Functions

Table:9.1.2, Table:9.1.3 and Table:9.1.4 show functions that can be used with each timer.

Table:9.1.2 Timer Functions (MN101EF76K)

	Timer 0 (8-bit)	Timer 1 (8-bit)	Timer 2 (8-bit)	Timer 3 (8-bit)	Timer 4 (8-bit)
Interrupt source	TM0IRQ	TM1IRQ	TM2IRQ	TM3IRQ	TM4IRQ
Timer operation	√	√	√	√	√
Event count	TM0IOA input/ TM0IOB input/ TM0IOC input	TM1IOA input/ TM1IOB input/ TM1IOC input	TM2IOA input/ TM2IOB input/ TM2IOC input	TM3IOA input/ TM3IOB input/ TM3IOC input	TM4IOA input/ TM4IOB input/ TM4IOC input
Timer pulse output	TM0IOA output/ TM0IOB output/ TM0IOC output	TM1IOA output/ TM1IOB output/ TM1IOC output	TM2IOA output/ TM2IOB output/ TM2IOC output	TM3IOA output/ TM3IOB output/ TM3IOC output	TM4IOA output/ TM4IOB output/ TM4IOC output
PWM output	TM0IOA output/ TM0IOB output/ TM0IOC output	-	TM2IOA output/ TM2IOB output/ TM2IOC output	-	TM4IOA output/ TM4IOB output/ TM4IOC output
PWM output with additional pulses	√	-	√	-	√
Synchronous output	-	Port 8	Port 8	-	-
Serial transfer clock output	Serial Interface 0, 1, 2, 3, 4	Serial Interface 0, 1, 2, 3, 4	Serial Interface 0, 1, 2, 3, 4	Serial Interface 0, 1, 2, 3, 4	Serial Interface 0, 1, 2, 3, 4
Pulse width measurement	P20/IRQ0A PC0/IRQ0B	-	P22/IRQ2A PC2/IRQ2B	-	P24/IRQ4A PC4/IRQ4B
Cascade connection	√		√		-
	√			-	-
	√				-
Clock source	fppll-div fppll-div/4 fppll-div/16 fppll-div/32 fppll-div/64 fppll-div/128 fs/2 fs/4 fs/8 fslow TM0IO input Synchronous fslow Synchronous TM0IO input Timer A output	fppll-div fppll-div/4 fppll-div/16 fppll-div/32 fppll-div/64 fppll-div/128 fs/2 fs/4 fs/8 fslow TM1IO input Synchronous fslow Synchronous TM1IO input Timer A output	fppll-div fppll-div/4 fppll-div/16 fppll-div/32 fppll-div/64 fppll-div/128 fs/2 fs/4 fs/8 fslow TM2IO input Synchronous fslow Synchronous TM2IO input Timer A output	fppll-div fppll-div/4 fppll-div/16 fppll-div/32 fppll-div/64 fppll-div/128 fs/2 fs/4 fs/8 fslow TM3IO input Synchronous fslow Synchronous TM3IO input Timer A output	fppll-div fppll-div/4 fppll-div/16 fppll-div/32 fppll-div/64 fppll-div/128 fs/2 fs/4 fs/8 fslow TM4IO input Synchronous fslow Synchronous TM4IO input Timer A output
fppll-div: Machine clock (High speed oscillation for peripheral functions) fslow: Machine clock (Low speed oscillation for peripheral functions) fs: System clock					

Table:9.1.3 Timer Functions (MN101EF57G)

	Timer 0 (8-bit)	Timer 1 (8-bit)	Timer 2 (8-bit)	Timer 3 (8-bit)	Timer 4 (8-bit)
Interrupt source	TM0IRQ	TM1IRQ	TM2IRQ	TM3IRQ	TM4IRQ
Timer operation	√	√	√	√	√
Event count	TM0IOA input/ TM0IOB input/	TM1IOA input/ TM1IOB input/	TM2IOA input/ TM2IOB input/	TM3IOA input/ TM3IOB input/	TM4IOA input/ TM4IOB input/
Timer pulse output	TM0IOA output/ TM0IOB output/	TM1IOA output/ TM1IOB output/	TM2IOA output/ TM2IOB output/	TM3IOA output/ TM3IOB output/	TM4IOA output/ TM4IOB output/
PWM output	TM0IOA output/ TM0IOB output/	-	TM2IOA output/ TM2IOB output/	-	TM4IOA output/ TM4IOB output/
PWM output with additional pulses	√	-	√	-	√
Synchronous output	-	Port 8	Port 8	-	-
Serial transfer clock output	Serial Interface 0, 1, 2, 4	Serial Interface 0, 1, 2, 4	Serial Interface 0, 1, 2, 4	Serial Interface 0, 1, 2, 4	Serial Interface 0, 1, 2, 4
Pulse width measurement	P20/IRQ0	-	P22/IRQ2	-	P24/IRQ4
Cascade connection	√	√		-	
	√	-			-
	√	-			
Clock source	fppll-div fppll-div/4 fppll-div/16 fppll-div/32 fppll-div/64 fppll-div/128 fs/2 fs/4 fs/8 fslow TM0IO input Synchronous fslow Synchronous TM0IO input Timer A output	fppll-div fppll-div/4 fppll-div/16 fppll-div/32 fppll-div/64 fppll-div/128 fs/2 fs/4 fs/8 fslow TM1IO input Synchronous fslow Synchronous TM1IO input Timer A output	fppll-div fppll-div/4 fppll-div/16 fppll-div/32 fppll-div/64 fppll-div/128 fs/2 fs/4 fs/8 fslow TM2IO input Synchronous fslow Synchronous TM2IO input Timer A output	fppll-div fppll-div/4 fppll-div/16 fppll-div/32 fppll-div/64 fppll-div/128 fs/2 fs/4 fs/8 fslow TM3IO input Synchronous fslow Synchronous TM3IO input Timer A output	fppll-div fppll-div/4 fppll-div/16 fppll-div/32 fppll-div/64 fppll-div/128 fs/2 fs/4 fs/8 fslow TM4IO input Synchronous fslow Synchronous TM4IO input Timer A output
	fppll-div: Machine clock (High speed oscillation for peripheral functions) fslow: Machine clock (Low speed oscillation for peripheral functions) fs: System clock				

Table:9.1.4 Timer Functions (MN101EF56K)

	Timer 0 (8-bit)	Timer 1 (8-bit)	Timer 2 (8-bit)	Timer 3 (8-bit)	Timer 4 (8-bit)
Interrupt source	TM0IRQ	TM1IRQ	TM2IRQ	TM3IRQ	TM4IRQ
Timer operation	√	√	√	√	√
Event count	TM0IOA input/ TM0IOB input/ TM0IOC input	TM1IOA input/ TM1IOB input/ TM1IOC input	TM2IOA input/ TM2IOB input/ TM2IOC input	TM3IOA input/ TM3IOB input/ TM3IOC input	TM4IOA input/ TM4IOB input/ TM4IOC input
Timer pulse output	TM0IOA output/ TM0IOB output/ TM0IOC output	TM1IOA output/ TM1IOB output/ TM1IOC output	TM2IOA output/ TM2IOB output/ TM2IOC output	TM3IOA output/ TM3IOB output/ TM3IOC output	TM4IOA output/ TM4IOB output/ TM4IOC output
PWM output	TM0IOA output/ TM0IOB output/ TM0IOC output	-	TM2IOA output/ TM2IOB output/ TM2IOC output	-	TM4IOA output/ TM4IOB output/ TM4IOC output
PWM output with additional pulses	√	-	√	-	√
Synchronous output	-	Port 8	Port 8	-	-
Serial transfer clock output	Serial Interface 0, 1, 2, 3, 4	Serial Interface 0, 1, 2, 3, 4	Serial Interface 0, 1, 2, 3, 4	Serial Interface 0, 1, 2, 3, 4	Serial Interface 0, 1, 2, 3, 4
Pulse width measurement	P20/IRQ0	-	P22/IRQ2	-	P24/IRQ4
Cascade connection	√	√			-
	√	-			-
	√	-			-
Clock source	fpll-div fpll-div/4 fpll-div/16 fpll-div/32 fpll-div/64 fpll-div/128 fs/2 fs/4 fs/8 fslow TM0IO input Synchronous fslow Synchronous TM0IO input Timer A output	fpll-div fpll-div/4 fpll-div/16 fpll-div/32 fpll-div/64 fpll-div/128 fs/2 fs/4 fs/8 fslow TM1IO input Synchronous fslow Synchronous TM1IO input Timer A output	fpll-div fpll-div/4 fpll-div/16 fpll-div/32 fpll-div/64 fpll-div/128 fs/2 fs/4 fs/8 fslow TM2IO input Synchronous fslow Synchronous TM2IO input Timer A output	fpll-div fpll-div/4 fpll-div/16 fpll-div/32 fpll-div/64 fpll-div/128 fs/2 fs/4 fs/8 fslow TM3IO input Synchronous fslow Synchronous TM3IO input Timer A output	fpll-div fpll-div/4 fpll-div/16 fpll-div/32 fpll-div/64 fpll-div/128 fs/2 fs/4 fs/8 fslow TM4IO input Synchronous fslow Synchronous TM4IO input Timer A output
fpll-div: Machine clock (High speed oscillation for peripheral functions) fslow: Machine clock (Low speed oscillation for peripheral functions) fs: System clock					

9.1.2 Block Diagram

■ Prescaler Block Diagram

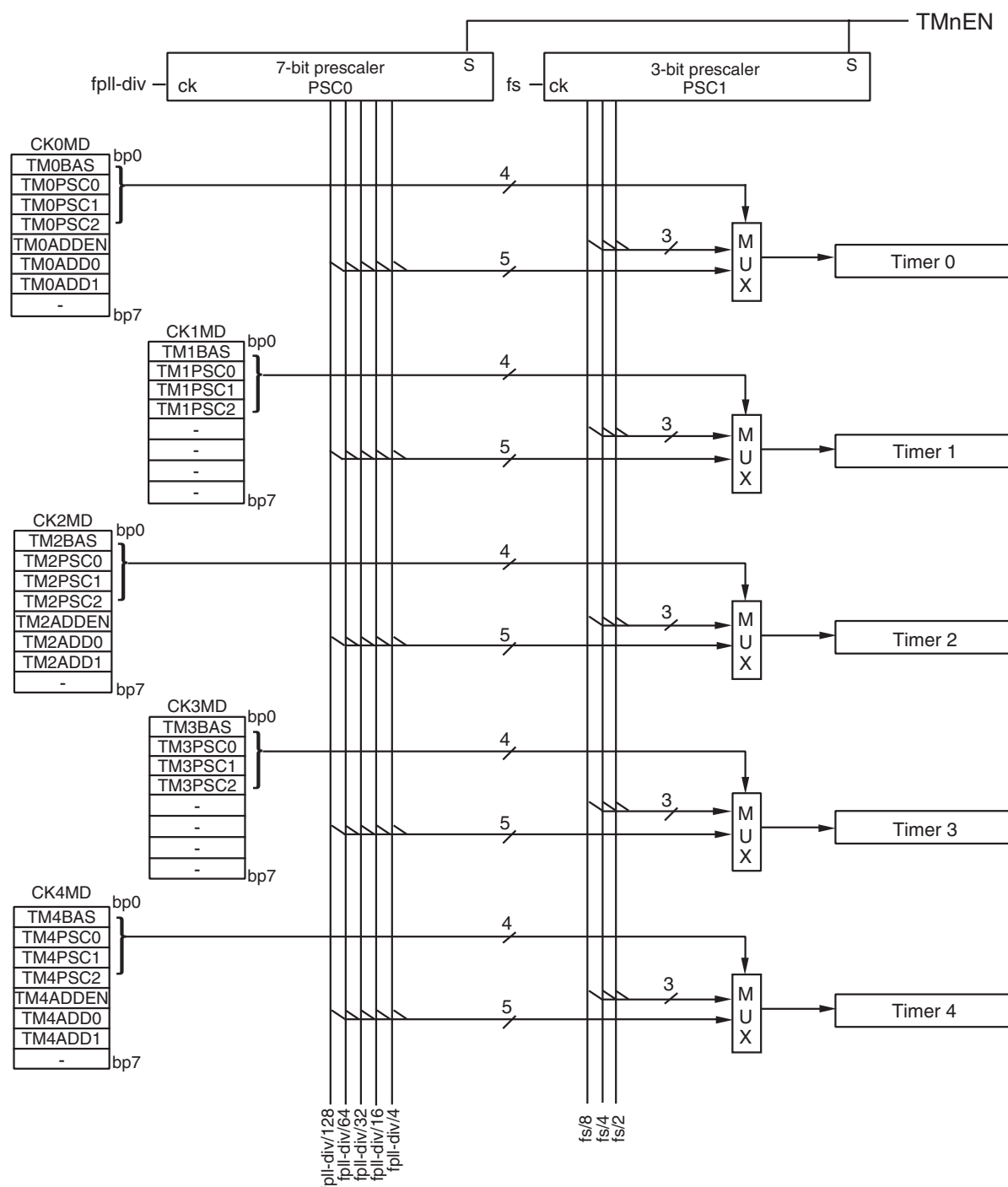


Figure:9.1.1 Prescaler Block Diagram

■ Timer 0 and Timer 1 Block Diagram

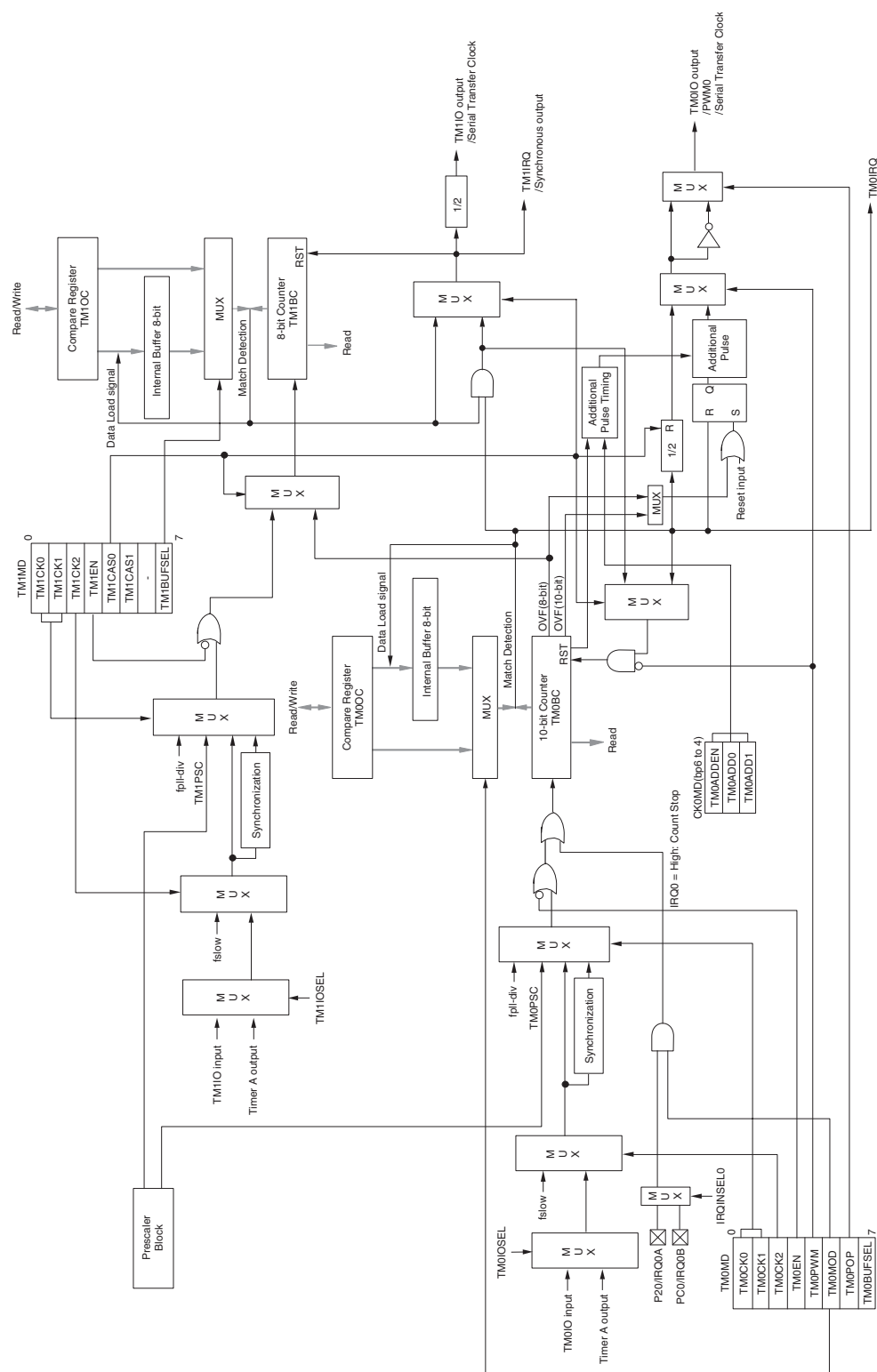


Figure:9.1.2 Timer 0 and Timer 1 Block Diagram (MN101EF76K)

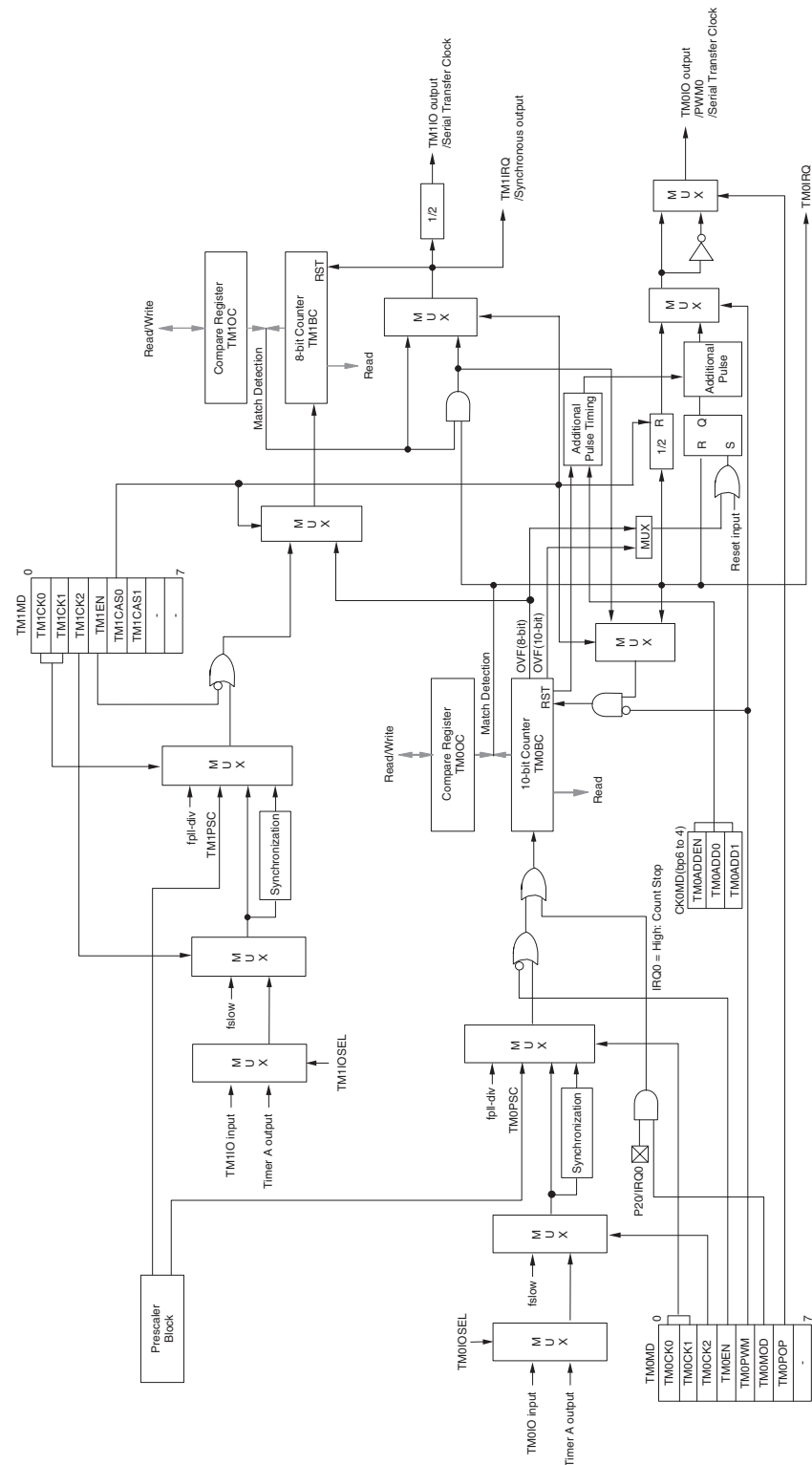


Figure:9.1.3 Timer 0 and Timer 1 Block Diagram (MN101EF57G)

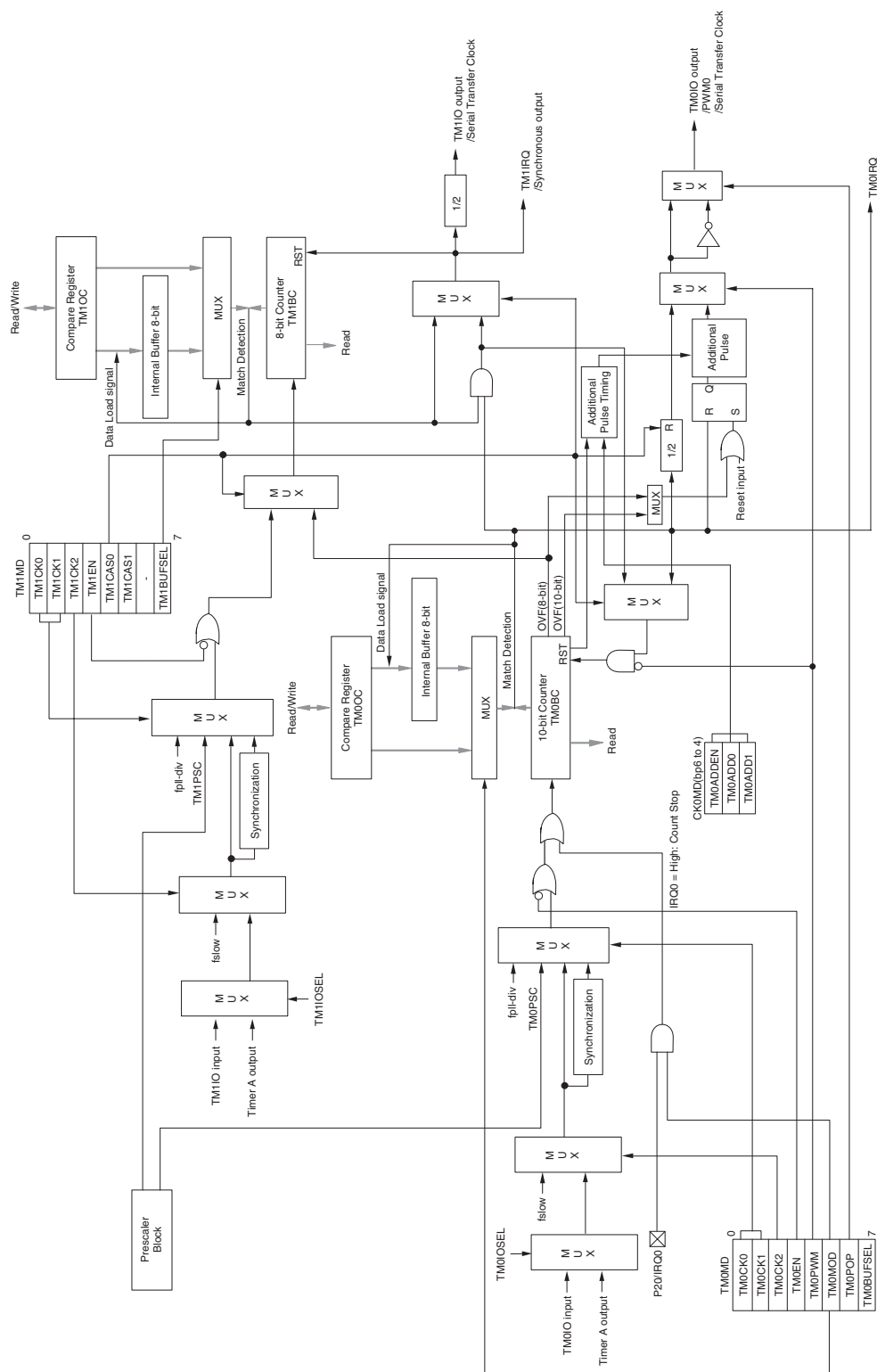


Figure:9.1.4 Timer 0 and Timer 1 Block Diagram (MN101EF56K)

■ Timer 2 and Timer 3 Block Diagram

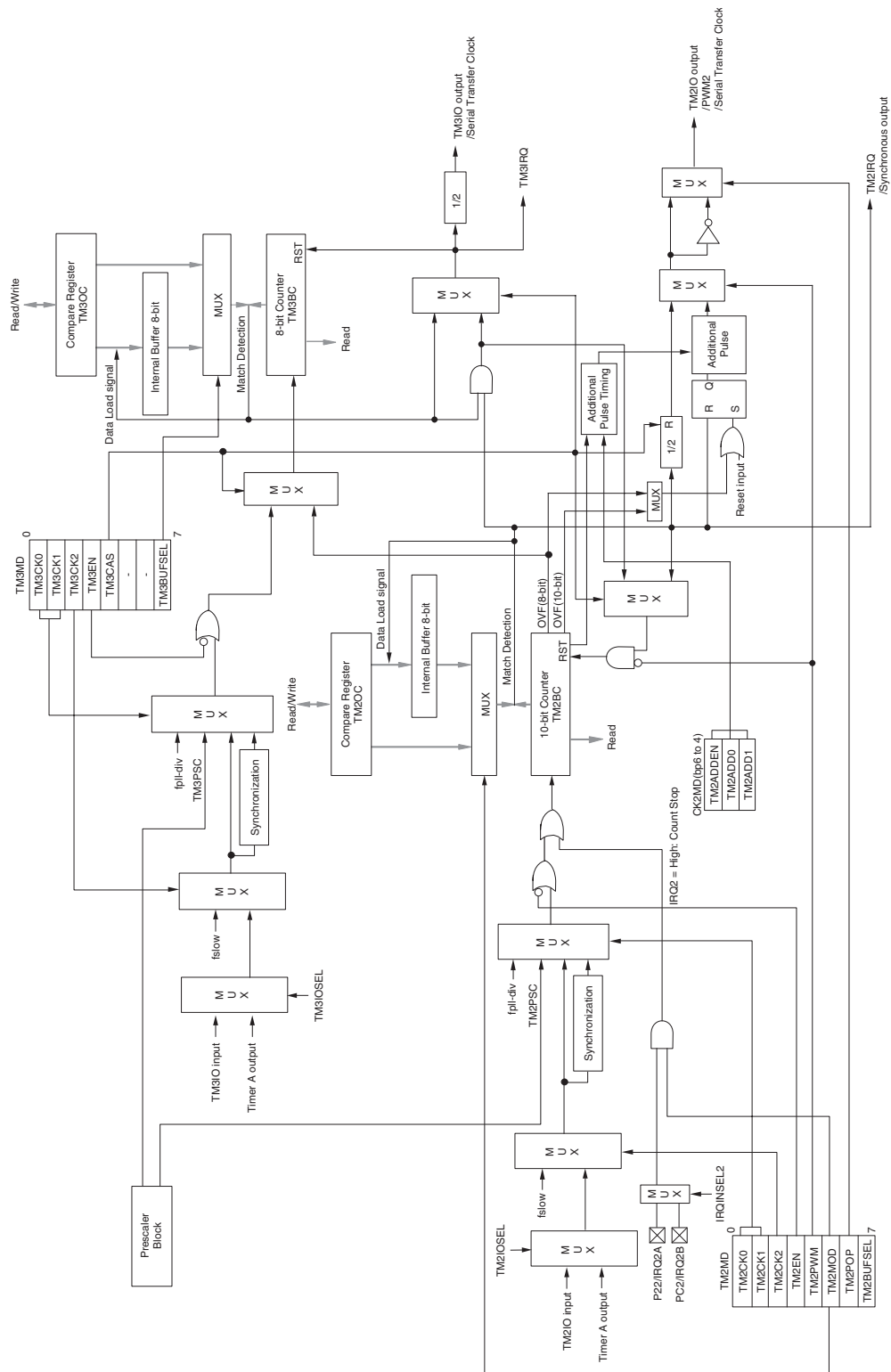


Figure:9.1.5 Timer 2 and Timer 3 Block Diagram (MN101EF76K)

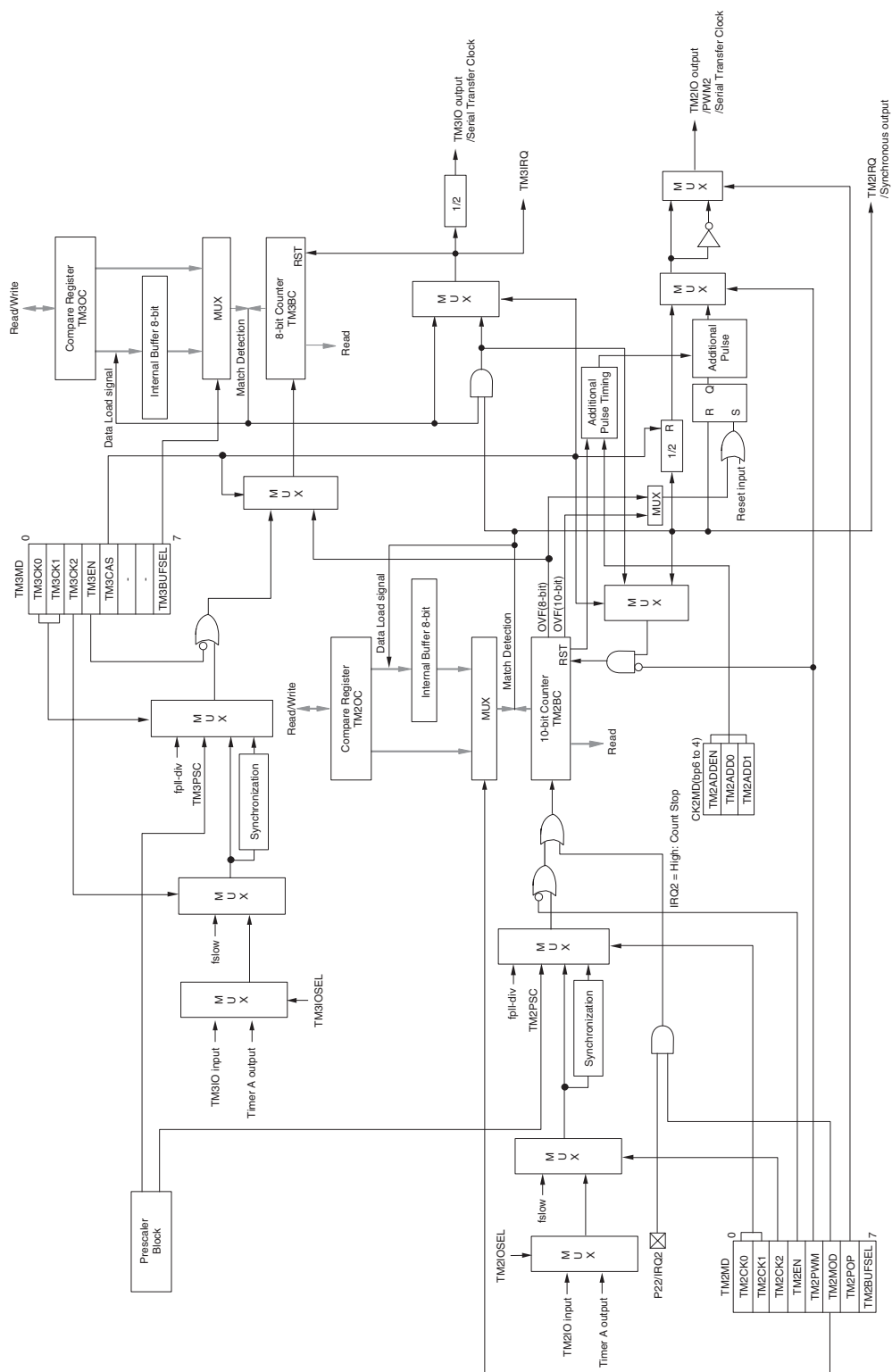


Figure:9.1.6 Timer 2 and Timer 3 Block Diagram (MN101EF57G/MN101EF56K)

■ Timer 4 Block Diagram

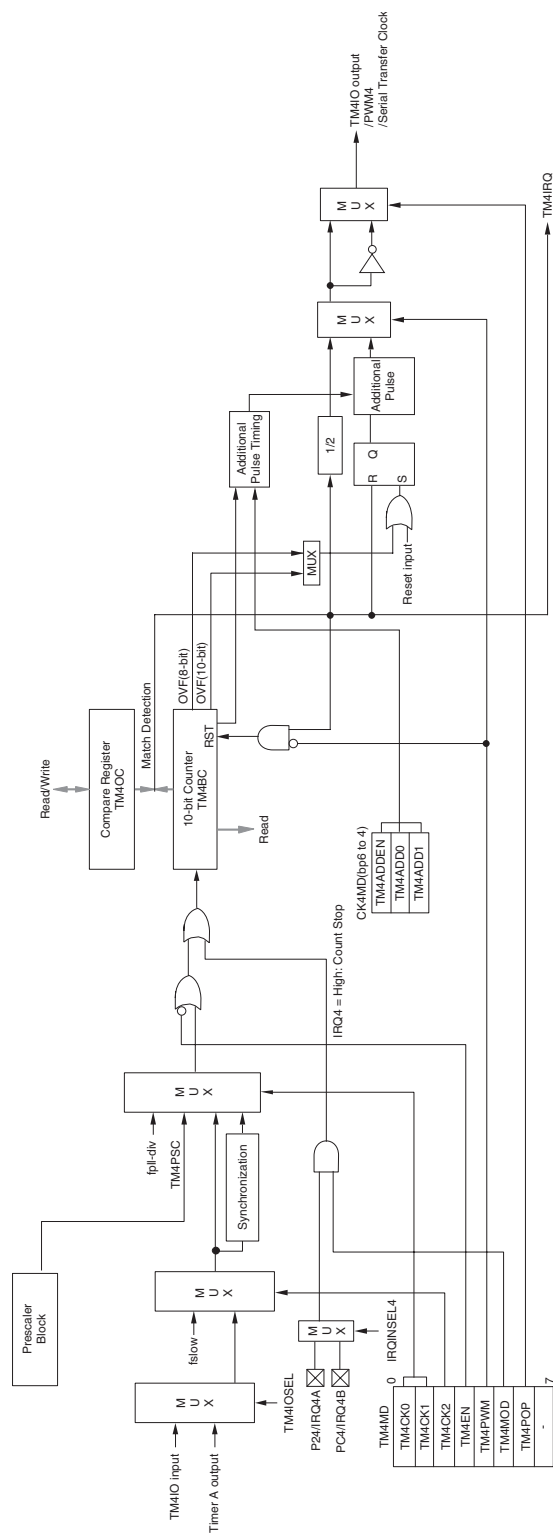


Figure:9.1.7 Timer 4 Block Diagram (MN101EF76K)

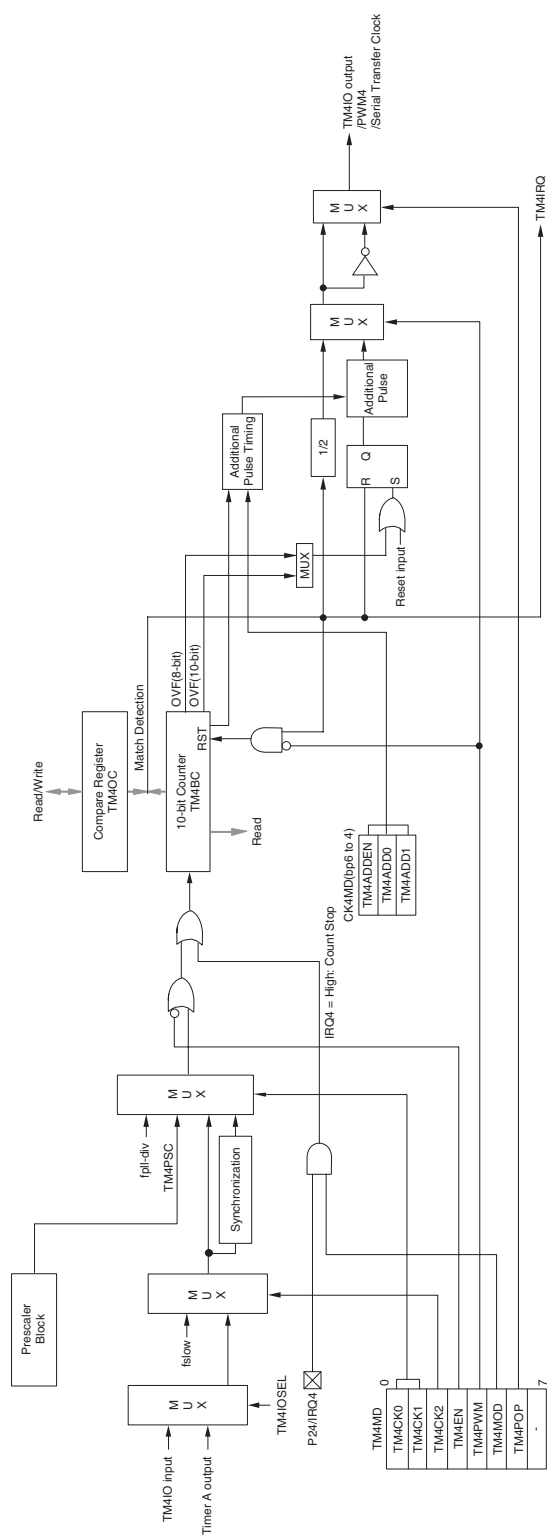


Figure:9.1.8 Timer 4 Block Diagram (MN101EF57G/MN101EF56K)

9.2 Control Registers

Timer 0 to 4 consist of binary counter (TMnBC) and compare register (TMnOC). And they are controlled by mode register (TMnMD).

When the prescaler output is selected as the count clock source for Timer 0 to 4, they should be controlled by prescaler selection register (CKnMD).

9.2.1 Registers

Table:9.2.1 shows registers that control Timer 0 to 4.

Table:9.2.1 8-bit Timer Control Registers

	Register	Address	R/W	Function	Page
Timer 0	TM0BC	0x03F60	R	Timer 0 binary counter	IX-22
	TM0OC	0x03F62	R/W	Timer 0 compare register	IX-20
	TM0MD	0x03F64	R/W	Timer 0 mode register	IX-25
	CK0MD	0x03F66	R/W	Timer 0 prescaler selection register	IX-17
	TM0ICR	0x03FE9	R/W	Timer 0 interrupt control register	VII-27
Timer 1	TM1BC	0x03F61	R	Timer 1 binary counter	IX-22
	TM1OC	0x03F63	R/W	Timer 1 compare register	IX-20
	TM1MD	0x03F65	R/W	Timer 1 mode register	IX-27
	CK1MD	0x03F67	R/W	Timer 1 prescaler selection register	IX-18
	TM1ICR	0x03FEA	R/W	Timer 1 interrupt control register	VII-27
Timer 2	TM2BC	0x03F68	R	Timer 2 binary counter	IX-22
	TM2OC	0x03F6A	R/W	Timer 2 compare register	IX-20
	TM2MD	0x03F6C	R/W	Timer 2 mode register	IX-28
	CK2MD	0x03F6E	R/W	Timer 2 prescaler selection register	IX-18
	TM2ICR	0x03FEB	R/W	Timer 2 interrupt control register	VII-27
Timer 3	TM3BC	0x03F69	R	Timer 3 binary counter	IX-22
	TM3OC	0x03F6B	R/W	Timer 3 compare register	IX-21
	TM3MD	0x03F6D	R/W	Timer 3 mode register	IX-30
	CK2MD	0x03F6F	R/W	Timer 3 prescaler selection register	IX-19
	TM3ICR	0x03FEC	R/W	Timer 3 interrupt control register	VII-27
Timer 4	TM4BC	0x03F70	R	Timer 4 binary counter	IX-22
	TM4OC	0x03F72	R/W	Timer 4 compare register	IX-21
	TM4MD	0x03F74	R/W	Timer 4 mode register	IX-31
	CK2MD	0x03F76	R/W	Timer 4 prescaler selection register	IX-19
	TM4ICR	0x03FED	R/W	Timer 4 interrupt control register	VII-27
	TMCKSEL1	0x03FB0	R/W	Timer clock selection register 1	IX-32
	TMINSEL1	0x03FB2	R/W	Timer input selection register 1	IX-34
	TMINSEL2	0x03FB3	R/W	Timer input selection register 2	IX-36

R/W: Readable / Writable

R: Read only

9.2.2 Timer Prescaler Registers

Timer prescaler selection register selects the count clock for 8-bit timer.

The register which selects prescaler output is included in timer prescaler selection register (CKnMD).

■ Timer 0 Prescaler Selection Register (CK0MD: 0x03F66)

bp	7	6	5	4	3	2	1	0
Flag	-	TM0ADD1	TM0ADD0	TM0ADDEN	TM0PSC2	TM0PSC1	TM0PSC0	TM0BAS
At reset	-	0	0	0	0	0	0	0
Access	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7	-	-
6 to 5	TM0ADD1 TM0ADD0	Positions of additional pulses (within 4 cycles of PWM basic waveform) 00: No pulse 01: At second cycle 10: At first and third cycle 11: At first, second and third cycle
4	TM0ADDEN	PWM output control with additional pulses 0: Prohibited (8-bit PWM output) 1: Permitted
3 to 0	TM0PSC2 TM0PSC1 TM0PSC0 TM0BAS	Clock source selection 0000: fpll-div/4 0010: fpll-div/16 0100: fpll-div/32 0110: fpll-div/64 1XX0: fpll-div/128 0X01: fs/2 0X11: fs/4 1XX1: fs/8

■ Timer 1 Prescaler Selection Register (CK1MD: 0x03F67)

bp	7	6	5	4	3	2	1	0
Flag	-	-	-	-	TM1PSC2	TM1PSC1	TM1PSC0	TM1BAS
At reset	-	-	-	-	0	0	0	0
Access	-	-	-	-	R/W	R/W	R/W	R/W

bp	Flag	Description
7 to 4	-	-
3 to 0	TM1PSC2 TM1PSC1 TM1PSC0 TM1BAS	Clock source selection 0000: fpll-div/4 0010: fpll-div/16 0100: fpll-div/64 0110: fpll-div/128 1XX0: fpll-div/32 0X01: fs/2 0X11: fs/8 1XX1: fs/4

■ Timer 2 Prescaler Selection Register (CK2MD: 0x03F6E)

bp	7	6	5	4	3	2	1	0
Flag	-	TM2ADD1	TM2ADD0	TM2ADDEN	TM2PSC2	TM2PSC1	TM2PSC0	TM2BAS
At reset	-	0	0	0	0	0	0	0
Access	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7	-	-
6 to 5	TM2ADD1 TM2ADD0	Positions of additional pulses (within 4 cycles of PWM basic waveform) 00: No pulse 01: At second cycle 10: At first and third cycle 11: At first, second and third cycle
4	TM2ADDEN	PWM output control with additional pulses 0: Prohibited (8-bit PWM output) 1: Permitted
3 to 0	TM2PSC2 TM2PSC1 TM2PSC0 TM2BAS	Clock source selection 0000: fpll-div/4 0010: fpll-div/16 0100: fpll-div/32 0110: fpll-div/64 1XX0: fpll-div/128 0X01: fs/2 0X11: fs/4 1XX1: fs/8

■ Timer 3 Prescaler Selection Register (CK3MD: 0x03F6F)

bp	7	6	5	4	3	2	1	0
Flag	-	-	-	-	TM3PSC2	TM3PSC1	TM3PSC0	TM3BAS
At reset	-	-	-	-	0	0	0	0
Access	-	-	-	-	R/W	R/W	R/W	R/W

bp	Flag	Description
7 to 4	-	-
3 to 0	TM3PSC2 TM3PSC1 TM3PSC0 TM3BAS	Clock source selection 0000: fpll-div/4 0010: fpll-div/16 0100: fpll-div/64 0110: fpll-div/128 1XX0: fpll-div/32 0X01: fs/2 0X11: fs/8 1XX1: fs/4

■ Timer 4 Prescaler Selection Register (CK4MD: 0x03F76)

bp	7	6	5	4	3	2	1	0
Flag	-	TM4ADD1	TM4ADD0	TM4ADDEN	TM4PSC2	TM4PSC1	TM4PSC0	TM4BAS
At reset	-	0	0	0	0	0	0	0
Access	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7	-	-
6 to 5	TM4ADD1 TM4ADD0	Positions of additional pulses (within 4 cycles of PWM basic waveform) 00: No pulse 01: At second cycle 10: At first and third cycle 11: At first, second and third cycle
4	TM4ADDEN	PWM output control with additional pulses 0: Prohibited (8-bit PWM output) 1: Permitted
3 to 0	TM4PSC2 TM4PSC1 TM4PSC0 TM4BAS	Clock source selection 0000: fpll-div/4 0010: fpll-div/16 0100: fpll-div/32 0110: fpll-div/64 1XX0: fpll-div/128 0X01: fs/2 0X11: fs/4 1XX1: fs/8

9.2.3 Programmable Timer Registers

Each of Timer 0 to 4 has 8-bit programmable timer registers.

Programmable timer register consists of compare register and binary counter.

Compare register is an 8-bit register which stores values to be compared with binary counter.

Timer compare register is double-buffer configuration by enabling the internal register when TM0BUFSEL, TM1BUFSEL, TM2BUFSEL and TM3BUFSEL of Timer 0, 1, 2 and 3 mode register are "1". In double-buffer configuration, comparison with the binary counter is operated by the internal register.

Compare register is changed while the counter is still counting.

However, the value in compare register only is loaded to the internal buffer when the counter is cleared after it reached the value stored in the internal buffer when compare register is selected as double-buffer configuration and in counting operation.

The internal register can not be read and written.

Table:9.2.2 Double Buffer Functions

Table remarks ✓: With function -: Without function

	Timer	MN101EF76K	MN101EF57G	MN101EF56K
Double buffer functions of compare register	Timer 0	✓	-	✓
	Timer 1	✓	-	✓
	Timer 2	✓	✓	✓
	Timer 3	✓	✓	✓

■ Timer 0 Compare Register (TM0OC: 0x03F62)

bp	7	6	5	4	3	2	1	0
Flag	TM0OC7	TM0OC6	TM0OC5	TM0OC4	TM0OC3	TM0OC2	TM0OC1	TM0OC0
At reset	X	X	X	X	X	X	X	X
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

■ Timer 1 Compare Register (TM1OC: 0x03F63)

bp	7	6	5	4	3	2	1	0
Flag	TM1OC7	TM1OC6	TM1OC5	TM1OC4	TM1OC3	TM1OC2	TM1OC1	TM1OC0
At reset	X	X	X	X	X	X	X	X
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

■ Timer 2 Compare Register (TM2OC: 0x03F6A)

bp	7	6	5	4	3	2	1	0
Flag	TM2OC7	TM2OC6	TM2OC5	TM2OC4	TM2OC3	TM2OC2	TM2OC1	TM2OC0
At reset	X	X	X	X	X	X	X	X
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

■ Timer 3 Compare Register (TM3OC: 0x03F6B)

bp	7	6	5	4	3	2	1	0
Flag	TM3OC7	TM3OC6	TM3OC5	TM3OC4	TM3OC3	TM3OC2	TM3OC1	TM3OC0
At reset	X	X	X	X	X	X	X	X
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

■ Timer 4 Compare Register (TM4OC: 0x03F72)

bp	7	6	5	4	3	2	1	0
Flag	TM4OC7	TM4OC6	TM4OC5	TM4OC4	TM4OC3	TM4OC2	TM4OC1	TM4OC0
At reset	X	X	X	X	X	X	X	X
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W



In double buffer configuration, when the clear timing of binary counter occurs at the same time as the write timing of compare register, the correct data may not be loaded to internal register.

Ensure the timing margin for rewriting the value of compare register to avoid the occurring of write timing and clear timing at the same time.



Under the following conditions, be sure to select a system clock (fs) as a clock source of timers to set compare register of Timer n.

<Conditions applicable to this note>

When one of the following conditions is met while Timer n is halted and double-buffer function is also selected:

Condition 1	Relevant Block	TMnMD (Timer n mode register)	The frequency of system clock (fs) and high-speed clock for peripheral functions (fppll-div)
		TMnCK2 to 0	
	Timer n	X00 (clock source: fppll-div)	The frequency of a system clock (fs) is faster than twice the frequency of a high-speed clock for peripheral functions (fppll-div).
Condition 2	Relevant Block	TMnMD (Timer n mode register)	CPUM (CPU mode control register)
		TMnCK2 to 0	STOP/HALT/OSC1/OSC0
	Timer n	010 (clock source: fslow)	0000 (Normal mode)
Condition 3	Relevant Block	TMnMD (Timer n mode register)	
		TMnCK2 to 0	
	Timer n	110 (clock source: TMnIO input)	

Binary counter is an 8-bit up counter. If any data is written to compare register when the counter is stopped, the binary counter is cleared to "0x00".

■ Timer 0 Binary Counter (TM0BC: 0x03F60)

bp	7	6	5	4	3	2	1	0
Flag	TM0BC7	TM0BC6	TM0BC5	TM0BC4	TM0BC3	TM0BC2	TM0BC1	TM0BC0
At reset	0	0	0	0	0	0	0	0
Access	R	R	R	R	R	R	R	R

■ Timer 1 Binary Counter (TM1BC: 0x03F61)

bp	7	6	5	4	3	2	1	0
Flag	TM1BC7	TM1BC6	TM1BC5	TM1BC4	TM1BC3	TM1BC2	TM1BC1	TM1BC0
At reset	0	0	0	0	0	0	0	0
Access	R	R	R	R	R	R	R	R

■ Timer 2 Binary Counter (TM2BC: 0x03F68)

bp	7	6	5	4	3	2	1	0
Flag	TM2BC7	TM2BC6	TM2BC5	TM2BC4	TM2BC3	TM2BC2	TM2BC1	TM2BC0
At reset	0	0	0	0	0	0	0	0
Access	R	R	R	R	R	R	R	R

■ Timer 3 Binary Counter (TM3BC: 0x03F69)

bp	7	6	5	4	3	2	1	0
Flag	TM3BC7	TM3BC6	TM3BC5	TM3BC4	TM3BC3	TM3BC2	TM3BC1	TM3BC0
At reset	0	0	0	0	0	0	0	0
Access	R	R	R	R	R	R	R	R

■ Timer 4 Binary Counter (TM4BC: 0x03F70)

bp	7	6	5	4	3	2	1	0
Flag	TM4BC7	TM4BC6	TM4BC5	TM4BC4	TM4BC3	TM4BC2	TM4BC1	TM4BC0
At reset	0	0	0	0	0	0	0	0
Access	R	R	R	R	R	R	R	R

9.2.4 Timer Mode Registers

Timer mode register is readable/writable register that controls Timer 0 to 4.

■ Timer 0 Mode Register (TM0MD: 0x03F64)

<MN101EF76K/MN101EF56K>

bp	7	6	5	4	3	2	1	0
Flag	TM0BUFSEL	TM0POP	TM0MOD	TM0PWM	TM0EN	TM0CK2	TM0CK1	TM0CK0
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7	TM0BUFSEL	Timer 0 compare register buffer selection 0: Single buffer 1: Double buffer
6	TM0POP	Output signal start polarity selection 0: Timer output Low→High, PWM High→Low 1: Timer output High→Low, PWM Low→High
5	TM0MOD	Pulse width measurement control 0: Normal timer operation 1: P20 pulse width measurement
4	TM0PWM	Timer 0 operation mode selection 0: Normal timer operation 1: PWM operation
3	TM0EN	Timer 0 count control 0: Halt the count 1: Operate the count
2 to 0	TM0CK2 TM0CK1 TM0CK0	Clock source selection X00: fpll-div X01: TM0PSC (prescaler output) 010: fslow 011: Synchronou fslow 110: TM0IO input 111: Synchronous TM0IO output



To load data from a compare register to an internal register while the counter is stopped, one count clock is required. The count clock should be input externally when the timer input is selected as the clock source.



Set both TM0BUFSEL and TM1BUFSEL to "1", when double buffering in 16-bit cascade connection of Timer 0 and Timer 1.

Set both TM2BUFSEL and TM3BUFSEL to "1", when double buffering in 16-bit cascade connection of Timer 2 and Timer 3.

Set each flag of TM0BUFSEL, TM1BUFSEL and TM2BUFSEL to "1", when double buffering in 24-bit cascade connection of Timer 0, Timer 1 and Timer 2.

Set each flag of TM0BUFSEL, TM1BUFSEL, TM2BUFSEL and TM3BUFSEL to "1", when double buffering in 32-bit cascade connection of Timer 0, Timer 1, Timer 2 and Timer 3.

■ Timer 0 Mode Register (TM0MD: 0x03F64)

<MN101EF57G>

bp	7	6	5	4	3	2	1	0
Flag	-	TM0POP	TM0MOD	TM0PWM	TM0EN	TM0CK2	TM0CK1	TM0CK0
At reset	-	0	0	0	0	0	0	0
Access	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7	-	-
6	TM0POP	Output signal start polarity selection 0: Timer output Low→High, PWM High→Low 1: Timer output High→Low, PWM Low→High
5	TM0MOD	Pulse width measurement control 0: Normal timer operation 1: P20 pulse width measurement
4	TM0PWM	Timer 0 operation mode selection 0: Normal timer operation 1: PWM operation
3	TM0EN	Timer 0 count control 0: Halt the count 1: Operate the count
2 to 0	TM0CK2 TM0CK1 TM0CK0	Clock source selection X00: fpll-div X01: TM0PSC (prescaler output) 010: fslow 011: Synchronous fslow 110: TM0IO input 111: Synchronous TM0IO output



To load data from a compare register to an internal register while the counter is stopped, one count clock is required. The count clock should be input externally when the timer input is selected as the clock source.

■ Timer 1 Mode Register (TM1MD: 0x03F65)

<MN101EF76K/MN101EF56K>

bp	7	6	5	4	3	2	1	0
Flag	TM1BUFSEL	Reserved	TM1CAS1	TM1CAS0	TM1EN	TM1CK2	TM1CK1	TM1CK0
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7	TM1BUFSEL	Timer 1 compare register buffer selection 0: Single buffer 1: Double buffer
6	Reserved	Always set to "0".
5 to 4	TM1CAS1 TM1CAS0	Operation mode of Timer 1 selection 00: Normal operation 01: 16-bit cascade connection 10: 24-bit cascade connection 11: 32-bit cascade connection
3	TM1EN	Timer 1 count control 0: Halt the count 1: Operate the count
2 to 0	TM1CK2 TM1CK1 TM1CK0	Clock source selection X00: fpll-div X01: TM1PSC (prescaler output) 010: fslow 011: Synchronous fslow 110: TM1IO input 111: Synchronous TM0IO input



To load data from a compare register to an internal register while the counter is stopped, one count clock is required. The count clock should be input externally when the timer input is selected as the clock source.



Set both TM0BUFSEL and TM1BUFSEL to "1", when double buffering in 16-bit cascade connection of Timer 0 and Timer 1.
Set both TM2BUFSEL and TM3BUFSEL to "1", when double buffering in 16-bit cascade connection of Timer 2 and Timer 3.
Set each flag of TM0BUFSEL, TM1BUFSEL and TM2BUFSEL to "1", when double buffering in 24-bit cascade connection of Timer 0, Timer 1 and Timer 2.
Set each flag of TM0BUFSEL, TM1BUFSEL, TM2BUFSEL and TM3BUFSEL to "1", when double buffering in 32-bit cascade connection of Timer 0, Timer 1, Timer 2 and Timer 3.

■ Timer 1 Mode Register (TM1MD: 0x03F65)

<MN101EF57G>

bp	7	6	5	4	3	2	1	0
Flag	Reserved	Reserved	TM1CAS1	TM1CAS0	TM1EN	TM1CK2	TM1CK1	TM1CK0
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7 to 6	Reserved	Always set to "0".
5 to 4	TM1CAS1 TM1CAS0	Operation mode of Timer 1 selection 00: Normal operation 01: 16-bit cascade connection 10: 24-bit cascade connection 11: 32-bit cascade connection
3	TM1EN	Timer 1 count control 0: Halt the count 1: Operate the count
2 to 0	TM1CK2 TM1CK1 TM1CK0	Clock source selection X00: fpll-div X01: TM1PSC (prescaler output) 010: fslow 011: Synchronous fslow 110: TM1IO input 111: Synchronous TM0IO input



To load data from a compare register to an internal register while the counter is stopped, one count clock is required. The count clock should be input externally when the timer input is selected as the clock source.

■ Timer 2 Mode Register (TM2MD: 0x03F6C)

bp	7	6	5	4	3	2	1	0
Flag	TM2BUFSEL	TM2POP	TM2MOD	TM2PWM	TM2EN	TM2CK2	TM2CK1	TM2CK0
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7	TM2BUFSEL	Timer 2 compare register buffer selection 0: Single buffer 1: Double buffer
6	TM2POP	Output signal start polarity selection 0: Timer output Low→High, PWM High→Low 1: Timer output High→Low, PWM Low→High
5	TM2MOD	Pulse width measurement control 0: Normal timer operation 1: P22 pulse width measurement
4	TM2PWM	Timer 2 operation mode selection 0: Normal timer operation 1: PWM operation
3	TM2EN	Timer 2 count control 0: Halt the count 1: Operate the count
2 to 0	TM2CK2 TM2CK1 TM2CK0	Clock source selection X00: fpll-div X01: TM2PSC (prescaler output) 010: fslow 011: Synchronou fslow 110: TM2IO input 111: Synchronous TM2IO output



To load data from a compare register to an internal register while the counter is stopped, one count clock is required. The count clock should be input externally when the timer input is selected as the clock source.



<MN101EF76K/MN101EF56K>

Set both TM0BUFSEL and TM1BUFSEL to "1", when double buffering in 16-bit cascade connection of Timer 0 and Timer 1.

Set both TM2BUFSEL and TM3BUFSEL to "1", when double buffering in 16-bit cascade connection of Timer 2 and Timer 3.

Set each flag of TM0BUFSEL, TM1BUFSEL and TM2BUFSEL to "1", when double buffering in 24-bit cascade connection of Timer 0, Timer 1 and Timer 2.

Set each flag of TM0BUFSEL, TM1BUFSEL, TM2BUFSEL and TM3BUFSEL to "1", when double buffering in 32-bit cascade connection of Timer 0, Timer 1, Timer 2 and Timer 3.



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Set both TM2BUFSEL and TM3BUFSEL to "1", when double buffering in cascade connection of Timer 2 and Timer 3.

Setting TM2BUFSEL and TM3BUFSEL to "1" is disabled in 24-bit or 32-bit cascade connection.

■ Timer 3 Mode Register (TM3MD: 0x03F6D)

bp	7	6	5	4	3	2	1	0
Flag	TM3BUFSEL	-	-	TM3CAS	TM3EN	TM3CK2	TM3CK1	TM3CK0
At reset	0	-	-	0	0	0	0	0
Access	R/W	-	-	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7	TM3BUFSEL	Timer 3 compare register buffer selection 0: Single buffer 1: Double buffer
6 to 5	-	-
4	TM3CAS	Operation mode of Timer 3 selection 0: Normal timer operation 1: PWM operation
3	TM3EN	Timer 3 count control 0: Halt the count 1: Operate the count
2 to 0	TM3CK2 TM3CK1 TM3CK0	Clock source selection X00: fpll-div X01: TM3PSC (prescaler output) 010: fslow 011: Synchronous fslow 110: TM3IO input 111: Synchronous TM3IO output



To load data from a compare register to an internal register while the counter is stopped, one count clock is required. The count clock should be input externally when the timer input is selected as the clock source.



<MN101EF76K/MN101EF56K>

Set both TM0BUFSEL and TM1BUFSEL to "1", when double buffering in 16-bit cascade connection of Timer 0 and Timer 1.

Set both TM2BUFSEL and TM3BUFSEL to "1", when double buffering in 16-bit cascade connection of Timer 2 and Timer 3.

Set each flag of TM0BUFSEL, TM1BUFSEL and TM2BUFSEL to "1", when double buffering in 24-bit cascade connection of Timer 0, Timer 1 and Timer 2.

Set each flag of TM0BUFSEL, TM1BUFSEL, TM2BUFSEL and TM3BUFSEL to "1", when double buffering in 32-bit cascade connection of Timer 0, Timer 1, Timer 2 and Timer 3.



<MN101EF57G>

Set both TM2BUFSEL and TM3BUFSEL to "1", when double buffering in cascade connection of Timer 2 and Timer 3.

Setting TM2BUFSEL and TM3BUFSEL to "1" is disabled in 24-bit or 32-bit cascade connection.

■ Timer 4 Mode Register (TM4MD: 0x03F74)

bp	7	6	5	4	3	2	1	0
Flag	-	TM4POP	TM4MOD	TM4PWM	TM4EN	TM4CK2	TM4CK1	TM4CK0
At reset	-	0	0	0	0	0	0	0
Access	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7	-	-
6	TM4POP	Output signal start polarity selection 0: Timer output Low→High, PWM High→Low 1: Timer output High→Low, PWM Low→High
5	TM4MOD	Pulse width measurement control 0: Normal timer operation 1: P24 pulse width measurement
4	TM4PWM	Timer 4 operation mode selection 0: Normal timer operation 1: PWM operation
3	TM4EN	Timer 4 count control 0: Halt the count 1: Operate the count
2 to 0	TM4CK2 TM4CK1 TM4CK0	Clock source selection X00: fpll-div X01: TM4PSC (prescaler output) 010: fslow 011: Synchronou fslow 110: TM4IO input 111: Synchronous TM2IO output

■ Timer Clock Selection Register 1 (TMCKSEL1: 0x03FB0)

bp	7	6	5	4	3	2	1	0
Flag	TM6CKSEL	-	-	TM4IOSEL	TM3IOSEL	TM2IOSEL	TM1IOSEL	TM0IOSEL
At reset	0	-	-	0	0	0	0	0
Access	R/W	-	-	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7	TM6CKSEL	Selection of Time Base Timer clock dividing ratio 0: Time base timer clock $\times (1/2^{13}$ or $1/2^{12}$) 1: Time base timer clock $\times (1/2^2$ or $1/2^3)$
6 to 5	-	-
4	TM4IOSEL	Timer 4 input selection 0: TM4IO 1: Timer A
3	TM3IOSEL	Timer 3 input selection 0: TM3IO 1: Timer A
2	TM2IOSEL	Timer 2 input selection 0: TM2IO 1: Timer A
1	TM1IOSEL	Timer 1 input selection 0: TM1IO 1: Timer A
0	TM0IOSEL	Timer 0 input selection 0: TM0IO 1: Timer A

■ Timer Input Selection Register 1 (TMINSEL1: 0x03FB2)

<MN101EF76K/MN101EF56K>

bp	7	6	5	4	3	2	1	0
Flag	TMINSEL 17	TMINSEL 16	TMINSEL 15	TMINSEL 14	TMINSEL 13	TMINSEL 12	TMINSEL 11	TMINSEL 10
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7 to 6	TMINSEL17 TMINSEL16	Timer 3 input selection (at port selection) 00: TM3IOA(PA3) 01: TM3IOB(P63) 10: TM3IOC(P13) 11: "1" input to timer
5 to 4	TMINSEL15 TMINSEL14	Timer 2 input selection (at port selection) 00: TM2IOA(PA2) 01: TM2IOB(P04) 10: TM2IOC(P11) 11: "0" input to timer
3 to 2	TMINSEL13 TMINSEL12	Timer 1 input selection (at port selection) 00: TM1IOA(PA1) 01: TM1IOB(P62) 10: TM1IOC(P12) 11: "0" input to timer
1 to 0	TMINSEL11 TMINSEL10	Timer 0 input selection (at port selection) 00: TM0IOA(PA0) 01: TM0IOB(P04) 10: TM0IOC(P10) 11: "1" input to timer

■ Timer Input Selection Register 1 (TMINSEL1: 0x03FB2)

<MN101EF57G>

bp	7	6	5	4	3	2	1	0
Flag	TMINSEL17	TMINSEL16	TMINSEL15	TMINSEL14	TMINSEL13	TMINSEL12	TMINSEL11	TMINSEL10
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7 to 6	TMINSEL17 TMINSEL16	Timer 3 input selection (at port selection) 00: TM3IOA(PA3) 01: TM3IOB(P63) 10: Setting prohibited 11: "1" input to timer
5 to 4	TMINSEL15 TMINSEL14	Timer 2 input selection (at port selection) 00: TM2IOA(PA2) 01: TM2IOB(P04) 10: Setting prohibited 11: "0" input to timer
3 to 2	TMINSEL13 TMINSEL12	Timer 1 input selection (at port selection) 00: TM1IOA(PA1) 01: TM1IOB(P62) 10: Setting prohibited 11: "0" input to timer
1 to 0	TMINSEL11 TMINSEL10	Timer 0 input selection (at port selection) 00: TM0IOA(PA0) 01: TM0IOB(P04) 10: Setting prohibited 11: "1" input to timer

■ Timer Input Selection Register 2 (TMINSEL2: 0x03FB3)

<MN101EF76K/MN101EF56K>

bp	7	6	5	4	3	2	1	0
Flag	TMINSEL 27	TMINSEL 26	TMINSEL 25	TMINSEL 24	TMINSEL 23	TMINSEL 22	TMINSEL 21	TMINSEL 20
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7 to 6	TMINSEL27 TMINSEL26	Timer 9 input selection (at port selection) 00: TM9IOA(PA7) 01: TM9IOB(P01) 10: "1" input to timer 11: "1" input to timer
5 to 4	TMINSEL25 TMINSEL24	Timer 8 input selection (at port selection) 00: TM8IOA(PA6) 01: TM8IOB(P03) 10: TM8IOC(P16) 11: "1" input to timer
3 to 2	TMINSEL23 TMINSEL22	Timer 7 input selection (at port selection) 00: TM7IOA(PA5) 01: TM7IOB(P02) 10: TM7IOC(P15) 11: "1" input to timer
1 to 0	TMINSEL21 TMINSEL20	Timer 4 input selection (at port selection) 00: TM4IOA(PA4) 01: TM4IOB(P64) 10: TM4IOC(P14) 11: "0" input to timer

■ Timer Input Selection Register 2 (TMINSEL2: 0x03FB3)

<MN101EF57G>

bp	7	6	5	4	3	2	1	0
Flag	TMINSEL 27	TMINSEL 26	TMINSEL 25	TMINSEL 24	TMINSEL 23	TMINSEL 22	TMINSEL 21	TMINSEL 20
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7 to 6	TMINSEL27 TMINSEL26	Timer 9 input selection (at port selection) 00: TM9IOA(PA7) 01: TM9IOB(P01) 10: "1" input to timer 11: "1" input to timer
5 to 4	TMINSEL25 TMINSEL24	Timer 8 input selection (at port selection) 00: TM8IOA(PA6) 01: TM8IOB(P03) 10: Setting prohibited 11: "1" input to timer
3 to 2	TMINSEL23 TMINSEL22	Timer 7 input selection (at port selection) 00: TM7IOA(PA5) 01: TM7IOB(P02) 10: Setting prohibited 11: "1" input to timer
1 to 0	TMINSEL21 TMINSEL20	Timer 4 input selection (at port selection) 00: TM4IOA(PA4) 01: TM4IOB(P64) 10: Setting prohibited 11: "0" input to timer

9.3 Prescaler

9.3.1 Prescaler Operation

■ Prescaler Operation (Prescaler 0 to 1)

Prescaler 0 and Prescaler 1 are 7-bit and 3-bit free-run counter respectively. They output the dividing clock of the reference clock. This count up operation starts automatically when any TMnEN flags of 8-bit timer are set to "1" and operate the Timer n counting. Also, it stops automatically when all TMnEN flags of 8-bit timer are set to "0" and stop all timer counting.

■ Count Timing of Prescaler Operation (Prescaler 0 to 1)

Prescaler 0 counts up at the rising edge of fp11-div.

Prescaler 1 counts up at the rising edge of fs.

■ Peripheral Functions

Peripheral functions which can use the prescaler output dividing clock, or registers which control the dividing clock selections are shown below.

Timer 0 Count Clock	CK0MD
Timer 1 Count Clock	CK1MD
Timer 2 Count Clock	CK2MD
Timer 3 Count Clock	CK3MD
Timer 4 Count Clock	CK4MD



Start the timer operation after the prescaler setup.

At timer side, the prescaler output should be set up by the timer mode register.

The prescaler starts counting at the start of the timer operation.

9.3.2 Setup Example

■ Prescaler Operation Setup Example

Clock $f_s/2$ which is output from the prescaler 1 is selected as the count clock of the timer 0.

The description below is an setup procedure example.

Setup Procedure	Description
(1) Select the prescaler output CK0MD(0x03F66) bp2 to 1: TM0PSC1 to 0 =X0 bp0: TM0BAS =1	(1) Select “fs/2” to the prescaler output by the TM0PSC1 to 0 and TM0BAS flags of Timer 0 prescaler selection register (CK0MD).

At timer side, prescaler output selection should be set up by the timer mode register.

9.4 8-bit Timer Count

9.4.1 8-bit Timer Operation

Timers have a function to generate interrupts regularly.

■ 8-bit Timer Operation (Timer 0, 1, 2, 3 and 4)

The generation cycle of timer interrupts is set by the clock source selection and the setting value of compare register (TMnOC), in advance. If the binary counter (TMnBC) reaches the setting value of compare register, an interrupt is generated at the next count clock, then binary counter is cleared and counting restarts from "0x00".

The table below shows clock sources that can be selected by timer.

Clock source	Time per Count	Timer 0 (8-bit)	Timer 1 (8-bit)	Timer 2 (8-bit)	Timer 3 (8-bit)	Timer 4 (8-bit)
fppll-div	100 ns	√	√	√	√	√
fppll-div/4	400 ns	√	√	√	√	√
fppll-div/16	1.6 μs	√	√	√	√	√
fppll-div/32	3.2 μs	√	√	√	√	√
fppll-div/64	6.4 μs	√	√	√	√	√
fppll-div/128	12.8 μs	√	√	√	√	√
fs/2	400 ns	√	√	√	√	√
fs/4	800 ns	√	√	√	√	√
fs/8	1.6 μs	√	√	√	√	√
fslow	30.5 μs	√	√	√	√	√
fppll-div = 10 MHz, fslow = 32.768 kHz fs = fppll-div/2 = 5 MHz						



When fslow is used as clock source, timer counter is counted at the falling of the count clock and when others are used, timer counter counted at the rising of the count clock.

When compare register is double-buffered, if the value of compare register is changed during Timer 2, 3 counting operation, the changed value is set to the internal register again with the binary counter timing.

By this function, cycles are not disrupted during timer operation, the compare value setting can be changed continuously. (reload function)

■ Count Timing of Timer Operation (Timer 0, 1, 2, 3 and 4)

Binary counter counts up with a selected clock source as a count clock.
The figure below shows the basic operation of all functions of 8-bit timer.

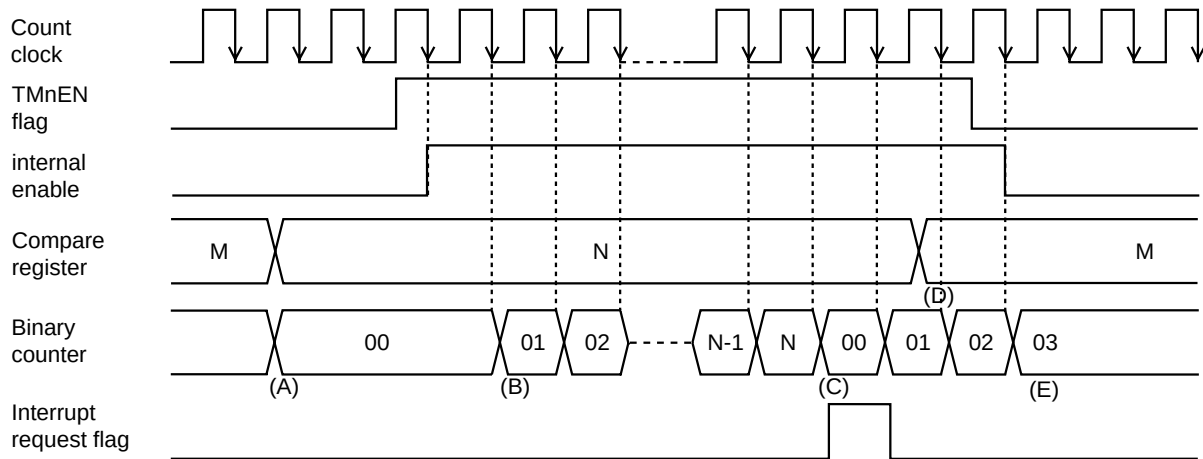


Figure:9.4.1 Count Timing of Timer Operation (Timer 0, 1, 2, 3 and 4)

- (A) If the value is written to compare register when TMnEN flag is stopped ("0"), the binary counter is cleared to 0x00.
- (B) When the TMnEN flag starts to operate ("1"), the Internal Enable will be turned on at the next Count Clock. Then the binary counter begins counting up.
- (C) If the binary counter reaches the value of compare register, the interrupt request flag is set at the next count clock, then the binary counter is cleared to 0x00 and the counting is restarted.
- (D) Even if compare register is rewritten when TMnEN flag is enabled ("1"), the binary counter is not changed.
- (E) When the TMnEN flag stops operating ("0"), the Internal Enable will be turned off at the next Count Clock. As a result, the binary counter stops counting.

■ Count Timing of Timer Operation When Compare Register is Double-buffered (Timer 0, 1, 2 and 3)

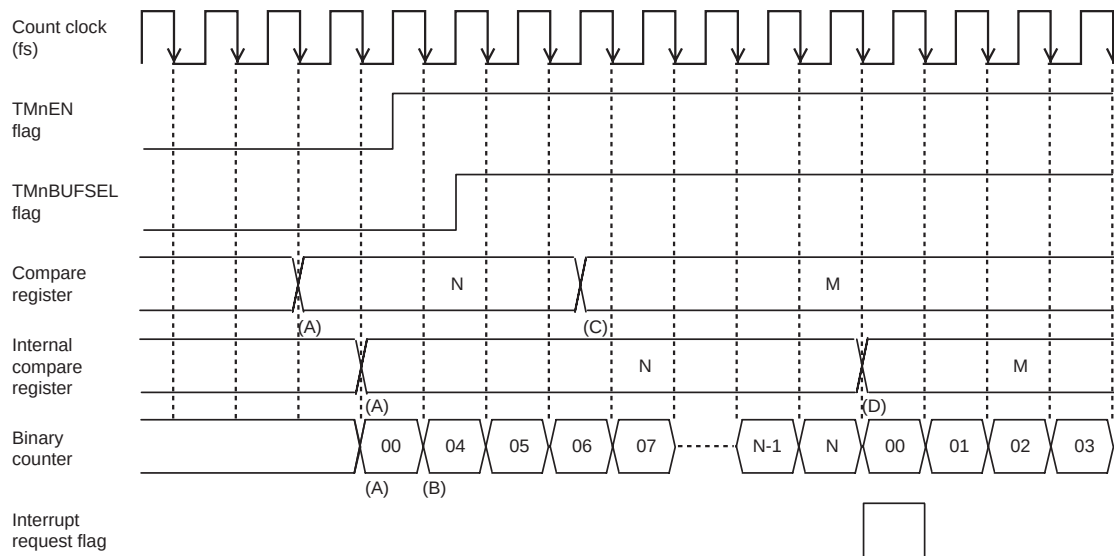


Figure:9.4.2 Count Timing of Timer Operation When compare register is Double-buffered (Timer 0, 1, 2 and 3)

- (A) If the value is written to the preset register when TMnEN flag is "1", the same value is loaded to the internal compare register at the next count clock of this writing cycle. Then the binary counter is cleared to 0x00.
- (B) When the TMnEN flag starts to operate ("1"), the binary counter starts counting. The counting is operated at the rising of the count clock.
- (C) Even if the preset register is rewritten when TMnEN flag is enabled ("1"), the binary counter is not changed.
- (D) If the binary counter reaches the value of the internal compare register, the value of the preset register is loaded again at the next count clock. Also, the interrupt request flag is set, then the binary counter is cleared to 0x00 and the counting is restarted.



MN101EF57G does not have double-buffer function of compare register in Timer 0 and 1.



Switch the count clock after the timer operation is stopped, since the counting is not generated correctly during the timer operation.



Do not change TMnEN flag of TMnMD register simultaneously with other bits to avoid errors in operation.



When the binary counter reaches the value in compare register, the interrupt request flag is set and the binary counter is cleared, at the next count clock. So set compare register as:
Compare register setting = (count till the interrupt request -1)
However, the interrupt generation cycle when compare register is set to "00" is the same as it is set to "01".



If compare register is set to have smaller number than the binary counter during the count operation, the binary counter counts up until it is overflow for the first time.



If the timer interrupt request flag may have already been set before timer is started, the timer interrupt request flag should be cleared.



The binary counter of the 8-bit timer is controlled by signals generated by sampling values of the TMnEN flag with the Count Clock. If the low-speed clock (fslow) is selected as a Count Clock source, it is important to remember the followings.

After the timer halts, read the binary counter as follows:
Wait until another count clock passes, then read the value. In other words, the value is one count less than the actual value if you don't wait. Another options is that read more than once to obtain accurate value by an appropriate program.

In case of modify the timer settings (selecting clocks or functions, for example) while the timer is suspended, let the Count Clock count one more after the flag has been turned off. Otherwise the results will not be guaranteed.



If CPU operation mode is changed (from NORMAL to SLOW) when the high-frequency oscillation clock (fpll-div) or the prescaler output (TMnPSC) is selected as clock source, the timer operation should be stopped before operation mode transition and should be reset to start the timer after operation mode transition.

In the SLOW/HALT1 mode, as timer clock source, do not select fpll-div or the clock generated from fpll-div.



If the low-speed clock (fslow) is selected as a Count Clock source, the delay produced by the binary counter may give a wrong value.

Do not write values into compare register (TMnOC) during counting. Selecting the synchronous low-speed clock (fslow) as a Count Clock source solves those problems; getting correct values and allowing to write into the register during counting.



After the timer interrupt request flag is generated, maximum 3 system clock is required until the next interrupt request flag is generated. During this period, the interrupt request flag is not generated even if the binary counter reaches the set value of compare register.



When compare register (TMnOC) is set to "00", clear the binary counter before starting the operation.

9.4.2 Setup Example

■ Timer Operation Setup Example (Timer 0,1, 2, 3 and 4)

Timer function can be set by using Timer 0 that generates interrupts constantly. An interrupt is generated in every 250 cycles (200 μ s) by selecting $f_s/2$ (at $f_s=2.5$ MHz operation) as a clock source.

A setup procedure example, with a description of each step is shown below.

Setup Procedure	Description
(1) Stop the counter TM0MD(0x03F64) bp3: TM0EN =0	(1) Set the TM0EN flag of Timer 0 mode register (TM0MD) to "0" to stop the counting of the timer 0.
(2) Disable the interrupt TM0ICR(0x03FE9) bp1: TM0IE =0	(2) Set the TM0IE flag of TM0ICR register to "0" to disable the interrupt.
(3) Select the normal timer operation TM0MD(0x03F64) bp4: TM0PWM =0 bp5: TM0MOD =0	(3) Set the TM0PWM flag and the TM0MOD flag of TM0MD register to "0" to select the normal timer operation.
(4) Select the count clock source TM0MD(0x03F64) bp2 to 0: TM0CK2 to 0 =X01	(4) Select the prescaler output to the clock source by the TM0CK2 to 0 flag of TM0MD register.
(5) Select and enable the prescaler output CK0MD(0x03F66) bp3 to 1: TM0PSC2 to 0 =0X0 bp0: TM0BAS =1	(5) Select $f_s/2$ to the prescaler output by the TM0PSC 2 to 0 flag and TM0BAS flag of Timer 0 prescaler selection register (CK0MD).
(6) Set the cycle of the interrupt generation TM0OC (0x03F62) =0xF9	(6) Set the value of the interrupt generation cycle to Timer 0 compare register (TM0OC). The cycle is 250, so that the setting value is set to 249 (0xF9). At that time, Timer 0 binary counter (TM0BC) is initialized to 0x00.
(7) Set the interrupt level TM0ICR(0x03FE9) bp7 to 6: TM0LV1 to 0 =10	(7) Set the interrupt level by the TM0LV1 to 0 flag of Timer 0 interrupt control register (TM0ICR). If the interrupt request flag may be already set, clear the request flag. [Chapter 7 7.1.4 Maskable Interrupt Control Register Setup]
(8) Enable the interrupt TM0ICR(0x03FE9) bp1: TM0IE =1	(8) Set the TM0IE flag of TM0ICR register to "1" to enable the interrupt.
(9) Start the timer operation TM0MD(0x03F64) bp3: TM0EN =1	(9) Set the TM0EN flag of TM0MD register to "1" to operate Timer 0.

TM0BC starts to count up from 0x00. When TM0BC reaches the setting value of TM0OC register, Timer 0 interrupt request flag is set at the next count clock, then the value of TM0BC becomes 0x00 and restarts to count up.

■ Timer Operation Setup Example When Compare Register is Double-buffered (Timer 0, 1, 2 and 3)

Timer function can be set by using Timer 2 that generates interrupts constantly. An interrupt is generated in every 250 cycles (200 μ s) by selecting $f_s/2$ (at $f_s=2.5$ MHz operation) as a clock source.

A setup procedure example, with a description of each step is shown below.

Setup Procedure	Description
(1) Stop the counter TM2MD(0x03F6C) bp3: TM2EN =0	(1) Set the TM2EN flag of Timer 0 mode register (TM2MD) to "0" to stop the counting of the timer 0.
(2) Disable the interrupt TM2ICR(0x03FEB) bp1: TM2IE =0	(2) Set the TM2IE flag of TM2ICR register to "0" to disable the interrupt.
(3) Select the normal timer operation TM2MD(0x03F6C) bp4: TM2PWM =0 bp5: TM2MOD =0	(3) Set the TM2PWM flag and the TM2MOD flag of TM2MD register to "0" to select the normal timer operation.
(4) Select the timer compare register buffer TM2MD(0x03F6C) bp7: TM2BUFSEL =1	(4) Set the TM2BUFSEL flag of TM2MD register to "1" to select double buffer.
(5) Select the count clock source TM2MD(0x03F6C) bp2 to 0: TM2CK2 to 0 =X01	(5) Select the prescaler output to the clock source by the TM2CK2 to 0 flag of TM2MD register.
(6) Select and enable the prescaler output CK2MD(0x03F6E) bp3 to 1: TM2PSC2 to 0 =0X0 bp0: TM2BAS =1	(6) Select $f_s/2$ to the prescaler output by the TM2PSC 2 to 0 and TM2BAS flags of Timer 2 prescaler selection register (CK2MD).
(7) Set the cycle of the interrupt generation TM2OC(0x03F6A) =11111001	(7) Set the value of the interrupt generation cycle to Timer 0 compare register (TM2OC). The cycle is 250, so that the setting value is set to 249 (0xF9). At that time, Timer 2 binary counter (TM2BC) is initialized to 0x00.
(8) Set the interrupt level TM2ICR(0x03FEB) bp7 to 6: TM2LV1 to 0 =10	(8) Set the interrupt level by the TM2LV1 to 0 flags of Timer 2 interrupt control register (TM2ICR). If the interrupt request flag may be already set, clear the request flag. [Chapter 7 7.1.4 Maskable Interrupt Control Register Setup]
(9) Enable the interrupt TM2ICR (0x03FEB) bp1: TM2IE =1	(9) Set the TM2IE flag of TM2ICR register to "1" to enable the interrupt.
(10) Start the timer operation TM2MD(0x03F6C) bp3: TM2EN =1	(10) Set the TM2EN flag of TM2MD register to "1" to operate the timer 2.

TM2BC starts to count up from 0x00. When TM2BC reaches the setting value of the internal compare register, Timer 2 interrupt request flag is set at the next count clock, then the value of TM2BC becomes 0x00 and restarts to count up.

9.5 8-bit Event Count

9.5.1 Operation

Event count operation has 2 types; TMnIO input and synchronous TMnIO input, according to the selected clock source.

■ 8-bit Event Count Operation (Timer 0, 1, 2, 3 and 4)

Event count operation means that binary counter (TMnBC) counts the input signal from external to the TMnIO pin. If the value of the binary counter reaches the setting value of compare register (TMnOC), interrupts can be generated at the next count clock.

Table:9.5.1 Event Count Input Clock

	Timer 0	Timer 1	Timer 2	Timer 3	Timer 4
Event input	TM0IOA input/ TM0IOB input/ TM0IOC input	TM1IOA input/ TM1IOB input/ TM1IOC input	TM2IOA input/ TM2IOB input/ TM2IOC input	TM3IOA input/ TM3IOB input/ TM3IOC input	TM4IOA input/ TM4IOB input/ TM4IOC input
	Synchronous TM0IO input	Synchronous TM1IO input	Synchronous TM2IO input	Synchronous TM3IO input	Synchronous TM4IO input
	Timer A output	Timer A output	Timer A output	Timer A output	Timer A output

■ Count Timing of TMnIO Input (Timer 0, 1, 2, 3 and 4)

When TMnIO input is selected, TMnIO is input to the count clock of Timer n.

The binary counter is started to count up at the falling edge of the TMnIO input signal.

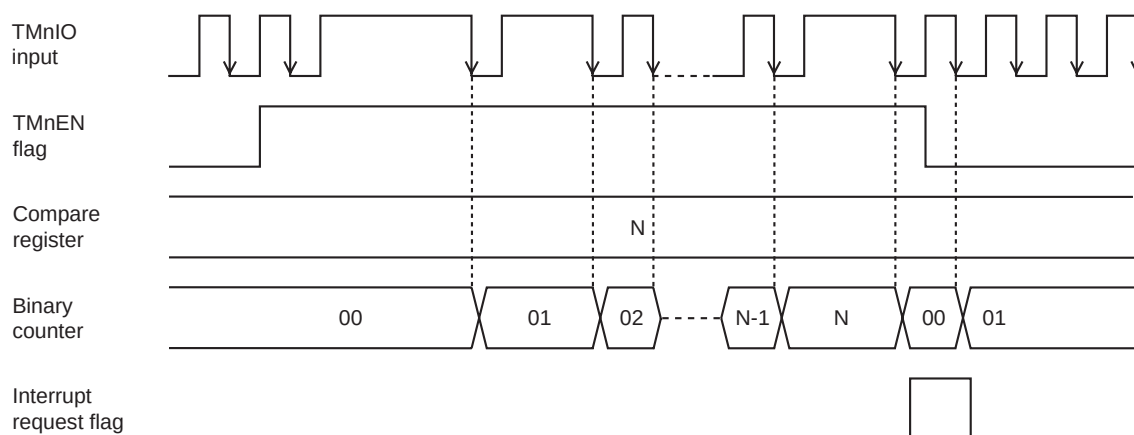


Figure:9.5.1 Count Timing of TMnIO Input (Timer 0, 1, 2, 3 and 4)



If the TMnIO Input is selected as a count clock source, the delay produced by the binary counter may give a wrong value.

Never write values into compare register (TMnOC) during counting. Selecting the event count of the synchronous TMnIO Input as a count clock source solves those problems; getting correct values and allowing to write into the register during counting.



The binary counter of the 8-bit timer is controlled by signals generated by sampling values of the TMnEN flag with the count clock. If the TMnIO Input is selected as a count clock source, it is important to remember the followings.

After the timer halts, read the binary counter as follows:

Wait until another count clock passes, then read the value. In other words, the value is one count less than the actual value if you do not wait. Another option is that read more than once to obtain accurate value by an appropriate program.

In case of modify the timer settings (selecting clocks or functions, for example) while the timer is suspended, let the count clock count one more after the TMnEN flag has been turned off. Otherwise the results will not be guaranteed.



When using TMnIO input in double-buffered configuration, set mode register and compare register after selecting frequency which is more than f_s as count clock.

Then select TMnIO input to start timer.

Do not write the value to compare register during operation.



When the event input is used, clear the binary counter before starting timer operation.



In double-buffer configuration, the clock other than f_s is selected as the count clock source, even if the setting value is input to compare register when timer is stopped, the same setting value cannot be loaded to the internal register.

To prevent this, set the count clock to be the system clock (f_s) before writing the setting value to compare register.

Then start the timer operation by selecting the event input as the clock source.

■ Count Timing of Synchronous TMnIO Input (Timer 0, 1, 2, 3 and 4)

If the synchronous TMnIO input is selected, f_s is input to Timer n count clock.
The count enable signal is synchronized with the falling edge of the system clock.

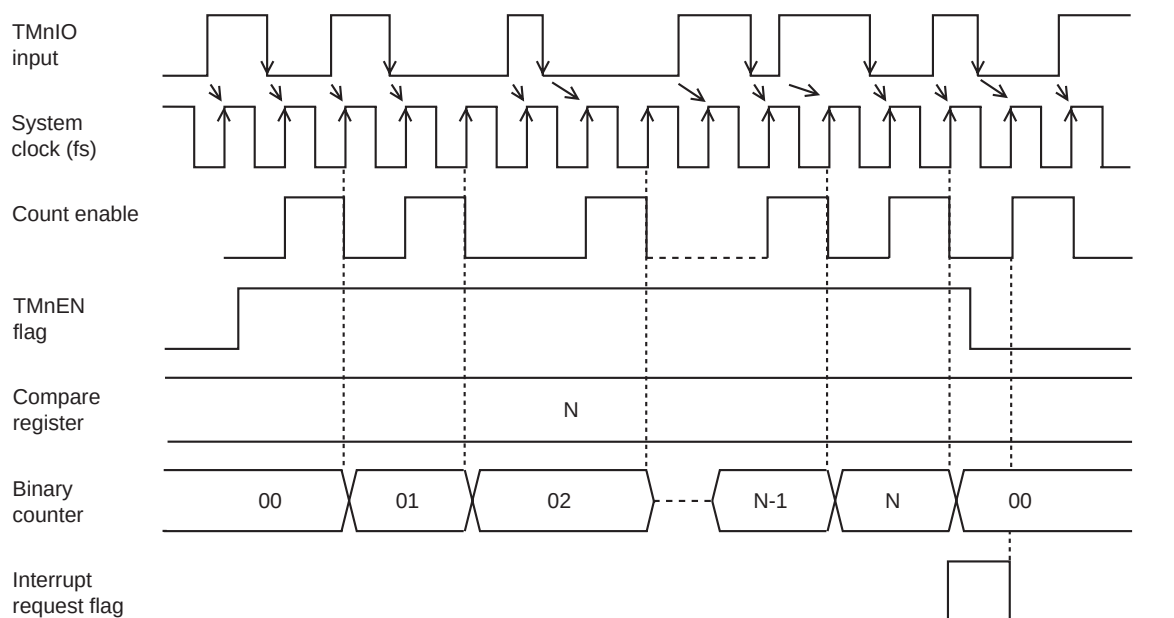


Figure:9.5.2 Count Timing of Synchronous TMnIO Input (Timer 0, 1, 2, 3 and 4)



When the synchronous TMnIO input is selected as the count clock source, Timer n counter counts up synchronous with system clock, therefore the correct value is always read out.



Input signal from TMnIO should be set with the cycle more than double the system clock (f_s). When other signals with a cycle shorter than this are input, the counting may not be performed correctly.

9.5.2 Setup Example

■ Event Count Setup Example (Timer 0, 1, 2, 3 and 4)

If the falling edge of the TM0IO input pin signal is detected for 5 times, an interrupt is generated.

A setup procedure example, with a description of each step is shown below.

Setup Procedure	Description
(1) Stop the counter TM0MD(0x03F64) bp3: TM0EN =0	(1) Set the TM0EN flag of Timer 0 mode register (TM0MD) to "0" to stop Timer 0 counting.
(2) Disable the interrupt TM0ICR(0x03FE9) bp1: TM0IE =0	(2) Set the TM0IE flag of TM0ICR register to "0" to disable the interrupt.
(3) Set the special function pin to input PADIR(0x03E9A) bp0: PADIR0 =0 TMINSEL1(0x03FB2) bp1 to 0: TMINSEL11 to 10 =00 TMCKSEL1(0x03FB0) bp0: TM0IOSEL =0	(3) Set the PADIR0 flag of Port A direction control register (PADIR) to "0", TMINSEL11 to 10 flag of TMINSEL1 register to "00" and TM0IOSEL of TMCKSEL1 register to "0" in order to set PA0 pin to input mode. [Chapter 8 I/O Port]
(4) Set the interrupt generation cycle TM0OC (0x03F62) =0x04	(4) Set the interrupt generation cycle to Timer 0 compare register (TM0OC). As a timer executes counting 5 times, the setting value should be 0x04. At the time, Timer 0 binary counter (TM0BC) is initializes to 0x00.
(5) Select the normal timer operation TM0MD(0x03F64) bp4: TM0PWM =0 bp5: TM0MOD =0	(5) Set the TM0PWM flag and the TM0MOD flag of TM0MD register to "0" to select the normal timer operation.
(6) Select the count clock source TM0MD(0x03F64) bp2 to 0: TM0CK2 to 0 =110	(6) Select the TM0IO input to the clock source by the TM0CK2 to 0 flags of TM0MD register.
(7) Set the interrupt level TM0ICR(0x03FE9) bp7 to 6: TM0LV1 to 0 =10	(7) Set the interrupt level by the TM0LV1 to 0 flags of Timer 0 interrupt control register (TM0ICR). If the interrupt request flag may be already set, clear the request flag. [Chapter 7 7.1.4 Maskable Interrupt Control Register Setup]
(8) Enable the interrupt TM0ICR(0x03FE9) bp1: TM0IE =1	(8) Set the TM0IE flag of TM0ICR register to "1" to enable the interrupt.
(9) Start the event count TM0MD(0x03F64) bp3: TM0EN =1	(9) Set the TM0EN flag of the TM0MD register to "1" to operate Timer 0.

Every time TM0BC detects a falling edge of TM0IO input, TM0BC counts up from 0x00. When TM0BC reaches the setting value of TM0OC register, Timer 0 interrupt request flag is set at the next count clock, then the value of TM0BC becomes 0x00 and counting up is restarted.

9.6 8-bit Timer Pulse Output

9.6.1 Operation

Timer pulse output function is output pulse signal through pin TMnIO with an arbitrary frequency.

■ Operation of Timer Pulse Output (Timer 0, 1, 2, 3 and 4)

Timers can output signals which are twice the cycle signal set to compare register (TMnOC).

Table:9.6.1 shows pins output the signals.

Table:9.6.1 Timer Pulse Output Pin

	Timer 0	Timer 1	Timer 2	Timer 3	Timer 4
Pulse output pin	TM0IOA output/ TM0IOB output/ TM0IOC output	TM1IOA output/ TM1IOB output/ TM1IOC output	TM2IOA output/ TM2IOB output/ TM2IOC output	TM3IOA output/ TM3IOB output/ TM3IOC output	TM4IOA output/ TM4IOB output/ TM4IOC output

■ Count Timing of Timer Pulse Output (Timer 0, 1, 2, 3 and 4)

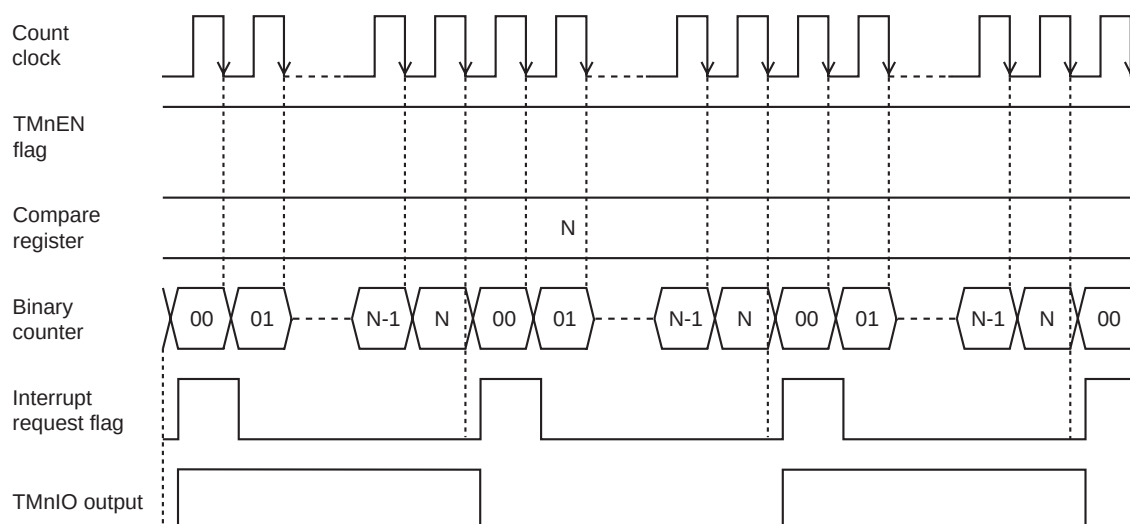


Figure:9.6.1 Count Timing of Timer Pulse Output (Timer 0, 1, 2, 3 and 4)

- The TMnIO pin outputs signals of $2 \times$ cycle of the setup value in compare register. If the binary counter reaches compare register, and the binary counter is cleared to 0x00, TMnIO output (timer output) is inverted.



When the synchronous TMnIO input is selected as the count clock source, Timer n counter counts up synchronous with system clock; therefore the correct value is always read out.

9.6.2 Setup Example

■ Timer Pulse Output Setup Example (Timer 0, 1, 2, 3 and 4)

TM0IO pin outputs 50 kHz pulse by using Timer 0. For this, select $f_s/2$ for clock source, and set a 1/2 cycle (100 kHz) for Timer 0 compare register (at $f_s = 10$ MHz).

An example setup procedure, with a description of each step is shown below.

Setup Procedure	Description
(1) Stop the counter TM0MD(0x03F64) bp3: TM0EN =0	(1) Set the TM0EN flag of Timer 0 mode register (TM0MD) to "0" to stop Timer 0 counting.
(2) Set the special function pin to the output mode PAOMD(0x03EBA) bp0: PAOMD0 =1 PADIR (0x03E9A) bp0: PADIR0 =1	(2) Set the PAOMD0 flag of Port A output mode register (PAOMD) to "1" to set PA0 pin to the special function pin. Set the PADIR0 flag of Port A direction control register (PADIR) to "1" to set the output mode. [Chapter 8 I/O Port]
(3) Select the normal timer operation TM0MD(0x03F64) bp4: TM0PWM =0 bp5: TM0MOD =0	(3) Set the TM0MOD flag of TM0MD register to "0" to select the normal timer operation.
(4) Select the count clock source TM0MD(0x03F64) bp2 to 0: TM0CK2 to 0 =X01	(4) Select the prescaler output to the clock source by the TM0CK2 to 0 flag of TM0MD register.
(5) Select and enable the prescaler output CK0MD(0x03F66) bp3 to 1: TM0PSC2 to 0 =0X0 bp0: TM0BAS =1	(5) Select $f_s/2$ to the prescaler output by the TM0PSC 2 to 0 flags and TM0BAS flag of Timer 0 prescaler selection register (CK0MD).
(6) Set the timer pulse output cycle TM0OC(0x03F62) =0x31	(6) Set Timer 0 compare register (TM0OC) to the 1/2 of Timer pulse output cycle. The setting value should be $50 \times 10^3 / 10^6 = 0.05$ (0x31), for 100 kHz to be divided by 5 MHz. At that time, Timer 0 binary counter (TM0BC) is initialized to 0x00.
(7) Start the timer operation TM0MD(0x03F64) bp3: TM0EN =1	(7) Set the TM0EN flag of TM0MD register to "1" to operate Timer 0.

TM0BC counts up from 0x00. If TM0BC reaches the setting value of TM0OC register, then TM0BC is cleared to 0x00, TM0IO output signal is inverted and TM0BC restarts to count up from 0x00.



If any data is written to compare register when the binary counter is stopped, the timer output turns to "Low".



[Calculation of compare register value]

Compare register = Timer pulse output / (Selection clock cycle × 2) - 1

9.7 8-bit PWM Output

The TMnIO pin outputs the PWM waveform, which is generated by match timing for values of both binary counter and compare register and by the overflow timing of the binary counter.

9.7.1 Operation

■ Operation of 8-bit PWM Output (Timer 0, 2 and 4)

The PWM waveform with an arbitrary duty cycle is generated by setting the duty cycle of PWM "High" period to compare register (TMnOC). The cycle is the period from the full count to the overflow of the 8-bit timer.

Table:9.7.1 shows PWM output pins;

Table:9.7.1 Output Pins of PWM Output

	Timer 0	Timer 2	Timer 4
PWM output pin	TM0IOA output/ TM0IOB output/ TM0IOC output	TM2IOA output/ TM2IOB output/ TM2IOC output	TM4IOA output/ TM4IOB output/ TM4IOC output

■ Count Timing of PWM Output (at Normal) (Timer 0, 2 and 4)

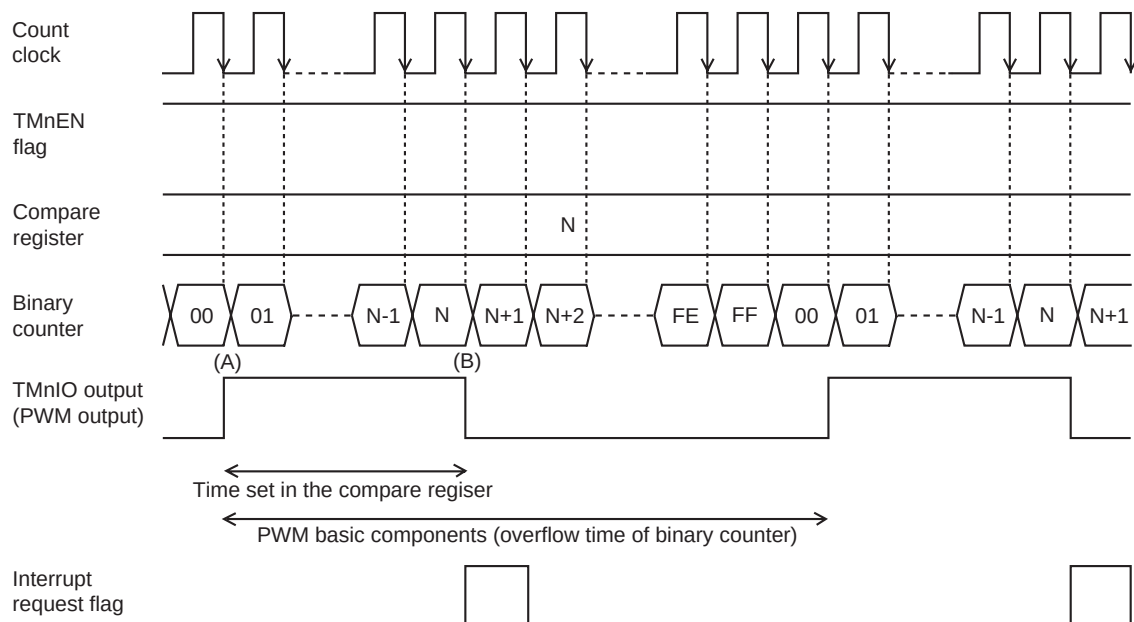


Figure:9.7.1 Count Timing of PWM Output (at Normal) (Timer 0, 2 and 4)

When TMnPOP flag is "0":

- (A) PWM output waveform is "High" while a binary counter is counting up from 0x01 to the value stored in compare register.
- (B) PWM output waveform is "Low" after the match to the value in the compare register, then binary counter continues counting up until it generates as overflow.



The initial setting of PWM output is changed from "Low" to "High" at the selection of PWM operation by the TMnPWM flag of TMnMD register (when TMnPOP flag = 0).

■ Count Timing of PWM Output (when compare register is 0x00) (Timer 0, 2 and 4)

Figure:9.7.2 shows the count timing when compare register is set to 0x00.

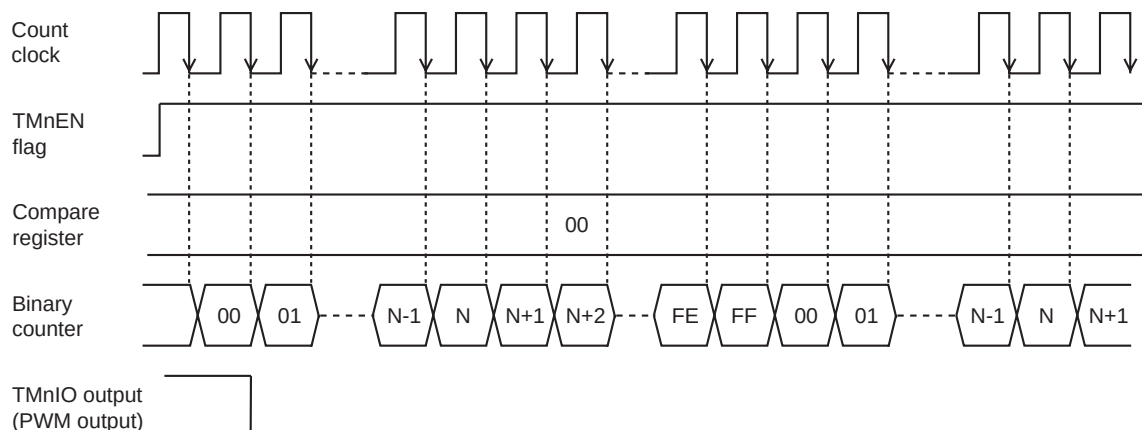


Figure:9.7.2 Count Timing of PWM Output (when compare register is 0x00) (Timer 0, 2 and 4)

When TMnEN flag is stopped ("0"), PWM output is "High".

■ Count Timing of PWM Output (when compare register is 0xFF) (Timer 0, 2 and 4)

Figure:9.7.3 shows the count timing when compare register is set to 0xFF.

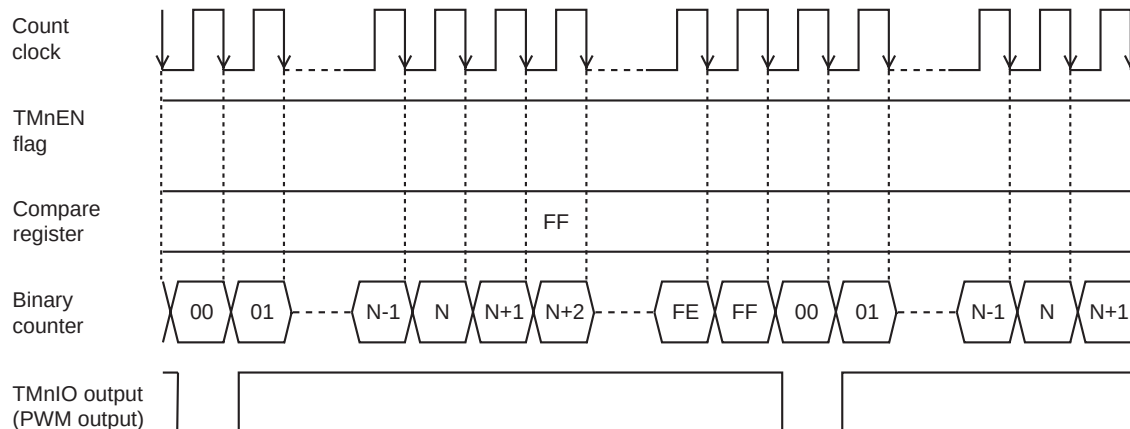


Figure:9.7.3 Count Timing of PWM Output (when compare register is 0xFF) (Timer 0, 2 and 4)

9.7.2 Setup Example

■ PWM Output Setup Example (Timer 0, 2 and 4)

The 1/4 duty cycle PWM output waveform is output from TM0IO output pin at 19.53 Hz by using Timer 0. Clock $f_s/2$ oscillates at 5 MHz. Cycle period of PWM output waveform is decided by the overflow of the binary counter. "High" period of the PWM output waveform is decided by the setting value of compare register.

An example setup procedure, with a description of each step is shown below.

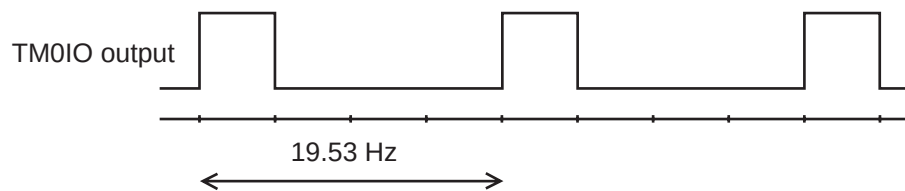


Figure:9.7.4 Output Waveform of TM0IO Output Pin

Setup Procedure	Description
(1) Stop the counter TM0MD(0x03F64) bp3: TM0EN =0	(1) Set the TM0EN flag of Timer 0 mode register (TM0MD) to "0" to stop Timer 0 counting.
(2) Set the special function pin to the output mode PAOMD(0x03EBA) bp0: PAOMD0 =1 PADIR (0x03E9A) bp0: PADIR0 =1	(2) Set the PAOMD0 flag of Port A output mode register 1 (PAOMD1) to "1" to set PA0 pin to the special function pin. Set the PADIR0 flag of Port A direction control register (PADIR) to "1" to set the output mode. [Chapter 8 I/O Port]
(3) Select the PWM operation TM0MD(0x03F64) bp4: TM0PWM =1 bp5: TM0MOD =0 bp6: TM0POP =0	(3) Set the TM0PWM flag of TM0MD register to "1" and the TM0MOD flag to "0" to select the PWM operation.
(4) Select the count clock source TM0MD(0x03F64) bp2 to 0: TM0CK2 to 0 =X01	(4) Select the prescaler output to the clock source by the TM0CK2 to 0 flags of TM0MD register.
(5) Select and enable the prescaler output CK0MD(0x03F66) bp3 to 1: TM0PSC2 to 0 =0X0 bp0: TM0BAS =1	(5) Select $f_s/2$ to the prescaler output by the TM0PSC2 to 0 and TM0BAS flags of Timer 0 prescaler selection register.
(6) Set the period of PWM "High" output TM0OC (0x03F62) =0x40	(6) Set the "High" period of PWM output to Timer 0 compare register (TM0OC). The setting value is set to $256/4=64$ (0x40), because it should be the 1/4 duty of the full count (256). At that time, Timer 0 binary counter (TM0BC) is initialized to 0x00.
(7) Start Timer operation TM0MD(0x03F64) bp3: TM0EN =1	(7) Set the TM0EN flag of TM0MD register to "1" to operate Timer 0.

9.7.3 PWM Outputs With Additional Pulses

■ PWM Output with Additional Pulses Method (Timer 0, 2 and 4)

This method allows the user to add one bit of pulse at a time into a PWM waveform. Up to 3 bits can be added in 4 cycles of the basic waveform.

To determine where to place, or not to place additional bits in the cycles is controlled by Timer 0 prescaler selection register (bits 5 and 6 of CK0MD), or Timer 2 prescaler selection register (bits 5 and 6 of CK2MD), or Timer 4 prescaler selection register (bits 5 and 6 of CK4MD).

■ How to add pulses

Timer 0 prescaler selection register (bits 5 and 6 of CK0MD), Timer 2 prescaler selection register (bits 5 and 6 of CK2MD) and Timer 4 prescaler selection register (bits 5 and 6 of CK4MD) control positions of pulses.

For example, if "00" is specified for CK0MD, CK2MD or CK4MD register, not a single pulse will be added to the PWM. However, if "11" is specified, total of 3 bits will be added in the 4 cycles of the PWM.

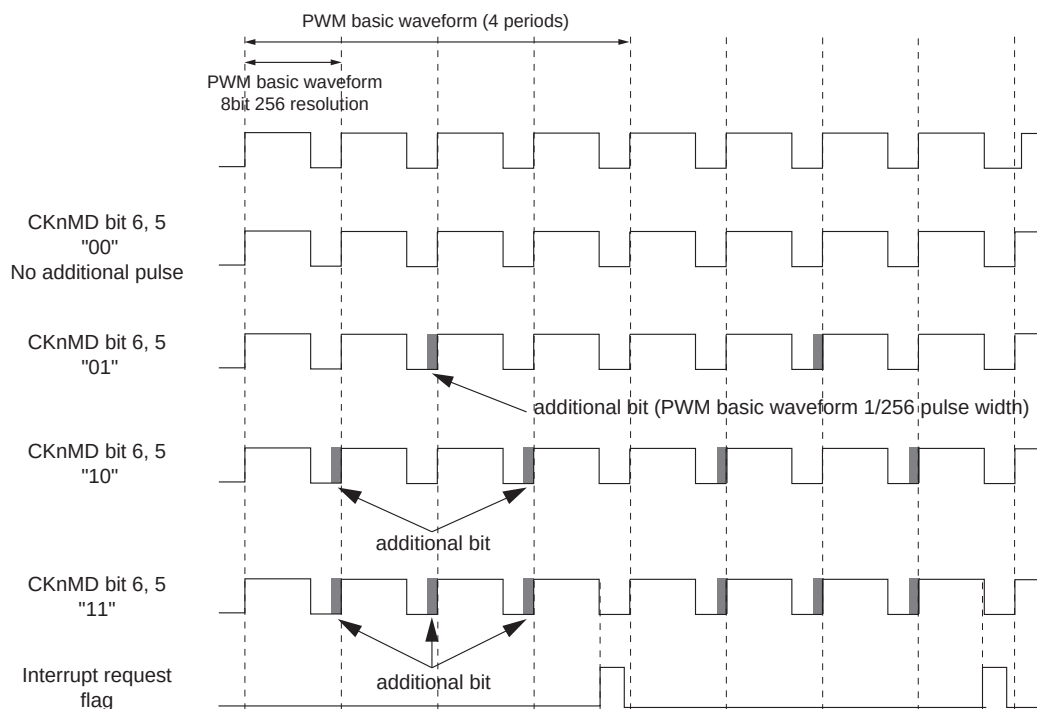
Table:9.7.2 shows the locations of additional pulses with the values of CK0MD/CK2MD/CK4MD register. Figure:9.7.5 shows the additional pulses and the PWM waveform.

Table:9.7.2 Additional pulses settings

CKnMD Register		Locations of additional pulses (Within the four cycles of the basic waveform)
bit6	bit5	
0	0	No pulse
0	1	At second cycle
1	0	At first and third cycle
1	1	At first, second and third cycle



An interrupt occurs at the 4th cycle of the PWM basic waveform.



During 4 cycles of the PWM basic waveform, additional pulses (1/256 pulse width of PWM basic waveform) can be added in any of the periods 0 to 3.

Figure:9.7.5 Additional pulses and the PWM waveform

9.8 Synchronous Output

9.8.1 Operation

When the binary counter of timer reaches the setting value of compare register, latch data is output from Port 8 at the next count clock.

■ Synchronous Output Operation by 8-bit Timer (Timer 1 and 2)

The value of Port 8 output register (P8OUT) is output from the output pin synchronous with the interrupt request generation by the match of the binary counter and compare register.

Only Port 8 can perform synchronous output operation, and individual bits can be set. Timer 1 and 2 are 8-bit timers that have synchronous output operation.

Table:9.8.1 Synchronous Output Port (Timer 1 and 2)

	Timer 1	Timer 2
Synchronous output Port	Port 8	Port 8

■ Timing of Synchronous Output (Timer 1 and 2)

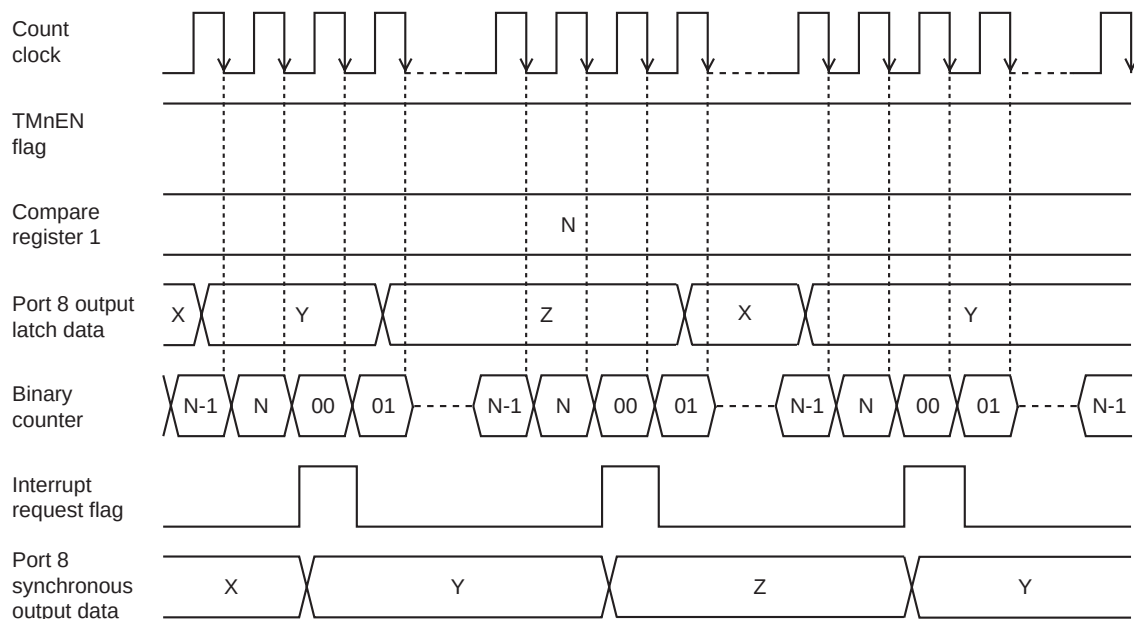


Figure:9.8.1 Timing of Synchronous Output (Timer 1 and 2)

- The value of P8OUT register is output from the output pin synchronous with the interrupt request generation by the match of binary counter and compare register.

9.8.2 Setup Example

■ Synchronous Output Setup Example (Timer 1 and 2)

Setup example that the value of Port 8 output register is output constantly (100 μ s) by using Timer 1 from the synchronous output pin is shown below. The clock source of Timer 1 is selected fs/2 (fs=2 MHz at operation).

A setup procedure example, with a description of each step is shown below.

Setup Procedure	Description
(1) Stop the counter TM1MD(0x03F65) bp3: TM1EN =0	(1) Set the TM1EN flag of Timer 1 mode register (TM1MD) to "0" to stop Timer 1 counting.
(2) Select the synchronous output event P8SEV(0x03EE9) bp1 to 0: P8SEV1 to 0 =11	(2) Set the P8SEV1 to 0 flag of pin control register (P8SEV) to "11" to set the synchronous output event to Timer 1 interrupt.
(3) Set the synchronous output pin P8SYO(0x03EF8) =0xFF P8DIR(0x03E98) =0xFF	(3) Set Port 8 synchronous output control register (P8SYO) to 0xFF to set the synchronous output pin. Set Port 8 direction control register (P8DIR) to 0xFF to set Port 8 to output mode. [Chapter 8 I/O Port]
(4) Select the normal timer operation TM1MD(0x03F65) bp4: TM1CAS =0	(4) Set the TM1CAS flag of TM1MD register to "0" to select the normal timer operation.
(5) Select the count clock source TM1MD(0x03F65) bp2 to 0: TM1CK2 to 0 =X01	(5) Select the prescaler output to the clock source by the TM1CK2 to 0 flags of TM1MD register.
(6) Select and enable the prescaler output CK1MD(0x03F67) bp3 to 1: TM1PSC2 to 0 =0X0 bp0: TM1BAS =1	(6) Select fs/2 to the prescaler output by the TM1PSC2 to 0 and TM1BAS flags of Timer 1 prescaler selection register (CK1MD).
(7) Set the synchronous output event TM1OC (0x03F63) =0x63	(7) Set the synchronous output generation cycle to Timer 1 compare register (TM1OC). The setting value is set to 100-1=99 (0x63), because 1 MHz is divided into 10 kHz. At that time, Timer 1 binary counter (TM1BC) is initialized to 0x00.
(8) Start Timer operation TM1MD(0x03F65) bp3: TM1EN =1	(8) Set the TM1EN flag of TM1MD register to "1" to operate Timer 1.

- TM1BC counts up from 0x00. If any data is written to Port 8 output register (P8OUT), the value of P8OUT register is output from the synchronous output pin in every time an interrupt request is generated by the match of TM1BC and the set value of TM1OC register.

9.9 Serial Transfer Clock Output

9.9.1 Operation

Serial transfer clock can be created by using Timer output signal.
Serial transfer clock operation by 8-bit timer (Timer 0, 1, 2, 3 and 4)

Table:9.9.1 Synchronous Output Port (Timer 1 and 2)

Timer	MN101EF76K/MN101EF56K	MN101EF57G
Timer 0	Serial Interface 0, 1, 2, 3, 4	Serial Interface 0, 1, 2, 4
Timer 1	Serial Interface 0, 1, 2, 3, 4	Serial Interface 0, 1, 2, 4
Timer 2	Serial Interface 0, 1, 2, 3, 4	Serial Interface 0, 1, 2, 4
Timer 3	Serial Interface 0, 1, 2, 3, 4	Serial Interface 0, 1, 2, 4
Timer 4	Serial Interface 0, 1, 2, 3, 4	Serial Interface 0, 1, 2, 4

■ Timing of Serial Transfer Clock (Timer 0, 1, 2, 3 and 4)

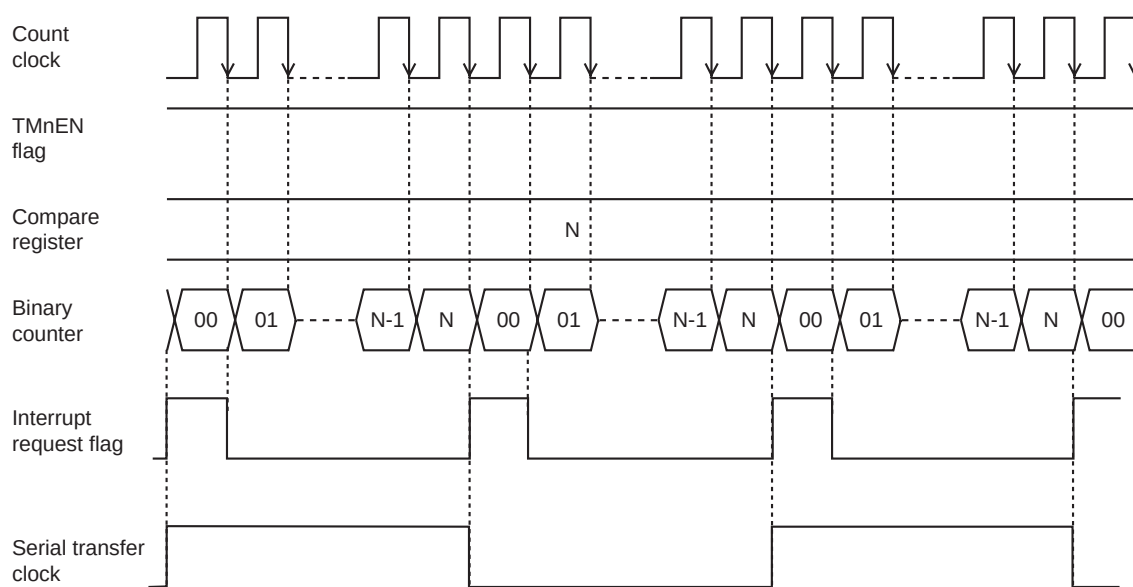


Figure:9.9.1 Timing of Serial Transfer Clock (Timer 0, 1, 2, 3 and 4)

- The timer frequency is $1/2$ of the set frequency set by compare register.
- For the baud rate calculation and Serial Interface setup, refer to [Chapter 18 Serial Interface].

9.9.2 Setup Example

■ Serial Transfer Clock Setup Example (Timer 0)

How to create a transfer clock for full duplex UART (Serial Interface 0) by using Timer 0 is shown below. The baud rate is selected to be 300 bps, the source clock of timer 0 is selected to be $f_s/2$ (at $f_s=2$ MHz).

An example setup procedure, with a description of each step is shown below.

Setup Procedure	Description
(1) Stop the counter TM0MD(0x03F64) bp3: TM0EN =0	(1) Set the TM0EN flag of Timer 0 mode register (TM0MD) to "0" to stop Timer 0 counting.
(2) Select the normal timer operation TM0MD(0x03F64) bp4: TM0PWM =0 bp5: TM0MOD =0	(2) Set the TM0PWM flag and the TM0MOD flag of TM0MD register to "0" to select the normal timer operation.
(3) Select the count clock source TM0MD(0x03F64) bp2 to 0: TM0CK2 to 0 =X01	(3) Select the prescaler output to the clock source by the TM0CK2 to 0 flags of TM0MD register.
(4) Select and enable the prescaler output CK0MD(0x03F66) bp3 to 1: TM0PSC2 to 0 =0X0 bp0: TM0BAS =1	(4) Select $f_s/2$ to the prescaler output by the TM0PSC2 to 0 flags and the TM0BAS flag of Timer 0 prescaler selection register.
(5) Set the baud rate TM0OC (0x03F62) =0xCF	(5) Set Timer 0 compare register (TM0OC) such a value that the baud rate comes to 300 bps. At that time, Timer 0 binary counter (TM0BC) is initialized to 0x00.
(6) Start Timer operation TM0MD(0x03F64) bp3: TM0EN =1	(6) Set the TM0EN flag of the TM0MD register to "1" to operate Timer 0.

- TM0BC counts up from 0x00. Timer 0 output is the clock of Serial Interface 0 at transmission and reception.
- For the setup value of compare register and the setup of Serial Interface operation, refer to [Chapter 18 Serial Interface].

9.10 Simple Pulse Width Measurement

9.10.1 Operation

Timer measures "Low" period of pulse signal input from the external interrupt pin.

■ Simple Pulse Width Measurement Operation by 8-bit Timer (Timer 0, 2 and 4)

Sample input signals of the external interrupt pins, which is used to measure the simple pulse width, at the Count Clock. The binary counter will count while the signals are "Low". Pulse width "Low" period can be measured by reading the count of the timer. 8-bit timers that have the simple pulse width measurement function are Timer 0, 2 and 4.

Table:9.10.1 Simple Pulse Width Measurable Pins

	Timer	MN101EF76K	MN101EF57G/MN101EF56K
Simple pulse width measurable pins	Timer 0	External Interrupt 0 (P20/IRQ0A, PC0/IRQ0B) UART data output (TXD0)	External Interrupt 0 (P20/IRQ0) UART data output (TXD0)
	Timer 2	External Interrupt 2 (P22/IRQ2A, PC2/IRQ2B)	External Interrupt 2 (P22/IRQ2)
	Timer 4	External Interrupt 4 (P24/IRQ4A, PC4/IRQ4B)	External Interrupt 4 (P24/IRQ4)

■ Count Timing of Simple Pulse Width Measurement (Timer 0, 2 and 4)

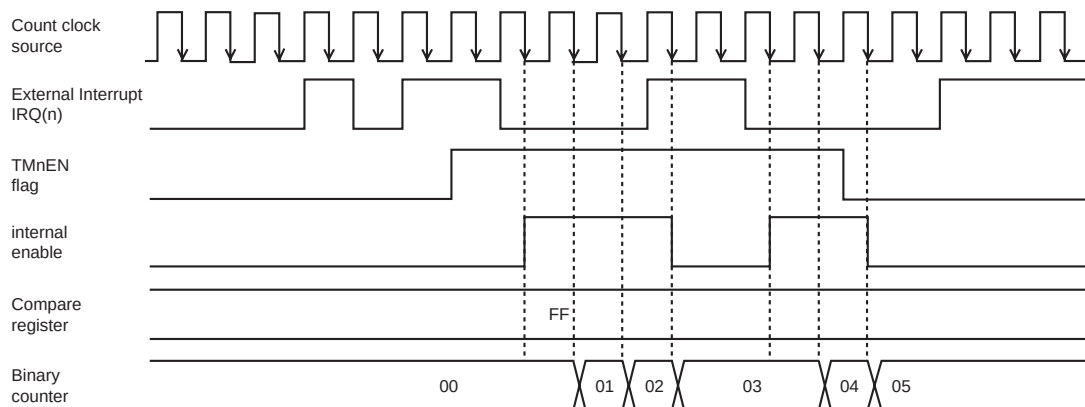


Figure:9.10.1 Count Timing of Simple Pulse Width Measurement (Timer 0 and 2)

- Internal enable signal is generated when the count clock executes samplings of the condition while the TMnEN flag is enable ("1") and the input signal of external interrupt pins for simple pulse width measurement is "Low". When the internal enable signal is "High", timer counts up.



Only when Serial Interface 0 is used as LIN function, Timer 0 can be used to measure the pulse width of UART data output (TXD0), Refer to [Chapter 19 Hardware LIN] for detailed.

9.10.2 Setup Example

■ Setup Example of Simple Pulse Width Measurement by 8-bit Timer (Timer 0, 2 and 4)

The pulse width of "Low" period of External Interrupt 0 (IRQ0) input signal is measured by Timer 0. The clock source of Timer 0 is selected to $f_s/2$. A setup procedure example, with a description of each step is shown below.

Setup Procedure	Description
(1) Stop the counter TM0MD(0x03F64) bp3: TM0EN =0	(1) Set the TM0EN flag of Timer 0 mode register (TM0MD) to "0" to stop Timer 0 counting.
(2) Set the pulse width measurement operation TM0MD(0x03F64) bp4: TM0PWM =0 bp5: TM0MOD =1	(2) Set the TM0PWM flag of TM0MD register to "0" and TM0MOD flag to "1" to enable the timer operation during "Low" period to be measured.
(3) Select the count clock source TM0MD(0x03F64) bp2 to 0: TM0CK2 to 0 =X01	(3) Select the prescaler output to the clock source by the TM0CK2 to 0 flags of TM0MD register.
(4) Select and enable the prescaler output CK0MD(0x03F66) bp3 to 1: TM0PSC2 to 0 =0X0 bp0: TM0BAS =1	(4) Select $f_s/2$ to the prescaler output by the TM0PSC2 to 0 flags and the TM0BAS flag of Timer 0 prescaler selection register (CK0MD).
(5) Set compare register TM0OC(0x03F62) =0xFF	(5) Set Timer 0 compare register (TM0OC) to larger value than the cycle of $f_s/2$ / "Low" period of measured pulse width. At that time, Timer 0 binary counter (TM0BC) is initialized to 0x00.
(6) Set the external interrupt IRQCNT(0x03FD0) bp0: P20EN =1	(6) Set the P20EN flag of External Interrupt pin setup register (IRQCNT) to "1" to set P20 to the external interrupt 0 (IRQ0).
(7) Set the interrupt level IRQ0ICR(0x03FE2) bp7 to 6: IRQ0LV1 to 0 =XX	(7) Set the interrupt level by the IRQ0LV1 to 0 flags of External Interrupt 0 control register (IRQ0ICR). If the interrupt request flag is already set, clear all interrupt request flags. [Chapter 7 7.1.4 Maskable Interrupt Control Register Setup]
(8) Set the interrupt valid edge IRQ0ICR(0x03FE2) bp5: REDG0 =1	(8) Set the REDG0 flag of IRQ0ICR register to "1" to specify the interrupt valid edge to the rising edge.
(9) Enable the interrupt IRQ0ICR(0x03FE2) bp1: IRQ0IE =1	(9) Set the IRQ0IE flag of IRQ0ICR register to "1" to enable the interrupt.
(10) Enable Timer operation TM0MD(0x03F64) bp3: TM0EN =1	(10) Set the TM0EN flag of TM0MD register to "1" to enable Timer 0 operation.

- The internal enable is set by sampling "Low" level of External Interrupt 0 (IRQ0) input with the count clock. TM0BC starts counting up from 0x00 after the internal enable is set. Timer 0 continues counting up while IRQ0 is "Low", and the counting is stopped after sampling a "High" level of IRQ0. At the same time, reading the value of TM0BC by an interrupt processing can detect "Low" period of IRQ.



Enable the external interrupt input by IRQCNT register when the external interrupt pin is used for the pulse width measurement.

9.11 Cascade Connection

9.11.1 Operation

Cascading Timer 0 and 1 or Timer 2 and 3 forms a 16-bit timer.

■ 8-bit timer Cascade Connection Operation (Timer 0 + Timer 1, Timer 2 + Timer 3)

Timer 0 and 1 or Timer 2 and 3 are combined to be a 16-bit timer.

Cascaded Timer is operated at the clock source of Timer 0 or 2 which are lower 8-bit.

Table:9.11.1 Timer Functions at Cascade Connection

	Timer 0 +Timer1 (16-bit)	Timer 2 + Timer3 (16-bit)
Interrupt source	TM1IRQ	TM3IRQ
Timer operation	√	√
Event count	TM0IO input	TM2IO input
Timer pulse output	TM1IO output	TM3IO output
PWM output	-	-
Synchronous output	√	-
Pulse width measurement	√	√
Clock source	fppll-div fppll-div/4 fppll-div/16 fppll-div/32 fppll-div/64 fppll-div/128 fs/2 fs/4 fs/8 fslow TM0IO input Synchronous fslow Synchronous TM0IO input Simple Timer A input	fppll-div fppll-div/4 fppll-div/16 fppll-div/32 fppll-div/64 fppll-div/128 fs/2 fs/4 fs/8 fslow TM2IO input Synchronous fslow Synchronous TM2IO input Simple Timer A input
fppll-div: Machine clock (High speed oscillation for peripheral functions) fslow: Machine clock (Low speed oscillation for peripheral functions) fs: System clock		

- At cascade connection, binary counter and compare register are operated as a 16-bit register.
At operation, set the TMnEN flag of the lower 8-bit timers to "1" to be operated.

The upper 8-bit timer outputs waveforms of Timer pulse output and interrupt request flags.
Also, select the clock source by the lower 8-bit timer.

Other setup and count timing is the same to the 8-bit timer at independently operation.



When Timer 0 and Timer 1 are used in cascade connection, Timer 1 is used for a Timer pulse output and an interrupt request flag. Timer pulse output of Timer 0 is "Low" fixed output. An interrupt request of Timer 0 is not generated, but Timer 0 interrupt should be disabled.



When Timer 2 and Timer 3 are used in cascade connection, Timer 3 is used for a Timer pulse output and an interrupt request flag. Timer pulse output of Timer 2 is "Low" fixed output. An interrupt request of Timer 2 is not generated, but Timer 2 interrupt should be disabled.



At 16-bit cascade connection, when the clear of binary counter is needed by rewriting compare register, set the TMnEN flag of both the upper 8-bit timer and the lower 8-bit timer to "0" to stop counting. Then rewrite all compare registers.



Use a 16-bit access instruction to set (TM1OC + TM0OC) register and (TM2OC + TM3OC) register.



During cascade connection, PWM output function cannot be used. When connecting cascade, always set TMPWM flag of timer mode register to "0".

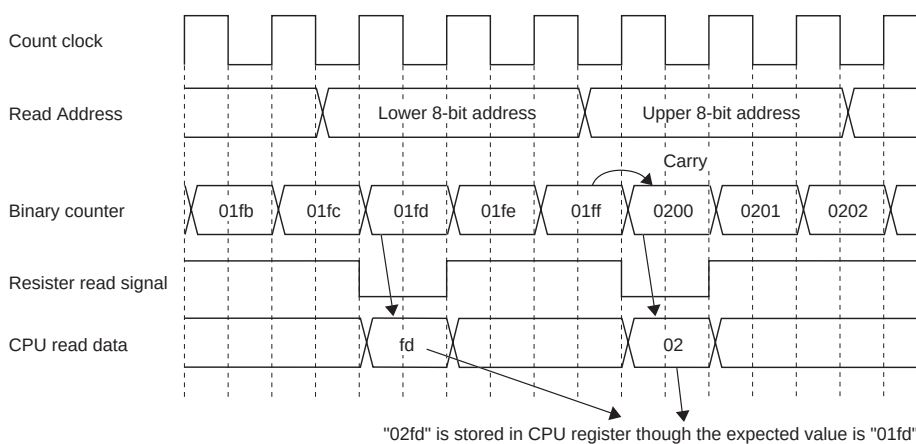


In reading out the value of binary counter during timer operation with 8-bit timer in cascade connection, the value of binary counter may not be correctly read out.

The extended binary counter cannot read all bits at a time when 8-bit timer in cascade connection is used as a 16-bit, or more (16, 24 and 32-bit setting are selectable depending on models). Inside LSI, the lower 8-bit is read out separately from the upper 8-bit, even if MOVW instruction of 16-bit access is used.

Therefore, when the value of binary counter is read out in timer operation, the correct value cannot be read if a carry from lower 8-bit to upper 8-bit is generated during read out.

Image chart when value of binary counter is read by two 8-bit accesses with 16-bit timer that connects 8-bit timer in cascade.



Stop the timer in order to read out the correct value of the timer in cascade connection.

9.11.2 Setup Example

■ Cascade Connection Timer Setup Example (Timer 0 + Timer 1)

Setting example of timer function that an interrupt is constantly generated by cascade connection of Timer 0 and Timer 1, as a 16-bit timer is shown. An interrupt is generated 2500 times every 1 ms by selecting source clock $f_s/2$ (at $f_s=5$ MHz). An example setup procedure, with a description of each step is shown below.

Setup Procedure	Description
(1) Stop the counter TM0MD(0x03F64) bp3: TM0EN =0 TM1MD(0x03F65) bp3: TM1EN =0	(1) Set the TM0EN flag of Timer 0 mode register (TM0MD) to "0", the TM1EN flag of Timer 1 mode register to "0" to stop Timer 0 and Timer 1 counting.
(2) Disable Timer interrupt TM0ICR(0x03FE9) bp1: TM0IE =0 TM1ICR(0x03FEA) bp1: TM1IE =0	(2) Set the TM0IE flag of Timer 0 interrupt control register (TM0ICR) to "0" to disable the interrupt. Set the TM1IE flag of Timer 1 interrupt control register (TM1ICR) to "0" to disable the interrupt.
(3) Select the normal lower Timer operation TM0MD(0x03F64) bp4: TM0PWM =0 bp5: TM0MOD =0	(3) Set the TM0PWM flag and the TM0MOD flag of TM0MD register to "0" to select the normal operation mode.
(4) Set the cascade connection TM1MD(0x03F65) bp4: TM1CAS =1	(4) Set the TM1CAS flag of TM1MD register to "0" to connect Timer 1 and Timer 0 to the cascade.
(5) Select the count clock source TM0MD(0x03F64) bp2 to 0: TM0CK2 to 0 =X01	(5) Select the prescaler to the clock source by the TM0CK2 to 0 flags of TM0MD register.
(6) Select and enable the prescaler output CK0MD(0x03F66) bp3 to 1: TM0PSC2 to 0 =0X0 bp0: TM0BAS =1	(6) Select $f_s/2$ to the prescaler output by the TM0PSC2 to 0 flags and the TM0BAS flag of Timer 0 prescaler selection register (CK0MD).
(7) Set the interrupt generation cycle TM1OC, TM0OC(0x03F63, 0x03F62) =0x09C3	(7) Set Timer 1 compare register + Timer 0 compare register (TM1OC + TM0OC) to the interrupt generation cycle (0x09C3:2500 cycles -1). At that time, Timer 1 binary counter + Timer 0 binary counter (TM1BC + TM0BC) are initialized to 0x0000.
(8) Set the level of the upper Timer interrupt TM1ICR(0x03FEA) bp7 to 6: TM1LV1 to 0 =10	(8) Set the interrupt level by the TM1LV1 to 0 flags of Timer 1 interrupt control register (TM1ICR). If any interrupt request flag may be already set, clear all request flags. [Chapter 7 7.1.4 Maskable Interrupt Control Register Setup]
(9) Enable the lower Timer interrupt TM1ICR(0x03FEA) bp1: TM1IE =1	(9) Set the TM1IE flag of TM1ICR register to "1" to enable the interrupt.
(10) Start the lower Timer operation TM0MD(0x03F64) bp3: TM0EN =1	(10) Set the TM0EN flag of TM0MD register to "1" to operate the 16-bit cascade connection.

- TM1BC + TM0BC counts up from 0x0000 as a 16-bit timer.

When TM1BC + TM0BC reaches the set value of TM1OC + TM0OC register, Timer 1 interrupt request flag is set at the next count clock, and the value of TM1BC + TM0BC becomes 0x0000 and restarts count up.

9.11.3 24-bit Cascade Connection Operation

Timer 0, 1 and 2 can be cascaded to form 24-bit timer in 24-bit cascade mode.

■ Operation (Timer 0 + Timer 1 + Timer 2)

Timer 0, 1 and 2 can be cascaded to form a 24-bit timer. Then, the timer runs by clock sources of Timer 0 for the lower 8-bit timer.

Table:9.11.2 Timer Functions in Cascade Mode

	Timer 0 + Timer 1 + Timer 2 (24-bit)
Interrupt source	TM2IRQ
Timer operation	√
Event count	TM0IO input
Timer pulse output	TM2IO output
PWM output	-
Synchronous output	√
Pulse width measurement	√
Clock sources	fp11-div fp11-div/4 fp11-div/16 fp11-div/32 fp11-div/64 fp11-div/128 fs/2 fs/4 fs/8 fslow TM0IO input Synchronous fslow Synchronous TM0IO input
fp11-div: Machine clock (High speed oscillation for peripheral functions) fslow: Machine clock (Low speed oscillation for peripheral functions) fs: System clock	

- At cascade connection, the binary counter and compare register are operated as 24-bit registers. At operation, set the TM0EN flag of Timer 0 to "1".

Timer 2 outputs waveforms of Timer pulse output and interrupt request flags.
Also, select the clock source by Timer 0.

Other setup and count timing are the same as the settings of 8-bit timer at independently operation.



When Timer 0, 1 and 2 are used in cascade connection, Timer 2 is used for a timer pulse output and an interrupt request flag. Timer pulse output of Timer 0 and 1 is fixed to "Low". An interrupt request of Timer 0 or Timer 1 is not generated, but Timer 0 and Timer 1 interrupts should be disabled.



At 24-bit cascade connection, when the clear of binary counter is needed by rewriting compare registers, set the TM0EN flag of TM0MD register to "0" to stop counting. Then rewrite all compare registers for Timer 0 to 2.



During cascade connection, PWM output function cannot be used. When connecting cascade, always set TMPWM flag of timer mode register to "0".



At 24-bit cascade connection, always set TM3CAS flag of TM3MD register to "0".



Stop the timer in order to read out the correct value of the timer in cascade connection.

9.11.4 32-bit Cascade Connection Operation

Timer 0, 1, 2 and 3 can be cascaded to form 32-bit timer in 32-bit cascade mode.

■ Operation (Timer 0 + Timer 1 + Timer 2 + Timer 3)

Timer 0, 1, 2 and 3 can be cascaded to form a 32-bit timer. Then, the timer runs by clock sources of Timer 0 for the lower 8-bit timer.

Table:9.11.3 Timer Functions in Cascade Mode

	Timer 0 + Timer 1 + Timer 2 + Timer 3 (32-bit)
Interrupt source	TM3IRQ
Timer operation	√
Event count	TM0IO input
Timer pulse output	TM2IO output
PWM output	-
Synchronous output	√
Pulse width measurement	√
Clock sources	fppll-div fppll-div/4 fppll-div/16 fppll-div/32 fppll-div/64 fppll-div/128 fs/2 fs/4 fs/8 fslow TM0IO input Synchronous fslow Synchronous TM0IO input
fppll-div: Machine clock (High speed oscillation for peripheral functions) fslow: Machine clock (Low speed oscillation for peripheral functions) fs: System clock	

- At cascade connection, binary counter and compare register are operated as 32-bit registers. At operation, set the TM0EN flag of Timer 0 to "1".

Timer 3 outputs waveforms of the timer pulse output and interrupt request flags. Also, select the clock source by Timer 0.

Other setup and count timing are the same as the settings of 8-bit timer at independently operation.



When Timer 0, 1, 2 and 3 are used in cascade connection, Timer 3 is used for a timer pulse output and an interrupt request flag. Timer pulse output of Timer 0, 1 and 2 is fixed to "Low". An interrupt request of Timer 0, 1 or 2 is not generated, but Timer 0, 1 and 2 interrupts should be disabled.



At 32-bit cascade connection, when the clear of binary counter is needed by rewriting compare registers, set the TM0EN flag of TM0MD register to "0" to stop counting. Then rewrite all compare registers for Timer 0 to 3.



During cascade connection, PWM output function cannot be used. When connecting cascade, always set TMPWM flag of Timer mode register to "0".



At 32-bit cascade connection, always set TM3CAS flag of TM3MD register to "1".



Stop the timer in order to read out the correct value of the timer in cascade connection.

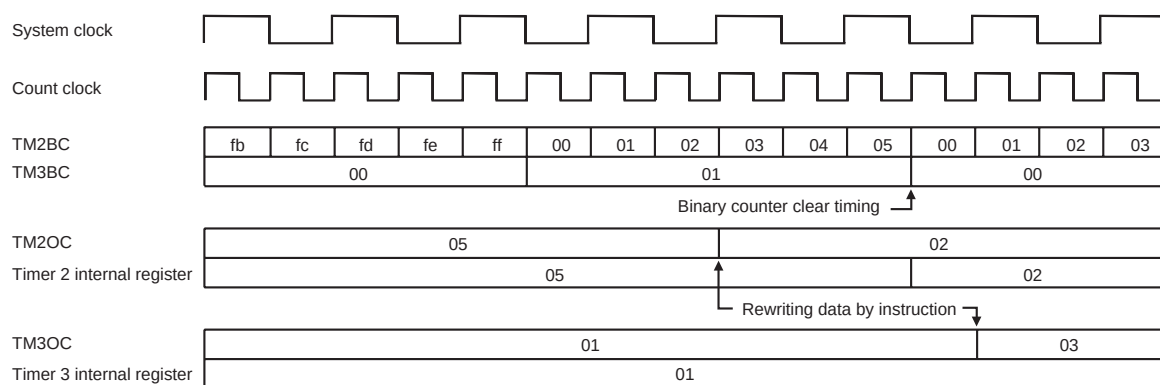


In cascade connection (16-bit, 24-bit, 32-bit) with double-buffered compare register, when the clearing of binary counter occurs at the same time as the writing to compare register, the correct data may not be loaded to internal register.

Inside LSI, the writing data is processed as 8-bit, even if MOVW instruction as 16-bit access is used to write to the extended compare register.

In 24-bit and 32-bit cascade connection, collectively writing by 24-bit or 32-bit cannot be executed.

Therefore, ensure the timing margin for rewriting the value of compare register to avoid the occurring of write timing and clear timing at the same time.



The clearing of binary counters (TM3BC, TM2BC) occur at the same time as the rewriting ("0x0105" to "0x0302") to compare registers (TM2OC, TM3OC) so that the incorrect data "0x0102" is loaded to internal registers.

Figure:9.11.1 Load Timing to Internal Registers in 16-bit Cascade Connection with Double Buffer

10.1 Overview

This timer is a 8-bit simple timer that can be used as a serial transfer clock or timer count clock.

8-bit simple timer is equipped with two prescalers. Each prescaler counts $f_{pll-div}$ and f_s as the base clock. Configurations of hardware are shown below.

Prescaler 0 (based on $f_{pll-div}$) 5-bit Prescaler
Prescaler 1 (based on f_s) 2-bit Prescaler

Prescaler 0 outputs $f_{pll-div}/2$, $f_{pll-div}/4$, $f_{pll-div}/8$, $f_{pll-div}/16$ and $f_{pll-div}/32$.

Prescaler 1 outputs $f_s/2$ and $f_s/4$.

The TMAMD1 register can select a clock source for timer from $f_{pll-div}$ and f_s by using the internal prescaler output.

10.1.1 Functions

Table:10.1.1 shows functions that can be used with each timer.

Table:10.1.1 8-bit Simple Timer Functions

	Timer A (8-bit)
Clock source	$f_{pll-div}$ $f_{pll-div}/2$ $f_{pll-div}/4$ $f_{pll-div}/8$ $f_{pll-div}/16$ $f_{pll-div}/32$ $f_s/2$ $f_s/4$
$f_{pll-div}$: Machine clock (High-speed oscillation for peripheral functions) f_s : System clock	



When changing the frequency of $f_{pll-div}$ by bp7 to bp4 of OSCCNT register, it should be executed after the 8-bit simple timer function is stopped.

10.1.2 Block Diagram

■ Timer A Block Diagram

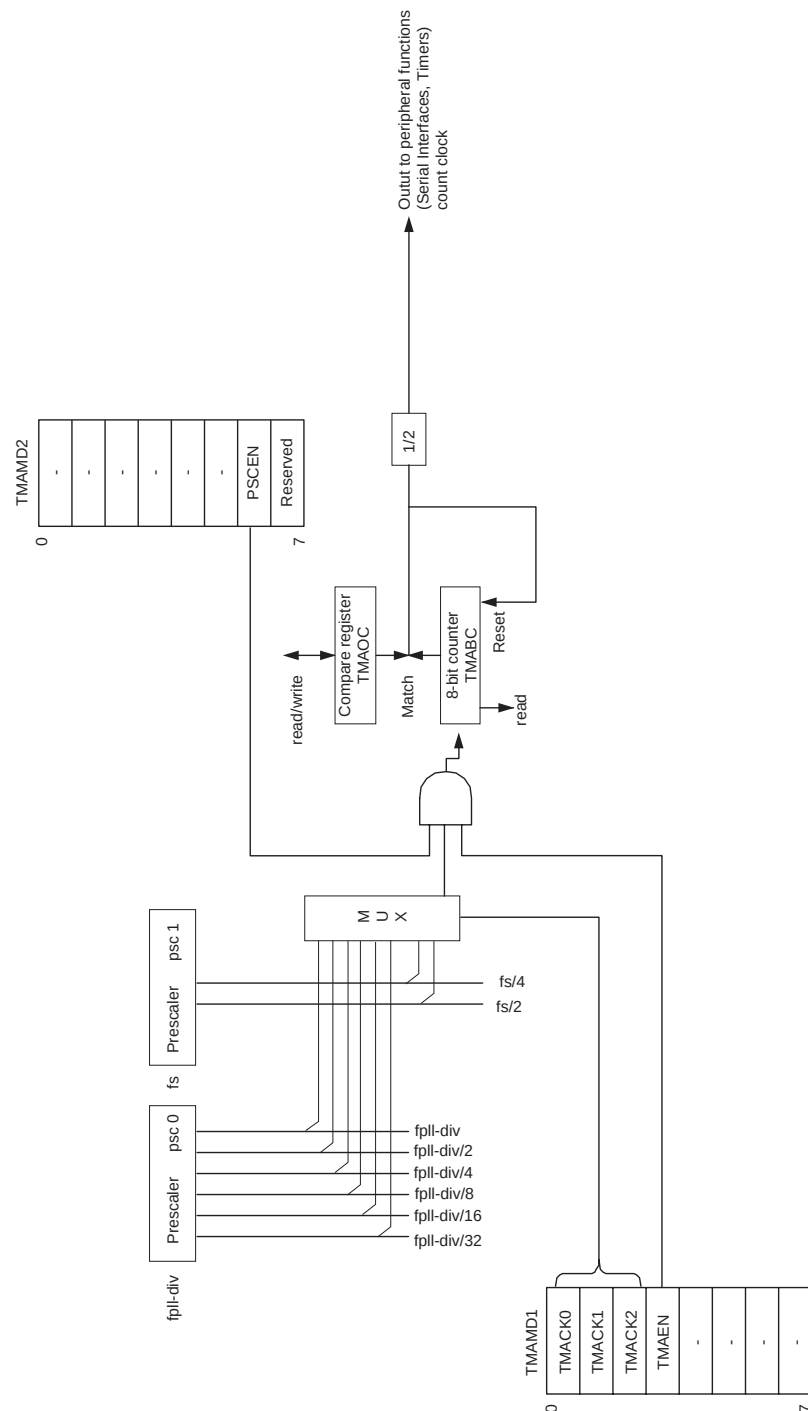


Figure:10.1.1 Timer A Block Diagram

10.2 Control Registers

Timer A consists of binary counter (TMABC) and compare register (TMAOC). It is controlled by mode registers (TMAMD1 and TMAMD2).

10.2.1 Registers

Table:10.2.1 shows registers that control Timer A.

Table:10.2.1 8-bit Timer Control Registers

	Register	Address	R/W	Function	Page
Timer A	TMABC	0x03F71	R	Timer A binary counter	X-5
	TMAOC	0x03F73	R/W	Timer A compare register	X-5
	TMAMD1	0x03F75	R/W	Timer A mode register 1	X-6
	TMAMD2	0x03F77	R/W	Timer A mode register 2	X-6

R/W: Readable/Writable

R: Read only

10.2.2 Programmable Timer Registers

Timer A has 8-bit programmable timer registers.

Programmable timer registers consist of a compare register and a binary counter.

The compare register is a 8-bit register which stores the value to be compared to the binary counter.

■ Timer A Compare Register (TMAOC: 0x03F73)

bp	7	6	5	4	3	2	1	0
Flag	TMAOC7	TMAOC6	TMAOC5	TMAOC4	TMAOC3	TMAOC2	TMAOC1	TMAOC0
At reset	X	X	X	X	X	X	X	X
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Binary counter is 8-bit up counter. If any data is written to compare register when the counting is stopped, binary counter is cleared to 0x00.

■ Timer A Binary Counter (TMABC: 0x03F71)

bp	7	6	5	4	3	2	1	0
Flag	TMABC7	TMABC6	TMABC5	TMABC4	TMABC3	TMABC2	TMABC1	TMABC0
At reset	0	0	0	0	0	0	0	0
Access	R	R	R	R	R	R	R	R

10.2.3 Timer Mode Registers

Timer mode registers are readable/writable registers that control Timer A.

■ Timer A Mode Register 1 (TMAMD1: 0x03F75)

bp	7	6	5	4	3	2	1	0
Flag	-	-	-	-	TMAEN	TMACK2	TMACK1	TMACK0
At reset	-	-	-	-	0	0	0	0
Access	-	-	-	-	R/W	R/W	R/W	R/W

bp	Flag	Description
7 to 4	-	-
3	TMAEN	Timer A count control 0: Halt the count 1: Operate the count
2 to 0	TMACK2 TMACK1 TMACK0	Clock source selection 000: fpll-div 001: fpll-div/2 010: fpll-div/4 011: fpll-div/8 100: fpll-div/16 101: fpll-div/32 110: fs/2 111: fs/4

■ Timer A Mode Register 2 (TMAMD2: 0x03F77)

bp	7	6	5	4	3	2	1	0
Flag	Reserved	PSCEN	-	-	-	-	-	-
At reset	0	0	-	-	-	-	-	-
Access	R/W	R/W	-	-	-	-	-	-

bp	Flag	Description
7	Reserved	Always set to "0".
6	PSCEN	Prescaler operation control 0: Disable 1: Enable
5 to 0	-	-

10.3 8-bit Simple Timer Count

10.3.1 8-bit Simple Timer Operation

8-bit simple timer contains one timer as an auxiliary function of [Chapter 9 8-bit Timers].

■ 8-bit Simple Timer Operation (Timer A)

The fundamental cycle of timer count is set by the clock source selection and the setting value of the compare register (TMAOC), in advance. If the binary counter (TMABC) reaches the setting value of the compare register, binary counter is cleared at the next count clock and counting is restarted from 0x00. This timer has no interrupt function.

The following table shows clock source that can be selected by timer.

Table:10.3.1 Clock Sources

Clock source	Time per Count	Timer A (8-bit)
fpll-div	50 ns	√
fpll-div/2	100 ns	√
fpll-div/4	200 ns	√
fpll-div/8	400 ns	√
fpll-div/16	0.8 μs	√
fpll-div/32	1.6 μs	√
fs/2	200 ns	√
fs/4	400 ns	√
fpll-div=20 MHz fs=fpll-div/2=10 MHz		

■ Count Timing of Timer Operation (Timer A)

Binary counter counts up with selected clock source as a count clock. The basic operation of the whole function of 8-bit timer is as follows:

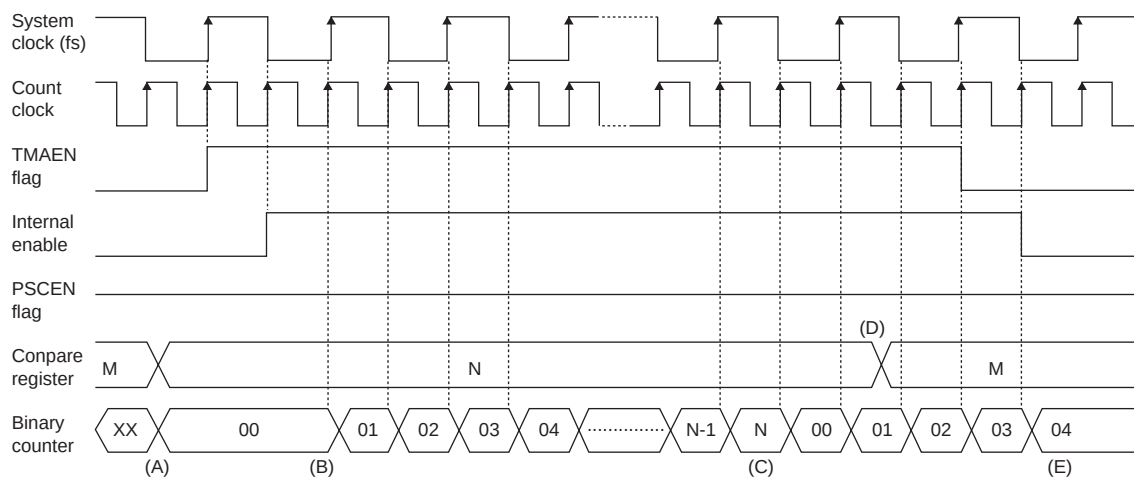


Figure:10.3.1 Count Timing of Timer Operation (Timer A)

- (A) If the value is written to compare register during the TMAEN flag is stopped ("0"), binary counter is cleared to 0x00.
- (B) When the TMAEN flag, the PSCEN flag start to operate ("1"), the internal enable will be turned on at the next Count Clock. Then binary counter begins counting up.
- (C) If binary counter reaches the value of compare register, binary counter is cleared to 0x00 at the next count clock and the counting is restarted.
- (D) Even if compare register is rewritten during the TMAEN flag is enabled ("1"), binary counter is not changed.
- (E) When the TMAEN flag stops operating ("0"), the internal enable will be turned off at the next count clock. As a result, binary counter stops counting.



Switch the count clock after the timer operation is stopped, since the counting is not generated correctly during the timer operation.



TMAEN flag of TMAMD and other bits should not be changed at the same time to operate correctly.



When binary counter reaches the value in compare register, the value of the internal count clock is inverted at the next count clock. So set compare register as:

Compare register setting = (count till the compare match -1)



If compare register is set to smaller number than binary counter during the count operation, binary counter counts up until overflow for the first time.



When compare register (TMAOC) is set to "00", clear binary counter before starting the operation.

10.4 Serial Transfer Clock Output

10.4.1 Operation

Serial transfer clock can be created by using the timer output signal.

Serial transfer clock operation by 8-bit timer (Timer A)

- Timer A: Serial Interface 0, 1, 2 and 4

■ Timing of Serial Transfer Clock (Timer A)

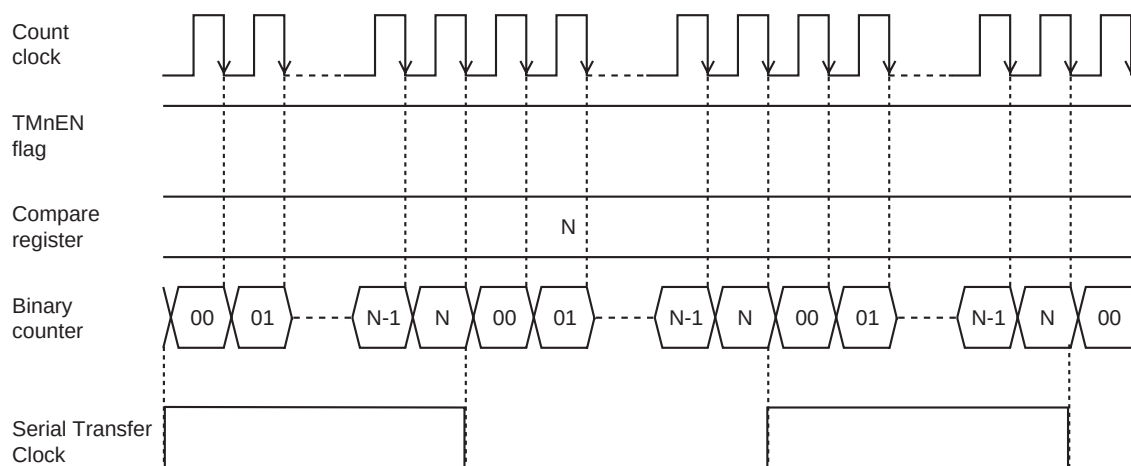


Figure:10.4.1 Timing of Serial Transfer Clock (Timer A)

- The timer frequency is 1/2 of the set frequency set by compare register.
- For the baud rate calculation and serial interface setup, refer to [Chapter 18 Serial Interface].

10.4.2 Setup Example

■ Serial Transfer Clock Setup Example (Timer A)

The setup and description described below show the procedures for generating transfer clock for full duplex UART (Serial Interface 0) using Timer A. In this case, select 300 bps and $f_s/2$ (at $f_s=2$ MHz) for baud rate and source clock of Timer A, respectively.

Setup Procedure	Description
(1) Stop the counter TMAMD1(0x03F75) bp3: TMAEN =0	(1) Set the TMAEN flag of Timer A mode register 1 (TMAMD1) to "0" to stop the timer A counting.
(2) Select the count clock source TMAMD1(0x03F75) bp2 to 0: TMACK2 to 0 =110	(2) Select the prescaler output to the clock source by the TMACK2 to 0 flags of TMAMD1 register.
(3) Set the baud rate TMAOC(0x03F73) =0xCF	(3) Set a value to set the baud rate for Timer A compare register (TMAOC) to be 300 bps. At that time, Timer A binary counter (TMABC) is initialized to 0x00.
(4) Enable the prescaler count TMAMD2(0x03F77) bp6: PSCEN =1	(4) Set the PSCEN flag of Timer A mode register 2 (TMAMD2) to "1" to operate the timer counting.
(5) Start the timer operation TMAMD1(0x03F75) bp3: TMAEN =1	(5) Set the TMAEN flag of TMAMD1 register to "1" to operate Timer A.

- TMABC counts up from 0x00. Timer A output is the clock of Serial Interface 0 at transmission and reception.
- For the setup of compare register and setup of serial interface operation, refer to [Chapter 18 Serial Interface].

11.1 Overview

This LSI consists of general-purpose 16-bit timer (Timer 7 and Timer 8). The 16-bit timer is equipped with double-buffered compare registers.

Timer n (high-precision 16-bit timer) has 2 sets of double-buffered compare register and 2 sets of independent interrupt functions such as Timer n interrupt and Timer n compare register 2 match interrupt.

Pins can be switched to TMnIOA/TMnIOB/TMnIOC.

Table:11.1.1 16-bit Timer Pin Functions

Table remarks ✓: With function -: Without function

Functions	Pin Name	MN101EF76K	MN101EF57G	MN101EF56K
TM7IOA	PA5	✓	✓	✓
TM7IOB	P02	✓	✓	✓
TM7IOC	P15	✓	-	✓
TM8IOA	PA6	✓	✓	✓
TM8IOB	P03	✓	✓	✓
TM8IOC	P16	✓	-	✓



On the text, if there is not much difference in the function between Pin A, B and C, "A", "B" and "C" of the pin names are omitted.



Be sure to halt the 16-bit timer function before changing the frequency of fpll-div by bp7 to bp4 of OSCCNT register.

11.1.1 Functions

Table:11.1.2 shows the functions of each timer.

Table:11.1.2 16-bit Timer Functions

	Timer 7 (High precision 16-bit timer)	Timer 8 (High precision 16-bit timer)
Interrupt source	TM7IRQ T70C2IRQ	TM8IRQ T80C2IRQ
Timer operation	√	√
Event count	TM7IOA input/ TM7IOB input/ TM7IOC input	TM8IOA input/ TM8IOB input/ TM8IOC input
Timer pulse output	TM7IOA output/ TM7IOB output/ TM7IOC output	TM8IOA output/ TM8IOB output/ TM8IOC output
Cascade connection	√	-
PWM output (duty is changeable)	TM7IOA output/ TM7IOB output/ TM7IOC output	TM8IOA output/ TM8IOB output/ TM8IOC output
High precision PWM output (duty/cycle are changeable)	TM7IOA output/ TM7IOB output/ TM7IOC output	TM8IOA output/ TM8IOB output/ TM8IOC output
IGBT output (duty is changeable)	TM7IOA output/ TM7IOB output/ TM7IOC output TM8IOA output/ TM8IOB output/ TM8IOC output	-
IGBT output (duty/cycle are changeable)	TM7IOA output/ TM7IOB output/ TM7IOC output TM8IOA output/ TM8IOB output/ TM8IOC output	-
Synchronous output	√	-
Real-time control	√	-
Capture function	√	√
High precision capture function	√	-
Pulse width measurement	√	√
Overflow detection (carry flag)	√	√
Clock source	fppll-div fppll-div/2 fppll-div/4 fppll-div/16 fs fs/2 fs/4 fs/16 TM7IO input TM7IO input/2 TM7IO input/4 TM7IO input/16 Synchronous TM7IO input Synchronous TM7IO input/2 Synchronous TM7IO input/4 Synchronous TM7IO input/16 Timer A output Timer A output/2 Timer A output/4 Timer A output/16 Serial Interface 0 transfer clock Serial Interface 0 transfer clock/2 Serial Interface 0 transfer clock/4 Serial Interface 0 transfer clock/16 Timer 6 compare match output Timer 6 compare match output/2 Timer 6 compare match output/4 Timer 6 compare match output/16	fppll-div fppll-div/2 fppll-div/4 fppll-div/16 fs fs/2 fs/4 fs/16 TM7IO input TM7IO input/2 TM7IO input/4 TM7IO input/16 Synchronous TM7IO input Synchronous TM7IO input/2 Synchronous TM7IO input/4 Synchronous TM7IO input/16 Timer A output Timer A output/2 Timer A output/4 Timer A output/16 Timer 6 compare match output Timer 6 compare match output/2 Timer 6 compare match output/4 Timer 6 compare match output/16
fppll-div: Machine clock (High speed oscillation for peripheral function) fs: System clock		

11.1.2 Block Diagram

■ Timer 7 Block Diagram (MN101EF76K)

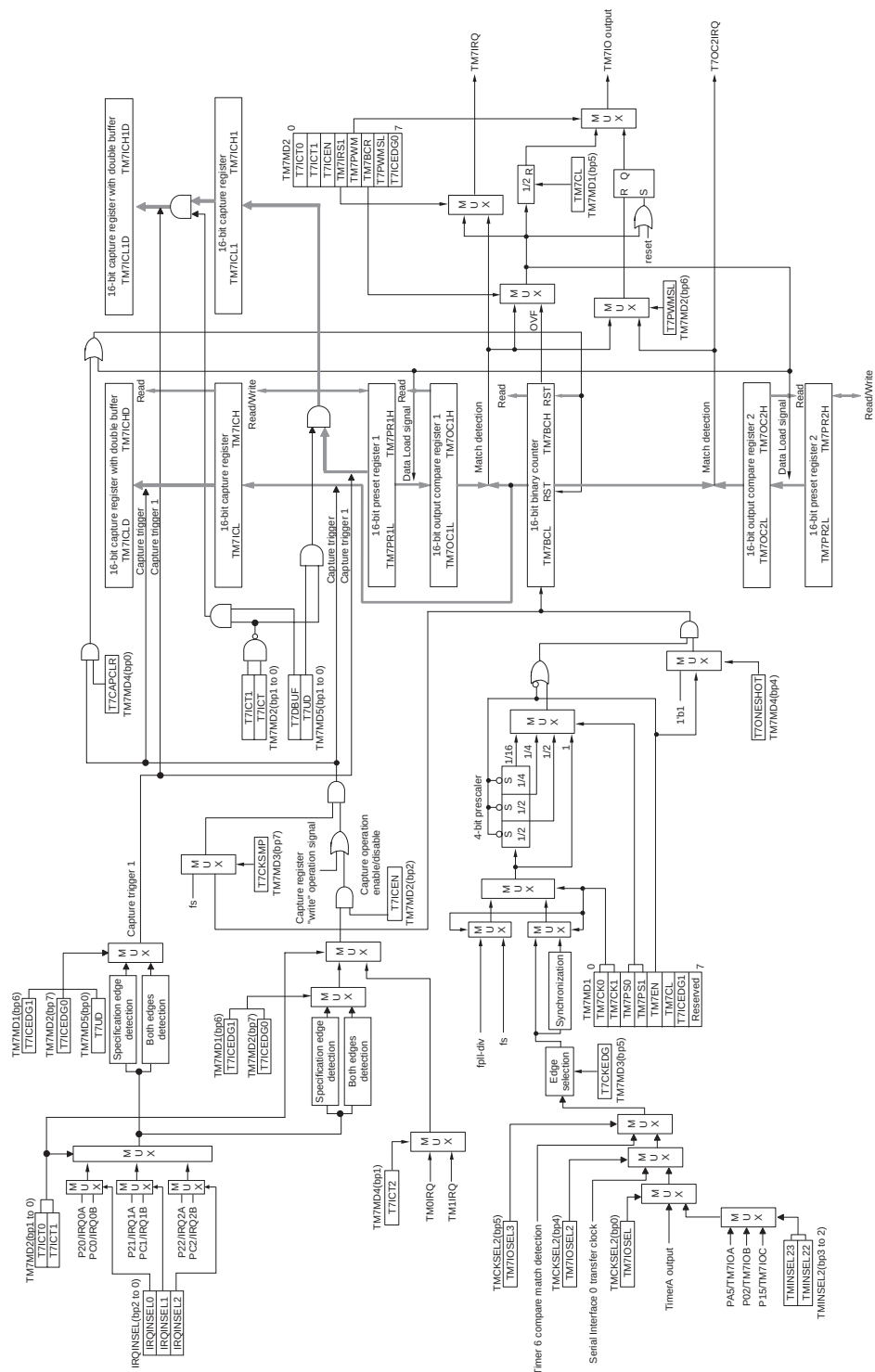


Figure:11.1.1 Timer 7 Block Diagram (MN101EF76K)

■ Timer 7 Block Diagram (MN101EF57G)

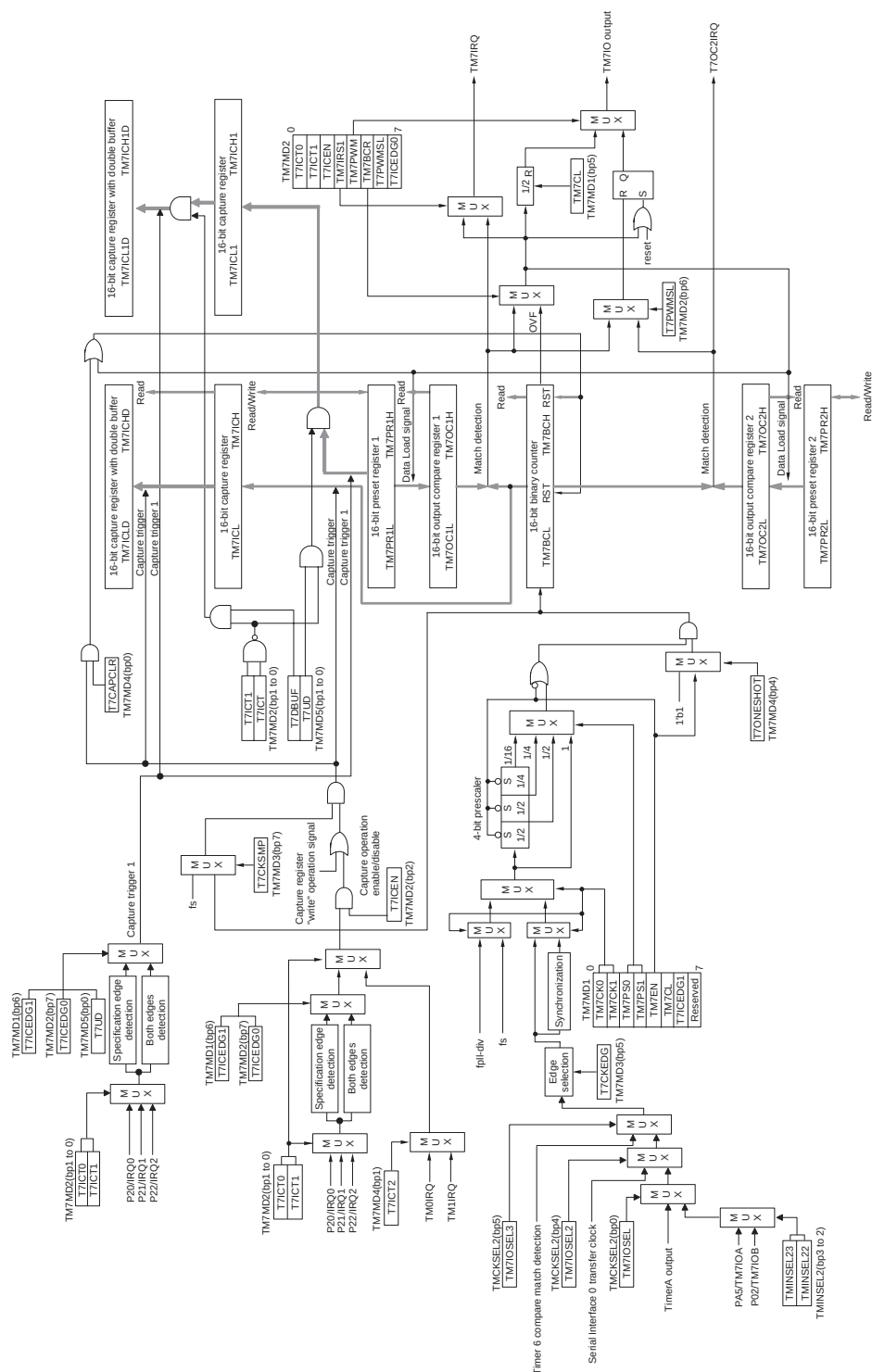


Figure:11.1.2 Timer 7 Block Diagram (MN101EF57G)

■ Timer 7 Block Diagram (MN101EF56K)

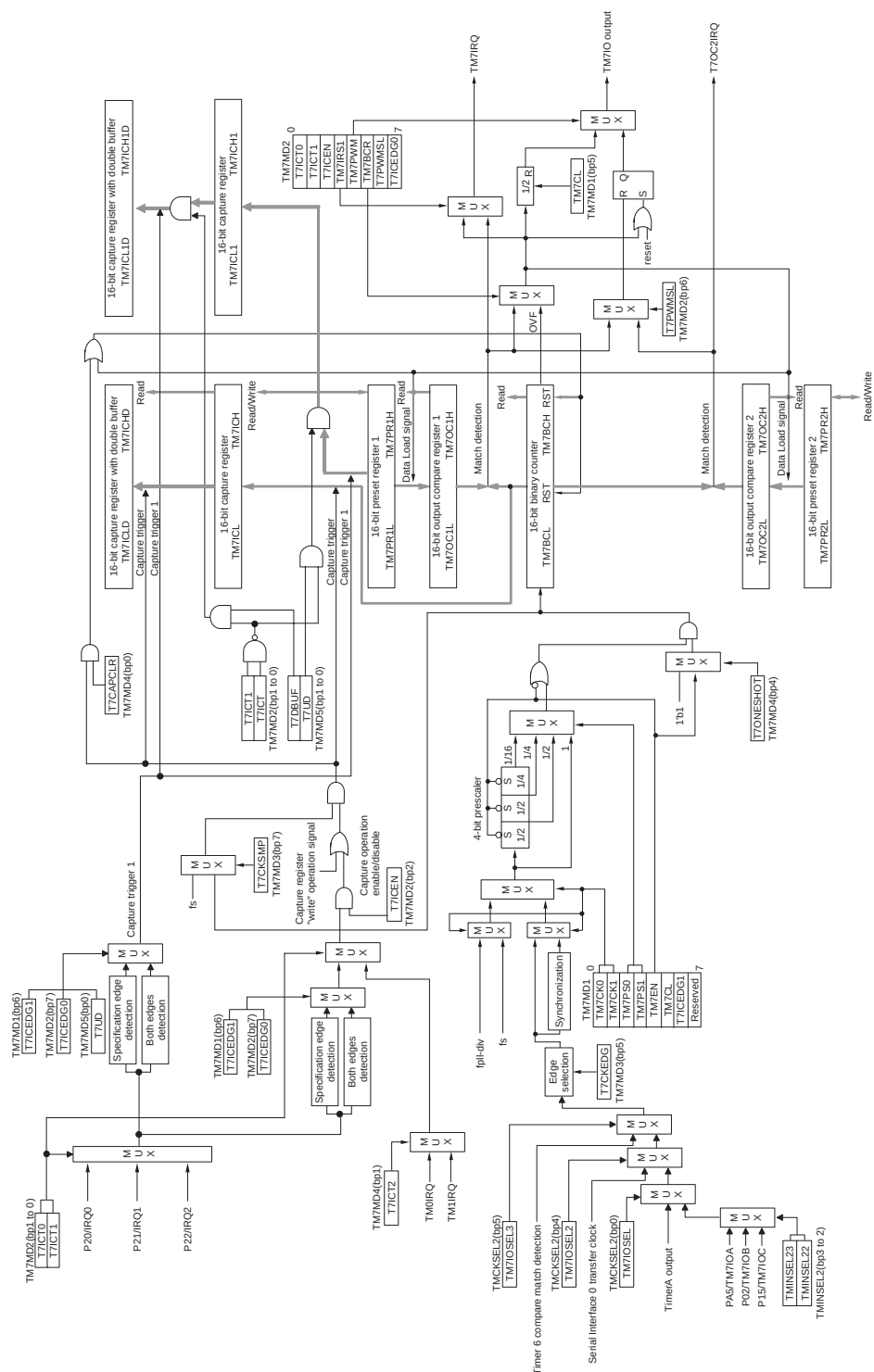


Figure:11.1.3 Timer 7 Block Diagram (MN101EF56K)

■ Timer 8 Block Diagram (MN101EF76K)

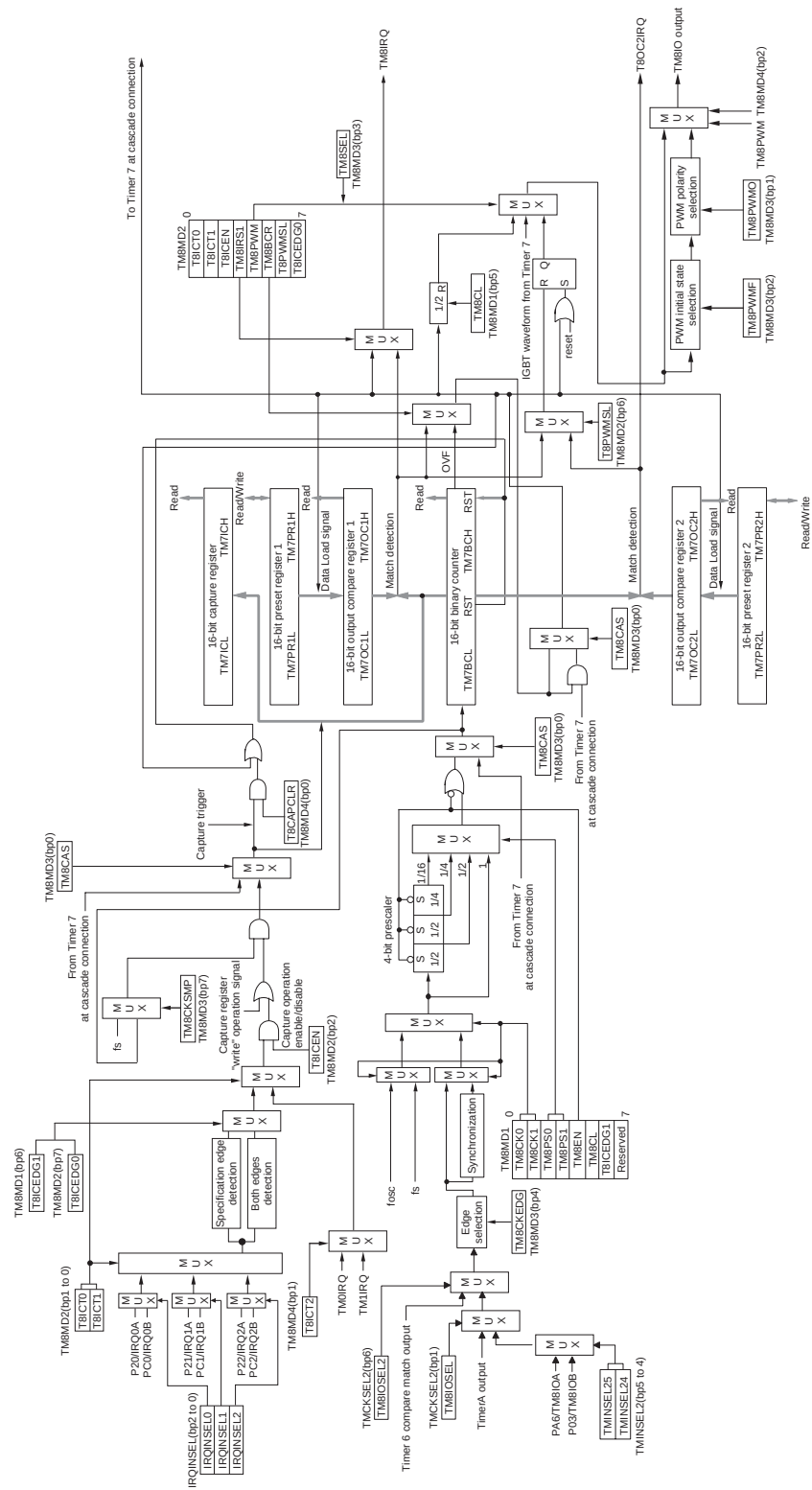


Figure:11.1.4 Timer 8 Block Diagram (MN101EF76K)

■ Timer 8 Block Diagram (MN101EF57G)

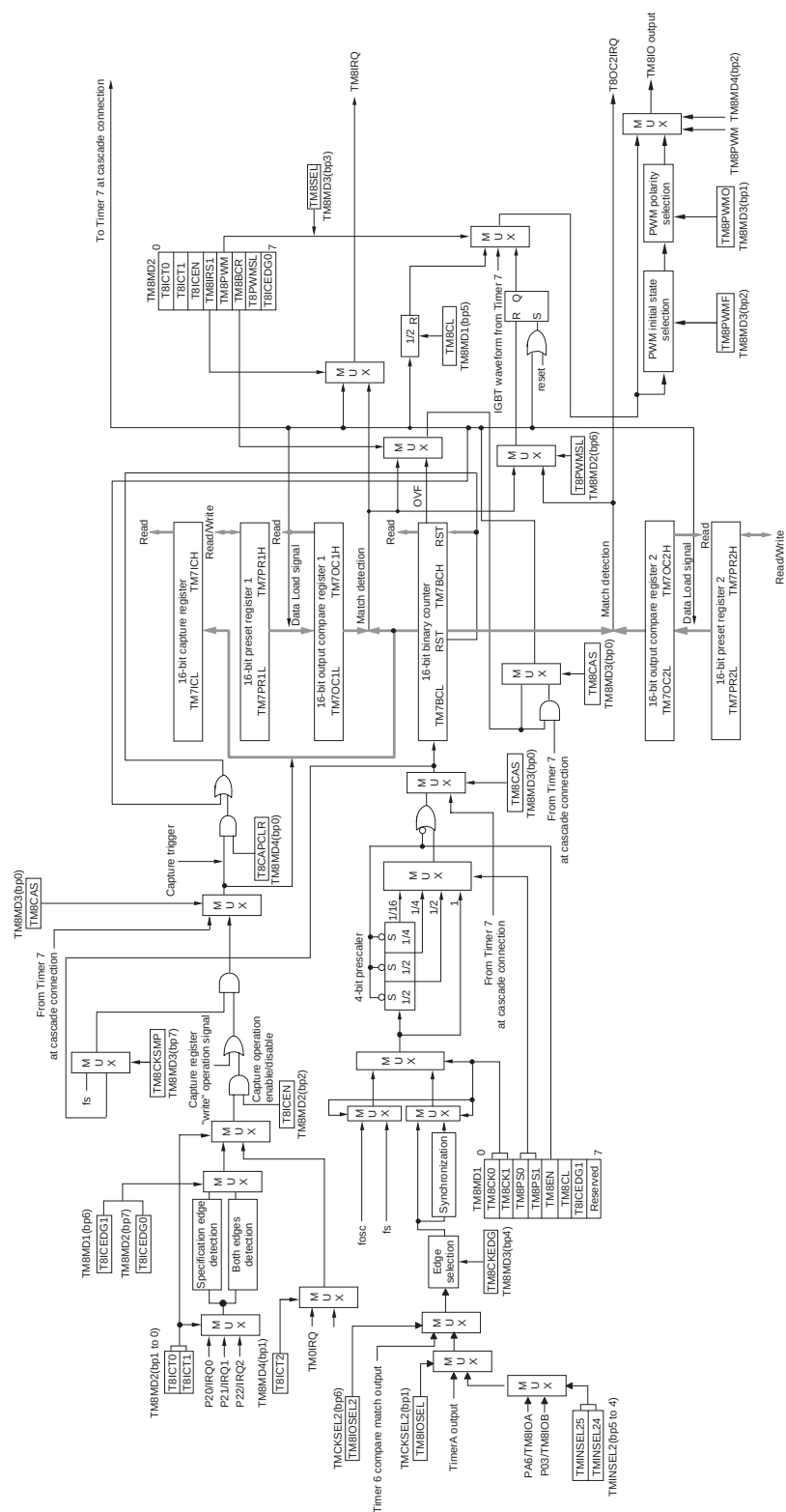


Figure:11.1.5 Timer 8 Block Diagram (MN101EF57G)

■ Timer 8 Block Diagram (MN101EF56K)

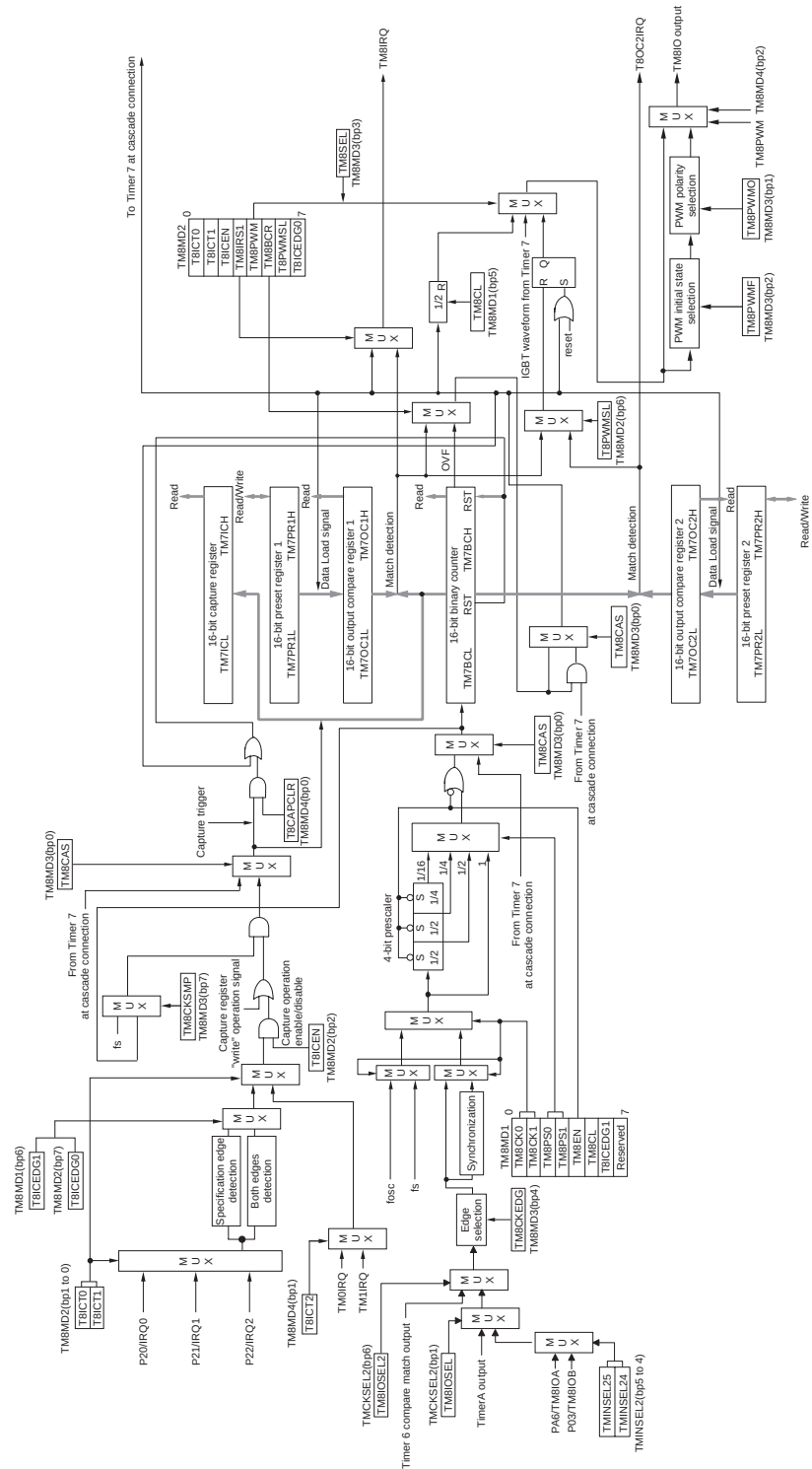


Figure:11.1.6 Timer 8 Block Diagram (MN101EF56K)

11.2 Control Registers

Timer 7 consists of binary counter (TM7BC), compare register 1 (TM7OC1) and its double buffer preset register 1 (TM7PR1), compare register 2 (TM7OC2) and its double buffer preset register 2 (TM7PR2), capture register (TM7IC) and its double buffer capture register D (TM7ICD), capture register 1 (TM7IC1) and its double buffer capture register D (TM7ICD1), dead time preset register 1 (TM7DPR1) and dead time preset register 2 (TM7DPR2).

Timer 7 is controlled by mode register 1 (TM7MD1), mode register 2 (TM7MD2), mode register 3 (TM7MD3), mode register 4 (TM7MD4) and mode register 5 (TM7MD5).

Timer 8 consists of binary counter (TM8BC), compare register 1 (TM8OC1) and its double buffer preset register 1 (TM8PR1), compare register 2 (TM8OC2) and its double buffer preset register 2 (TM8PR2) and capture register (TM8IC).

Timer 8 is controlled by mode register 1 (TM8MD1), mode register 2 (TM8MD2), mode register 3 (TM8MD3) and mode register 4 (TM8MD4).

11.2.1 Registers

Table:11.2.1 shows the registers that control Timer 7 and 8.

Table:11.2.1 16-bit Timer Control Registers

	Register	Address	R/W	Function	Page
Timer 7	TM7BCL	0x03F80	R	Timer 7 binary counter (lower 8 bits)	XI-14
	TM7BCH	0x03F81	R	Timer 7 binary counter (upper 8 bits)	XI-14
	TM7OC1L	0x03F82	R	Timer 7 compare register 1 (lower 8 bits)	XI-12
	TM7OC1H	0x03F83	R	Timer 7 compare register 1 (upper 8 bits)	XI-12
	TM7PR1L	0x03F84	R/W	Timer 7 preset register 1 (lower 8 bits)	XI-13
	TM7PR1H	0x03F85	R/W	Timer 7 preset register 1 (upper 8 bits)	XI-13
	TM7ICL	0x03F86	R	Timer 7 input capture register (lower 8 bits)	XI-15
	TM7ICH	0x03F87	R	Timer 7 input capture register (upper 8 bits)	XI-15
	TM7ICLD	0x03FA2	R	Timer 7 input capture register D (lower 8 bits)	XI-16
	TM7ICHd	0x03FA3	R	Timer 7 input capture register D (upper 8 bits)	XI-16
	TM7ICL1	0x03FA0	R	Timer 7 input capture register 1 (lower 8 bits)	XI-15
	TM7ICH1	0x03FA1	R	Timer 7 input capture register 1 (upper 8 bits)	XI-15
	TM7ICLD1	0x03FA4	R	Timer 7 input capture register D1 (lower 8 bits)	XI-16
	TM7ICHd1	0x03FA5	R	Timer 7 input capture register D1 (upper 8 bits)	XI-16
	TM7MD1	0x03F88	R/W	Timer 7 mode register 1	XI-21
	TM7MD2	0x03F89	R/W	Timer 7 mode register 2	XI-22
	TM7OC2L	0x03F8A	R	Timer 7 compare register 2 (lower 8 bits)	XI-12
	TM7OC2H	0x03F8B	R	Timer 7 compare register 2 (upper 8 bits)	XI-12
	TM7PR2L	0x03F8C	R/W	Timer 7 preset register 2 (lower 8 bits)	XI-13
	TM7PR2H	0x03F8D	R/W	Timer 7 preset register 2 (upper 8 bits)	XI-13
	TM7MD3	0x03F9E	R/W	Timer 7 mode register 3	XI-23
	TM7MD4	0x03F7E	R/W	Timer 7 mode register 4	XI-24
	TM7MD5	0x03FA6	R/W	Timer 7 mode register 5	XI-25
	TM7DPR1	0x03F8E	R/W	Timer 7 dead time preset register 1	XI-17
	TM7DPR2	0x03F8F	R/W	Timer 7 dead time preset register 2	XI-17
Timer 8	TM8BCL	0x03F90	R	Timer 8 binary counter (lower 8 bits)	XI-14
	TM8BCH	0x03F91	R	Timer 8 binary counter (upper 8 bits)	XI-14
	TM8OC1L	0x03F92	R	Timer 8 compare register 1 (lower 8 bits)	XI-12
	TM8OC1H	0x03F93	R	Timer 8 compare register 1 (upper 8 bits)	XI-12
	TM8PR1L	0x03F94	R/W	Timer 8 preset register 1 (lower 8 bits)	XI-13
	TM8PR1H	0x03F95	R/W	Timer 8 preset register 1 (upper 8 bits)	XI-13
	TM8ICL	0x03F96	R	Timer 8 input capture register (lower 8 bits)	XI-15
	TM8ICH	0x03F97	R	Timer 8 input capture register (upper 8 bits)	XI-15
	TM8MD1	0x03F98	R/W	Timer 8 mode register 1	XI-21
	TM8MD2	0x03F99	R/W	Timer 8 mode register 2	XI-22
	TM8OC2L	0x03F9A	R	Timer 8 compare register 2 (lower 8 bits)	XI-12
	TM8OC2H	0x03F9B	R	Timer 8 compare register 2 (upper 8 bits)	XI-12
	TM8PR2L	0x03F9C	R/W	Timer 8 preset register 2 (lower 8 bits)	XI-13
	TM8PR2H	0x03F9D	R/W	Timer 8 preset register 2 (upper 8 bits)	XI-13
	TM8MD3	0x03F9F	R/W	Timer 8 mode register 3	XI-23
	TM8MD4	0x03F7F	R/W	Timer 8 mode register 4	XI-24
	TMCKSEL2	0x03FB1	R/W	Timer clock selection register 2	XI-30
	TMINSEL2	0x03FB3	R/W	Timer input selection register 2	XI-32

11.2.2 Programmable Timer Registers

Timer 7 has a set of 16-bit programmable timer registers, which contains a compare register, a preset register, a binary counter and a capture register. Each register has 2 sets of 8-bit register. They can be operated by 16-bit access.

A compare register is a 16-bit register which stores the values which will use to compare with the binary counter.

■ Timer 7 Compare Register 1 Lower 8 bits (TM7OC1L: 0x03F82)

bp	7	6	5	4	3	2	1	0
Flag	TM7OC1L7	TM7OC1L6	TM7OC1L5	TM7OC1L4	TM7OC1L3	TM7OC1L2	TM7OC1L1	TM7OC1L0
At reset	X	X	X	X	X	X	X	X
Access	R	R	R	R	R	R	R	R

■ Timer 7 Compare Register 1 Upper 8 bits (TM7OC1H: 0x03F83)

bp	7	6	5	4	3	2	1	0
Flag	TM7OC1H7	TM7OC1H6	TM7OC1H5	TM7OC1H4	TM7OC1H3	TM7OC1H2	TM7OC1H1	TM7OC1H0
At reset	X	X	X	X	X	X	X	X
Access	R	R	R	R	R	R	R	R

■ Timer 7 Compare Register 2 Lower 8 bits (TM7OC2L: 0x03F8A)

bp	7	6	5	4	3	2	1	0
Flag	TM7OC2L7	TM7OC2L6	TM7OC2L5	TM7OC2L4	TM7OC2L3	TM7OC2L2	TM7OC2L1	TM7OC2L0
At reset	X	X	X	X	X	X	X	X
Access	R	R	R	R	R	R	R	R

■ Timer 7 Compare Register 2 Upper 8 bits (TM7OC2H: 0x03F8B)

bp	7	6	5	4	3	2	1	0
Flag	TM7OC2H7	TM7OC2H6	TM7OC2H5	TM7OC2H4	TM7OC2H3	TM7OC2H2	TM7OC2H1	TM7OC2H0
At reset	X	X	X	X	X	X	X	X
Access	R	R	R	R	R	R	R	R

Timer 7 preset registers 1 and 2 are buffer registers of compare register 1 and 2 of Timer 7. If the set value is written to Timer 7 preset registers 1 and 2 when the counting is stopped, the same set value is loaded to Timer 7 compare register. If the set value is written to Timer 7 preset registers 1 and 2 during counting, the set value of Timer 7 preset registers 1 and 2 is loaded to Timer 7 compare registers 1 and 2 at the timing when Timer 7 binary counter is cleared.

Also, if the set value is written to Timer 7 preset registers 1 and 2 during IGBT operation, the set value of Timer 7 preset registers 1 and 2 is loaded to Timer 7 compare registers 1 and 2 at the timing that IGBT is disabled.

■ Timer 7 Preset Register 1 Lower 8 bits (TM7PR1L: 0x03F84)

bp	7	6	5	4	3	2	1	0
Flag	TM7PR1L7	TM7PR1L6	TM7PR1L5	TM7PR1L4	TM7PR1L3	TM7PR1L2	TM7PR1L1	TM7PR1L0
At reset	X	X	X	X	X	X	X	X
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

■ Timer 7 Preset Register 1 Upper 8 bits (TM7PR1H: 0x03F85)

bp	7	6	5	4	3	2	1	0
Flag	TM7PR1H7	TM7PR1H6	TM7PR1H5	TM7PR1H4	TM7PR1H3	TM7PR1H2	TM7PR1H1	TM7PR1H0
At reset	X	X	X	X	X	X	X	X
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

■ Timer 7 Preset Register 2 Lower 8 bits (TM7PR2L: 0x03F8C)

bp	7	6	5	4	3	2	1	0
Flag	TM7PR2L7	TM7PR2L6	TM7PR2L5	TM7PR2L4	TM7PR2L3	TM7PR2L2	TM7PR2L1	TM7PR2L0
At reset	X	X	X	X	X	X	X	X
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

■ Timer 7 Preset Register 2 Upper 8 bits (TM7PR2H: 0x03F8D)

bp	7	6	5	4	3	2	1	0
Flag	TM7PR2H7	TM7PR2H6	TM7PR2H5	TM7PR2H4	TM7PR2H3	TM7PR2H2	TM7PR2H1	TM7PR2H0
At reset	X	X	X	X	X	X	X	X
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W



Under the following conditions, be sure to select a system clock (fs) as a clock source of timer to set preset register 1 and 2 of Timer 7 and Timer 8.

<Conditions>

- The frequency of a system clock (fs) is faster than twice the frequency of a high-speed clock for peripheral functions (fppll-div), and
- one of the following conditions is met.

	Relevant Block	TM7MD1 (Timer 7 mode register 1)		TM7MD3 (Timer 7 mode register 3)	
		TM7CK1 to 0	TM7EN	T7IGBTEN	TM7BUFSEL
Condition 1	Timer 7	00 (Clock source: fppll-div)	0 (Stop counter)	0 (Disable IGBT operation)	0 (Double buffer)
Condition 2	Timer 7	00 (Clock source: fppll-div)	-	1 (Enable IGBT operation)	0 (Double buffer)

	Relevant Block	TM8MD1 (Timer 8 mode register 1)		TM8MD3 (Timer 8 mode register 3)	
		TM8CK1 to 0	TM8EN	TM8BUFSEL	
Condition 3	Timer 8	00 (Clock source: fppll-div)	0 (Stop counter)	0 (Double buffer)	

Binary counter is a 16-bit up counter. If the set value is written to preset register when the counting is stopped, binary counter is cleared to 0x0000. At IGBT setting, when IGBT operation is stopped, binary counter is cleared to 0x0000. Also, by setting T7CAPCLR, binary counter is cleared to 0x0000 at capture.

■ Timer 7 Binary Counter Lower 8 bits (TM7BCL: 0x03F80)

bp	7	6	5	4	3	2	1	0
Flag	TM7BCL7	TM7BCL6	TM7BCL5	TM7BCL4	TM7BCL3	TM7BCL2	TM7BCL1	TM7BCL0
At reset	X	X	X	X	X	X	X	X
Access	R	R	R	R	R	R	R	R

■ Timer 7 Binary Counter Upper 8 bits (TM7BCH: 0x03F81)

bp	7	6	5	4	3	2	1	0
Flag	TM7BCH7	TM7BCH6	TM7BCH5	TM7BCH4	TM7BCH3	TM7BCH2	TM7BCH1	TM7BCH0
At reset	X	X	X	X	X	X	X	X
Access	R	R	R	R	R	R	R	R

Input capture register is a register that holds the value loaded from the binary counter by the capture trigger. The capture trigger is generated by the external interrupt, Timer 0 interrupt, Timer 1 interrupt and when an arbitrary value is written to the input capture register. (Directly writing to the register by program is disabled.)

■ Timer 7 Input Capture Register Lower 8 bits (TM7ICL: 0x03F86)

bp	7	6	5	4	3	2	1	0
Flag	TM7ICL7	TM7ICL6	TM7ICL5	TM7ICL4	TM7ICL3	TM7ICL2	TM7ICL1	TM7ICL0
At reset	X	X	X	X	X	X	X	X
Access	R	R	R	R	R	R	R	R

■ Timer 7 Input Capture Register Upper 8 bits (TM7ICH: 0x03F87)

bp	7	6	5	4	3	2	1	0
Flag	TM7ICH7	TM7ICH6	TM7ICH5	TM7ICH4	TM7ICH3	TM7ICH2	TM7ICH1	TM7ICH0
At reset	X	X	X	X	X	X	X	X
Access	R	R	R	R	R	R	R	R

■ Timer 7 Input Capture Register 1 Lower 8 bits (TM7ICL1: 0x03FA0)

bp	7	6	5	4	3	2	1	0
Flag	TM7ICL17	TM7ICL16	TM7ICL15	TM7ICL14	TM7ICL13	TM7ICL12	TM7ICL11	TM7ICL10
At reset	X	X	X	X	X	X	X	X
Access	R	R	R	R	R	R	R	R

■ Timer 7 Input Capture Register 1 Upper 8 bits (TM7ICH1: 0x03FA1)

bp	7	6	5	4	3	2	1	0
Flag	TM7ICH17	TM7ICH16	TM7ICH15	TM7ICH14	TM7ICH13	TM7ICH12	TM7ICH11	TM7ICH10
At reset	X	X	X	X	X	X	X	X
Access	R	R	R	R	R	R	R	R



Timer 7 input capture register 1 is valid only in case when T7UD flag of TM7MD5 register is set to "High".



The capture trigger for Timer 7 input capture register 1 is used as an input signal of the External Interrupt 0, 1 and 2, or used as the writing to software. Timer 0, 1 interrupts can not be used.

Timer 7 input capture D register is a buffer register for Timer 7 input capture register. Timer 7 input capture register is updated when T7DBUF of Timer 7 mode register 5 is "High", the value of Timer 7 input capture register is loaded to Timer 7 input capture register D.

■ Timer 7 Input Capture Register D Lower 8 bits (TM7ICLD: 0x03FA2)

bp	7	6	5	4	3	2	1	0
Flag	TM7ICLD7	TM7ICLD6	TM7ICLD5	TM7ICLD4	TM7ICLD3	TM7ICLD2	TM7ICLD1	TM7ICLD0
At reset	X	X	X	X	X	X	X	X
Access	R	R	R	R	R	R	R	R

■ Timer 7 Input Capture Register D Upper 8 bits (TM7ICHD: 0x03FA3)

bp	7	6	5	4	3	2	1	0
Flag	TM7ICHD7	TM7ICHD6	TM7ICHD5	TM7ICHD4	TM7ICHD3	TM7ICHD2	TM7ICHD1	TM7ICHD0
At reset	X	X	X	X	X	X	X	X
Access	R	R	R	R	R	R	R	R

■ Timer 7 Input Capture Register D1 Lower 8 bits (TM7ICLD1: 0x03FA4)

bp	7	6	5	4	3	2	1	0
Flag	TM7ICLD 17	TM7ICLD 16	TM7ICLD 15	TM7ICLD 14	TM7ICLD 13	TM7ICLD 12	TM7ICLD 11	TM7ICLD 10
At reset	X	X	X	X	X	X	X	X
Access	R	R	R	R	R	R	R	R

■ Timer 7 Input Capture Register D1 Upper 8 bits (TM7ICHD1: 0x03FA5)

bp	7	6	5	4	3	2	1	0
Flag	TM7ICHD 17	TM7ICHD 16	TM7ICHD 15	TM7ICHD 14	TM7ICHD 13	TM7ICHD 12	TM7ICHD 11	TM7ICHD 10
At reset	X	X	X	X	X	X	X	X
Access	R	R	R	R	R	R	R	R



Timer 7 input capture register D is valid only when T7DBUF flag of TM7MD5 register is set to "1".



Timer 7 input capture register D1 is valid only when T7UD flag and T7DBUF flag of TM7MD5 register is set to "1".



External Interrupt 0, 1, 2 or the writing to software is used as the capture trigger for Timer 7 input capture register 1. Timer 0 or 1 interrupt can not be used.



When Timer 7 input capture register is valid, Timer 7 input capture register executes capturing only at the rising edge of the capture trigger and Timer 7 input capture register 1 executes capturing only at the falling edge of the capture trigger.

■ Timer 7 Dead Time Preset Register 1 (TM7DPR1: 0x03F8E)

bp	7	6	5	4	3	2	1	0
Flag	TM7DPR 17	TM7DPR 16	TM7DPR 15	TM7DPR 14	TM7DPR 13	TM7DPR 12	TM7DPR 11	TM7DPR 10
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

■ Timer 7 Dead Time Preset Register 2 (TM7DPR2: 0x03F8F)

bp	7	6	5	4	3	2	1	0
Flag	TM7DPR 27	TM7DPR 26	TM7DPR 25	TM7DPR 24	TM7DPR 23	TM7DPR 22	TM7DPR 21	TM7DPR 20
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Timer 8 has a set of 16-bit programmable timer registers, which contains a compare register, a preset register, a binary counter and a capture register. Each register has 2 sets of 8-bit register. They can be operated by 16-bit access.

A compare register is a 16-bit register which stores the values which use to compare with the binary counter. The compare value which have been written to the preset register is loaded.

■ Timer 8 Compare Register 1 Lower 8 bits (TM8OC1L: 0x03F92)

bp	7	6	5	4	3	2	1	0
Flag	TM8OC1 L7	TM8OC1 L6	TM8OC1 L5	TM8OC1 L4	TM8OC1 L3	TM8OC1 L2	TM8OC1 L1	TM8OC1 L0
At reset	X	X	X	X	X	X	X	X
Access	R	R	R	R	R	R	R	R

■ Timer 8 Compare Register 1 Upper 8 bits (TM8OC1H: 0x03F93)

bp	7	6	5	4	3	2	1	0
Flag	TM8OC1 H7	TM8OC1 H6	TM8OC1 H5	TM8OC1 H4	TM8OC1 H3	TM8OC1 H2	TM8OC1 H1	TM8OC1 H0
At reset	X	X	X	X	X	X	X	X
Access	R	R	R	R	R	R	R	R

■ Timer 8 Compare Register 2 Lower 8 bits (TM8OC2L: 0x03F9A)

bp	7	6	5	4	3	2	1	0
Flag	TM8OC2 L7	TM8OC2 L6	TM8OC2 L5	TM8OC2 L4	TM8OC2 L3	TM8OC2 L2	TM8OC2 L1	TM8OC2 L0
At reset	X	X	X	X	X	X	X	X
Access	R	R	R	R	R	R	R	R

■ Timer 8 Compare Register 2 Upper 8 bits (TM8OC2H: 0x03F9B)

bp	7	6	5	4	3	2	1	0
Flag	TM8OC2 H7	TM8OC2 H6	TM8OC2 H5	TM8OC2 H4	TM8OC2 H3	TM8OC2 H2	TM8OC2 H1	TM8OC2 H0
At reset	X	X	X	X	X	X	X	X
Access	R	R	R	R	R	R	R	R

Timer 8 preset registers 1 and 2 are buffer registers of compare registers 1 and 2 of Timer 8. If the set value is written to Timer 8 preset registers 1 and 2 when the counting is stopped, the same set value is loaded to Timer 8 compare register. If the set value is written to Timer 8 preset registers 1 and 2 during counting, the set value of Timer 8 preset registers 1 and 2 is loaded to Timer 8 compare registers 1 and 2 at the timing when Timer 8 binary counter is cleared.

■ Timer 8 Preset Register 1 Lower 8 bits (TM8PR1L: 0x03F94)

bp	7	6	5	4	3	2	1	0
Flag	TM8PR1 L7	TM8PR1 L6	TM8PR1 L5	TM8PR1 L4	TM8PR1 L3	TM8PR1 L2	TM8PR1 L1	TM8PR1 L0
At reset	X	X	X	X	X	X	X	X
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

■ Timer 8 Preset Register 1 Upper 8 bits (TM8PR1H: 0x03F95)

bp	7	6	5	4	3	2	1	0
Flag	TM8PR1 H7	TM8PR1 H6	TM8PR1 H5	TM8PR1 H4	TM8PR1 H3	TM8PR1 H2	TM8PR1 H1	TM8PR1 H0
At reset	X	X	X	X	X	X	X	X
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

■ Timer 8 Preset Register 2 Lower 8 bits (TM8PR2L: 0x03F9C)

bp	7	6	5	4	3	2	1	0
Flag	TM8PR2 L7	TM8PR2 L6	TM8PR2 L5	TM8PR2 L4	TM8PR2 L3	TM8PR2 L2	TM8PR2 L1	TM8PR2 L0
At reset	X	X	X	X	X	X	X	X
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

■ Timer 8 Preset Register 2 Upper 8 bits (TM8PR2H: 0x03F9D)

bp	7	6	5	4	3	2	1	0
Flag	TM8PR2 H7	TM8PR2 H6	TM8PR2 H5	TM8PR2 H4	TM8PR2 H3	TM8PR2 H2	TM8PR2 H1	TM8PR2 H0
At reset	X	X	X	X	X	X	X	X
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Binary counter is a 16-bit up counter. If the set value is written to preset register when the counting is stopped, binary counter is cleared to 0x0000. Also, by setting T8CAPCLR, binary counter is cleared to 0x0000 at capture.

■ Timer 8 Binary Counter Lower 8 bits (TM8BCL: 0x03F90)

bp	7	6	5	4	3	2	1	0
Flag	TM8BC L7	TM8BC L6	TM8BC L5	TM8BC L4	TM8BC L3	TM8BC L2	TM8BC L1	TM8BC L0
At reset	X	X	X	X	X	X	X	X
Access	R	R	R	R	R	R	R	R

■ Timer 8 Binary Counter Upper 8 bits (TM8BCH: 0x03F91)

bp	7	6	5	4	3	2	1	0
Flag	TM8BC H7	TM8BC H6	TM8BC H5	TM8BC H4	TM8BC H3	TM8BC H2	TM8BC H1	TM8BC H0
At reset	X	X	X	X	X	X	X	X
Access	R	R	R	R	R	R	R	R

Input capture register is a register that holds the value loaded from the binary counter by the capture trigger. The capture trigger is generated by the external interrupt, Timer 0 interrupt, Timer 1 interrupt and when an arbitrary value is written to the input capture register. (Directly writing to the register by program is disabled.)

■ Timer 8 Input Capture Register Lower 8 bits (TM8ICL: 0x03F96)

bp	7	6	5	4	3	2	1	0
Flag	TM8ICL7	TM8ICL6	TM8ICL5	TM8ICL4	TM8ICL3	TM8ICL2	TM8ICL1	TM8ICL0
At reset	X	X	X	X	X	X	X	X
Access	R	R	R	R	R	R	R	R

■ Timer 8 Input Capture Register Upper 8 bits (TM8ICH: 0x03F97)

bp	7	6	5	4	3	2	1	0
Flag	TM8ICH7	TM8ICH6	TM8ICH5	TM8ICH4	TM8ICH3	TM8ICH2	TM8ICH1	TM8ICH0
At reset	X	X	X	X	X	X	X	X
Access	R	R	R	R	R	R	R	R

11.2.3 Timer Mode Registers

The following are readable/writable registers which control Timer 7.

■ Timer 7 Mode Register 1 (TM7MD1: 0x03F88)

bp	7	6	5	4	3	2	1	0
Flag	Reserved	T7ICEDG1	TM7CL	TM7EN	TM7PS1	TM7PS0	TM7CK1	TM7CK0
At reset	0	0	1	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7	Reserved	Set always to "0".
6	T7ICEDG1	Capture trigger edge selection 0: Falling edge 1: Rising edge
5	TM7CL	Timer output reset signal 0: Enable timer output 1: Disable timer output (reset)
4	TM7EN	Timer count control 0: Halt the count 1: Start the count
3 to 2	TM7PS1 TM7PS0	Count clock selection 00: 1/1 of clock 01: 1/2 of clock 10: 1/4 of clock 11: 1/16 of clock
1 to 0	TM7CK1 TM7CK0	Clock source selection 00: fpll-div 01: fs 10: TMIO input 11: Synchronous TMIO input

■ Timer 7 Mode Register 2 (TM7MD2: 0x03F89)

bp	7	6	5	4	3	2	1	0
Flag	T7ICEDG0	T7PWMSL	TM7BCR	TM7PWM	TM7IRS1	T7ICEN	T7ICT1	T7ICT0
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7	T7ICEDG0	Capture trigger edge selection 0: Select the both edges 1: Select the specified edge
6	T7PWMSL	PWM mode selection 0: Set duty by OC1 1: Set duty by OC2
5	TM7BCR	Timer 7 count clear factor selection 0: Full count OVF 1: Match of TM7BC and TM7OC1
4	TM7PWM	Timer output waveform selection 0: Output timer 1: Output PWM
3	TM7IRS1	Timer 7 interrupt factor selection 0: Counter clear 1: Match of TM7BC and TM7OC1
2	T7ICEN	Input capture operation enable select flag 0: Disable capture operation 1: Enable capture operation
1 to 0	T7ICT1 T7ICT0	Capture trigger selection 00: External Interrupt 0 input signal 01: External Interrupt 1 input signal 10: External Interrupt 2 input signal 11: Timer interrupt



When T7ICT1 to 0 flags select “timer interrupt”, the timer interrupt set by T7ICT2 flag of TM7MD4 register is used as a capture trigger.

■ Timer 7 Mode Register 3 (TM7MD3: 0x03F9E)

bp	7	6	5	4	3	2	1	0
Flag	TM7CK SMP	TM7BUF SEL	TM7CK EDG	T7IGBT TR	T7IGBT DT	T7IGBT EN	T7IGBT1	T7IGBT0
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7	TM7CKSMP	Capture sampling selection 0: Count clock 1: fs
6	TM7BUFSEL	Buffer selection 0: Double buffer 1: Single buffer
5	TM7CKEDG	TM7IO count edge selection 0: Rising edge 1: Both edges
4	T7IGBTTR	IGBT trigger level selection 0: High 1: Low
3	T7IGBTDT	IGBT dead time inserting timing 0: Falling edge standard 1: Rising edge standard
2	T7IGBTEN	IGBT operation enable 0: Disable 1: Enable
1 to 0	T7IGBT1 T7IGBT0	IGBT/Timer start up factor selection 00: Timer 7 counting operation 01: External Interrupt 0 input signal 10: External Interrupt 1 input signal 11: External Interrupt 2 input signal



When IGBT output is not selected, set as;
T7IGBTEN = 0
T7IGBT1 to 0 = 00



TM7CKEDG flag must be set to "0" when Timer 6 compare match output is selected as a count clock by TM7IOSEL and TM7IOSEL2 to 3 flags of TMCKSEL2 register.

■ Timer 7 Mode Register 4 (TM7MD4: 0x03F7E)

bp	7	6	5	4	3	2	1	0
Flag	Reserved	T7IGBT SFT	T7IGBT CNT	T7ONE SHOT	T7ONED	-	T7ICT2	T7CAP CLR
At reset	0	0	0	0	0	-	0	0
Access	R/W	R/W	R/W	R/W	R/W	-	R/W	R/W

bp	Flag	Description
7	Reserved	Always set to "0"
6	T7IGBTSFT	IGBT software start up flag 0: Disable 1: Enable
5	T7IGBTCNT	BC operation at IGBT disable is invalid 0: Disable 1: Enable
4	T7ONESHOT	One shot pulse selection 0: Continuous pulse 1: One shot pulse
3	T7ONED	Dead time selection 0: Dead time enable 1: Dead time disable
2	-	-
1	T7ICT2	Capture trigger selection 0: Timer 0 interrupt 1: Timer 1 interrupt
0	T7CAPCLR	TM7BC clearing at capture 0: Not clear 1: Clear



T7CAPCLR flag is valid when timer is operating.
Note that binary counter is not cleared when data is being captured during timer stop.

■ Timer 7 Mode Register 5 (TM7MD5: 0x03FA6)

bp	7	6	5	4	3	2	1	0
Flag	-	-	-	-	-	-	T7DBUF	T7UD
At reset	-	-	-	-	-	-	0	0
Access	-	-	-	-	-	-	R/W	R/W

bp	Flag	Description
7 to 2	-	-
1	T7DBUF	Input capture register buffer selection 0: Single buffer 1: Double buffer
0	T7UD	Input capture register 1 selection 0: Invalid 1: Valid



When T7DBUF and T7UD are set to "0" after these flags are set to "1", the register that becomes invalid keeps holding last captured value.



Be sure to set T7DBUF and T7UD to "0" when operating by cascade connection.



A startup factor of ATC is valid only in a capture trigger of TM7IC.
ATC can not startup by a capture trigger to TM7IC1.

The following are readable/writable registers which control Timer 8.

■ Timer 8 Mode Register 1 (TM8MD1: 0x03F98)

bp	7	6	5	4	3	2	1	0
Flag	Reserved	T8ICEDG1	TM8CL	TM8EN	TM8PS1	TM8PS0	TM8CK1	TM8CK0
At reset	0	0	1	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7	Reserved	Always set to "0"
6	T8ICEDG1	Capture trigger edge selection 0: Falling edge 1: Rising edge
5	TM8CL	Timer output reset signal 0: Enable timer output 1: Disable timer output (reset)
4	TM8EN	Timer count control 0: Halt the count 1: Start the count
3 to 2	TM8PS1 TM8PS0	Count clock selection 00: 1/1 of clock 01: 1/2 of clock 10: 1/4 of clock 11: 1/16 of clock
1 to 0	TM8CK1 TM8CK0	Clock source selection 00: fpll-div 01: fs 10: TMIO input 11: Synchronous TMIO input

■ Timer 8 Mode Register 2 (TM8MD2: 0x03F99)

bp	7	6	5	4	3	2	1	0
Flag	T8ICEDG0	T8PWMSL	TM8BCR	TM8PWM	TM8IRS1	T8ICEN	T8ICT1	T8ICT0
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7	T8ICEDG0	Capture trigger edge selection 0: Select the both edges 1: Select the specified edge
6	T8PWMSL	PWM mode selection 0: Set duty by TM8OC1 1: Set duty by TM8OC2
5	TM8BCR	Timer 8 count clear factor selection 0: Full count OVF 1: Match of TM8BC and TM8OC1
4	TM8PWM	Timer output waveform selection 0: Output timer 1: Output PWM
3	TM8IRS1	Timer 8 interrupt factor selection 0: Counter clear 1: Match of TM8BC and TM8OC1
2	T8ICEN	Input capture operation enable select flag 0: Disable capture operation 1: Enable capture operation
1 to 0	T8ICT1 T8ICT0	Capture trigger selection 00: External Interrupt 0 input signal 01: External Interrupt 1 input signal 10: External Interrupt 2 input signal 11: Timer interrupt



When T8ICT1 to 0 flags select “timer interrupt”, the timer interrupt set by T8ICT2 flag of TM8MD4 register is used as a capture trigger.

■ Timer 8 Mode Register 3 (TM8MD3: 0x03F9F)

bp	7	6	5	4	3	2	1	0
Flag	TM8CKSMP	TM8BUFSEL	-	TM8CKEDG	-	TM8PWMF	TM8PWMO	TM8CAS
At reset	0	0	-	0	-	0	0	0
Access	R/W	R/W	-	R/W	-	R/W	R/W	R/W

bp	Flag	Description
7	TM8CKSMP	Capture sampling selection 0: Count clock 1: fs
6	TM8BUFSEL	Buffer selection 0: Double buffer 1: Single buffer
5	-	-
4	TM8CKEDG	TM8IO count edge selection 0: Falling edge 1: Both edges
3	-	-
2	TM8PWMF	PWM output value selection while Timer 8 is stopped 0: Low 1: High
1	TM8PWMO	Timer 8 PWM output polarity selection 0: Normal output 1: Inverted output
0	TM8CAS	Cascade selection 0: Timer 7 and Timer 8 are independent 1: Timer 7 and Timer 8 are connected in cascade



TM8CKEDG flag must be set to "0" when Timer 6 compare match output is selected as a count clock by TM8IOSEL and TM8IOSEL2 flags of TMCKSEL2 register.

■ Timer 8 Mode Register 4 (TM8MD4: 0x03F7F)

<MN101EF76K/MN101EF56K>

bp	7	6	5	4	3	2	1	0
Flag	Reserved	Reserved	Reserved	TM8SEL_C	Reserved	TM8SEL_A	T8ICT2	T8CAPCLR
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7 to 5	Reserved	Always set to "0".
4	TM8SEL_C	Timer output selection (TM8IOC) 0: Timer 8 output 1: IGBT output
3	Reserved	Always set to "0".
2	TM8SEL_A	Timer output selection (TM8IOA/TM8IOB) 0: Timer 8 output 1: IGBT output
1	T8ICT2	Capture trigger selection 0: Timer 0 interrupt 1: Timer 1 interrupt
0	T8CAPCLR	TM8BC clearing at capture 0: Not cleared 1: Cleared

<MN101EF57G>

bp	7	6	5	4	3	2	1	0
Flag	Reserved	Reserved	Reserved	Reserved	Reserved	TM8SEL	T8ICT2	T8CAPCLR
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7 to 3	Reserved	Always set to "0".
2	TM8SEL	Timer output selection 0: Timer 8 output 1: IGBT output
1	T8ICT2	Capture trigger selection 0: Timer 0 interrupt 1: Timer 1 interrupt
0	T8CAPCLR	TM8BC clearing at capture 0: Not clear 1: Clear



T8CAPCLR flag is valid when timer is operating.
Note that binary counter is not cleared when data is being captured during timer stop.

■ Timer Clock Selection Register 2 (TMCKSEL2: 0x03FB1)

bp	7	6	5	4	3	2	1	0
Flag	-	TM8 IOSEL2	TM7 IOSEL3	TM7 IOSEL2	-	TM9 IOSEL	TM8 IOSEL	TM7 IOSEL
At reset	-	0	0	0	-	0	0	0
Access	-	R/W	R/W	R/W	-	R/W	R/W	R/W

bp	Flag	Description
7	-	-
6	TM8IOSEL2	Timer 8 input selection2 0: External input/Timer A 1: Timer 6 compare match output
5	TM7IOSEL3	Timer 7 input selection3 0: Serial Interface 0 transfer clock/ Timer A/ External input 1: Timer 6 compare match output
4	TM7IOSEL2	Timer 7 input selection2 0: External input/ Timer A 1: Serial Interface 0 transfer clock
3	-	-
2	TM9IOSEL	Timer 9 input selection 0: External input 1: Timer A
1	TM8IOSEL	Timer 8 input selection 0: External input 1: Timer A
0	TM7IOSEL	Timer 7 input selection 0: External input 1: Timer A

Table:11.2.2 Timer 7 and 8 Input Selection

Timer 7 input selection	TM7IOSEL3	TM7IOSEL2	TM7IOSEL
External input	0	0	0
Timer A	0	0	1
Serial Interface 0 transfer clock	0	1	X
Timer 6 compare match output	1	X	X

Timer 8 input selection	TM8IOSEL2	TM8IOSEL2
External input	0	0
Timer A	0	1
Timer 6 compare match output	1	X

■ Timer Input Selection Register 2 (TMINSEL2: 0x03FB3)

<MN101EF76K/MN101EF56K>

bp	7	6	5	4	3	2	1	0
Flag	TMINSEL 27	TMINSEL 26	TMINSEL 25	TMINSEL 24	TMINSEL 23	TMINSEL 22	TMINSEL 21	TMINSEL 20
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7 to 6	TMINSEL27 TMINSEL26	Timer 9 input selection (at port selection) 00: TM9IOA (PA7) 01: TM9IOB (P01) 10: "1" input to timer 11: "1" input to timer
5 to 4	TMINSEL25 TMINSEL24	Timer 8 input selection (at port selection) 00: TM8IOA (PA6) 01: TM8IOB (P03) 10: TM8IOC (P16) 11: "1" input to timer
3 to 2	TMINSEL23 TMINSEL22	Timer 7 input selection (at port selection) 00: TM7IOA (PA5) 01: TM7IOB (P02) 10: TM7IOC(P15) 11: "1" input to timer
3 to 2	TMINSEL21 TMINSEL20	Timer 4 input selection (at port selection) 00: TM4IOA (PA4) 01: TM4IOB (P64) 10: TM4IOC(P14) 11: "0" input to timer

■ Timer Input Selection Register 2 (TMINSEL2: 0x03FB3)

<MN101EF57G>

bp	7	6	5	4	3	2	1	0
Flag	TMINSEL 27	TMINSEL 26	TMINSEL 25	TMINSEL 24	TMINSEL 23	TMINSEL 22	TMINSEL 21	TMINSEL 20
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7 to 6	TMINSEL27 TMINSEL26	Timer 9 input selection (at port selection) 00: TM9IOA (PA7) 01: TM9IOB (P01) 10: "1" input to timer 11: "1" input to timer
5 to 4	TMINSEL25 TMINSEL24	Timer 8 input selection (at port selection) 00: TM8IOA (PA6) 01: TM8IOB (P03) 10: Setting prohibited 11: "1" input to timer
3 to 2	TMINSEL23 TMINSEL22	Timer 7 input selection (at port selection) 00: TM7IOA (PA5) 01: TM7IOB (P02) 10: Setting prohibited 11: "1" input to timer
3 to 2	TMINSEL21 TMINSEL20	Timer 4 input selection (at port selection) 00: TM4IOA (PA4) 01: TM4IOB (P64) 10: Setting prohibited 11: "0" input to timer

11.3 Operation

11.3.1 Operation

Timer has a function to generate interrupts constantly in a certain period of time.

■ 16-bit Timer Operation (Timer 7 and Timer 8)

The generation cycle of an timer interrupt is set by the clock source selection and the set value of compare register 1 (TMnOC1), in advance. When binary counter (TMnBC) reaches the set value of compare register 1, an interrupt is generated at the next count clock. There are 2 factors to be selected to clear binary counter; TMnOC1 compare match or the full count overflow. After binary counter is cleared, the counting up is restarted from 0x0000.

Table:11.3.1 16-bit Timer Interrupt Factor and Binary Counter Clear Factor (Timer 7 and Timer 8)

TM7MD2 register		Interrupt factor	Binary counter clear factor
TM7IRS1	TM7BCR		
1	1	TM7OC1 compare match	TM7OC1 compare match
0	1	TM7OC1 compare match	TM7OC1 compare match
1	0	TM7OC1 compare match	Full count overflow
0	0	Full count overflow	Full count overflow

TM8MD2 register		Interrupt factor	Binary counter clear factor
TM8IRS1	TM8BCR		
1	1	TM8OC1 compare match	TM8OC1 compare match
0	1	TM8OC1 compare match	TM8OC1 compare match
1	0	TM8OC1 compare match	Full count overflow
0	0	Full count overflow	Full count overflow

Timer n can generate another set of an independent interrupt (Timer n compare register 2 match interrupt) by the set value of Timer n compare register 2 (TMnOC2). At the time of the interrupt, binary counter is cleared as the above setup.

Compare register is double buffer type. When a value of preset registers is changed during counting operation, the value changed is stored to compare register when binary counter is cleared. This function can change compare register value constantly, without disturbing the cycle during timer operation (Reload function).



When CPU reads 16-bit binary counter (TMnBC), the read data is handled in 8-bit units even if it is a 16-bit MOVW instruction. As a result, it will read the data incorrectly if a carry from the lower 8 bits to the upper 8-bit occurs during counting operation.

To read the correct value of 16-bit counting (TMnBC), use the write program function to write to input capture register (TMnIC). By writing to TMnIC, counting data of TMnBC are stored to TMnIC so that the values can be read out correctly during operation.

[Chapter 11 11.9.1 Operation]



To count properly, do not switch the count clock during the timer operation.
To switch the count clock, stop the timer operation.



Set Timer n mode register when the TMnEN flag of TMnMD1 register is set to "0" to stop counting.



To change CPU operation mode from NORMAL mode to SLOW mode when high-speed oscillation clock (fppll-div) is selected as clock source, stop the timer at first. And then, set CPU again to start the timer.

In SLOW/HALT1 mode, do not select the high-speed oscillation clock (fppll-div) for the clock source.



Use the MOVW instruction of 16-bit access to write the data to preset register (TMnPR1 and TMnPR2) when 16-bit timer is stopped.

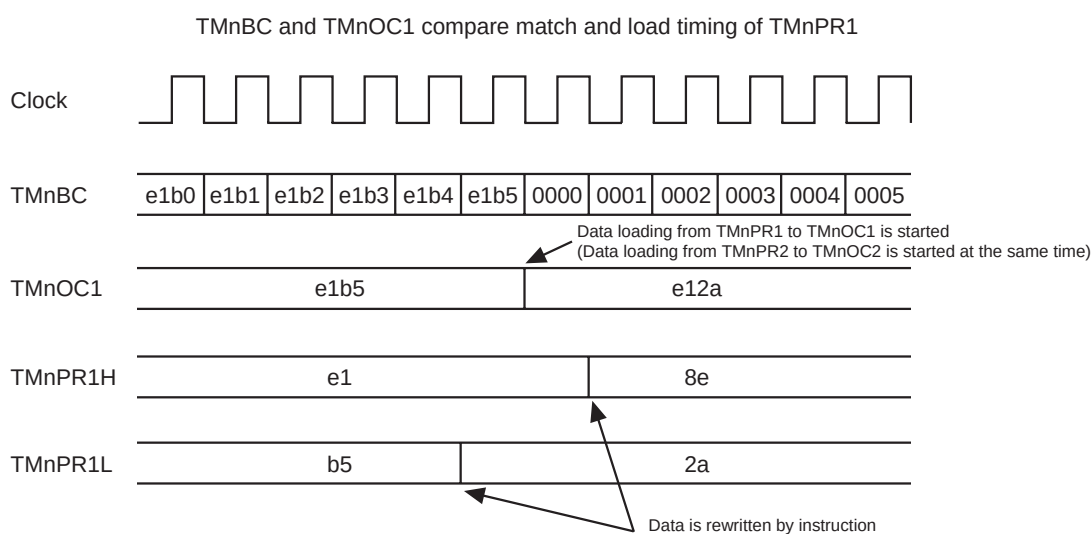
If lower 8-bit is written after upper 8-bit is written to preset register by using the MOV instruction of 8-bit access, the rewritten data is loaded to upper 8-bit and data which is not written is loaded to lower 8-bit of compare register (TMnOC1 and TMnOC2).



When data is written to preset register (TMnPR1, TMnPR2) of 16-bit timer, the data is recognized as 8-bit unit data inside LSI even if it is a 16-bit access MOVW instruction. When data is loaded from preset register to compare register (TMnOC1, TMnOC2) before writing upper 8-bit data but after writing lower 8-bit data to preset register, upper 8-bit data before rewriting and lower 8-bit data after writing are loaded to compare register respectively.

Therefore, writing data to preset register (TMnPR1, TMnPR2) need to be completed before data loading from preset register to compare register is started.

The following chart shows the timing of rewriting to TMnPR1 and TMnOC1 and loading. Note that the same phenomenon could occur at data writing to TMnPR2 and incorrect data might be written to TMnOC2.



Data "e12a" is loaded to TMnOC1 as TMnPR1 writing ("e1b5" to "8e2a") and loading to TMnOC1 are operated at the same time.

Table:11.3.2 shows the clock source that can be selected.

Table:11.3.2 Clock Source at Timer Operation (Timer 7 and Timer 8)

Clock source	Time per count
fpll-div	100 ns
fpll-div/2	200 ns
fpll-div/4	400 ns
fpll-div/16	1.6 ms
fs	200 ns
fs/2	400 ns
fs/4	800 ns
fs/16	3.2 ms
fpll-div = 10 MHz (PLL is not used) fs = fpll-div/2 = 5 MHz	

■ Count Timing of Timer Operation (Timer 7 and Timer 8)

Binary counter counts up with a selected clock source as a count clock.
The basic operation of whole 16-bit timer functions is as below.

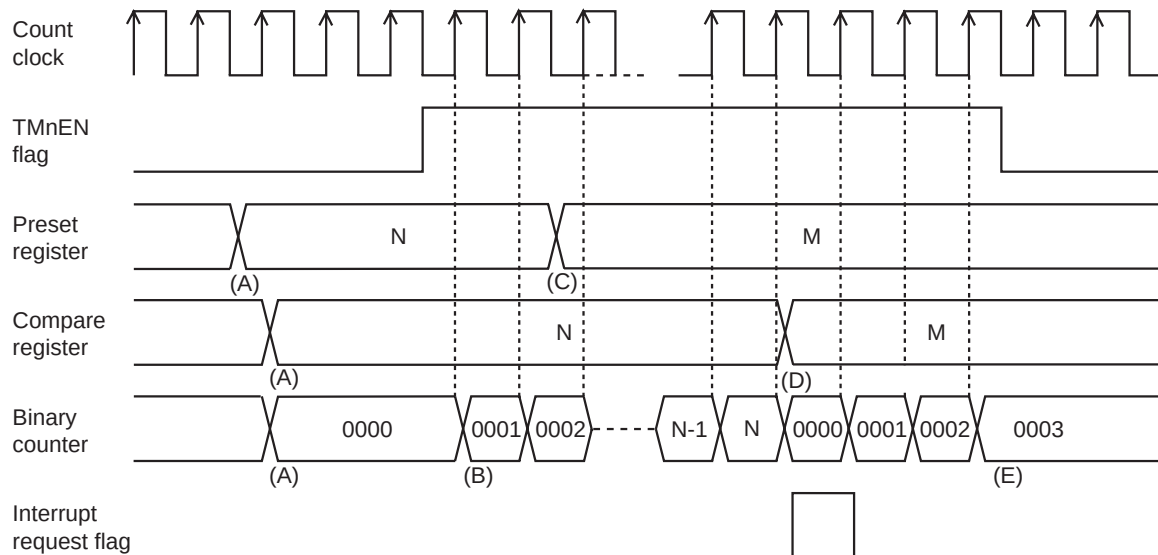


Figure:11.3.1 Count Timing of Timer Operation (Timer 7 and Timer 8)

- (A) When a data is written to preset register while the TMnEN flag is stopped "0", the same value is loaded during the writing cycle and binary counter is cleared to 0x0000.
- (B) When TMnEN flag is "1", binary counter starts counting. The counting starts at the rising edge of the count clock.
- (C) Even if preset register is rewritten when the TMnEN flag is "1", binary counter is not changed.
- (D) When binary counter reaches value of compare register 1, the set value of preset register is loaded to compare register at the next count clock. And the interrupt request flag is set at the next count clock, and binary counter is cleared to 0x0000 to restart counting up.
- (E) When the TMnEN flag is "0", binary counter is stopped.



When the binary counter reaches the value of the compare register, the interrupt request flag is set at the next count clock, and the binary counter is cleared.

So, set the compare register as:

(the set value of the compare register) = (the counts till the interrupt generation - 1)

However, if "00" is specified for the compare register, an interrupt timing is the same as if you set it to "01".



After Timer n interrupt request flag is generated, up to 3 system clock are needed until the next interrupt request flag is generated. During this period, an interrupt request flag is not generated even if a compare match occurs.



When Timer n compare register 2 match interrupt is generated and TM7OC1 compare match is selected as a binary counter clear source, the set value of compare register 2 should be smaller than the set value of compare register 1.



When timer interrupt request flag has been already set, be sure to clear timer interrupt request flag before starting the timer.



When the binary counter is used as a free-counter that counts 0x0000 to 0xFFFF, set 0xFFFF to the compare register or set the TMnBCR flag of TMnMD2 register to "0".



Do not change TMnEN flag of TMnMD register simultaneously with other bits to avoid any error in operation.



Be sure to set count clock of 16-bit timer while timer interrupt is disabled.



When binary counter is read during the timer operation, the data is processed as 8-bit unit data in LSI. If there occurs a carry-out from lower 8-bit to upper 8-bit during reading, correct values cannot be read out. Stop timer to read correct values.

11.3.2 Setup Example

■ Timer Operation Setup Example

Timer 7 generates an interrupt constantly for timer function. fppll-div/2 (fppll-div=10 MHz at operation) is selected as a clock source to generate an interrupt every 1000 cycles (200 ms).

An example setup procedure, with a description of each step is shown below.

Setup Procedure	Description
(1) Stop the counter TM7MD1(0x03F78) bp4: TM7EN =0	(1) Set the TM7EN flag of Timer 7 mode register 1 (TM7MD1) to "0" to stop Timer 7 counting.
(2) Disable the interrupt TM7ICR(0x03FEF) bp1: TM7IE =0	(2) Set the TM7IE flag of TM7ICR register to "0" to disable the interrupt.
(3) Select the timer clear source TM7MD2(0x03F79) bp5: TM7BCR =1	(3) Set the TM7BCR flag of Timer 7 mode register 2 (TM7MD2) to "1" to select the compare match to the binary counter clear source.
(4) Select the count clock source TM7MD1 (0x03F78) bp1 to 0: TM7CK1 to 0 =00 bp3 to 2: TM7PS1 to 0 =01	(4) Select fppll-div to the clock source by the TM7CK1 to 0 flags of TM7MD1 register. Besides, select 1/2 fppll-div to the count clock source by the TM7PS1 to 0 flags.
(5) Select the IGBT/Timer startup factor TM7MD3 (0x03F8E) bp1 to 0: T7IGBT1 to 0 =00	(5) Set the IGBT/Timer startup factor to Timer 7 counting operation.
(6) Set the interrupt generation cycle TM7PR1(0x03F75, 0x03F74) =0x03E7	(6) Set the interrupt generation cycle to Timer 7 preset register 1 (TM7PR1). The set value should be 1000-1=999 (0x03E7) because the cycle is 1000. At the time, the same value is loaded to Timer 7 compare register 1 (TM7OC1), and Timer 7 binary counter (TM7BC) is initialized to 0x0000.
(7) Set the interrupt level TM7ICR(0x03FEF) bp7 to 6: TM7LV1 to 0 =10	(7) Set the interrupt level by the TM7LV1 to 0 flags of Timer 7 interrupt control register (TM7ICR). If the interrupt request flag has already been set, clear the request flag. [Chapter 7 7.1.4 Maskable Interrupt Control Register Setup]
(8) Enable the interrupt TM7ICR (0x03FEF) bp1: TM7IE =1	(8) Set the TM7IE flag of TM7ICR register to "1" to enable the interrupt.
(9) Start the timer operation TM7MD1 (0x03F78) bp4: TM7EN =1	(9) Set the TM7EN flag of TM7MD1 register to "1" to operate.

TM7BC counts up from 0x0000. When TM7BC reaches the set value of TM7OC1 register, Timer 7 interrupt request flag is set at the next count clock and TM7BC becomes 0x0000 and counts up again.

11.4 16-bit Event Count

11.4.1 Operation

Event count operation has 2 types; TMnIO input and synchronous TMnIO input. These can be selected as the count clock. Each type can select 1/1, 1/2, 1/4 or 1/16 as a count clock source. Also, it is possible to select the count edge. (the falling edge and the both edge at the normal operation are selectable)

■ 16-bit Event Count Operation (Timer 7 and Timer 8)

Binary counter (TMnBC) counts the external signal input to the TMnIO pin. If binary counter reaches the set value of compare register (TMnOC), an interrupt can be generated at the next count clock.

Table:11.4.1 Event Count Input Clock Source

	Timer 7	Timer 8
Event input	TM7IOA input TM7IOB input TM7IOC input	TM8IOA input TM8IOB input TM8IOC input
	Synchronous TM7IO input	Synchronous TM8IO input
	Timer A output Serial 0 transfer clock Timer 6 compare match output	Timer A output Timer 6 compare match output

■ Count Timing of TMnIO Input (Timer 7 and Timer 8)

When TMnIO input is selected, TMnIO input signal is input to Timer n count clock.

Binary counter counts up at the falling edge of TMnIO input signal or TMnIO input signal that passed the divider.

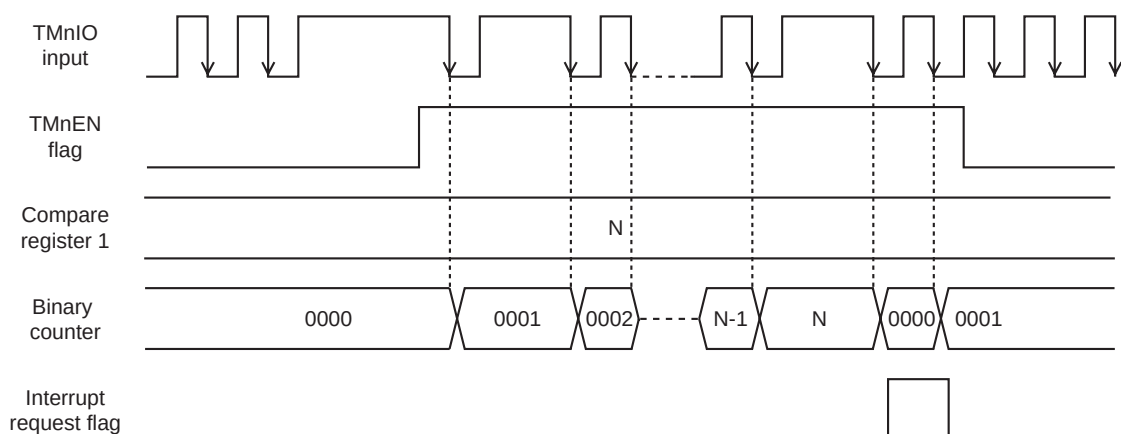


Figure:11.4.1 Count Timing TMnIO Input (Timer 7 and Timer 8)



When event input (TMnIO input) is selected as a count clock source, do not read out values of Timer n binary counter. To read out values of Timer n binary counter, use an event counter with synchronous TMnIO input which is indicated in the following page.



When using TMnIO input, be sure to set each mode register and preset register after selecting fs as a count clock source. Then, select TMnIO input to start a timer. Do not write any data to the preset register during operation. Only TMnIO input can recover from STOP mode in 16-bit timer.



When using the event input (TMnIO input), clear binary counter before starting the timer operation. Also, when setting 0x0000 to compare register, use the event count by synchronous TMnIO input.



When event input (TMnIO input) is selected as a count clock source, binary counter may reach an unexpected value by stopping timer operation. When event input (TMnIO input) is selected as a count clock source, do not read binary counter.

■ Count Timing of Synchronous TMnIO Input (Timer 7 and Timer 8)

If the synchronous TMnIO input is selected, the synchronizing circuit output signal is input to Timer n count clock. The synchronization circuit output signal is changed at the rising edge of the system clock after the TMnIO input signal is changed. The binary counter counts up at the falling edge of the synchronization circuit output signal or the synchronization circuit output signal that passed through the division circuit.

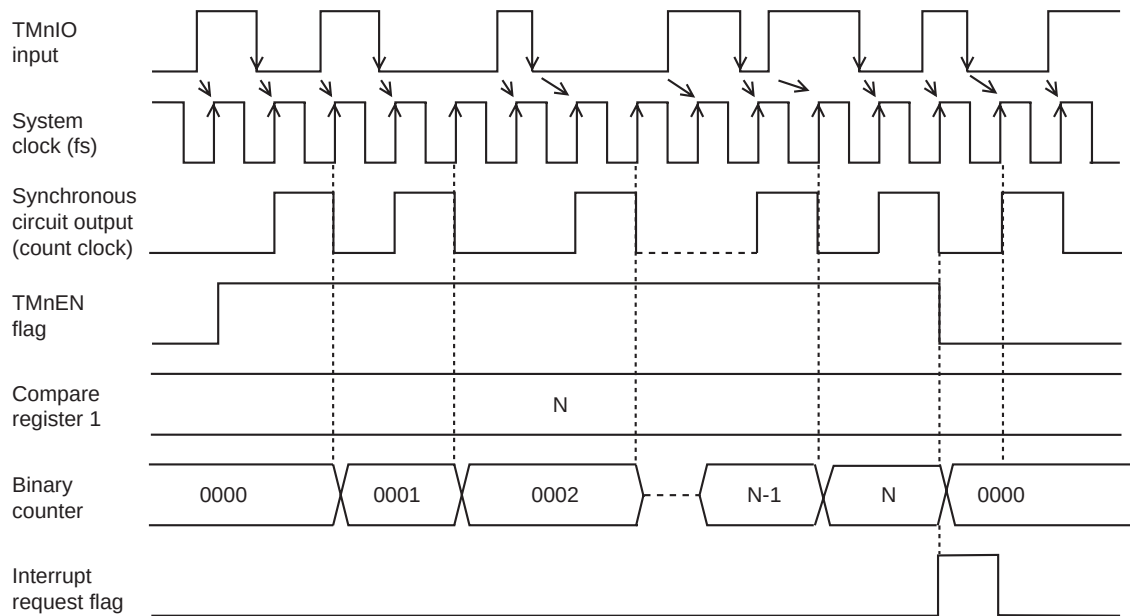


Figure:11.4.2 Count Timing of Synchronous TMnIO Input (Timer 7 and Timer 8)



As Timer n counts up binary counter using signals synchronized to system clock, correct values are always read out when Timer n binary counter is read out.



Input signal from TMnIO should be set with the cycle more than twice the system clock (fs). When other signals with a cycle shorter than this are input, the counting may not be performed correctly.

■ Count Timing of TMnIO Input (Both edges selected)

When TMnIO input is selected, TMnIO input signal is input to Timer n count clock. Binary counter counts up at the rising or the falling edge of TMnIO input signal that passed the divider or TMnIO input signal.

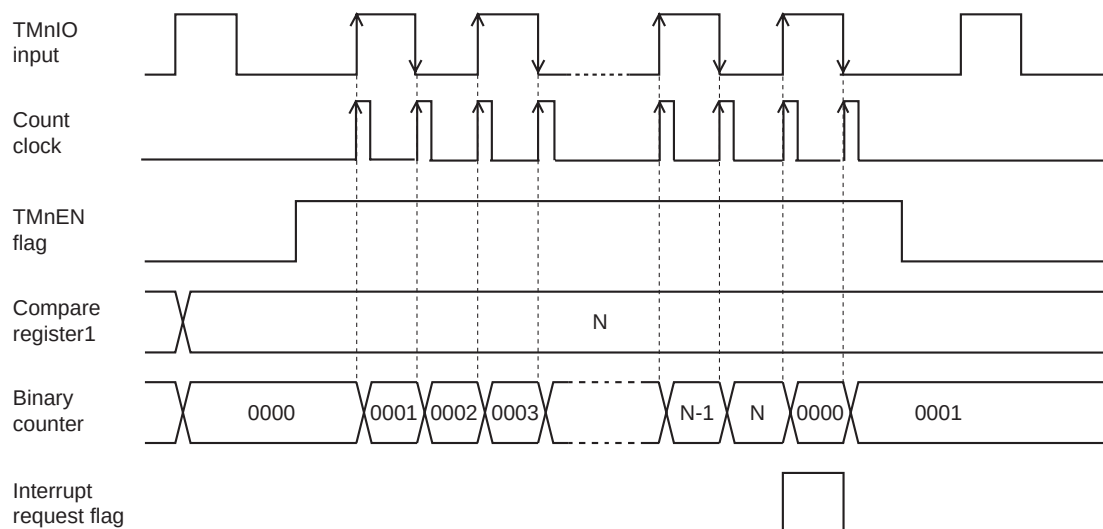


Figure:11.4.3 Count Timing TMnIO Input (Timer 7 and Timer 8)



When both edges are selected, they are counted only at the normal operation (high-speed oscillation). The counting can not be operated when synchronous TMnIO is set as input. Input from TMnIO should be done the waveform which has more than 2 times cycle than $f_{pll-div}$. If less than the above waveforms are input, it may not be counted correctly.



Do not set the both edges counting when Timer 6 compare match output is set as a count clock.

■ Count Timing of Synchronous Timer 6 Compare Match Output

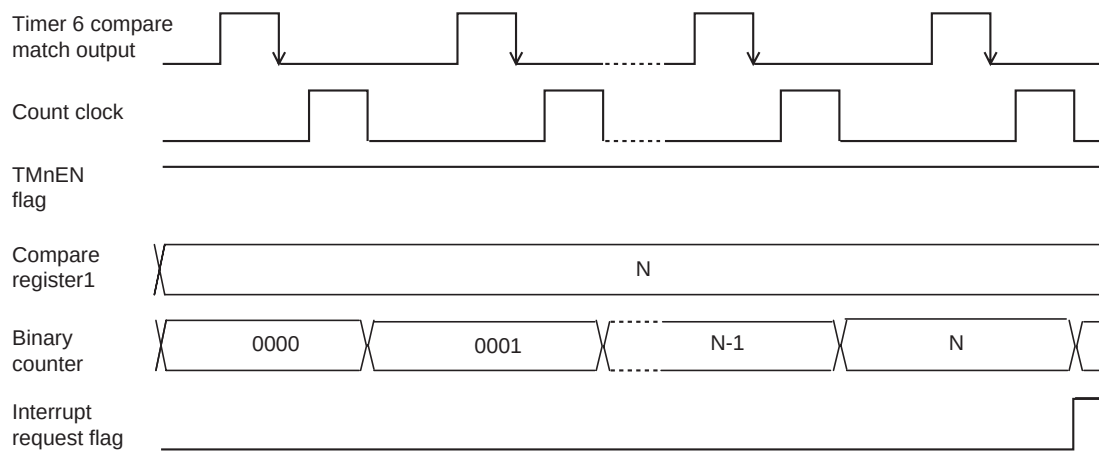


Figure:11.4.4 Count Timing of Synchronous Timer 6 Compare Match Output (Timer 7 and Timer 8)



Do not set the both edges counting when Timer 6 compare match output is set as a count clock.

11.4.2 Setup Example

■ Event Count Setup Example

When the falling edge of TM7IO input pin signal is detected 5 times using Timer 7, an interrupt is generated.

An example setup procedure, with a description of each step is shown below.

Setup Procedure	Description
(1) Stop the counter TM7MD1 (0x03F78) bp4: TM7EN =0	(1) Set the TM7EN flag of Timer 7 mode register 1 (TM7MD1) to "0" to stop Timer 7 counting.
(2) Disable the interrupt TM7ICR(0x03FEF) bp1: TM7IE =0	(2) Set the TM7IE flag of TM7ICR register to "0" to disable the interrupt.
(3) Select the timer clock TMCKSEL2(0x03F43) bp0: TM7IOSEL =0	(3) Select the external input by setting the TM7IOSEL flag of timer clock selection register 2 (TMCKSEL2) to "0".
(4) Select the timer input TMINSEL2(0x03E45) bp3 to 2: TMINSEL23 to 2 =10	(4) Set P15 to timer input by setting the TMINSEL23-2 flags of timer input selection register 2 (TMINSEL2) to "10".
(5) Set the special function pin to input PADIR (0x03E3A) bp6: PADIR5 =0	(5) Set the PADIR5 flag of Port A direction control register (PADIR) to "0" to set PA5 pin to the input mode. Add pull-up/pull-down resistor, if necessary. [Chapter 8 I/O Port]
(6) Select the count clock source TM7MD1(0x03F78) bp1 to 0: TM7CK1 to 0 =01 bp3 to 2: TM7PS1 to 0 =00	(6) Select fs to the clock source by the TM7CK1 to 0 flags of TM7MD1 register. Besides, select 1/1 to the count clock source by the TM7PS1 to 0 flags.
(7) Select the IGBT/Timer 7 startup factor TM7MD3(0x03F8E) bp1 to 0: T7IGBT1 to 0 =00	(7) Set the IGBT/Timer 7 startup factor to Timer 7 counting operation.
(8) Set the interrupt generation cycle TM7PR1(0x03F75, 0x03F74) =0x0004	(8) Set the interrupt generation cycle to Timer 7 preset register 1 (TM7PR1). As counting is executed 5 times, the set value should be 0x0004. At that time, the same value is loaded to Timer 7 compare register 1 (TM7OC1), and Timer 7 binary counter (TM7BC) is initialized to 0x0000.
(9) Select the timer clear source TM7MD2 (0x03F79) bp5: TM7BCR =1	(9) Set the TM7BCR flag of Timer 7 mode register 2 (TM7MD2) to "1" to select the compare match as a binary counter clear source.
(10) Select the count clock source TM7MD1 (0x03F78) bp1 to 0: TM7CK1 to 0 =10 bp3 to 2: TM7PS1 to 0 =00	(10) Select TM7IO to the clock source by the TM7CK1 to 0 flags of TM7MD1 register. Besides, select 1/1 (no dividing) to the count clock source by the TM7PS1 to 0 flags.

Setup Procedure	Description
(11) Set the interrupt level TM7ICR (0x03FEF) bp7 to 6: TM7LV1 to 0 =10	(11) Set the interrupt level by the TM7LV1 to 0 flags of Timer 7 interrupt control register (TM7ICR). If the interrupt request flag is already set, clear the request flag. [Chapter 7 7.1.4 Maskable Interrupt Control Register Setup]
(12) Enable the interrupt TM7ICR (0x03FEF) bp1: TM7IE =1	(12) Set the TM7IE flag of TM7ICR register to "1" to enable the interrupt.
(13) Start the event count TM7MD1 (0x03F78) bp4: TM7EN =1	(13) Set the TM7EN flag of TM7MD1 register to "1" to operate Timer 7.

Every time TM7BC reaches the falling edge of TM7IO input, it counts up from 0x0000.

When TM7BC reaches the set value of TM7OC1 register, Timer 7 interrupt request flag is set at the next count clock, and the value of TM7BC is 0x0000 to restart counting up.

Be sure to follow the procedures from (6) to (10) to avoid any error in operation.

11.5 16-bit Timer Pulse Output

11.5.1 Operation

TMnIO pin can output a pulse signal with a arbitrary frequency.

■ 16-bit Timer Pulse Output Operation (Timer 7 and Timer 8)

These timers can output twice the cycle signal set to compare register 1 (TM7OC1) and output twice the cycle signal of the 16-bit full count.

Table:11.5.1 shows timer pulse output pin.

Table:11.5.1 Timer Pulse Output Pin

	Timer 7	Timer 8
Pulse output pin	TM7IOA output TM7IOB output TM7IOC output	TM8IOA output TM8IOB output TM8IOC output

Table:11.5.2 shows timer interrupt generation sources and the flags that control timer pulse output cycle.

Table:11.5.2 16-bit Timer Interrupt Generation Source and Timer Pulse Output Cycle (Timer 7)

TM7MD2 register		Interrupt source	Timer pulse output cycle
TM7IRS1	TM7BCR		
1	1	TM7OC1 compare match	Set value of TM7OC1 $\times$ 2
0	1	TM7OC1 compare match	Set value of TM7OC1 $\times$ 2
1	0	TM7OC1 compare match	Full count of TM7BC $\times$ 2
0	0	Full count over flow	Full count of TM7BC $\times$ 2

TM8MD2 register		Interrupt source	Timer pulse output cycle
TM8IRS1	TM8BCR		
1	1	TM8OC1 compare match	Set value of TM8OC1 $\times$ 2
0	1	TM8OC1 compare match	Set value of TM8OC1 $\times$ 2
1	0	TM8OC1 compare match	Full count of TM8BC $\times$ 2
0	0	Full count over flow	Full count of TM8BC $\times$ 2

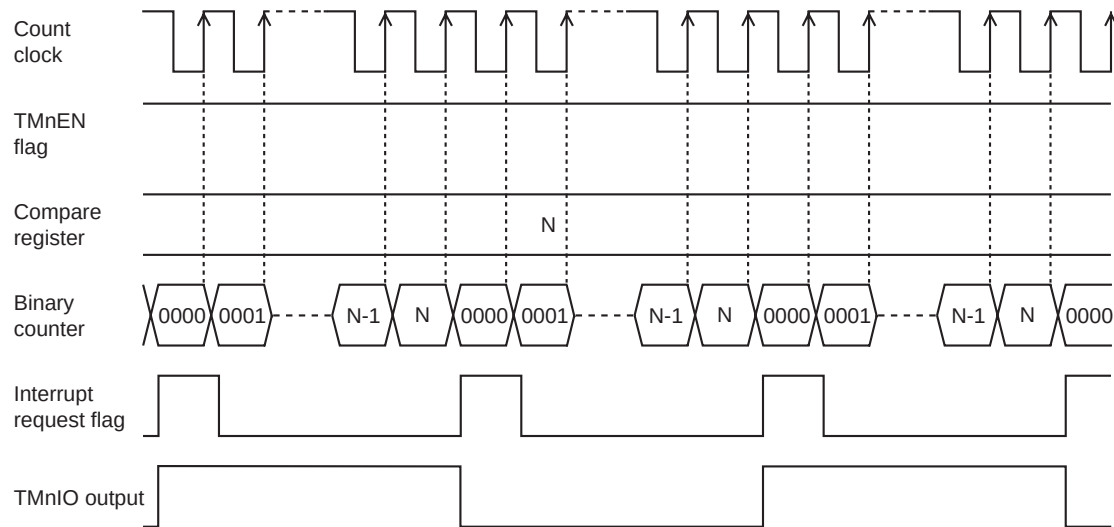


Figure:11.5.1 Count Timing of Timer Pulse Output (Timer 7 and Timer 8)

TMnIO output pin outputs twice the cycle signal set to compare register 1. If binary counter reaches the compare value or full count overflow occurs, binary counter is cleared to 0x0000, and TMnIO output is inverted.



In the initial state after releasing reset, the timer pulse output is reset, and "Low" output is fixed. Therefore, release the reset of the timer pulse output by setting the TMnCL flag of TMnMD1 register to "0".



Regardless of whether the binary counter is stopped or operated, the timer output is "Low", when the TMnCL flag of TMnMD2 register is set to "1".



Reset release of the timer pulse output should be done when the timer count is stopped.



To start timer pulse output with the divided clock source, set dividing ratio after releasing the reset of timer pulse output. Dividing ratio can be set by TMnPS1 to TMnPS0 of TMnMD1 register.

11.5.2 Setup Example

■ Timer Pulse Output Setup Example

TM7IO pin outputs a 25 kHz pulse using Timer 7. Select fppll-div as the clock source and set 1/2 cycle (25 kHz) to Timer 7 compare register (at fppll-div=10 MHz).

An example setup procedure, with a description of each step is shown below.

Setup Procedure	Description
(1) Stop the counting TM7MD1 (0x03F78) bp4: TM7EN =0	(1) Set the TM7EN flag of Timer 7 mode register 1 (TM7MD1) to "0" to stop Timer 7 counting.
(2) Set the special function pin PAOMD (0x03EE6) bp5: PAOMD5 =1 PADIR (0x03F3A) bp5: PADIR5 =1	(2) Set the PAOMD5 flag of Port A output mode register (PAOMD) to "1" to set PA5 as the special function pin. Set the PADIR5 flag of Port A direction control register (PADIR) to "1" to set the output mode. [Chapter 8 I/O Port]
(3) Set the timer pulse TM7MD2 (0x03F79) bp4: TM7PWM =0	(3) Set the TM7PWM flag of Timer 7 mode register 2 (TM7MD2) to "0" to select the timer pulse output.
(4) Select the timer clear source TM7MD2 (0x03F79) bp5: TM7BCR =1	(4) Set the TM7BCR flag of TM7MD2 register to "1" to select the compare match as the binary counter clear source.
(5) Release the reset of the timer pulse TM7MD1 (0x03F78) bp5: TM7CL =0	(5) Set the TM7CL flag of TM7MD1 register to "0" to enable the pulse output.
(6) Select the count clock source TM7MD1 (0x03F78) bp1 to 0: TM7CK1 to 0 =00 bp3 to 2: TM7PS1 to 0 =00	(6) Select fppll-div as the clock source by the TM7CK1 to 0 flags of TM7MD1 register. Also, select 1/1 dividing as the clock source by the TM7PS1 to 0 flags.
(7) Select the IGBT/Timer 7 startup factor TM7MD3(0x03F8E) bp1 to 0: T7IGBT1 to 0 =00	(7) Set the IGBT/Timer 7 startup factor to Timer 7 counting operation.
(8) Set the timer pulse output generation cycle TM7PR1(0x03F75, 0x03F74) =0x00C7	(8) Set 1/2 of the timer pulse output cycle to Timer 7 preset register 1 (TM7PR1). To set 50 kHz by dividing 10 MHz, set as; 200-1=199 (0xC7) At the same time, the same value is loaded to Timer 7 compare register 1 (TM7BC) and Timer 7 binary counter (TM7BC) is initialized to 0x0000.
(9) Start the timer operation TM7MD1 (0x03F78) bp4: TM7EN =1	(9) Set the TM7EN flag of TM7MD1 register to "1" to operate Timer 7.

TM7BC counts up from 0x0000. If TM7BC reaches the set value of TM7OC1 register, TM7BC is cleared to 0x0000, the signal of TM7IO output is inverted and TM7BC counts up from 0x0000 again.

11.6 16-bit Standard PWM Output (Only duty can be changed continuously)

TMnIO pin outputs the standard PWM waveform, which is generated when values of both binary counter of timer and compare register match and when binary counter overflows.

11.6.1 Operation

■ 16-bit Standard PWM Output (Timer 7 and Timer 8)

PWM waveform with an arbitrary duty is generated by setting a duty of PWM "High" period to the compare register 1 (TMnOC1). Its cycle is the time when 16-bit timer full count overflow.

Table:11.6.1 shows the PWM output pin.

Table:11.6.1 PWM Output Pin

	Timer 7	Timer 8
PWM output pin	TM7IOA output TM7IOB output TM7IOC output	TM8IOA output TM8IOB output TM8IOC output

■ Count Timing of Standard PWM Output (at Normal) (Timer 7 and Timer 8)

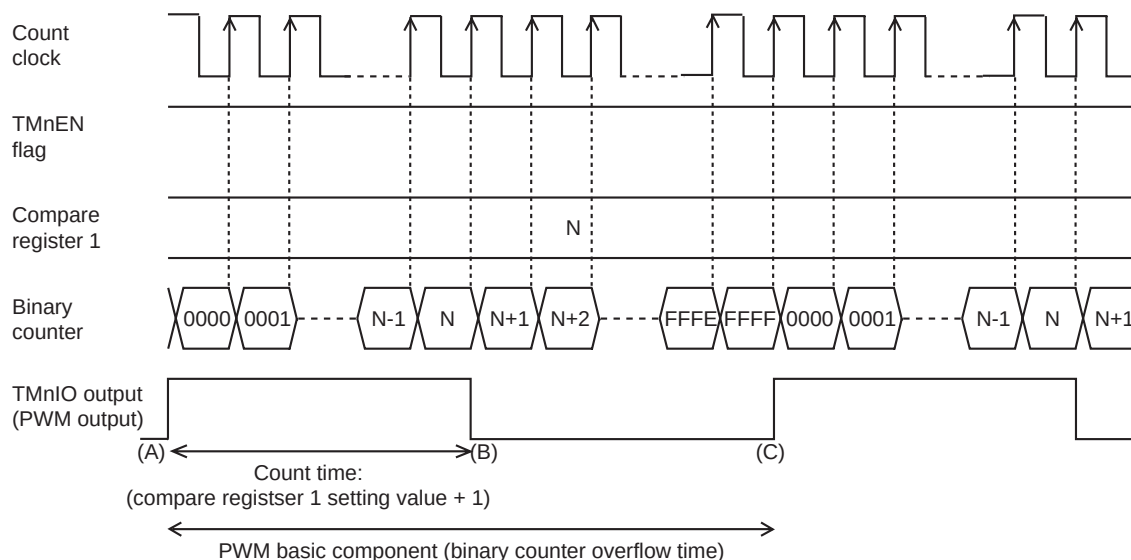


Figure:11.6.1 Count Timing of Standard PWM Output (at Normal)

■ The Polarity Selection at PWM Waveform Stops (Timer 8)

When PWM is stopped, select TM8IO output waveform by TM8PWMF of TM8MD3 register. Select the polarity of PWM output by TM8PWM0.



Before starting the second PWM or later, clear binary counter and PWM waveform by writing to preset register as the PWM output waveform of the first cycle cannot be guaranteed.

- (A) shows "High" until the binary counter reaches the compare register value from 0x0000.
- (B) shows "Low" after the compare match, then the binary counter counts up till the overflow.
- (C) shows "High" again if the binary counter overflow.

■ Count Timing of Standard PWM Output (when compare register 1 is 0x0000) (Timer 7 and Timer 8)

The count timing at setting 0x0000 to compare register 1 is shown below.

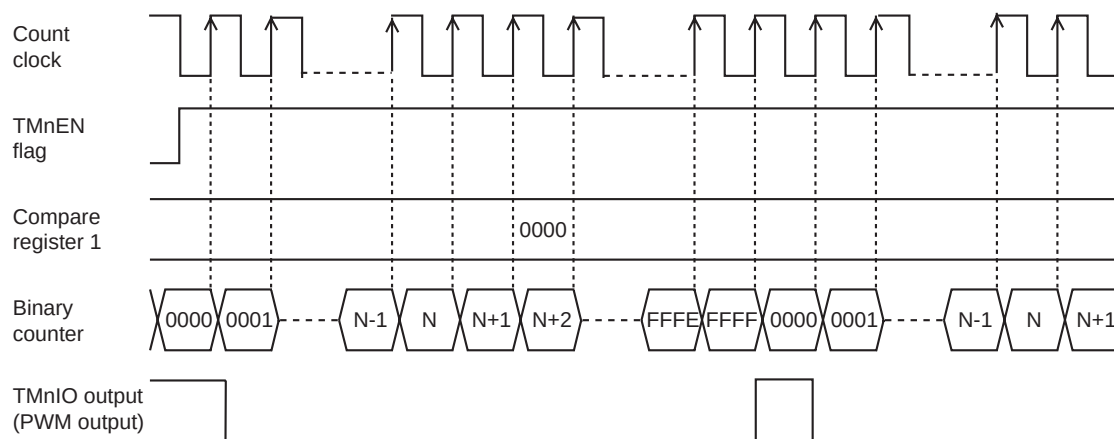


Figure:11.6.2 Count Timing of Standard PWM Output (when compare register 1 is 0x0000)

PWM output shows "High", when TMnEN flag is stopped (at "0").

■ Count Timing of Standard PWM Output (when compare register 1 is 0xFFFF) (Timer 7 and Timer 8)

The count timing at setting 0xFFFF to compare register 1 is shown below.

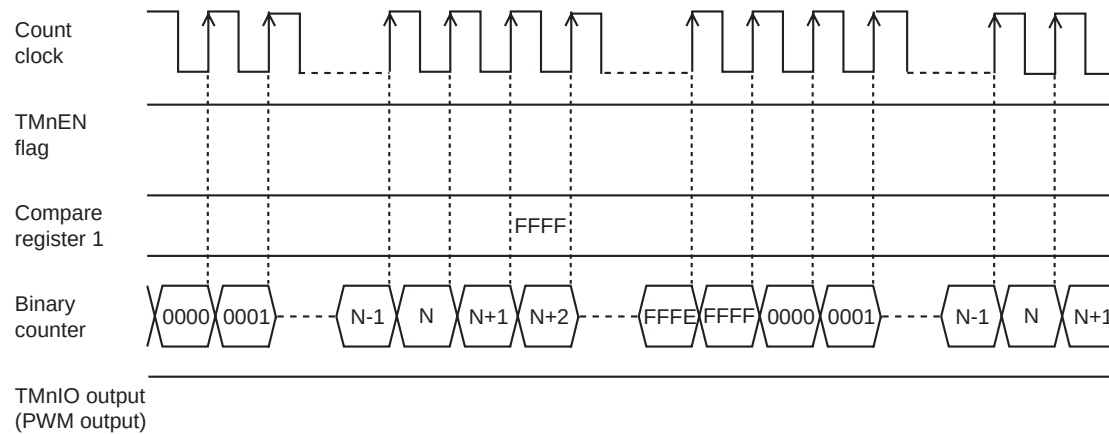


Figure:11.6.3 Count Timing of Standard PWM Output (when compare register 1 is 0xFFFF)



To output the standard PWM output, set the TMnBCR flag of TMnMD2 register to "0" to select the full count overflow as the binary counter clear source and the PWM output set ("High" output) source.



The TMnOC1 compare match or the TMnOC2 compare match can be selected as a PWM output reset ("Low" output) source with the TnPWMSL flag of TMnMD2 register.



In the initial state of the PWM output, it is changed to "High" output from "Low" output at the timing that the PWM operation is selected by the TMnPWM flag of TMnMD2 register.



To guarantee the PWM waveform of the first cycle, after PWM operation is stopped, write to the preset register to clear the binary counter and the PWM waveform when restarting the PWM operation.

11.6.2 Setup Example

■ Standard PWM Output Setup Example

The TM7IOA output pin outputs the 1/4 duty PWM output waveform at 152.6 Hz with Timer 7 (at the high frequency oscillation, $f_{pll-div}=10$ MHz). One cycle of the PWM output waveform is decided by the overflow of binary counter. "High" period of the PWM output waveform is decided by the set value of compare register 1. An example setup procedure, with a description of each step is shown below.

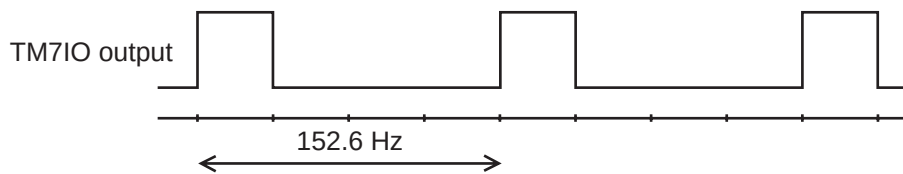


Figure:11.6.4 Output Waveform of TM7IO Output Pin

Setup Procedure	Description
(1) Stop the counter TM7MD1 (0x03F78) bp4: TM7EN =0	(1) Set the TM7EN flag of Timer 7 mode register 1 (TM7MD1) to "0" to stop Timer 7 counting.
(2) Set the special function pin to output PAOMD (0x03EE6) bp5: PAOMD5 =1 PADIR (0x03F3A) bp5: PADIR5 =1	(2) Set the PAOMD5 flag of Port A output mode register (PAOMD) to "1" to set the PA5 pin as a special function pin. Set the PADIR5 flag of Port A direction control register (PADIR) to "1" to set the output mode. [Chapter 8 I/O Port]
(3) Set the PWM output TM7MD2 (0x03F79) bp4: TM7PWM =1	(3) Set the TM7PWM flag of Timer 7 mode register 2 (TM7MD2) to "1" to select the PWM output.
(4) Set the standard PWM output TM7MD2 (0x03F79) bp5: TM7BCR =0	(4) Set the TM7BCR flag of TM7MD2 register to "0" to select the full count overflow as the binary counter clear source.
(5) Select the count clock source TM7MD1 (0x03F78) bp1 to 0: TM7CK1 to 0 =00 bp3 to 2: TM7PS1 to 0 =00	(5) Select $f_{pll-div}$ as the clock source by the TM7CK1 to 0 flags of TM7MD1 register. Also, select 1/1 dividing as the count clock source by the TM7PS1 to 0 flags.
(6) Select the IGBT/Timer 7 startup factor TM7MD3(0x03F8E) bp1 to 0: T7IGBT1 to 0 =00	(6) Set the IGBT/Timer 7 startup factor to Timer 7 counting operation.
(7) Set "High" period of the PWM output TM7PR1(0x03F75, 0x03F74) =0x3FFF	(7) Set "High" period of the PWM output to Timer 7 preset register 1 (TM7PR1). To set 1/4 duty of the full count 65536, set as; $65536/4-1=16383$ (0x03FFF) At the same time, the same value is loaded to Timer 7 compare register 1 (TM7OC1) and Timer 7 binary counter (TM7BC) is initialized to 0x0000.
(8) Start the timer operation TM7MD1 (0x03F78) bp4: TM7EN =1	(8) Set the TM7EN flag of TM7MD1 register to "1" to operate Timer 7.

TM7BC counts up from 0x0000. The PWM source waveform outputs "High" until TM7BC reaches the set value of TM7OC1 register, then after the match it outputs "Low". After that, TM7BC continues to count up. Once a overflow occurs, the PWM source waveform outputs "High" again, and TM7BC counts up from 0x0000, again.

11.7 16-bit High Precision PWM Output (Cycle/Duty can be changed continuously)

The TMnIO pin outputs high precision PWM waveform, which is determined when the values of binary counter and compare register 1 match and when binary counter and compare register 2 match.

11.7.1 Operation

■ 16-bit High Precision PWM Output Operation (Timer 7 and Timer 8)

The PWM waveform of any cycle/duty is generated by setting the cycle of PWM to compare register 1 (TM7OC1) and setting the duty of the "High" period to compare register 2 (TM7OC2).

■ Count Timing of High Precision PWM Output (at Normal) (Timer 7 and Timer 8)

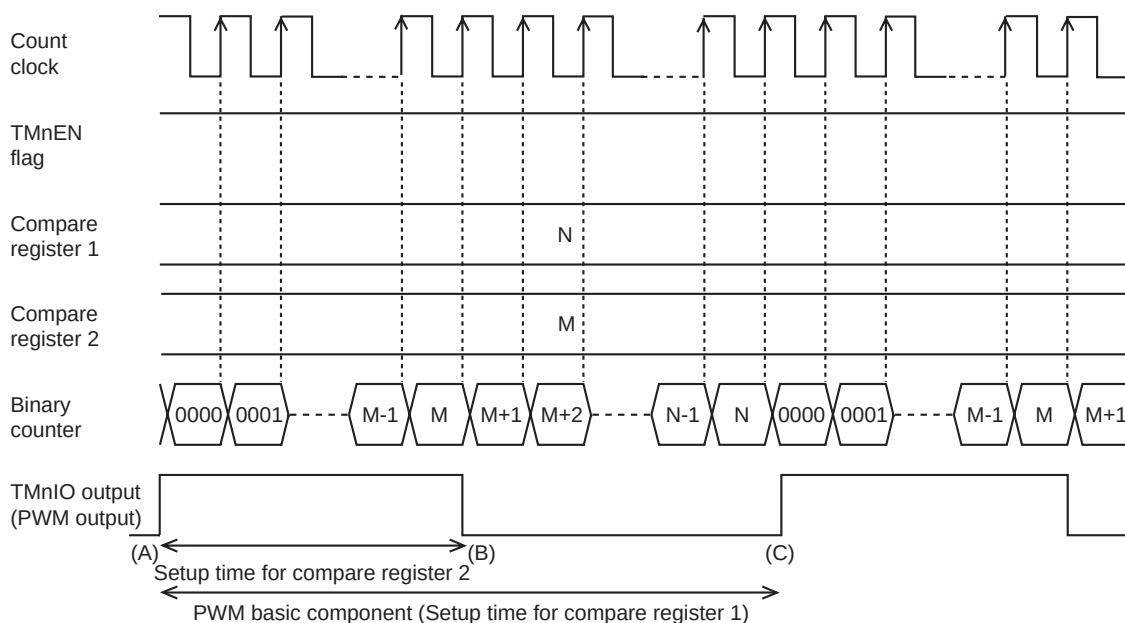


Figure:11.7.1 Count Timing of High Precision PWM Output (at Normal)

PWM source waveform:

(A) shows "High" until binary counter reaches the compare register from 0x0000.

(B) shows "Low" after TMnOC2 compare match, binary counter then counts up until binary counter reaches TMnOC1 compare register is cleared.

(C) shows "High" again, when the binary counter is cleared.

■ Count Timing of High Precision PWM Output
(When compare register 2 is 0x0000) (Timer 7 and Timer 8)

Figure:11.7.2 shows count timing as compare register 2 is set to 0x0000.

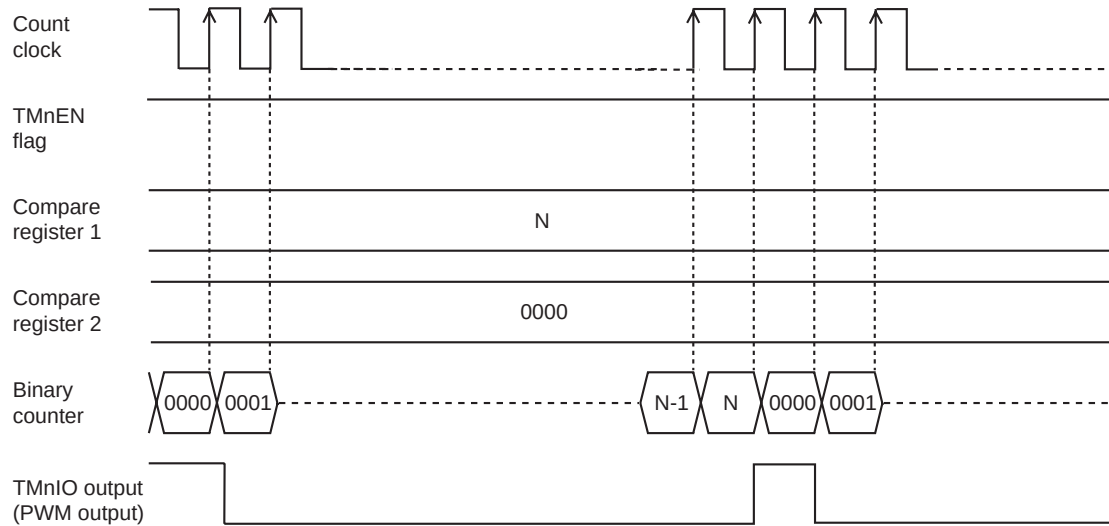


Figure:11.7.2 Count Timing of High Precision PWM Output
(When compare register 2 is 0x0000)

When the TMnEN flag is stopped (at "0"), the PWM output shows "High".

■ Count Timing of High Precision PWM Output
(When compare register 2 = compare register 1 - 1) (Timer 7 and Timer 8)

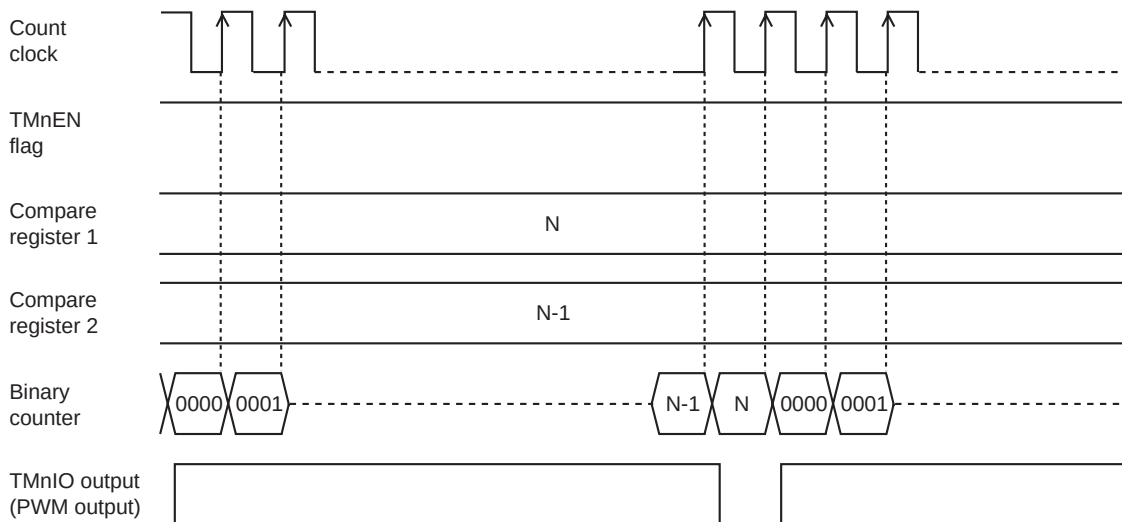


Figure:11.7.3 Count Timing of High Precision PWM Output
(When compare register 2 = compare register 1 - 1)



To execute the high precision PWM output, set the TMnBCR flag of TMnMD2 register to "1" to select TMnOC1 compare match as the clear source for the binary counter, and the set ("High" output) source of the PWM output.

Also, set the TnPWMLS flag to "1" to select TMnOC2 compare match as the reset ("Low" output) source of the PWM output.



In the initial state of the PWM output, it is changed to "High" output from "Low" output at the timing that the PWM operation is selected by the TMnPWM flag of TMnMD register.



Set as the set value of TMnOC2 < the set value of TMnOC1.

If it is set as the set value of TMnOC2 $\geq$ the set value of TMnOC1, the PWM output is a "High" fixed output.

11.7.2 Setup Example

■ High Precision PWM Output Setup Example

The TM7IOA pin outputs the 1/4 duty PWM output waveform at 400 Hz with Timer 7.

Select fpll-div/2 (at fpll-div=10 MHz) as the clock source. One cycle of the PWM output waveform is decided by the set value of compare register 1. "High" period of the PWM output waveform is decided by the set value of compare register 2.

An example setup procedure, with a description of each step is shown below.

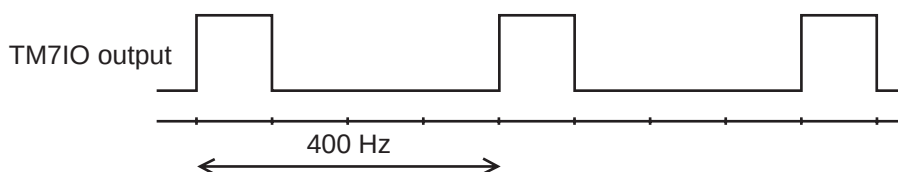


Figure:11.7.4 Output Waveform of TM7IO Output Pin

Setup Procedure	Description
(1) Stop the counter TM7MD1(0x03F78) bp4: TM7EN =0	(1) Set the TM7EN flag of Timer 7 mode register 1 (TM7MD1) to "0" to stop Timer 7 counting.
(2) Set the special function pin to output PAOMD (0x03EE6) bp5: PAOMD5 =1 PADIR (0x03F3A) bp5: PADIR5 =1	(2) Set the PAOMD5 flag of Port A output mode register (PAOMD) to "1" to set the PA5 pin as a special function pin. Set the PADIR5 flag of Port A direction control register (PADIR) to "1" to set the output mode. [Chapter 8 I/O Port]
(3) Set the PWM output TM7MD2(0x03F79) bp4: TM7PWM =1	(3) Set the TM7PWM flag of Timer 7 mode register 2 (TM7MD2) to "1" to select the PWM output.
(4) Set the high precision PWM output TM7MD2(0x03F79) bp5: TM7BCR =1 bp6: T7PWMSL =1	(4) Set the TM7BCR flag of TM7MD2 register to "1" to select TM7OC1 compare match as the clearing factor of the binary counter. Besides, set the T7PWMSL flag to "1" to select TM7OC2 compare match as the duty determination factor of PWM output.
(5) Select the count clock source TM7MD1(0x03F78) bp1 to 0: TM7CK1 to 0 =00 bp3 to 2: TM7PS1 to 0 =00	(5) Select fpll-div as the clock source by the TM7CK1 to 0 flags of TM7MD1 register. Also, select 1/1 dividing as the count clock source by the TM7PS1 to 0 flags.
(6) Select the IGBT/Timer 7 startup factor TM7MD3(0x03F8E) bp1 to 0: T7IGBT1 to 0 =00	(6) Set the IGBT/Timer 7 startup factor to Timer 7 counting operation.
(7) Set the PWM output cycle TM7PR1(0x03F75, 0x03F74) =0x61a7	(7) Set the PWM output cycle to Timer 7 preset register 1 (TM7PR1). To set 400 Hz by dividing 10 MHz, set as; 25000-1=24999 (0x61a7) At the same time, the same value is loaded to Timer 7 compare register 1 (TM7OC1), Timer 7 binary counter is initialized to 0x0000.

Setup Procedure	Description
(8) Set the "High" period of the PWM output TM7PR2(0x03F7D, 0x03F7C) =0x1869	(8) Set "High" period of the IGBT output to Timer 7 preset register 2 (TM7PR2). To set 1/4 duty of 25000 dividing, set as; 25000/4=6250 (0x1869) At the same time, the same value is loaded Timer 7 compare register 2 (TM7OC2).
(9) Start the timer operation TM7MD1(0x03F78) bp4: TM7EN =1	(9) Set the TM7EN flag of TM7MD1 register to "1" to operate Timer 7.

TM7BC counts up from 0x0000. The PWM source waveform outputs "High" until TM7BC matches the set value of TM7OC2 register. Once they matches, it outputs "Low". After that, TM7BC continues to count up. Once TM7BC matches TM7OC1 register to be cleared, the PWM output waveform outputs "High" again and TM7BC counts up from 0x0000 again.

11.8 16-bit Timer Synchronous Output

11.8.1 Operation

If binary counter of the timer reaches the set value of the compare register, Port 8 outputs the output latched data at the next count clock.

■ 16-bit Timer Synchronous Output Operation (Timer 7)

Port 8 outputs the latched data at the match of TM7OC1 register and binary counter or at the interrupt request generation by the full count overflow. Only Port 8 can be used in this operation, and each bit can be set individually.

■ Count Timing of Synchronous Output (Timer 7)

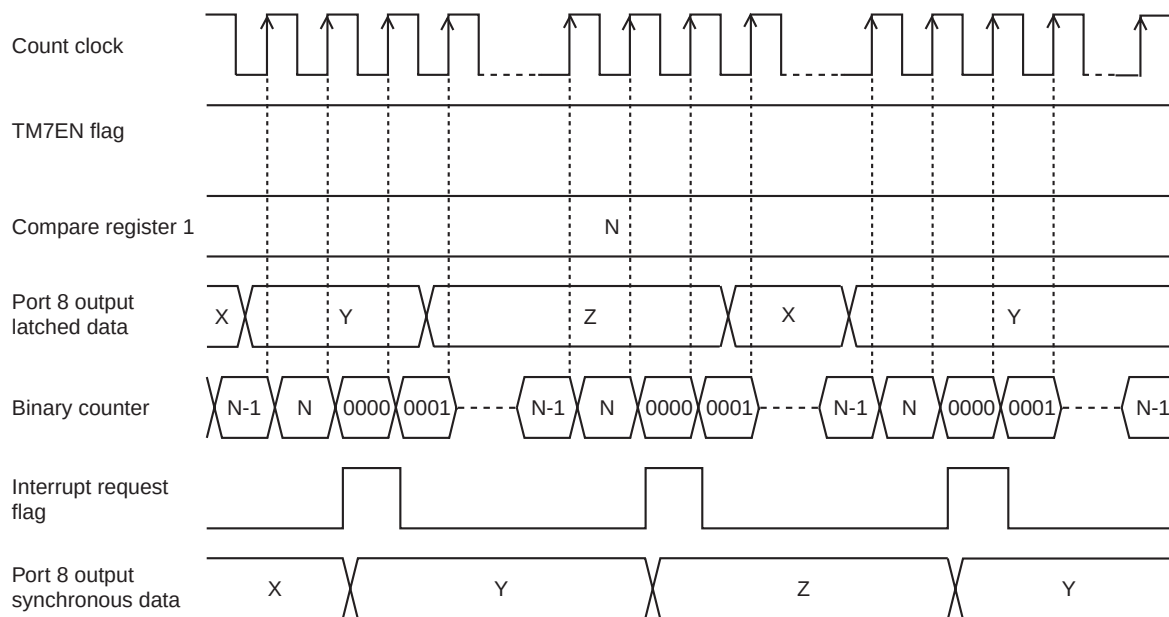


Figure:11.8.1 Count Timing of Synchronous Output (Timer 7)

Output pin outputs Port 8 output latch data at the interrupt request generation by the match of binary counter and compare register 1.

11.8.2 Setup Example

■ Synchronous Output Setup Example

The following is an example to output the data of P8OUT register from the synchronous output pin constantly (in every 100 ms) with Timer 7. As the clock source of Timer 7, $f_s/4$ (at $f_{osc}=4$ MHz) is selected.

Setup Procedure	Description
(1) Stop the counter TM7MD1 (0x03F78) bp4: TM7EN =0	(1) Set the TM7EN flag of Timer 7 mode register 1 (TM7MD1) to "0" to stop Timer 7 counting.
(2) Select the synchronous output event P8SEV (0x03EF8) bp1 to 0: P8SEV1 to 0 =01	(2) Set the P8SEV1 to 0 flags of the pin control register (P8SEV) to "01" to set the synchronous output event to Timer 7 interrupt.
(3) Set the synchronous output pin P8SYO (0x03EF7) =0xFF P8DIR (0x03F38) =0xFF	(3) Set Port 8 synchronous output control register (P8SYO) to 0xFF to set the synchronous output pin. (P87 to P80: synchronous output pin) Set Port 8 direction control register (P8DIR) to 0xFF to set Port 8 to the output pin. [Chapter 8 I/O Port]
(4) Select the timer clear source TM7MD2 (0x03F79) bp5: TM7BCR =1	(4) Set the TM7BCR flag of TM7MD2 register to "1" to select the compare match as the binary counter clear source.
(5) Select the count clock source TM7MD1 (0x03F78) bp1 to 0: TM7CK1 to 0 =01 bp3 to 2: TM7PS1 to 0 =10	(5) Select f_s as the clock source by the TM7CK1 to 0 flags of TM7MD1 register. Also, select 4 frequency dividing as the count clock source by the TM7PS1 to 0 flags.
(6) Set the synchronous output event generation cycle TM7PR1 (0x03F75, 0x03F74) =0x0063	(6) Set the synchronous output event generation cycle to Timer 7 preset register 1 (TM7PR1). To set 10 kHz by dividing 1 MHz, set as; $100-1=99$ (0x0063) At the same time, the same value is loaded to Timer 7 compare register 1 (TM7OC1), and Timer 7 binary counter (TM7BC) is initialized to 0x0000.
(7) Start the timer operation TM7MD1 (0x03F78) bp4: TM7EN =1	(7) Set the TM7EN flag of TM7MD1 register to "1" to operate Timer 7.

TM7BC counts up from 0x0000. If any data is written to Port 8 output register (P8OUT), TM7BC is set to the set value of TM7OC1 register and the synchronous output pin outputs data of Port 8 every time the interrupt request is generated.

11.9 16-bit Timer Capture

11.9.1 Operation

The value of binary counter is read during occurrence of the external interrupt input signal which is synchronized to fpll-div, fs or external event signal, Timer 0 or Timer 1 interrupts or the writing any value to the capture register. The capture register can be double-buffered by setting TM7MD5.

■ Capture Operation with External Interrupt Signal as the Trigger (Timer 7 and Timer 8)

Input capture trigger is generated at the external interrupt input signal. The capture trigger is selected by Timer n mode register 1 (TMnMD1) and Timer n mode register 2 (TMnMD2).

The selectable types of capture trigger and the interrupt flag settings are shown below.

Table:11.9.1 Capture Trigger

Capture trigger source	TM7MD2		TM7MD1	TM8MD2		TM8MD1
	T7ICT1 to 0	T7ICEDG0	T7ICEDG1	T8ICT1 to 0	T8ICEDG0	T8ICEDG1
IRQ0 falling edge	00 (IRQ0)	1	0	00 (IRQ0)	1	0
IRQ0 rising edge	00 (IRQ0)	1	1	00 (IRQ0)	1	1
IRQ0 both edges	00 (IRQ0)	0	X	00 (IRQ0)	0	X
IRQ1 falling edge	01 (IRQ1)	1	0	01 (IRQ1)	1	0
IRQ1 rising edge	01 (IRQ1)	1	1	01 (IRQ1)	1	1
IRQ1 both edges	01 (IRQ1)	0	X	01 (IRQ1)	0	X
IRQ2 falling edge	10 (IRQ2)	1	0	10 (IRQ2)	1	0
IRQ2 rising edge	10 (IRQ2)	1	1	10 (IRQ2)	1	1
IRQ2 both edges	10 (IRQ2)	0	X	10 (IRQ2)	0	X



If the system clock (fs) is selected as the capture clock and the capture operation is done during TMnIO input or operation with fpll-div, an incomplete value at the count up of the binary counter may be written to the input capture register.

To prevent this, use fx or synchronous TMnIO input as the count clock.
[Chapter 11 11.4.1 Operation].



Capture trigger signals of the 16-bit Timer 7 and Timer 8 are generated by sampling the rising edge of the capture clock selected by the TMnCKSMP flag of TMnMD3 register.

Therefore, even capture trigger is input, the value of the binary counter is not loaded to the capture register until the rising edge of the next capture clock.

If the clock which is slower than CPU operation speed (fs) is set as the timer source clock, set the TMnCKSMP of TMnMD3 register to fs.

Also, the interval of each capture trigger should be set more than 2 cycles of the clock which is set at the TMnCKSMP of TMnMD3 register.



If the capture clock frequency is longer against the system clock, the value of the capture register may be read out before capturing.

■ Capture Count Timing as Both Edges of External Interrupt Signal are selected as Trigger (Timer 7 and Timer 8)

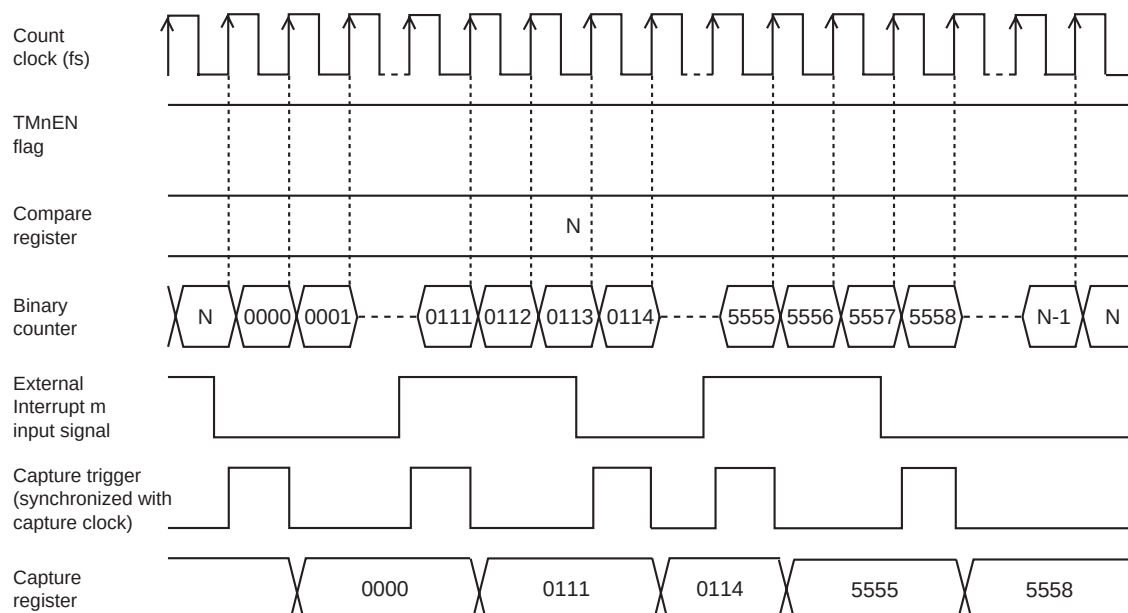


Figure:11.9.1 Capture Count Timing as External Interrupt Signal is selected as Trigger (Timer 7 and Timer 8)

A capture trigger is generated at the both edges of External Interrupt m input signal. In synchronized with this capture trigger, the value of binary counter is loaded to input capture register. The value loaded to capture register is depending on the value of binary counter at the falling edge of the capture trigger. When the specified edge is selected as the capture trigger source, the capture trigger is generated only at that edge. The other count timing is the same as the count timing of the timer operation.

When the capture trigger is started at both edges of External Interrupt input signal, if it is engaged with data automatic transfer (ATC1), it is possible to measure the precision width which can measure the input signal between "High" and "Low" continuously. Set the address of input capture register TMnICL to the memory pointer 1 by the transfer mode 5 of ATC1. Transferring the value of input capture register (TMnICL, TMnICH) to the memory sequentially with every generation of the capture trigger make it possible to measure the input signal between "High" and "Low" continuously.

■ Capture Count Timing as Both Edges are selected as Trigger with Double Buffered (Timer 7)

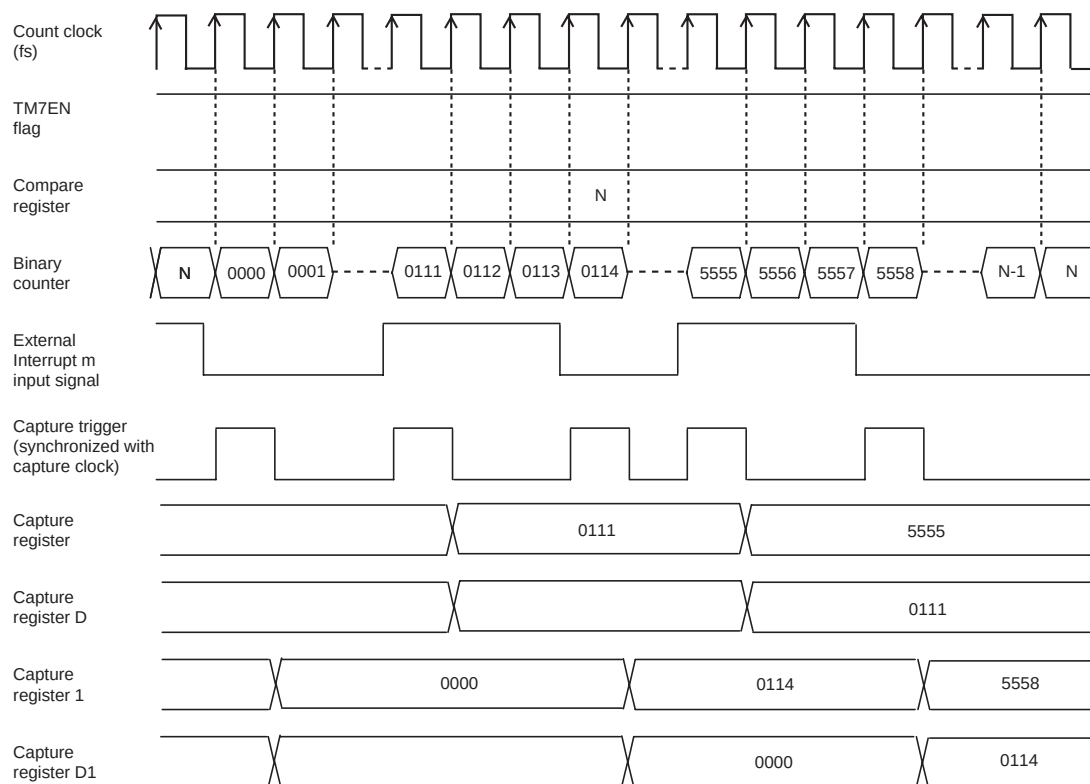


Figure:11.9.2 Capture Count Timing as External Interrupt Signal are selected as Trigger (with Double Buffered)

A capture trigger is generated at both edges of External Interrupt m input signal.

In synchronized with the rising edge of the capture trigger of the external interrupt input signal, the value of binary counter is loaded to input capture register.

In synchronized with the falling edge of the capture trigger of the external interrupt input signal, the value of binary counter is loaded to input capture register 1.

The value of capture register is loaded to capture register D when updating capture register.

When the specified edge is selected as the capture trigger source, the capture trigger is generated only at that edge.

The other count timing is the same as the count timing of the timer operation.



When binary counter is used as a free counter which counts 0x0000 to 0xFFFF, set compare register 1 to 0xFFFF, or set the TMnBCR flag of TMnMD2 to "0".



Even if an event is generated before the value of input capture register is read out, the value of input capture register can be rewritten.

If the writing and reading of the value of the input capture register are operated at the same time, an uncompleted value may be read.



In the initial state after releasing the reset, the generation of trigger by External Interrupt signal is disabled. Set the TnICEN flag of TMnMD2 register to "1" to enable the trigger generation.



When using External Interrupt signal as capture trigger, enable External Interrupt input by IRQCNT register.



When Timer 7 input capture register is valid, it executes capturing only at the rising edge of the capture trigger and Timer 7 input capture register 1 executes capturing only at the falling edge of the capture trigger.

■ Capture Operation Triggered by Writing Software (Timer 7 and Timer 8)

A capture trigger is generated by writing an arbitrary value to input capture register (TMnIC). When writing a value to the register, the capture trigger synchronized with the clock which is set by the TMnCKSMP flag of timer mode register 3 is generated. At the timing of the capture trigger falling, the value of binary counter can be loaded into input capture register.

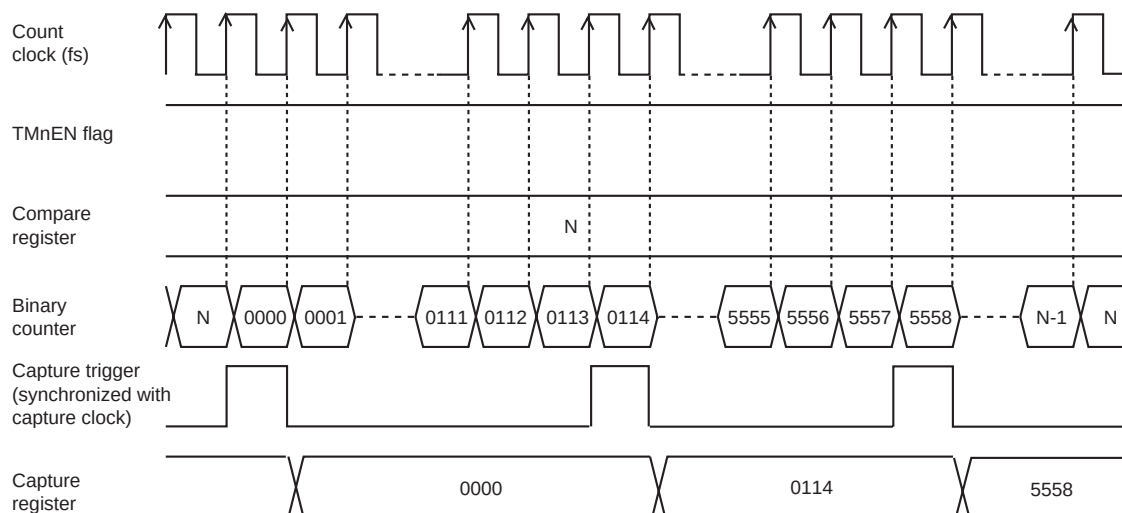


Figure:11.9.3 Capture Count Timing Triggered by Writing Software (Timer 7 and Timer 8)

The capture trigger is generated at the write signal to input capture register. The write signal is generated at the last cycle of the write instruction. In synchronized with this capture trigger, the value of binary counter is loaded to input capture register. The value is depending on the value of binary counter at the falling edge of the capture trigger. The other timing is the same as the timer operation.



The writing to the input capture to generate the capture trigger should be done with 8-bit access instruction to TMnICL register or TMnICH register. At this time, data is not actually written to TMnIC register.



On hardware, there is no flag to disable the capture operation triggered by software writing. Capture operation is enabled regardless of the TnICEN flag of TMnMD2 register.

■ Capture Operation as Timer 0 and Timer 1 interrupts are selected as Trigger (Timer 7 and Timer 8)

A capture trigger of the input capture function is generated by Timer 0 and Timer 1 interrupts signals. Select the capture trigger by timer mode register 2 (TMnMD2) and timer mode register 4 (TMnMD4). When Timer 0 and Timer 1 interrupt signals are selected as the capture trigger, the edges of the capture trigger are disabled.

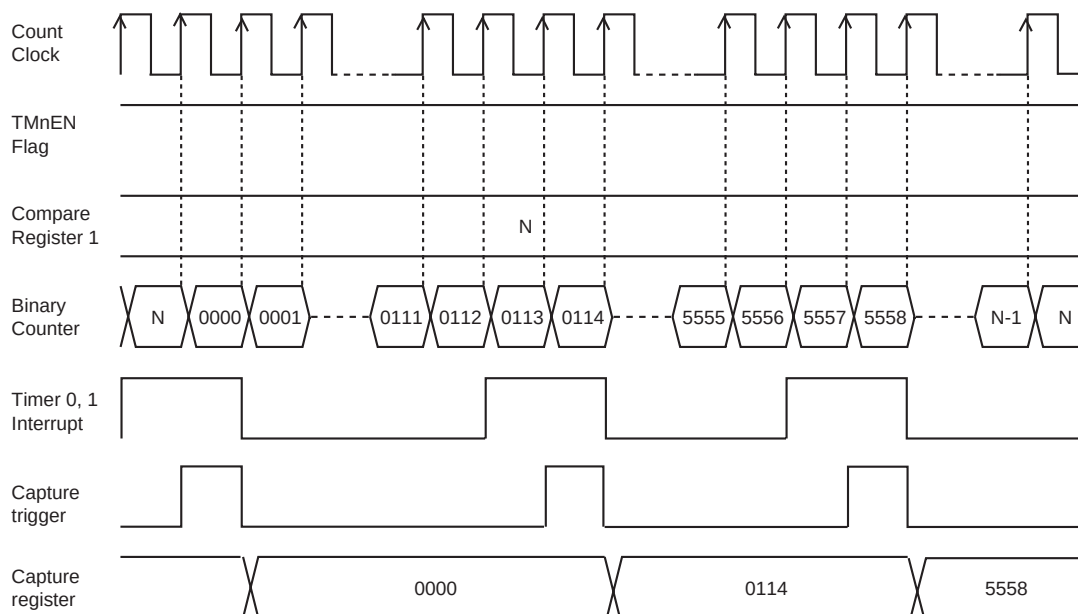


Figure:11.9.4 Capture Operation as Timer 0 and Timer 1 interrupts are selected as Trigger (Timer 7 and Timer 8)



When the TnICT1 to 0 flags of timer mode register 2 is set to "11", a capture trigger of the input capture function is generated by Timer 0 and Timer 1 interrupt signals. Select the capture trigger by timer mode register 2 (TMnMD2) and timer mode register 4 (TMnMD4). When Timer 0 and Timer 1 interrupt signals are selected as the capture trigger, the edges of the capture trigger are disabled.

When setting the TMnCKSMP flag to "Count clock" to execute the event count operation, Timer 0 and Timer 1 interrupt signals may not be recognized. To prevent this, select the synchronous TMnIO input as the clock source.

■ Binary Counter Clearance at the Timing of Capture (Timer 7 and Timer 8)

When selecting the external interrupt input signal or Timer 0 and Timer 1 interrupts as the capture trigger, binary counter can be cleared during capture operation by setting the TnCAPCLR flag of timer mode register 4 (TMnMD4) to "1". Binary counter can only be cleared during timer count operation.

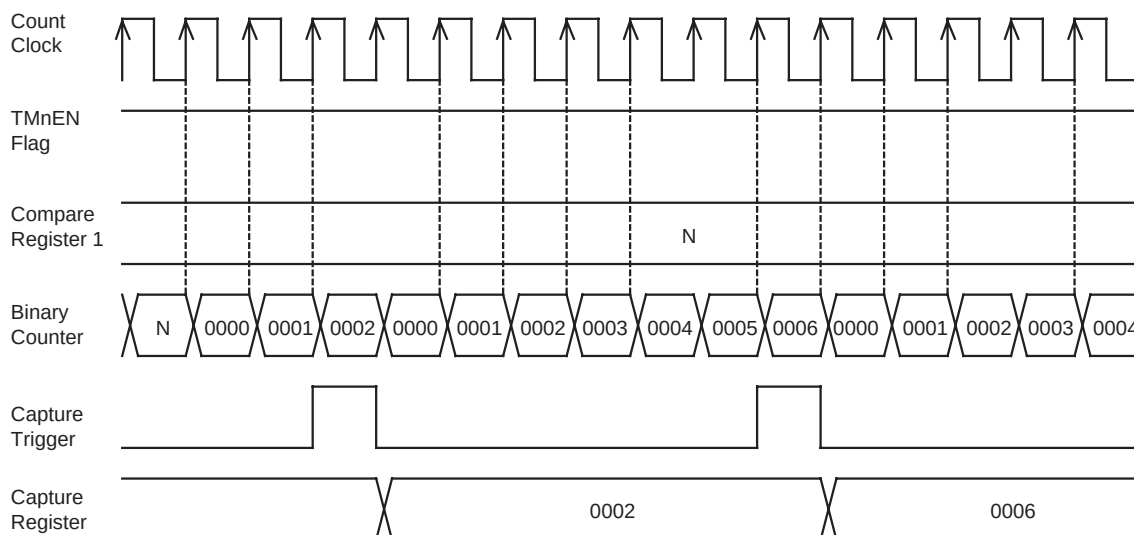


Figure:11.9.5 Binary Counter Clearance at the Timing of Capture (Timer 7 and Timer 8)

■ Binary Counter Clearance at the Timing of Capture

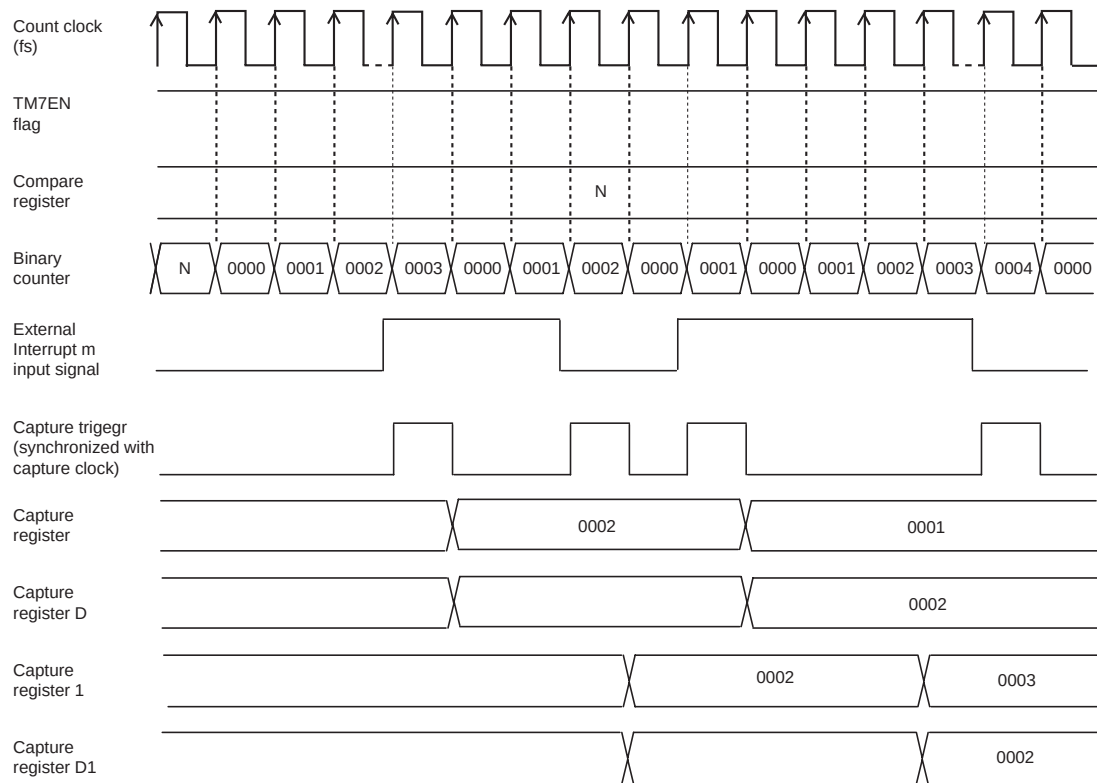


Figure:11.9.6 Binary Counter Clearance at the Timing of Capture (with Double Buffered)



Only external interrupts and software interrupt can be selected as capture triggers when double buffered is configured. Do not set Timer 0 and Timer 1.

11.9.2 Setup Example

■ Capture Function Setup Example

Pulse width measurement is enabled by storing the value of the binary counter to the capture register at the interrupt generation edge of the External Interrupt 0 signal with Timer 7. The interrupt generation edge is specified to be the rising edge.

An example setup procedure, with a description of each step is shown below.

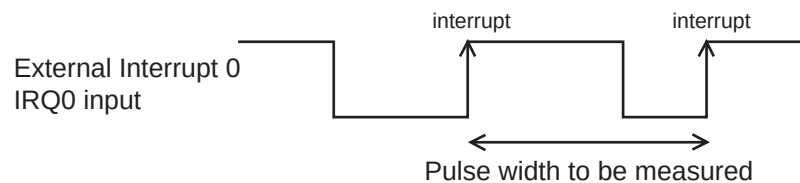


Figure:11.9.7 Pulse Width Measurement of External Interrupt 0 Input Signal

Setup Procedure	Description
(1) Stop the counter TM7MD1(0x03F78) bp4: TM7EN =0	(1) Set the TM7EN flag of Timer 7 mode register 1 (TM7MD1) to "0" to stop Timer 7 counting.
(2) Disable the interrupt IRQ0ICR(0x03FE2) bp1: IRQ0IE =0	(2) Set the IRQIE flag of IRQ0ICR register to "0" to disable the interrupt.
(3) Select the timer clear source TM7MD2(0x03F79) bp5: TM7BCR =1	(3) Set the TM7BCR flag of Timer 7 mode register 2 (TM7MD2) to "1" to select the compare match as binary counter clear source.
(4) Select the count clock source TM7MD1(0x03F78) bp1 to 0: TM7CK1 to 0 =00 bp3 to 2: TM7PS1 to 0 =00	(4) Select fpll-div as the clock source by the TM7CK1 to 0 flags of TM7MD1 register. Also, select 1/1 dividing of fpll-div as the count clock source by the TM7PS1 to 0 flags.
(5) Select the IGBT/Timer startup factor TM7MD3(0x03F8E) bp1 to 0: T7IGBT1 to 0 =00	(5) Set the IGBT/Timer startup factor to Timer 7 counting operation.
(6) Set the compare register TM7PR1(0x03F75, 0x03F74) =0xFFFF	(6) Set 0xFFFF to Timer 7 preset register 1(TM7PR1). At that time, the same value is loaded to Timer 7 compare register 1 (TM7OC1), Timer 7 binary counter (TM7BC) is initialized to 0x0000.
(7) Select the capture trigger generation interrupt source TM7MD2(0x03F79) bp1 to 0: T7ICT1 to 0 =00	(7) Select the External Interrupt 0 (IRQ0) input as the capture trigger generation source by the T7ICT1 to 0 flags of TM7MD2 register.

Setup Procedure	Description
(8) Select the capture trigger generation edge TM7MD1(0x03F78) bp6: T7ICEDG1 =1 TM7MD2 (0x03F79) bp7: T7ICEDG0 =1	(8) Set the T7ICEDG1 flag of TM7MD1 register to "1" to select the rising edge as the capture trigger generation edge. Also, set the T7ICEDG0 flag of TM7MD2 register to "1" to enable the specify edge as the capture trigger generation source.
(9) Select the capture sampling TM7MD3(0x03F8E) bp7: TM7CKSMP =0	(9) Select the capture sampling as the count clock.
(10) Select the interrupt generation valid edge IRQ0ICR(0x03FE2) bp5: REDG0 =1	(10) Set the REDG0 flag of the External Interrupt 0 control register (IRQ0ICR) to "1" to select the rising edge as the interrupt generation valid edge.
(11) Set the interrupt level IRQ0ICR(0x03FE2) bp7 to 6: IRQ0LV1 to 0 =10	(11) Set the interrupt level by the IRQ0LV1 to 0 flag of IRQ0ICR register. If the interrupt request flag has already been set, clear the request flag. [Chapter 7 7.1.4 Maskable Interrupt Control Register Setup]
(12) Enable the interrupt IRQ0ICR(0x03FE2) bp1: IRQ0IE =1	(12) Set the IRQ0IE flag of IRQ0ICR register to "1" to enable the interrupt.
(13) Enable the capture trigger generation TM7MD2(0x03F79) bp2: T7ICEN =1	(13) Set the T7ICEN flag of TM7MD2 register to "1" to enable the capture trigger generation.
(14) Start the timer operation TM7MD1(0x03F78) bp4: TM7EN =1	(14) Set the TM7EN flag of TM7MD1 register to "1" to operate Timer 7.

TM7BC counts up from 0x0000. At the timing of the rising edge of the External Interrupt 0 input signal, the value of TM7BC is loaded to TM7IC register. At that time, the pulse width between rising edge of the external interrupt input signal can be measured by reading the value of TM7IC register through interrupt service routine, and calculating the difference between the capture values.

■ Capture Function Setup Example when Double buffered

Pulse width measurement is enabled by storing the value of the binary counter to the capture register at the interrupt generation edge of the External Interrupt 0 signal with Timer 7. The interrupt generation edge is specified to be the rising edge.

An example setup procedure, with a description of each step is shown below.

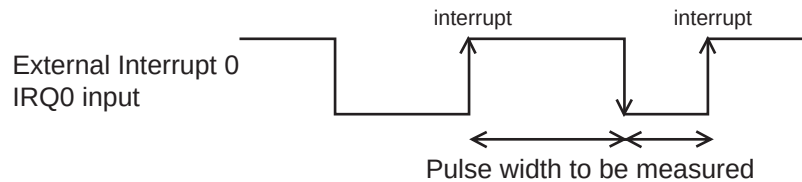


Figure:11.9.8 Pulse Width Measurement of External Interrupt 0 Input Signal

Setup Procedure	Description
(1) Stop the counter TM7MD1(0x03F78) bp4: TM7EN =0	(1) Set the TM7EN flag of Timer 7 mode register 1 (TM7MD1) to "0" to stop Timer 7 counting.
(2) Disable the interrupt IRQ0ICR(0x03FE2) bp1: IRQ0IE =0	(2) Set the IRQIE flag of IRQ0ICR register to "0" to disable the interrupt.
(3) Select the timer clear source TM7MD2(0x03F79) bp5: TM7BCR =1	(3) Set the TM7BCR flag of Timer 7 mode register 2 (TM7MD2) to "1" to select the compare match as the binary counter clear source.
(4) Select the count clock source TM7MD1(0x03F78) bp1 to 0: TM7CK1 to 0 =00 bp3 to 2: TM7PS1 to 0 =00	(4) Select fpll-div as the clock source by the TM7CK1 to 0 flags of TM7MD1 register. Also, select 1/1 dividing of fpll-div as the count clock source by the TM7PS1 to 0 flags.
(5) Select the IGBT/Timer startup factor TM7MD3(0x03F8E) bp1 to 0: T7IGBT1 to 0 =00	(5) Set the IGBT/Timer startup factor to Timer 7 counting operation.
(6) Set the compare register TM7PR1(0x03F75, 0x03F74) =0xFFFF	(6) Set 0xFFFF to Timer 7 preset register 1(TM7PR1). At that time, the same value is loaded to Timer 7 compare register 1 (TM7OC1), Timer 7 binary counter (TM7BC) is initialized to 0x0000.
(7) Select the capture trigger generation interrupt source TM7MD2(0x03F79) bp1 to 0: T7ICT1 to 0 =00	(7) Select the External Interrupt 0 (IRQ0) input as the capture trigger generation source by the T7ICT1 to 0 flags of TM7MD2 register.
(8) Select the capture trigger generation edge TM7MD2 (0x03F79) bp7: T7ICEDG0 =1	(8) Set the T7ICEDG0 flag of TM7MD2 register to "1" to enable the both edges as the capture trigger generation source.

Setup Procedure	Description
(9) Select the capture sampling TM7MD3(0x03F8E) bp7: TM7CKSMP =0	(9) Select the capture sampling as the count clock.
(10) BC clear at capture TM7MD4(0x03F7F) bp0: T7CAPCLR =1	(10) Clear the binary counter at capture.
(11) Set the input capture register 1 TM7MD5(0x03F6E) bp0: T7UD =1	(11) Set input capture register 1 to valid. By this operation, the value of the rising edge is captured to the input capture register, and the value of the falling edge is captured to input capture register 1.
(12) Select the interrupt generation valid edge IRQ0ICR(0x03FE2) bp5: REDG0 =1	(12) Set the REDG0 flag of External Interrupt 0 control register (IRQ0ICR) to "1" to select the rising edge as the interrupt generation valid edge.
(13) Set the interrupt level IRQ0ICR(0x03FE2) bp7 to 6: IRQ0LV1 to 0 =10	(13) Set the interrupt level by the IRQ0LV1 to 0 flag of IRQ0ICR register. If the interrupt request flag is already set, clear the request flag. [Chapter 7 7.1.4 Maskable Interrupt Control Register Setup]
(14) Enable the interrupt IRQ0ICR(0x03FE2) bp1: IRQ0IE =1	(14) Set the IRQ0IE flag of IRQ0ICR register to "1" to enable the interrupt.
(15) Enable the capture trigger generation TM7MD2(0x03F79) bp2: T7ICEN =1	(15) Set the T7ICEN flag of TM7MD2 register to "1" to enable the capture trigger generation.
(16) Start the timer operation TM7MD1(0x03F78) bp4: TM7EN =1	(16) Set the TM7EN flag of TM7MD1 register to "1" to operate Timer 7.

TM7BC counts up from 0x0000. At the timing of the rising edge of the External Interrupt 0 input signal, the value of TM7BC is loaded to TM7IC register and TM7BC is cleared.

Next, TM7BC counts up from 0x0000. At the timing of the rising edge of the External Interrupt 0 input signal, the value of TM7BC is loaded to TM7IC register.

At that time, the pulse width of "High" and "Low" period of the external interrupt input signal can be measured by reading the value of TM7IC register and TM7IC1 register by the interrupt service routine.

11.1016-bit High Precision IGBT Output (Cycle/Duty can be changed continuously)

High precision IGBT output starts counting by treating the external interrupt input signal as trigger and output from TM7I0 and TM8I0. Startup trigger can be selected by External Interrupt 0, 1 and 2 or starting of Timer 7 count operation. When counting starts, the operation is the same as the high precision PWM output.

11.10.1 16-bit High Precision IGBT Output Operation

■ IGBT Trigger Selection

IGBT trigger can be selected from IRQ0, IRQ1, IRQ2 or the start of Timer 7 count operation. Setup should be done at the T7IGBT0 and T7IGBT1 flags of TM7MD3 register.

When the startup is controlled from external of the microcontroller, either IRQ0 or IRQ2 should be selected.

IGBT starts after a trigger input level is detected. Either "High" or "Low" level can be selected with the T7IGBTTR flag of TM7MD3 register.

When "1" (the rising edge) is selected, count operation continues while the trigger pin is "High". When "0" (the falling edge) is selected, count operation continues while the trigger pin is "Low".

To control the startup by commands, TM7EN count operation should be selected. In this case, timer count operation or IGBT output are controlled by the TM7EN flag of TM7MD1 register.

When "1" (count operation) is selected, count continues counting until "0" (count stop) is set. Make sure to set the TM7IGBT0 and TM7IGBT1 of TM7MD3 register before operating the TM7EN flag of TM7MD1 register. In this case, setup of T7IGBTTR is neglected. The binary counter is cleared as the counting stops. The value is loaded from the preset register to the compare register in synchronization with the counter clock.

■ 16-bit High Precision IGBT Output (Timer 7)

The IGBT waveform of any cycle/duty is generated by setting the cycle of IGBT to TM7OC1 register and setting the duty of the "High" period to TM7OC2 register. Timer 7 can be used for high precision IGBT output.

Table:11.10.1 IGBT Output Pin

	Timer 7	Timer 8
IGBT output pin	TM7IOA output TM7IOB output TM7IOC output	TM8IOA output TM8IOB output TM8IOC output

Table:11.10.2 IGBT Trigger

	Timer 7 mode register 3	
	T7IGBT1 to T7IGBT0	T7IGBTTR
IRQ0 falling edge	01 (IRQ0)	1
IRQ0 rising edge	01 (IRQ0)	0
IRQ1 falling edge	10 (IRQ1)	1
IRQ1 rising edge	10 (IRQ1)	0
IRQ2 falling edge	11 (IRQ2)	1
IRQ2 rising edge	11 (IRQ2)	0
TM7EN counting operation	00	-

■ One Shot Pulse Output Setup

One shot pulse output can be done by setting the T7ONESHOT flag of TM7MD4 register to "1".

■ Count Timing of High Precision IGBT Output (At Normal) (Timer 7)

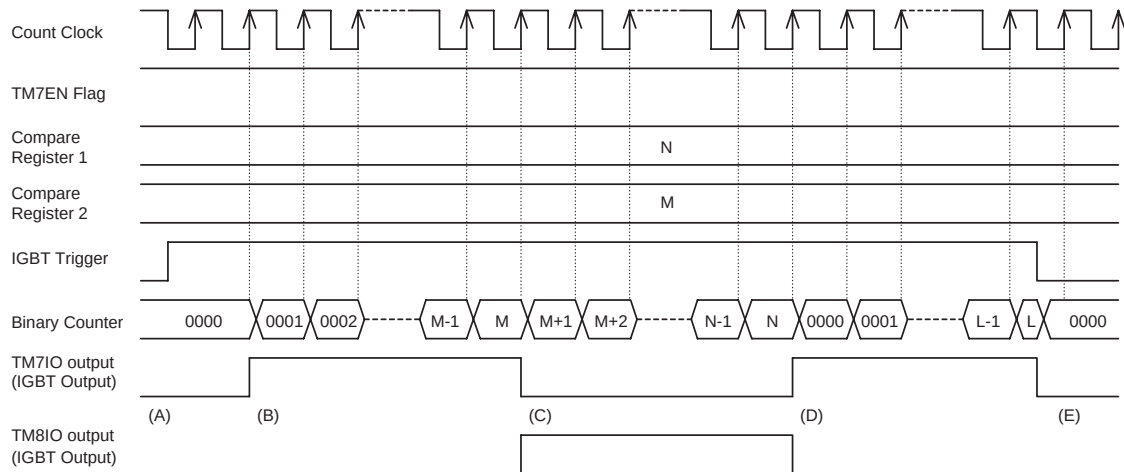


Figure:11.10.1 Count Timing of High Precision IGBT Output (At Normal)

- (A) When IGBT trigger is input, IGBT operation becomes valid after 1 count clock.
After IGBT output is valid, it is "Low" until the next count clock.
- (B) When IGBT trigger is valid, it is "High" during the period when the value of the binary counter reaches that of TM7OC2 from 0x0000. ("High" output from 0x0001 at the first operation cycle)
- (C) After the TM7OC2 compare match, it is "Low" and the binary counter counts up until the counter reaches TM7OC1 compare register to be cleared.
- (D) When the binary counter is cleared, it becomes "High" again.
- (E) When IGBT trigger becomes invalid, the timer is initialized and IGBT output forcibly becomes "Low".

■ Count Timing of High Precision IGBT Output
(When compare register 2 is 0x0000) (Timer 7)

The following shows the count timing as compare register 2 is set to 0x0000.

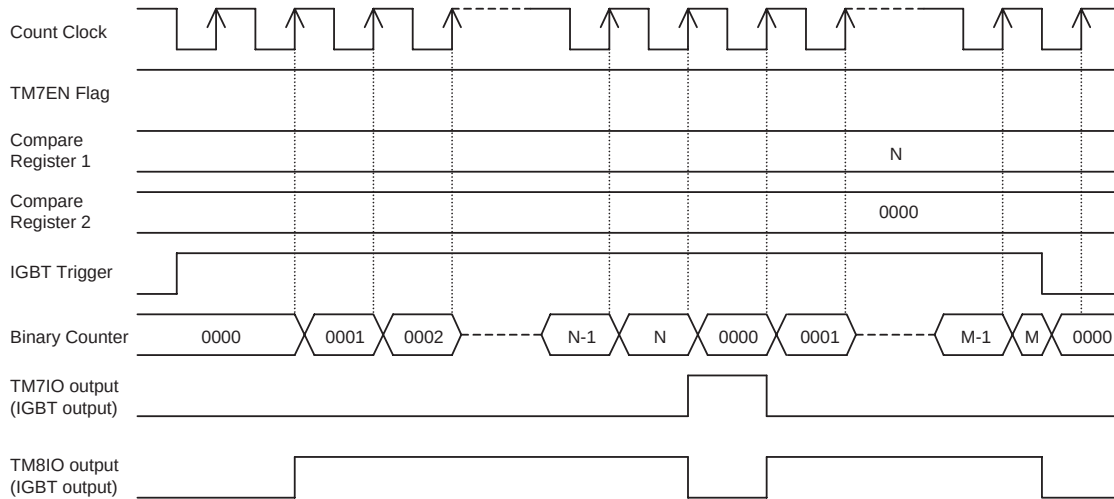


Figure:11.10.2 Count Timing of High Precision IGBT Output
(When compare register 2 is 0x0000)

When the TM7EN flag is set to "0" (stop status) and the T7IGBTDT of TM7MD3 register is set to "0", both TM7IO and TM8IO output "Low".

■ Count Timing of High Precision IGBT Output
(When compare register 2 = compare register 1) (Timer 7)

The following shows the count timing when the value of compare register 1 is set to compare register 2.

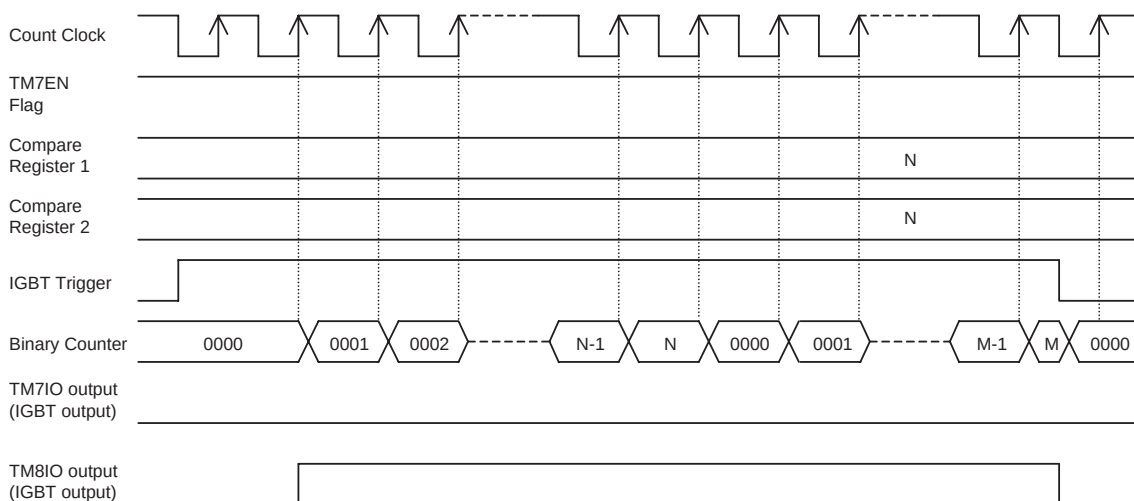


Figure:11.10.3 Count Timing of High Precision IGBT Output
(When compare register 2 = compare register 1)



For high precision IGBT output, set the TM7BCR flag of TM7MD2 register to "1" and select TM7OC1 compare match as the binary counter clear factor and IGBT output set ("High" output) factor. Also, set the TM7PWMSL flag of TM7MD2 register to "1" and select the TM7OC2 compare match as the IGBT output reset ("Low" output) factor.



In the initial state of the IGBT output (TM7IO), when the IGBT output is selected by the T7IGBTEN of TM7MD3 register, it is "Low" output. After the trigger is input, it changes to "High" at the second cycle.



Set as $TM7OC2 \text{ value} \leq TM7OC1 \text{ value}$.
When $TM7OC2 \text{ value} > TM7OC1 \text{ value}$, the IGBT output waveform is fixed to "High".

■ One Shot Pulse Output of High Precision IGBT Output (At Normal) (Timer 7)

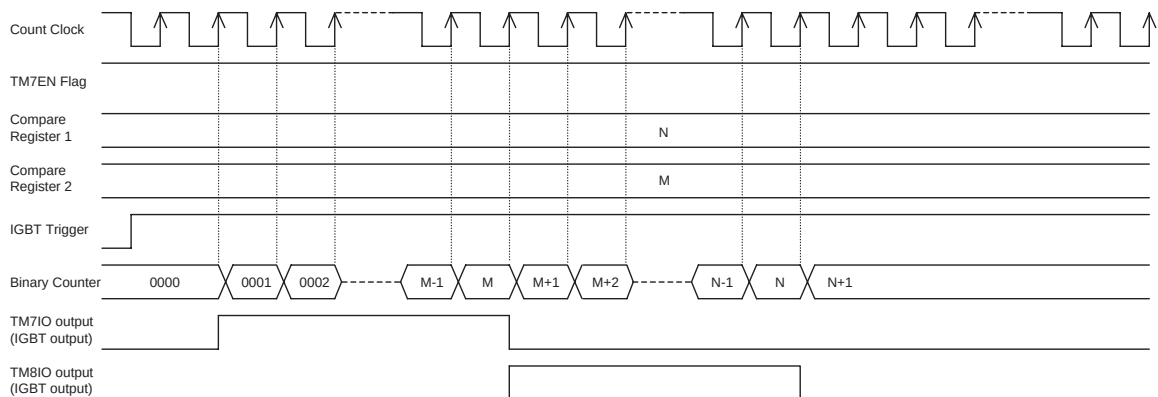


Figure:11.10.4 One Shot Pulse Output of High Precision IGBT Output (At Normal) (Timer 7)

■ One Shot Pulse Output of High Precision IGBT Output
(When compare register 2 is 0x0000) (Timer 7)

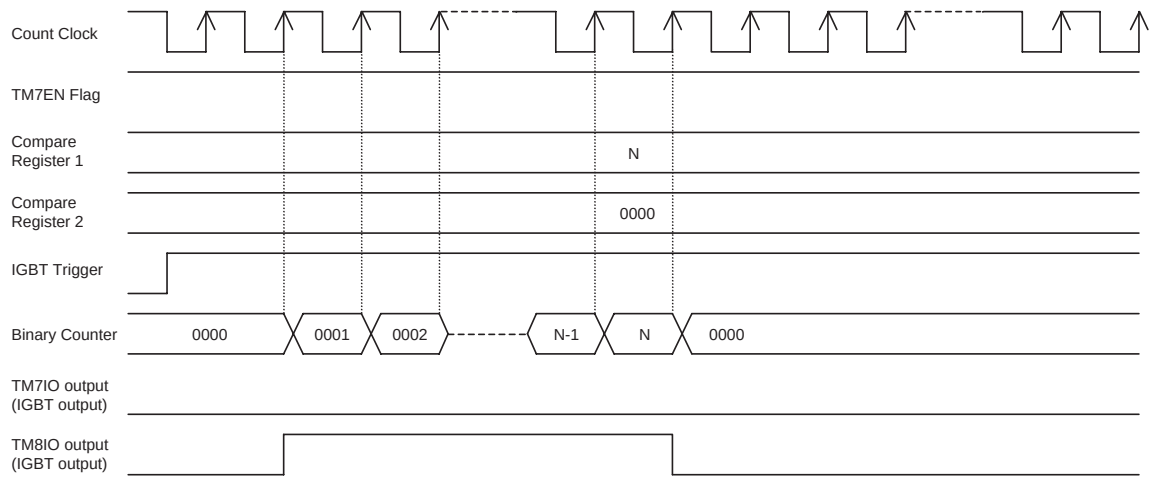


Figure:11.10.5 One Shot Pulse Output of High Precision IGBT Output
(When compare register 2 is 0x0000) (Timer 7)

■ One Shot Pulse Output of High Precision IGBT Output
(When compare register 2 = compare register 1) (Timer 7)

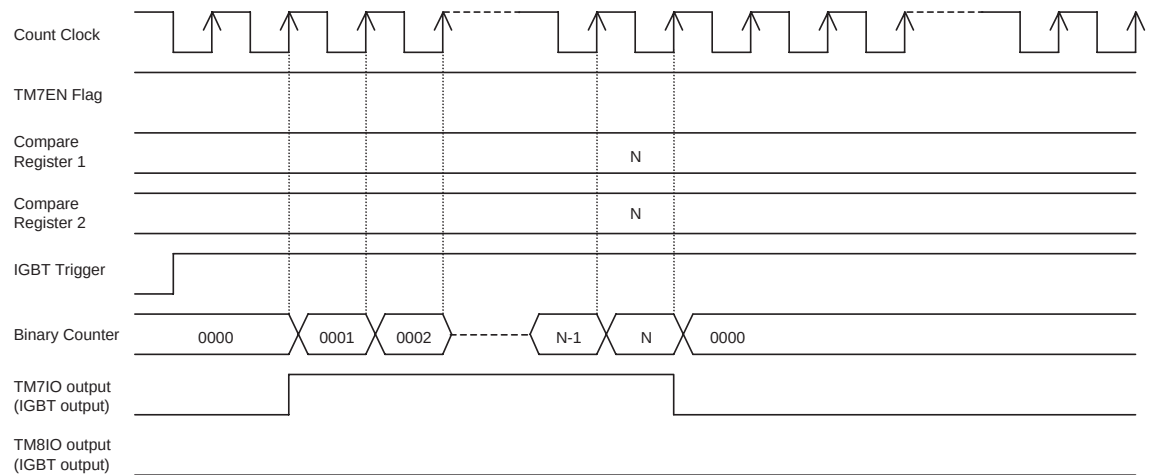


Figure:11.10.6 One Shot Pulse Output of High Precision IGBT Output
(When compare register 2 = compare register 1) (Timer 7)

11.10.2 16-bit High Precision IGBT Output Setup Example

■ High precision IGBT Output Setup Example (Timer 7)

At the interrupt generation edge of the External Interrupt 0 input signal, TM7IOA output pin outputs the waveform of 1/4 duty IGBT output waveform at 400 Hz using Timer 7. Select fpll-div (at fpll-div=10 MHz) as the clock source. Required period for one IGBT output waveform cycle depends on the set value of compare register 1. "High" period of IGBT output waveform depends on the set value of compare register 2.

An example setup procedure, with a description of each step is shown below.

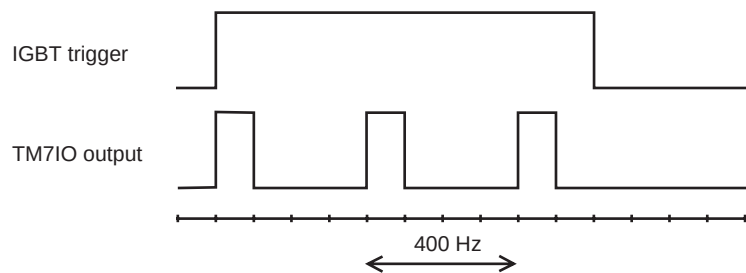


Figure:11.10.7 Output Waveform of TM7IO Output Pin

Setup Procedure	Description
(1) Stop the counter TM7MD1(0x03F78) bp4: TM7EN =0	(1) Set the TM7EN flag of Timer 7 mode register 1 (TM7MD1) to "0" to stop Timer 7 counting.
(2) Set the output of special function pins PAOMD(0x03EE6) bp5: PAOMD5 =1 bp6: PAOMD6 =1 PADIR(0x03F3A) bp5: PADIR5=1 bp6: PADIR6=1	(2) Set the PAOMD5 flag of Port A output mode register (PAOMD) to "1" to set the PA5 pin to the special function pin. Set the PAOMD6 flag of Port A output mode register (PAOMD) to "1" to set the PA6 pin to the special function pin. Set the PADIR5 flag of Port A direction control register (PADIR) to "1", the PADIR6 flag to "1" to set the output mode. Add pull-up/pull-down resistor if necessary. [Chapter 8 I/O Port]
(3) Set Timer 8 output pin TM8MD4(0x03F6F) bp2: TM8SEL_A =1	(3) Set TM8SEL_A flag of Timer 8 mode register 4 (TM8MD4) to "1" to select the output from PA6 pin to IGBT output.
(4) Set IGBT output TM7MD3(0x03F8E) bp2: TM7IGBTEN =1	(4) Set the T7IGBTEN flag of Timer 7 mode register 3 (TM7MD3) to "1" to select IGBT output.
(5) Set the high precision IGBT output operation TM7MD2(0x03F79) bp5: TM7BCR =1 bp6: T7PWMSL =1 TM7MD4(0x03F6E) bp3: T7NODED =1	(5) Set the TM7BCR flag of TM7MD2 register to "1" to select the TM7OC1 compare match as the clear factor of the binary counter. Set the T7PWMSL flag to "1" to select the TM7OC2 compare match as the duty determination factor of IGBT output. Set the T7NODED flag of TM7MD4 register to "1" to select the IGBT waveform without dead time.

Setup Procedure	Description
(6) Select IGBT trigger generation interrupt source TM7MD3(0x03F8E) bp1 to 0: T7IGBT1 to 0 =01	(6) Set External Interrupt 0 (IRQ0) input as IGBT trigger generation factor by the T7IGBT1 to 0 flags of TM7MD3 register.
(7) Select the interrupt generation valid edge IRQ0ICR(0x03FE2) bp5: REDG0 =1	(7) Set the REDG0 flag of External Interrupt 0 control register (IRQ0ICR) to "1" to select the rising edge as the interrupt generation valid edge.
(8) Select IGBT trigger generation level TM7MD3(0x03F8E) bp4: T7IGBTTR=0	(8) Set the T7IGBTTR flag of TM7MD3 register to "0" to set the IGBT trigger level to "High".
(9) Select IGBT trigger generation edge TM7MD2(0x03F79) bp7: T7ICEDG=1	(9) Set the T7ICEDG flag of TM7MD2 register to "1" to select the external interrupt specified edge as the IGBT trigger generation factor.
(10) Set the interrupt level IRQ0ICR(0x03FE2) bp7 to 6: IRQ0LV1 to 0 =10	(10) Set the interrupt level by the IRQ0LV1 to 0 flag of IRQ0ICR register. If any interrupt request flag is already set, clear it. [Chapter 7 7.1.4 Maskable Interrupt Control Register Setup]
(11) Enable the interrupt IRQ0ICR(0x03FE2) bp1: IRQ0IE =1	(11) Set the IRQ0IE flag of IRQ0ICR register to "1" to enable the External Interrupt 0.
(12) Select the count clock source TM7MD1(0x03F78) bp1 to 0: TM7CK1 to 0 =00 bp3 to 2: TM7PS1 to 0 =00	(12) Select fosc as the clock source by the TM7CK1 to 0 flags of TM7MD1 register. Also, select 1/1 dividing as the count clock source by the TM7PS1 to 0 flags.
(13) Set the IGBT output cycle TM7PR1(0x03F75, 0x03F74) =0x61a7	(13) Set the IGBT output cycle to Timer 7 preset register 1 (TM7PR1). To set 400 Hz by dividing 10 MHz, set as; 25000-1=24999 (0x61a7) At the same time, the same value is loaded to Timer 7 compare register 1 (TM7OC1), and Timer 7 binary counter (TM7BC) is initialized to 0x0000.
(14) Set "High" period of the IGBT output TM7PR2(0x03F7D, 0x03F7C) =0x1869	(14) Set the "High" period of the IGBT waveform to Timer 7 preset register 2 (TM7PR2). To set 1/4 duty of 25000 dividing, set as; 25000/4-1=6249(0x1869) At the same time, the same value is loaded to Timer 7 compare register 2 (TM7OC2).
(15) Start the timer operation TM7MD1(0x03F78) bp4: TM7EN= 1	(15) Set the TM7EN flag of TM7MD1 register to "1" to operate Timer 7.

TM7BC counts up from 0x0000 at the interrupt generation edge of the External Interrupt 0 input signal. The IGBT output waveform outputs "High" until TM7BC matches the set value of TM7OC2 register. Once they match, it outputs "Low". After that, TM7BC continues to count up. Once TM7BC value matches TM7OC1 register value and it is cleared, the IGBT output waveform outputs "High" and TM7BC counts up from 0x0000 again.

11.11 16-bit Standard IGBT Output (Only duty can be changed continuously)

Startup trigger can be selected by External Interrupt 0, 1 and 2 or starting of Timer 7 count operation. When counting starts, the operation is the same as 16-bit standard precision PWM output.

11.11.1 16-bit Standard IGBT Output Operation

■ IGBT Trigger Selection

IGBT trigger can be selected from IRQ0, IRQ1, IRQ2 or the start of Timer 7 count operation. Setup should be done at the T7IGBT0 and T7IGBT1 flags of TM7MD3 register.

When the startup is controlled from external of the microcontroller, either IRQ0 or IRQ2 should be selected.

IGBT starts when input level is detected. Either "High" or "Low" level can be selected with the T7IGBTTR flag of TM7MD3 register.

When "1" (the rising edge) is selected, count operation continues while the trigger pin is "High". When "0" (the falling edge) is selected, count operation continues while the trigger pin is "Low".

To control the startup by commands, TM7EN count operation should be selected. In that case, timer count operation or IGBT output are controlled by the TM7EN flag of TM7MD1 register.

When "1" (count operation) is selected, count continues counting until "0" (count stop) is set. Make sure to set the T7IGBT0 and T7IGBT1 of TM7MD3 register before operating the TM7EN flag of TM7MD1 register.

■ 16-bit Standard IGBT Output (Timer 7)

The IGBT waveform of any duty is generated by setting the duty of "High" period to TM7OC1 register and detecting the trigger that is generated by the external interrupt after the external interrupt interface block has passed. The cycle is the full count overflow time of the 16-bit timer. Timer 7 can be used for standard IGBT output. The following shows IGBT output pins, types of the selectable IGBT trigger, and the setting of the interrupt flags.

Table:11.11.1 IGBT Output Pin

	Timer 7	Timer 8
IGBT output pin	TM7IOA output TM7IOB output	TM8IOA output TM8IOB output

Table:11.11.2 IGBT Trigger

	Timer 7 mode register 3	
	T7IGBT1 to T7IGBT0	T7IGBTTR
IRQ0 falling edge	01 (IRQ0)	1
IRQ0 rising edge	01 (IRQ0)	0
IRQ1 falling edge	10 (IRQ1)	1
IRQ1 rising edge	10 (IRQ1)	0
IRQ2 falling edge	11 (IRQ2)	1
IRQ2 rising edge	11 (IRQ2)	0
TM7EN counting operation	00	-

■ Count Timing of Standard IGBT Output (At Normal)

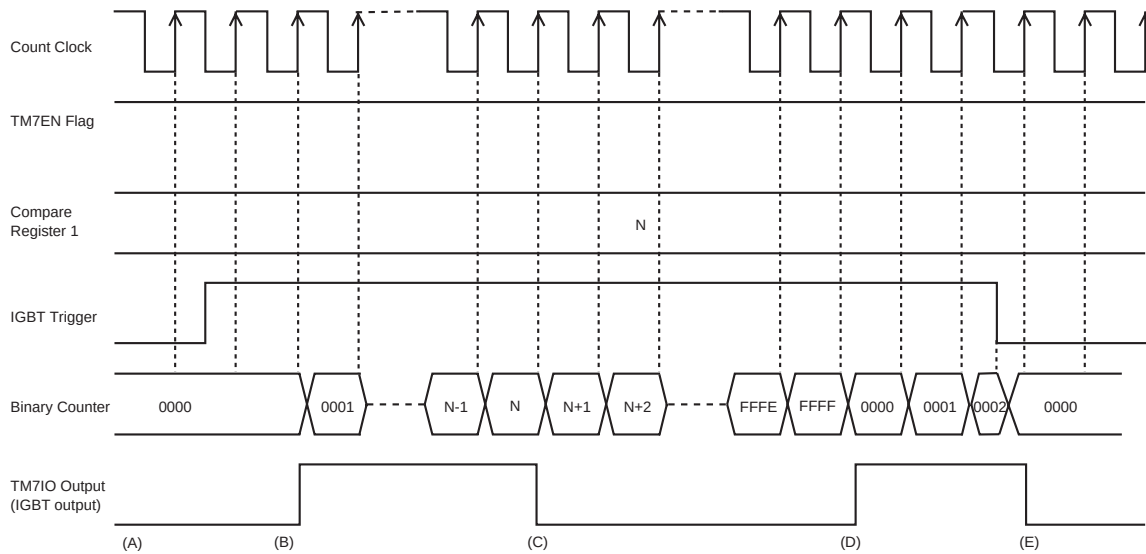


Figure:11.11.1 Count Timing of Standard IGBT Output (At Normal)

- (A) When IGBT trigger is input, IGBT operation becomes valid after 1 count clock. After IGBT output is valid, it is "Low" until the next count clock.
- (B) When IGBT trigger is valid, it is "High" during the period when the value of the binary counter reaches TM7OC1 from 0X0000. ("High" output from 0X0001 at the first operation cycle)
- (C) After TM7OC1 compare match, it is "Low" and binary counter counts up till overflow.
- (D) When binary counter is cleared, it becomes "High" again.
- (E) When IGBT trigger becomes invalid, the timer is initialized and IGBT output is forced to "Low".

■ Count Timing of Standard IGBT Output (When compare register 1 is 0x0000) (Timer 7)

The following shows the count timing as compare register 1 is set to 0x0000.

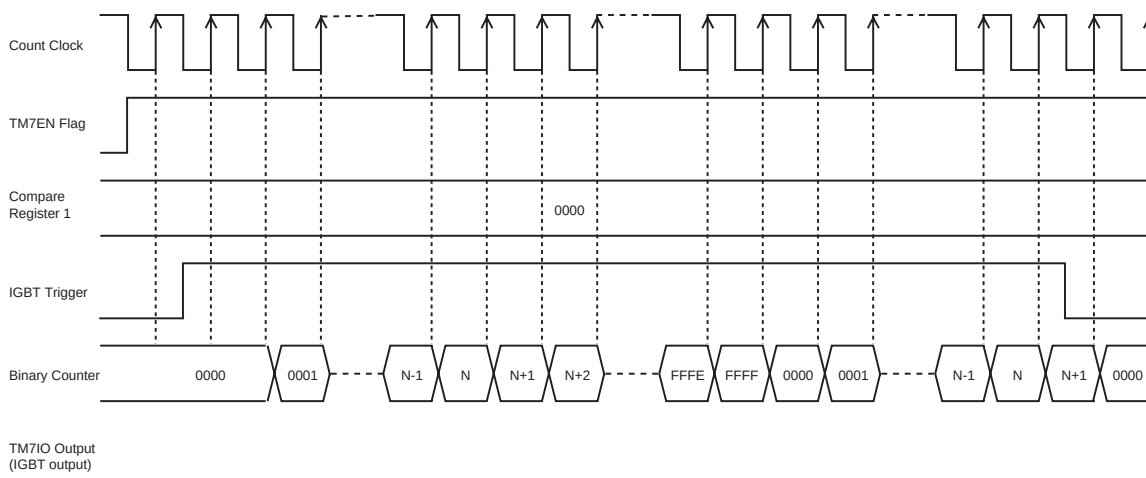


Figure:11.11.2 Count Timing of Standard IGBT Output (When compare register 1 is 0x0000)

When TM7EN flag is set to "0" (stop status), IGBT output is "Low".

■ Count Timing of Standard IGBT Output (When compare register 1 is 0xFFFF) (Timer 7)

The following shows the count timing as compare register 1 is set to 0xFFFF.

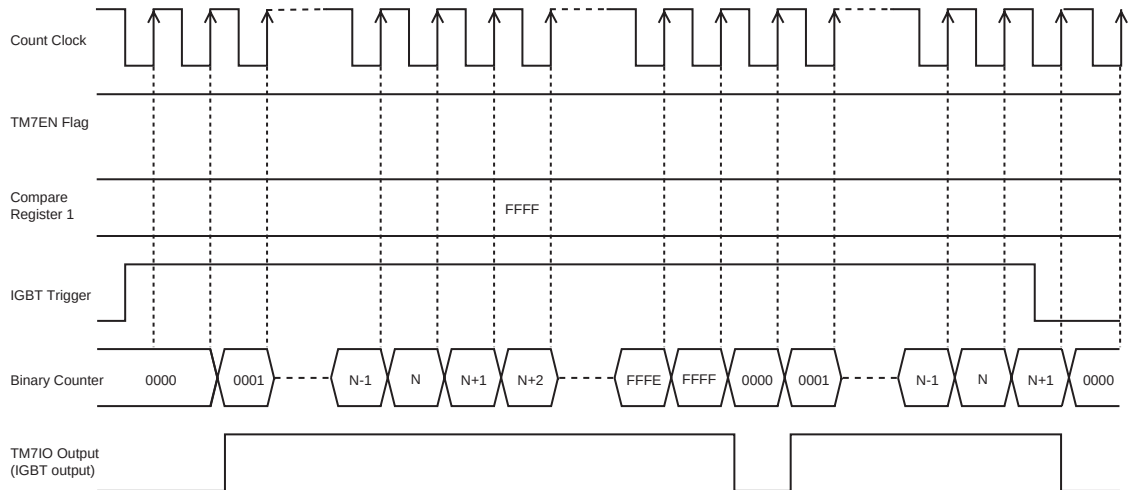


Figure:11.11.3 Count Timing of Standard IGBT Output (When compare register 1 is 0xFFFF)



For standard IGBT output, set the TM7BCR flag of TM7MD2 register to "0" to select the full count overflow as the binary counter clear factor and the IGBT output set ("High" output) factor.



TM7OC1 compare match or TM7OC2 compare match can be selected as the IGBT output reset ("Low" output) factor by the TM7PWMSL flag of TM7MD2 register.

11.11.2 16-bit Standard IGBT Output Setup Example

■ Standard IGBT Output Setup Example (Timer 7)

At the interrupt generation edge of External Interrupt 0 input signal, TM7IOA output pins output the waveform of 1/4 duty IGBT waveform at 152.59 Hz using Timer 7. Frequency for high-speed operation (fp11-div) is 10 MHz. Required period for one IGBT output waveform cycle depends on the overflow time of binary counter. "High" period of IGBT output waveform depends on the set value of compare register 1. An example setup procedure, with a description of each step is shown below.

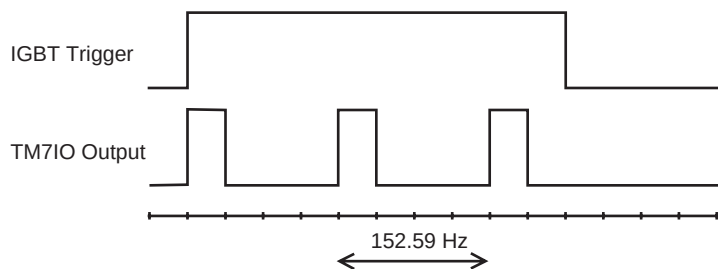


Figure:11.11.4 Output Waveform of TM7IO Output Pin

Setup Procedure	Description
(1) Stop the counter TM7MD1(0x03F78) bp4: TM7EN =0	(1) Set the TM7EN flag of Timer 7 mode register 1 (TM7MD1) to "0" to stop Timer 7 counting.
(2) Set the special function pin to output PAOMD(0x03EE6) bp5: PAOMD5 =1 PADIR(0x03F3A) bp5: PADIR5 =1	(2) Set the PAOMD5 flag of Port A output mode register (PAOMD) to "1" to set PA5 pin as a special function pin. Set the PADIR5 flag of Port A direction control register (PADIR) to "1" to set the output mode. [Chapter 8 I/O Port]
(3) Set IGBT output TM7MD3(0x03F8E) bp2: T7IGBTEN =1 TM7MD2(0x03F79) bp4: PM7PWM =1	(3) Set the T7IGBTEN flag of Timer 7 mode register 3 (TM7MD3) to "1", the TM7PWM flag of Timer 7 mode register 2 (TM7MD2) to "1" to select IGBT output.
(4) Set the standard IGBT output operation TM7MD2(0x03F79) bp5: TM7BCR =0	(4) Set the TM7BCR flag of TM7MD2 register to "0" to select the full count overflow as the clear factor of the binary counter.
(5) Select IGBT trigger generation interrupt source TM7MD3(0x03F8E) bp1 to 0: T7IGBT1 to 0 =01	(5) Set External Interrupt 2 (IRQ2) input as IGBT trigger generation factor by the T7IGBT1 to 0 flags of TM7MD3 register.
(6) Select the trigger level TM7MD3(0x03F8E) bp4: T7IGBTTR=1	(6) Set the T7IGBTTR flag of TM7MD3 register to "0" to set the IGBT trigger level to "High".
(7) Select the dead time TM7MD4(0x03F6E) bp3: T7NODED =1	(7) Set the T7NODED flag of TM7MD4 register to "1" to select without dead time.

Setup Procedure	Description
(8) Select the count clock source TM7MD1(0x03F78) bp1 to 0: TM7CK1 to 0 =00 bp3 to 2: TM7PS1 to 0 =00 (9) Set "High" period of the IGBT output TM7PR1(0x03F75, 0x03F74) = 0x3FFF (10) Start Timer operation TM7MD1(0x03F78) bp4: TM7EN =1	(8) Select fosc as the clock source by the TM7CK1 to 0 flags of TM7MD1 register. Also, select 1/1 dividing as the count clock source by the TM7PS1 to 0 flags. (9) Set the "High" period value of the IGBT output on Timer 7 preset register 1 (TM7PR1). Set the value as $65536 / 4 - 1 = 16383$ (0x3FFF) to make the IGBT output 1/4 duty of the full count 65536. (10) Set the TM7EN flag of TM7MD1 register to "1" to enable the operation of Timer 7. After "High" is input to P20, IGBT is output from PA5.

11.12 Dead Time IGBT Output

IGBT output with dead time generates the waveform, inclusive ON or OFF time delay, during the base IGBT signal inversion. The waveform is output through TM7IO and TM8IO pins. Startup trigger can be selected by External Interrupt 0, 1 and 2 or starting of Timer 7 count operation.

11.12.1 Dead Time IGBT Output Operation

■ Dead Time IGBT Output Operation (Timer 7)

Dead time IGBT output can be selected by the T7IODED of Timer 7 mode register 4 (TM7MD4). Also, dead time can be set to dead time preset register 1 and 2 (TM7DEADPR1, 2). Only Timer 7 can use for IGBT dead time output functions.

■ IGBT Trigger Selection

IGBT trigger can be selected from IRQ0, IRQ1, IRQ2 or the start of Timer 7 count operation.

Setup should be done at the T7IGBT0 and T7IGBT1 flags of TM7MD3 register.

When the startup is controlled from external of the microcontroller, either IRQ0 or IRQ2 should be selected.

IGBT starts by this trigger input level is detected. Either "High" or "Low" level can be selected with the T7IGBTTR flag of TM7MD3 register.

When "1" is selected, count operation continues while the trigger pin is "High". When "0" is selected, count operation continues while the trigger pin is "Low".

To control the startup by commands, TM7EN count operation should be selected. In this case, timer count operation or IGBT output are controlled by the TM7EN flag of TM7MD1 register.

When "1" (count operation) is selected, count continues counting until "0" (count stop) is set. Make sure to set the TM7IGBT0 and 1 of TM7MD3 register before operating the TM7EN flag of TM7MD1 register. In this case, setup of T7IGBTTR is neglected. The binary counter is cleared as the counting stops. The value is loaded from preset register to compare register in synchronization with the count clock.

■ Dead Time Count

Dead time counter counts the timer clock source. When the dead time insert is set as falling standard, set the period from the falling of TM8IO to the rising of TM7IO to dead time preset register 1 (TM7DPR1) and the period from the falling of TM7IO to the rising of TM8IO to dead time preset register 2 (TM7DPR2).

Dead time is inserted for the period of the set value + 1. Only for the period from the IGBT output is enabled by the IGBT trigger to the first rising of TM7IO (in the case of the IGBT falling standard), the set value of the TM7DPR1 + 2 is inserted to the dead time. (for 1 count clock, the dead time should be longer than usually.)

■ Count Timing of Dead Time IGBT Output (Timer 7)

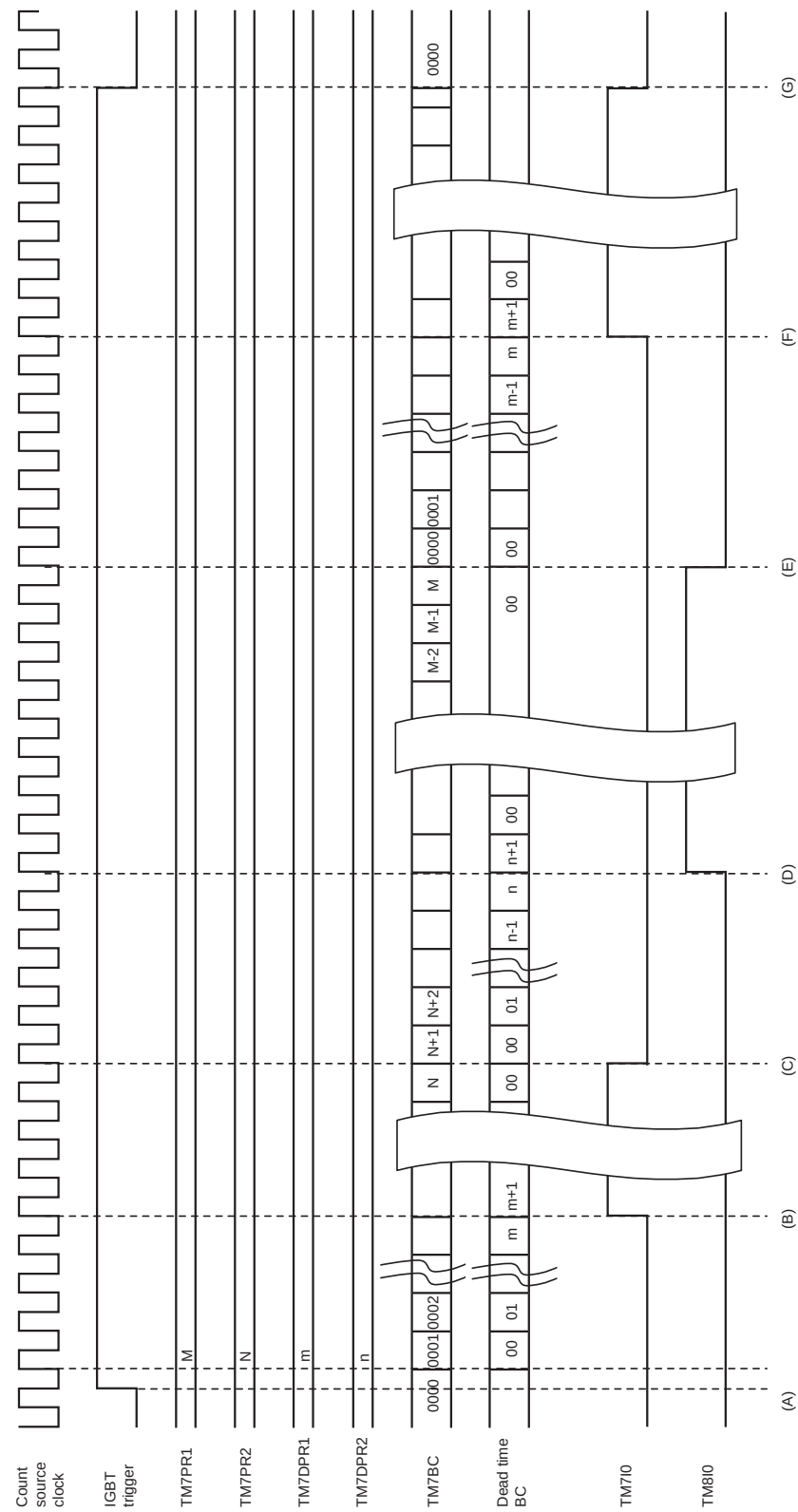


Figure:11.12.1 Count Timing of Dead Time IGBT Output (Timer 7)

Output waveform of the IGBT with dead time (at falling-edge standard)

- (A) TM7IO="Low", TM8IO="Low" until the IGBT trigger is input and become valid.
- (B) After the trigger is input, after next count clock rising edge and after 1 count clock + count clock $\times$ (value of the dead time preset register 1 + 1), TM7IO outputs "High".
- (C) After the compare matching (the value of the binary counter matches that of TM7OC2), and after 1 count clock, TM7IO outputs "Low".
- (D) After TM7IO outputs "Low" and after count clock $\times$ (value of the dead time preset register 2 + 1), TM8IO outputs "High".
- (E) After the compare matching (the value of the binary counter matches that of TM7OC1) and after 1 count clock, TM8IO transit from "High" to "Low".
- (F) After TM8IO outputs "Low" and after count clock $\times$ (value of the dead time preset register 1 + 1), TM7IO outputs "High".
- (G) When IGBT trigger becomes invalid, both TM7IO and TM8IO become "Low" right away.



Set as TM7OC2 value $\leq$ TM7OC1 value.
When TM7OC2 value $>$ TM7OC1 value, the IGBT output waveform is fixed as TM7IO="Low", TM8IO="Low" at falling edge standard.



If IGBT trigger is enabled within 2 cycles of count clock after IGBT trigger is disabled, the following cases may occur:

- the value set in the preset register during IGBT operation may not be loaded to the compare register.
- the value set in the dead time preset register during IGBT operation may not be reflected.



If the event input (TM7IO) is selected as the count clock source, the following cases may occur when IGBT trigger is disabled:

- the value set in the preset register during IGBT operation may not be loaded to the compare register.
- the value set in the dead time preset register during IGBT operation may not be reflected.



When the event input (TM7IO) is selected as the count clock source, the value which is entered on the dead time preset register while IGBT operation halt may not be reflected. To prevent this, select the system clock (fs) as the count clock source and enter the value on the dead time preset register. Then, select the event input (TM7IO) as the count clock source and start the IGBT operation.



When IRQ0, IRQ1 or IRQ2 is selected as the IGBT trigger, the timing of IGBT operation start may delay up to 1 count clock.

■ One Shot Pulse Output Setup

One shot pulse can be output by setting the T7ONESHOT flag of TM7MD4 register to "1".

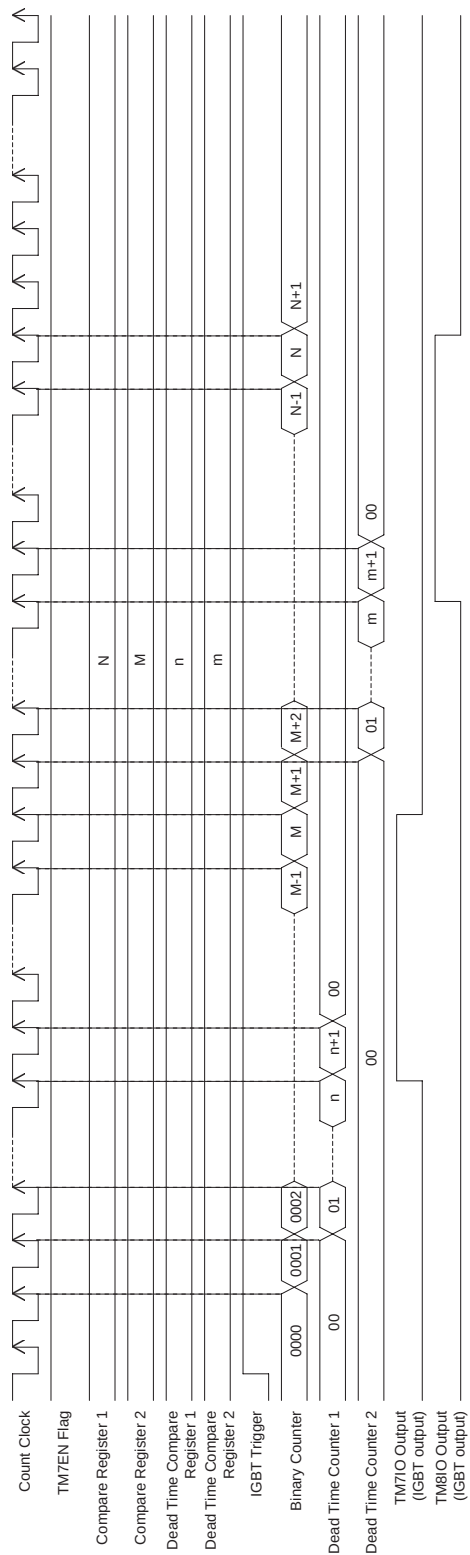


Figure:11.12.2 Count Timing of Dead Time IGBT One Shot Pulse Output (Timer 7)

■ Count Timing of Dead Time High Precision IGBT Output
(At continuous counting: T7IGBTCNT = 1) (Timer 7)

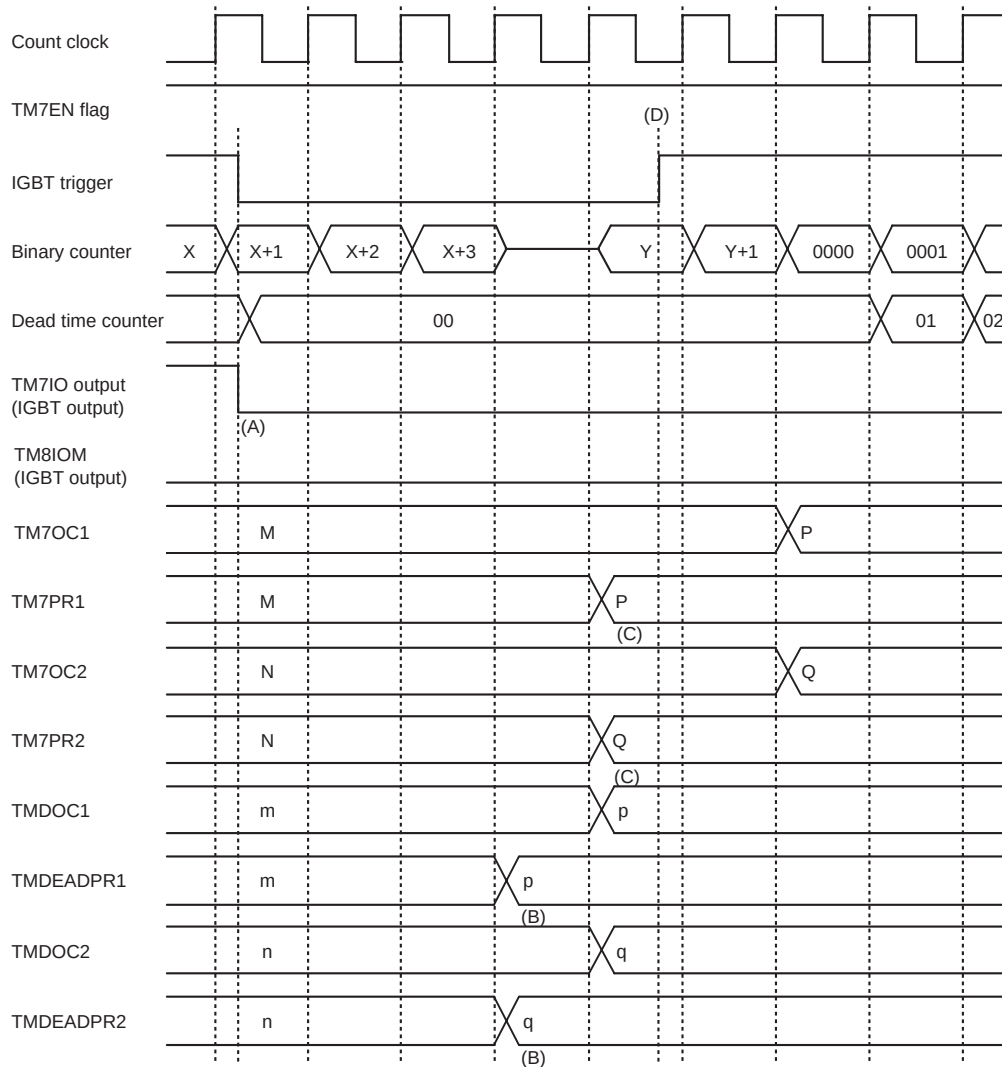


Figure:11.12.3 Count Timing of Dead Time High Precision IGBT Output
(At continuous counting: T7IGBTCNT = 1) (Timer 7)

Dead time high precision IGBT output (at continuous counting/ falling-edge standard)

- (A) When the IGBT trigger becomes invalid, TM7IO = "Low", TM8IO = "Low" and the dead time counter is cleared to "00". At that time, binary counter continues counting.
- (B) If values are loaded to the dead time preset register when the IGBT trigger is invalid, values will be loaded to dead time compare register at the rising edge of the next count clock.
- (C) Even if values are written to preset register when the IGBT trigger is invalid, because binary counter is in operation, values will not be loaded to compare register until the next compare match or the IGBT trigger becomes valid.
- (D) When the IGBT trigger becomes valid again, binary counter is cleared to 0x0000 at the rising edge after two count clock. Also, values are loaded from preset register to compare register.

■ Count Timing of Dead Time High Precision IGBT Output
(Trigger edge reception is disabled during active output: T7TRGACT = 1)

Setting the T7TRGACT flag of Timer 7 mode register 4 (TM7MD4) to "1" enables IGBT output without disrupting PWM waveform even if IGBT trigger is disabled during IGBT output operation.

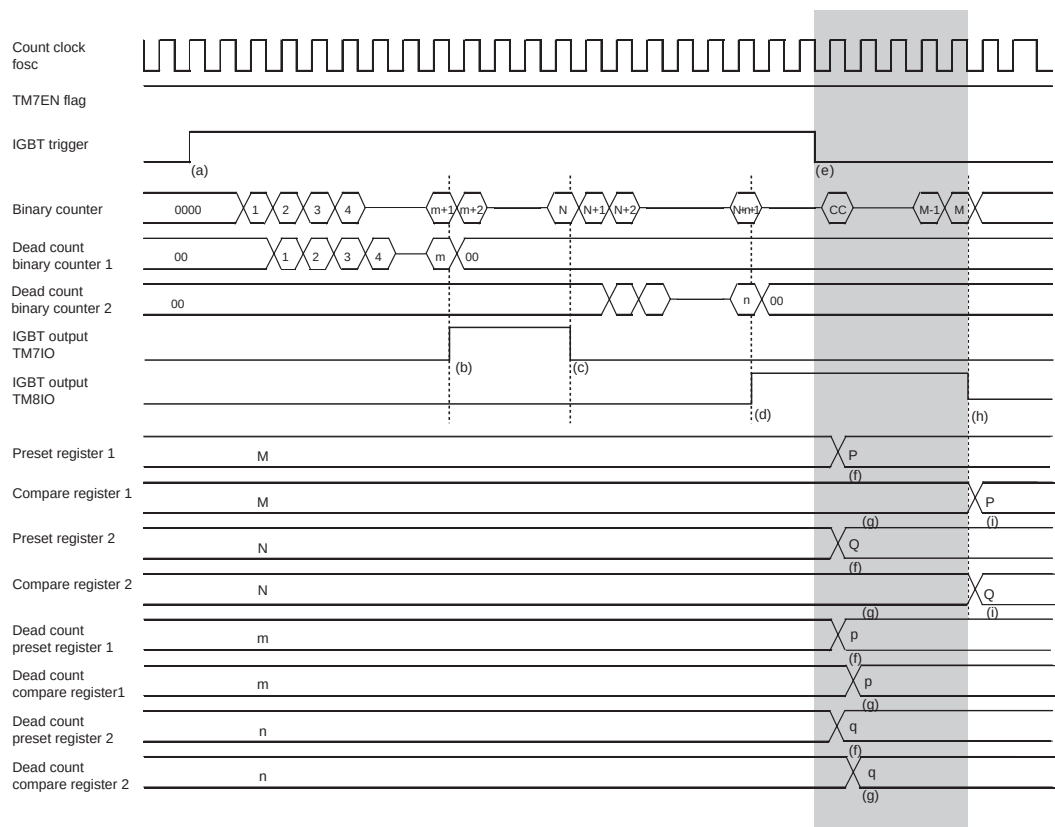


Figure:11.12.4 Count Timing of Dead Time High Precision IGBT Output
(Trigger edge reception is disabled during active output: T7TRGACT = 1)

- (A) The IGBT trigger becomes valid and the count operation starts.
- (B) The value of dead count binary counter 1 and dead count compare register 1 match, and "High" is output to TM7IO.
- (C) The value of binary counter and compare register 2 match, and TM7IO = "Low". Also, dead count binary counter 2 starts count operation.
- (D) The value of dead count binary counter 2 and dead count compare register 2 match, and "High" is output to TM8IO.
- (E) The IGBT trigger becomes invalid but binary counter continues count operation because IGBT output is valid.
- (F) When the IGBT trigger is "Low" and IGBT output is operation, values are written to preset register 1 and 2, and dead count preset register 1 and 2.
- (G) Since binary counter is in operation, values are not loaded to compare register 1 and 2. However, the values same as dead count preset register 1 and 2 are loaded to dead count compare register 1 and 2 at the next count clock, as usual.
- (H) The value of binary counter and compare register 1 match, and TM8IO becomes "Low". Also, the value of preset register 1 and 2 are loaded to compare register 1 and 2.

11.12.2 Dead Time IGBT Output Setup Example

■ Dead Time IGBT Output Setup Example (Timer 7)

At the interrupt generation edge of External Interrupt 0 input signal, TM7IO and TM8IO output pins output the waveform of 1/4 duty IGBT waveform at 200 Hz with 0.01 ms, 0.02 ms dead time by the falling standard using Timer 7. Select fpll-div/1 (at fpll-div=8.0 MHz) as the clock source. Required period for one IGBT output waveform cycle depends on the set value of compare register 1. "High" period of IGBT output standard waveform depends on the set value of compare register 2. Dead time period depends on the value of dead time preset register 1 and 2.

An example setup procedure, with a description of each step is shown below.

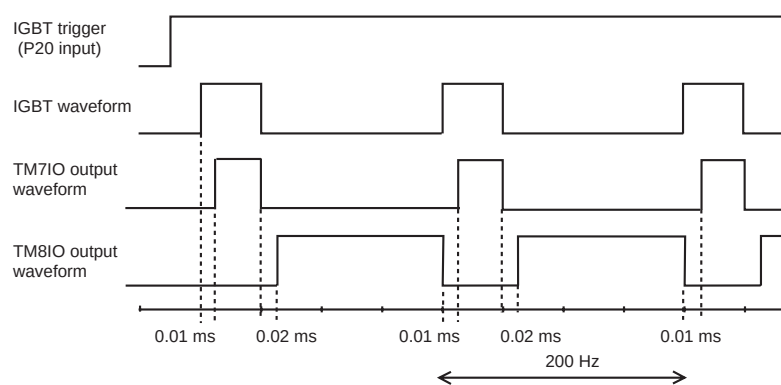


Figure:11.12.5 Output Waveform of TM7IO Output Pin and TM8IO Output Pin

Setup Procedure	Description
(1) Stop the counter TM7MD1(0x03F78) bp4: TM7EN =0	(1) Set the TM7EN flag of Timer 7 mode register 1 (TM7MD1) to "0" to stop Timer 7 counting.
(2) Set the dead time IGBT output operation TM7MD2(0x03F79) bp5: TM7BCR =1 bp6: T7PWMSL =1	(2) Set the TM7BCR flag of TM7MD2 register to "1" to select TM7OC1 compare register match as the binary counter clear source. Also, set the T7PWMSL flag to "1" to select the TM7OC2 compare match as the duty determination source of the IGBT output.
(3) Select IGBT/timer startup factor TM7MD3(0x03F8E) bp1 to 0: T7IGBT1 to 0 =01	(3) Select External Interrupt 0 (IRQ0) input as the IGBT/timer startup factor by the T7IGBT1 to 0 flags of TM7MD3 register.
(4) Select the interrupt generation valid edge TM7MD3(0x03F8E) bp4: T7IGBTTR =0	(4) Set the T7IGBTTR flag of TM7MD3 register to "0" to select the rising edge as the interrupt generation valid edge.
(5) Set the dead time edge TM7MD3(0x03F6C) bp4: T7IGBTDT =1	(5) Set the T7IGBTDT flag of Timer 7 mode register 3 (TM7MD3) to "0" to select the falling standard as the dead time edge.
(6) Set the interrupt level IRQ0ICR(0x03FE2) bp0: IRQ1LV1 to 0 =10	(6) Set the interrupt level by the IRQ1LV1 to 0 flags of IRQ0ICR register. If any interrupt request flag is already set, clear it.
(7) Enable the interrupt IRQ0ICR(0x03FE2) bp1: IRQ1IE =1	(7) Set the IRQ0IE flag of IRQ0ICR register to "1" to enable the interrupt.

Setup Procedure	Description
(8) Select the count clock source TM7MD1(0x03F78) bp1 to 0: TM7CK1 to 0 =00 bp3 to 2: TM7PS1 to 0 =00 (9) Set the IGBT output cycle TM7PR1(0x03F75, 0x03F74) =0x9C3F bp2: T7ICEN =1 (10) Set "High" period of the IGBT output TM7PR2(0x03F7D, 0x03F7C) =0x270F (11) Set the dead time TM7DPR1(0x03F7E) = 0x4F TM7DPR2(0x03F7F) = 0x9F (12) Set the IGBT output TM7MD2 (0x03F79) bp4: TM7PWM =1 TM7MD3 (0x03F6C) bp2: T7IGBTEN =1 TM8MD4 (0x03F6F) bp2: TM8SEL_A =1 (13) Set the output of special function pins PAOMD(0x03EE6) bp5: PAOMD5 =1 bp6: PAOMD6 =1 PADIR(0x03F3A) bp5: PADIR5=1 bp6: PADIR6=1 (14) Start the timer operation TM7MD1(0x03F78) bp4: TM7EN= 1	(8) Select fosc as the clock source by the TM7CK1 to 0 flags of TM7MD1 register. Also, select 1/1 dividing as the count clock source by the TM7PS1 to 0 flags. (9) Set the IGBT output cycle to Timer 7 preset register 1 (TM7PR1). To set 200 Hz by dividing 8.0 MHz, set as; 40000-1=39999 (0x9C3F) At the same time, the same value is loaded to Timer 7 compare register 1 (TM7OC1), and Timer 7 binary counter (TM7BC) is initialized to 0x0000. (10) Set the "High" period of the IGBT waveform to Timer 7 preset register 2 (TM7PR2). To set 1/4 duty of 40000 dividing, set as; 40000/4-1=9999 (0x270F) At the same time, the same value is loaded to Timer 7 compare register 2 (TM7OC2). (11) Set the period from the falling of TM8IO to the rising of the TM7IO to Timer 7 preset register 1 (TM7DPR1) and from the falling of TM7IO to the rising of TM8IO to Timer 7 preset register 2 (TM7DPR2). To make dead time which is from TM7IO falling to TM8IO rising is 0.02 ms and from TM8IO falling to TM7IO rising is 0.01 ms, set 0x4F to Timer 7 dead time preset register 1 and 0x9F to Timer 7 dead time preset register 2. (12) Set TM7PWM flag of Timer 7 mode register 2 (TM7MD2) to "1" and set T7IGBTEN flag of Timer 7 mode register 3 (TM7MD3) to "1" to select the dead time IGBT output. Set TM8SEL_A flag of Timer 8 mode register 4 (TM8MD4) to "1" to select Timer 7 IGBT output. (13) Set the PAOMD5 flag of Port A output mode register (PAOMD) to "1" to set PA5 pin to the special function pin. Set the PAOMD6 flag of Port A output mode register (PAOMD) to "1" to set PA6 pin to the special function pin. Set the PADIR5 flag of Port A direction control register (PADIR) to "1", the PADIR6 flag to "1" to set the output mode. Add pull-up resistor if necessary. [Chapter 8 I/O Port] Set PA5 and PA6 to the special function pin after setting of (1) to (13). (14) Set the TM7EN flag of TM7MD1 register to "1" to operate Timer 7. After "High" is input to P20 pin, IGBT is output from P14 and P15.

TM7BC counts up from 0x0000. The IGBT output waveform outputs "High" until TM7BC matches the set value of TM7OC2 register. Once they match, it outputs "Low". After that, TM7BC continues to count up. Once TM7BC value matches TM7OC1 register value and it is cleared, the IGBT output waveform outputs "High" and TM7BC counts up from 0x0000 again. TM7IO pin outputs IGBT output waveform with dead time. TM8IO pin outputs inverted IGBT output waveform with dead time.

11.1316-bit Timer Cascade Connection

11.13.1 16-bit Timer Cascade Connection Operation

Cascading Timer 7 and 8 forms a 32-bit timer.

■ 16-bit Timer Cascade Connection Operation (Timer 7 + Timer 8)

Timer 7 and Timer 8 are combined to be a 32-bit timer.

Cascading timer is operated at clock source of Timer 7 which are lower 16 bits.

Table:11.13.1 Timer Function at Cascade Connection

	Timer 7 + Timer 8 (32-bit)
Interrupt source	TM8IRQ1, TM8IRQ2
Timer operation	√
Timer pulse output	TM8IO output
PWM output	√
Synchronous output	-
Capture function	√
Pulse width measurement	√
Clock source	fp11-div fp11-div/2 fp11-div/4 fp11-div/16 fs fs/2 fs/4 fs/16 TM7IO input TM7IO input/2 TM7IO input/4 TM7IO input/16
fp11-div: Machine clock (High speed oscillation for peripheral functions) fs: System clock	

* At cascade connection, Timer 8 interrupt factor is only counter-clear.

At cascade connection, the binary counter and the compare register are operated as a 32-bit register.

At operation, set the TM7EN flag of the lower 16-bit timers to "1" to be operated. Also, select the clock source with the lower 16-bit timer. Other setup and count timing are the same at independently operation as the 16-bit timer.



When Timer 7 and Timer 8 are used in the cascade connection, the interrupt request flag of Timer 8 is used as the interrupt request flag. Timer pulse output of Timer 7 is fixed to "Low". Timer 7 interrupt should be disabled to prevent the interrupt request of Timer 7 from being generated.



The preset registers (TM7PR1 and TM8PR1, TM7PR2 and TM8PR2) cannot be written at once. If the loading timing from the preset register to the compare register occurs at the same time as the writing timing of the preset register, the correct data may not be loaded.

11.13.2 16-bit Timer Cascade Connection Setup Example (Timer Operation)

■ Cascade Connection Timer Setup Example (Timer 7 + Timer 8)

Setting example of timer function that the interrupt is constantly generated by cascade connection of Timer 7 and Timer 8, as 32-bit timer is shown. An interrupt is generated in every 100000 cycles (40 ms) by selecting source clock to $f_s/2$ ($f_{pll-div}=10$ MHz, $f_s=f_{pll-div}/2$).

An example setup procedure, with a description of each step is shown below.

Setup Procedure	Description
(1) Stop the counter TM7MD1 (0x03F78) bp4: TM7EN= 0 TM8MD1 (0x03F88) bp4: TM8EN = 0	(1) Set the TM7EN flag of Timer 7 mode register (TM7MD1) to "0" and the TM8EN flag of Timer 8 mode register (TM8MD1) to "0" to stop Timer 7 and Timer 8 counting.
(2) Select the timer clear source TM7MD2 (0x03F79) bp5: TM7BCR =1	(2) Set the TM7BCR flag of TM7MD2 register to "1" to select the compare match as the binary counter clear source.
(3) Select the normal lower operation TM7MD2 (0x03F79) bp2: T7ICEN= 0 TM7MD3 (0x03F6C) bp1 to 0: TM7OUT1 to 0= 00	(3) Set the T7ICEN flag of TM7MD2 register to "0" and set the TM7OUT1 to 0 flags of TM7MD3 register to "00" to select the normal timer operation.
(4) Set the cascade connection TM8MD3 (0x03F8F) bp0: TM8CAS=1	(4) Set the TM8CAS flag of TM8MD3 register to "1" to connect Timer 7 and Timer 8 to the cascade.
(5) Select the count clock source TM7MD1 (0x03F78) bp1 to 0: TM7CK1 to 0=01 bp3 to 2: TM7PS1 to 0=01	(5) Select f_s as the clock source by the TM7CK1 to 0 of TM7MD1 register. Also, select $1/2$ of f_s as the count clock source by the TM7PS1 to 0 flags.
(6) Select IGBT / timer activation factor TM7MD3 (0x03F8E) bp1 to 0: T7IGBT1 to 0=00	(6) Set the IGBT / timer activation factor to Timer 7 count.
(7) Set the interrupt generation cycle TM7PR1 (0x03F75, 0x03F74)=0x869F TM8PR1 (0x03F85, 0x03F84)=0x0001	(7) Set the interrupt generation cycle to Timer 7 preset register 1 (TM7PR1) and Timer 8 preset register (TM8PR1). (100000 cycles-1). At the same time, the same values as the preset registers are loaded to Timer 7 compare register 1 (TM7OC1) and Timer 8 compare register (TM8OC1), and binary counters (TM7BC, TM8BC) are initialized to 0x00000.
(8) Disable the lower timer interrupt TM7ICR (0x03FEF) bp1: TM7IE=0	(8) Set the TM7IE flag of Timer 7 interrupt control register (TM7ICR) to "0" to disable the interrupt.

Setup Procedure	Description
(9) Clear the peripheral function group interrupt request flag IRQEXPDT(0x03F4F) bp1: IRQEXPDT1	(9) Read the peripheral function group interrupt factor retention register (IRQEXPDT) and write the contents to clear the peripheral function group interrupt request flag. Refer to [Chapter 7 7.1.5 Peripheral Function Group Interrupt Control Register Setup].
(10) Set the interrupt level PERIILR(0x03FEF) bp7 to 6: PERIILV1 to 0=10	(10) Set the interrupt level by the PERIILV1 to 0 of peripheral function group interrupt level setting register.
(11) Set the peripheral function group interrupt input IRQEXPEN(0x03F4E) bp1: IRQEXPEN1=1	(11) Set the IRQEXPEN1 flags of peripheral function group interrupt input enable register (IRQEXPEN) to enable the extended interrupt
(12) Start the lower timer operation TM7MD1 (0x03F78) bp4: TM7EN=1	(12) Set the TM7EN flag of TM7MD1 register to "1" to operate Timer 7.

TM7BCL+TM7BCH+TM8BCL+TM8BCH counts up from 0x00000000 as a 32-bit timer.

When TM7BCL+TM7BCH+TM8BCL+TM8BCH reaches the set value of TM7OC1L+TM7OC1H+TM8OC1L+TM8OC1H register, Timer 8 interrupt request flag is set at the next count clock and the value of TM7BCL+TM7BCH+TM8BCL+TM8BCH becomes 0x00000000 to restart count up.

11.13.3 16-bit Timer Cascade Connection Setup Example (PWM Operation)

■ Cascade Connection PWM Output Setup Example (Timer 7 + Timer 8)

TM8IOA output pin outputs the 1/10 duty PWM output waveform at 1/60 Hz with the cascade connection of Timer 7 and Timer 8, as a 32-bit timer. Select fpll-div/1 (fpll-div=8 MHz, at operation) as the clock source. One cycle of the PWM output waveform is depending on the set value of compare register 1. "High" period of the PWM output waveform is depending of the set value of compare register 2. An example setup procedure, with a description of each step is shown below.

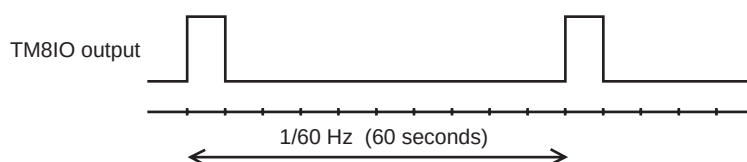


Figure:11.13.1 Output Waveform of TM8IO Output Pin

Setup Procedure	Description
(1) Stop the counter TM7MD1 (0x03F78) bp4: TM7EN=0 TM8MD1 (0x03F88) bp4: TM8EN=0	(1) Set the TM7EN flag of Timer 7 mode register (TM7MD1) to "0", the TM8EN flag of Timer 8 mode register (TM8MD1) to "0" to stop Timer 7 and Timer 8 counting.
(2) Set the special function pin to output PAOMD (0x03EE6) bp6: PAOMD6=1 PADIR (0x03F3A) bp6: PADIR6=1	(2) Set the PAOMD6 flag of Port A output mode register (PAOMD) to "1" to set PA6 pin as the special function pin. Set the PADIR6 flag of Port A direction control register (PADIR) to "1" to set the output mode. Add pull-up/pull-down register, if necessary. [Chapter 8 I/O Port]
(3) Set the cascade connection TM8MD3 (0x03F8F) bp0: TM8CAS=1	(3) Set the TM8CAS flag of TM8MD3 register to "1" to connect Timer 7 and Timer 8 to the cascade.
(4) Set the PWM output TM7MD2 (0x03F79) bp4: TM7PWM=1	(4) Set the TM7PWM flag of Timer 7 mode register 2 (TM7MD2) to "1" to select the PWM output.
(5) Set the high precision PWM output operation TM7MD2 (0x03F79) bp5: TM7BCR=1 bp6: T7PWMSL=1	(5) Set the TM7BCR flag of TM7MD2 register to "1" to select the TM7OC1 compare match as the binary counter clear source. Also, set the TM7PWMSL flag to "1" to select the TM7OC2 compare match as the duty determination factor of the PWM output.
(6) Select the count clock source TM7MD1 (0x03F78) bp1 to 0: TM7CK1 to 0=00 bp3 to 2: TM7PS1 to 0=00	(6) Select fosc as the clock source by the TM7CK1 to 0 flags of TM7MD1 register. Also, select 1/1 dividing as the count clock source by the TM7PS1 to 0 flags.

Setup Procedure	Description
(7) Select IGBT / timer activation factor TM7MD3 (0x03F8E) bp1 to 0: T7IGBT1 to 0=00 (8) Set the PWM output cycle TM7PR1 (0x03F75,0x03F74) = 0x37FF TM8PR1 (0x03F85,0x03F84) = 0x1C9C (9) Set "High" period of the PWM output TM7PR2 (0x03F7D,0x03F7C) = 0x6BFF TM8PR2 (0x03F8D,0x03F8C) = 0x02DC (10) Start the timer operation TM7MD1 (0x03F78) bp4: TM7EN=1	(7) Set the IGBT / timer activation factor to Timer 7 count. (8) Set the PWM output cycle to Timer 7 preset register 1 (TM7PR1) and Timer 8 preset register 1 (TM8PR1). To set 1/60 Hz by dividing 8 MHz, set as; 480.000.000-1=479.999.999 (0x01C9C37FF) At the same time, the same values are loaded to Timer 7 compare register 1 (TM7OC1) and Timer 8 compare register 1 (TM8OC1), and Timer 7 binary counter (TM7BC) and Timer 8 binary counter (TM8BC) are initialized each to 0x0000. (9) Set "High" period of the PWM output to Timer 7 preset register 2 (TM7PR2) and Timer 8 preset register 2 (TM8PR2). To set 1/10 duty of 480.000.000 dividing, set as; 480.000.000/10-1=47.999.999 (0x002DC6BFF) At the same time, the same values are loaded to Timer 7 compare register 2 (TM7OC2) and Timer 8 compare register 2 (TM8OC2). (10) Set the TM7EN flag of TM7MD1 register to "1" to operate Timer 7 and Timer 8.

TM7BCL+TM7BCH+TM8BCL+TM8BCH counts up from 0x00000000 as a 32-bit timer.
TM8IO outputs "High" until TM7BCL+TM7BCH+TM8BCL+TM8BCH reaches the set value of TM7OC2L+TM7OC2H+TM8OC2L+TM8OC2H register. Once they mach, it outputs "Low".
After that, TM7BCL+TM7BCH+TM8BCL+TM8BCH continues to count up.
Once TM7BCL+TM7BCH+TM8BCL+TM8BCH reaches TM7OC1L+TM7OC1H+TM8OC1L+TM8OC1H register, the value of TM7BCL+TM7BCH+TM8BCL+TM8BCH becomes 0x00000000.
TM8IO outputs "High" again and TM7BCL+TM7BCH+TM8BCL+TM8BCH counts up from 0x00000000 again.



In the initial state of the PWM output, "Low" output is changed to "High" output as the PWM output is selected by the TM7PWM flag of TM7MD2 register. (when TM8PWMF=1)



Set value should be set as;

$$TM7OC2L + TM7OC2H + TM8OC2L + TM8OC2H \leq TM7OC1L + TM7OC1H + TM8OC1L + TM8OC1H.$$

If it is set as;

$$TM7OC2L + TM7OC2H + TM8OC2L + TM8OC2H > TM7OC1L + TM7OC1H + TM8OC1L + TM8OC1H,$$
the PWM output becomes a "High" fixed output.

12.1 Overview

This LSI consists of a motor control 16-bit timer function (Timer 9).

In addition to the normal 16-bit timer function, Timer 9 of this LSI can be used as complementary 3 phases PWM and 4 phases PWM for motor control applications.

Table:12.1.1 shows the list of Timer 9 used pins. The used pins can be switched between A type or B type. For A type, "A" is added to the end of the pin name, and for B type, "B" is added.

Table:12.1.1 Timer 9 Pin Functions

Table remarks √/: With function -/: Without function

Functions	Pin Name	MN101EF76K	MN101EF57G	MN101EF56K
TM9IOA	PA7	√	√	√
TM9IOB	P01	√	√	√
TM9OD0A (MN101EF76K) TM9OD0 (MN101EF57G) TM9OD0 (MN101EF56K)	P80	√	√	√
TM9OD1A (MN101EF76K) TM9OD1 (MN101EF57G) TM9OD1 (MN101EF56K)	P81	√	√	√
TM9OD2A (MN101EF76K) TM9OD2 (MN101EF57G) TM9OD2 (MN101EF56K)	P82	√	√	√
TM9OD3A (MN101EF76K) TM9OD3 (MN101EF57G) TM9OD3 (MN101EF56K)	P83	√	√	√
TM9OD4A (MN101EF76K) TM9OD4 (MN101EF57G) TM9OD4 (MN101EF56K)	P84	√	√	√
TM9OD5A (MN101EF76K) TM9OD5 (MN101EF57G) TM9OD5 (MN101EF56K)	P85	√	√	√
TM9OD0B	PD0	√	-	-
TM9OD1B	PD1	√	-	-
TM9OD2B	PD2	√	-	-
TM9OD3B	PD3	√	-	-
TM9OD4B	PD4	√	-	-
TM9OD5B	PD5	√	-	-

Table:12.1.2 Timer 9 Used Pins

	MN101EF76K		MN101EF57G/MN101EF56K	
16-bit timer function	TM9IOA (PA7)	TM9IOB (P01)	TM9IOA (PA7)	TM9IOB (P01)
Complementary 3 phases PWM function	TM9OD0A (P80) TM9OD1A (P81) TM9OD2A (P82) TM9OD3A (P83) TM9OD4A (P84) TM9OD5A (P85)	TM9OD0B (PD0) TM9OD1B (PD1) TM9OD2B (PD2) TM9OD3B (PD3) TM9OD4B (PD4) TM9OD5B (PD5)	TM9OD0 (P80) TM9OD1 (P81) TM9OD2 (P82) TM9OD3 (P83) TM9OD4 (P84) TM9OD5 (P85)	
4 phases PWM function	TM9OD0A (P80) TM9OD2A (P82) TM9OD4A (P84) TM9OD1A (P81)	TM9OD0B (PD0) TM9OD2B (PD2) TM9OD4B (PD4) TM9OD1B (PD1)	TM9OD0 (P80) TM9OD2 (P82) TM9OD4 (P84) TM9OD1 (P81)	First phase Second phase Third phase Forth phase



In this manual, if there is not much difference in the function between Pin A and B, “A” and “B” of the pin names are omitted.



When changing fppll-div frequency by bp7 to bp4 of OSCCNT register, it should be executed after the motor control 16-bit timer is stopped.

12.1.1 Functions

Table:12.1.3 Motor Control 16-bit Timer Functions

		Timer 9 (16-bit timer for motor control)
16-bit timer	Interrupt factor	PWMOVIRQ, PWMUDIRQ, TM9OC2IRQ
	Timer operation	√
	Event count	TM9IOA input/ TM9IOB input
	Timer pulse output	TM9IOA output/ TM9IOB output
Complementary 3 phases PWM	Interrupt factor	PWMOVIRQ, PWMUDIRQ, TM9OC2IRQ
	Complementary 3 phases PWM output	U-phase, V-phase, W-phase
	Waveform mode	Triangular wave, Saw-tooth wave
	Dead Time setting	√
	High/Low level output	√
4 phases PWM	Interrupt factor	PWMOVIRQ, PWMUDIRQ
	4 phases PWM output	Stepping motor 1-phase excitation 2-phase excitation 1-phase to 2-phase excitation PWM output
	Waveform mode	Triangular wave, Saw-tooth wave
	Dead Time setting	√
	Output order inversion	√
Clock source		fppll-div fppll-div/2 fppll-div/4 fppll-div/16 fs fs/2 fs/4 fs/16 Synchronous TM9IO input Synchronous TM9IO input/2 Synchronous TM9IO input/4 Synchronous TM9IO input/16
fppll-div: High-speed clock fs: System clock [Chapter 3 Clock Control]		

12.1.2 Block Diagram

■ 16-bit Timer for Motor Control Block Diagram (MN101EF76K)

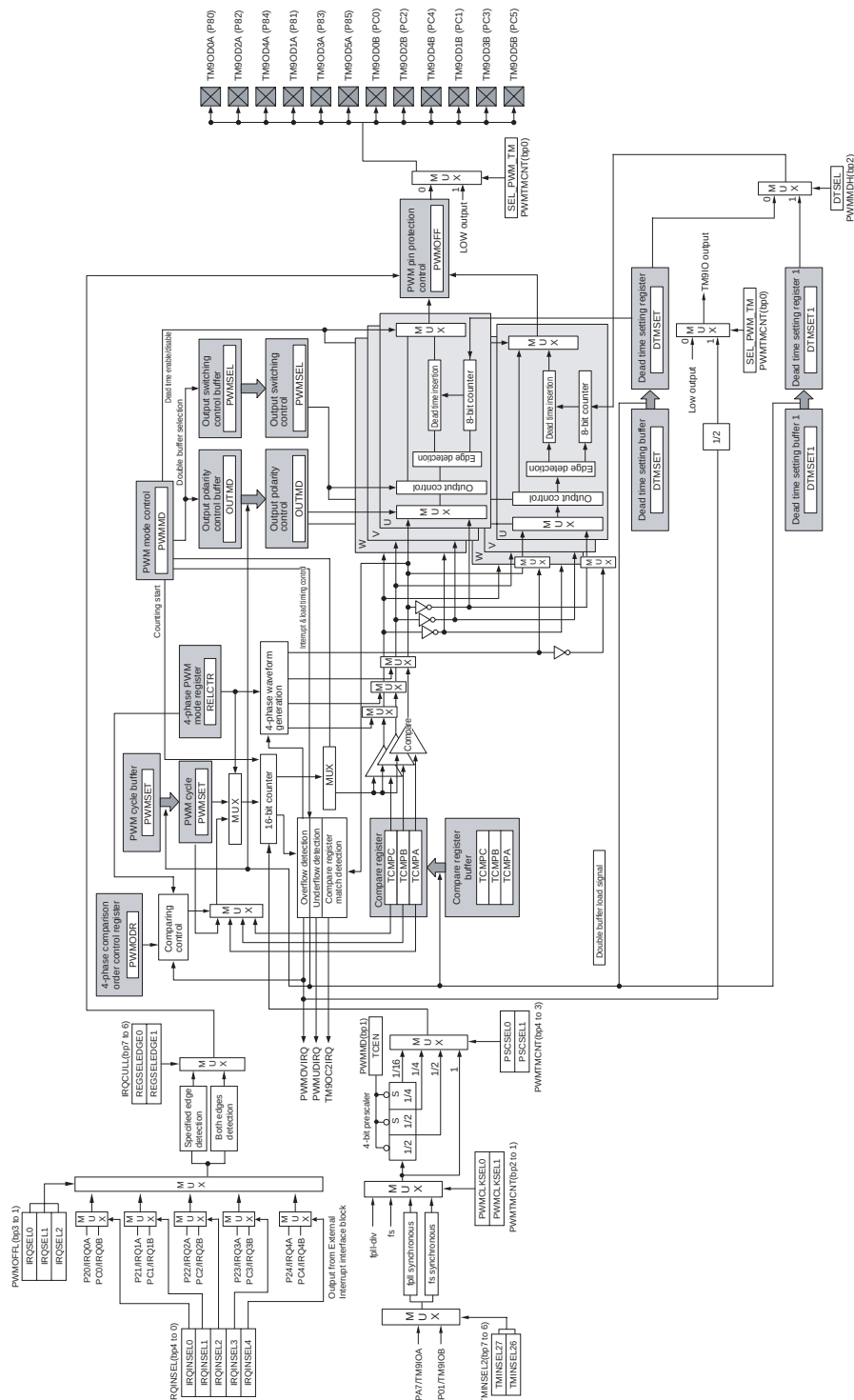


Figure:12.1.1 16-bit Timer for Motor Control Block Diagram (MN101EF76K)

■ 16-bit Timer for Motor Control Block Diagram (MN101EF57G/MN101EF56K)

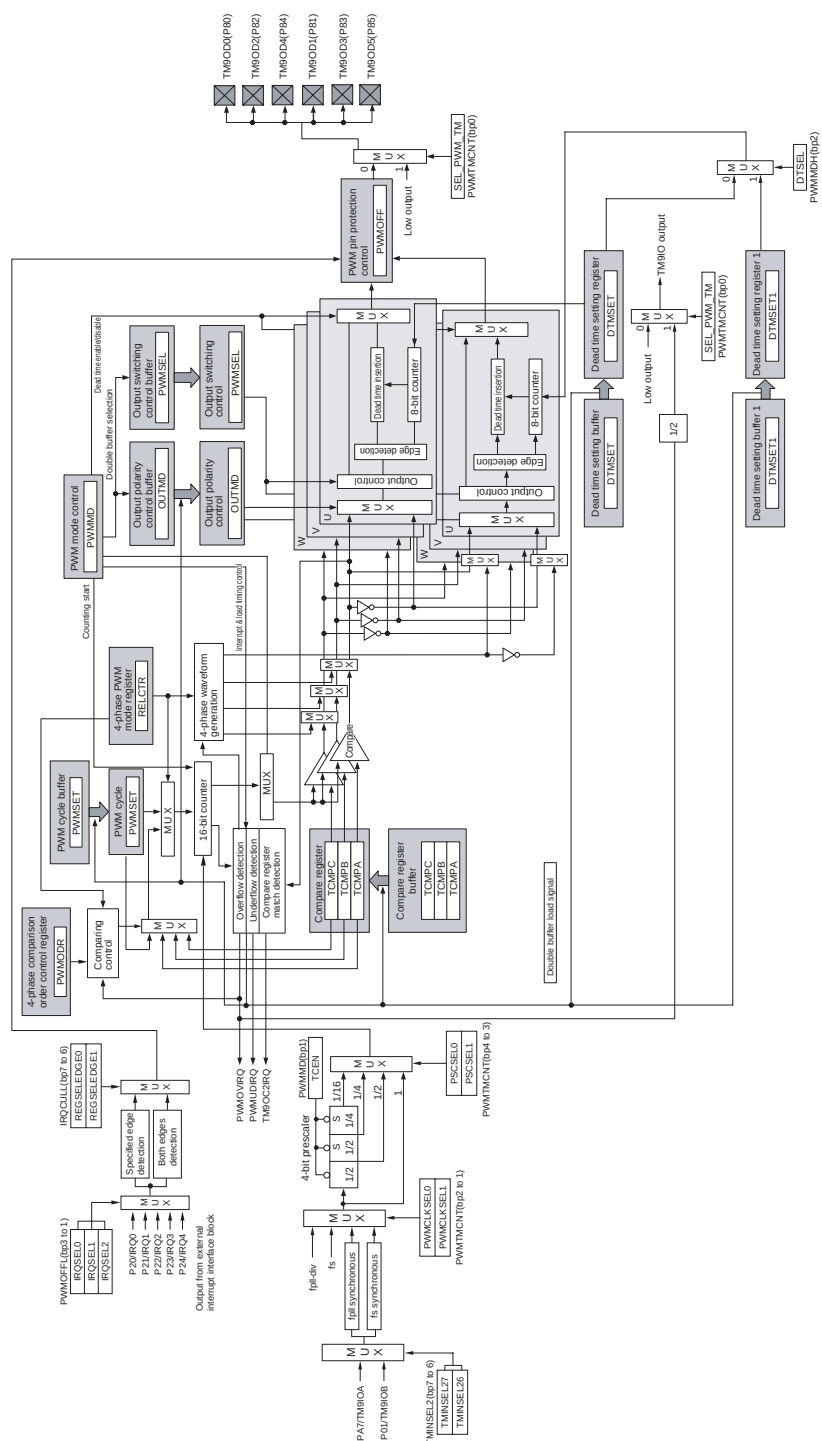


Figure:12.1.2 16-bit Timer for Motor Control Block Diagram (MN101EF57G/MN101EF56K)

12.2 Control Registers

12.2.1 Control Registers for Motor Control 16-bit Timer

Table:12.2.1 shows registers which controls motor control 16-bit timer applications.

Table:12.2.1 Control Registers for Motor Control PWM

Registers	Address	R/W	Function	Page
PWMMDL	0x03E00	R/W	PWM mode control register lower 8 bits	XII-8
PWMMDH	0x03E01	R/W	PWM mode control register upper 8 bits	XII-9
PWMSELL	0x03E02	R/W	PWM output control register lower 8 bits	XII-11
PWMSELH	0x03E03	R/W	PWM output control register upper 8 bits	XII-11
PWMSETL	0x03E04	R/W	PWM cycle setting register lower 8 bits	XII-12
PWMSETH	0x03E05	R/W	PWM cycle setting register upper 8 bits	XII-12
TCMPAL	0x03E06	R/W	PWM U-phase comparison setting register lower 8 bits	XII-13
TCMPAH	0x03E07	R/W	PWM U-phase comparison setting register upper 8 bits	XII-13
TCMPBL	0x03E08	R/W	PWM V-phase comparison setting register lower 8 bits	XII-13
TCMPBH	0x03E09	R/W	PWM V-phase comparison setting register upper 8 bits	XII-14
TCMPCL	0x03E0A	R/W	PWM W-phase comparison setting register lower 8 bits	XII-14
TCMPCH	0x03E0B	R/W	PWM W-phase comparison setting register upper 8 bits	XII-14
OUTMD	0x03E0C	R/W	PWM output polarity control register	XII-10
DTMSET	0x03E0D	R/W	PWM dead time setting register	XII-15
DTMSET1	0x03E0E	R/W	PWM dead time setting register 1	XII-15
PWMBCL	0x03E10	R	PWMBC value read lower 8 bits	XII-16
PWMBCH	0x03E11	R	PWMBC value read upper 8 bits	XII-16
BCSTR	0x03E12	R	PWMBC status read register	XII-17
PWMOFFL	0x03E13	R/W	PWM pin protection control register lower 8 bits	XII-18
PWMOFFH	0x03E14	R/W	PWM pin protection control register upper 8 bits	XII-19
IRQCULL	0x03E15	R/W	PWM interrupt output control register	XII-20
PWMTMCNT	0x03E16	R/W	PWM timer operation control register	XII-21
RELCTR	0x03E17	R/W	4 phases PWM mode register	XII-22
PWMODR	0x03E18	R/W	4 phases PWM output order control register	XII-23
RELSTAT	0x03E19	R	4 phases PWM output status register	XII-23
PWMCMP1	0x03E1A	W	4 phases PWM cycle/comparison collective setting register 1	XII-24
PWMCMP2	0x03E1B	W	4 phases PWM cycle/comparison collective setting register 2	XII-24

R/W Readable / Writable

R Read only

W Write only

12.2.2 PWM Mode Control Registers

PWM mode control registers are used to set various modes for Timer 9.

■ PWM Mode Control Register Lower 8 bits (PWMMMDL: 0x03E00)

bp	7	6	5	4	3	2	1	0
Flag	PCRAEN	PCRBEN	INTAEN	INTBEN	DTEN	ORMD	TCEN	WAVEMD
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description	Setting condition
7	PCRAEN	Double buffer mode load timing enable (PWM binary counter underflow)	0: Disable 1: Enable
6	PCRBEN	Double buffer mode load timing enable (PWM binary counter overflow)	0: Disable 1: Enable
5	INTAEN	Timer 9 underflow interrupt (PWMUDIRQ) enable	0: Disable 1: Enable
4	INTBEN	Timer 9 overflow interrupt (PWMOVIRQ) enable	0: Disable 1: Enable
3	DTEN	Dead Time insertion	0: Not insert 1: Insert
2	ORMD	Dead Time insertion logic	0: Positive logic (High active) 1: Negative logic (Low active)
1	TCEN	PWM counting operation enable	0: Disable 1: Enable
0	WAVEMD	PWM waveform mode	0: Triangular wave 1: Saw-tooth wave

■ PWM Mode Control Register Upper 8 bits (PWMMDH: 0x03E01)

bp	7	6	5	4	3	2	1	0
Flag	-	INTCEN	-	Reserved	-	DTSEL	SDSELA	SDSELB
At reset	-	0	-	0	-	0	0	0
Access	-	R/W	-	R	-	W	R/W	R/W

bp	Flag	Description	Setting condition
7	-	-	-
6	INTCEN	Timer 9 TCMPA compare match interrupt (TM9OC2IRQ) enable	0: Disable 1: Enable
5	-	-	-
4	Reserved	Always set to "0".	
3	-	-	-
2	DTSEL	Dead time insertion mode selection	0: Insert same dead time value in all PWM outputs (The value of DTMSET register is valid only.) 1: Insert different dead time value in PWM output of TM9OD0,2,4 and TM9OD1,3,5. (The value of DTMSET, DTMSET1 register is valid only.)
1	SDSELA	OUTMD buffer mode	0: Single-buffer mode 1: Double-buffer mode
0	SDSELB	PWMSEL buffer mode	0: Single-buffer mode 1: Double-buffer mode



When using Timer 9 as a 16-bit timer, set the WAVEMD flag to "1" to select saw-tooth wave.



To set PWM mode control register, set the TCEN flag of PWMMDL register to "0" when the counting is stopped.



DTSEL flag can only write it. The value of DTSEL flag is read to be "0".

12.2.3 PWM Output Polarity Control Registers

PWM output polarity control register is used to select polarity for each of the PWM outputs.

This register can select double-buffer or single-buffer mode by the SDSELA flag of PWM mode control register (PWMMDL/H). When double-buffer mode is selected, the value of OUTMD is loaded into the register at the timing selected with PWMMDL/H register. When the PWM counter is not running, the double-buffer value is loaded into the register as is regardless of the specified read timing; thus, ensuring smooth use of double buffer from the initial state where the PWM counter starts.

■ PWM Output Polarity Control Register (OUTMD: 0x03E0C)

bp	7	6	5	4	3	2	1	0
Flag	-	-	PXD TNW	PXD TW	PXD TNV	PXD TV	PXD TNU	PXD TU
At reset	-	-	0	0	0	0	0	0
Access	-	-	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description	Setting condition
7 to 6	-	-	-
5	PXD TNW	Output polarity for TM9OD5	0: Negative phase 1: Positive phase
4	PXD TW	Output polarity for TM9OD4	0: Positive phase 1: Negative phase
3	PXD TNV	Output polarity for TM9OD3	0: Negative phase 1: Positive phase
2	PXD TV	Output polarity for TM9OD2	0: Positive phase 1: Negative phase
1	PXD TNU	Output polarity for TM9OD1	0: Negative phase 1: Positive phase
0	PXD TU	Output polarity for TM9OD0	0: Positive phase 1: Negative phase

12.2.4 PWM Output Control Registers

PWM output control register is used to switch between 2 output sources, PWM output or High/Low level output.

This register can select double-buffer or single-buffer mode by the SDSELB flag of PWM mode control register (PWMMDL/H). When double-buffer mode is selected, the value of PWMSELL/H is loaded into the register at the timing selected with PWMMDL/H register. When the PWM counter is not running, the double-buffer value is loaded into the register as is regardless of the specified read timing.

■ PWM0 Output Control Register Lower 8 bits (PWMSELL: 0x03E02)

bp	7	6	5	4	3	2	1	0
Flag	PSELN00	PSEL00	OTLVN02	OTLV02	OTLVN01	OTLV01	OTLVN00	OTLV00
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description	Setting condition
7	PSELN00	TMPOD1 output source	0: PWM output 1: High/Low level output
6	PSEL00	TMPOD0 output source	0: PWM output 1: High/Low level output
5	OTLVN02	TMPOD5 High/Low level output	0: Low level output 1: High level output
4	OTLV02	TMPOD4 High/Low level output	0: Low level output 1: High level output
3	OTLVN01	TMPOD3 High/Low level output	0: Low level output 1: High level output
2	OTLV01	TMPOD2 High/Low level output	0: Low level output 1: High level output
1	OTLVN00	TMPOD1 High/Low level output	0: Low level output 1: High level output
0	OTLV00	TMPOD0 High/Low level output	0: Low level output 1: High level output

■ PWM0 Output Control Register Upper 8 bits (PWMSELH: 0x03E03)

bp	7	6	5	4	3	2	1	0
Flag	-	-	-	-	PSELN02	PSEL02	PSELN01	PSEL01
At reset	-	-	-	-	0	0	0	0
Access	-	-	-	-	R/W	R/W	R/W	R/W

bp	Flag	Description	Setting condition
7 to 4	-	-	-
3	PSELN02	TMPOD5 output source	0: PWM output 1: High/Low level output
2	PSEL02	TMPOD4 output source	0: PWM output 1: High/Low level output
1	PSELN01	TMPOD3 output source	0: PWM output 1: High/Low level output
0	PSEL01	TMPOD2 output source	0: PWM output 1: High/Low level output

12.2.5 PWM Cycle Setting Registers

PWM cycle setting register is used to determine the cycle of 16-bit timer and PWM output.

This register needs to be set only when double-buffer mode is selected. The value of PWMSETL/H is loaded into the register at the timing selected with PWM mode control register (PWMMDL/H). When the PWM counter is not running, the double-buffer value is loaded into the register as is regardless of the specified read timing.

■ PWM Cycle Setting Register Lower 8 bits (PWMSETL: 0x0x03E04)

bp	7	6	5	4	3	2	1	0
Flag	PMSET07	PMSET06	PMSET05	PMSET04	PMSET03	PMSET02	PMSET01	PMSET00
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description	Setting condition
7 to 0	PMSET07 to PMSET00	PWM cycle setting (except 4 phases PWM) First phase output period (4 phases PWM)	Set the cycle in the PWM 16-bit counter.

■ PWM Cycle Setting Register Upper 8 bits (PWMSETH: 0x0x03E05)

bp	7	6	5	4	3	2	1	0
Flag	PMSET0F	PMSET0E	PMSET0D	PMSET0C	PMSET0B	PMSET0A	PMSET09	PMSET08
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description	Setting condition
7 to 0	PMSET0F to PMSET08	PWM cycle setting (except 4 phases PWM) First phase output period (4 phases PWM)	Set the cycle in the PWM 16-bit counter.



When accessing PWMSETL and PWMSETH registers, make sure to access the lower 8 bits (PWMSETL register) by the MOVW instruction.

12.2.6 PWM Phase Comparison Setting Registers

PWM phase comparison setting register is used to determine the timing when the outputs of 3 phases(U/V/W) of PWM change. This register needs to be set only when double-buffer mode is selected. The value of TCMPn is loaded into the register at the timing selected with PWM mode control register (PWMMDL/H). When the PWM counter is not running, the double-buffer value is loaded into the register as is regardless of the specified read timing.

■ PWM U-Phase Comparison Setting Register Lower 8 bits (TCMPAL: 0x03E06)

bp	7	6	5	4	3	2	1	0
Flag	TCPA07	TCPA06	TCPA05	TCPA04	TCPA03	TCPA02	TCPA01	TCPA00
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description	Setting condition
7 to 0	TCPA07 to TCPA00	Timing setting for PWM U-phase output change (complementary 3 phases) Second phase output period (4 phases PWM)	Set the value of PWM U-phase to compare with the binary counter

■ PWM U-Phase Comparison Setting Register Upper 8 bits (TCMPAH: 0x03E07)

bp	7	6	5	4	3	2	1	0
Flag	TCPA0F	TCPA0E	TCPA0D	TCPA0C	TCPA0B	TCPA0A	TCPA09	TCPA08
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description	Setting condition
7 to 0	TCPA0F to TCPA08	Timing setting for PWM U-phase output change (complementary 3 phases) Second phase output period (4 phases PWM)	Set the value of PWM U-phase to compare with the binary counter

■ PWM V-Phase Comparison Setting Register Lower 8 bits (TCMPBL: 0x03E08)

bp	7	6	5	4	3	2	1	0
Flag	TCPB07	TCPB06	TCPB05	TCPB04	TCPB03	TCPB02	TCPB01	TCPB00
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description	Setting condition
7 to 0	TCPB07 to TCPB00	Timing setting for PWM V-phase output change (complementary 3 phases) Third phase output period (4 phases PWM)	Set the value of PWM V-phase to compare with the binary counter

■ PWM V-Phase Comparison Setting Register Upper 8 bits (TCMPBH: 0x03E09)

bp	7	6	5	4	3	2	1	0
Flag	TCPB0F	TCPB0E	TCPB0D	TCPB0C	TCPB0B	TCPB0A	TCPB09	TCPB08
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description	Setting condition
7 to 0	TCPB0F to TCPB08	Timing setting for PWM V-phase output change (complementary 3 phases) Third phase output period (4 phases PWM)	Set the value of PWM V-phase to compare with the binary counter

■ PWM W-Phase Comparison Setting Register Lower 8 bits (TCMPCL: 0x03E0A)

bp	7	6	5	4	3	2	1	0
Flag	TCPC07	TCPC06	TCPC05	TCPC04	TCPC03	TCPC02	TCPC01	TCPC00
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description	Setting condition
7 to 0	TCPC07 to TCPC00	Timing setting for PWM W-phase output change (complementary 3 phases) Fourth phase output period (4 phases PWM)	Set the value of PWM W-phase to compare with the binary counter

■ PWM W-Phase Comparison Setting Register Upper 8 bits (TCMPCH: 0x03E0B)

bp	7	6	5	4	3	2	1	0
Flag	TCPC0F	TCPC0E	TCPC0D	TCPC0C	TCPC0B	TCPC0A	TCPC09	TCPC08
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description	Setting condition
7 to 0	TCPC0F to TCPC08	Timing setting for PWM W-phase output change (complementary 3 phases) Fourth phase output period (4 phases PWM)	Set the value of PWM W-phase to compare with the binary counter



When accessing TCMPA, TCMPB and TCMPCL registers, make sure to access to the lower 8 bits (TCMPnL register) by the MOVW instruction.



Each PWM phase comparison setting register should be set under the setting value of the PWM cycle setting register.

12.2.7 Dead Time Setting Registers

Dead time setting register is used to set dead time of PWM output. Dead time is designed to insert an ON-time delay into each of the upper and lower phases when the signal is inverted at PWM output. The dead time counter functions in synchronization with clock set by PWM mode control register (PWMMMD) and counts 1 every 2 clock cycles. Delay time of the dead time is calculated based on "setting $\times 2 + 1$ ". Thus, when "00" is set, 1 clock cycle of dead time is inserted if dead time is enabled. This register needs to be set only when double-buffer mode is selected. The value of DTMSET is loaded into the register at the timing selected with PWM mode control register (PWMMMD). When the PWM counter is not running, the double-buffer value is loaded into the register as is regardless of the specified read timing. Also, different dead times can be set in TM9OD0, 2, 4 and TM9OD1, 3, 5 by the DTSEL flag of PWM mode control register (PWMMMDL/H).

■ PWM Dead Time Setting Register (DTMSET: 0x03E0D)

bp	7	6	5	4	3	2	1	0
Flag	DTST07	DTST06	DTST05	DTST04	DTST03	DTST02	DTST01	DTST00
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description	Setting condition
7 to 0	DTST07 to DTST00	PWM dead time setting	When DTSEL="0" Set the dead time value to the 8-bit dead time counter. When DTSEL="1" Set the dead time value to the PWM output of TM9OD0, 2, 4.

■ PWM Dead Time Setting Register 1 (DTMSET1: 0x03E0E)

bp	7	6	5	4	3	2	1	0
Flag	DTST17	DTST16	DTST15	DTST14	DTST13	DTST12	DTST11	DTST10
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description	Setting condition
7 to 0	DTST17 to DTST10	PWM dead time setting	When DTSEL="1" Set the dead time value to the PWM output of TM9OD1, 3, 5.

12.2.8 BC Value Read Registers

BC value read register is used to read binary counter value of Timer 9.

■ PWM BC Value Read Register Lower 8 bits (PWMBCL: 0x03E10)

bp	7	6	5	4	3	2	1	0
Flag	PWMBC07	PWMBC06	PWMBC05	PWMBC04	PWMBC03	PWMBC02	PWMBC01	PWMBC00
At reset	0	0	0	0	0	0	0	0
Access	R	R	R	R	R	R	R	R

bp	Flag	Description	Setting condition
7 to 0	PWMBC07 to PWMBC00	PWM binary counter value read	Read binary counter value of PWM

■ PWM BC Value Read Register Upper 8 bits (PWMBCH: 0x03E11)

bp	7	6	5	4	3	2	1	0
Flag	PWMBC15	PWMBC14	PWMBC13	PWMBC12	PWMBC11	PWMBC10	PWMBC09	PWMBC08
At reset	0	0	0	0	0	0	0	0
Access	R	R	R	R	R	R	R	R

bp	Flag	Description	Setting condition
7 to 0	PWMBC15 to PWMBC08	PWM binary counter value read	Read binary counter value of PWM



When the CPU reads binary counter, the read data is handled in 8 bits units in the LSI even if it is a 16-bit MOVW instruction. Thus, it will read the data incorrectly if a carry from the lower 8 bits to the upper 8 bits occurs during counting operation.

12.2.9 BC Status Read Registers

BC status read register is used to read binary counter's counting status of Timer 9.

■ PWM BC Status Read Register (BCSTR: 0x03E12)

bp	7	6	5	4	3	2	1	0
Flag	-	-	-	-	-	-	-	PWMSTR
At reset	-	-	-	-	-	-	-	1
Access	-	-	-	-	-	-	-	R

bp	Flag	Description	Setting condition
7 to 1	-	-	-
0	PWMSTR	PWM binary counter's counting status read	0: Down-count 1: Up-count

12.2.10 PWM Pin Protection Control Registers

This register is used to automatically bring the PWM output pins into Hi-Z state or inactive output by the specified external interrupt generation.

■ PWM Pin Protection Control Register Lower 8 bits (PWMOFFL: 0x03E13)

bp	7	6	5	4	3	2	1	0
Flag	PRTANU1	PRTANU0	PRTAU1	PRTAU0	IRQSEL2	IRQSEL1	IRQSEL0	OUTEN0
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description	Setting condition
7 to 6	PRTANU1 PRTANU0	TM9OD1 output protection function	00: Unused 01: Hi-Z output 10: Inactive output 11: Setting prohibited
5 to 4	PRTAU1 PRTAU0	TM9OD0 output protection function	00: Unused 01: Hi-Z output 10: Inactive output 11: Setting prohibited
3 to 1	IRQSEL2 IRQSEL1 IRQSEL0	PWM pin protection external interrupt selection	000: IRQ00 001: IRQ01 010: IRQ02 011: IRQ03 100: IRQ04 101: Setting prohibited 110: Setting prohibited 111: Setting prohibited
0	OUTEN0	Hi-Z output enable	0: Disabled 1: Enabled



Be sure to set the IRQSEL2 to IRQSEL0 flags before setting the other flags in PWMOFFL register.

■ PWM Pin Protection Control Register Upper 8 bits (PWMOFFH: 0x03E14)

bp	7	6	5	4	3	2	1	0
Flag	PRTANW1	PRTANW0	PRTAW1	PRTAW0	PRTANV1	PRTANV0	PRTAV1	PRTAV0
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description	Setting condition
7 to 6	PRTANW1 PRTANW0	TM9OD5 output protection function	00: Unused 01: Hi-Z output 10: Inactive output 11: Setting prohibited
5 to 4	PRTAW1 PRTAW0	TM9OD4 output protection function	00: Unused 01: Hi-Z output 10: Inactive output 11: Setting prohibited
3 to 1	PRTANV1 PRTANV0	TM9OD3 output protection function	00: Unused 01: Hi-Z output 10: Inactive output 11: Setting prohibited
1 to 0	PRTAV1 PRTAV0	TM9OD2 output protection function	00: Unused 01: Hi-Z output 10: Inactive output 11: Setting prohibited



When the PWM output pin is set to Hi-Z output by the PRTAxx flag of PWMOFF register, the OUTEN0 flag should be set to "1". If the OUTEN0 flag is not set to "1", Hi-Z output is not enabled.

12.2.11 Interrupt Output Control Register

This register is used to control the interrupt factor signals PWMOVIRQ, PWMUDIRQ and TM9OC2IRQ, and also used to select edge of PWM pin protection function interrupt factor signals.

■ Interrupt Output Control Register (IRQCULL: 0x03E15)

bp	7	6	5	4	3	2	1	0
Flag	REGSEL EDGE1	REGSEL EDGE0	REGCULL CMP1	REGCULL CMP0	REGCULL UDF1	REGCULL UDF0	REGCULL OVF1	REGCULL OVF0
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description	Setting condition
7 to 6	REGSELEDGE1 REGSELEDGE0	PWMOFF interrupt factor input edge selection	00: Both edges detected 01: Both edges detected 10: Falling edge detected 11: Rising edge detected
5 to 4	REGCULLCMP1 REGCULLCMP0	TM9OC2IRQ output control	00: Normal output 01: 4-for-1 output 10: 2-for-1 output 11: Setting prohibited
3 to 2	REGCULLUDF1 REGCULLUDF0	PWMUDIRQ output control	00: Normal output 01: 4-for-1 output 10: 2-for-1 output 11: Setting prohibited
1 to 0	REGCULLOVF1 REGCULLOVF0	PWMOVIRQ output control	00: Normal output 01: 4-for-1 output 10: 2-for-1 output 11: Setting prohibited

12.2.12 PWM Timer Operation Control Register

This register is used to determine the clock of 16-bit counter and also used to select motor control PWM operation or 16-bit timer operation.

■ PWM Timer Operation Control Register (PWMTMCNT: 0x03E16)

bp	7	6	5	4	3	2	1	0
Flag	-	-	-	PSCSEL1	PSCSEL0	PWMCKSEL1	PWMCKSEL0	SEL_PWM_TM
At reset	-	-	-	0	0	0	0	0
Access	-	-	-	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description	Setting condition
7 to 5	-	-	-
4 to 3	PSCSEL1 PSCSEL0	Count clock division selection	00: 1/1 clock 01: 1/2 clock 10: 1/4 clock 11: 1/16 clock
2 to 1	PWMCKSEL1 PWMCKSEL0	Count clock source selection	00: fs 01: Synchronous TMIO input (fs synchronous) 10: fpll-div 11: Synchronous TMIO input (fpll-div synchronous)
0	SEL_PWM_TM	Operation selection	0: PWM operation 1: 16-bit timer operation



Set the SEL_PWM_TM flag to "1" to use 16-bit timer operation. In order to set PWM operation, set the SEL_PWM_TM flag to "0" and select complementary 3 phases PWM output or 4 phases PWM output by output mode selection of RELCTR register.

12.2.13 4 phases PWM Mode Register

This register is used to perform 4 phases PWM operation mode control setting.

■ 4 phases PWM Mode Register (RELCTR: 0x03E17)

bp	7	6	5	4	3	2	1	0
Flag	RELIRQA	RELIRQB	RELIRQC	RELIRQD	-	MD1CTR	RELMD	
At reset	0	0	0	0	-	0	0	0
Access	R/W	R/W	R/W	R/W	-	R/W	R/W	R/W

bp	Flag	Description	Setting condition
7	RELIRQA	First phase interrupt event enable control	0: Disabled 1: Enabled
6	RELIRQB	Second phase interrupt event enable control	0: Disabled 1: Enabled
5	RELIRQC	Third phase interrupt event enable control	0: Disabled 1: Enabled
4	RELIRQD	Forth phase interrupt event enable control	0: Disabled 1: Enabled
3	-	-	-
2	MD1CTR	PWM output order control in 1-phase mode	0: In inversion setting, inverted after output phase active period 1: In inversion setting, immediately switched to inversion
1 to 0	RELMD	PWM output mode selection	00: Complementary 3 phases PWM output 01: 4 phases PWM output (1-phase mode) 10: 4 phases PWM output (2-phase mode) 11: 4 phases PWM output (1-phase to 2-phase mode)



Be sure to set bp0 of PWMTMCNT register to "0" to select PWM output mode.

12.2.14 4 phases PWM Output Order Control Register

This register is used to set 4 phases PWM output order.

■ 4 phases PWM Output Order Control Register (PWMODR: 0x03E18)

bp	7	6	5	4	3	2	1	0
Flag	-	-	-	-	-	-	-	ORDER
At reset	-	-	-	-	-	-	-	0
Access	-	-	-	-	-	-	-	R/W

bp	Flag	Description	Setting condition
7 to 1	-	-	-
0	ORDER	4 phases PWM output order	0: Normal rotation 1: Reverse rotation

12.2.15 4 phases PWM Output Status Register

This register is used to show the active interval in 4 phases PWM mode.

■ 4 phases PWM Output Status Register (RELSTAT: 0x03E19)

bp	7	6	5	4	3	2	1	0
Flag	-	-	-	-	-	-	STAT1	STAT0
At reset	-	-	-	-	-	-	0	0
Access	-	-	-	-	-	-	R	R

bp	Flag	Description	Setting condition
7 to 2	-	-	-
1 to 0	STAT1 STAT0	4 phases PWM output status	00: During 1-phase output 01: During 2-phase output 10: During 3-phase output 11: During 4-phase output

12.2.16 PWM Cycle/Compare Collective Setting Register 1

This register is used to collectively set the lower 8 bits of PWM cycle setting register and PWM phase comparison setting register at equivalence. This register is only for writing and the read data is undefined.

■ PWM Cycle/Compare Collective Setting Register 1 (PWMCMP1: 0x03E1A)

bp	7	6	5	4	3	2	1	0
Flag	PWMCMP1 7	PWMCMP1 6	PWMCMP1 5	PWMCMP1 4	PWMCMP1 3	PWMCMP1 2	PWMCMP1 1	PWMCMP1 0
At reset	0	0	0	0	0	0	0	0
Access	W	W	W	W	W	W	W	W

bp	Flag	Description	Setting condition
7 to 0	PWMCMP17 to PWMCMP10	An equivalence is written to the lower 8-bits of PWM cycle setting register and PWM phase comparison setting register.	

12.2.17 PWM Cycle/Compare Collective Setting Register 2

This register is used to collectively set the upper 8 bits of PWM cycle setting register and PWM phase comparison setting register at equivalence. This register is only for writing and the read data is undefined.

■ PWM Cycle/Compare Collective Setting Register 2 (PWMCMP2: 0x03E1B)

bp	7	6	5	4	3	2	1	0
Flag	PWMCMP2 7	PWMCMP2 6	PWMCMP2 5	PWMCMP2 4	PWMCMP2 3	PWMCMP2 2	PWMCMP2 1	PWMCMP2 0
At reset	0	0	0	0	0	0	0	0
Access	W	W	W	W	W	W	W	W

bp	Flag	Description	Setting condition
7 to 0	PWMCMP27 to PWMCMP20	An equivalence is written to the upper 8-bits of PWM cycle setting register and PWM phase comparison setting register.	



When accessing PWMCMP1 and 2 registers, make sure to access the lower 8 bits (PWMCMP1 register) by word instruction.

12.3 16-bit Timer Operation

12.3.1 Operation

16-bit timer operation is the function which can repeatedly generate interrupts at regular time intervals.

■ 16-bit Timer Operation

The generation cycle of timer interrupts is set by the clock source selection and the set value of PWM cycle setting register (PWMSET), in advance. When the binary counter reaches the set value of PWM cycle setting register, an interrupt is generated at the next count clock. The binary counter is cleared at the time of PWMSET compare match. After the binary counter is cleared, the counting up is restarted from 0x0000.

Timer 9 can generate another set of an independent interrupt (Timer 9 TCMPI compare match interrupt) by the set value of PWM U-phase compare register (TCMPA). At that time, the binary counter is cleared as the above setup.

Table:12.3.1 16-bit Timer Interrupt Generation Factor and Binary Counter Clear Factor

PWMMDL, PWMMDH register	Interrupt generation factor	Interrupt factor	Binary counter clear factor
INTAEN1	PWMSET compare match	PWMOVIRQ	PWMSET compare match
INTBEN1	Binary counter clear	PWMUDIRQ	
INTCEN1	TCMPA compare match	PWMOC2IRQ	

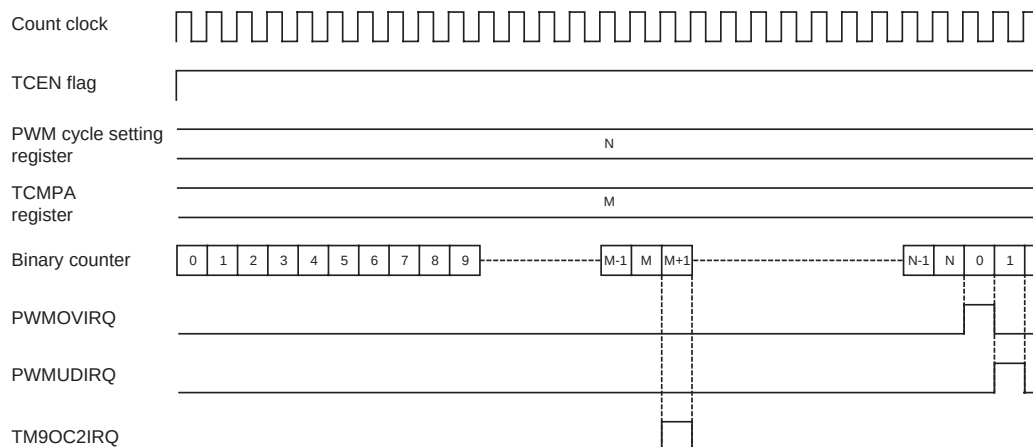


Figure:12.3.1 Interrupt Generation Timing of Motor Control 16-bit Timer

PWM cycle setting register is double buffer type. Thus, when the value of preset registers is changed during the timer counting, the changed value is set to PWM cycle setting register once again at the timing of the binary counter is cleared. This function can change the compare value constantly, without disturbing the cycle even during timer operation (Reload function).



When the CPU reads the 16-bit binary counter, the read data is handled in 8-bits units in the LSI even if it is a 16-bit MOVW instruction. As a result, it will read the data incorrectly if a carry from the lower 8 bits to the upper 8 bits occurs during counting operation.



To count properly, do not switch the count clock on the timer operation. To switch the count clock, stop the timer operation.



Under the following conditions, be sure to set PWMCKSL1 to 0 flags of PWM timer operation control register to "00" (clock source: fs) before updating the value of PWM cycle setting register.

Then start the timer operation by selecting the used clock source.

<Conditions>

- The frequency of a system clock (fs) is faster than twice the frequency of a high-speed clock for peripheral functions (fppll-div), and
- The following conditions are met.

	Relevant Block	PWMMDL (PWM mode control register)	PWMTMCNT (PWM timer operation control register)	
		TCEN	SEL_PWM_TM	PWMCKSEL1 to 0
Condition	Timer 9	0 (Disable operation)	1 (16-bit timer operation)	10 (fppll-div)

Clock source can be selected as follows.

Table:12.3.2 Clock Source at Timer Operation

Clock source	Time per count
fppll-div	100 ns
fppll-div/2	200 ns
fppll-div/4	400 ns
fppll-div/16	1.6 μ s
fs	200 ns
fs/2	400 ns
fs/4	800 ns
fs/16	3.2 μ s
fppll-div = 10 MHz (PLL not used) fs = fppll-div/2 = 5 MHz	

■ Count Timing of Timer Operation (Timer 9)

The binary counter counts up with the selected clock source as a count clock. The basic operation of whole 16-bit timer functions is as below.

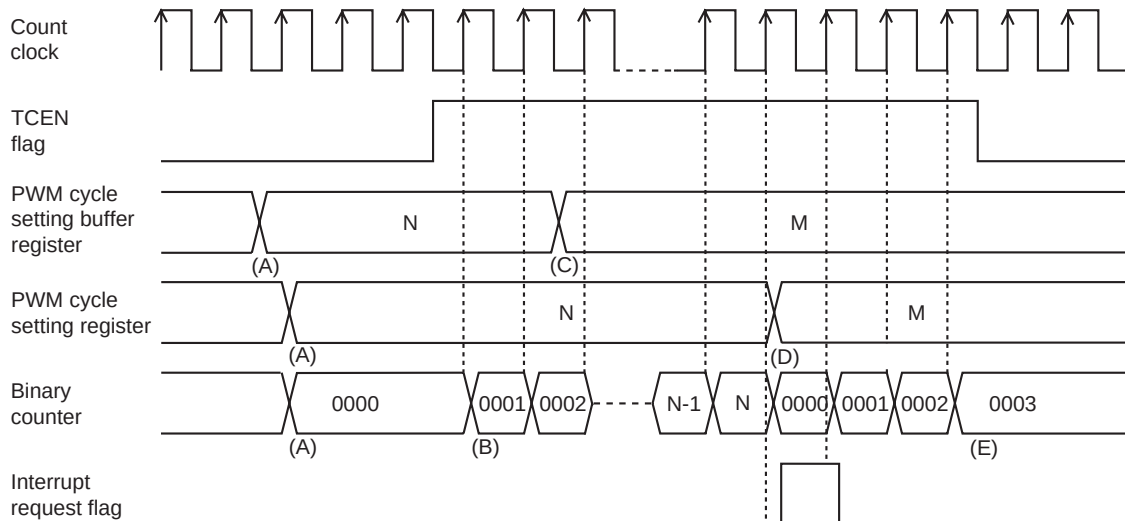


Figure:12.3.2 Count Timing of Timer Operation (Timer 9)

- (A) When a value is written to PWM cycle setting buffer register while the TCEN flag is in stop state ("0"), the same value is loaded to PWM cycle setting register at the next count clock of the writing cycle and the binary counter is cleared to 0x0000.
- (B) When the TCEN flag is in operating state ("1"), the binary counter starts counting. The counting is executed at the rising edge of the count clock.
- (C) Even if PWM cycle setting buffer register is rewritten when the TCEN flag is in operating state ("1"), the binary counter is not changed.
- (D) When the binary counter reaches the value of PWM cycle setting register, the set value of PWM cycle setting buffer register is loaded to PWM cycle setting register at the next count clock. In addition, the interrupt request flag is set, the binary counter is cleared to 0x0000 and counting up starts again.
- (E) When the TCEN flag is in stop state ("0"), the binary counter is stopped.



When the binary counter reaches the value of PWM cycle setting register, the interrupt request flag is set and the binary counter is cleared at the next count clock. Thus, set PWM cycle setting register as:
 $(\text{Set value of PWM cycle setting register}) = (\text{Number of counting until an interrupt request is generated} - 1)$



After a timer interrupt request generation, up to 3 system clocks are needed to generate the next interrupt request flag. Even if compare match occurs during this period, no interrupt request flag is generated.



When Timer 9 TCMPI compare match interrupt is used, change the set value of PWM U-phase compare register (TCMPI) smaller than the set value of PWM cycle setting register (PWMSET).



On the interrupt service routine, clear the timer interrupt request flag before the timer is started.



When the TCEN flag of PWMMD register is changed with other bits at the same time, the binary counter may count up by switching operation.



Timer 9 count clock should be set when the timer interrupt is disabled.



When the CPU reads the binary counter, the read data is handled in 8-bits units in the LSI. As a result, it will read the data incorrectly if a carry from the lower 8 bits to the upper 8 bits occurs during counting operation.
To read the correct value, stop the timer before reading.



Under the following conditions, be sure to set PWMCKSL1 to 0 flags of PWM timer operation control register to "00" (clock source: fs) before updating the value of PWM cycle setting register.
Then start the timer operation by selecting the used clock source.

<Conditions>

- The frequency of a system clock (fs) is faster than twice the frequency of a high-speed clock for peripheral functions (fppll-div), and
- The following conditions are met.

	Relevant Block	PWMMDL (PWM mode control register)	PWMTMCNT (PWM timer operation control register)	
		TCEN	SEL_PWM_TM	PWMCKSEL1 to 0
Condition	Timer 9	0 (Disable operation)	1 (16-bit timer operation)	10 (fppll-div)

12.3.2 Setup Example

■ Timer Operation Setup Example

Timer 9 generates an interrupt at regular time intervals to realize the clock function. Select $f_{pll-div}/2$ ($f_{pll}=10$ MHz at operation) as the clock source to generate an interrupt every 1000 division (200 μ s).

An example setup procedure, with a description of each step is shown below.

Setup Procedure	Description
(1) Stop the counter, select saw to tooth wave PWMMDL(0x03E00) bp1: TCEN =0 bp0: WAVEMD =1	(1) Set the TCEN flag of PWM mode control register (PWMMD) to "0" to stop Timer 9 counting. In addition, set the WAVEMD flag to "1" to select the saw-tooth wave.
(2) Select the count clock source and timer operation PWMTMCNT(0x03E16) bp4 to 3: PSCSEL1 to 0 =01 bp2 to 1: PWMCKSEL1 to 0 =10 bp0: SEL_PWM_TM =1	(2) Select $f_{pll-div}$ as the clock source by the PWMCKSEL1 to 0 flags of PWMTMCNT register, and select 1/2 of $f_{pll-div}$ as the count clock source by the PSCSEL1 to 0 flags. Also, set the SEL_PWM_TM flag to "1" to select timer operation.
(3) Set the interrupt generation cycle PWMSET(0x03E05, 0x03E04)=0x03E7	(3) Set the value of the interrupt generation cycle in Timer 9 PWM cycle setting register (PWMSET). The cycle is 1000 divisions. As a result, the set value should be 1000-1=999. At the time, Timer 9 binary counter is initialized to 0x0000.
(4) Enable the interrupt PWMMDL(0x03E00) bp4: INTBEN =1	(4) Set the INTBEN flag of PWMMDL register to "1" to enable interrupt output.
(5) Clear the peripheral function group interrupt request flag IRQEXPDT(0x03F4F) bp2: IRQEXPDT2	(5) Read peripheral function group interrupt factor retention register (IRQEXPDT) and write the contents to clear the peripheral function group interrupt request flag, refer to [Chapter 7 7.1.5 Peripheral Function Group Interrupt Control Register Setup].
(6) Set the interrupt level PERIILR(0x03FEF) bp7 to 6: PERIILV1 to 0=10	(6) Set the interrupt level by the PERIILV1 to 0 of peripheral function group interrupt level setting register.
(7) Set the peripheral function group interrupt input IRQEXPEN(0x03F4E) bp2: IRQEXPEN2=1	(7) Set the IRQEXPEN2 flags of peripheral function group interrupt input enable register (IRQEXPEN) to enable the extended interrupt
(8) Start the timer operation PWMMDL(0x03E00) bp1: TCEN=1	(8) Set the TCEN flag of PWMMDL register to "1" to operate Timer 9.

Binary counter counts up from 0x0000. When the binary counter reaches the set value of PWMSET register, Timer 9 interrupt request flag is set at the next count clock and the value of binary counter becomes 0x0000 to start counting up again.

12.4 16-bit Event Count

12.4.1 Operation

Event count operation can be performed with 1/1 (no division), 1/2, 1/4 and 1/16 of the synchronous TM9IO input as the clock source by selection.

■ 16-bit Event Count Operation (Timer 9)

The operation of the event count synchronizes an external signals input into TM9IO pin with fs or fpll-div, and counts the signals by binary counter. If the binary counter reaches a preset value of compare register (PWMSET), an interrupt can be generated at the next count clock.

Table:12.4.1 Event Count Input Clock Source

	Timer 9
Event input	Synchronous TM9IOA input
	Synchronous TM9IOB input

■ Count Timing of Synchronous TM9IO Input (Timer 9)

If the synchronous TM9IO input is selected, a synchronous circuit output signal is input to Timer 9 count clock. The synchronous circuit output signal is changed at the rising edge of the system clock (f_s or $f_{pll-div}$) after TM9IO input signal is changed. The binary counter counts up at the falling edge of the synchronous circuit output signal or the synchronous circuit output signal that passed through the division circuit.

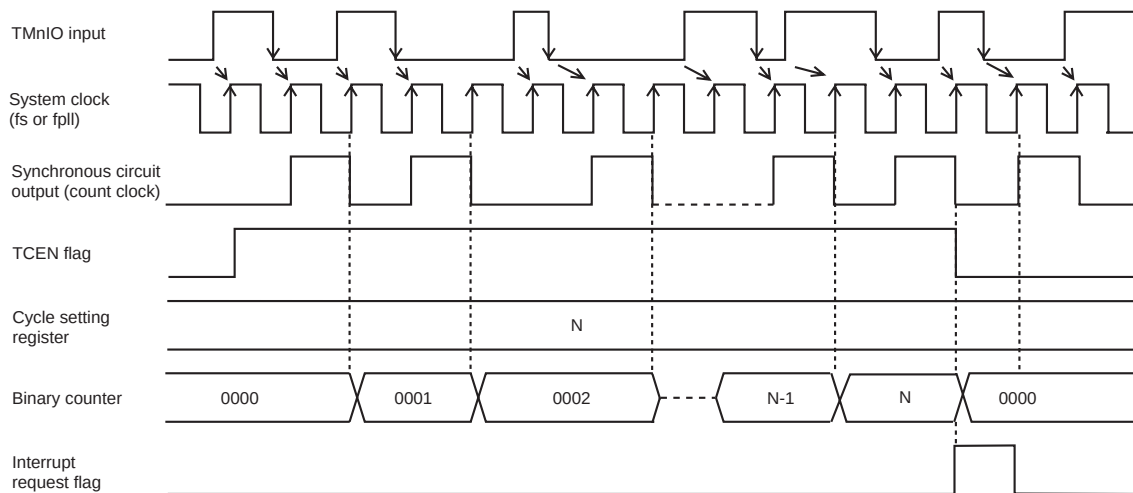


Figure:12.4.1 Count Timing of Synchronous TM9IO Input (Timer 9)



Timer 9 counts up the binary counter by a signal in synchronization with the system clock (f_s or $f_{pll-div}$). Thus, collect value is always read when the binary counter is read.



Input from TM9IO should be done the waveform which has more than 2 times cycle than synchronous clock ($f_{pll-div}$ or f_s). If less than the above waveforms are input, it may not be counted correctly.

12.4.2 Setup Example

■ Event Count Setup Example

When the falling edge of TM9IOA input pin signal is detected 5 times with using Timer 9, an interrupt is generated. An example setup procedure, with a description of each step is shown below.

Setup Procedure	Description
(1) Stop the counter, select saw to tooth wave PWMMDL(0x03E00) bp1: TCEN =0 bp0: WAVEMD =1	(1) Set the TCEN flag of PWM mode control register (PWMMD) to "0" to stop Timer 9 counting. In addition, set the WAVEMD flag to "1" to select the saw-tooth wave.
(2) Select the timer input TMINSEL2(0x03FB3) bp7 to 6: TMINSEL27 to 26 =00	(2) Set the TMINSEL27 to 26 flags of timer input selection register 2 (TMINSEL2) to "00" to set PA7 pin as timer input.
(3) Set the special function pin to input PADIR(0x03E9A) bp7: PADIR7 =0	(3) Set the PADIR7 flag of Port A direction control register (PADIR) to "0" to set PA7 pin to input mode. Add a pull-up resistor, if necessary. [Chapter 8 I/O Port]
(4) Select the count clock source PWMTMCNT(0x03E16) bp4 to 3: PSCSEL1 to 0 =00 bp2 to 1: PWMCKSEL1 to 0 =11 bp0: SEL_PWM_TM =1	(4) Select synchronous TMIO input (fpll-div synchronous) as the clock source by the PWMCKSEL1 to 0 flags of PWMTMCNT register. Besides, select 1/1 as the count clock source by the PSCSEL1 to 0 flags. Set the SEL_PWM_TM flag to "1" to select timer operation.
(5) Set the interrupt generation cycle PWMSET(0x03E05,0x03E04) =0x0004	(5) Set the value of the interrupt generation cycle in Timer 9 cycle setting register (PWMSET). The set value should be 0x0004 to count 5 times. At that time, Timer 9 binary counter is initialized to 0x0000.
(6) Enable the interrupt PWMMDL(0x03E00) bp4: INTBEN =1	(6) Set the INTBEN flag of PWMMDL register to "1" to enable interrupt output.
(7) Clear the peripheral function group interrupt request flag IRQEXPDT(0x03F4F) bp2: IRQEXPDT2	(7) Read peripheral function group interrupt factor retention register (IRQEXPDT) and write the contents to clear the peripheral function group interrupt request flag, refer to [Chapter 7 7.1.5 Peripheral Function Group Interrupt Control Register Setup].
(8) Set the interrupt level PERIILR(0x03FEF) bp7 to 6: PERIILV1 to 0 =00	(8) Set the interrupt level by the PERIILV1 to 0 of peripheral function group interrupt level setting register.
(9) Set the peripheral function group interrupt input IRQEXPEN(0x03F4E) bp2: IRQEXPEN2 =1	(9) Set the IRQEXPEN2 flags of peripheral function group interrupt input enable register (IRQEXPEN) to enable the extended interrupt
(10) Start the event count PWMMDL(0x03E00) bp1: TCEN =1	(10) Set the TCEN flag of PWMMDL register to "1" to start Timer 9 operation.

Binary counter counts up from 0x0000 every time a falling edge of TM9IO input is detected. When the binary counter reaches the set value of PWMSET register, Timer 9 interrupt request flag is set at the next count clock and the value of the binary counter becomes 0x0000 to start counting up again.

If the above procedures (4) to (7) are not followed, it can cause improper operation.

12.5 16-bit Timer Pulse Output

12.5.1 Operation

Timer pulse output function can output pulse signal with an arbitrary frequency from TM9IO pin. Also, PWM output operation can output an arbitrary pulse width periodically from TM9OD pin. Switching between 16-bit timer operation and PWM operation can be set by the SEL_PWM_TM flag of PWM timer operation control register.

Pulse output in 16-bit timer operation setting and PWM output in PWM operation setting are shown below.

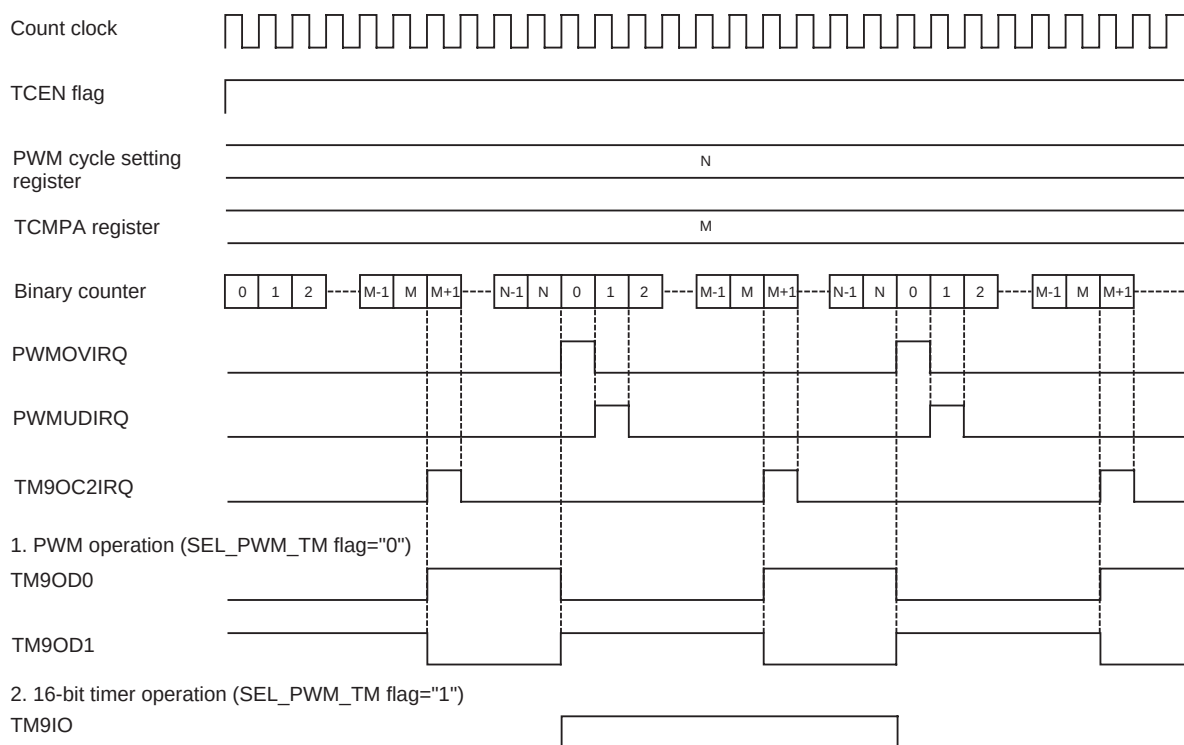


Figure:12.5.1 Timing Chart

12.5.2 Setup Example

■ Timer Pulse Output Setup Example

TM9IOA output pin outputs a 25 kHz pulse with using Timer 9. Select fp11-div as the clock source and set 1/2 cycle (25 kHz) to PWM cycle setting register to output 25kHz pulse (at fp11-div=10 MHz).

An example setup procedure, with a description of each step is shown below.

Setup Procedure	Description
(1) Stop the counter, select saw-tooth wave PWMMDL(0x03E00) bp1: TCEN =0 bp0: WAVEMD =1 (2) Set the special function pin PAOMD1(0x03EBA) bp7: PAOMD7 =1 PADIR(0x03E9A) bp7: PADIR7 =1 (3) Select the count clock source PWMTMCNT(0x03E16) bp4 to 3: PSCSEL1 to 0 =00 bp2 to 1: PWMCKSEL1 to 0 =11 bp0: SEL_PWM_TM =1 (4) Set the timer pulse output generation cycle PWMSET(0x03E05, 0x03E04) =0x00C7 (5) Start the timer operation PWMMDL(0x03E00) bp1: TCEN =1	(1) Set the TCEN flag of PWM mode control register (PWMMD) to "0" to stop Timer 9 counting. In addition, set the WAVEMD flag to "1" to select the saw-tooth wave. (2) Set the PAOMD7 flag of Port A output mode register 1 (PAOMD1) to "1" to set PA7 pin as a special function pin. In addition, set the PADIR7 flag of Port A direction control register (PADIR) to "1" to set PA7 pin to output mode. [Chapter 8 I/O Port] (3) Select synchronous TMIO input (fp11-div synchronous) as the clock source by the PWMCKSEL1 to 0 flags of PWMTMCNT register. Besides, select 1/1 (no division) as the count clock source by the PSCSEL1 to 0 flags. Set the SEL_PWM_TM flag to "1" to select timer operation. (4) Set 1/2 of the timer pulse output cycle to Timer 9 cycle setting register (PWMSET). To set 50 kHz by dividing 10 MHz, set as; 200-1=199 (0xC7) At the same time, Timer 9 binary counter (TM9BC) is initialized to 0x0000. (5) Set the TCEN flag of PWMMDL register to "1" to start Timer 9 operation.

Binary counter counts up from 0x0000. If the binary counter reaches the set value of PWMSET register and is cleared to 0x0000, the signal of TM9IO output is inverted and the binary counter counts up from 0x0000 again.

12.6 Complementary 3 phases PWM

12.6.1 Operation

■ Complementary 3 phases PWM Waveform Mode

Waveform mode can be set by the WAVEMD flag of PWM mode control register (PWMMMD). When the flag is set to "0", triangular waves are specified. When the flag is set to "1", saw-tooth waves are specified.

Table:12.6.1 shows the output waveform logical operation expression and output level.

Table:12.6.1 Logical Operation Expression and Output Level

Logical operation expression	TM9OD0, 2, 4	TM9OD1, 3, 5
Value to be compared $\leq$ Counter value	High	Low
Value to be compared $>$ Counter value	Low	High

When triangular waves are set, the counter counts up and down. It counts the cycle setting value twice at the end of counting up, and counts "0" twice at the end of counting down. When saw-tooth waves are set, the counter counts up. After the counter counts up until the cycle setting value, it becomes "0" at the next counting. Triangular and saw-tooth wave output figures are shown below.

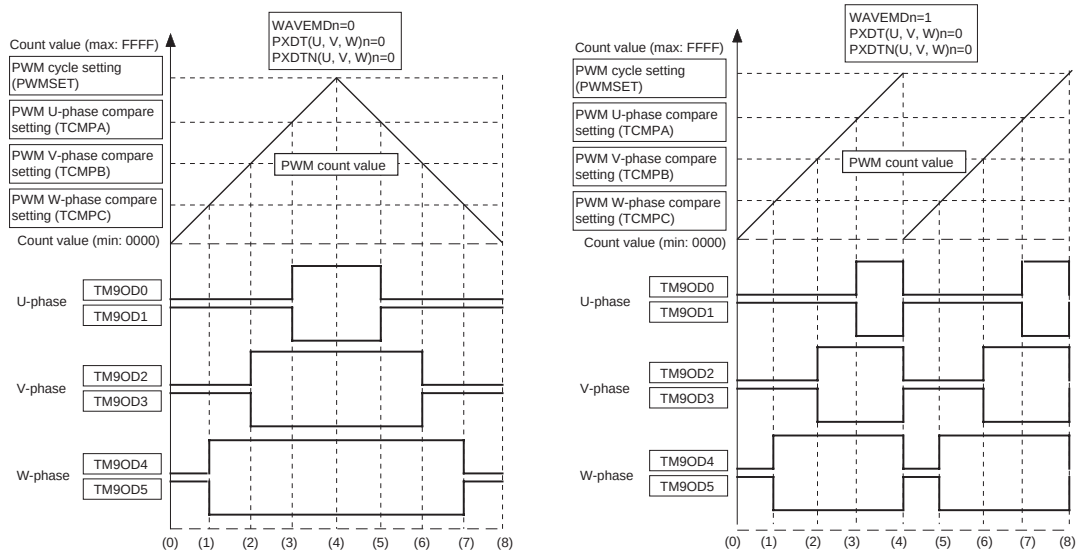


Figure:12.6.1 Triangular and Saw-tooth Waves



Set TCMPC, TCMPB and TCMPC to meet the following condition.

$$TCMPA/TCMPB/TCMPC \leq PWMSET$$

■ Setting PWM Period

The 3 phases cycle for PWM is set by PWM cycle setting register (PWMSET). PWM counting is operated by the PWM binary counter (PWMBC). The expression of the PWM cycle is as follows. The count clock of the PWMBC counter is set by the PWMTMCNT flag.

Waveform mode	PWM period
Triangular wave	Count clock period $\times$ (PWMSETn set value +1) $\times$ 2
Saw-tooth wave	Count clock period $\times$ (PWMSETn set value +1)

■ Starting and Stopping PWM Output

PWM output starts when the TCEN flag of PWMMMD register is set to "1", and stops when the flag is set to "0". Table:12.6.2 shows Timer 9 status when stopping.

Table:12.6.2 PWM Status When Counting Operation is Disabled

PWM block		Status
Phase output	When the output polarity positive phase is selected	Low
	When the output polarity negative phase is selected	High
PWM binary counter (16-bit counter)		PWM operation: 0, timer operation: retained
Dead Time counter		Reset status
PWM control register (double-buffer)		Double-buffer data loaded
PWM control register (single-buffer)		Retained



When the timer operation is selected, the binary counter retains the count value without clearing.
To clear binary counter during timer operation selection, set the TCEN to "0" once, and then access cycle setting register.

■ Output Waveform Polarity

PWM output polarity control register (OUTMD) can be used to control the polarity of the PWM output waveform.

Table:12.6.3 shows the relationship between flags and set values. When the PXDT flag is set to "1", PWM output is switched to NPWM output.

Table:12.6.3 Flags and Settings

	Flag	Setting value	
		"0"	"1"
TM9OD0	PXDTU	Positive phase	Negative phase
TM9OD1	PXDTNU	Negative phase	Positive phase
TM9OD2	PXDTV	Positive phase	Negative phase
TM9OD3	PXDTNV	Negative phase	Positive phase
TM9OD4	PXDTW	Positive phase	Negative phase
TM9OD5	PXDTNW	Negative phase	Positive phase

Figure:12.6.2 shows the flag value and the output waveform.

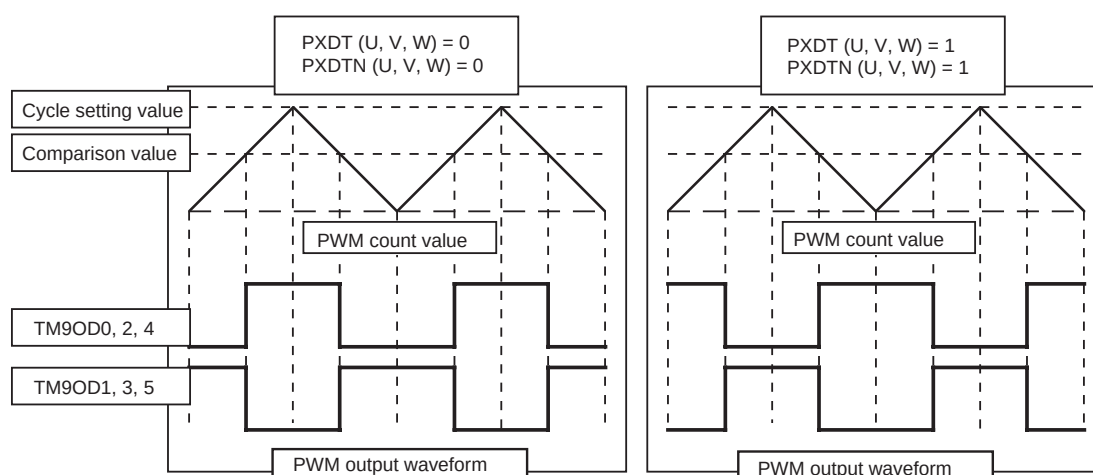


Figure:12.6.2 Flag Value and Output Waveform

■ Double Buffer

Each of Timer 9 is double-buffered to allow data changes during operation. Registers read from and written to by the microcontroller are independent of registers referenced by Timer 9. This makes it possible for microcontroller's register values to be loaded into PWM's registers in synchronization with Timer 9 cycle. However, mode control register (PWMMMD) is single-buffered as it is a basic register that controls Timer 9 operation mode. In addition, PWMOFF, IRQCULL, PWMTMCNT, RELCTR and PWMODR registers are also single-buffered. The other PWM control registers are double-buffered or switchable between double-buffered and single-buffered. Therefore, the configuration that best suits the application can be selected. Check PWM control register list for buffer configuration. Double buffer load timing can be set by overflow and underflow of PWM cycle. PWMMMD register can be used to enable or disable either of these timings.

In addition, if the binary counter is stopped with counting operation disabled, the double buffer values are directly loaded into Timer 9 registers.

Table:12.6.4 Buffer Configuration Available with PWM Control Registers

Register	Double-buffered	Single-buffered	Remarks
PWMMMD	-	√	
OUTMD	√	√	Switching by the SDSLAN flag in the PWMMMD register
PWMSEL	√	√	Switching by the SDSLBn flag in the PWMMMD register
PWMSET	√	-	
TCMPA	√	-	
TCMPB	√	-	
TCMPC	√	-	
DTMSET	√	-	
DTMSET1	√	-	
PWMOFF	-	√	
IRQCULL	-	√	
PWMTMCNT	-	√	
RELCTR	-	√	
PWMODR	-	√	
PWMCMP1	√	-	
PWMCMP2	√	-	



The values of single-buffered registers are reflected immediately.

■ Double Buffer Load Timing

Double buffer load can be enabled by the PCRAEN flag and PCRBEN flag of PWMMD register. Table:12.6.5 shows the relationship between the double buffer load timing and enable setting flag.

Table:12.6.5 Double Buffer Load Timing and Enable Setting Flag

Load timing	Flag (Register)	Description	
At underflow	PCRAEN (PWMMD)	0	Disabled
		1	Enabled
At overflow	PCRBEN (PWMMD)	0	Disabled
		1	Enabled

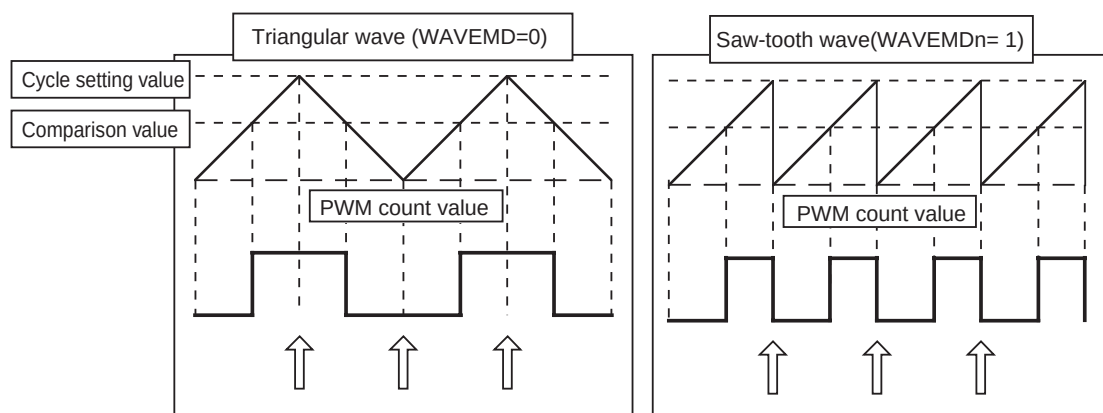


Figure:12.6.3 Double Buffer Load Timing

■ Setting Interrupt Timing

Interrupt signal can be generated in synchronization with Timer 9 cycle. Interrupt signal is generated when an underflow or overflow is generated or when PWM U-phase comparison setting register reaches the set value of binary counter. Table:12.6.6 shows the relationship between the interrupt timing and the enable setting flags.

Table:12.6.6 Interrupt Timing and Enable Setting Flag

Load timing	Flag (Register)	Description		Interrupt source
At underflow	INTAEN(PWMMD)	0	Disabled	PWMUDIRQ
		1	Enabled	
At overflow	INTBEN(PWMMD)	0	Disabled	PWWOVIRQ
		1	Enabled	
TCMPA register reaches the value of binary counter	INTCEN(PWMMD)	0	Disabled	TM9OC2IRQ
		1	Enabled	

■ Dead Time

Dead Time is designed to insert ON time delay into each of the upper and lower phases when the signal is inverted at each phase of PWM output. The DTEN flag of PWMMD register is used to select whether to enable or disable dead time. The ORMD flag of PWMMD register is used to select output logic at dead time insertion. Dead time setting register (DTMSET) is used to specify delay time inserted as dead time.

Any of "00" to "FF" can be selected as dead time with 8-bit data. The dead time counter functions in synchronization with the clock source set by the PWMCKSEL1 to 0 flags of PWMTMCNT register, and counts by 1 every 2 clock cycles. Calculate the dead time or delay time based on "set value $\times 2 + 1$ ".

Thus, when "00" is specified, 1 clock cycle of dead time is inserted if dead time is enabled.

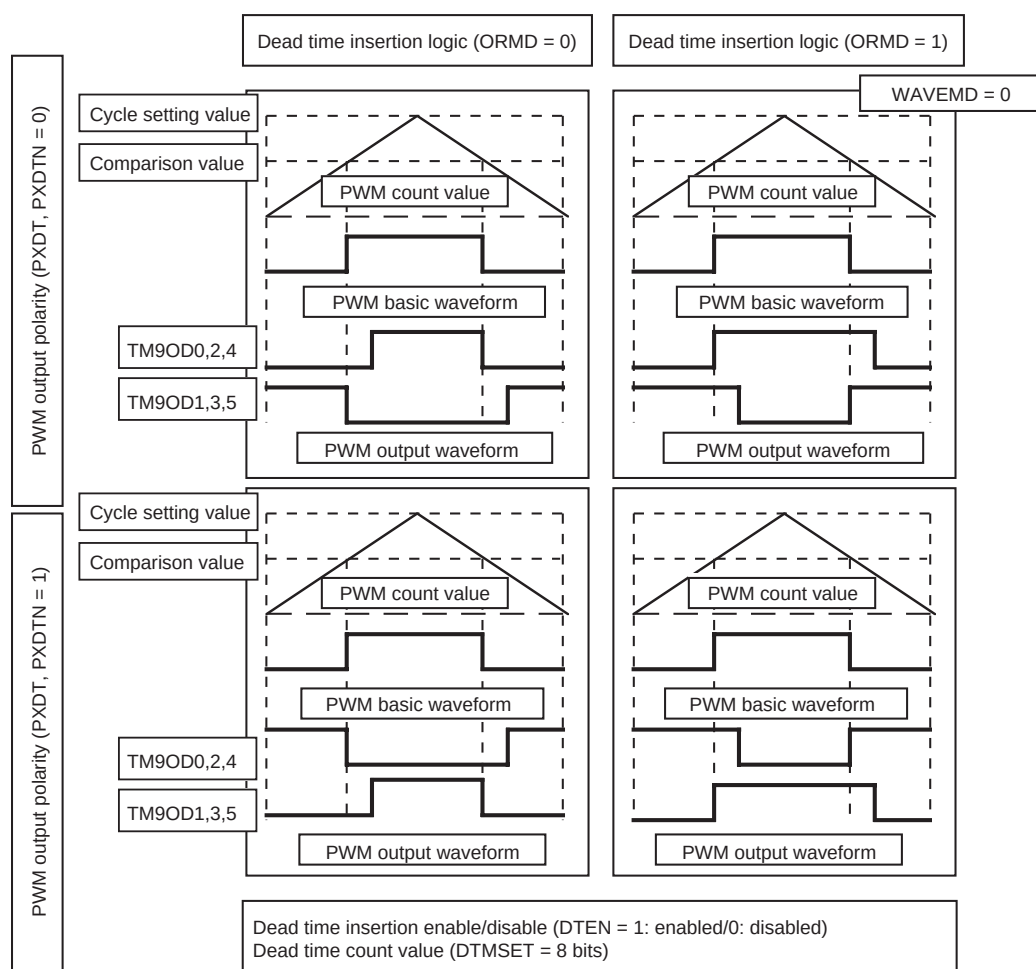


Figure:12.6.4 Dead Time



Though only a single-phase of PWM basic waveform is shown in Figure:12.6.4, PWM 3 phases can be output respectively.

■ Dead Time Insertion Mode Setting

Different dead times can be set in TM9OD0, 2, 4 and TM9OD1, 3, 5 of the PWM output by the DTSEL flag of PWMMD register.

Delay time inserted in dead time insertion mode is set by dead time setting register (DTMSET) for TM9OD0, 2, 4. It is also set by dead time setting register 1 (DTMSET1) for TM9OD1, 3, 5.

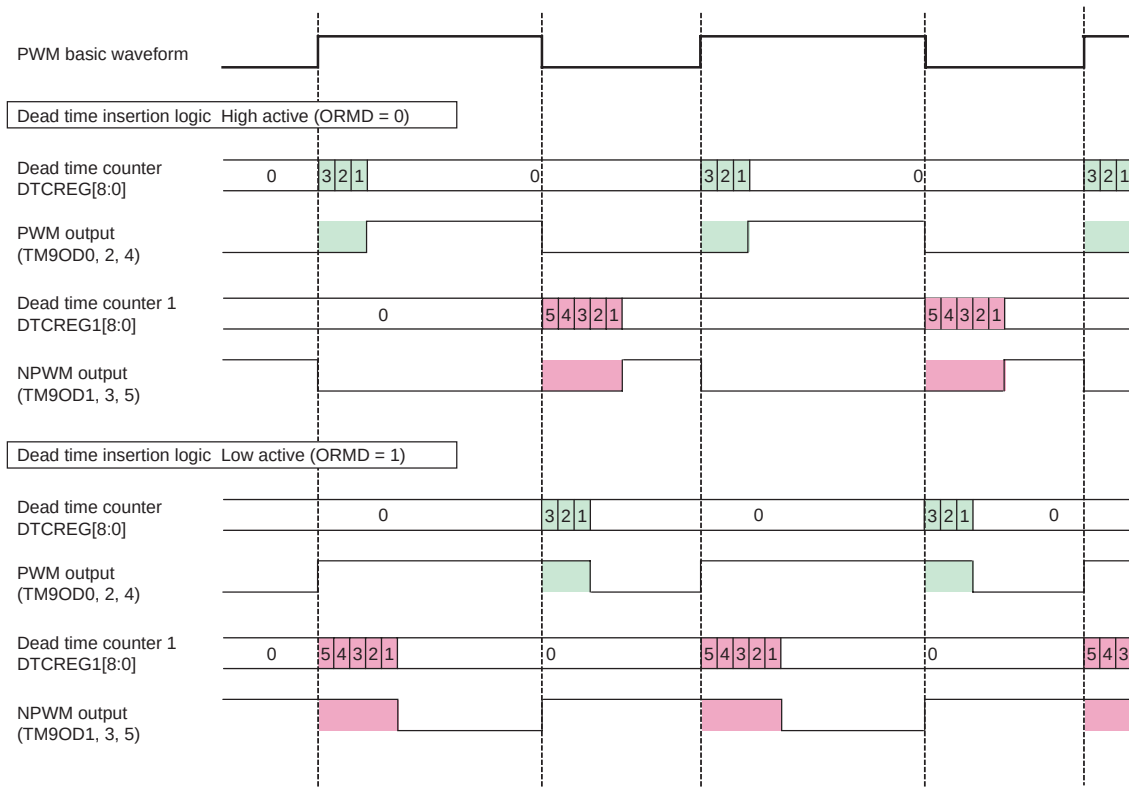


Figure:12.6.5 Dead Time Insertion Mode Setting



Though only a single-phase of PWM basic waveform is shown in Figure:12.6.5, PWM 3 phases can be output respectively.

■ High/Low Level Output Setting

PWM output or High/Low level output can be selected for each of six PWM pins by the PSELN02 to 00 flags and the PSEL02 to 00 flags of PWM output control register (PWMSEL). When High/Low level output is selected, Low level output or High level output can be selected by the OTLVN02 to 00 and OTLV02 to 00. The output timing of High/Low level output is shown below.

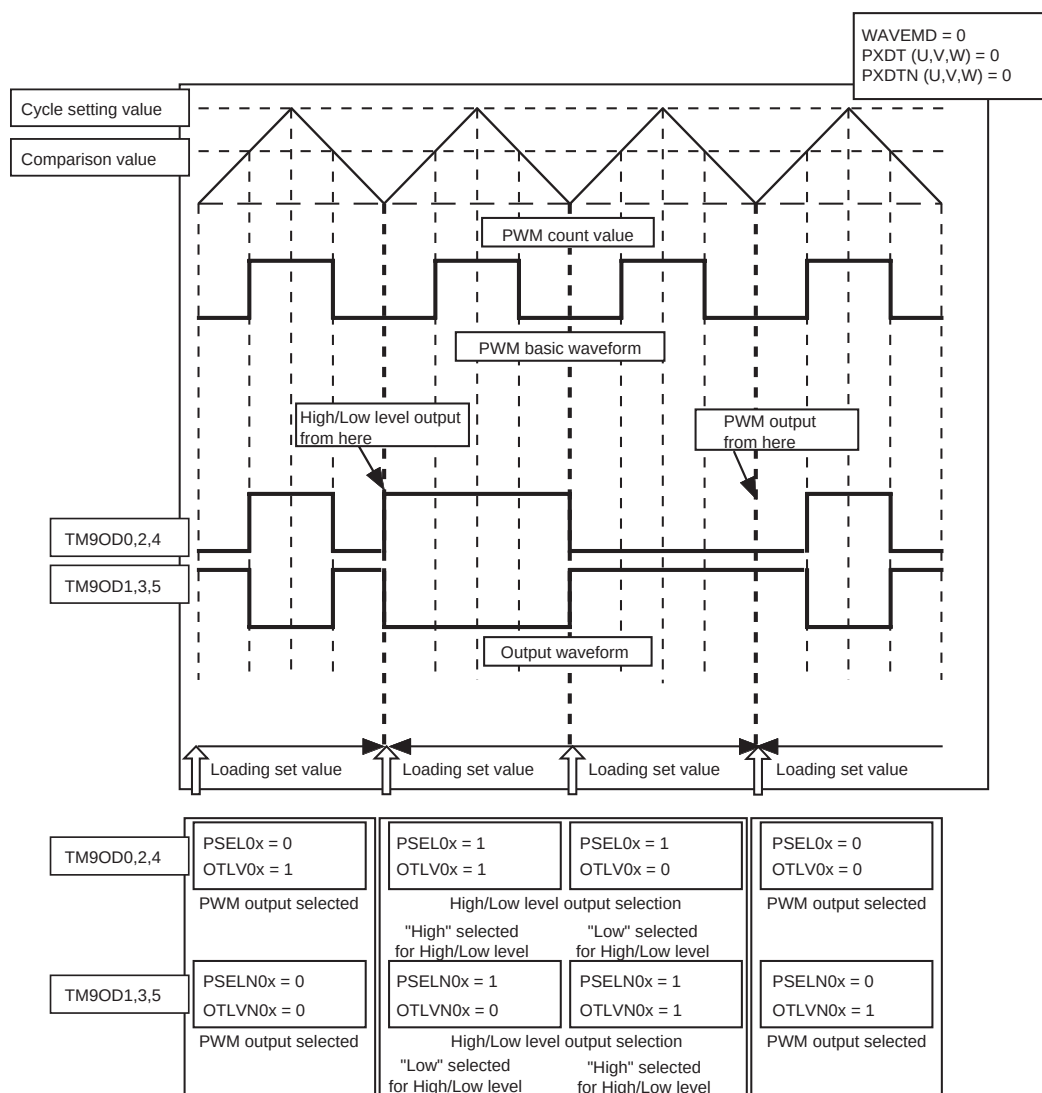


Figure:12.6.6 High/Low Level Output Timing



Though only a single-phase of PWM basic waveform is shown in Figure:12.6.6, PWM 3 phases can be output respectively.

■ Dead Time Insertion at High/Low Level Output

Dead time is inserted as delay time when signals are switched. As a result, dead time is also inserted when PWM output is switched over to High/Low level output. The timing of dead time insertion is shown below.

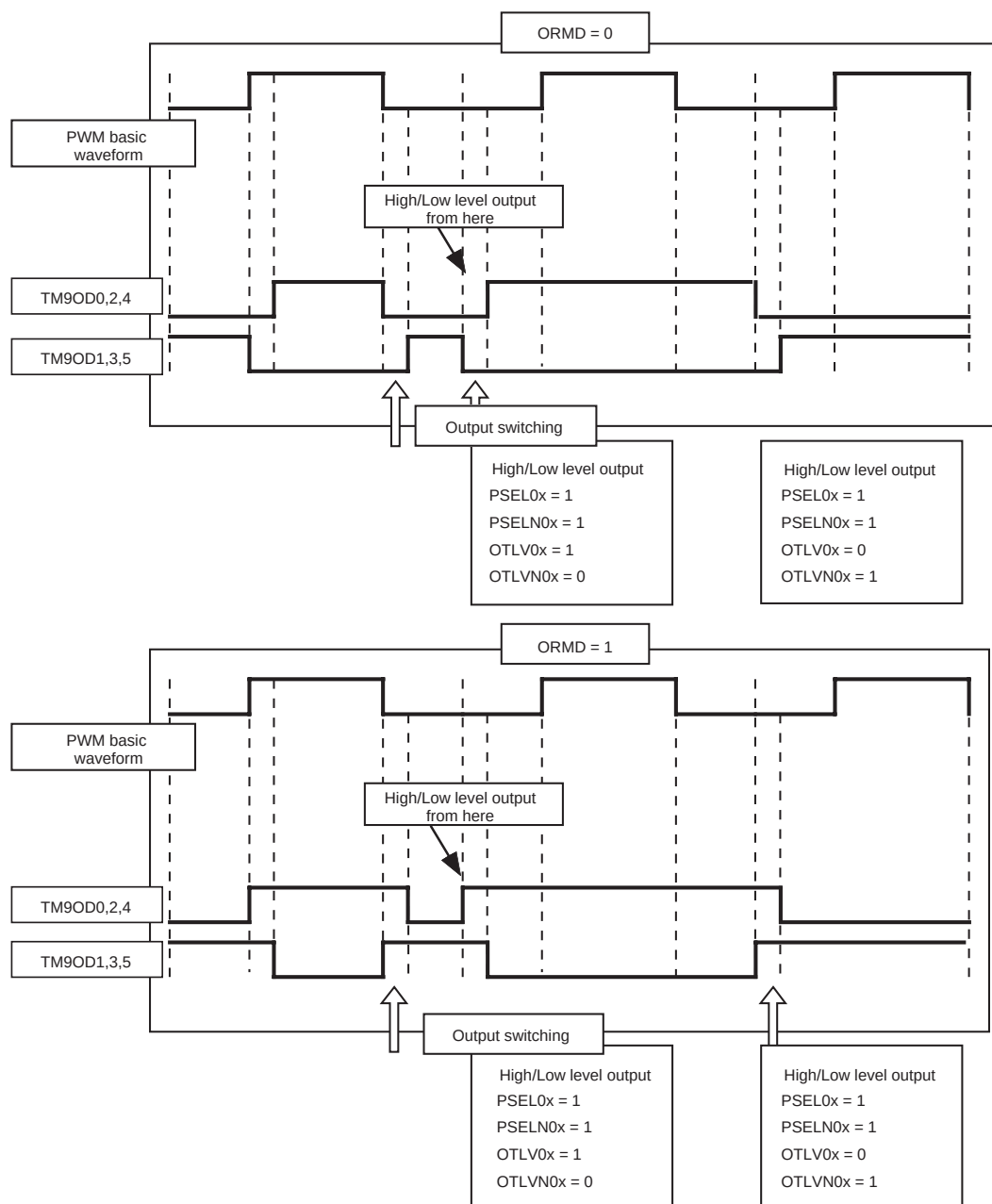


Figure:12.6.7 Dead Time Insertion Timing at High/Low Level Output



Though only a single-phase of PWM basic waveform is shown in Figure:12.6.7, PWM 3 phases can be output respectively.

■ PWM Output Protection Function

Each PWM can change the pin output forcibly by using external interrupt as a trigger.

The changeable pin outputs can be selected from Hi-Z output and inactive output, and all outputs of the six phases can be set individually.

Inactive output follows ORMD (dead time insertion logic) of PWM mode control register (PWMMD). In order to cancel the output protection function, set not to use the output protection.

Setup example is shown below.

Example)

ORMD = 0 (Inactive output is Low output)

IRQSEL = 0x1 (External interrupt IRQ01 is selected)

PRTAU = 0x1 (Hi-Z output)

PRTANU = 0x2 (Inactive output)

PRTAV = 0x0 (Unused)

PRTANV = 0x1 (Hi-Z output)

PRTAW = 0x2 (Inactive output)

PRTANW = 0x0 (Unused)

OUTEN0 = 1 (Hi-Z output enabled)

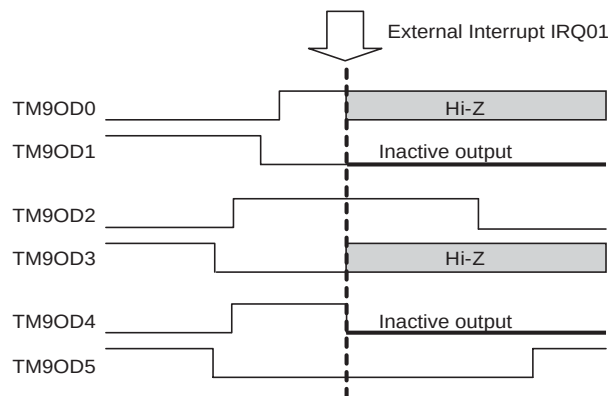


Figure:12.6.8 PWM Output Protection Function



By PWM output protection function, maximum 2 machine cycles are required from inputting External Interrupt as a trigger to changing the pin output level.

■ Interrupt Factor Control Function

Interrupt factor control function can control interrupt factors, which is output when the binary counter underflows, overflows, or reaches the value of PWM U-phase comparison setting register, to output once every two times or once every four times. The relationship between interrupt factor timing and setting value are shown below.

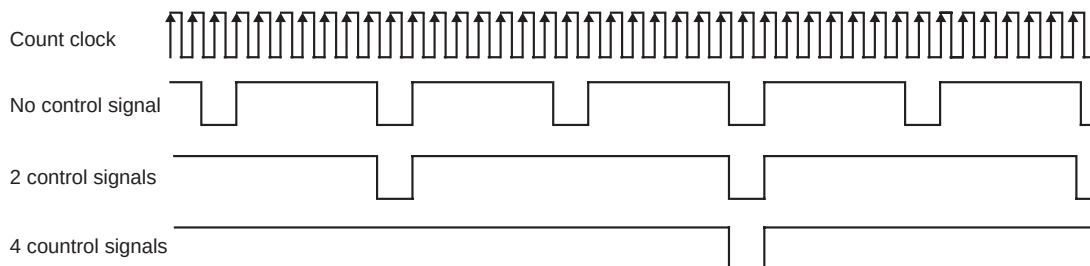


Figure:12.6.9 Relationship between Interrupt Factor Timing and Setting Value

12.7 4 phases PWM

12.7.1 Operation

4 phases PWM can generate multiple phases of PWM waveform. This function can realize 4 phases PWM waveform for ultrasonic monitor driving, stepping motor 1-phase excitation, 2-phase excitation and 1- to 2- phases excitation PWM waveform output.

■ Basic Waveform of 4 phases PWM

Basic waveform generated by 4 phases PWM is shown below.

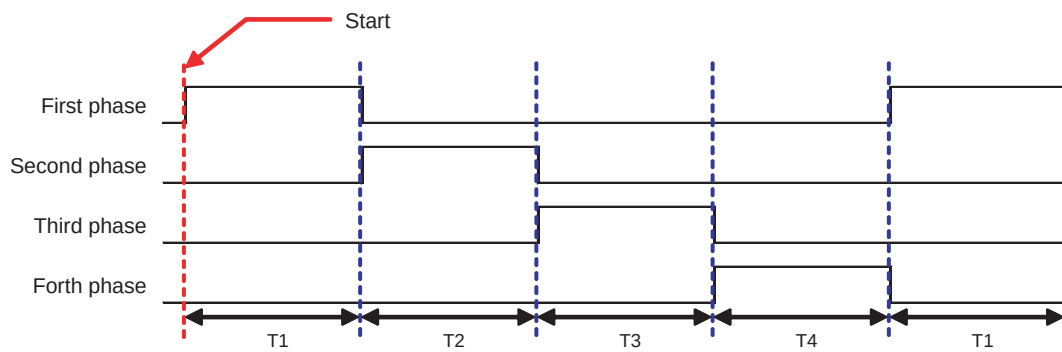


Figure:12.7.1 Basic Waveform 1 (1-phase Mode, MD1CTR = 1)

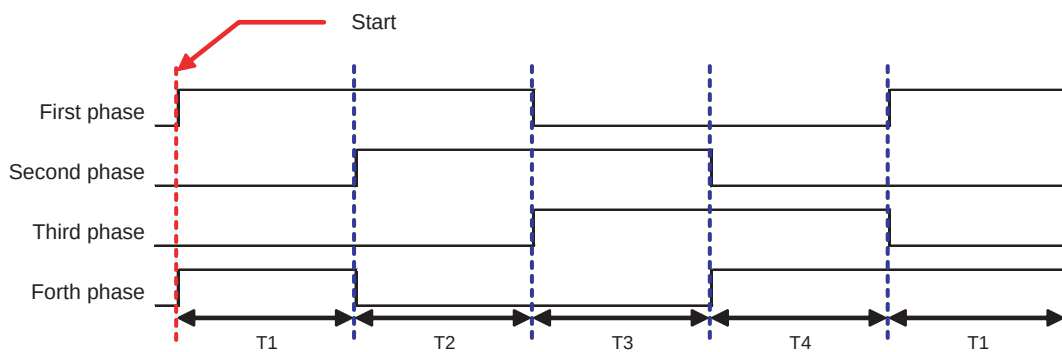


Figure:12.7.2 Basic Waveform 2 (2-phase Mode)

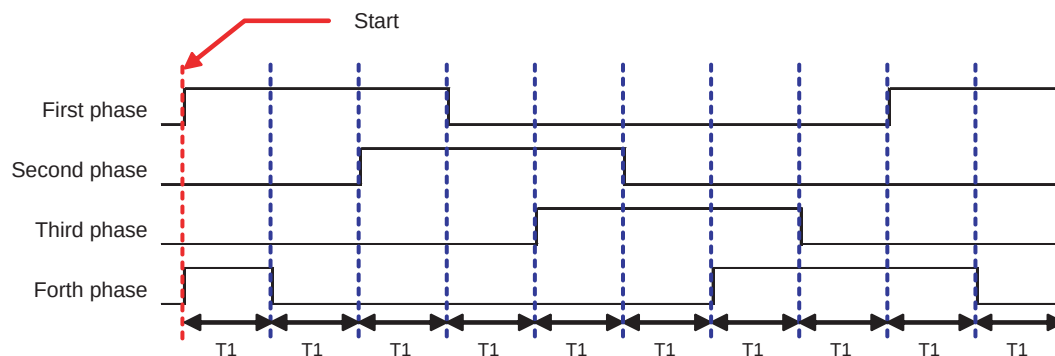


Figure:12.7.3 Basic Waveform 3 (1- to 2-phases Mode)

Each interval of T1, T2, T3 and T4 shown in Figure:12.7.1 to Figure:12.7.3 is determined by registers. Also, the registers can invert output order of PWM waveform. The first mode's waveform is shown below.

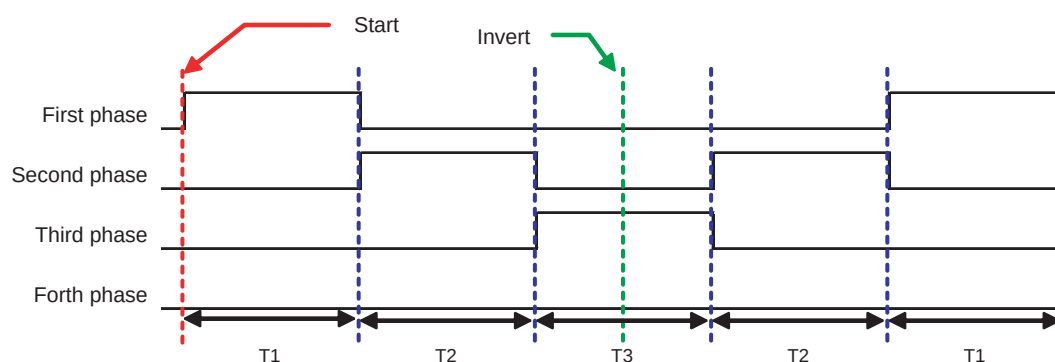


Figure:12.7.4 Basic Waveform 1 Dynamic Inversion (First Mode, MD1CTR = 0)

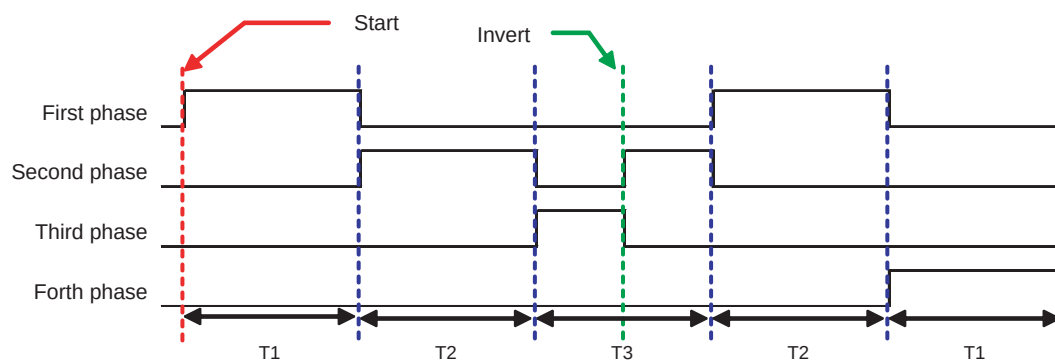


Figure:12.7.5 Basic Waveform 1 Dynamic Inversion (First Mode, MD1CTR = 1)

■ Dead Time Insertion

Dead time inserts delay to the starting point of active intervals of each phase.
The waveform when the dead time is inserted is shown below.

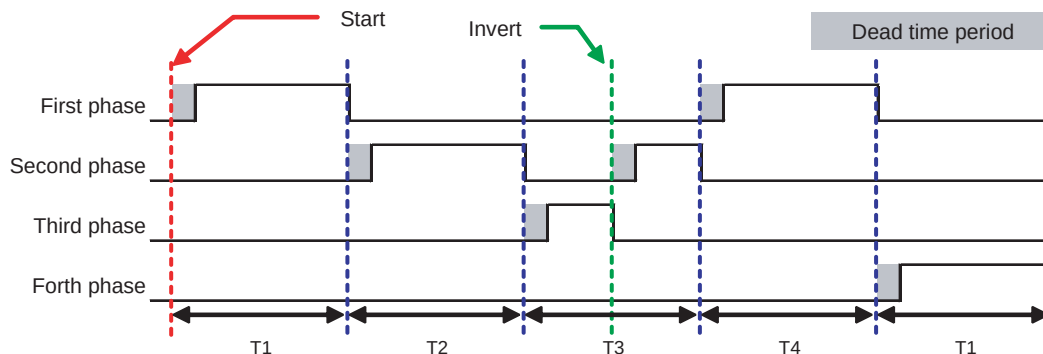


Figure:12.7.6 Dead Time Insertion Basic Waveform 1 (1-phase Mode)

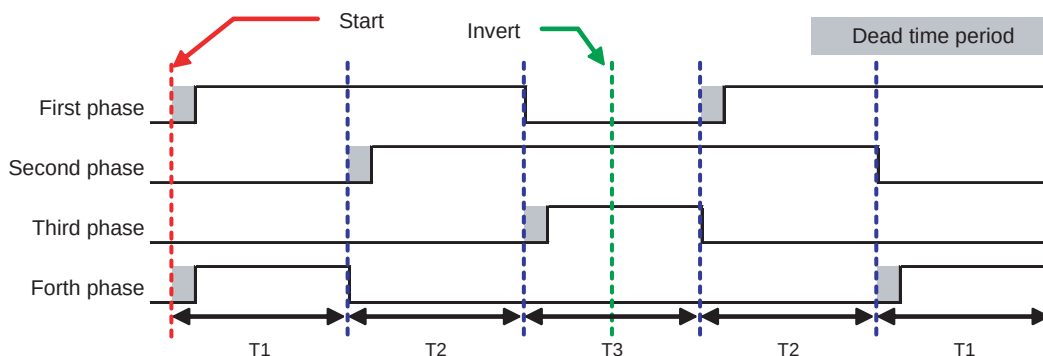


Figure:12.7.7 Dead Time Insertion Basic Waveform 2 (2-phase Mode)



PWM waveform from the first phase to the forth phase are assigned to TM9OD0, 2, 4 and 1.

12.7.2 Setup Example

An example setup procedure, with a description of 4 phases PWM 2-phase mode is shown below.

Setup Procedure	Description
(1) Stop the counter, select saw-tooth wave PWMMDL(0x03E00) bp1: TCEN =0 bp6: PCRBEN =1 bp0: WAVEMD =1 (2) Set the special function pin output P8OMD(0x03EB8) bp0: P8OMD0 =1 bp1: P8OMD1 =1 bp2: P8OMD2 =1 bp4: P8OMD4 =1 P8DIR(0x03E98) bp0: P8DIR0 =1 bp1: P8DIR1 =1 bp2: P8DIR2 =1 bp4: P8DIR4 =1 (3) Set the 2-phase PWM output PWMTMCNT(0x03E16) bp0: SEL_PWM_TM =0 RELCTR(0x03E17) bp1 to 0: RELMD =10 (4) Set the PWM output cycle PWMCMP =0x0031 (5) Set the PWM dead time DTMSET0 =0x0000 PWMMDL(0x03E00) bp3: DTEN =1 (6) Start the timer operation PWMMDL(0x03E00) bp1: TCEN =0	(1) Set the TCEN flag of PWM mode control register (PWMMD) to "0" to stop Timer 9 counting. In addition, set the PCRBEN flag to "1" to set load timing enable set the WAVEMD flag to "1" to select the saw-tooth wave. (2) Set the P8OMD1,2, and 4 flags of Port 8 output mode register 1 (P8OMD1) to "1" to select TM9OD1, 2 and 4. Also, set the P8DIR1, 2 and 4 flags of Port 8 direction control register (P8DIR) to "1" to set the output mode. [Chapter 8 I/O Port] (3) Set the SEL_WOM_TM flag of PWM timer operation control register to "0" to select the PWM operation. In addition, set the RELMD flag of 4 phases PWM mode register to "10" to select the 4 phases PWM 2-phase mode. (4) Set the value of the PWM output cycle in PWM cycle/compare collective setting register (PWMCMP). To set 50 kHz by dividing 10 MHz, set as; (200/4) - 1=49(0x0031). At the same time, Timer 9 binary counter (TM9BC) is initialized to 0x0000. (5) Set the dead time value of the PWM output phase change in dead time setting register (setting value n - 1). In addition, set the DTEN of PWM mode control register to "1" to enable dead time insertion. (6) Set the TCEN flag of PWM mode control register to "1" to start Timer 9 operation.

13.1 Overview

This LSI contains a timer called 24H timer which can count 24 hours at 0.5 second resolution. Also, this timer can generate interrupts at any time by setting an alarm in advance.

As a clock source of 24H timer, $4 \times n$, $5 \times n$, and $4.19 \times n$ MHz ($n=1, 2, 4$) for the high-speed oscillation and 32.768 kHz for the low-speed oscillation can be used.

When using the high-speed oscillation, 2, 4, and 8 multiplication (up to 40 MHz) of the PLL function can be set.

13.1.1 Functions

Table:13.1.1 24H Timer Functions

Timer operation		√
Timer resolution		0.5 second
Clock source		
High-speed oscillation input	(External high-speed oscillation)	4, 4.19, 5, 8, 8.39, 10 MHz
	(Internal high-speed oscillation)	20 MHz, 16 MHz
Low-speed oscillation input	(External low-speed oscillation)	32.768 kHz
	(Internal low-speed oscillation)	(30 kHz * Handle the interval interrupts as a guide because the timer cannot count accurately for precision.)
Interrupt type		2 types (Interval, alarm match)
Interrupt cycle (Interval)		0.5 second
		1 second
		1 minute
		1 hour
		24 hour
Operating mode		PLL
		NORMAL
		SLOW
		HALT0
		HALT1
		HALT2
Counter correction		√ (Can be set by 0.5 second when the counter stops)
Counter initialization		Register writing
Alarm function		√ (Alarm match interrupt)

13.1.2 Block Diagram

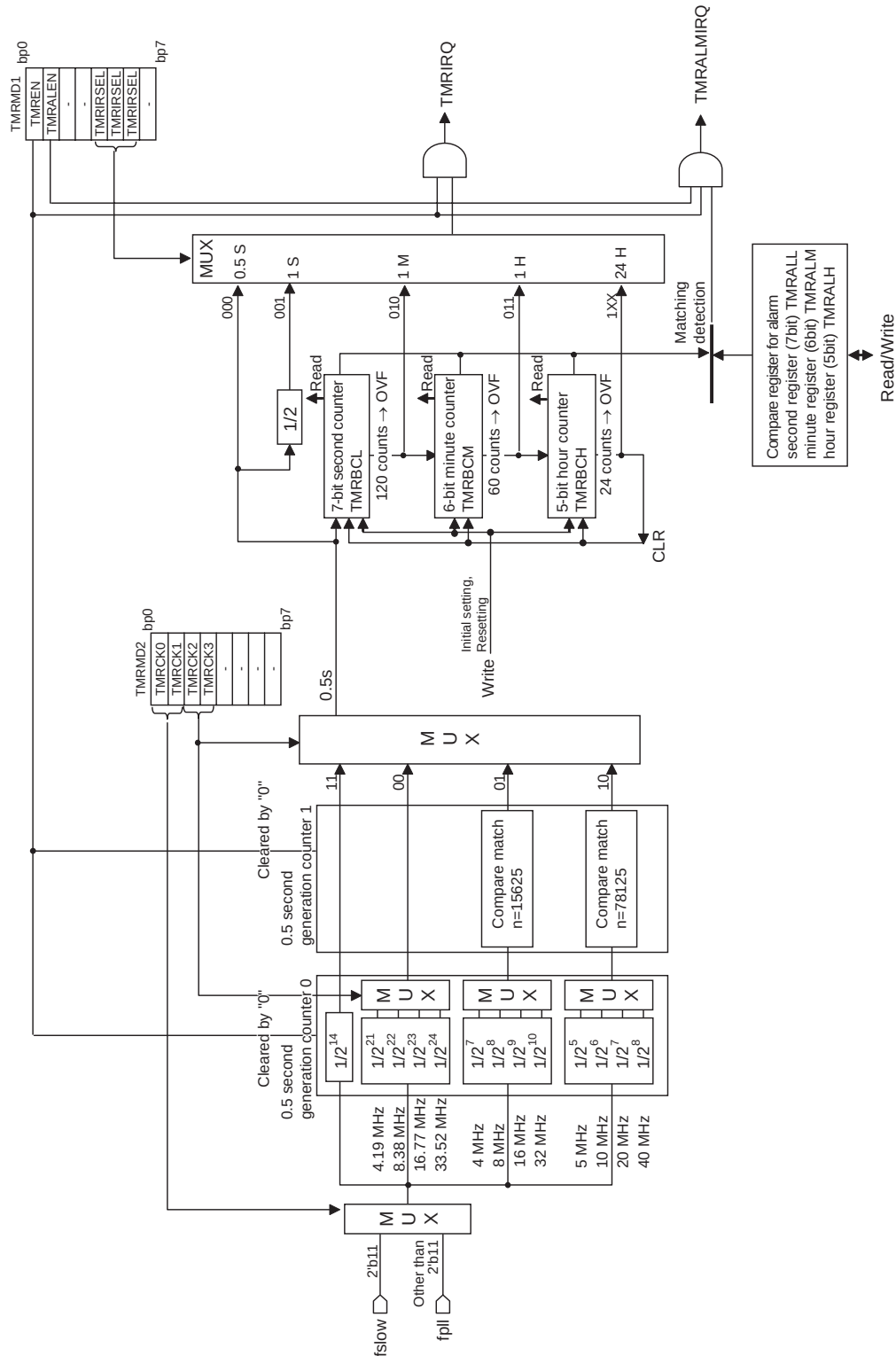


Figure:13.1.1 24H Timer Block Diagram

13.2 Registers

24H timer consists of three programmable 24H timer binary counters (TMRBCL/M/H) corresponding to seconds, minutes, and hours, and can be controlled by 24H timer mode register 0 (TMRMD0) and 24H timer mode register 1 (TMRMD1).

24H timer alarm compare registers (TMRALL/M/H) can generate alarm match interrupts by matching detection with 24H timer binary counters (TMRBCL/M/H).

13.2.1 Control Registers

Table:13.2.1 24H Timer Control Registers

	Register	Address	R/W	Function	Page
24H timer	TMRMD1	0x03FA8	R/W	24H timer mode register 1	XIII-5
	TMRMD2	0x03FA9	R/W	24H timer mode register 2	XIII-6
	TMRBCL	0x03FAA	R/W	24H timer binary conter (7-bit second counter)	XIII-7
	TMRBCM	0x03FAB	R/W	24H timer binary conter (6-bit minute counter)	XIII-7
	TMRBCH	0x03FAC	R/W	24H timer binary conter (5-bit hour counter)	XIII-7
	TMRALL	0x03FAD	R/W	24H timer alarm compare register (7-bit second setting)	XIII-9
	TMRALM	0x03FAE	R/W	24H timer alarm compare register (6-bit minute setting)	XIII-9
	TMRALH	0x03FAF	R/W	24H timer alarm compare register (5-bit hour setting)	XIII-9
Interrupt	TMRICR	0x03FEE	R/W	24H timer interrupt control register	VII-28
	PERIILR	0x03FFE	R/W	Peripheral function group interrupt level setting register	VII-37
	IRQEXPEN	0x03F4E	R/W	Peripheral function group interrupt input enable register	VII-38
	IRQEXPDT	0x03F4F	R/W	Peripheral function group interrupt factor retention register	VII-39

13.2.2 24H Timer Control Registers

These registers are used to set interrupts, interrupt input frequency, and enable or disable of 24H timer.

■ 24H Timer Mode Register 1 (TMRMD1: 0x03FA8)

bp	7	6	5	4	3	2	1	0
Flag	-	TMRIRSEL2	TMRIRSEL1	TMRIRSEL0	-	-	TMRALMEN	TMREN
At reset	-	0	0	0	-	-	0	0
Access	-	R/W	R/W	R/W	-	-	R/W	R/W

bp	Flag	Description
7	-	-
6 to 4	TMRIRSEL2 TMRIRSEL1 TMRIRSEL0	Interrupt generation cycle selection 000: Generated every 0.5 second 001: Generated every 1 second 010: Generated every 1 minute 011: Generated every 1 hour 1XX: Generated every 24 hour
3 to 2	-	-
1	TMRALMEN	Alarm function operation enable 0: Disabled 1: Enabled
0	TMREN	24H timer operation enable 0: Disabled 1: Enabled



When TMREN=1, TMRMD1 register (except TMRALMEN) must not be switched.



To use the alarm function, TMRALL/M/H must be set before starting 24H timer.



After setting 24H timer to disable (TMREN=0), 24H timer stops after 1 count clock (1 fpll or 1 fslow) .



When specifying the generation cycle of the interval interrupts in 1 minute, 1 hour, and 24 hours in TMRIRSEL2 to 0, the interrupts are generated at clear time of binary counters of 1 minute: TMRBCL, 1 hour: TMRBCM, and 24 hours: TMRBCH.

■ 24H Timer Mode Register 2(TMRMD2: 0x03FA9)

bp	7	6	5	4	3	2	1	0
Flag	Reserved	Reserved	Reserved	Reserved	TMRCK3	TMRCK2	TMRCK1	TMRCK0
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7 to 4	Reserved	Always set to "0".
3 to 2	TMRCK3 TMRCK2	For 24H timer, frequency specification of fpll, TMRCK1 to 0 setting frequency $\times$ n * When the high-speed clk (except fslow) is used 00: n=1 01: n=2 10: n=4 11: n=8
1 to 0	TMRCK1 TMRCK0	fpll clock specification 00: 4.19 MHz 01: 4 MHz 10: 5 MHz 11: 32.768 kHz



Writing to TMRMD register must be performed when TMREN=0.



When the value of the frequency of fpll for 24H timer (TMRCK1 to 0 setting frequency $\times$ n) is not same as the clock source, 24H timer cannot count correctly.

13.2.3 Programmable Counter

These registers are used as binary counter to store programmable seconds, minutes, and hours.

Set each value of seconds, minutes, and hours to TMRBCL/M/H in binary. After setting, they cannot be cleared by reset. Therefore, reset them within the initial routine as required.

■ 24H Timer Binary Counter L (Seconds Counter) (TMRBCL: 0x03FAA)

Binary counter by 0.5 seconds (binary).

bp	7	6	5	4	3	2	1	0
Flag	-	TMRBC6	TMRBC5	TMRBC4	TMRBC3	TMRBC2	TMRBC1	TMRBC0
At reset	-	X	X	X	X	X	X	X
Access	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W

■ 24H Timer Binary Counter M (Minutes Counter) (TMRBCM: 0x03FAB)

Binary counter by minutes (binary).

bp	7	6	5	4	3	2	1	0
Flag	-	-	TMRBC12	TMRBC11	TMRBC10	TMRBC9	TMRBC8	TMRBC7
At reset	-	-	X	X	X	X	X	X
Access	-	-	R/W	R/W	R/W	R/W	R/W	R/W

■ 24H Timer Binary Counter H (Hour Counter)(TMRBCH: 0x03FAC)

Binary counter by hours (binary).

bp	7	6	5	4	3	2	1	0
Flag	-	-	-	TMRBC17	TMRBC16	TMRBC15	TMRBC14	TMRBC13
At reset	-	-	-	X	X	X	X	X
Access	-	-	-	R/W	R/W	R/W	R/W	R/W



Writing must be performed when 24H timer is stopped (TMREN=0). When 24H timer is operating (TMREN=1), writing cannot be performed.



When reading at the same timing as counting up, the value may not be read accurately. To avoid it, fix the read value by more than three times reading or majority vote.



When reading, the counter value is applied after 2 count clocks ($2 \times f_{pll}$ or $2 \times f_{slow}$).



Set read data within the range of TMRBCL=0h00 to 0h77 (0 to 119), TMRBCM=0h00 to 0h3B (0 to 59), TMRBCH=0h00 to 0h17 (0 to 23).

13.2.4 Alarm Compare Registers

These registers are alarm compare registers of seconds, minutes, and hours, which can generate interrupts by perfect matching of seconds, minutes, and hours with binary counters.

Set the value of seconds, minutes, and hours to TMRALL/M/H in binary. After setting, the values cannot be cleared by reset. Therefore, reset them within the initial routine as required.

■ 24H Timer Alarm Compare Register L (Second) (TMRALL: 0x03FAD)

Alarm compare register by 0.5 seconds (binary)

bp	7	6	5	4	3	2	1	0
Flag	-	TMRAL6	TMRAL5	TMRAL4	TMRAL3	TMRAL2	TMRAL1	TMRAL0
At reset	-	X	X	X	X	X	X	X
Access	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W

■ 24H Timer Alarm Compare Register M (Minute) (TMRALM: 0x03FAE)

Alarm compare register by minutes (binary)

bp	7	6	5	4	3	2	1	0
Flag	-	-	TMRAL12	TMRAL11	TMRAL10	TMRAL9	TMRAL8	TMRAL7
At reset	-	-	X	X	X	X	X	X
Access	-	-	R/W	R/W	R/W	R/W	R/W	R/W

■ 24H Timer Alarm Compare Register H (Hour) (TMRALH: 0x03FAF)

Alarm compare register by hours (binary)

bp	7	6	5	4	3	2	1	0
Flag	-	-	-	TMRAL17	TMRAL16	TMRAL15	TMRAL14	TMRAL13
At reset	-	-	-	X	X	X	X	X
Access	-	-	-	R/W	R/W	R/W	R/W	R/W



The counter will not be cleared at reset. Therefore, reset them within initial routine as needed.



Set write data within the range of TMRALL=0h00 to 0h77 (0 to 119), TMRALM=0h00 to 0h3B (0 to 59), TMRALH=0h00 to 0h17 (0 to 23).



When setting the same value as TMRBCL/M/H, an interrupt is generated after counting 24 hours.

13.3 Operation

13.3.1 Normal Operation

The clock count of 0.5 seconds resolution can be performed by the clock of the PLL circuit output clock f_{pll} or the low-speed clock for peripheral function f_{slow} .

Set the initial value (initial time) when 24H timer stops, and then start counting.

When the operation is stopped (TMREN flag = 0), the counter which generates 0.5 seconds (0.5 seconds base binary counter) is cleared.

When the operation is enabled again (TMREN flag = 1), the 0.5 seconds base binary counter starts counting after synchronizing with 1 count clock (1 f_{pll} or 1 f_{slow}).

In addition, for the interval interrupt function, 24H timer can generate interrupts in the cycle of 0.5 seconds, 1 second, 1 minute, 1 hour and 24 hours.

■ Counter Operation Timing 1

Binary counter counts up the selected clock source as count timing.

EX) $f_{slow}=32.768\text{ kHz}$ is selected (0.5 seconds base compare register = 2^{14})
Initial value of TMRBCL is 0, 0.5 seconds interval interrupt is ON

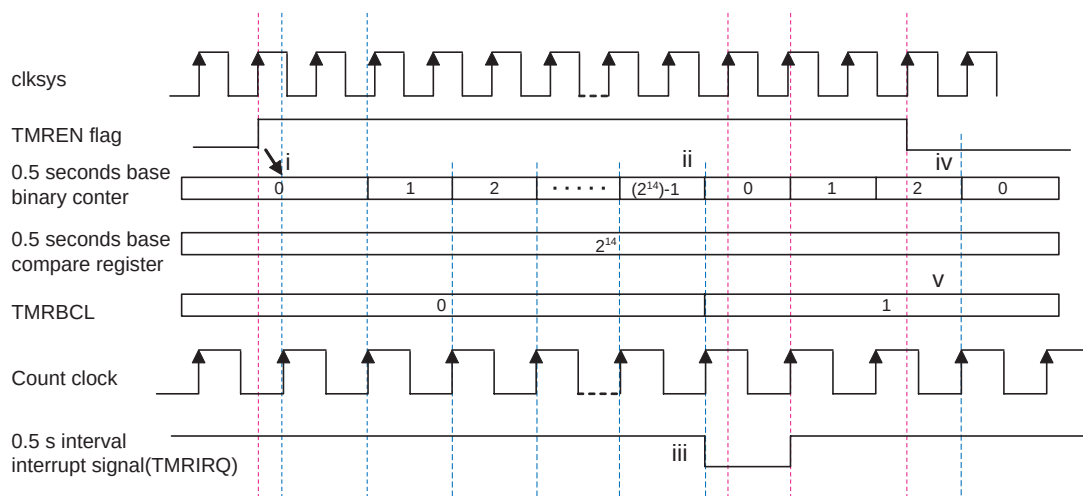


Figure:13.3.1 Count Up Timing of 24H Timer Operation

- Counting starts at the timing of the count clock synchronization after setting the TMREN flag.
- The counter of TMRBCL (in 0.5 second) increments at the timing when the 0.5 seconds base binary counter reaches the 0.5 base compare register.
- Interval interrupts generate at the timing when the 0.5 seconds base binary counter counts up from " $(0.5\text{ second base compare register})-1$ ".
- The count value of the 0.5 seconds base counter is cleared when counting is stopped (TMREN flag = 0).
- The count value of 24H timer binary counter (TMRBCL/M/H) is not cleared even when counting is stopped (TMREN flag = 0) or resetting. Write in the registers to change the value.

■ Counter Operation Timing 2

Binary counter counts up the selected clock source as count timing.

EX) fslow=32.768 kHz is selected (0.5 second base compare register = 2^{14})
Initial value of TMRBCL: 119 (decimal), TMRBCM: 59 (decimal), TMRBCH: 23 (decimal),
24 hours interval interrupt ON

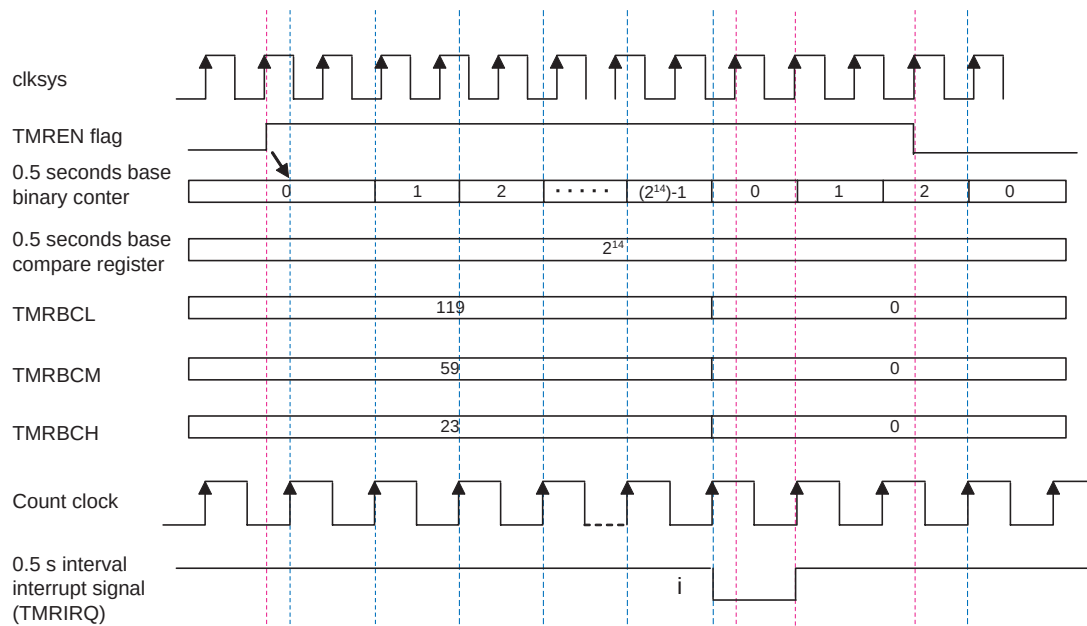


Figure:13.3.2 Count Up Timing of 24H Timer Binary Counter

- i. 24 hours interval interrupt is generated at the same time as TMRBCL/M/H are cleared at the timing when the 0.5 seconds base binary counter reaches the 0.5 base compare register.

■ Alarm Match Interrupt Timing

Alarm setting is operating.

EX) fslow=32.768 kHz is selected (0.5 second base compare register = 2^{14})
Initial value of TMRBCL: 20 (decimal), TMRBCM: 10 (decimal), TMRBCH: 5 (decimal),
24 hours interval interrupt ON

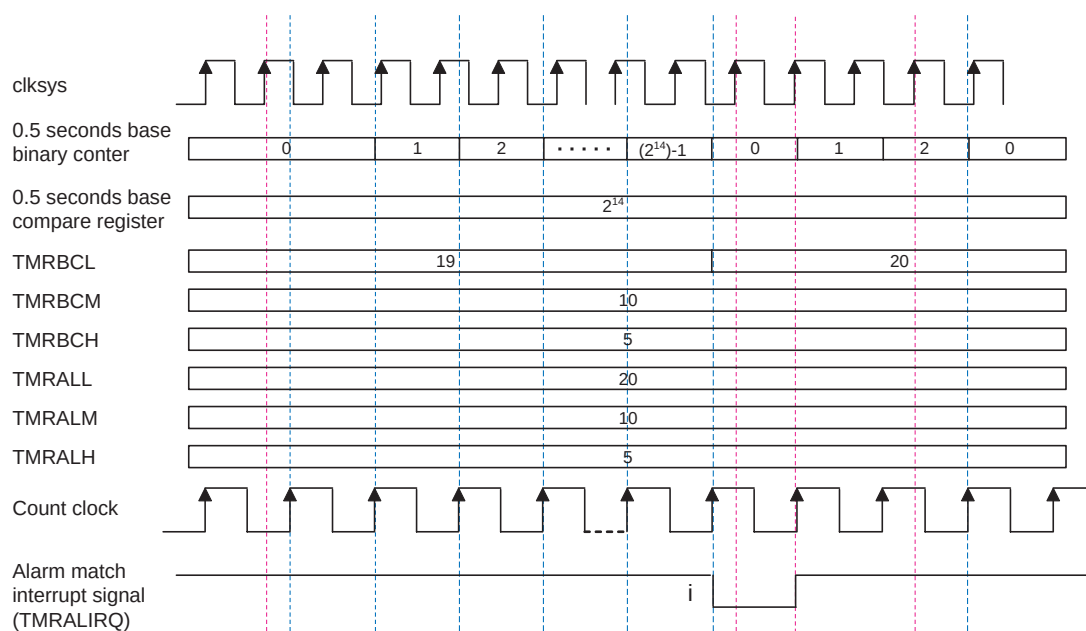


Figure:13.3.3 Alarm Match Interrupt Generation Timing

- i. Alarm match interrupt is generated at the timing when TMRBCL/M/H is incremented to the set value of TMRALL/M/H when the 0.5 seconds base binary counter reaches the 0.5 base compare register.

13.3.2 Setup Example 1

24H timer is used to generate an interrupt at regular time intervals (1 minute) to realize clock functions. Count up with fpll (32 MHz) as clock source from the initial time 10:23:35.

An example setup procedure, with a description of each step is shown below.

Setup Procedure	Description
(1) Select clock source TMRMD2(0x03FA9) bp3 to 2: TMRCK3 to 0=1101	(1) Set fpll=4 MHz*8 as clock source.
(2) Disable interrupts TMRICR(0x03FEE) bp1: TMRIE = 0	(2) Set the TMRIE flag of TMRICR register to "0" to disable interrupts.
(3) Set the initial time of 24H timer TMRBCL(0x03FAA) bp6 to 0: TMRBC6 to 0 = 1000110 TMRBCM(0x03FAB) bp5 to 0: TMRBC12 to 7 = 010111 TMRBCH(0x03FAC) bp4 to 0: TMRBC17 to 13 = 01010	(3) Input initial time to 24H binary counter TMRBCL/M/H. Set TMRBC6 to 0 flags to "35 seconds" (binary: 1000110). Set TMRBC12 to 7 flags to "23 minutes" (binary: 010111). Set TMRBC17 to 13 flags to "10 hours" (binary: 01010).
(4) Set interrupt level TMRICR(0x03FEE) bp7 to 6: TMRLV1 to 0 = 01	(4) Set interrupt level by the TMRLV1 to 0 flags of 24H timer interrupt control register. If any interrupt request flags are already set, clear them. [Chapter 1 7.1.4 Maskable Interrupt Control Register Setup]
(5) Set 24H timer interrupt cycle TMRMD1(0x03FA8) bp6 to 4: TMRIRSEL2 to 0 = 010	(5) Set the TMRIRSEL2 to 0 flags of 24H timer mode register 1 to select interrupt generation cycle.
(6) Set interrupts enable TMRICR(0x03FEE) bp1: TMRIE = 1	(6) Set the TMRIE flag of TMICR register to "1" to enable interrupts.
(7) Set 24H timer enable TMRMD1(0x03FA8) bp0: TMREN = 1	(7) Set the TMREN flag of 24H timer mode register 1 to enable 24H timer operation.

13.3.3 Setup Example 2

24H timer is used to perform operation with alarm function.

Count up with fpll (32 MHz) as clock source from initial time 10:23:35, and generate interrupts as alarm match interrupts at 21:56:15.

An example setup procedure, with a description of each step is shown below.

Setup Procedure	Description
(1) Select clock source TMRMD2(0x03FA9) bp3 to 2: TMRCK3 to 0=1101	(1) Set fpll=4 MHz*8 as clock source.
(2) Disable all maskable interrupts PSW bp6: MIE=0	(2) Clear the MIE flag of PSW to disable all maskable interrupts.
(3) Clear the extended interrupt request flag IRQEXPDT(0x03F4F) bp4: IRQEXPDT4	(3) Clear the extended interrupt request flag by reading peripheral function group interrupt factor retention register to write value in it. For register settings, refer to [Chapter 1 7.1.5 Peripheral Function Group Interrupt Control Register Setup].
(4) Set the extended interrupt input IRQEXPEN(0x03F4E) bp4: IRQEXPEN4	(4) Set the IRQEXPEN6 to 0 flags of peripheral function group interrupt input enable register (IRQEXPEN) to enable the extended interrupts.
(5) Set the initial time of 24H timer TMRBCL(0x03FAA) bp6 to 0: TMRBC6 to 0 = 1000110 TMRBCM(0x03FAB) bp5 to 0: TMRBC12 to 7 = 010111 TMRBCH(0x03FAC) bp4 to 0: TMRBC17 to 13 = 01010	(5) Input initial time to 24H binary counter TMRBCL/M/H. Set TMRBC6 to 0 flags to "35 seconds" (binary: 1000110). Set TMRBC12 to 7 flags to "23 minutes" (binary: 010111). Set TMRBC17 to 13 flags to "10 hours" (binary: 01010).
(6) Set the alarm interrupt time TMRALL(0x03FAD) bp6 to 0: TMRAL6 to 0 = 0011110 TMRALM(0x03FAE) bp5 to 0: TMRAL12 to 7 = 111000 TMRALH(0x03FAF) bp4 to 0: TMRAL17 to 13 = 10101	(6) Input initial time to 24H timer alarm compare register TMRALL/M/H. Set TMRAL6 to 0 flags to "15 seconds" (binary: 0011110). Set TMRAL12 to 7 flags to "56 minutes" (binary: 111000). Set TMRAL17 to 13 flags to "21 hours" (binary: 10101).
(7) Set interrupt level PERIILR(0x03FEF) bp7 to 6: PERIILV1 to 0 = 01	(7) Set interrupt level by the PERIILV1 to 0 flags of peripheral function group interrupt control register. If any interrupt request flags are already set, clear them. [Chapter 1 7.1.4 Maskable Interrupt Control Register Setup]
(8) Set 24H timer interrupt cycle TMRMD1(0x03FA8) bp6 to 4: TMRIRSEL2 to 0 = 010	(8) Set the TMRIRSEL2 to 0 flags of 24H timer mode register 1 to select interrupt generation cycle.
(9) Enable alarm match interrupt TMRMD1(0x03FA8) bp1: TMRALEN = 1	(9) Set the TMRALEN flag of 24H timer mode register to enable 24H timer alarm interrupt operation.
(10) Enable all maskable interrupts PSW bp6: MIE=1	(10) Set the MIE flag of PSW to enable maskable interrupts.
(11) Set 24H timer enabled TMRMD1(0x03FA8) bp0: TMREN = 1	(11) Set the TMREN flag of 24H timer mode register 1 to enable 24H timer operation.

14.1 Overview

This LSI has a time base timer and a 8-bit free-run timer (Timer 6). Time Base Timer is a 15-bit timer counter.

14.1.1 Functions

Table:14.1.1 shows the clock source and the interrupt generation cycle that Timer 6 and Time Base Timer can use.

Table:14.1.1 Clock Source and Generation Cycle

	Time Base Timer	Timer 6 (8-bit free-run)
8-bit timer operation	-	√
Interrupt	TBIRQ	TM6IRQ
Clock source	fppll-div fslow	fppll-div fs fslow $\text{fppll-div} \times 1/2^2 *1$ $\text{fppll-div} \times 1/2^3 *1$ $\text{fppll-div} \times 1/2^{12} *1$ $\text{fppll-div} \times 1/2^{13} *1$ $\text{fslow} \times 1/2^2 *2$ $\text{fslow} \times 1/2^3 *2$ $\text{fslow} \times 1/2^{12} *2$ $\text{fslow} \times 1/2^{13} *2$ synchronous fslow synchronous $\text{fppll-div} \times 1/2^2 *1$ synchronous $\text{fppll-div} \times 1/2^3 *1$ synchronous $\text{fppll-div} \times 1/2^{12} *1$ synchronous $\text{fppll-div} \times 1/2^{13} *1$ synchronous $\text{fslow} \times 1/2^2 *2$ synchronous $\text{fslow} \times 1/2^{13} *2$ synchronous $\text{fslow} \times 1/2^{12} *2$ synchronous $\text{fslow} \times 1/2^{13} *2$
Interrupt generation cycle	$\text{fppll-div} \times 1/2^7$ $\text{fppll-div} \times 1/2^8$ $\text{fppll-div} \times 1/2^9$ $\text{fppll-div} \times 1/2^{10}$ $\text{fppll-div} \times 1/2^{13}$ $\text{fppll-div} \times 1/2^{15}$ $\text{fslow} \times 1/2^7$ $\text{fslow} \times 1/2^8$ $\text{fslow} \times 1/2^9$ $\text{fslow} \times 1/2^{10}$ $\text{fslow} \times 1/2^{13}$ $\text{fslow} \times 1/2^{15}$	The interrupt generation cycle is decided by the arbitrary value written to TM6OC.
fppll-div: High-speed clock for peripheral function fslow: Low-speed clock for peripheral function fs: System clock *1 Can be used when fppll-div is selected as a clock source of Time Base Timer *2 Can be used when fslow is selected as a clock source of Time Base Timer		



When changing the frequency of fppll-div by bp7 to bp4 of OSCCNT register, it should be performed after Time Base Timer / Free-run Timer function are stopped.

14.1.2 Block Diagram

■ Timer 6 and Time Base Timer Block Diagram

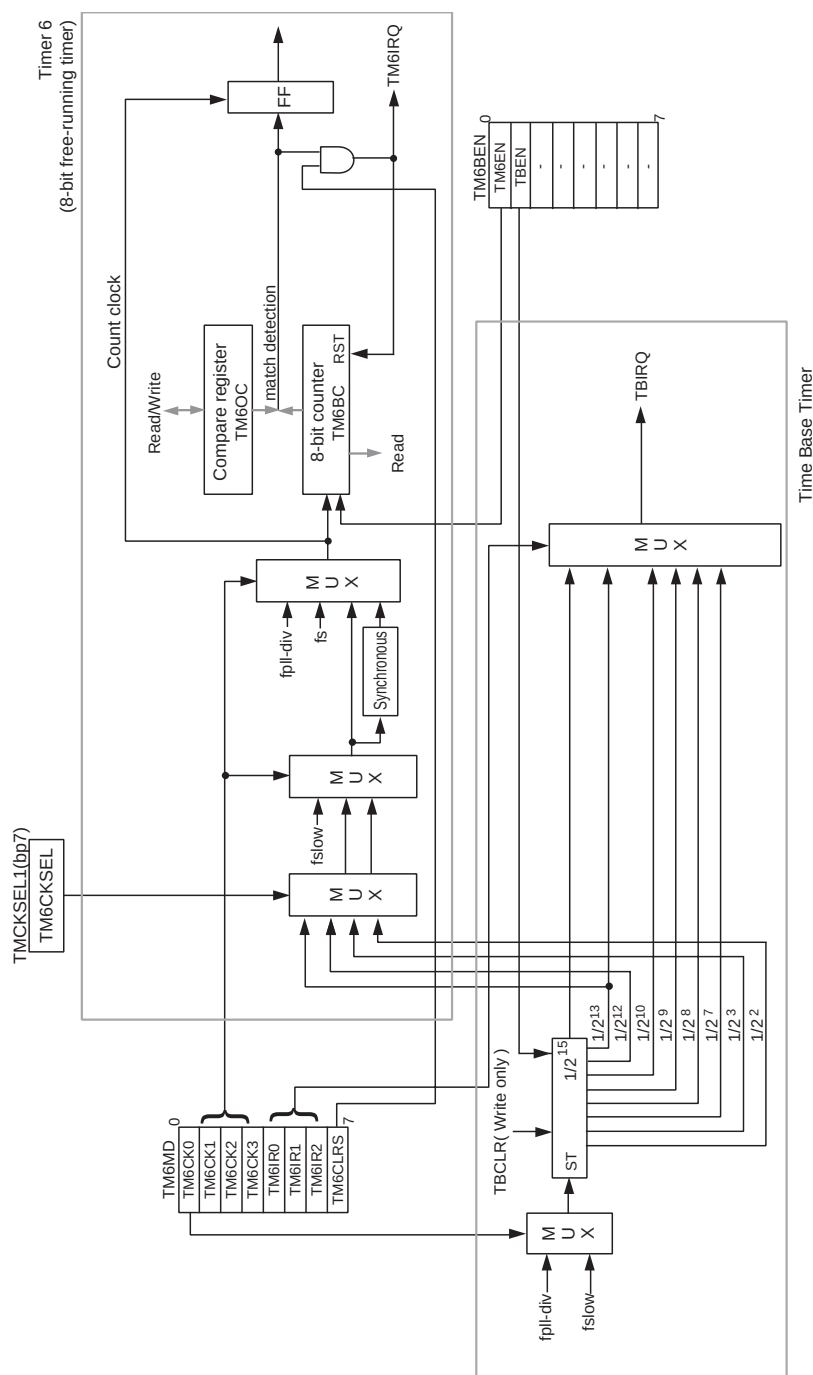


Figure:14.1.1 Block Diagram (Timer 6 and Time Base Timer)

14.2 Control Registers

Timer 6 consists of binary counter (TM6BC), compare register (TM6OC), and is controlled by mode register (TM6MD). Time Base Timer is controlled by mode register (TM6MD) and Time Base Timer clear register (TBCLR). Both timers are operated by an enable signal of Timer 6 enable register (TM6BEN).

14.2.1 Control Registers

Table:14.2.1 shows the registers that control Timer 6 and Time Base Timer.

Table:14.2.1 Control Registers

	Register	Address	R/W	Function	Page
Timer 6	TM6BC	0x03F78	R	Timer 6 binary counter	XIV-6
	TM6OC	0x03F79	R/W	Timer 6 compare register	XIV-6
	TM6MD	0x03F7A	R/W	Timer 6 mode register	XIV-8
	TM6BEN	0x03F7C	R/W	Timer 6 enable register	XIV-7
	TM6ICR	0x03FEE	R/W	Timer 6 interrupt control register	VII-27
Time Base Timer	TM6MD	0x03F7A	R/W	Timer 6 mode register	XIV-8
	TBCLR	0x03F7B	W	Time Base Timer clear control register	XIV-6
	TBICR	0x03FEF	R/W	Time base interrupt control register	VII-29
	TMCKSEL1	0x03EB0	R/W	Timer clock selection register 1	XIV-9

14.2.2 Programmable Timer Registers

Timer 6 is a 8-bit programmable counter.

Programmable counter consists of compare register (TM6OC) and binary counter (TM6BC).

Binary counter is a 8-bit up-counter. When the TM6CLRS flag of Timer 6 mode register (TM6MD) is "0" and the interrupt cycle data is written to compare register (TM6OC), Timer 6 binary counter (TM6BC) is cleared to 0x00.

■ Timer 6 Binary Counter (TM6BC: 0x03F78)

bp	7	6	5	4	3	2	1	0
Flag	TM6BC7	TM6BC6	TM6BC5	TM6BC4	TM6BC3	TM6BC2	TM6BC1	TM6BC0
At reset	0	0	0	0	0	0	0	0
Access	R	R	R	R	R	R	R	R

■ Timer 6 Compare Register (TM6OC: 0x03F79)

bp	7	6	5	4	3	2	1	0
Flag	TM6OC7	TM6OC6	TM6OC5	TM6OC4	TM6OC3	TM6OC2	TM6OC1	TM6OC0
At reset	X	X	X	X	X	X	X	X
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Time Base Timer can be reset its operation by the software. Time Base Timer can be cleared by writing an arbitrary value to Time Base Timer clear control register (TBCLR).

■ Time Base Timer Clear Control Register (TBCLR: 0x03F7B)

bp	7	6	5	4	3	2	1	0
Flag	TBCLR7	TBCLR6	TBCLR5	TBCLR4	TBCLR3	TBCLR2	TBCLR1	TBCLR0
At reset	-	-	-	-	-	-	-	-
Access	W	W	W	W	W	W	W	W

14.2.3 Timer 6 Enable Register

This register controls the starting operation of Timer 6 and Time Base Timer.

■ Timer 6 Enable Register (TM6BEN: 0x03F7C)

bp	7	6	5	4	3	2	1	0
Flag	-	-	-	-	-	Reserved	TBEN	TM6EN
At reset	-	-	-	-	-	0	0	0
Access	-	-	-	-	-	R/W	R/W	R/W

bp	Flag	Description
7 to 3	-	-
2	Reserved	Always set to "0".
1	TBEN	Time Base Timer operation control 0: Disabled 1: Enabled
0	TM6EN	Timer 6 operation control 0: Disabled 1: Enabled



The TM6EN flag of TM6BEN register must be set to "1" to start Timer 6 operation.



The the TBEN flag of TM6BEN register must be set to "1" to start Time Base Timer operation.

14.2.4 Timer Mode Register

This is readable/writable register that controls Timer 6 and Time Base Timer.

■ Timer 6 Mode Register (TM6MD: 0x03F7A)

bp	7	6	5	4	3	2	1	0
Flag	TM6CLRS	TM6IR2	TM6IR1	TM6IR0	TM6CK3	TM6CK2	TM6CK1	TM6CK0
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description		
7	TM6CLRS	Timer 6 binary counter clear selection flag 0: TM6BC initialization enabled when writing to TM6OC. 1: TM6BC initialization disabled when writing to TM6OC. * TM6IRQ is disable when TM6CLRS = 0, TM6IRQ is enabled when TM6CLRS = 1.		
6 to 4	TM6IR2 TM6IR1 TM6IR0	Time Base Timer interrupt cycle selection 000: Time base selection clock $\times 1/2^7$ 001: Time base selection clock $\times 1/2^8$ 010: Time base selection clock $\times 1/2^9$ 011: Time base selection clock $\times 1/2^{10}$ 10X: Time base selection clock $\times 1/2^{13}$ 11X: Time base selection clock $\times 1/2^{15}$		
3 to 1	TM6CK3 TM6CK2 TM6CK1	Timer 6 clock source selection 000: fpll-div 001: fs 010: fslow 011: Synchronous fslow		
			TM6CKSEL	
		0	1	
		100: Time base selection clock 101: Synchronous time base selection clock 110: Time base selection clock 111: Synchronous time base selection clock	$\times 1/2^{13}$ $\times 1/2^{13}$ $\times 1/2^{12}$ $\times 1/2^{12}$	$\times 1/2^2$ $\times 1/2^2$ $\times 1/2^3$ $\times 1/2^3$
0	TM6CK0	Time Base Timer clock source selection 0: fpll-div 1: fslow		



When the TM6CKSEL flag of TMCKSEL1 register is set to "1", $1/2^2$ and $1/2^3$ of both time base selection clock and synchronous time base selection clock can be selected.

When the TM6CKSEL flag is set to "0", $1/2^{12}$ and $1/2^{13}$ of both time base selection clock and synchronous time base selection clock can be selected.

■ Timer Clock Selection Register (TMCKSEL1: 0x03FB0)

bp	7	6	5	4	3	2	1	0
Flag	TM6CKSEL	-	-	TM4IOSEL	TM3IOSEL	TM2IOSEL	TM1IOSEL	TM0IOSEL
At reset	0	-	-	0	0	0	0	0
Access	R/W	-	-	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7	TM6CKSEL	Selection of Time Base Timer clock dividing ratio 0: Time base timer clock $\times (1/2^{13}$ or $1/2^{12})$ 1: Time base timer clock $\times (1/2^2$ or $1/2^3)$
6 to 5	-	-
4	TM4IOSEL	Timer 4 input selection 0: TM4IO 1: Timer A
3	TM3IOSEL	Timer 3 input selection 0: TM3IO 1: Timer A
2	TM2IOSEL	Timer 2 input selection 0: TM2IO 1: Timer A
1	TM1IOSEL	Timer 1 input selection 0: TM1IO 1: Timer A
0	TM0IOSEL	Timer 0 input selection 0: TM0IO 1: Timer A

14.3 8-bit Free-run Timer

14.3.1 Operation

■ 8-bit Free-run Timer (Timer 6)

The interrupt generation cycle should be set in advance, by the set value of compare register (TM6OC) and the clock source selection. When binary counter (TM6BC) reaches the set value of compare register, an interrupt request is generated at the next count clock and binary counter is cleared to restart count up from 0x00.

Table:14.3.1 shows selectable clock source.

Table:14.3.1 Clock Source at Timer Operation (Timer 6)

Clock source	Time per count		
	At fpll-div=10 MHz	At fpll-div=8.39 MHz	At fpll-div=2 MHz
fpll-div	100 ns	119.1 ns	500 ns
fslow	30.5 μs		
fs	200 ns	238.3 ns	1000 ns
fpll-div × 1/2 ²	400 ns	476.7 ns	2000 ns
fpll-div × 1/2 ³	800 ns	953.5 ns	4000 ns
fpll-div × 1/2 ¹²	409.6 μs	487.4 μs	2048 μs
fpll-div × 1/2 ¹³	819.2 μs	976.4 μs	4096 μs
fslow × 1/2 ²	122 μs		
fslow × 1/2 ³	244 μs		
fslow × 1/2 ¹²	125 ms		
fslow × 1/2 ¹³	250 ms		
fpll-div = 10 MHz, 8.39 MHz, 2 MHz fslow = 32.768 kHz fs = fpll-div/2			

■ 8-bit Free-run Timer as a 1 Minute-timer, a 1 Second-timer

Table:14.3.2 shows the clock source selection and TM6OC register setup, when a 8-bit free-run timer is used as a 1 minute-timer, a 1 second-timer.

Table:14.3.2 1 Minute-timer, 1 Second-timer (Timer 6) Setup

Interrupt Generation Cycle	Clock source	TM6OC register
1 minute	$f_{\text{slow}} \times 1/2^{13}$	0xEF
1 second	$f_{\text{slow}} \times 1/2^{13}$	0x03
$f_{\text{slow}} = 32.768 \text{ kHz}$		

When the 1 minute-timer (1 minute) is set on Table:14.3.2, the bp2 waveform frequency (cycle) of TM6BC register is 1 Hz (1 second). So, that can be used for adjusting the seconds.

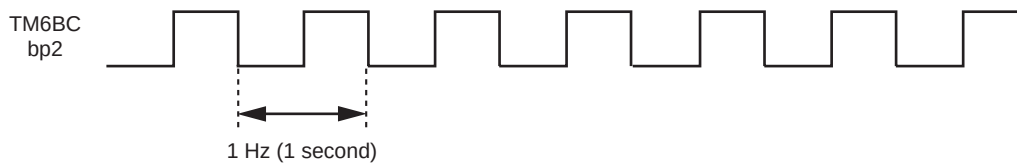


Figure:14.3.1 Waveform of bp2 in TM6BC Register (Timer 6)



Switch the count clock after the timer operation is stopped, as the counting is not performed correctly during the timer operation.

■ Count Timing of Timer Operation (Timer 6)

Binary counter counts up with the selected clock source as a count clock.

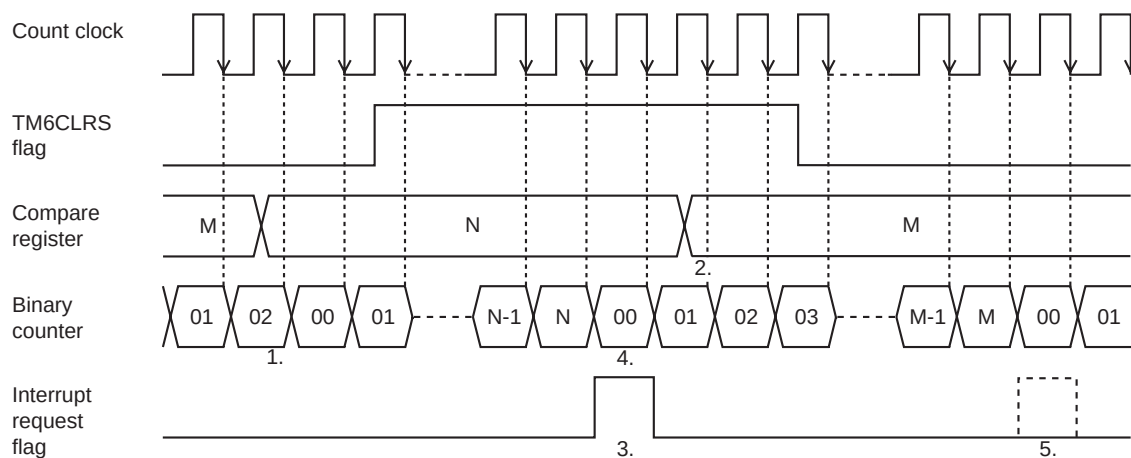


Figure:14.3.2 Count Timing of Timer Operation (Timer 6)

1. If any data is written to compare register when the TM6CLRS flag is "0", binary counter is cleared to 0x00.
2. If any data is written to compare register when the TM6CLRS flag is "1", binary counter is not cleared.
3. If binary counter reaches the value of compare register when the TM6CLRS flag is "1", an interrupt request flag is set at the next count clock.
4. When an interrupt request flag is set, binary counter is cleared to 0x00 and restarts the counting.
5. If binary counter reaches the value of compare register when the TM6CLRS flag is "0", no interrupt request flag is set.



When binary counter reaches the value in compare register, the interrupt request flag is set and binary counter is cleared at the next count clock.

So set compare register as:

(Compare register setting) = (count till the interrupt request - 1)



If fslow input is selected as a clock source in Timer 6, uncertain value may be read when binary counter is read on the operation.

To prevent this, select the synchronous fslow input.



If fslow is used as a clock source, binary counter should be cleared before starting the timer operation. Also, to set 0x00 to compare register, the synchronous fslow should be used.



If the smaller value than binary counter is set to compare register during the counting operation, binary counter continues counting till overflow. (When TM6CLRS flag is "1".)



Up to 3 system clocks are needed from the timer n interrupt request flag till the next interrupt request flag. During the period, no interrupt request flag is generated even if a compare match occurs.



When the fslow is selected for the count clock source and the value of compare register is rewritten, the operation cannot be ensured during the transition to the circle of the setting value.

To operate the timer in the expected circle from the beginning, stop the timer before rewriting compare register, then start the timer operation.



When fslow is used as a clock source, it is counted at "falling" of the count clock. When the other is used, it is counted at "rising" of the count clock.



Count clock source should be changed when the timer interrupt is disabled.



If the TM6CLRS flag of TM6MD register is set to "0", TM6BC can be initialized at every rewriting of TM6OC register, but in that state, Timer 6 interrupt is disabled. To use Timer 6 interrupt, set the TM6CLRS flag to "1" after rewriting TM6OC register.



To select time base selection clock or synchronous time base selection clock, be sure to set a clock source for Time Base Timer.



For Timer 6 and Time Base Timer on this LSI, operation or stop of binary counter is controlled by the signal sampling the value of the TMnEN flag by the count clock. Therefore, if fslow is selected as a count clock source, use binary counter with attention to the following two points:

1. When reading binary counter value after timer halts, put the TMnEN flag down, wait 1 count cycle, and read the value.
To read the value, program to read binary counter multiple times. However, if reading the value without waiting for 1 count cycle, the read value is [count value - 1].
 2. When halting the timer and changing its setting (clock selection, function switching, etc.), wait 1 count clock after the timer is disabled before setting the timer.
If the setting is switched during timer operation, proper operation is not guaranteed.
-

14.3.2 Setup Example

■ Timer Operation Setup (Timer 6)

Timer 6 generates interrupts constantly for timer function. Interrupts are generated in every 250 dividing (25 μ s) by selecting fs (fpll-div = 10 MHz at operation) as a clock source.

An example setup procedure, with a description of each step is shown below.

Setup Procedure	Description
(1) Enable binary counter initialization TM6MD(0x03F7A) bp7: TM6CLRS =0	(1) Set the TM6CLRS flag of Timer 6 mode register (TM6MD) to "0". At this time, the initialization of Timer 6 binary counter (TM6BC) is enabled.
(2) Disable the interrupt TM6ICR(0x03FEF) bp1: TM6IE =0	(2) Set the TM6IE flag of TM6ICR register to "0" to disable the interrupt.
(3) Select the clock source TM6MD(0x03F7A) bp3 to 1: TM6CK3 to 1 =001	(3) Clock source can be selected by the TM6CK3 to 1 flag of TM6MD register. Actually, fs is selected.
(4) Set the interrupt generation cycle TM6OC(0x03F79) =0xF9	(4) Set the interrupt generation cycle to Timer 6 compare register (TM6OC). At that time, TM6BC is initialized to 0x00.
(5) Enable the interrupt request TM6MD(0x03F7A) bp7: TM6CLRS =1	(5) Set the TM6CLRS flag of TM6MD register to "1" to enable the interrupt request generation.
(6) Set the interrupt level TM6ICR(0x03FEF) bp7 to 6: TM6LV1 to 0 =01	(6) Set the interrupt level by the TM6LV1 to 0 flag of Timer 6 interrupt control register (TM6ICR). If the interrupt request flag may be already set, clear them. [Chapter 7 7.1.4 Maskable Interrupt Control Register Setup]
(7) Enable the interrupt TM6ICR(0x03FEF) bp1: TM6IE =1	(7) Set the TM6IE flag of TM6ICR register to "1" to enable the interrupt.
(8) Start the TM6 operation TM6BEN(0x03F7C) bp0: TM6EN =1	(8) Set the TM6EN flag of TM6BEN register to "1" to start Timer 6.

As TM6OC is set, TM6BC is initialized to 0x00.

When TM6BC matches TM6OC, Timer 6 interrupt request flag is set at the next count clock and TM6BC is cleared to 0x00 to restart counting.

14.4 Time Base Timer

14.4.1 Operation

■ Time Base Timer (Time Base Timer)

Interrupt is constantly generated by a selected clock source and a interrupt generation cycle. Table:14.4.1 shows the interrupt cycle is combination with the clock source;

Table:14.4.1 Selection of Time Base Timer Interrupt Generation Cycle

Selected clock source	Interrupt generation cycle	
fppll-div	$\text{fppll-div} \times 1/2^7$	12.8 μs
	$\text{fppll-div} \times 1/2^8$	25.6 μs
	$\text{fppll-div} \times 1/2^9$	51.2 μs
	$\text{fppll-div} \times 1/2^{10}$	102.4 μs
	$\text{fppll-div} \times 1/2^{13}$	819.2 μs
	$\text{fppll-div} \times 1/2^{15}$	3.27 ms
fslow	$\text{fslow} \times 1/2^7$	3.9 ms
	$\text{fslow} \times 1/2^8$	7.8 ms
	$\text{fslow} \times 1/2^9$	15.6 ms
	$\text{fslow} \times 1/2^{10}$	31.2 ms
	$\text{fslow} \times 1/2^{13}$	250 ms
	$\text{fslow} \times 1/2^{15}$	1 s
fppll-div =10 MHz, fslow =32.768 kHz		

■ Count Timing Timer Operation (Time Base Timer)

The counter counts up with the selected clock source as a counter clock.

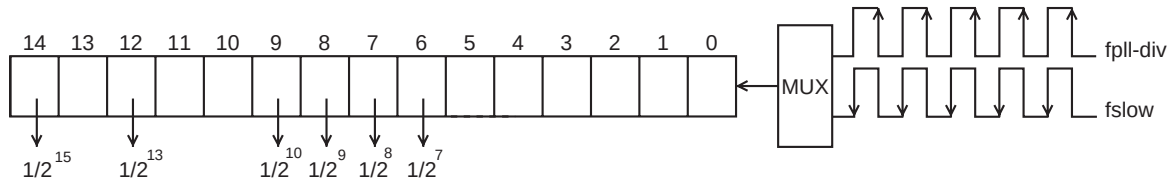


Figure:14.4.1 Count Timing of Timer Operation (Time Base Timer)

- When the selected interrupt cycle is passed, the interrupt request flag of time base interrupt control register (TBICR) is set.



An interrupt may be generated at switching of the clock source. Enable the interrupt after switching the clock source.



The initialization can be done by writing an arbitrary value to Time Base Timer clear control register (TBCLR).

14.4.2 Setup Example

■ Timer Operation Setup (Time Base Timer)

An interrupt can be generated constantly with Time Base Timer in the selected interrupt cycle. The interrupt generation cycle is $\text{fp11-div} \times 1/2^{13}$ (1 ms: $\text{fp11-div} = 8.192 \text{ MHz}$) to generate interrupts.

An example setup procedure, with a description of each step is shown below

Setup Procedure	Description
(1) Select the clock source TM6MD(0x03F7A) bp0: TM6CK0 =0	(1) Select fp11-div as a clock source by the TM6CK0 flag of Timer 6 mode register (TM6MD).
(2) Disable the interrupt TBICR(0x03FF0) bp1: TBIE =0	(2) Set the TBIE flag of TBICR register to "0" to disable the interrupt.
(3) Select the interrupt generation cycle TM6MD(0x03F7A) bp6 to 4: TM6IR2 to 0 =100	(3) Select the selected clock $\times 1/2^{13}$ as an interrupt generation cycle by the TM6IR2 to 0 flags of TM6MD register.
(4) Initialize Time Base Timer TBCLR(0x03F7B) =0x00	(4) Write value to Time Base Timer clear control register (TBCLR) to initialize Time Base Timer.
(5) Set the interrupt level TBICR(0x03FF0) bp7 to 6: TBLV1 to 0 =01	(5) Set the interrupt level by the TBLV1 to 0 flags of time base interrupt control register (TBICR). If any interrupt request flag may be already set, clear them. [Chapter 7 7.1.4 Maskable Interrupt Control Register Setup]
(6) Enable the interrupt TBICR(0x03FF0) bp1: TBIE =1	(6) Set the TBIE flag of TBICR register to "1" to enable the interrupt.
(7) Start Time Base Timer operation TM6BEN(0x03F7C) bp1: TBEN =1	(7) Set the TBEN flag of TM6BEN register to "1" to start Time Base Timer.

- When the selected interrupt generation cycle is passed, the interrupt request flag of time base interrupt control register (TBICR) is set to "1".

Chapter 15 Watchdog Timer

15

15.1 Overview

This LSI has a watchdog timer. This watchdog timer is used to detect software processing errors. It is controlled by watchdog timer control register (WDCTR). Once the timer generates an overflow, a watchdog timer interrupt (WDIRQ) is generated. After watchdog timer interrupts are generated second times in a row, software cannot execute in the intended sequence, thus forced reset is executed by the hardware.

15.1.1 Functions

Table:15.1.1 shows the functions of watchdog timer.

Table:15.1.1 Watchdog Timer Functions

Watchdog time-out cycle setup selection	$2^{16} \times$ system clock cycle $2^{18} \times$ system clock cycle $2^{20} \times$ system clock cycle
Watchdog timer enable	Disable Enable

15.1.2 Block Diagram

■ Watchdog Timer Block Diagram

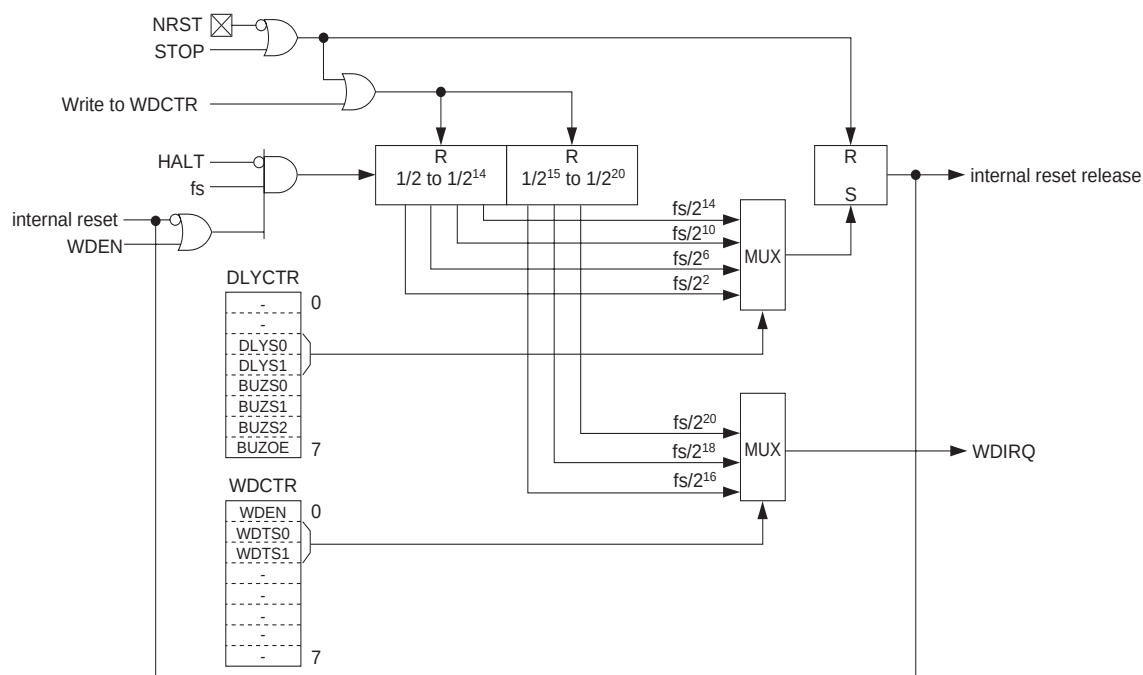


Figure:15.1.1 Block Diagram (Watchdog Timer)

Watchdog timer is also used as a timer to count oscillation stabilization wait time when returning from STOP mode or releasing the reset. Otherwise, it is used as a timer to detect software processing errors.

Watchdog timer is initialized during reset or in STOP mode. It starts counting using system clock (**fs**) as a clock source from the initial value (0x0000). The oscillation stabilization wait time is set by oscillation stabilization wait time control register (**DLYCTR**).

15.2 Control Register

The watchdog timer consists of watchdog timer control register (WDCTR).

15.2.1 Control Registers

Table:15.2.1 shows the registers that control the watchdog timer.

Table:15.2.1 Watchdog Timer Control Register Functions

Register	Address	R/W	Function	Page
WDCTR	0x03F02	R/W	Watchdog timer control register	XV-5
DLYCTR	0x03F03	R/W	Oscillation stabilization wait time control register	XV-6
PRTKEY	0x03E50	R/W	Register protect control register	V-4

15.2.2 Watchdog Timer Control Register

The watchdog timer is controlled by watchdog timer control register (WDCTR).

■ Watchdog Timer Control Register (WDCTR: 0x03F02)

bp	7	6	5	4	3	2	1	0
Flag	-	-	Reserved	Reserved	Reserved	WDTS1	WDTS0	WDEN
At reset	-	-	0	0	0	1	1	0
Access	-	-	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7 to 6	-	-
5 to 3	Reserved	Always set to "0".
2 to 1	WDTS1 WDTS0	Watchdog error detect cycle setup 00: $2^{16} \times$ system clock cycle 01: $2^{18} \times$ system clock cycle 1X: $2^{20} \times$ system clock cycle
0	WDEN	Watchdog timer enable 0: Disabled 1: Enabled



When changing above registers, always access register protect control register (PRTKEY) and enable writing.
Refer to [Chapter 5 Register Protect].



Set the watchdog timer error detect cycle to " $2^{20} \times$ system clock cycle", when the flash memory is rewritten by the command library.

■ Oscillation Stabilization Wait Time Control Register (DLYCTR: 0x03F03)

bp	7	6	5	4	3	2	1	0
Flag	BUZOE	BUZS2	BUZS1	BUZS0	DLYS1	DLYS0	-	-
At reset	0	0	0	0	0	1	-	-
Access	R/W	R/W	R/W	R/W	R/W	R/W	-	-

bp	Flag	Description
7	BUZOE	Buzzer output selection(Buzzer Function) 0: Disabled 1: Enabled
6 to 4	BUZS2 BUZS1 BUZS0	Buzzer output frequency(Buzzer Function) 000: $f_{pll-div}/2^{14}$ 001: $f_{pll-div}/2^{13}$ 010: $f_{pll-div}/2^{12}$ 011: $f_{pll-div}/2^{11}$ 100: $f_{pll-div}/2^{10}$ 101: $f_{pll-div}/2^9$ 110: $f_{slow}/2^4$ 111: $f_{slow}/2^3$
3 to 2	DLYS1 DLYS0	Oscillation stabilization wait period selection 00: $2^{14} \times$ system clock cycle 01: $2^{10} \times$ system clock cycle 10: $2^6 \times$ system clock cycle 11: $2^2 \times$ system clock cycle
1 to 0	-	-



For the oscillation stabilization wait cycle required for high-speed/low-speed oscillation by the settings of DLYS1 to DLYS0 flags, it is recommended to consult the oscillator manufacturer for determining the appropriate values.



When returning from STOP mode, more than 100 μ s of the oscillation stabilization wait cycle must be set for the internal regulator to stabilize.



Refer to [Chapter 17 Buzzer] for buzzer function.

15.3 Operation

15.3.1 Operation

Watchdog timer counts using system clock (fs) as a clock source. When the counter of watchdog timer overflows, watchdog interrupt (WDIRQ) is generated as a non-maskable interrupt (NMI). When LSI is reset, watchdog timer is disabled. However, once watchdog timer starts operating, it cannot be stopped except when LSI is reset. Use watchdog timer control register (WDCTR) to disable watchdog timer and to set the error detect cycle.

In the event that an watchdog interrupt (WDIRQ) is generated twice consecutively, the software cannot execute in the intended sequence, thus forced reset is executed by the hardware.



Once the watchdog timer starts operating, it cannot be stopped.
However, watchdog timer stops in STOP/HALT mode.

■ Usage of Watchdog Timer

When watchdog timer is used, constant clear in program is needed to prevent an overflow of the watchdog timer. In the event of software failure, the software cannot execute in the intended sequence, thus watchdog timer will overflow due to the errors.



Programming of the watchdog is generally performed in the last step of its programming.

■ How to Detect Incorrect Code Execution

The watchdog timer is executed to be cleared in the certain cycle on the correct code execution. In this LSI, watchdog timer detects errors when watchdog timer overflows.

When watchdog timer detects any error, the watchdog interrupt (WDIRQ) is generated as a non-maskable interrupt (NMI).

■ How to Clear Watchdog Timer

Watchdog timer can be cleared by writing any value to watchdog timer control register (WDCTR). It is recommended to use the bit set (BSET) instructions and etc., which do not change the value in WDCTR register.

■ Watchdog Time-out Period

The watchdog time-out period is decided by the bp2 and bp1 (WDTS1 to 0) of watchdog timer control register (WDCTR) and the system clock (fs). When watchdog timer is not cleared before it reaches the set value, it is regarded as an error and the watchdog interrupt (WDIRQ) of the non-maskable interrupt (NMI) is generated.

The system clock is decided by CPU mode control register (CPUM).

Refer to [Chapter 3 3.2.2 Oscillation Control Registers]

The watchdog time-out period is generally decided based on an execution time for main routine of program. Be sure to set a time-out period with a value longer than the value calculated by dividing the execution time of a main routine by a natural number (1, 2, . . .).

And insert a command to clear watchdog timer at a regular intervals for the same time as the natural number.

■ Watchdog Timer and CPU Mode



Generally, in the system use STOP mode, it branches whether STOP mode is used or not in the execution of program. However, in this case, the counting value of the watchdog timer is different.

Table:15.3.1 Watchdog Timer Condition in Each CPU Operation Mode

CPU OPERATION mode	Watchdog timer condition
NORMAL	Counting up with the system clock. * The counting is continued regardless of switching to NORMAL, IDLE or SLOW mode.
IDLE	
SLOW	
HALT	Counting stops. (the counting value is retained.)
STOP	Counting stops. (the counting value is cleared.) * Watchdog interrupts cannot be generated in STOP mode.
After returning from STOP mode	Counting continues after the oscillation stabilization wait time had passed and watchdog timer is enabled. Counting stops after the oscillation stabilization wait time had passed and watchdog timer is disabled.
After reset is released	Counting stops.

■ Protect function of Watchdog Timer Control Register

In order to avoid error writing by processing error, watchdog timer control register (WDCTR) has protect function.

When changing WDCTR, always access register protect control register (PRTKEY) and enable writing.

Refer to [Chapter 5 Register Protect]

15.3.2 Setup Example

The watchdog timer detects errors. On the following example, the time-out period is set to $2^{18} \times$ system clock.

An example setup procedure, with a description of each step is shown below.

■ Initial Setup Program (Watchdog Timer Initial Setup Example)

Setup Procedure	Description
(1) Set the time-out period PRTKEY(0x03E50) bp7 to 0: PRTK7 to 0 =0x44 WDCTR(0x03F02) bp2 to 1: WDTS1 to 0 =01 (2) Start the watchdog timer operation PRTKEY(0x03E50) bp7 to 0: PRTK7 to 0 =0x44 WDCTR(0x03F02) bp0: WDEN =1 (3) Set the register protect PRTKEY(0x03E50) bp7 to 0: PRTK7 to 0 =0xFF	(1) "0x44" must be written to PRTKEY to enable writing. Set the WDTS1 to 0 flag of watchdog timer control register (WDCTR) to "01" to select the time-out period to $2^{18} \times$ system clock. (2) "0x44" must be written to PRTKEY to enable writing. Set the WDEN flag of WDCTR register to "1" to start the watchdog timer operation. (3) Set any value (0xFF) to PRTKEY to set the protect again.

■ Main Routine Program (Watchdog Timer Constant Clear Setup Example)

Setup Procedure	Description
(1) Set the watchdog timer for the constant clear Writing to WDCTR(0x03F02) PRTKEY(0x03E50) bp7 to 0: PRTK7 to 0 =0x44 (c.f.) BSET (WDCTR) WDEN (bp0: WDEN=1) PRTKEY(0x03E50) bp7 to 0: PRTK7 to 0 =0xFF	(1) "0x44" must be written to PRTKEY to enable writing. Clear the watchdog timer by the cycle from $2^{18} \times$ system clock. The watchdog timer clear should be inserted in the main routine, with the same cycle, and to be the set cycle. It is recommended that the bit set (BSET) instructions and etc., which do not change WDCTR register's value. Set any value (0xFF) to PRTKEY to set the protect again.

■ Interrupt Service Routine Setup

Setup Procedure	Description
(1) Set the watchdog interrupt service routine NMICR(0x03FE1) TBNZ (NMICR), WDIR, WDPRO	(1) If the watchdog timer overflows, the non maskable interrupt is generated. Confirm that the WDIR flag of non-maskable interrupt control register (NMICR) is "1" in the interrupt processing routine and execute the appropriate processing for system.



The operation, just before the watchdog interrupt may be executed wrongly. In that case, proper operation is not guaranteed.

Chapter 16 Watchdog Timer 2

16.1 Overview

This LSI has a watchdog timer 2, separately from the watchdog timer to operate a system clock as a clock source, is used to operate an internal low-speed oscillation as a clock source.

Watchdog timer 2 is not the errors detection function synchronized with the system operation like a watchdog timer, but an asynchronous function to detect errors to use a stable internal clock. If an overflow happens without periodic counter clear during the errors detection cycle, NMI interrupt generates. Additionally, if another count clear is not be executed, a reset is executed by the hardware.

Watchdog timer 2 can change the operation object mode by setting a flash option. (Refer to [Chapter 2 2.2.6 Flash Option]). Please select the operation mode suited for the system.

16.1.1 Functions

Table:16.1.1 shows watchdog timer 2 functions.

Table:16.1.1 Watchdog Timer 2 Functions

Functions	System errors detection
Operation clock	Internal low-speed oscillation (frcs: 30 kHz)
Operation object mode	Operation target mode is selectable by users. (When setting flash option) - Operating (default setting) - Disabled in STOP mode - Disabled in HALT mode - Disabled in STOP/HALT mode - Disabled in all modes
Operation in the overflow	First overflow: NMI interrupt Second overflow: Reset by the hardware
Errors detection cycle	Selectable from the below range frcs/2 ⁴ to frcs/2 ¹⁵



Set the flash option [Chapter 2 2.2.6 Flash Option] to change the operation target mode. It is always operating if it is not setup.

16.1.2 Block Diagram

■ Watchdog Timer 2 Block Diagram

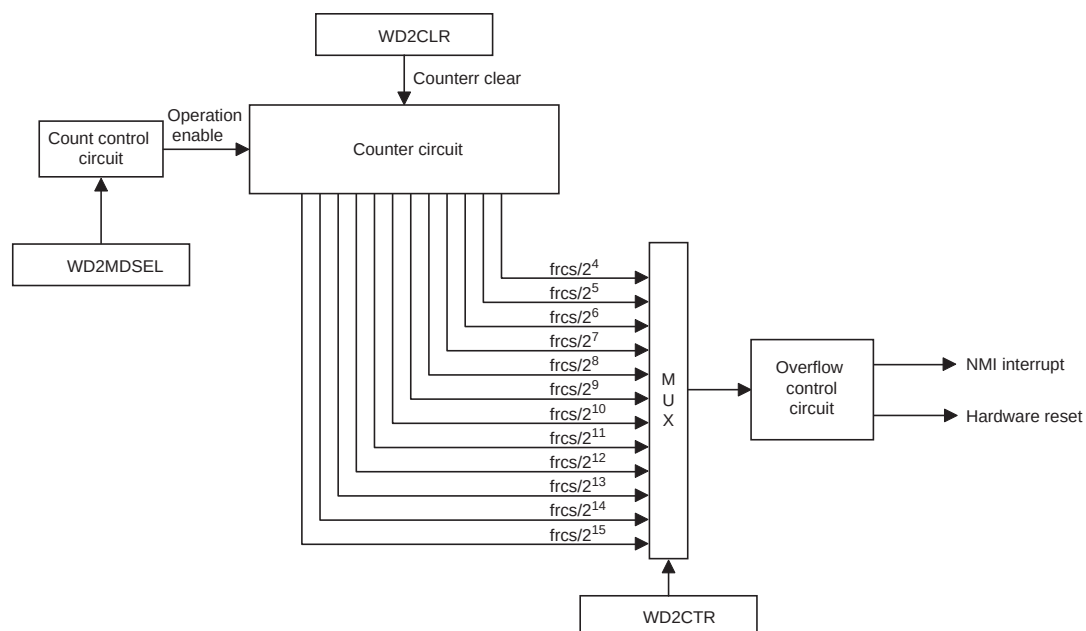


Figure:16.1.1 Block Diagram (Watchdog Timer 2)

16.2 Control Registers

16.2.1 Control Registers

Table:16.2.1 shows the registers that control watchdog timer 2.

Table:16.2.1 Watchdog Timer 2 Control Register Functions

Register	Address	R/W	Function	Page
WDCTR2	0x03F05	R/W	Watchdog timer 2 control register	XVI-5
WD2MDSEL	0x03F4C	R/W	Watchdog timer 2 operation mode setup register	XVI-6
WD2CLR	0x03F4D	R/W	Watchdog timer 2 clear register	XVI-8
PRTKEY	0x03E50	R/W	Register protect control register	V-4

R/W: Readable/Writable

16.2.2 Watchdog Timer 2 Control Register

Watchdog timer 2 control register is a register to set the errors detection cycle of watchdog timer 2.

■ Watchdog Timer 2 Control Register (WDCTR2: 0x03F05)

bp	7	6	5	4	3	2	1	0
Flag	Reserved	Reserved	-	-	WD2TS3	WD2TS2	WD2TS1	WD2TS0
At reset	0	0	-	-	1	0	1	1
Access	R/W	R/W	-	-	R/W	R/W	R/W	R/W

bp	Flag	Description
7 to 6	Reserved	Always set to "0".
5 to 4	-	-
3 to 0	WD2TS3 WD2TS2 WD2TS1 WD2TS0	Watchdog timer 2 cycle setup 0000: $\text{frcs}/2^4$ 0001: $\text{frcs}/2^5$ 0010: $\text{frcs}/2^6$ 0011: $\text{frcs}/2^7$ 0100: $\text{frcs}/2^8$ 0101: $\text{frcs}/2^9$ 0110: $\text{frcs}/2^{10}$ 0111: $\text{frcs}/2^{11}$ 1000: $\text{frcs}/2^{12}$ 1001: $\text{frcs}/2^{13}$ 1010: $\text{frcs}/2^{14}$ 1011: $\text{frcs}/2^{15}$ (default) 1100: Setting prohibited 1101: Setting prohibited 1110: Setting prohibited 1111: Setting prohibited



When changing the above register, always access register protect control register (PRTKEY) and enable writing.
Refer to [Chapter 5 Register Protect].



When changing the above register, always clear the counter in WD2CLR register. If not, unintended overflow might be generated.



Set the cycle of watchdog timer 2 to " $\text{frcs}/2^{11}$ " or more, when the flash memory is rewritten by the command library.

16.2.3 Watchdog Timer 2 Operation Mode Register

Watchdog timer 2 operation mode control register selects the operation mode of watchdog timer 2.

Writing to watchdog timer 2 operation mode register is possible only once, even if flash option 0 register (FLOP0) is set "101" to select "Setting enabled for WD2MDSEL register".

■ Watchdog Timer 2 Operation Mode Register (WD2MDSEL: 0x03F4C)

bp	7	6	5	4	3	2	1	0
Flag	WD2MD7	WD2MD6	WD2MD5	WD2MD4	WD2MD3	WD2MD2	WD2MD1	WD2MD0
At reset	1	0	1	0	1	0	1	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7 to 0	WD2MD7 to WD2MD0	Watchdog timer 2 operation mode setup 0x00: Operation deactivated *1 0x33: Count deactivated in HALT mode *2 *3 0x55: Count deactivated in STOP mode *2 *4 0xAA: Always operating (Default setup) *2 *3 0xCC: Count deactivated in HALT/STOP mode *2 *4 others: Always operating *2 *3



*1 When watchdog timer 2 operation is deactivated, the internal low-speed oscillation (frcs) can be controlled by internal low-speed oscillation control register (RCSCNT) [Chapter 3 Clock Control]
But when setting stop mode forcible deactivation is executed by the hardware.



*2 When watchdog timer 2 operation is activated, the internal low-speed oscillation (frcs) can not be controlled in internal low-speed oscillation control register (RCSCNT). Writing is disabled.
[Chapter 3 Clock Control]



*3 When watchdog timer 2 is operating in STOP mode, the internal low-speed oscillation (frcs) is always operating.



*4 When watchdog timer 2 is deactivated in STOP mode, the internal low-speed oscillation (frcs) is forcibly halted by the hardware.



The above register can not be changed without setting the flash option.
Refer to [Chapter 2 2.2.6 Flash Option].
But even if the flash option is set, writing is possible only once.



When changing the above register, always access register protect control register (PRTKEY) and enable writing.
Refer to [Chapter 5 Register Protect].

16.2.4 Watchdog Timer 2 Clear Register

Watchdog timer 2 clear register is a register to clear the counter of watchdog timer 2.

■ Watchdog Timer 2 Clear Register (WD2CLR: 0x03F4D)

bp	7	6	5	4	3	2	1	0
Flag	WD2CL7	WD2CL6	WD2CL5	WD2CL4	WD2CL3	WD2CL2	WD2CL1	WD2CL0
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7 to 0	WD2CL7 to WD2CL0	Watchdog timer 2 counter clear 0x55: Counter clear others: NMI interrupt



When changing the above register, always access register protect control register (PRTKEY) and enable writing.
Refer to [Chapter 5 Register Protect].



The above register can not clear the counter without writing "0x55". If writing any value other than that, NMI interrupt is generated.



The above register is always read out to "0x00".

16.3 Operation

16.3.1 Operation

Watchdog timer 2 counts the internal low-speed oscillation (frcs) as a clock source. If watchdog timer 2 is overflowed, NMI interrupt is generated. And if counter clear is not executed after that, the forcible reset is executed by the hardware.

Watchdog timer 2 is always operating on start-up. The following operation mode can be changed by watchdog timer 2 operation mode setup register (WD2MDSEL) if the flash option is set. Each operation availability is selectable in continuous operating, deactivating, or HALT mode/STOP mode.

The errors detection cycle of watchdog timer 2 is set in watchdog timer 2 control register (WD2CTR) and the clear of the counter is set in watchdog timer 2 clear register (WD2CLR).

All control registers of watchdog timer 2 include register protect function. (Refer to [Chapter 5 Register Protect]) The specified value should be written in register protect control register (PRTKEY) and the writing of the control register should be enabled.

■ How to Detect Errors

Watchdog timer 2 is operated with expectation of a constant frequency when the program normally operates. When the program operates incorrectly, a constant frequency of clear does not generate and the counter overflows.

In the overflow, NMI interrupt generates at first. If this program is in the condition to be able to operate the program, the system control can be transferred to the safe state in the interrupt routine. If this LSI is in the incorrect state not to control program, errors state continues. In this case, the forcible reset should be executed by the hardware in the second overflow.

■ How to Clear the Watchdog Timer 2

The clear of watchdog timer 2 can be executed by writing in watchdog timer 2 clear register (WD2CLR). In this case, always write "0x55". If any value other than that is written, NMI interrupt generates.

■ Errors Detection Cycle

The errors detection cycle is determined by watchdog timer 2 control register (WD2CTR). When the clear of the counter is not executed until this setting value, the state is considered incorrect state and the forcible reset is executed by the hardware.

The errors detection cycle is normally determined by the execution time of program main routine. We recommend the longer errors detection cycles than the execution time divided by arbitrary counting number.

■ Watchdog Timer 2 Function and the Operation Mode

If the flash option (Refer to [Chapter 2 2.2.6 Flash Option]) is set, the operation mode of watchdog timer 2 function is selectable from the following.

1. Continuous operation mode (Default mode)

Regardless of CPU state, continues counting.

Full attention is required to use the mode (HALT/STOP mode) keeping CPU to wait.

2. All disabled mode

Count is deactivated.

As the operation mode is in the continuous operation mode on start-up, set watchdog timer 2 operation mode setup register (WD2MDSEL).

3. HALT disabled mode

Count is disabled only in the HALT mode. (The count value is not initialized).

As the operation mode is in the continuous operation mode on start-up, set watchdog timer 2 operation mode setup register (WD2MDSEL).

Full attention is required to use STOP mode.

4. STOP disabled mode

Count is disabled only in the STOP mode. (The count value is not initialized).

As the operation mode is in the continuous operation mode on start-up, set in watchdog timer 2 operation mode setup register (WD2MDSEL).

Full attention is required to use HALT mode.

5. HALT/STOP disabled mode

Count is disabled in keeping CPU to wait (HALT/STOP mode). (The count value is not initialized).

As the operation mode is in the continuous operation mode on start-up, set in watchdog timer 2 operation mode setup register (WD2MDSEL).

16.3.2 Setup Example

An example setup procedure of watchdog timer 2 function is shown below.

Flash option: Valid

The operation mode: HALT/STOP deactivated mode

Errors detection cycles: $\text{frcs}/2^{11}$

Setup Procedure	Description
(1) The setup of the operation mode PRTKEY (0x3E50) bp7 to 0: PRTK7 to 0 =0x88 WD2MDSEL(0x3F4C) bp7 to 0: WD2MD7 to 0 =0xCC	(1) HALT/STOP disabled mode is set. Set "0x88" to register protect control register (PRTKEY) and enable the writing of watchdog timer 2 operation mode setup register (WD2MDSEL). Set "0xCC" to WD2MDSEL and HALT/STOP disabled mode.
(2) Counter clear PRTKEY (0x3E50) bp7 to 0: PRTK7 to 0 =0xAA WD2CLR(0x3F4D) bp7 to 0: WD2CL7 to 0 =0x55	(2) Always clear the counter if changing the errors detection cycle. Set "0xAA" to register protect control register (PRTKEY) and enable the writing of watchdog timer 2 clear register (WD2CLR). Set "0x55" to WD2CLR and clear the counter.
(3) Setup the errors detection cycle PRTKEY (0x3E50) bp7 to 0: PRTK7 to 0 =0x66 WD2CTR(0x3F05) bp3 to 0: WD2TS3 to 0 =0x7	(3) Set the errors detection cycle to $\text{frcs}/2^{11}$. Set "0x66" to register protect control register (PRTKEY) and enable the writing of watchdog timer 2 control register (WD2CTR). Set "0x07" to WD2CTR and to $\text{frcs}/2^{11}$.
(4) Setup register protect PRTKEY (0x3E50) bp7 to 0: PRTK7 to 0 =0xFF	(4) Set register protect. Set arbitrate value "0xFF" to register protect control register (PRTKEY) and again set protect.
(5) Periodic counter clear PRTKEY (0x3E50) bp7 to 0: PRTK7 to 0 =0xAA WD2CLR(0x3F4D) bp7 to 0: WD2CL7 to 0 =0x55 PRTKEY(0x3E50) bp7 to 0: PRTK7 to 0 =0xFF	(5) The counter is periodically cleared at a constant frequency in main routine. Set "0xAA" to register protect control register (PRTKEY) and enable the writing of watchdog timer 2 clear register (WD2CLR). Set "0x55" to WD2CLR and clear the counter. Clear not to pass the errors detection cycle. Set arbitrate value "0xFF" to register protect control register (PRTKEY) and again set protect.



When the flash option is not set, the operation mode of watchdog timer 2 does not change even if setting the above setting.
(The operation is in the continuous operation mode)



The programming of watchdog timer 2 function is executed in the last stage of the program debug.

Chapter 17 Buzzer

17.1 Overview

This LSI has a buzzer. It can output the square wave that multiply by $1/2^9$ to $1/2^{14}$ of the high frequency oscillation clock, or by $1/2^3$ to $1/2^4$ of the low frequency oscillation clock.

17.1.1 Functions

Table:17.1.1 shows buzzer functions.

Table:17.1.1 Buzzer Functions

output selection	Disable buzzer output Enable buzzer output
buzzer reverse output	Polarity reverse output
output pin selection	P56, P57 output (P56 is the reverse output of P57)
Buzzer output frequency selection	$fp\text{ll-div}/2^{14}$ $fp\text{ll-div}/2^{13}$ $fp\text{ll-div}/2^{12}$ $fp\text{ll-div}/2^{11}$ $fp\text{ll-div}/2^{10}$ $fp\text{ll-div}/2^9$ $fs\text{low}/2^4$ $fs\text{low}/2^3$
Oscillation stabilization wait cycle selection	$fs/2^{14}$ $fs/2^{10}$ $fs/2^6 *1$ $fs/2^2 *1$

*1: Use this function at slow-speed operation (SLOW mode), not at high-speed operation (NORMAL mode).



Immediately after the BUZOE flag is set to "0", the output of both BUZZER and NBUZZER is turned to "Low".



To change the frequency of $fp\text{ll-div}$ by bp7 to 4 of OSCCNT register, stop buzzer function beforehand.

17.1.2 Block Diagram

■ Buzzer Block Diagram

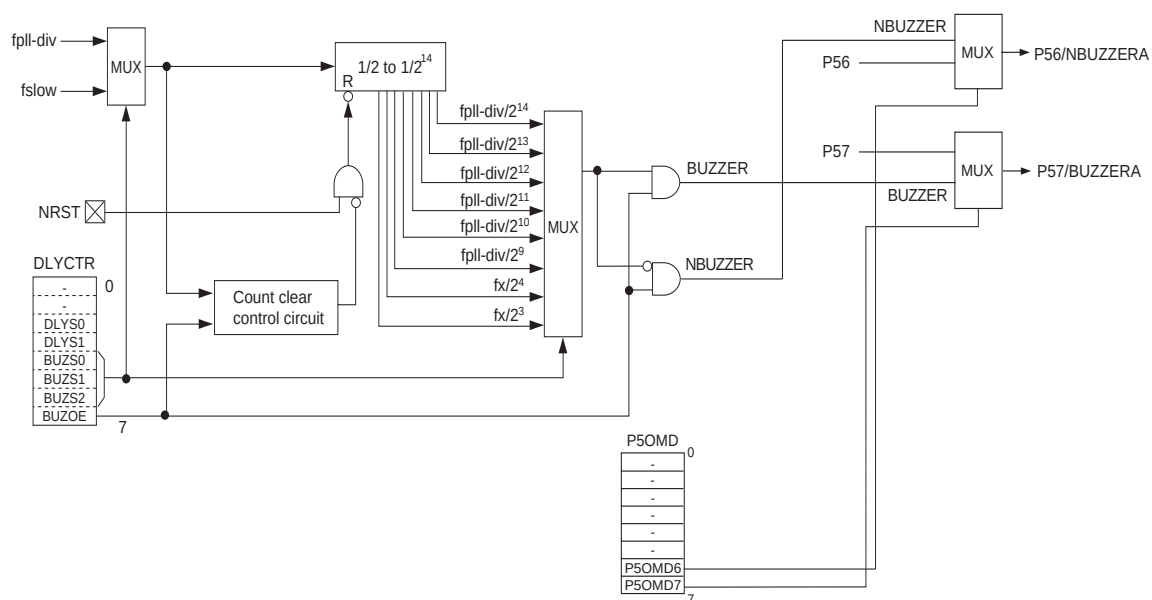


Figure:17.1.1 Buzzer Block Diagram

17.2 Control Registers

Buzzer function is controlled by upper 4 bits of oscillation stabilization wait time control register (DLYCTR).

17.2.1 Registers

Table:17.2.1 shows oscillation stabilization wait time control register.

Table:17.2.1 Buzzer Control Register

Register	Address	R/W	Function	Page
DLYCTR	0x03F03	R/W	Oscillation Stabilization Wait Time Control Register	XVII-5
P5DIR	0x03F95	R/W	Port 5 direction control register	VIII-94
P5OMD	0x03EB5	R/W	Port 5 output mode register	XVII-4

17.2.2 Buzzer Control Registers

■ Port 5 Output Mode Register (P5OMD: 0x03EB5)

bp	7	6	5	4	3	2	1	0
Flag	P5OMD7	P5OMD6	-	-	-	-	-	-
At reset	0	0	-	-	-	-	-	-
Access	R/W	R/W	-	-	-	-	-	-

bp	Flag	Description
7	P5OMD7	Buzzer output selection 0: P57 1: BUZZERA
6	P5OMD6	Buzzer reverse output selection 0: P56 1: NBUZZERA
5 to 0	-	-

■ Oscillation Stabilization Wait Time Control Register (DLYCTR: 0x03F03)

bp	7	6	5	4	3	2	1	0
Flag	BUZOE	BUZS2	BUZS1	BUZS0	DLYS1	DLYS0	-	-
At reset	0	0	0	0	0	1	-	-
Access	R/W	R/W	R/W	R/W	R/W	R/W	-	-

bp	Flag	Description
7	BUZOE	Buzzer output selection(Buzzer Function) 0: Disabled 1: Enabled
6 to 4	BUZS2 BUZS1 BUZS0	Buzzer output frequency(Buzzer Function) 000: $f_{pll-div}/2^{14}$ 001: $f_{pll-div}/2^{13}$ 010: $f_{pll-div}/2^{12}$ 011: $f_{pll-div}/2^{11}$ 100: $f_{pll-div}/2^{10}$ 101: $f_{pll-div}/2^9$ 110: $f_{slow}/2^4$ 111: $f_{slow}/2^3$
3 to 2	DLYS1 DLYS0	Oscillation stabilization wait period selection (Watchdog Timer Function) 00: $2^{14} \times$ system clock cycle 01: $2^{10} \times$ system clock cycle 10: $2^6 \times$ system clock cycle 11: $2^2 \times$ system clock cycle
1 to 0	-	-



Do not set BUZOE flag and BUZS2 to BUZS0 flags simultaneously.



DLYS1 to DLYS0 flags are for watchdog timer function.
Refer to [Chapter 15 Watchdog Timer] for details.

17.3 Operation

17.3.1 Operation

■ Buzzer

Buzzer outputs square wave, with frequency $1/2^9$ to $1/2^{14}$ of the high oscillation clock (fppll-div), or $1/2^3$ to $1/2^4$ of the low oscillation clock (fslow). The BUZS2 to BUZS0 flags of oscillation stabilization wait time control register (DLYCTR) set the frequency of buzzer output. The BUZOE flag of DLYCTR register enables/disables buzzer output.

■ Buzzer Output Frequency

The frequency of buzzer output is decided by the frequency of the high oscillation clock (fppll-div) or the low oscillation clock (fslow) and the BUZS2 to BUZS0 flags of DLYCTR register.

Table:17.3.1 Buzzer Output Frequency

fppll-div	fslow	BUZS2	BUZS1	BUZS0	Buzzer output frequency
10 MHz	-	0	1	0	2.44 kHz
10 MHz	-	0	1	1	4.88 kHz
4 MHz	-	0	1	1	1.95 kHz
4 MHz	-	1	0	0	3.91 kHz
-	32 kHz	1	1	0	2 kHz
-	32 kHz	1	1	1	4 kHz



Immediately after the BUZOE flag is set to "0", the output of both BUZZER and NBUZZER is turned to "Low".

17.3.2 Setup Example

■ Setup Example

Buzzer outputs square wave of 2.44 kHz from pin P57. This function uses 10 MHz as the high oscillation clock (fppll-div).

An example of setup procedure, with a description of each step is shown below.

Setup Procedure	Description
(1) Set buzzer frequency DLYCTR (0x03F03) bp6 to 4: BUZS2 to 0 =010	(1) Set BUZS2 to BUZS0 flags of oscillation stabilization wait time control register (DLYCTR) to "010" to select fppll-div/2 ¹² to buzzer frequency. When the high oscillation clock fppll-div is 10 MHz, buzzer output frequency is 2.44 kHz.
(2) Set pin P57 P5OMD (0x03EB5) bp7: P5OMD7 =1 P5DIR (0x03F95) bp7: P5DIR7 =1	(2) Set P5OMD7 flag of Port 5 output mode register (P5OMD) to "1" to set P57 pin to a special function pin. Set P5DIR7 flag of Port 5 direction control register (P5DIR) to "1" to set output mode; then, "Low-level" is output from the pin P57.
(3) Set Buzzer output ON DLYCTR (0x03F03) bp7: BUZOE =1	(3) Set the BUZSE flag of DLYCTR register to "1" to output the square wave of buzzer output frequency set by the P57 pin.
(4) Set Buzzer output OFF DLYCTR (0x03F03) bp7: BUZOE =0	(4) Set the BUZOE flag of DLYCTR register to "0" to disable buzzer, and P57 pin outputs "Low-level".



Be sure to set buzzer output to "enable" after setting buzzer frequency. When low-speed clock (fslow) dividing is selected for buzzer output frequency, buzzer dividing counter is only cleared at one low-speed oscillation clock cycle after buzzer output is disabled. Wait for one low-speed oscillation clock cycle or longer before enabling buzzer output again.

Chapter 18 Serial Interface

18.1 Overview

MN101EF76K and MN101EF56K has 5 serial interfaces (Serial Interface 0, 1, 2, 3 and 4).
MN101EF57G has 4 serial interfaces (Serial Interface 0, 1, 2 and 4).

Serial Interface 0, 1, 2 and 3 can be used for clock synchronous and UART (duplex) communication.
Serial Interface 4 can be used for clock synchronous and IIC (multi-master) communication.

Table:18.1.1 Serial Interface Communication Types

	Serial Interface 0	Serial Interface 1	Serial Interface 2	Serial Interface 3	Serial Interface 4
Clock synchronous	√	√	√	√	√
UART(duplex)	√	√	√	√	-
Multi master IIC	-	-	-	-	√

Table:18.1.2 and Table:18.1.3 show the pins used for serial interfaces. Serial interfaces can switch pins to A type or B type. For A-type pin, "A" is added to the end of its name. For B-type pin, "B" is added to the end of its name.

Table:18.1.2 Serial Interface Pins (MN101EF76K/MN101EF56K)

MN101EF76K/MN101EF56K		Serial Interface 0		Serial Interface 1		Serial Interface 2		Serial Interface 3		Serial Interface 4	
Pin switching function		√		√		√		√		√	
Clock synchronous	Data I/O pin	SBO0A (P03)	SBO0B (P43)	SBO1A (P50)	SBO1B (P75)	SBO2A (P65)	SBO2B (P30)	SBO3A (P05)	SBO3B (P40)	SBO4A (P71)	SBO4B (P33)
	Data input pin	SBI0A (P02)	SBI0B (P44)	SBI1A (P51)	SBI1B (P76)	SBI2A (P66)	SBI2B (P31)	SBI3A (P06)	SBI3B (P41)	SBI4A (P70)	SBI4B (P35)
	Clock I/O pin	SBT0A (P04)	SBT0B (P45)	SBT1A (P52)	SBT1B (P77)	SBT2A (P67)	SBT2B (P32)	SBT3A (P07)	SBT3B (P42)	SBT4A (P72)	SBT4B (P34)
UART(duplex)	Data I/O pin	TXD0A (P03)	TXD0B (P43)	TXD1A (P50)	TXD1B (P75)	TXD2A (P65)	TXD2B (P30)	TXD3A (P05)	TXD3B (P40)	-	
	Data input pin	RXD0A (P02)	RXD0B (P44)	RXD1A (P51)	RXD1B (P76)	RXD2A (P66)	RXD2B (P31)	RXD3A (P06)	RXD3B (P41)	-	
Multi master IIC	Data I/O pin	-		-		-		-		SDA4A (P71)	SDA4B (P33)
	Clock I/O pin	-		-		-		-		SCL4A (P72)	SCL4B (P34)

Table:18.1.3 Serial Interface Pins (MN101EF57G)

MN101EF57G		Serial Interface 0		Serial Interface 1		Serial Interface 2		Serial Interface 4	
Pin switching function		√		√		√		√	
Clock synchronous	Data I/O pin	SBO0A (P03)	SBO0B (P43)	SBO1A (P50)	SBO1B (P75)	SBO2A (P65)	SBO4A (P71)	SBO4B (P33)	
	Data input pin	SBI0A (P02)	SBI0B (P44)	SBI1A (P51)	SBI1B (P76)	SBI2A (P66)	SBI4A (P70)	SBI4B (P35)	
	Clock I/O pin	SBT0A (P04)	SBT0B (P45)	SBT1A (P52)	SBT1B (P77)	SBT2A (P67)	SBT4A (P72)	SBT4B (P34)	
UART(duplex)	Data I/O pin	TXD0A (P03)	TXD0B (P43)	TXD1A (P50)	TXD1B (P75)	TXD2A (P65)	-		
	Data input pin	RXD0A (P02)	RXD0B (P44)	RXD1A (P51)	RXD1B (P76)	RXD2A (P66)	-		
Multi master IIC	Data I/O pin	-		-		-	SDA4A (P71)	SDA4B (P33)	
	Clock I/O pin	-		-		-	SCL4A (P72)	SCL4B (P34)	



In this LSI manual, if there is not much difference in functions for pins A or B, "A" and "B" of the pin names are omitted.



Be sure to change the frequency of fpll-div by bp7 to 4 of OSCCNT register after stopping the serial interface functions.

18.1.1 Functions

Table:18.1.4 shows clock synchronous serial interface functions.

Table:18.1.5 shows UART (duplex) serial interface functions.

Table:18.1.6 shows single master IIC interface functions.

Table:18.1.4 Clock Synchronous Serial Interface Functions

	Serial Interface 0	Serial Interface 1	Serial Interface 2	Serial Interface 3	Serial Interface 4
Interrupt	SC0TIRQ SC0RIRQ	SC1TIRQ SC1RIRQ	SC2TIRQ	SC3TIRQ	SC4IRQ
PIns to be used	SBO0 SBI0 SBT0	SBO1 SBI1 SBT1	SBO2 SBI2 SBT2	SBO3 SBI3 SBT3	SBO4 SBI4 SBT4
3-wire system	√	√	√	√	√
2-wire system	√ (SBO0,SBT0)	√ (SBO1,SBT1)	√ (SBO2,SBT2)	√ (SBO3,SBT3)	√ (SBO4,SBT4)
Specification of transfer bit count (1 to 8 bits)	√	√	√	√	√
Selection of start condition	√	√	√	√	√
Specification of the first transfer bit	√	√	√	√	√
Input edge/output edge	√	√	√	√	√
SBO output control after transmission (High/Low/Last data hold)	√	√	√	√	√
Communication function in STANDBY mode (only slave reception is available)	√	√	√	√	√
Continuous operation (with ATC1)	√	√	√	√	√
Selection of transfer clock dividing	√	√	√	√	-
Selection of transfer clock division ratio	No dividing Divide by 8 Divide by16	No dividing Divide by 8 Divide by16	No dividing Divide by 8 Divide by16	No dividing Divide by 8 Divide by16	-
Clock source	fppll-div/2 fppll-div/4 fppll-div/16 fppll-div/64 fs/2 fs/4 External clock Timer 0 to 4 or A output divided by 1, 2, 4, 8 or 16	fppll-div/2 fppll-div/4 fppll-div/16 fppll-div/64 fs/2 fs/4 External clock Timer 0 to 4 or A output divided by 1, 2, 4, 8 or 16	fppll-div/2 fppll-div/4 fppll-div/16 fppll-div/64 fs/2 fs/4 External clock Timer 0 to 4 or A output divided by 1, 2, 4, 8 or 16	fppll-div/2 fppll-div/4 fppll-div/16 fppll-div/64 fs/2 fs/4 External clock Timer 0 to 4 or A output divided by 1, 2, 4, 8 or 16	fppll-div/2 fppll-div/4 fppll-div/16 fppll-div/32 fs/2 fs/4 External clock Timer 0 to 4 or A output divided by 1, 2, 4, 8 or 16
Maximum transfer rate	5.0 MHz	5.0 MHz	5.0 MHz	5.0 MHz	5.0 MHz
fppll-div: high-speed clock for peripheral functions fs: system clock					



Set the transfer rate which is slower than the system clock (fs).

Table:18.1.5 UART (duplex) Serial Interface Functions

		Serial Interface0	Serial Interface1	Serial Interface2	Serial Interface3
Interrupt	(at transmission completion) (at reception completion)	SC0TIRQ SC0RIRQ	SC1TIRQ SC1RIRQ	SC2TIRQ SC2RIRQ	SC3TIRQ SC3RIRQ
Pins to be used		TXD0 RXD0	TXD1 RXD1	TXD2 RXD2	TXD3 RXD3
2-wire communication		√			
1-wire communication		√ (TXD0)	√ (TXD1)	√ (TXD2)	√ (TXD3)
Specification of transfer bit count/frame selection		7 bits + 1STOP 7 bits + 2STOP 8 bits + 1STOP 8 bits + 2STOP			
Selection of parity bit		√			
Parity bit control		0 parity 1 parity Odd parity Even parity			
Specification of the first transfer bit		√			
Selection of transfer clock dividing ratio		Divide by 8/ divide by16			
Continuous operation (with ATC1)		√			
Clock source		fppll-div/2 fppll-div/4 fppll-div/16 fppll-div/64 fs/2 fs/4 Timer 0 to 4 or A output divided by 1, 2, 4, 8 or 16			
Maximum transfer rate		300 kbps			
fppll-div: high-speed clock for peripheral functions fs: system clock In UART communication, a transfer clock is obtained by dividing a clock source internally.					



Set the transfer rate which is slower than the system clock (fs).

Table:18.1.6 Multi Master IIC Serial Interface Functions

		Serial Interface 4
Interrupt		SC4IRQ (communication completion interrupt) SC4STPCIRQ (stop condition detection interrupt)
Pins to be used		SDA4A/SDA4B, SCL4A/SCL4B
Communication mode	Master transmission	√
	Master reception	√
	Slave transmission	√
	Slave reception	√
Transfer format	Addressing format	√
	Free data format	Available only in master communication
Address format	7-bit address	√
	10-bit address	Available only in master transmission/reception or slave reception
	General call	√
Communication format	Standard mode (100 bps)	√
	High-speed mode (400 bps)	√
Specification of first transfer bit		1 to 8 bits (in master communication) 8 bits (in slave communication)
Specification of transfer first bit		√
ACK bit selection		Available only in master communication
ACK bit level selection		√
Continuous operation (with ATC1)		√
Maximum transfer rate		Normal mode: 100 kHz High-speed mode: 400 kHz
Clock sources		fpll-div/2 fpll-div/4 fpll-div/16 fpll-div/32 fs/2 fs/4 Timer 0 to 4 or A output divided by 1, 2, 4, 8 or 16
fpll-div: high-speed clock for peripheral functions fs: system clock In IIC communication, a transfer clock is obtained by dividing a clock by 8.		



Set the transfer rate which is slower than the system clock (fs).

18.1.2 Block Diagram

Serial Interface 0 Block Diagram (MN101EF76K/MN101EF56K)

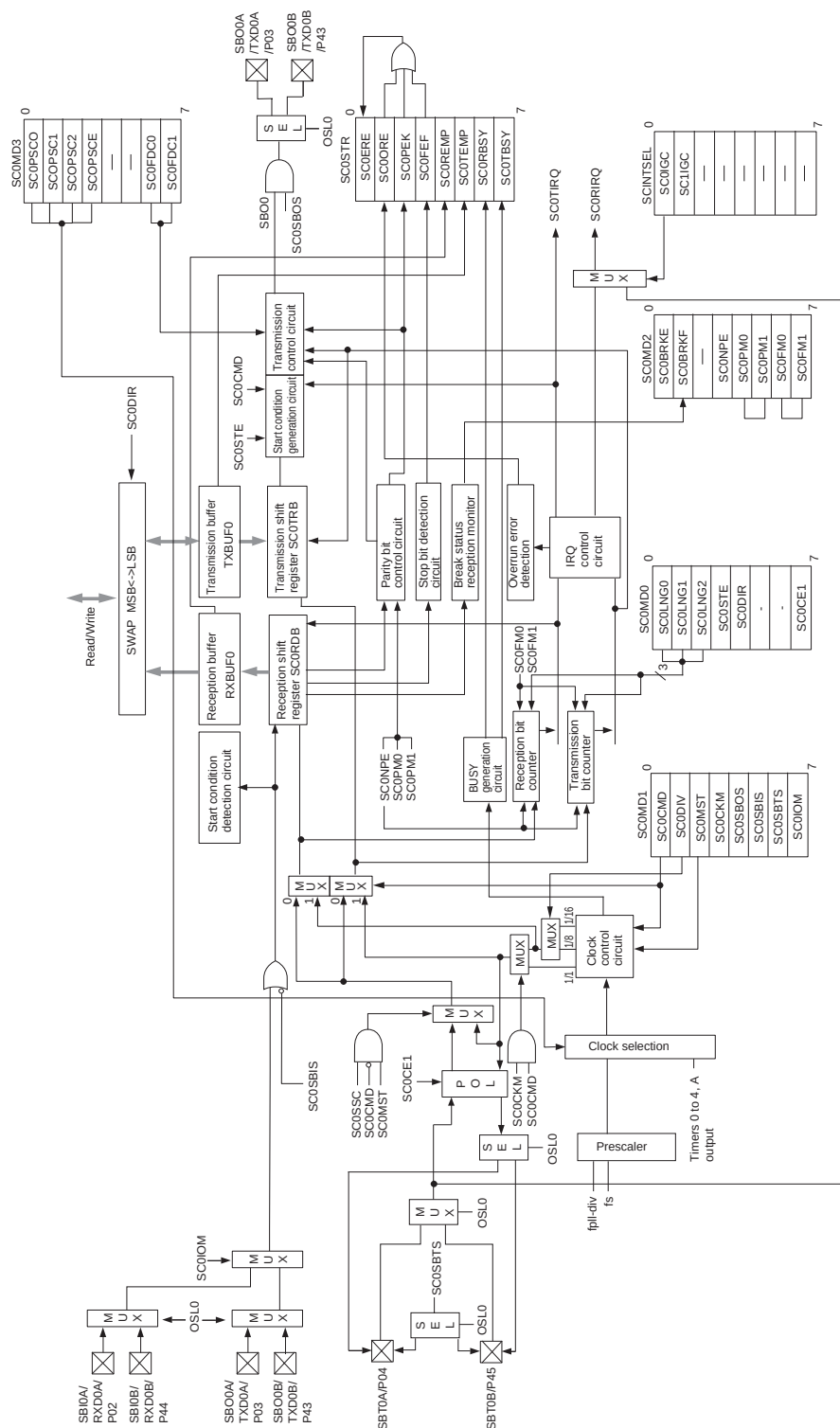


Figure:18.1.1 Serial Interface 0 Block Diagram (MN101EF76K/MN101EF56K)

■ Serial Interface 0 Block Diagram (MN101EF57G)

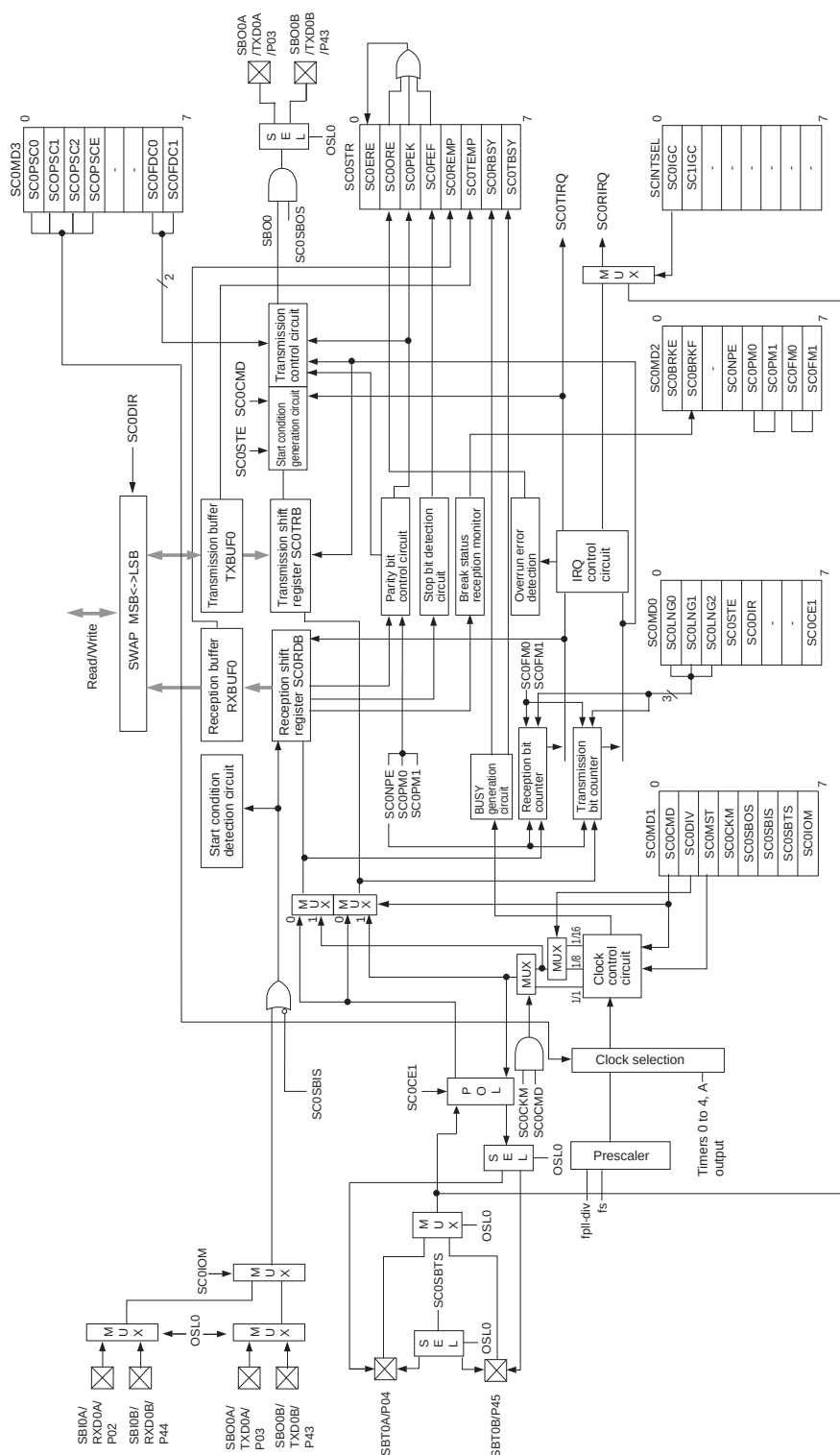


Figure:18.1.2 Serial Interface 0 Block Diagram (MN101EF57G)

■ Serial Interface 1 Block Diagram (MN101EF76K/MN101EF56K)

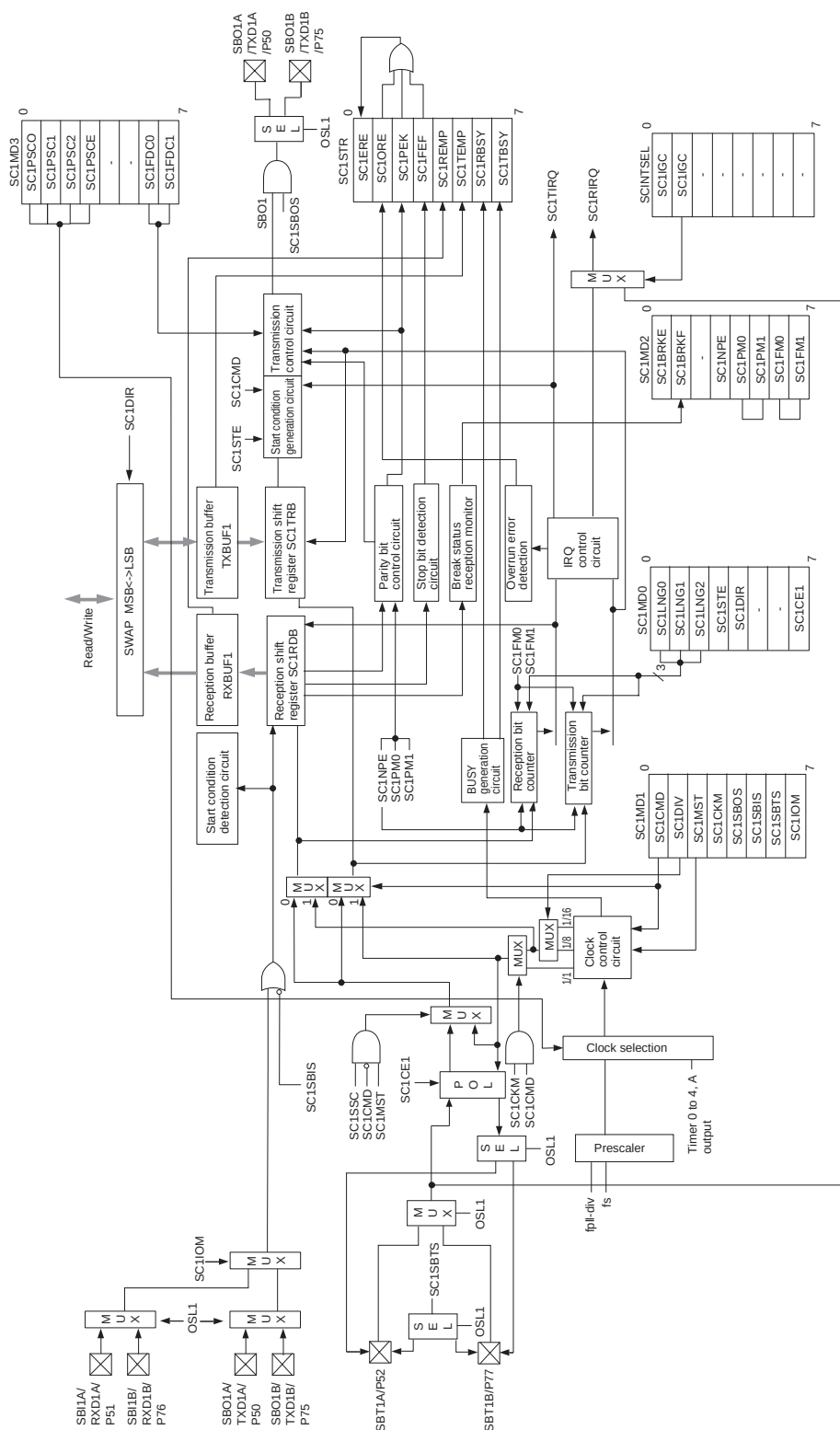


Figure:18.1.3 Serial Interface 1 Block Diagram (MN101EF76K/MN101EF56K)

■ Serial Interface 1 Block Diagram (MN101EF57G)

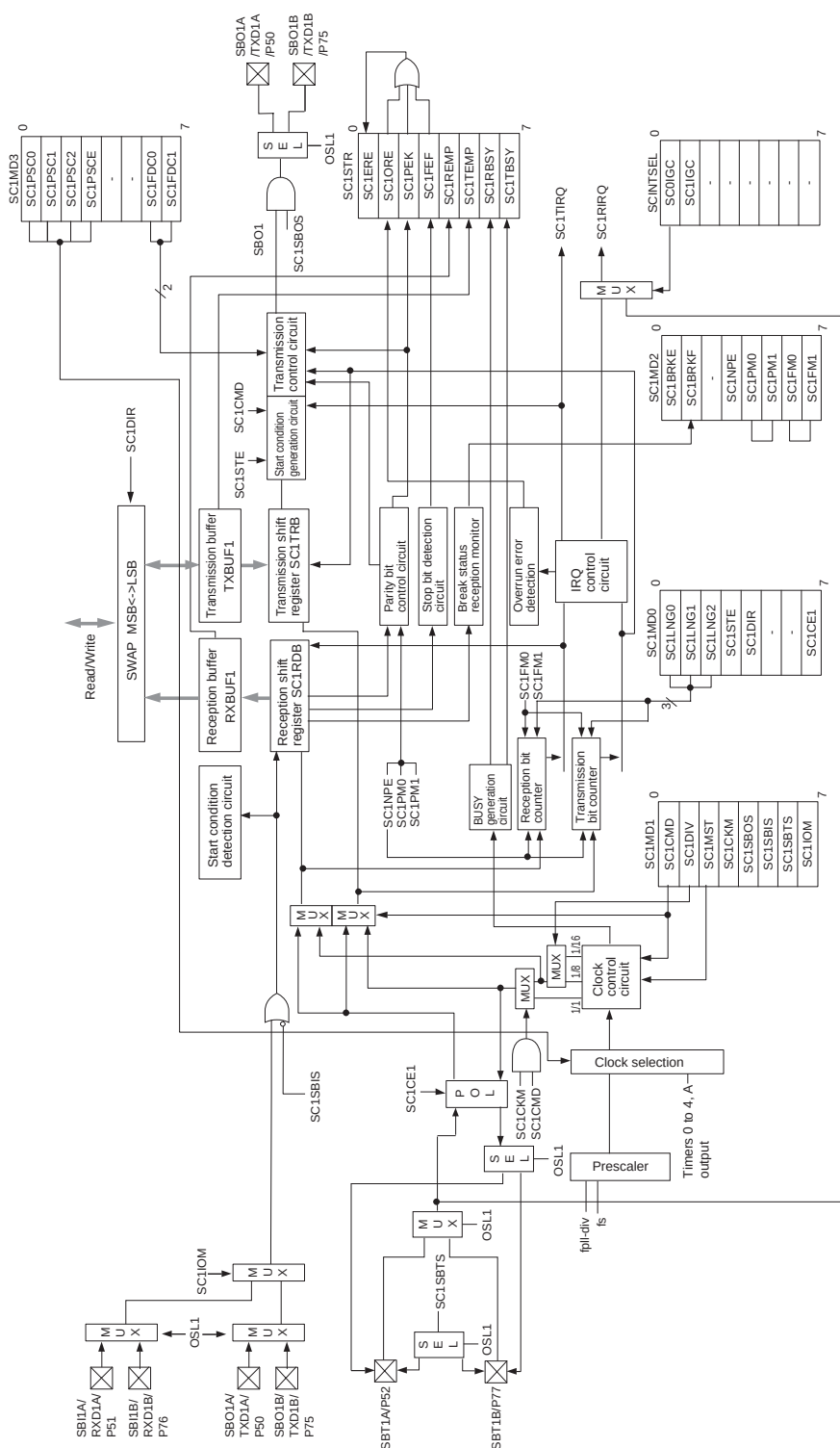


Figure:18.1.4 Serial Interface 1 Block Diagram (MN101EF57G)

■ Serial Interface 2 Block Diagram (MN101EF76K/MN101EF56K)

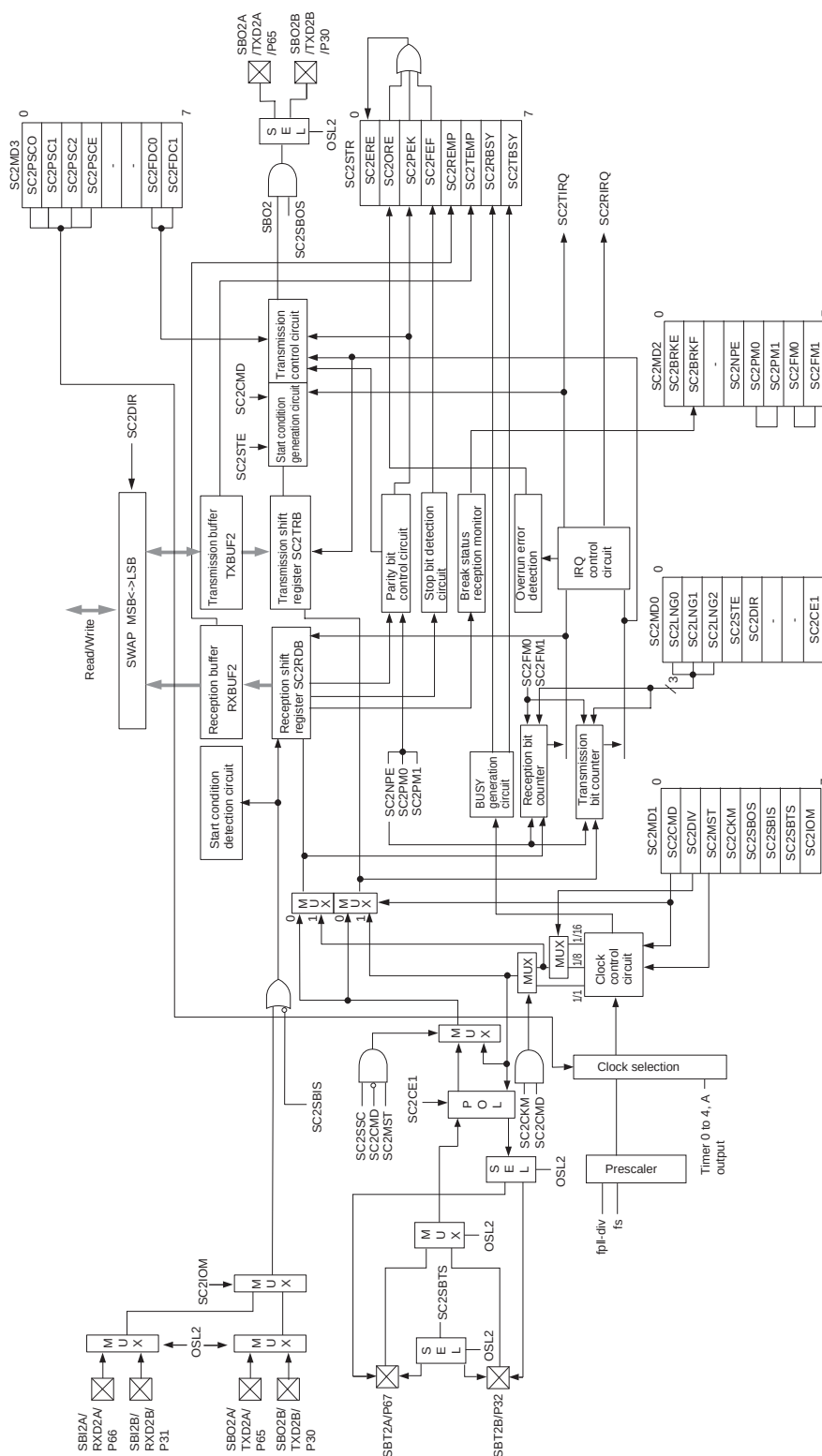


Figure:18.1.5 Serial Interface 2 Block Diagram (MN101EF76K/MN101EF56K)

■ Serial Interface 2 Block Diagram (MN101EF57G)

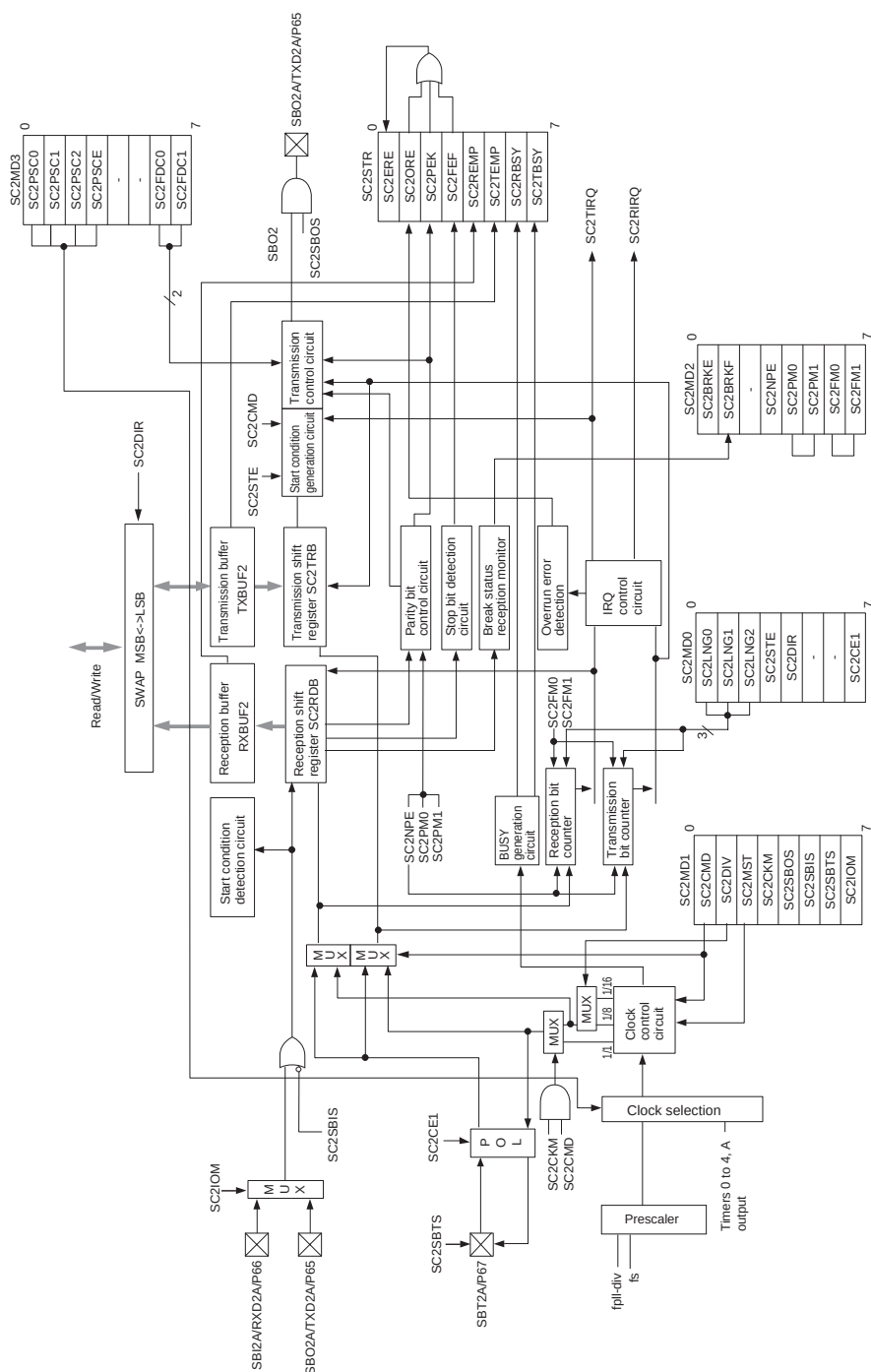


Figure:18.1.6 Serial Interface 2 Block Diagram (MN101EF57G)

■ Serial Interface 3 Block Diagram (MN101EF76K/MN101EF56K)

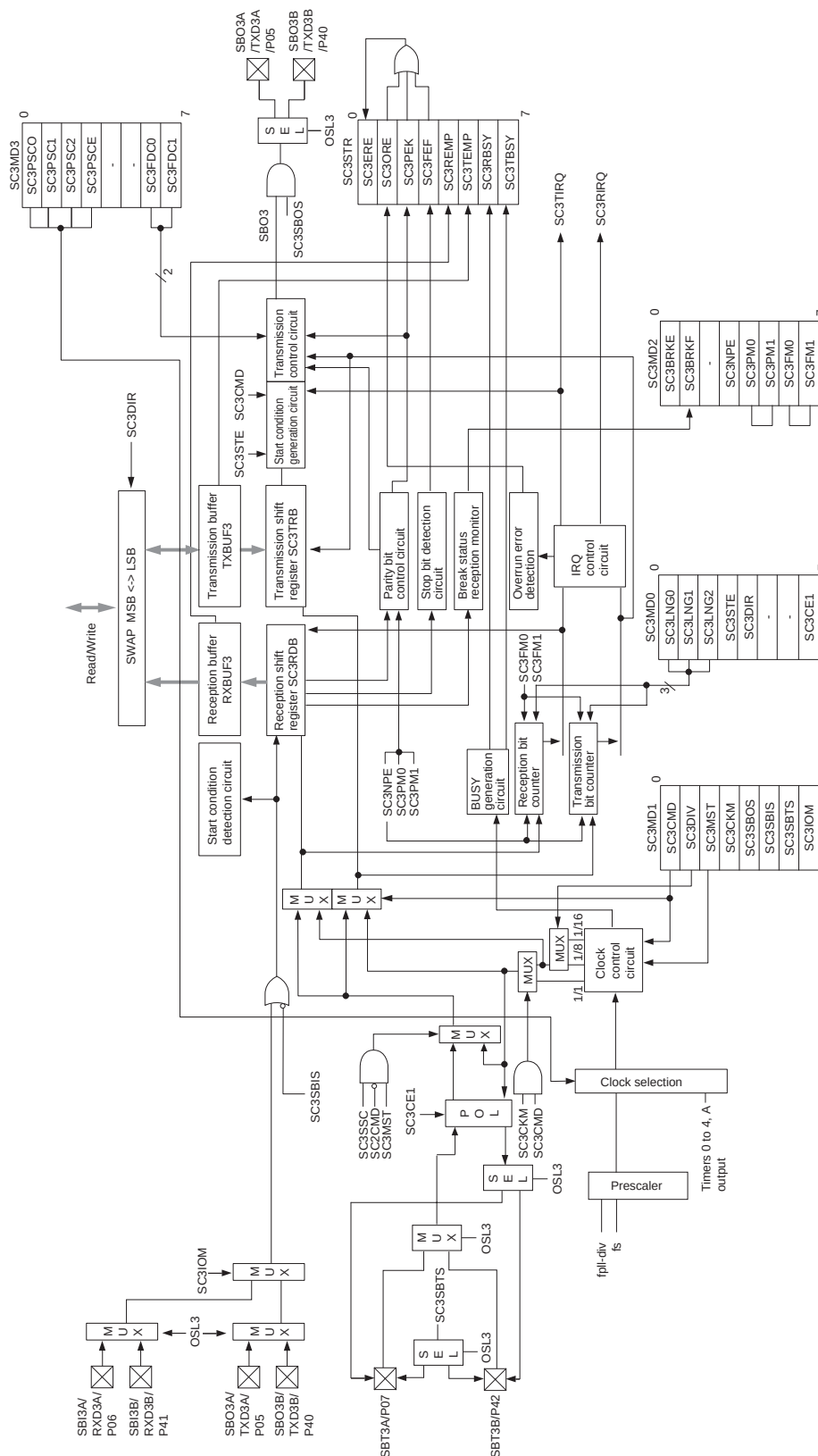


Figure:18.1.7 Serial Interface 3 Block Diagram (MN101EF76K/MN101EF56K)

■ Serial Interface 4 Block Diagram (MN101EF76K/MN101EF56K)

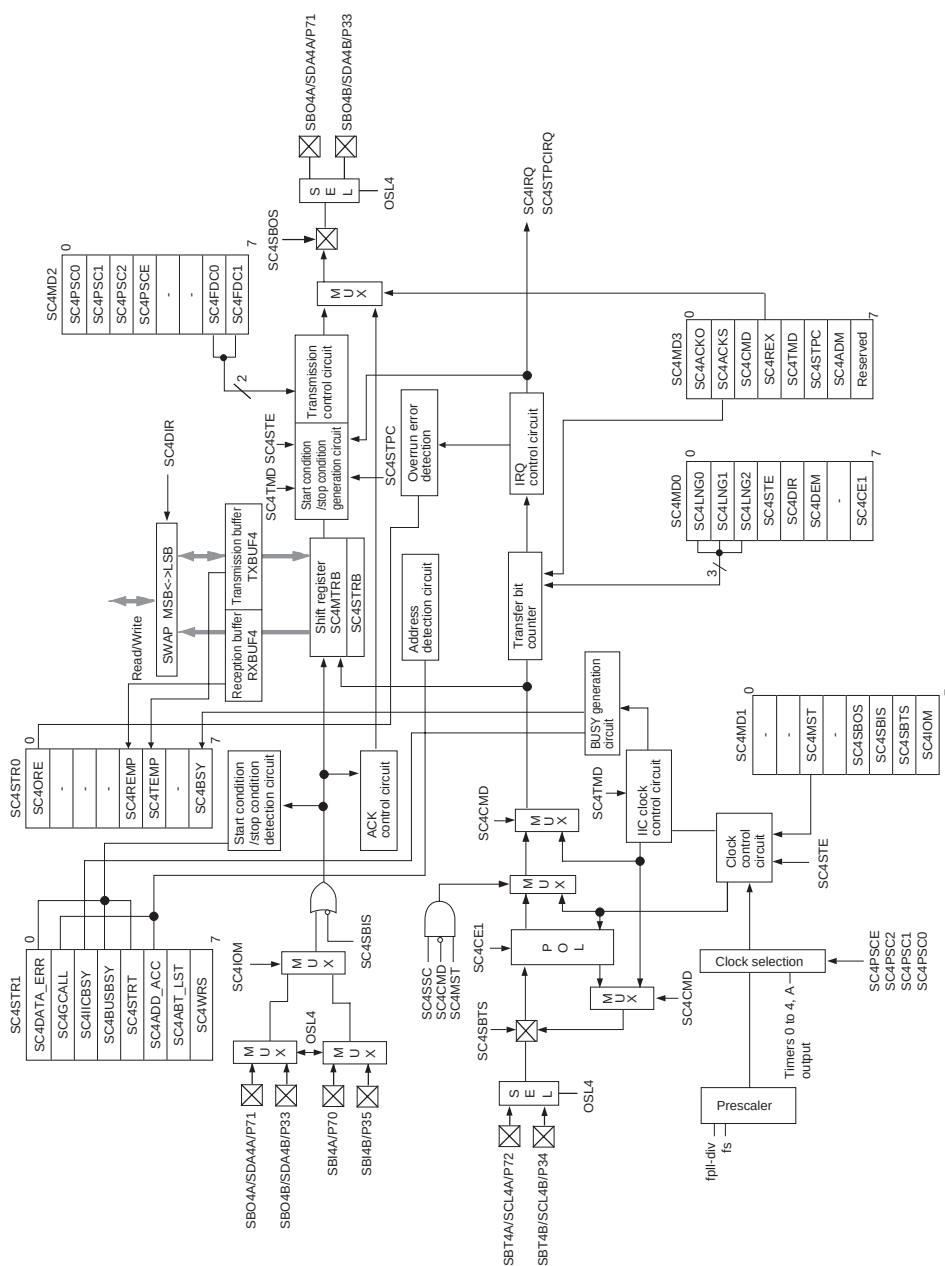


Figure:18.1.8 Serial Interface 4 Block Diagram (MN101EF76K/MN101EF56K)

■ Serial Interface 4 Block Diagram (MN101EF57G)

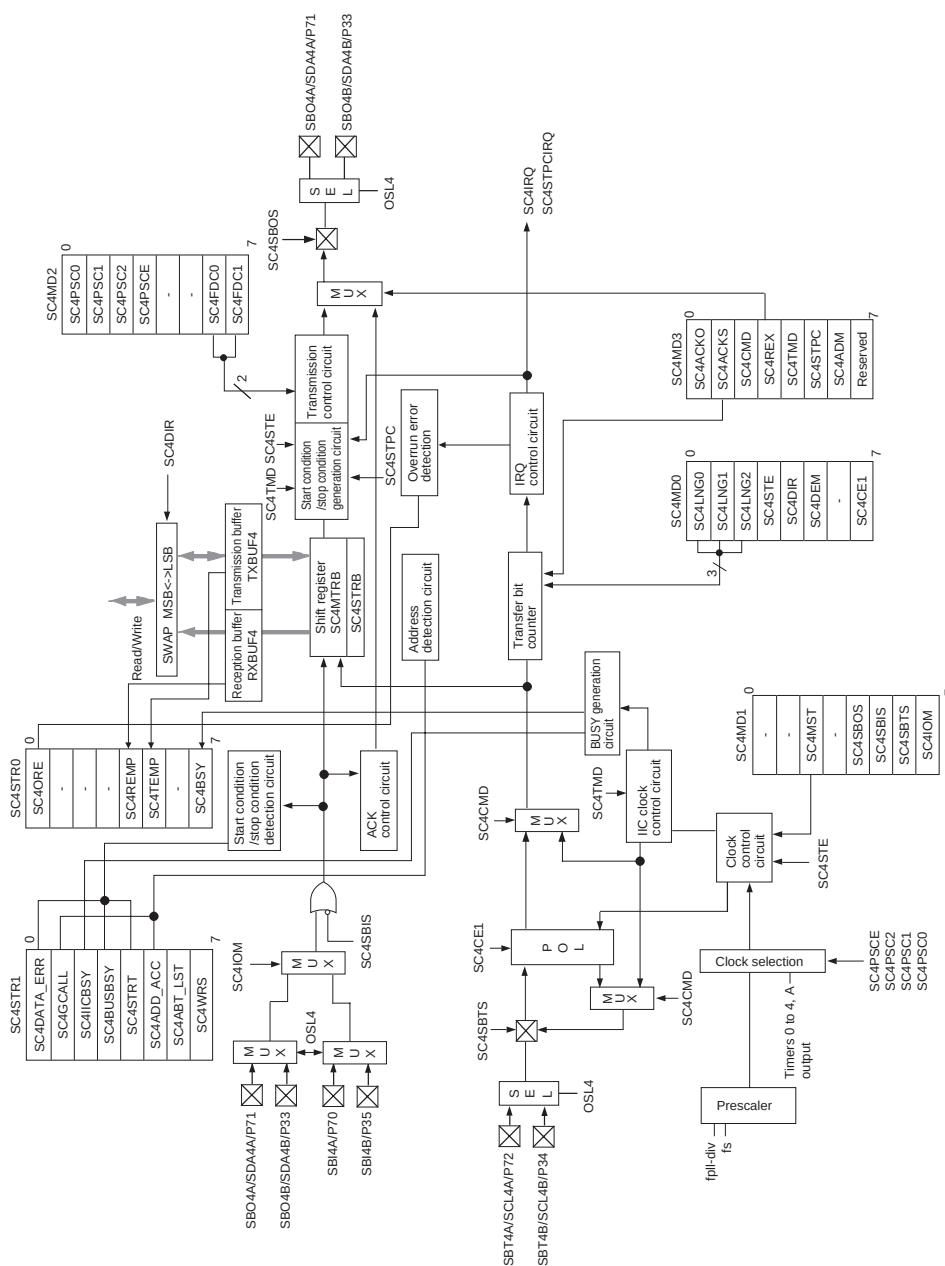


Figure:18.1.9 Serial Interface 4 Block Diagram (MN101EF57G)

18.2 Control Registers

18.2.1 Registers

Table:18.2.1 shows registers to control serial interface.

Table:18.2.1 Serial Interface Control Registers

	Register	Address	R/W	Function	Page
Serial Interface 0	SC0MD0	0x03F11	R/W	Serial Interface 0 mode register 0	XVIII-27
	SC0MD1	0x03F12	R/W	Serial Interface 0 mode register 1	XVIII-28
	SC0MD2	0x03F13	R/W	Serial Interface 0 mode register 2	XVIII-29
	SC0MD3	0x03F14	R/W	Serial Interface 0 mode register 3	XVIII-30
	SC0STR	0x03F15	R	Serial Interface 0 status register	XVIII-31
	RXBUF0	0x03F16	R	Serial Interface 0 reception data buffer	XVIII-25
	TXBUF0	0x03F17	R/W	Serial Interface 0 transmission data buffer	XVIII-25
	SC0SEL	0x03F10	R/W	Serial 0 I/O pin switching control register	XVIII-19
	LINCTR	0x03F18	R/W	LIN control register	XVIII-32
	P0ODC	0x03EF0	R/W	Port 0 Nch open-drain control register	VIII-11
	P0DIR	0x03E90	R/W	Port 0 direction control register	VIII-9
	P0PLUD	0x03EA0	R/W	Port 0 pull-up/pull-down resistor control register	VIII-9
	P4ODC	0x03EF4	R/W	Port 4 Nch open-drain control register	VIII-76
	P4DIR	0x03E94	R/W	Port 4 direction control register	VIII-74
	P4PLUD	0x03EA4	R/W	Port 4 pull-up/pull-down resistor control register	VIII-75
	SC0RICR	0x03FF5	R/W	Serial 0 reception interrupt control register	VII-34
	SC0TICR	0x03FF4	R/W	Serial 0 UART transmission interrupt control register	VII-33
	SCINTSEL	0x03F28	R/W	Clock synchronous serial reception interrupt control register	XVIII-33

	Register	Address	R/W	Function	Page
Serial Interface 1	SC1MD0	0x03F21	R/W	Serial Interface 1 mode register 0	XVIII-27
	SC1MD1	0x03F22	R/W	Serial Interface 1 mode register 1	XVIII-28
	SC1MD2	0x03F23	R/W	Serial Interface 1 mode register 2	XVIII-29
	SC1MD3	0x03F24	R/W	Serial Interface 1 mode register 3	XVIII-30
	SC1STR	0x03F25	R	Serial Interface 1 status register	XVIII-31
	RXBUF1	0x03F26	R	Serial Interface 1 reception data buffer	XVIII-25
	TXBUF1	0x03F27	R/W	Serial Interface 1 transmission data buffer	XVIII-25
	SC1SEL	0x03F20	R/W	Serial 1 I/O pin switching control register	XVIII-20
	P5ODC	0x03EF5	R/W	Port 5 Nch open-drain control register	VIII-94
	P5DIR	0x03F95	R/W	Port 5 direction control register	VIII-94
	P5PLUD	0x03EA5	R/W	Port 5 pull-up/pull-down resistor control register	VIII-94
	P7ODC	0x03EF7	R/W	Port 7 Nch open-drain control register	VIII-133
	P7DIR	0x03F97	R/W	Port 7 direction control register	VIII-132
	P7PLUD	0x03EA7	R/W	Port 7 pull-up/pull-down resistor control register	VIII-132
	SC1RICR	0x03FF7	R/W	Serial 1 UART reception interrupt control register	VII-34
	SC1TICR	0x03FF6	R/W	Serial 1 interrupt control register	VII-33
	SCINTSEL	0x03F28	R/W	Clock synchronous serial reception interrupt control register	VII-33
Serial Interface 2	SC2MD0	0x03F31	R/W	Serial Interface 2 mode register 0	XVIII-27
	SC2MD1	0x03F32	R/W	Serial Interface 2 mode register 1	XVIII-28
	SC2MD2	0x03F33	R/W	Serial Interface 2 mode register 2	XVIII-29
	SC2MD3	0x03F34	R/W	Serial Interface 2 mode register 3	XVIII-30
	SC2STR	0x03F35	R	Serial Interface 2 status register	XVIII-31
	RXBUF2	0x03F36	R	Serial Interface 2 reception data buffer	XVIII-25
	TXBUF2	0x03F37	R/W	Serial Interface 2 transmission data buffer	XVIII-25
	SC2SEL	0x03F30	R/W	Serial 2 I/O pin switching control register	XVIII-21
	P6ODC	0x03EF6	R/W	Port 6 Nch open-drain control register	VIII-115
	P6DIR	0x03E96	R/W	Port 6 direction control register	VIII-113
	P6PLU	0x03EA6	R/W	Port 6 pull-up resistor control register	VIII-114
	SC2RICR	0x03FF9	R/W	Serial 2 UART reception interrupt control register	VII-34
	SC2TICR	0x03FFA	R/W	Serial 2 interrupt control register	VII-33

	Register	Address	R/W	Function	Page
Serial Interface 3	SC3MD0	0x03F41	R/W	Serial Interface 3 mode register 0	XVIII-27
	SC3MD1	0x03F42	R/W	Serial Interface 3 mode register 1	XVIII-28
	SC3MD2	0x03F43	R/W	Serial Interface 3 mode register 2	XVIII-29
	SC3MD3	0x03F44	R/W	Serial Interface 3 mode register 3	XVIII-30
	SC3STR	0x03F45	R	Serial Interface 3 status register	XVIII-31
	RXBUF3	0x03F46	R	Serial Interface 3 reception data buffer	XVIII-25
	TXBUF3	0x03F47	R/W	Serial Interface 3 transmission data buffer	XVIII-25
	SC3SEL	0x03F40	R/W	Serial 3 I/O pin switching control register	XVIII-23
	P0ODC	0x03EF0	R/W	Port 0 Nch open-drain control register	VIII-11
	P0DIR	0x03E90	R/W	Port 0 direction control register	VIII-9
	P0PLUD	0x03EA0	R/W	Port 0 pull-up/pull-down resistor control register	VIII-9
	P4ODC	0x03EF4	R/W	Port 4 Nch open-drain control register	VIII-76
	P4DIR	0x03E94	R/W	Port 4 direction control register	VIII-74
	P4PLUD	0x03EA4	R/W	Port 4 pull-up/pull-down resistor control register	VIII-75
	SC3RICR	0x03FFB	R/W	Serial 3 UART reception interrupt control register	VII-34
	SC3TICR	0x03FFA	R/W	Serial 3 interrupt control register	VII-33
Serial Interface 4	SC4MD0	0x03F50	R/W	Serial Interface 4 mode register 0	XVIII-35
	SC4MD1	0x03F51	R/W	Serial Interface 4 mode register 1	XVIII-36
	SC4MD2	0x03F52	R/W	Serial Interface 4 mode register 2	XVIII-37
	SC4MD3	0x03F53	R/W	Serial Interface 4 mode register 3	XVIII-38
	SC4AD0	0x03F54	R/W	Serial Interface 4 address setting register 0	XVIII-39
	SC4AD1	0x03F55	R/W	Serial Interface 4 address setting register 1	XVIII-39
	SC4STR0	0x03F56	R/W	Serial Interface 4 status register 0	XVIII-40
	SC4STR1	0x03F57	R/W	Serial Interface 4 status register 1	XVIII-41
	RXBUF4	0x03F58	R	Serial Interface 4 reception data buffer	XVIII-34
	TXBUF4	0x03F59	R/W	Serial Interface 4 transmission data buffer	XVIII-34
	SC4SEL	0x03F5A	R/W	Serial 4 I/O pin switching control register	XVIII-24
	P7DIR	0x03E97	R/W	Port 7 direction control register	VIII-132
	P3DIR	0x03E93	R/W	Port 3 direction control register	VIII-57
	SC4ICR	0x03FFC	R/W	Serial 4 interrupt control register	VII-35
	SC4STPCICR	0x03FFD	R/W	Serial 4 stop condition interrupt control register	VII-36



To change setting values of mode registers, set forced reset for serial interface at first by setting both SCnSBIS and SCnSBOS flags of SCnMD1 to "0". (n = 0 to 3)



Set forced reset for serial interface by setting both SC4SBIS and SC4SBOS flags of SC4MD1 to "0", before changing setting values of Serial Interface 4 mode registers (except for SC4STE, SC4STPC, SC4REX and SC4ACK0 flags) and address setting registers.

18.2.2 Serial I/O Pin Switching Control Registers

Pins are shared between serial interfaces.

SCnSEL registers are used for switching pins.

■ Serial 0 I/O Pin Switching Control Register (SC0SEL: 0x03F10)

bp	7	6	5	4	3	2	1	0
Flag	SBO0SEL	SC0BRP2	SC0BRP1	SC0BRP0	OSL0	SC0SEL2	SC0SEL1	SC0SEL0
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7	SBO0SEL	UART reverse output selection 0: UART output 1: UART reverse output
6 to 4	SC0BRP2 SC0BRP1 SC0BRP0	Timer clock output selection XX0: Timer output 001: Timer output/2 011: Timer output/4 101: Timer output/8 111: Timer output/16
3	OSL0	Serial output port selection 0: A type 1: B type
2 to 0	SC0SEL2 SC0SEL1 SC0SEL0	Timer selection X00: Timer 0 X01: Timer 1 010: Timer 2 011: Timer 3 110: Timer 4 111: Timer A

■ Serial 1 I/O Pin Switching Control Register (SC1SEL: 0x03F20)

bp	7	6	5	4	3	2	1	0
Flag	SBO1SEL	SC1BRP2	SC1BRP1	SC1BRP0	OSL1	SC1SEL2	SC1SEL1	SC1SEL0
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7	SBO1SEL	UART reverse output selection 0: UART output 1: UART reverse output
6 to 4	SC1BRP2 SC1BRP1 SC1BRP0	Timer clock output selection XX0: Timer output 001: Timer output/2 011: Timer output/4 101: Timer output/8 111: Timer output/16
3	OSL1	Serial output port selection 0: A type 1: B type
2 to 0	SC1SEL2 SC1SEL1 SC1SEL0	Timer selection X00: Timer 0 X01: Timer 1 010: Timer 2 011: Timer 3 110: Timer 4 111: Timer A

■ Serial 2 I/O Pin Switching Control Register (SC2SEL: 0x03F30)

<MN101EF76K/MN101EF56K>

bp	7	6	5	4	3	2	1	0
Flag	SBO2SEL	SC2BRP2	SC2BRP1	SC2BRP0	OSL2	SC2SEL2	SC2SEL1	SC2SEL0
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7	SBO2SEL	UART reverse output selection 0: UART output 1: UART reverse output
6 to 4	SC2BRP2 SC2BRP1 SC2BRP0	Timer clock output selection XX0: Timer output 001: Timer output/2 011: Timer output/4 101: Timer output/8 111: Timer output/16
3	OSL2	Serial output port selection 0: A type 1: B type
2 to 0	SC2SEL2 SC2SEL1 SC2SEL0	Timer selection X00: Timer 0 X01: Timer 1 010: Timer 2 011: Timer 3 110: Timer 4 111: Timer A

■ Serial 2 I/O Pin Switching Control Register (SC2SEL: 0x03F30)

<MN101EF57G>

bp	7	6	5	4	3	2	1	0
Flag	SBO2SEL	SC2BRP2	SC2BRP1	SC2BRP0	Reserved	SC2SEL2	SC2SEL1	SC2SEL0
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7	SBO2SEL	UART reverse output selection 0: UART output 1: UART reverse output
6 to 4	SC2BRP2 SC2BRP1 SC2BRP0	Timer clock output selection XX0: Timer output 001: Timer output/2 011: Timer output/4 101: Timer output/8 111: Timer output/16
3	Reserved	Always set to "0".
2 to 0	SC2SEL2 SC2SEL1 SC2SEL0	Timer selection X00: Timer 0 X01: Timer 1 010: Timer 2 011: Timer 3 110: Timer 4 111: Timer A

■ Serial 3 I/O Pin Switching Control Register (SC3SEL: 0x03F40)

<MN101EF76K/MN101EF56K>

bp	7	6	5	4	3	2	1	0
Flag	SBO3SEL	SC3BRP2	SC3BRP1	SC3BRP0	OSL3	SC3SEL2	SC3SEL1	SC3SEL0
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7	SBO3SEL	UART reverse output selection 0: UART output 1: UART reverse output
6 to 4	SC3BRP2 SC3BRP1 SC3BRP0	Timer clock output selection XX0: Timer output 001: Timer output/2 011: Timer output/4 101: Timer output/8 111: Timer output/16
3	OSL3	Serial output port selection 0: A type 1: B type
2 to 0	SC3SEL2 SC3SEL1 SC3SEL0	Timer selection X00: Timer 0 X01: Timer 1 010: Timer 2 011: Timer 3 110: Timer 4 111: Timer A

■ Serial 4 I/O Pin Switching Control Register (SC4SEL: 0x03F5A)

bp	7	6	5	4	3	2	1	0
Flag	Reserved	SC4BRP2	SC4BRP1	SC4BRP0	OSL4	SC4SEL2	SC4SEL1	SC4SEL0
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7	Reserved	Always set to "0".
6 to 4	SC4BRP2 SC4BRP1 SC4BRP0	Timer clock output selection XX0: Timer output 001: Timer output/2 011: Timer output/4 101: Timer output/8 111: Timer output/16
3	OSL4	Serial output port selection 0: A type 1: B type
2 to 0	SC4SEL2 SC4SEL1 SC4SEL0	Timer selection X00: Timer 0 X01: Timer 1 010: Timer 2 011: Timer 3 110: Timer 4 111: Timer A

18.2.3 Serial Interface 0, 1, 2 and 3 Control Registers

Serial Interface 0, 1, 2 and 3 can be used for clock synchronous and UART (duplex) communication.

Each serial interface consists of 2 buffers and 6 registers.

- Serial Interface n reception data buffer (RXBUF_n)
- Serial Interface n transmission data buffer (TXBUF_n)
- Serial Interface n mode register 0 (SCnMD0)
- Serial Interface n mode register 1 (SCnMD1)
- Serial Interface n mode register 2 (SCnMD2)
- Serial Interface n mode register 3 (SCnMD3)
- Serial Interface n status register (SCnSTR)
- LIN control register (LINCTR)



"n" = 0, 1, 2 and 3 for Serial Interface 0, 1, 2 and 3 respectively in [Chapter 18 18.2.3 Serial Interface 0, 1, 2 and 3 Control Registers].

- Serial Interface n Reception Data Buffer
(RXBUF0: 0x03F16, RXBUF1: 0x03F26, RXBUF2: 0x03F36, RXBUF3: 0x03F46)

bp	7	6	5	4	3	2	1	0
Flag	RXBUF _n 7	RXBUF _n 6	RXBUF _n 5	RXBUF _n 4	RXBUF _n 3	RXBUF _n 2	RXBUF _n 1	RXBUF _n 0
At reset	X	X	X	X	X	X	X	X
Access	R	R	R	R	R	R	R	R

- Serial Interface n Transmission Data Buffer
(TXBUF0: 0x03F17, TXBUF1: 0x03F27, TXBUF2: 0x03F37, TXBUF3: 0x03F47)

bp	7	6	5	4	3	2	1	0
Flag	TXBUF _n 7	TXBUF _n 6	TXBUF _n 5	TXBUF _n 4	TXBUF _n 3	TXBUF _n 2	TXBUF _n 1	TXBUF _n 0
At reset	X	X	X	X	X	X	X	X
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

- Serial Interface n Mode Register 0
(SC0MD0: 0x03F11, SC1MD0: 0x03F21, SC2MD0: 0x03F31, SC3MD0: 0x03F41)

<MN101EF76K/MN101EF56K>

bp	7	6	5	4	3	2	1	0
Flag	SCnCE1	SCnSSC	Reserved	SCnDIR	SCnSTE	SCnLNG2	SCnLNG1	SCnLNG0
At reset	0	0	0	0	0	1	1	1
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7	SCnCE1	Transmission data output edge 0: Falling 1: Rising Reception data input edge 0: Rising 1: Falling
6	SCnSSC	Internal operating clock source 0: Input via SBT 1: Internal generation clock
5	Reserved	Always set to "0".
4	SCnDIR	Transfer bit specification 0: MSB first 1: LSB first
3	SCnSTE	Start condition selection 0: Disable 1: Enable
2 to 0	SCnLNG2 SCnLNG1 SCnLNG0	Synchronous serial transfer bit count 000: 1 bit 001: 2 bits 010: 3 bits 011: 4 bits 100: 5 bits 101: 6 bits 110: 7 bits 111: 8 bits



To change setting values of mode registers, set forced reset for serial interface at first by setting both SCnSBIS and SCnSBOS flags of SCnMD1 to "0". (n = 0 to 3)

- Serial Interface n Mode Register 0
(SC0MD0: 0x03F11, SC1MD0: 0x03F21, SC2MD0: 0x03F31, SC3MD0: 0x03F41)

<MN101EF57G>

bp	7	6	5	4	3	2	1	0
Flag	SCnCE1	-	Reserved	SCnDIR	SCnSTE	SCnLNG2	SCnLNG1	SCnLNG0
At reset	0	-	0	0	0	1	1	1
Access	R/W	-	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7	SCnCE1	Transmission data output edge Reception data input edge 0: Falling 0: Rising 1: Rising 1: Falling
6	-	-
5	Reserved	Always set to "0".
4	SCnDIR	Transfer bit specification 0: MSB first 1: LSB first
3	SCnSTE	Start condition selection 0: Disable 1: Enable
2 to 0	SCnLNG2 SCnLNG1 SCnLNG0	Synchronous serial transfer bit count 000: 1 bit 001: 2 bits 010: 3 bits 011: 4 bits 100: 5 bits 101: 6 bits 110: 7 bits 111: 8 bits



To change setting values of mode registers, set forced reset for serial interface at first by setting both SCnSBIS and SCnSBOS flags of SCnMD1 to "0". (n = 0 to 3)

- Serial Interface n Mode Register 1
(SC0MD1: 0x03F12, SC1MD1: 0x03F22, SC2MD1: 0x03F32, SC3MD1: 0x03F42)

bp	7	6	5	4	3	2	1	0
Flag	SCnIOM	SCnSBTS	SCnSBIS	SCnSBOS	SCnCKM	SCnMST	SCnDIV	SCnCMD
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description	
7	SCnIOM	Serial data input pin selection 0: Data input from SBIn (RXDn) 1: Data input from SBO n (TXDn)	
6	SCnSBTS	SBT pin function selection 0: Port 1: Transfer clock I/O	
5	SCnSBIS	Serial input control selection 0: "1" input 1: Serial data input	
4	SCnSBOS	SBO (TXD) pin function selection 0: Port 1: Serial data output	
3	SCnCKM	Clock synchronous (SCnCMD=0) Transfer clock division selection 0: Not divided 1: Divided	UART (SCnCMD=1) Transfer clock is divided regardless of the setting value. The data read out is the setting value.
2	SCnMST	Clock master/salve selection 0: Clock slave 1: Clock master	
1	SCnDIV	Transfer clock division value selection 0: Divide by 8 1: Divide by 16	
0	SCnCMD	Clock synchronous/UART selection 0: Clock synchronous 1: UART	



To set serial interface communication mode to "UART", set Serial Interface n mode register 1 (SCnMD1) while the serial data input pin is at "High".



To change setting values of mode registers, set forced reset for serial interface at first by setting both SCnSBIS and SCnSBOS flags of SCnMD1 to "0". (n = 0 to 3)



If setting both the SCnSBIS flag and the SCnSBOS flag to "0", the serial interface functions are forced to reset and stopped.

- Serial Interface n Mode Register 2
(SC0MD2: 0x03F13, SC1MD2: 0x03F23, SC2MD2: 0x03F33, SC3MD2: 0x03F43)

bp	7	6	5	4	3	2	1	0
Flag	SCnFM1	SCnFM0	SCnPM1	SCnPM0	SCnNPE	Reserved	SCnBRKF	SCnBRKE
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R	R/W

bp	Flag	Description
7 to 6	SCnFM1 SCnFM0	Frame mode specification 00: Data 7 bits + stop 1 bit 01: Data 7 bits + stop 2 bits 10: Data 8 bits + stop 1 bit 11: Data 8 bits + stop 2 bits
5 to 4	SCnPM1 SCnPM0	Additional bit specification At transmission 00: Add "0" 01: Add "1" 10: Add odd parity 11: Add even parity At reception Check for "0" Check for "1" Check for odd parity Check for even parity
3	SCnNPE	Parity enable 0: Enable parity bit 1: Disable parity bit
2	Reserved	Always set to "0".
1	SCnBRKF	Break status receive monitor 0: Data reception 1: Break reception
0	SCnBRKE	Break status transmit control 0: Data transmission 1: Break transmission

- Serial Interface n Mode Register 3
(SC0MD3: 0x03F14, SC1MD3: 0x03F24, SC2MD3: 0x03F34, SC3MD3: 0x03F44)

bp	7	6	5	4	3	2	1	0
Flag	SCnFDC1	SCnFDC0	-	-	SCnPSCE	SCnPSC2	SCnPSC1	SCnPSC0
At reset	0	0	-	-	0	0	0	0
Access	R/W	R/W	-	-	R/W	R/W	R/W	R/W

bp	Flag	Description
7 to 6	SCnFDC1 SCnFDC0	Output selection after SBO final data transmit 00: Fixed at "High" output 01: Hold the final data 10: Fixed at "Low" output 11: Setting prohibited
5 to 4	-	-
3	SCnPSCE	Prescaler count control 0: Disable count 1: Enable count
2 to 0	SCnPSC2 SCnPSC1 SCnPSC0	Clock selection 000: fpll-div/2 001: fpll-div/4 010: fpll-div/16 011: fpll-div/64 100: fs/2 101: fs/4 11X: Timer output * Timer 0 to 4 or A can be selected by the SCnSEL2 to 0 flags of SCnSEL register.

- Serial Interface n Status Register
(SC0STR: 0x03F15, SC1STR: 0x03F25, SC2STR: 0x03F35, SC3STR: 0x03F45)

bp	7	6	5	4	3	2	1	0
Flag	SCnTBSY	SCnRBSY	SCnTEMP	SCnREMP	SCnFEF	SCnPEK	SCnORE	SCnERE
At reset	0	0	0	0	0	0	0	0
Access	R	R	R	R	R	R	R	R

bp	Flag	Description
7	SCnTBSY	Serial bus status 0: Serial transmission NOT in progress 1: Serial transmission in progress
6	SCnRBSY	Serial bus status 0: Serial reception NOT in progress 1: Serial reception in progress
5	SCnTEMP	Transmission buffer empty flag 0: Empty 1: Full
4	SCnREMP	Reception buffer empty flag 0: Empty 1: Full
3	SCnFEF	Framing error detection 0: No error 1: Error
2	SCnPEK	Parity error detection 0: No error 1: Error
1	SCnORE	Overrun error detection 0: No error 1: Error
0	SCnERE	Error monitor flag 0: No error 1: Error

■ LIN Control Register (LINCTR: 0x03F18)

bp	7	6	5	4	3	2	1	0
Flag	SFBSY	SBFBSY	SFREC	SBFREC	SBFEN	SBFSEL2	SBFSEL1	SBFSEL0
At reset	0	0	0	0	0	0	0	0
Access	R	R	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7	SFBSY	SF reception busy flag 0: Except during SF reception 1: During SF reception
6	SBFBSY	SBF reception busy flag 0: Except during SBF reception 1: During SBF reception
5	SFREC	SF reception enable 0: Normal operation 1: SF reception
4	SBFREC	SBF reception enable 0: Normal operation 1: SBF reception
3	SBFEN	SBF transmission trigger/ monitor 0: Except during SBF transmission 1: During SBF transmission
2 to 0	SBFSEL2 SBFSEL1 SBFSEL0	SBF transmission width selection 000: 12 bits 001: 13 bits 010: 14 bits 011: 15 bits 100: 16 bits 101: 17 bits 110: 18 bits 111: 19 bits

■ Clock Synchronous Serial Reception Interrupt Control Register (SCINTSEL: 0x03F28)

bp	7	6	5	4	3	2	1	0
Flag	-	-	-	-	-	-	SC1IGC	SC0IGC
At reset	-	-	-	-	-	-	0	0
Access	-	-	-	-	-	-	R/W	R/W

bp	Flag	Description
7 to 2	-	-
1	SC1IGC	Serial 1 reception interrupt generation timing at clock synchronous serial data reception 0: Reception is completed 1: SBT1 falling
0	SC0IGC	Serial 0 reception interrupt generation timing at clock synchronous serial data reception 0: Reception is completed 1: SBT0 falling



The SCnIGC flag of SCINTSEL register can only be set to "1" during clock synchronous slave reception.
If other communication setting is used, the SCnIGC flag must be set to "0".

18.2.4 Serial Interface 4 Control Registers

Serial Interface 4 can be used for clock synchronous and multi master IIC communication.

This serial interface consists of 2 buffers and 8 registers.

- Serial Interface 4 reception data buffer (RXBUF4)
- Serial Interface 4 transmission data buffer (TXBUF4)
- Serial Interface 4 mode register 0 (SC4MD0)
- Serial Interface 4 mode register 1 (SC4MD1)
- Serial Interface 4 mode register 2 (SC4MD2)
- Serial Interface 4 mode register 3 (SC4MD3)
- Serial Interface 4 address setting register 0 (SC4AD0)
- Serial Interface 4 status register 0 (SC4STR0)
- Serial Interface 4 status register 1 (SC4STR1)

■ Serial Interface 4 Reception Data Buffer (RXBUF4: 0x03F58)

bp	7	6	5	4	3	2	1	0
Flag	RXBUF47	RXBUF46	RXBUF45	RXBUF44	RXBUF43	RXBUF42	RXBUF41	RXBUF40
At reset	X	X	X	X	X	X	X	X
Access	R	R	R	R	R	R	R	R

■ Serial Interface 4 Transmission Data Buffer (TXBUF4: 0x03F59)

bp	7	6	5	4	3	2	1	0
Flag	TXBUF47	TXBUF46	TXBUF45	TXBUF44	TXBUF43	TXBUF42	TXBUF41	TXBUF40
At reset	X	X	X	X	X	X	X	X
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

18.2.5 Serial Interface 4 Mode Register

■ Serial Interface 4 Mode Register 0 (SC4MD0: 0x03F50)

bp	7	6	5	4	3	2	1	0
Flag	SC4CE1	-	SC4DEM	SC4DIR	SC4STE	SC4LNG2	SC4LNG1	SC4LNG0
At reset	0	-	0	0	0	1	1	1
Access	R/W	-	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7	SC4CE1	Transmission data output edge Reception data input edge 0: Falling 0: Rising 1: Rising 1: Falling
6	-	-
5	SC4DEM	Operation selection after detecting communication data instability in IIC slave communication mode 0: Start condition is not detected 1: Start condition is detected
4	SC4DIR	Transfer bit specification 0: MSB first 1: LSB first
3	SC4STE	Start condition selection 0: Disable 1: Enable
2 to 0	SC4LNG2 SC4LNG1 SC4LNG0	Transfer bit count 000: 1 bit 001: 2 bits 010: 3 bits 011: 4 bits 100: 5 bits 101: 6 bits 110: 7 bits 111: 8 bits



Set forced reset for serial interface by setting both SC4SBIS and SC4SBOS flags of SC4MD1 to "0", before changing setting values of Serial Interface 4 mode registers (except for SC4STE, SC4STPC, SC4REX and SC4ACK0 flags) and address setting registers.

■ Serial Interface 4 Mode Register 1 (SC4MD1: 0x03F51)

bp	7	6	5	4	3	2	1	0
Flag	SC4IOM	SC4SBTS	SC4SBIS	SC4SBOS	-	SC4MST	-	-
At reset	0	0	0	0	-	0	-	-
Access	R/W	R/W	R/W	R/W	-	R/W	-	-

bp	Flag	Description
7	SC4IOM	Serial data input pin selection 0: Data input from SBI4 1: Data input from SBO4 (SDA4)
6	SC4SBTS	SBT4 pin function selection 0: Port 1: Transfer clock I/O
5	SC4SBIS	Serial input control selection 0: "1" input 1: Serial input
4	SC4SBOS	SBO4 (SDA4) pin function selection 0: Port 1: Serial data output
3	-	-
2	SC4MST	Clock master/Clock slave selection 0: Clock slave 1: SC4CMD=0: Clock master SC4CMD=1: Clock master/Clock slave
1 to 0	-	-



Set forced reset for serial interface by setting both SC4SBIS and SC4SBOS flags of SC4MD1 to "0", before changing setting values of Serial Interface 4 mode registers (except for SC4STE, SC4STPC, SC4REX and SC4ACK0 flags) and address setting registers.

■ Serial Interface 4 Mode Register 2 (SC4MD2: 0x03F52)

bp	7	6	5	4	3	2	1	0
Flag	SC4FDC1	SC4FDC0	-	-	SC4PSCE	SC4PSC2	SC4PSC1	SC4PSC0
At reset	0	0	-	-	0	0	0	0
Access	R/W	R/W	-	-	R/W	R/W	R/W	R/W

bp	Flag	Description
7 to 6	SC4FDC1 SC4FDC0	Output selection after SBO final data transmit 00: Fixed to "High" output 01: Hold the final data 10: Fixed to "Low" output 11: Setting prohibited
5 to 4	-	-
3	SC4PSCE	Prescaler count control 0: Disable count 1: Enable count
2 to 0	SC4PSC2 SC4PSC1 SC4PSC0	Clock selection 000: fpll-div/2 001: fpll-div/4 010: fpll-div/16 011: fpll-div/32 100: fs/2 101: fs/4 11X: Timer output * Timer 0 to 4 or A can be selected by the SC4SEL2 to 0 flags of SC4SEL register.

■ Serial Interface 4 Mode Register 3 (SC4MD3: 0x03F53)

bp	7	6	5	4	3	2	1	0
Flag	Reserved	SC4ADM	SC4STPC	SC4TMD	SC4REX	SC4CMD	SC4ACKS	SC4ACKO
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description	
7	Reserved	Always set to "0".	
6	SC4ADM	Address format selection 0: 7-bit address 1: 10-bit address	
5	SC4STPC	Stop condition generation request flag in IIC communication 0: Disable 1: Stop condition is generated	
4	SC4TMD	Communication mode selection in IIC communication 0: Standard mode 1: High-speed mode	
3	SC4REX	Transmission/reception mode selection in IIC communication 0: Transmission 1: Reception	
2	SC4CMD	Clock synchronous/IIC selection 0: Clock synchronous 1: IIC	
1	SC4ACKS	ACK bit enable 0: Enable 1: Disable	
0	SC4ACKO	At transmission ACK/NACK bit detection flag 0: ACK detection 1: NACK detection	At reception ACK/NACK bit transmission flag 0: ACK transmission 1: NACK transmission



SC4STPC and SC4ACKO flags are updated by hardware. Thus, when values of these flags are read out, the values might be different from the values which are previously written. When the SC4STPC flag is set to "1", the circuit will hold the status (flag:1) for a certain period of time. And then, the flag is cleared to "0". The written value of the SC4ACKO flag will be reloaded when the communication starts. With the features above, when changing SC4MD3 register, set the value of the register with careful attention to settings of the SC4STPC flag and the SC4ACKO flag. In particular, do not execute Read-Modify-Write operation, including bit manipulation instructions such as BSET/BCLR. Otherwise, multiple accesses may rewrite the value of the SC4MD3 register to an unexpected value.



Set forced reset for serial interface by setting both SC4SBIS and SC4SBOS flags of SC4MD1 to "0", before changing setting values of Serial Interface 4 mode registers (except for SC4STE, SC4STPC, SC4REX and SC4ACKO flags) and address setting registers.

18.2.6 Serial Interface 4 Address Setting Register

Serial Interface 4 has address setting registers for 4-bit data.

■ Serial Interface 4 Address Setting Register 0 (SC4AD0: 0x03F54)

bp	7	6	5	4	3	2	1	0
Flag	SC4ADR7	SC4ADR6	SC4ADR5	SC4ADR4	SC4ADR3	SC4ADR2	SC4ADR1	SC4ADR0
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

■ Serial Interface 4 Address Setting Register 1 (SC4AD1: 0x03F55)

bp	7	6	5	4	3	2	1	0
Flag	-	-	-	-	-	-	SC4ADR9	SC4ADR8
At reset	-	-	-	-	-	-	0	0
Access	-	-	-	-	-	-	R/W	R/W

18.2.7 Serial Interface 4 Status Register

■ Serial Interface 4 Status Register 0 (SC4STR0: 0x03F56)

bp	7	6	5	4	3	2	1	0
Flag	SC4BSY	-	SC4TEMP	SC4REMP	-	-	-	SC4ORE
At reset	0	-	0	0	-	-	-	0
Access	R	-	R	R	-	-	-	R/W

bp	Flag	Description
7	SC4BSY	Clock synchronous communication serial bus status 0: Serial transmission NOT in progress 1: Serial transmission in progress
6	-	-
5	SC4TEMP	Transmission buffer empty flag 0: Empty 1: Full
4	SC4REMP	Reception buffer empty flag 0: Empty 1: Full
3 to 1	-	-
0	SC4ORE	Overrun error detection 0: No error 1: Error



Regarding SC4ORE flag:
when "1" is written, the value of this flag does not change, and
when "0" is written, the value of this flag is cleared to "0".

■ Serial Interface 4 Status Register 1 (SC4STR1: 0x03F57)

bp	7	6	5	4	3	2	1	0
Flag	SC4 WRS	SC4 ABT_LST	SC4 ADD_ACC	SC4 STRT	SC4 BUSBSY	SC4 IICBSY	SC4 GCALL	SC4 DATA_ERR
At reset	0	0	0	0	0	0	0	0
Access	R	R/W	R	R	R	R	R	R/W

bp	Flag	Description
7	SC4WRS	Data transfer determination flag in slave communication 0: Master to Slave 1: Slave to Master
6	SC4ABT_LST	Arbitration lost detection flag 0: Not detected 1: Detected
5	SC4ADD_ACC	Slave address comparison flag 0: Address is unmatched 1: Address is matched
4	SC4STRT	Start condition detection flag 0: Not detected 1: Detected
3	SC4BUSBSY	Bus busy flag 0: Bus free status 1: Bus busy status
2	SC4IICBSY	Serial bus status in IIC communication 0: Serial transmission NOT in progress 1: Serial transmission in progress
1	SC4GCALL	General call detection flag 0: Not detected 1: Detected
0	SC4DATA_ERR	Communication error detection flag 0: Not detected 1: Detected



Regarding SC4ABT_LST flag and SC4DATA_ERR flag:
when "1" is written, the values of these flags do not change, and
when "0" is written, the values of these flags are cleared to "0".

18.3 Clock Synchronous Serial Interface

18.3.1 Operation

This section describes the clock synchronous communication method of Serial Interface 0, 1, 2, 3 and 4.



"n" = 0, 1, 2, 3 and 4 for Serial Interface 0, 1, 2, 3 and 4 respectively in [Chapter 18 18.3 Clock Synchronous Serial Interface].

■ Communication Type

The communication mode can be selected from 3-wire type (clock pin (SBTn pin), data output pin (SBO pin), data input pin (SBI pin)) or 2-wire type (clock pin (SBTn pin), data I/O pin (SBO pin)). Set the communication mode by the SCnIOM flag of SCnMD1 register. In 2-wire reception, select "serial data input" by setting the SCnSBIS flag of SCnMD1 register to "1". SBI pin can be used as a general-purpose port.

■ Activation Factor for Communication

Table:18.3.1 shows activation factors for communication. In master communication, the transfer clock is generated by communication activation factors. In slave communication, except during communication, the signal input from SBTn pin is masked in serial interface to prevent errors by noise or so; thus input an external clock after releasing the mask by communication activation factors.

In addition, if "set transmission data" or "set dummy data" is performed as a communication activation factor, input the external clock after 3.5 transfer clock interval or longer after the data is set to TXBUFn. This wait time is needed to load the data from TXBUFn to internal shift register.

Table:18.3.1 Synchronous Serial Interface Activation Factors

	Activation factors	
	Transmission	Reception
Master communication	Set transmission data	Set dummy data
		Input start condition
Slave communication	Input clock after transmission data is set	Input clock after dummy data is set
		Input clock after start condition is input



SBTn pin is masked in serial interface to prevent errors caused by noise except during communication. In slave communication, set data to TXBUFn or input a start condition before inputting clock to SBTn pin.



Wait 3.5 transfer clocks or longer before inputting the external clock after the data set to TXBUFn. Otherwise, proper operation cannot be guaranteed.

■ Transfer Bit Setup

The transfer bit count can be selected from 1 to 8 bits. Set the SCnLNG2 to 0 flag of SCnMD0 register (at reset: 111). The SCnLNG2 to 0 flags hold the previous set value until other value is set.

■ Start Condition Setup

The SCnSTE flag of SCnMD0 register sets whether a start condition is enabled or disabled. When the data (SBIn (3-wire) or SBOIn (2-wire)) pin changes from "High" to "Low" during the clock (SBTn) pin = "High", setting the SCnCE1 flag of SCnMD0 register to "0" allows a start condition to be recognized. When the data (SBIn (3-wire) or SBOIn pin (2-wire)) pin changes from "High" to "Low" during the clock (SBTn) pin = "Low", setting the SCnCE1 flag to "1" allows a start condition to be recognized. Set the SCnSBOS flag and the SCnSBIS flag of SCnMD1 register to "0" respectively to change the start condition enable/disable setting. When operating transmission and reception at the same time, select "start condition disable"; otherwise, proper operation cannot be guaranteed.

■ First Transfer Bit Setup

Either MSB first or LSB first at transfer can be selected. Set the first bit by the SCnDIR flag of SCnMD0 register.

■ Transmission Data Buffer

The transmission data buffer TXBUF_n is a spare buffer which stores data to be loaded into internal shift register. Store the data to be transmitted into transmission data buffer TXBUF_n. The data is automatically loaded into internal shift register. The data loading period of 3 transfer clocks is needed for loading data. During the data loading period, setting another data in TXBUF_n may cause an error. The transmission buffer empty flag SCnTEMP of SCnSTR register can be monitored to determine whether it is within a data loading period. When data is set in TXBUF_n, the SCnTEMP flag is set to "1" until the data loading is finished. When communication is restarted, the SCnTEMP flag is automatically cleared to "0".

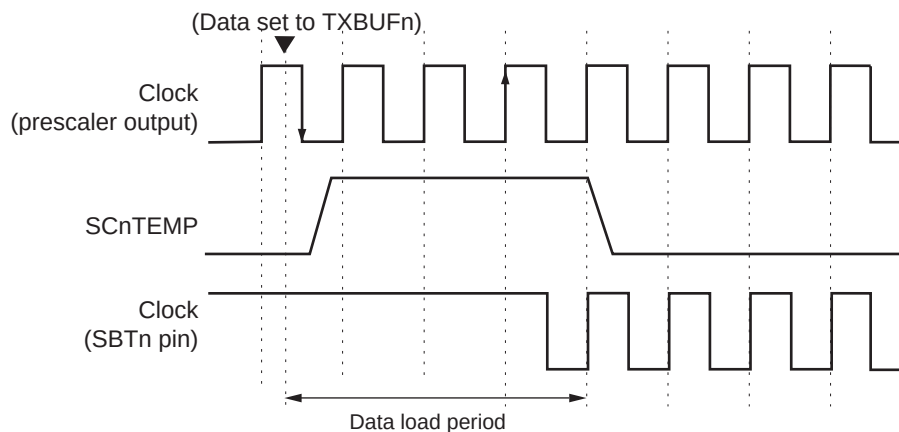


Figure:18.3.1 Transmission Data Buffer

■ Reception Data Buffer

The reception data buffer RXBUF_n are the spare buffers which save the data received by reception shift register. After Serial Interface *n* interrupt SCn(T)IRQ is generated, data in internal shift register are automatically stored in the reception data buffers RXBUF_n. RXBUF_n can store data up to 1 byte. Read the data in RXBUF_n before the next reception is completed since RXBUF_n is rewritten every time the reception is completed. The reception data buffer empty flags SCnREMP is set to "1" after SCn(T)IRQ is generated. SCnREMP is cleared to "0" when reading RXBUF_n.



If a start condition is input to restart during reception, the transmission data is invalid. Set the transmission data in TXBUF_n again to restart communication.



RXBUF_n is rewritten every time the reception is completed. In continuous communication, read the data in RXBUF_n before the next reception is completed.

■ Transmit Bit Count and First Transfer Bit

In transmission, when the transfer bit is 1 bit to 7 bits, data storage method for the transmission data buffer differs depending on the first transfer bit specification. At MSB first, store data in the upper bits of TXBUF_n. When there are 6 bits to be transferred, from "F" to "A", data "A" to "F" are stored in bp2 to bp7 of TXBUF_n, as shown in Figure:18.3.2. At LSB first, store data in the lower bits of TXBUF_n. When there are 6 bits to be transferred, from "F" to "A", data "A" to "F" are stored in bp0 to bp5 of TXBUF_n, as shown in Figure:18.3.3.

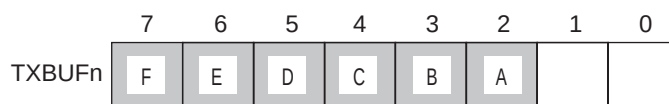


Figure:18.3.2 Transfer Bit Count and First Transfer Bit (At MSB First)



Figure:18.3.3 Transfer Bit Count and First Transfer Bit (At LSB First)

■ Reception Bit Count and First Transfer Bit

In reception, when the transfer bit is 1 to 7 bits, data storage method for the reception data buffer RXBUF_n differs depending on the first transfer bit specification. At MSB first, reception data are stored in the lower bits of RXBUF_n. When there are 6 bits to be transferred, with "A" to be the first bit to transfer and "F" to be the last bit to transfer, data "A" to "F" are stored in bp5 to bp0 of RXBUF_n, as shown in Figure:18.3.4. At LSB first, data are stored in the upper bits of RXBUF_n. When there are 6 bits to be transferred, with "A" to be the first bit to transfer and "F" to be the last bit to transfer, data "A" to "F" are stored in bp2 to bp7 of RXBUF_n, as shown in Figure:18.3.5.

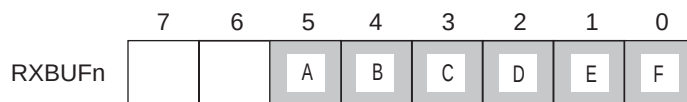


Figure:18.3.4 Reception Bit Count and Transfer First Bit (At MSB First)

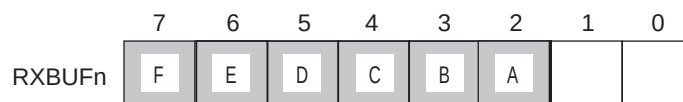


Figure:18.3.5 Reception Bit Count and Transfer First Bit (At LSB First)



When the reception transfer bit count shows 1 to 7 bits, data except the reception data of the specified transfer bit is undefined. The reception data should be masked to use with and instruction or others.

■ Continuous Transmission

This serial interface has a continuous communication function. If data is set in the transmission data buffer TXBUF_n during transmission, the transmission buffer empty flag SCnTEMP is set, and transmission is automatically continued. Set data in TXBUF_n before Serial Interface n interrupt SCn(T)IRQ is generated after data is loaded into transmission shift register. Communication blanks, from SCn(T)IRQ generation to the next transfer clock output, are 4 transfer clocks.

■ Automatic Continuous Transfer by ATC

This serial interface can start up by ATC_n, the automatic data transfer function built-in this LSI. When starting by ATC_n, data up to 255 byte can be transferred continuously. For the start-up method by ATC_n, refer to transfer modes 8 to 9 in [Chapter 23 23.1 Automatic Transfer Controller].

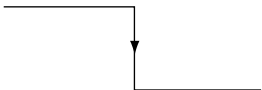
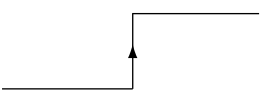
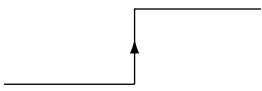
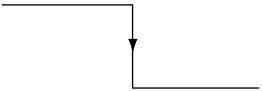


When the slave reception is performed with the start condition "enable" in the continuous communication, the system configuration is needed to notify the master of the read completion. Without the notification, the data which is read previously may be overwritten.

■ Input Edge/Output Edge Setup

An output edge of transmission data and an input edge of reception data can be set by the SCnCE1 flag of SCnMD0 register. When the SCnCE1 flag = "0", transmission data is output synchronously at the falling edge of the clock; when the SCnCE1 flag = "1", the data is output synchronously at the rising edge. When the SCnCE1 flag = "0", reception data is loaded synchronously at the rising edge of the clock; when the SCnCE1 flag = "1", the data is loaded synchronously at the falling edge.

Table:18.3.2 Transmission Data Output Edge and Reception Data Input Edge

SCnCE1	Transmission data output edge	Reception data input edge
0		
1		

■ Clock Setup

The clock source can set with a dedicated prescaler or timer (2 systems) output. Set it by the SCnPSC2 to 0 of SCnMD3 register. The dedicated prescaler starts to operate when "prescaler operation" is selected by the SCnP-SCE flag of SCnMD3 register. Either the internal clock (clock master) or the external clock (clock slave) can be selected by the SCnMST flag of SCnMD1 register. If selecting the external clock, set the internal clock that has the same clock cycle or lower to the external clock by SCnMD3 register. Table:18.3.3 shows the internal clock source that can be set by SCnMD3 register.

Table:18.3.3 Synchronous Serial Interface Clock Source

		Serial Interface 0	Serial Interface 1	Serial Interface 2	Serial Interface 3	Serial Interface 4
Clock source (internal clock)	fpll-div/2	√	√	√	√	√
	fpll-div/4	√	√	√	√	√
	fpll-div/8	-	-	-	-	-
	fpll-div/16	√	√	√	√	√
	fpll-div/32	-	-	-	-	√
	fpll-div/64	√	√	√	√	-
	fs/2	√	√	√	√	√
	fs/4	√	√	√	√	√
	Timer 0 output	√	√	√	√	√
	Timer 1 output	√	√	√	√	√
	Timer 2 output	√	√	√	√	√
	Timer 3 output	√	√	√	√	√
	Timer 4 output	√	√	√	√	√
	Timer A output	√	√	√	√	√

■ Serial Interface Internal Operating Clock Source Selection (MN101EF76K/MN101EF56K)

Serial interface internal clock source can be selected at clock synchronous master communication.

When the SCnSSC flag is set to "0", it is possible to communicate by operating the serial interface internal logic with the communication clock output from SBTn pin on the condition similar to the clock input to the slave device.

When the SCnSSC flag is "1", the influence of the exogenous noise on the communication clock line can be decreased by operating the Serial Interface internal logic directly with the clock generated in the communication clock control circuit.

The SCnSSC flag is not available on the clock synchronous slave communication, UART, or multi master IIC mode.

Table:18.3.4 Clock Source Selection Function for Internal Circuit Operations

Table remarks ✓: With function -: Without function

Serial Interfaces	MN101EF76K MN101EF56K	MN101EF57G
Serial Interface 0	✓	-
Serial Interface 1	✓	-
Serial Interface 2	✓	-
Serial Interface 3	✓	-
Serial Interface 4	✓	-

Without function, the operation is in condition of "SCnSSC=0".



The transfer speed should be up to 5.0 MHz. If the transfer clock exceeds 5.0 MHz, the transfer data may not be sent correctly.



Be sure to set the SCnSBIS flag and SC0SBOS flag of SCnMD1 register to "0" to switch clock setting.

■ Interrupt

When the clock synchronous serial interface is used, only one type interrupt, SCn(T)IRQ is available. SCn(T)IRQ is generated at the completion of transmission/reception.

■ Data Input Pin Setup

3-wire communication (clock pin (SBTn), data output pin (SBO_n) and data input pin (SBI_n)) or 2-wire type (clock pin (SBTn) and data I/O pin (SBO_n)) can be selected as communication mode. Set it by the SCnIOM flag of SCnMDI register. SBI_n pin can be used only for serial data input. SBO_n pin can select to be a serial data input or output. Selecting "data input from SBO_n pin" sets 2-wire communication; I/O mode selection of port direction control register controls direction of SBO_n pin to switch between transmission/reception. At this time, SBI_n pin can be used as a general-purpose port.



In reception, if SCnIOM of SCnMD1 register is set to "1" to select "serial data input from SBO_n pin", SBI_n pin can be used as a general-purpose port.

■ Reception Buffer Empty Flag

After completion of reception (Serial Interrupt n interrupt SCnTIRQ is generated), data is automatically stored into RXBUF_n from internal shift register. If data is stored into shift register RXBUF_n when the SCnSBIS flag of SC0MD_n register is set to "serial input", the reception buffer empty flag SCnREMP of SCnSTR register is set to "1". This indicates that the received data is waiting to be read. SCnREMP is cleared to "0" by reading data of RXBUF_n.

■ Transmission Buffer Empty Flag

During communication (until Serial Interrupt n interrupt SCnTIRQ is generated since data is loaded into internal shift register) if data is set in TXBUF_n, the transmission buffer empty flag SCnTEMP of SCnSTR register is set to "1". This indicates that the next transmission is waiting to be loaded. When data is loaded to internal shift register from TXBUF_n after SCnTIRQ is generated and SC0TEMP is cleared to "0", the next transfer starts automatically.

■ Overrun Error and Error Monitor Flag (Serial Interface 0, 1, 2 and 3)

When the next data reception is completed before reading the data of reception data buffer RXBUF_n received in previous communication, an overrun error is generated and the SCnORE flag of SCnSTR register is set to "1". At the same time, the error monitor flag SCnERE is set indicating that the reception has an error. The SCnORE flag is cleared when the next Serial Interrupt n interrupt SCnTIRQ is generated the data of RXBUF_n is read. SCnERE is also cleared after the SCnORE flag is cleared. These error flags have no effects on the communication operation.

■ Reception BUSY Flag Serial Interface 0, 1, 2 and 3)

If data is set in RXBUF_n or a start condition is recognized when the SCnSBIS flag of SCnMD1 register is set to "serial data input", the BUSY flag of SCnSTR register (SCnRBSY) is set to "1". The SCnRBSY flag is cleared to "0" after Serial Interface n interrupt SCnTIRQ is generated. During continuous communication, the set SCnRBSY flag is retained. If the transmission buffer empty flag SCnTEMP has already been cleared to "0" when Serial Interface n interrupt SCnTIRQ is generated, SCnRBSY is cleared to "0". If the SCnSBIS flag is set to "0" during communication, the SCnRBSY flag is cleared to "0".

■ Transmission BUSY Flag Serial Interface 0, 1, 2 and 3)

If data is set in TXBUF_n or a start condition is recognized when the SCnSBOS flag of SCnMD1 register is set to "serial data output", the transmission Busy SCnTBSY flag of SCnSTR register is set. After Serial Interface n interrupt SCnTIRQ is generated, the flag is cleared to "0". During continuous communication, the set SCnTBSY flag is retained. If the transmission buffer empty flag SCnTEMP is cleared to "0" when Serial Interface n interrupt SCnTIRQ is generated, SCnTBSY flag has been already cleared to "0". If the SCnSBOS flag is set to "0" during communication, the SCnTBSY flag is reset to "0".

■ BUSY Flag (Serial Interface 4)

If the data is set in TXBUF₄ or a start condition is recognized, SC4BSY flag of the SC4STR0 is set. It is cleared to "0" after Serial Interface 4 interrupt SC4IRQ is generated. During continuous communication, the set SC4BSY flag is retained. If the transmission buffer empty flag SC4TEMP has already been cleared to "0" when Serial Interface 4 interrupt SC4IRQ is generated, SC4BSY is cleared to "0".

■ Emergency Reset

This serial interface has an emergency reset function for malfunction. The SCnSBOS flag and the SCnSBIS flag of SCnMD1 register should be set to "0" (SBOn pin function: port, input data: "1" input) to operate an emergency reset.

At emergency reset, status register (the SCnBRKF flag of SCnMD2 register, all flags of SC0STR, SC1STR and SC2STR registers, the bp6 to 0 flags of SC4STR1 register and bp5 (SC4STPC flag) of SC4MD3 register) is initialized to the reset value, but the set value of the any other control registers are held.

■ Last Bit of Transmission Data

The following table shows the data output hold period for the last bit in transmission and the required minimum data input period for the last bit in reception. The internal clock should be set at slave to keep the data hold time in transmission.

Table:18.3.5 Last Bit Data Length of Transfer Data

	The last bit data hold period at transmission	The last bit data input period at reception
At master	1 bit data length	1 bit data length (minimum)
At slave	$[1 \text{ bit data length of external clock} \times 1/2] + [\text{internal clock cycle} \times (1/2 \text{ to } 3/2)]$	

When the start condition is disabled (SCnSTE flag = 0), SBO0 output after the data output hold period of the last bit can be set by the SCnFDC1 to 0 flags of SCnMD3 register as indicated in Figure:18.3.6.

After the reset is released, the output before a serial transfer is "High" despite the set value of the SCnFDC1 to 0 flags. When the start condition is enabled (SCnSTE flag = 1), "High" is output despite the set values of the SCnFDC1 to 0 flags.

Table:18.3.6 SBO0 Output after Data Output Hold Period of Last Bit (Without Start Condition)

SCnFDC1 flag	SCnFDC0 flag	SBO0 output after data output hold period of last bit
0	0	Fixed to "High" output
1	0	Fixed to "Low" output
0	1	Hold the final data
1	1	Setting prohibited

■ Other Control Flag Setup (Serial Interface 0, 1, 2 and 3)

The flags shown below are not needed to be set or monitored since they are not used at clock synchronous communication.

Table:18.3.7 Other Control Flags

Serial Interface 0		Serial Interface 1		Serial Interface 2		Serial Interface 3		Detail
Register	Flag	Register	Flag	Register	Flag	Register	Flag	
SC0MD2	SC0BRKE	SC1MD2	SC1BRKE	SC2MD2	SC2BRKE	SC3MD2	SC3BRKE	Break status transmission control
	SC0BRKF		SC1BRKF		SC2BRKF		SC3BRKF	Break status reception monitor
	SC0NPE		SC1NPE		SC2NPE		SC3NPE	Parity enable
	SC0PM1 SC0PM0		SC1PM1 SC1PM0		SC2PM1 SC2PM0		SC3PM1 SC3PM0	Additional bit specification
	SC0FM1 SC0FM0		SC1FM1 SC1FM0		SC2FM1 SC2FM0		SC3FM1 SC3FM0	Frame mode specification
SC0STR	SC0PEK	SC1STR	SC1PEK	SC2STR	SC2PEK	SC3STR	SC3PEK	Parity error detection
	SC0FEF		SC1FEF		SC2FEF		SC3FEF	Frame error detection

18.3.2 Clock Synchronous Operation Timing

■ Transmission Timing

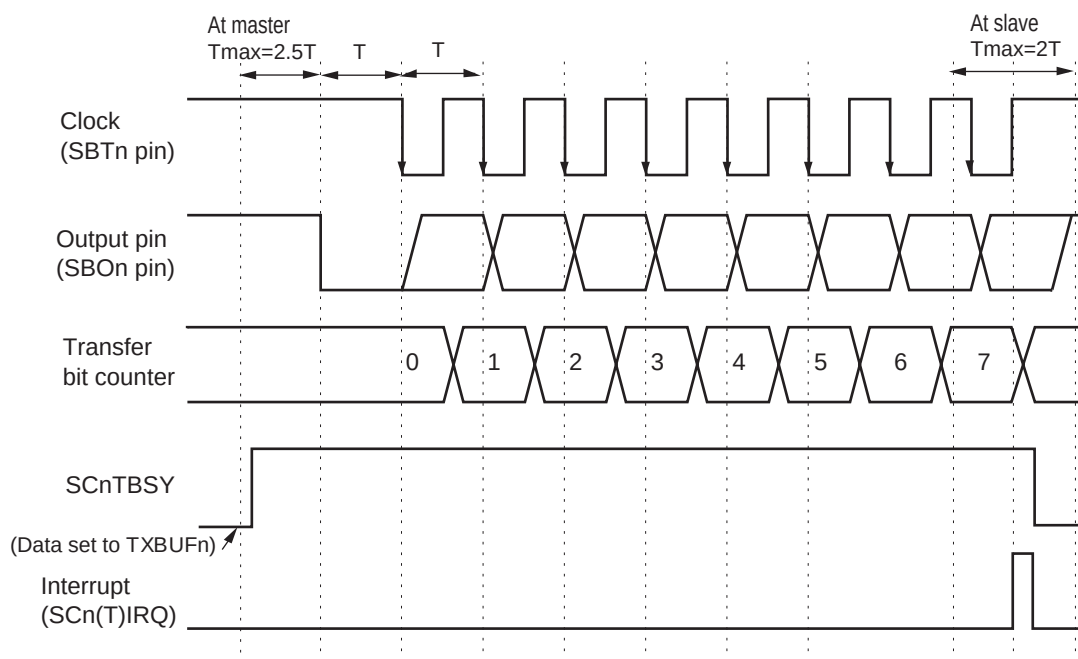


Figure:18.3.6 Transmission Timing (At Falling Edge, Start Condition is Enabled)

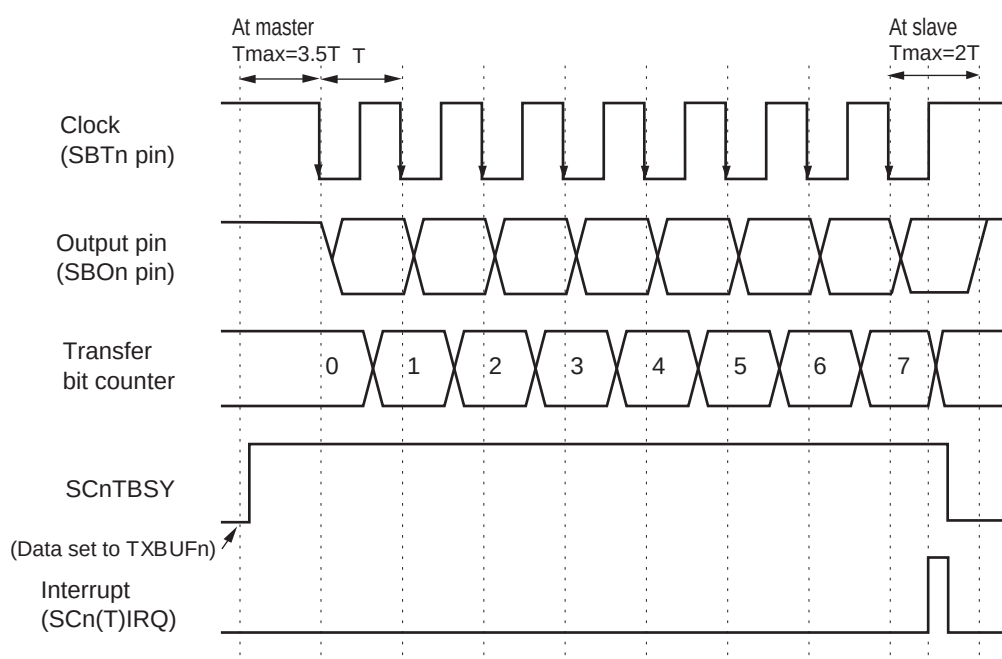


Figure:18.3.7 Transmission Timing (At Falling Edge, Start Condition is Disabled)

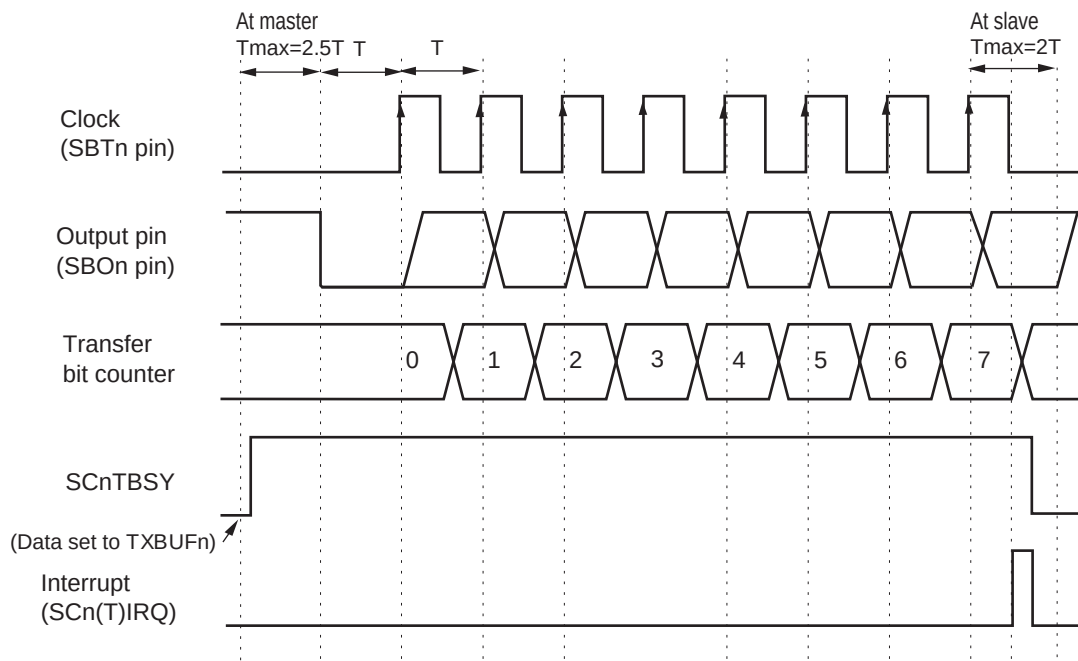


Figure:18.3.8 Transmission Timing (At Rising Edge, Start Condition is Enabled)

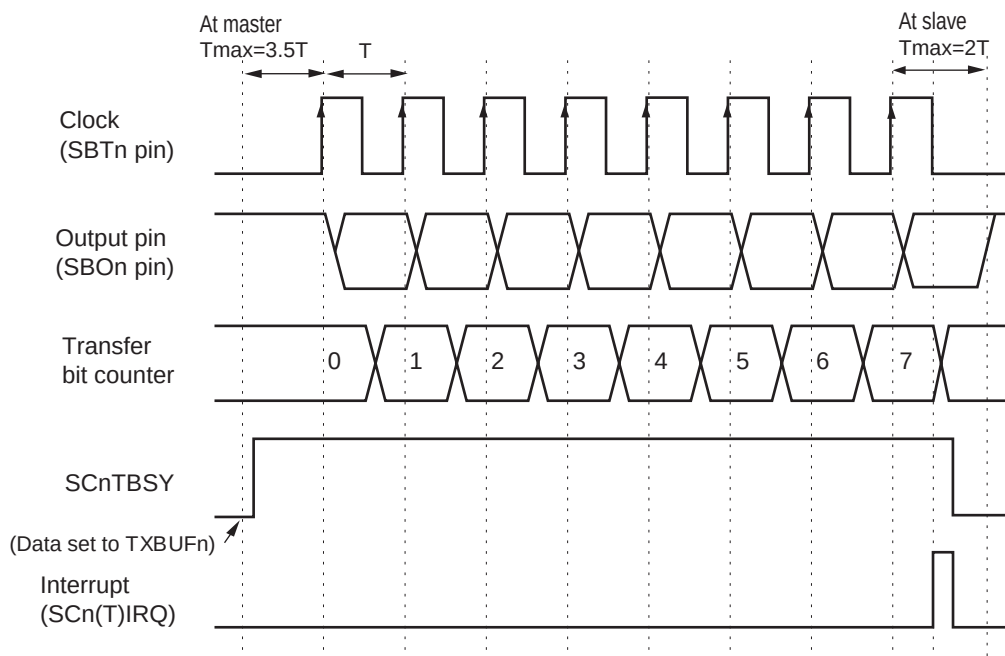


Figure:18.3.9 Transmission Timing (At Rising Edge, Start Condition is Disabled)

■ Reception Timing

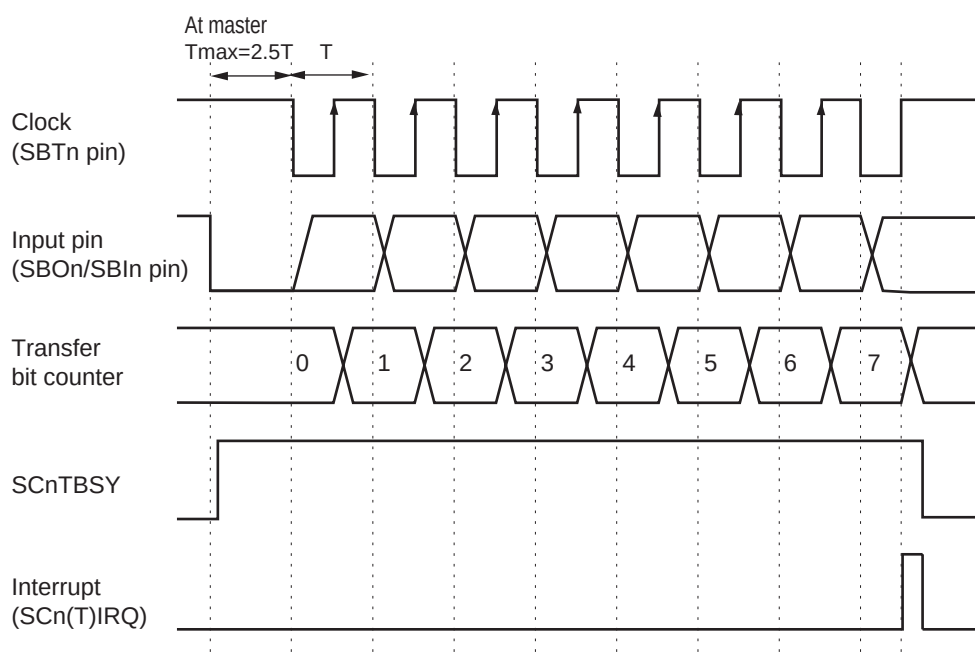


Figure:18.3.10 Reception Timing (At Rising Edge, Start Condition is Enabled)

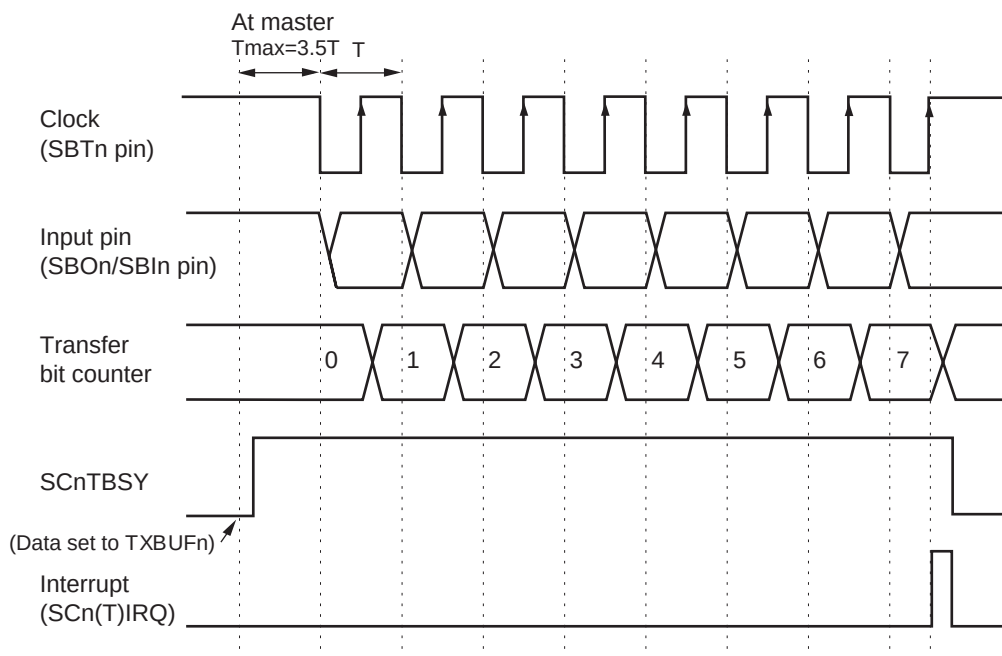


Figure:18.3.11 Reception Timing (At Rising Edge, Start Condition is Disabled)

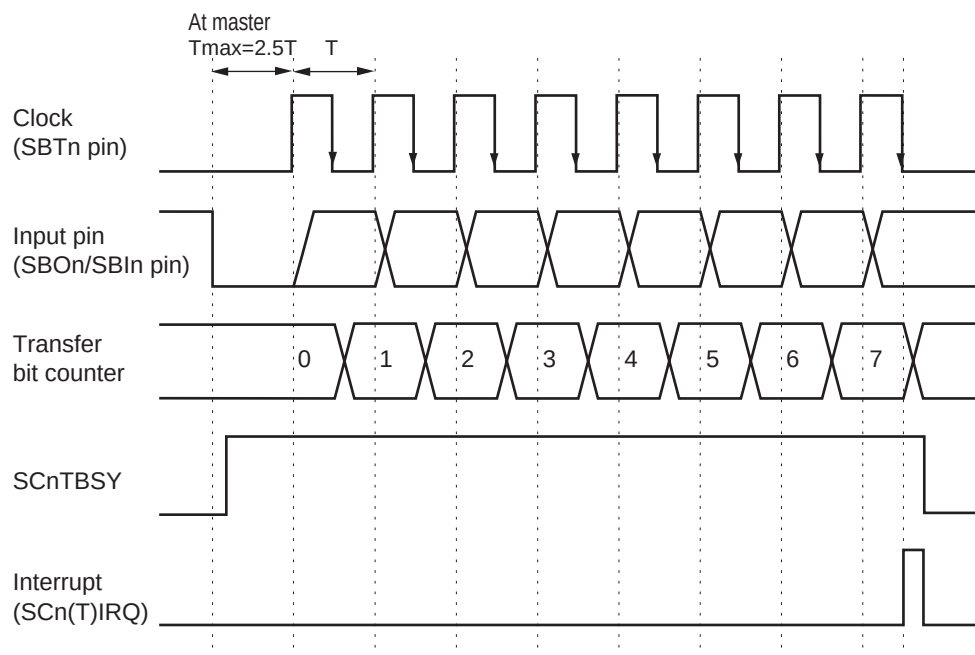


Figure:18.3.12 Reception Timing (At Falling Edge, Start Condition is Enabled)

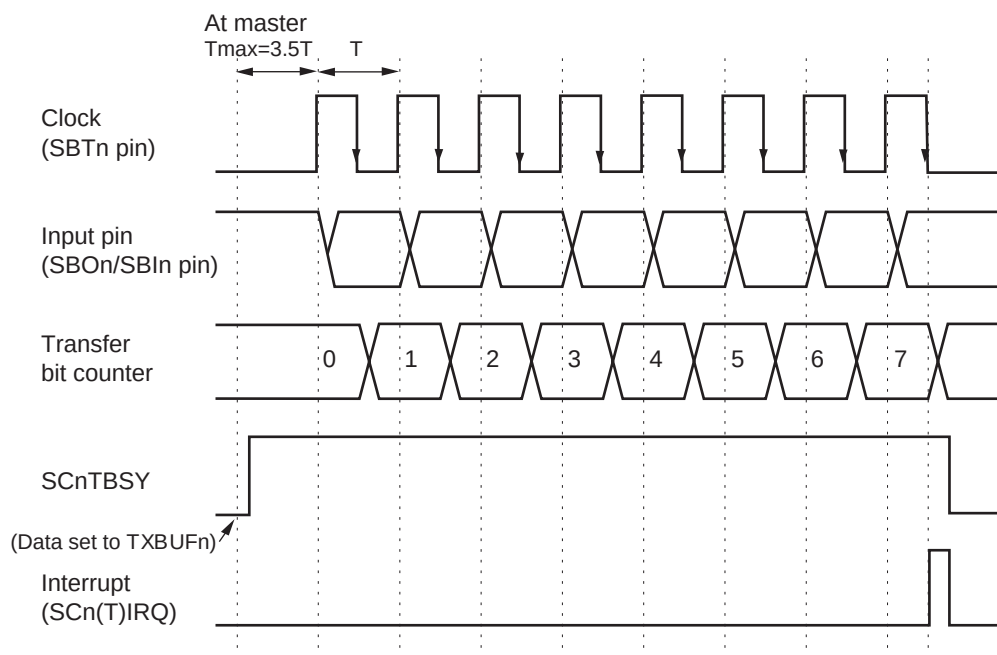


Figure:18.3.13 Reception Timing (At Falling Edge, Start Condition is Disabled)

■ Transmission/Reception Timing 1

When transmission and reception are performed at the same time, the SCnCE1 flag of SCnMD0 register should be set to "0" or "1". Data is received at the opposite edge timing of the output edge of transmission data. Set the polarity of the reception data input edge to be opposite to the output edge of the target transmission data.

Select "start condition disable" when transmission and reception are performed simultaneously.

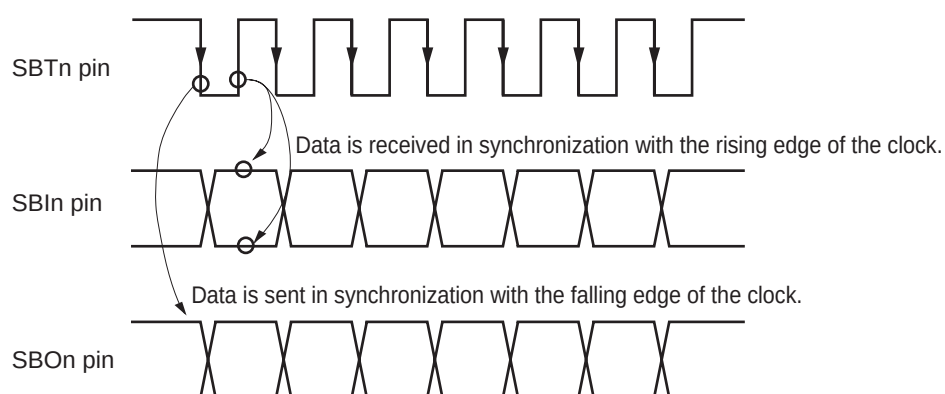


Figure:18.3.14 Transmission/ Reception Timing
(Reception: at Rising Edge, Transmission: at Falling Edge)

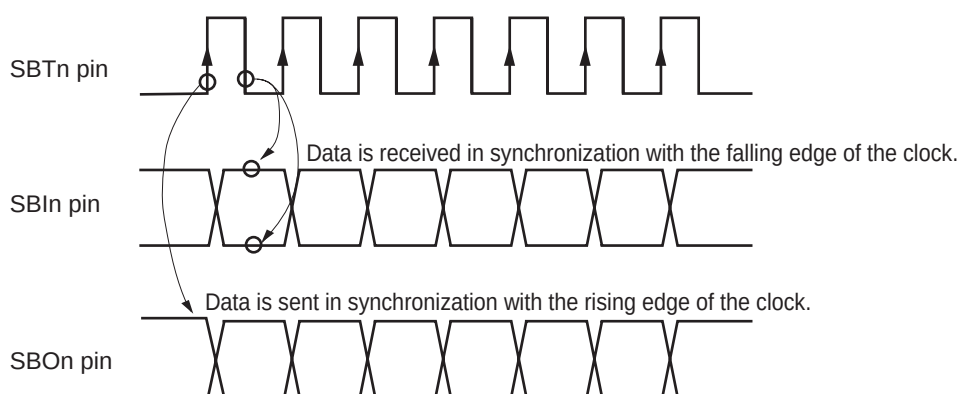


Figure:18.3.15 Transmission/Reception Timing
(Reception: at Falling Edge, Transmission: at Rising Edge)



Select "disable start condition" when transmission and reception are performed simultaneously. Otherwise, proper operation cannot be guaranteed.

18.3.3 Communication in STANDBY mode

■ Recovery from STANDBY Mode by Communication Completion Interrupt

This serial interface has the following method for recovering from STANDBY mode.

This serial interface can perform slave reception in STANDBY mode. CPU operation status can be recovered from STANDBY mode to CPU OPERATION mode by Serial Interface n interrupt SCnTIRQ generated after slave reception.

(In STANDBY mode, continuous reception is not available since the next data can not be accepted after the data of the transfer bit count, set by the SCnLNG2 to 0 flags of SCnMD0 register, is received once.) Read the received data from the reception data buffer RXBUF_n after recovering to CPU OPERATION mode.

During reception in STANDBY mode, communication with start condition enabled is not available. Disable the start condition. The dummy data should be set in the transmission data buffer TXBUF_n before the transition to STANDBY mode.

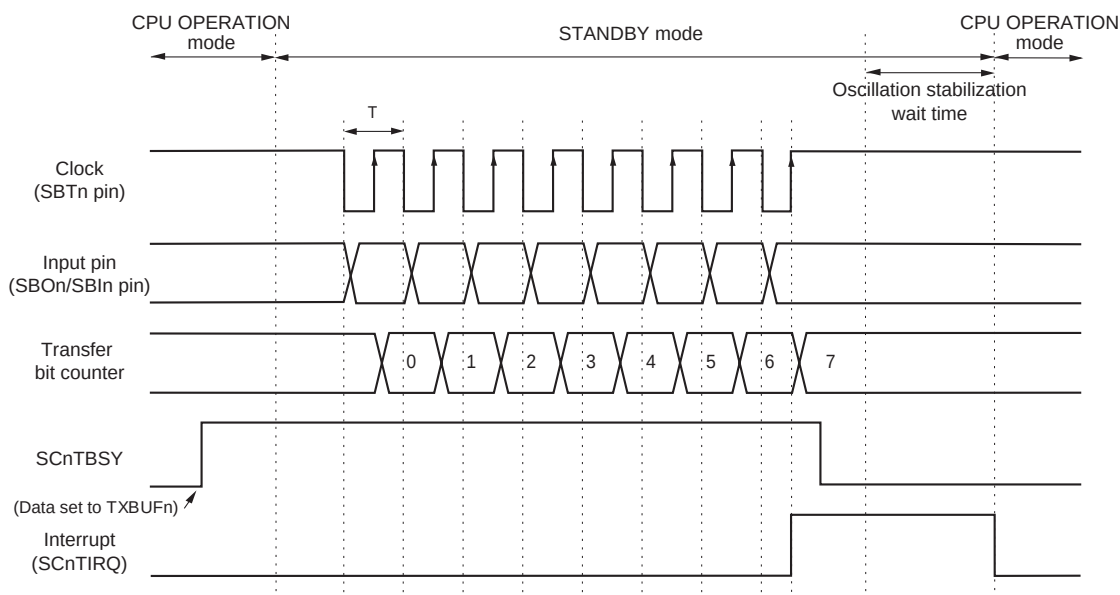


Figure:18.3.16 Reception Timing in STANDBY Mode
(Reception: at Rising Edge, Start Condition is Disabled)

■ Recovery from STANDBY Mode by Reception Interrupt (Serial Interface 0 and 1)

On Serial Interface 0 and 1, a Serial n UART reception interrupt (SCnRIRQ) is generated at the falling edge of the serial clock I/O pin (SBTn) by setting the SCnIGC flag of SCINTSEL register to "1".

By this function, the transition to CPU OPERATION mode can be faster than the recovery by a communication completion interrupt when recovering from STANDBY mode to CPU OPERATION mode by serial slave reception.

For example, the differences of operations in recovery from STOP mode are shown below.

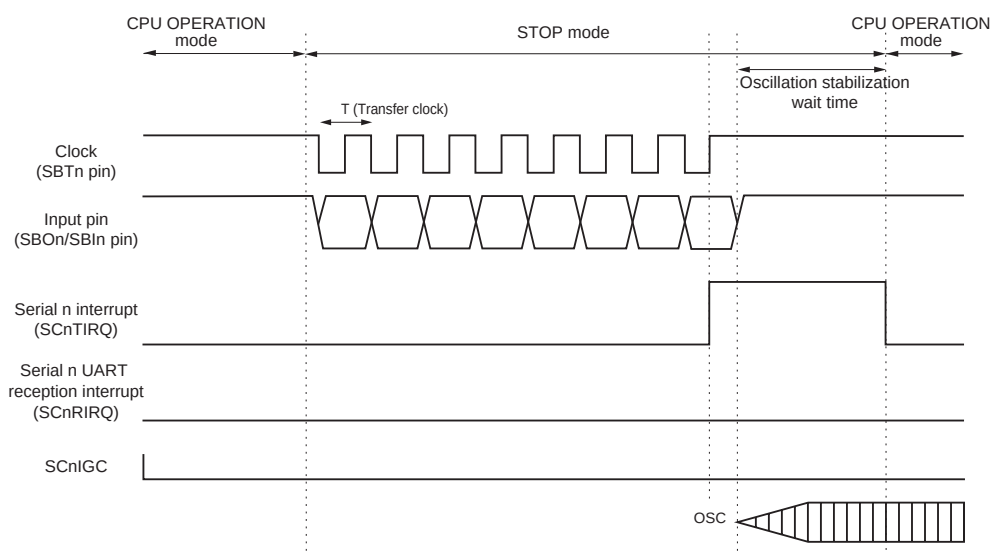


Figure:18.3.17 Recovery Timing from STOP mode
(Serial n UART Reception Interrupt is Not Generated: SCnIGC=0)

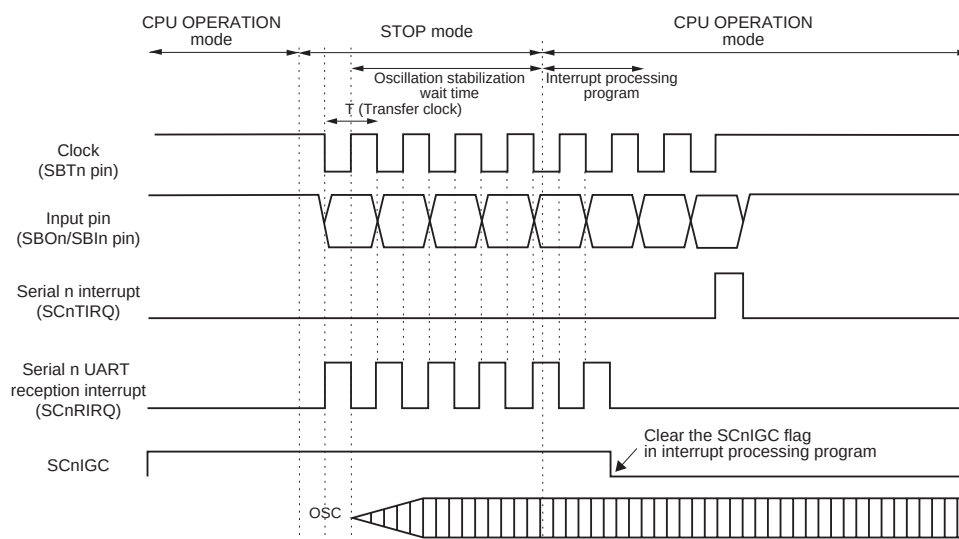


Figure:18.3.18 Recovery Timing from STOP mode
(Serial n UART Reception Interrupt is Generated: SCnIGC=1)



Figure:18.3.17 and Figure:18.3.18 indicate the examples when the serial transfer period is longer than the oscillation stabilization wait time.



To return from STANDBY mode by using Serial Interface n UART reception interrupts, execute the following operation in accordance with the procedures below.

1. Clear the SCnICG flag of SCINTSEL register to "0".
2. Clear the MIE flag of PSW register to "0".
3. Set the IRWE flag of MEMCTR register to "1".
4. Clear the SCnRIR flag of SCnRICR register to "0".
(Do not rewrite the SCnRLV1 to 0 flags.)
5. Clear the IRWE flag of MEMCTR register to "0".

Executing the procedures above can avoid multiple executions of interrupt processing program.



SCnIGC flag of clock synchronous serial reception interrupt control register (SCINTSEL) is set to "1" during slave reception in OPERATION mode.

A Serial n UART reception interrupt is generated at the falling edge of SBTn before the transition to STANDBY mode from OPERATION mode causing the SCnIGC flag to be cleared. As a result, The CPU will be stuck in STANDBY mode if the transition to STANDBY mode happened after the Serial Interface n UART reception interrupt where the SCnIGC flag is cleared.

To avoid this operation, check program counter (PC) which is saved to the stack during interrupt processing. If this improper operation could occur, rewrite PC of the stack to prevent the transition to STANDBY mode or save the SCnIGC flag and recover it after the interrupt processing.



Recovery from STANDBY mode is available only with synchronous functions. It cannot be performed with UART function.

18.3.4 PIn Setup

■ Synchronous Serial Interface 0 Pin Setup

Port	Wire system	Type	Pin	Pin setup (flag setup)								
				SC0SEL register	PnDIR register	PnODC register	PnPLUD register	SC0MD1 register				
				Serial 0 I/O pin switching	I/O mode selection	Nch open-drain output selection	Pull-up/pull-down resistor selection	Serial data input selection	SBT0 pin function selection	Serial input control selection	SBO0 (TXD0) pin function selection	
						Arbitrary setting	Arbitrary setting					
				0: P02 to P04 1: P43 to P45	0: input mode 1: output mode	0: push/pull 1: Nch open-drain	0: not added 1: added	0: data input from SBI0 1: data input from SBO0	0: port 1: transfer clock I/O	0: "1" input 1: serial input	0: port 1: serial data output	
		OSL0	PnDIRm	PnODCm	PnPLUDm	SC0IOM	SC0SBTS	SC0SBIS	SC0SBOS			
Port 0	3-wire	Transmission only	P03/SBO0A		0	P0DIR3:1	P0ODC3	P0PLUD3	0	1	0	1
			-			-	-	-				
			P04/SBT0A	Master		P0DIR4:1	P0ODC4	P0PLUD4				
				Slave		P0DIR4:0	-					
		Reception only	-		0	P0DIR2:0	-	-	0	1	1	0
			P02/SBI0A			P0DIR4:1	P0ODC4	P0PLUD4				
			P04/SBT0A	Master		-						
				Slave		-						
		Transmission/Reception	P03/SBO0A		0	P0DIR3:1	P0ODC3	P0PLUD3	0	1	1	1
			P02/SBI0A			P0DIR2:0	-	-				
			P04/SBT0A	Master		P0DIR4:1	P0ODC4	P0PLUD4				
				Slave		P0DIR4:0	-					
	2-wire	Transmission	P03/SBO0A		0	P0DIR3:1	P0ODC3	P0PLUD3	1	1	0	1
			-			-	-	-				
			P04/SBT0A	Master		P0DIR4:1	P0ODC4	P0PLUD4				
				Slave		P0DIR4:0	-					
Reception		P03/SBO0A		0	P0DIR3:0	-	-	1	1	1	0	
		-			-	-	-					
		P04/SBT0A	Master		P0DIR4:1	P0ODC4	P0PLUD4					
			Slave		P0DIR4:0	-						
Port 4	3-wire	Transmission only	P43/SBO0B		1	P4DIR3:1	P4ODC3	P4PLUD3	0	1	0	1
			-			-	-	-				
			P45/SBT0B	Master		P4DIR5:1	P4ODC5	P4PLUD5				
				Slave		P4DIR5:0	-					
		Reception only	-		1	P4DIR4:0	-	-	0	1	1	0
			P44/SBI0B			P4DIR5:1	P4ODC5	P4PLUD5				
			P45/SBT0B	Master		-						
				Slave		-						
		Transmission/Reception	P43/SBO0B		1	P4DIR3:1	P4ODC3	P4PLUD3	0	1	1	1
			P44/SBI0B			P4DIR4:0	-	-				
			P45/SBT0B	Master		P4DIR5:1	P4ODC5	P4PLUD5				
				Slave		P4DIR5:0	-					
	2-wire	Transmission	P43/SBO0B		1	P4DIR3:1	P4ODC3	P4PLUD3	1	1	0	1
			-			-	-	-				
			P45/SBT0B	Master		P4DIR5:1	P4ODC5	P4PLUD5				
				Slave		P4DIR5:0	-					
Reception		P43/SBO0B		1	P4DIR3:0	-	-	1	1	1	0	
		-			-	-	-					
		P45/SBT0B	Master		P4DIR5:1	P4ODC5	P4PLUD5					
			Slave		P4DIR5:0	-						

■ Synchronous Serial Interface 1 Pin Setup

Port	Wire system	Type	Pin	Pin setup (flag setup)														
				SC1SEL register	PnDIR register	PnODC register	PnPLUD register	SC1MD1 register										
				Serial 1 I/O pin switching	I/O mode selection	Nch open-drain output selection Arbitrary setting	Pull-up/pull-down resistor selection Arbitrary setting	Serial data input selection	SBT1 pin function selection	Serial input control selection	SBO1(TXD1) pin function selection							
				0: P50 to P52 1: P75 to P77	0: input mode 1: output mode	0: push/pull 1: Nch open-drain	0: not added 1: added	0: data input from SBI1 1: data input from SBO1	0: port 1: transfer clock I/O	0: "1" input 1: serial input	0: port 1: serial data output							
				OSL1	PnDIRm	PnODCm	PnPLUDm	SC1IOM	SC1SBTS	SC1SBIS	SC1SBOS							
Port 5	3-wire	Transmission only	P50/SBO1A		0	P5DIR0:1		P5ODC0	P5PULD0	0	1	0	1					
			-			-		-										
			P52/SBT1A	Master Slave		P5DIR2:1		P5ODC2	P5PULD2									
		P5DIR2:0		-														
		Reception only	-		0	-		-	-	0	1	1	0					
			P51/SBI1A			P5DIR1:0		-	-									
	P52/SBT1A		Master Slave	P5DIR2:1		P5ODC2	P5PULD2											
	P5DIR2:0		-															
	Transmission/reception	P50/SBO1A		0	P5DIR0:1		P5ODC0	P5PULD0	0	1	1	1						
		P51/SBI1A			P5DIR1:0		-	-										
		P52/SBT1A	Master Slave		P5DIR2:1		P5ODC2	P5PULD2										
					P5DIR2:0		-											
2-wire		Transmission	P50/SBO1A		0	P5DIR0:1		P5ODC0					P5PULD0	1	1	0	1	
			-			-		-										
	P52/SBT1A		Master Slave	P5DIR2:1		P5ODC2	P5PULD2											
	P5DIR2:0		-															
	Reception	P50/SBO1A		0	P5DIR0:0		-	-	1	1	1	0						
		-			-		-											
P52/SBT1A		Master Slave	P5DIR2:1		P5ODC2	P5PULD2												
P5DIR2:0		-																
Port 7	3-wire	Transmission only	P75/SBO1B		1	P7DIR5:1		P7ODC5	P7PLUD5	0	1	0	1					
			-			-		-										
			P77/SBT1B	Master Slave		P7DIR7:1		P7ODC7	P7PLUD7									
		P7DIR7:0			-													
		Reception only	-		1	-		-	-	0	1	1	0					
			P76/SBI1B			P7DIR6:0		-	-									
	P77/SBT1B		Master Slave	P7DIR7:1		P7ODC7	P7PLUD7											
	P7DIR7:0		-															
	Transmission/reception	P75/SBO1B		1	P7DIR5:1		P7ODC5	P7PLUD5	0	1	1	1						
		P76/SBI1B			P7DIR6:0		-	-										
		P77/SBT1B	Master Slave		P7DIR7:1		P7ODC7	P7PLUD7										
					P7DIR7:0		-											
2-wire		Transmission	P75/SBO1B		1	P7DIR5:1		P7ODC5					P7PLUD5	1	1	0	1	
			-			-		-										
	P77/SBT1B		Master Slave	P7DIR7:1		P7ODC7	P7PLUD7											
	P7DIR7:0		-															
	Reception	P75/SBO1B		1	P7DIR5:0		-	-	1	1	1	0						
		-			-		-											
P77/SBT1B		Master Slave	P7DIR7:1		P7ODC7	P7PLUD7												
P7DIR7:0		-																

■ Synchronous Serial Interface 2 Pin Setup

Port	Wire system	Type	Pin	Pin setup (flag setup)								
				SC2SEL register	PnDIR register	PnODC register	PnPLU register	SC2MD1 register				
				Serial 2 I/O pin switching	I/O mode selection	Nch open-drain output selection	Pull-up/ resistor selection	Serial data input selection	SBT2 pin function selection	Serial input control selection	SBO2(TXD2) pin function selection	
						Arbitrary setting	Arbitrary setting					
				0: P65 to P67 1: P30 to P32	0: input mode 1: output mode	0: push/pull 1: Nch open-drain	0: not added 1: added	0: data input from SBI2 1: data input from SBO2	0: port 1: transfer clock I/O	0: "1" input 1: serial input	0: port 1: serial data output	
		OSL2	PnDIRm	PnODCm	PnPLUm	SC2IOM	SC2SBTS	SC2SBIS	SC2SBOS			
Port 6	3-wire	Transmission only	P65/SBO2A		P6DIR5:1	P6DIR5:1	P6ODC5	P6PLU5	0	1	0	1
			-		-	-	-	-				
			P67/SBT2A	Master	P6DIR7:1	P6DIR7:1	P6ODC7	P6PLU7				
				Slave	P6DIR7:0	P6DIR7:0	-					
		Reception only	-		-	-	-	-	0	1	1	0
			P66/SBI2A		P6DIR6:0	P6DIR6:0	-	-				
			P67/SBT2A	Master	P6DIR7:1	P6DIR7:1	P6ODC7	P6PLU7				
				Slave	P6DIR7:0	P6DIR7:0	-					
	Transmission/reception	P65/SBO2A		P6DIR5:1	P6DIR5:1	P6ODC5	P6PLU5	0	1	1	1	
		P66/SBI2A		P6DIR6:0	P6DIR6:0	-	-					
		P67/SBT2A	Master	P6DIR7:1	P6DIR7:1	P6ODC7	P6PLU7					
			Slave	P6DIR7:0	P6DIR7:0	-						
	2-wire	Transmission	P65/SBO2A		P6DIR5:1	P6DIR5:1	P6ODC5	P6PLU5	1	1	0	1
			-		-	-	-	-				
			P67/SBT2A	Master	P6DIR7:1	P6DIR7:1	P6ODC7	P6PLU7				
				Slave	P6DIR7:0	P6DIR7:0	-					
Reception		P65/SBO2A		P6DIR5:0	P6DIR5:0	-	-	1	1	1	0	
		-		-	-	-	-					
		P67/SBT2A	Master	P6DIR7:1	P6DIR7:1	P6ODC7	P6PLU7					
			Slave	P6DIR7:0	P6DIR7:0	-						
Port 3	3-wire	Transmission only	P30/SBO2B		1	P3DIR0:1	P3ODC0	P3PLU0	0	1	0	1
			-			-	-	-				
			P32/SBT2B	Master		P3DIR2:1	P3ODC2	P3PLU2				
				Slave		P3DIR2:0	-					
		Reception only	-		1	-	-	-	0	1	1	0
			P31/SBI2B			P3DIR1:0	-	-				
			P32/SBT2B	Master		P3DIR2:1	P3ODC2	P3PLU2				
				Slave		P3DIR2:0	-					
	Transmission/reception	P30/SBO2B		1	P3DIR0:1	P3ODC0	P3PLU0	0	1	1	1	
		P31/SBI2B			P3DIR1:0	-	-					
		P32/SBT2B	Master		P3DIR2:1	P3ODC2	P3PLU2					
			Slave		P3DIR2:0	-						
	2-wire	Transmission	P30/SBO2B		1	P3DIR0:1	P3ODC0	P3PLU0	1	1	0	1
			-			-	-	-				
			P32/SBT2B	Master		P3DIR2:1	P3ODC2	P3PLU2				
				Slave		P3DIR2:0	-					
Reception		P30/SBO2B		1	P3DIR0:0	-	-	1	1	1	0	
		-			-	-	-					
		P32/SBT2B	Master		P3DIR2:1	P3ODC2	P3PLU2					
			Slave		P3DIR2:0	-						

■ Synchronous Serial Interface 3 Pin Setup (MN101EF76K/MN101EF56K)

Port	Wire sys-tem	Type	Pin	Pin setup (flag setup)														
				SC3SEL register	PnDIR register	PnODC register	PnPLUD register	SC3MD1 register										
				Serial 3 I/O pin switching	I/O mode selection	Nch open-drain output selection Arbitrary setting	Pull-up/- pull-down resistor selection Arbitrary setting	Serial data input selection	SBT3 pin function selection	Serial input control selection	SBO3(TXD3) pin function selection							
				0: P05 to P07 1: P40 to P42	0: input mode 1: output mode	0: push/pull 1: Nch open-drain	0: not added 1: added	0: data input from SBI3 1: data input from SBO3	0: port 1: transfer clock I/O	0: "1" input 1: serial input	0: port 1: serial data output							
				OSL3	PnDIRm	PnODCm	PnPLUDm	SC3IOM	SC3SBTS	SC3SBIS	SC3SBOS							
Port 0	3-wire	Transmission only	P05/SBO3A		0	P0DIR5:1		P0ODC5	P0PULD5	0	1	0	1					
			-			-		-										
			P07/SBT3A	Master Slave		P0DIR7:1		P0ODC7	P0PULD7									
		P0DIR7:0		-														
		Reception only	-		0	-		-	-	0	1	1	0					
			P06/SBI3A			P0DIR6:0		-	-									
	P07/SBT3A		Master Slave	P0DIR7:1		P0ODC7	P0PULD7											
	P0DIR7:0		-															
	Transmission/reception	P05/SBO3A		0	P0DIR5:1		P0ODC5	P0PULD5	0	1	1	1						
		P06/SBI3A			P0DIR6:0		-	-										
		P07/SBT3A	Master Slave		P0DIR7:1		P0ODC7	P0PULD7										
					P0DIR7:0		-											
2-wire		Transmission	P05/SBO3A		0	P0DIR5:1		P0ODC5					P0PULD5	1	1	0	1	
			-			-		-					-					
	P07/SBT3A		Master Slave	P0DIR7:1		P0ODC7	P0PULD7											
	P0DIR7:0		-															
	Reception	P05/SBO3A		0	P0DIR5:0		-	-	1	1	1	0						
		-			-		-	-										
P07/SBT3A		Master Slave	P0DIR7:1		P0ODC7	P0PULD7												
P0DIR7:0		-																
Port 4	3-wire	Transmission only	P40/SBO3B		1	P4DIR0:1		P4ODC0	P4PLUD0	0	1	0	1					
			-			-		-	-									
			P42/SBT3B	Master Slave		P4DIR2:1		P4ODC2	P4PLUD2									
		P4DIR2:0			-													
		Reception only	-		1	-		-	-	0	1	1	0					
			P41/SBI3B			P4DIR1:0		-	-									
	P42/SBT3B		Master Slave	P4DIR2:1		P4ODC2	P4PLUD2											
		P4DIR2:0		-														
	Transmission/reception	P40/SBO3B		1	P4DIR0:1		P4ODC0	P4PLUD0	0	1	1	1						
		P41/SBI3B			P4DIR1:0		-	-										
		P42/SBT3B	Master Slave		P4DIR2:1		P4ODC2	P4PLUD2										
					P4DIR2:0		-											
2-wire		Transmission	P40/SBO3B		1	P4DIR0:1		P4ODC0					P4PLUD0	1	1	0	1	
			-			-		-					-					
	P42/SBT3B		Master Slave	P4DIR2:1		P4ODC2	P4PLUD2											
		P4DIR2:0		-														
	Reception	P40/SBO3B		1	P4DIR0:0		-	-	1	1	1	0						
		-			-		-	-										
P42/SBT3B		Master Slave	P4DIR2:1		P4ODC2	P4PLUD2												
	P4DIR2:0		-															

■ Synchronous Serial Interface 4 Pin Setup

Port	Wire system	Type	Pin	Pin setup (flag setup)							
				SC4SEL register	PnDIR register	PnODC register	PnPLUD register	SC3MD1 register			
				Serial 4 I/O pin switching	I/O mode selection	Nch open-drain output selection Arbitrary setting	Pull-up/pull-down resistor selection Arbitrary setting	Serial data input selection	SBT4 pin function selection	Serial input control selection	SBO4(TXD4) pin function selection
				0: P70 to P72 1: P33 to P35 OSL4	0: input mode 1: output mode PnDIRm	0: push/pull 1: Nch open-drain PnODCm	0: not added 1: added PnPLUDm	0: data input from SBI4 1: data input from SBO4 SC4IOM	0: port 1: transfer clock I/O SC4SBTS	0: "1" input 1: serial input SC4SBIS	0: port 1: serial data output SC4SBOS
Port 7	3-wire	Transmission only	P71/SBO4A	0	P7DIR1:1	P7ODC1	P7PLUD1	0	1	0	1
			-		-	-	-				
			P72/SBT4A Master		P7DIR2:1	P7ODC2	P7PLUD2				
		Reception only	P72/SBT4A Slave	0	P7DIR2:0	-	-	0	1	1	0
			-		P7DIR0:0	-	-				
			P70/SBI4A Master		P7DIR2:1	P7ODC2	P7PLUD2				
	2-wire	Transmission / reception	P72/SBT4A Slave	0	P7DIR2:0	-	-	0	1	1	1
			P71/SBO4A		P7DIR1:1	P7ODC1	P7PLUD1				
			-		P7DIR0:0	-	-				
		Transmission	P72/SBT4A Master	0	P7DIR2:1	P7ODC2	P7PLUD2	1	1	0	1
			P71/SBO4A		P7DIR1:1	P7ODC1	P7PLUD1				
			-		P7DIR2:0	-	-				
Port 3	3-wire	Transmission only	P72/SBT4A Slave	1	P7DIR2:0	-	-	0	1	0	1
			P33/SBO4B		P3DIR3:1	P3ODC3	P3PLU3				
			-		-	-	-				
		Reception only	P34/SBT4B Master	1	P3DIR4:1	P3ODC4	P3PLU4	0	1	1	0
			P35/SBI4B		P3DIR5:0	-	-				
			-		P3DIR4:1	P3ODC4	P3PLU4				
	2-wire	Transmission / reception	P34/SBT4B Slave	1	P3DIR4:0	-	-	0	1	1	1
			P33/SBO4B		P3DIR3:1	P3ODC3	P3PLU3				
			-		P3DIR5:0	-	-				
		Transmission	P34/SBT4B Master	1	P3DIR4:1	P3ODC4	P3PLU4	1	1	0	1
			P33/SBO4B		P3DIR3:1	P3ODC3	P3PLU3				
			-		-	-	-				
	Reception	Reception	P34/SBT4B Master	1	P3DIR4:1	P3ODC4	P3PLU4	1	1	1	0
			P33/SBO4B		P3DIR3:0	-	-				
			-		P3DIR4:1	P3ODC4	P3PLU4				

18.3.5 Setup Example

■ Transmission/Reception Setup Example

The setup example for clock synchronous serial communication using Serial Interface 1 is shown. Table:18.3.8 shows the conditions in transmission/reception. The basic setup procedures are the same in Serial Interface 0, 2 and 4. However, pin settings (4) and (5) differ in each serial interface.

Table:18.3.8 Setup Examples for Synchronous Serial Interface Transmission/Reception

Setup item	Set to
Serial data input selection	SBI1
Transfer bit count	8 bits
Start condition	None
First transfer bit	MSB
Input edge	Falling edge
Output edge	Rising edge
Clock	Clock master
Clock source	fs/2
Clock source divide-by-8 (Serial Interface 0, 1 and 2)	Not divided by 8
SBT1/SBO1 pin style	Nch open-drain
SBT1 pin pull-up resistor	Added
SBO1 pin pull-up resistor	Added
Serial 1 interrupt	Enabled
SBO1 output after last data output	Fixed to "High"

An example setup procedure, with a description of each step is shown below

Setup Procedure	Description
(1) Select the prescaler operation SC1MD3(0x03F24) bp3: SC1PSCE =1 (2) Select the clock source SC1MD3(0x03F24) bp2 to 0: SC1PSC2 to 0 =100 (3) Control the SBO1 output after the last data output SC1MD3(0x03F24) bp7 to 6: SC1FDC1 to 0 =00	(1) Set the SC1PSCE flag of SC1MD3 register to "1" to select "Enable count" for prescaler count control. (2) Set the SC1PSC2 to 0 flags of SC1MD3 register to "100" to select fs/2 to the clock source. (3) Set the SC1FDC1 to 0 flags of SC1MD3 register to "0, 0" to select output fixed to "High" after the SBO1 last data output.

Setup Procedure	Description
(4) Control the pin style [Set the pin corresponding to each serial interface] P5ODC(0x03EF5) bp0: P5ODC0 =1 bp2: P5ODC2 =1 P5PLUD(0x03FA5) bp0: P5PLU0 =1 bp2: P5PLU2 =1 (5) Control the pin direction [Set the pin corresponding to each serial interface] P5DIR(0x03F95) bp0: P5DIR0 =1 bp2: P5DIR2 =1 (6) Set SC1MD0 register Select the transfer bit count SC1MD0(0x03F21) bp2 to 0: SC1LNG2 to 0 =111 Select the start condition SC1MD0(0x03F21) bp3: SC1STE =0 Select the transfer first bit SC1MD0(0x03F21) bp4: SC1DIR =0 Select the transfer edge SC1MD0(0x03F21) bp7: SC1CE1 =1 (7) Set SC1MD1 register Select the communication type SC1MD1(0x03F22) bp0: SC1CMD =0 Select the transfer clock SC1MD1(0x03F22) bp2: SC1MST =1 bp3: SC1CKM =0 (Serial Interface 0, 1 and 2) Control the pin function SC1MD1(0x03F22) bp4: SC1SBOS =1 bp5: SC1SBIS =1 bp6: SC1SBTS =1 bp7: SC1IOM =0 (8) Set the interrupt level PSW bp6: MIE =0 SC1TICR(0x03FF8) bp7 to 6: SC1LV1 to 0 =10	(4) Set the P5ODC0 and 2 flags of P5ODC register to "11" to select Nch open-drain to SBO1/SBT1 pin style. Set the P5PLUD0 and 2 flags of P5PLUD register to "11" to enable the pull-up resistor. (Set the pin corresponding to each serial interface) (5) Set the P5DIR0 and 2 flags of Port 5 direction control register (P5DIR) to "11" to set P50 and P52 to output mode and P51 to input mode. (Set the pin corresponding to each serial interface) (6) Set the SC1LNG2 to 0 flags of Serial Interface 1 mode register 0 (SC1MD0) to "111" to set the transfer bit count to 8 bits. Set the SC1STE flag of SC1MD0 register to "0" to disable the start condition. Set the SC1DIR flag of SC1MD0 register to "0" to set MSB as the transfer first bit. Set the SC1CE1 flag of SC1MD0 register to "1" to set the transmission data input edge to "rising" and the reception data output edge to "falling". (7) Set the SC1CMD flag of SC1MD1 register to "0" to select the clock synchronous. Set the SC1MST flag of SC1MD1 register to "1" to select the clock master (internal clock); and, set the SC1CKM flag to "0" to select "the source clock not divided by 8" for Serial Interface 0, 1 and 2. Set the SC1SBOS, SC1SBIS and SC1SBTS flags of SC1MD1 register to "1" to set SBO1 pin to the serial data output, SBI1 pin to the serial data input and SBT1 pin to the serial clock input/output. Set the SC1IOM flag to "0" to set the serial data input from SBI1 pin. (8) Clear the MIE flag of PSW to "0" to disable all maskable interrupts. Set the interrupt level by the SC1LV1 to 0 flags of Serial 1 interrupt control register (SC1TICR).

Setup Procedure	Description
(9) Enable the interrupt SC1TICR (0x03FF8) bp1: SC1TIE =1 PSW bp6: MIE =1 (10) Start the serial transmission Transmission data→TXBUF1 (0x03F27) Reception data→Input to SBI1 pin	(9) Set the SC1TIE flag of SC1TICR register to "1" to enable the interrupt. If any interrupt request flag (SC1TIR of SC1TICR register) is already set, clear SC1TIR prior to enabling the interrupt. Set the MIE flag of PSW to "1" to enable maskable interrupts. Set the transmission data to the serial transmission data buffer TXBUF1. The transfer clock is generated and transmission or reception is started. When transmission is finished, the Serial Interface 1 interrupt SC1TIRQ is generated. [Chapter 7 7.1.4 Maskable Interrupt Control Register Setup]

*Each setup in (1) to (3), (6) to (7) and (8) to (9) can be set at the same time.



Set the SCnSBIS of SCnMD1 register to "0" and select a port in order to operate only transmission with 3-wire system. Set the SCnSBOS of SCnMD1 register to "0" and select a port in order to operate only reception.



In a 2-wire system communicating using SBO pin, serial data is input and output via SBO pin. Port direction control register controls the switching between input and output. SCnSBIS of SCnMD1 register must be set to "1" to select "serial data input". SBIn pin can be used as a general-purpose port.



This serial interface has an emergency reset function. If the communication is needed to be stopped by force, set SCnSBOS and SCnSBIS of SCnMD1 register to "0".



Set each flag in accordance with the procedures indicated in the table above. Be sure to start communication after the settings of all control registers (refer to Table:18.2.1 but exclude TXBUF_n, RXBUF_n and SCnTRB) are completed.



The transfer rate must be under 5.0 MHz for setting a transfer clock by SCnMD3 register.

■ Reception Setup Example (STANDBY Mode Reception)

The following shows the setup example for STANDBY mode reception of clock synchronous serial communication using Serial Interface 1. Table:18.3.9 shows the condition at reception. The basic procedures are the same in Serial Interface 0, 2 and 4. Pin settings (4) and (5) differ in each serial interface.

Table:18.3.9 Setup Examples for Synchronous Serial Interface Reception

Setup item	Set to
Serial data input selection	SBI1
Transfer bit count	8 bits
Start condition	None
First transfer bit	MSB
Input edge	Falling edge
Output edge	Rising edge
Clock	Clock slave
Clock source	fs/2
Clock source divide-by-8 (Serial Interface 0, 1 and 2)	Not divided by 8
SBT1/SBO1 pin style	Nch open-drain
SBT1 pin pull-up resistor	Added
SBO1 pin pull-up resistor	Added
Serial 1 interrupt	Enabled
SBO1 output after last data output	Fixed to "High"

An example setup procedure, with a description of each step is shown below

Setup Procedure	Description
(1) Select the prescaler operation SC1MD3 (0x03F24) bp3: SC1PSCE =1 (2) Select the clock source SC1MD3 (0x03F24) bp2 to 0: SC1PSC2 to 0 =100 (3) Control the SBO0 output after the last data output SC1MD3 (0x03F24) bp7 to 6: SC1FDC1 to 0 =00 (4) Control the pin style [Set the pin corresponding to each serial interface] P5ODC(0x03EF5) bp0: P5ODC0 =1 bp2: P5ODC2 =1 P5PLU(0x05FA5) bp0: P5PLUD0 =1 bp2: P5PLUD2 =1	(1) Set the SC1PSCE flag of SC1MD3 register to "1" to select "Enable count" for prescaler count control. (2) Set the SC1PSC2 to 0 flags of SC1MD3 register to "100" to select fs/2 to the clock source. (3) Set the SC1FDC1 to 0 flags of SC1MD3 register to "0, 0" to select output fixed at "High" after the SBO1 last data output. (4) Set the P5ODC0 and 2 flags of P5ODC register to "11" to select Nch open-drain to SBO1/SBT1 pin style. Set the P5PLU0 and 2 flags of P5PLUD register to "11" to enable the pull-up resistor. (Set the pin corresponding to each serial interface)

Setup Procedure	Description
(5) Control the pin direction [Set the pin corresponding to each serial interface] P5DIR(0x03F95) bp2: P5DIR2 =0 bp1: P5DIR1 =0 (6) Set SC1MD0 register Select the transfer bit count SC1MD0 (0x03F21) bp2 to 0: SC1LNG2 to 0 =111 Select the start condition SC1MD0 (0x03F21) bp3: SC1STE =0 Select the transfer first bit SC1MD0 (0x03F21) bp4: SC1DIR =0 Select the transfer edge SC1MD0 (0x03F21) bp7: SC1CE1 =1 (7) Set SC1MD1 register Select the communication type SC1MD1 (0x03F22) bp0: SC1CMD =0 Select the transfer clock SC1MD1 (0x03F22) bp2: SC1MST =0 bp3: SC1CKM =0 (Serial Interface 0, 1 and 2) Control the pin function SC1MD1 (0x03F22) bp4: SC1SBOS =0 bp5: SC1SBIS =1 bp6: SC1SBTS =1 bp7: SC1IOM =0 (8) Set the interrupt level PSW bp6: MIE =0 SC1TICR (0x03FF6) bp7 to 6: SC1LV1 to 0 =10 (9) Enable the interrupt SC1TICR (0x03FF6) bp1: SC1TIE =1 PSW bp6: MIE =1 (10) Start the serial transmission Dummy data → TXBUF1 (0x03F27)	(5) Set the P5DIR1 flag of Port 5 direction control register (P5DIR) to "0" and set the P5DIR2 flag to "0" to set P51 and P52 to input mode (Set the pin corresponding to each serial interface) (6) Set the SC1LNG2 to 0 flags of Serial Interface 1 mode register 0 (SC1MD0) to "111" to set the transfer bit count to 8 bits. Set the SC1STE flag of SC1MD0 register to "0" to disable the start condition. Set the SC1DIR flag of SC1MD0 register to "0" to select MSB as the transfer first bit. Set the SC1CE1 flag of SC1MD0 register to "1" to set the transmission data input edge to "rising" and the reception data output edge to "falling". (7) Set the SC1CMD flag of SC1MD1 register to "0" to select the clock synchronous. Set the SC1MST flag of SC1MD1 register to "0" to select the clock slave (external clock). Set the SC1CKM flag to "0" to select "the source clock not divided by 8" for Serial Interface 0, 1 and 2. Set the SC1SBIS and SC1SBTS flags of SC1MD1 register to "1" to set SBI1 pin to the serial input and SBT1 pin to the transfer clock input/output. Set the SC1IOM flag "0" to set the serial data input from SBI1 pin. Set the SC1SBOS flag to "0" to select the port as SBO1 pin function. (8) Clear the MIE flag of PSW to "0" to disable all maskable interrupts. Set the interrupt level by the SC1LV1 to 0 flags of Serial 1 interrupt control register (SC1TICR). (9) Set the SC1TIE flag of SC1TICR register to "1" to enable the interrupt. If any interrupt request flag (SC1TIR of SC1TICR register) is already set, clear SC1TIR prior to enabling the interrupt. Set the MIE flag of PSW to "1" to enable maskable interrupts. (10) Set dummy data to the serial transmission data buffer TXBUF1.

Setup Procedure	Description
(11) Transfer to STOP mode CPUM(0x03F00) bp3: STOP =1	(11) Set the STOP flag of CPUM register to "1" to transfer to STOP mode.
(12) Start the serial communication Transmission clock → input SBT1 pin Received data → input SBI1 pin	(12) Input the transfer clock to SBT1 pin and transfer data to SBI1 pin.
(13) Return from the STANDBY mode	(13) The Serial Interface 1 interrupt SC1TIRQ is generated at the same time of the 8 th bits data reception. CPU is then returned from STOP mode to NORMAL mode after the oscillation stabilization wait.

*Each setup (1) to (3) and (6) to (8) can be set at the same time.



Disable the start condition during slave reception in STANDBY mode. If the start condition is enabled, proper reception may not be guaranteed.



In a 2-wire system communicating using SBOn pin, serial data is input and output via SBOn pin. Port direction control register controls the switching between input and output. SCnSBIS of SCnMD1 register must be set to "1" to select "serial data input". SBIn pin can be used as a general-purpose port.



This serial interface has an emergency reset function. If the communication is needed to be stopped by force, set SCnSBOS and SCnSBIS of SCnMD1 register to "0".



Set each flag in accordance with the procedures indicated in the table above. Be sure to start communication after the settings of all control registers (refer to Table:18.2.1 but exclude TXBUF_n, RXBUF_n and SCnTRB) are completed.



The transfer rate must be under 5.0 MHz for setting a transfer clock by SCnMD3 register.



Insert three NOP instructions immediately after the transition to HALT or STOP mode.

18.4 Duplex UART Serial Interface

18.4.1 Operation

Serial Interface 0, 1, 2 and 3 can be used for duplex UART communication.



When setting this serial interface communication mode to "UART", set Serial Interface n mode register 1 (SCnMD1) while the serial data input pin is at "High".



"n" = 0, 1, 2, 3 for Serial Interface 0, 1, 2 and 3 respectively in section [Chapter 18 18.4.1 Operation].

■ Communication Type

The communication mode can be selected from 2-wire type or 1-wire type. Set the communication mode by the SCnIOM flag of SCnMD1 register. In 1-wire type, a single pin of the data I/O pin (SBO_n) is used, and the data input pin (SBI_n) can be used as a general-purpose port. In 1-wire reception, select "serial data input" by setting the SCnSBIS flag of SCnMD1 register to "1".

■ Activation Factor for Communication

Setting data in the transmission data buffer TXBUF_n generates a start condition, and transmission will start. Receiving a start condition starts reception. In reception, when the data length of "Low" for the start bit is longer than 0.5 bit, a start condition is recognized.

■ Transmission

When data is set in the transmission data buffer TXBUF_n, transmission is automatically started. When the transmission is completed, the Serial n transmission complete interrupt SCnTIRQ is generated.

■ Reception

When a start condition is recognized, reception is started after the transfer bit counter that counts transfer bits is cleared. When reception is completed, the Serial n reception complete interrupt SCnRIRQ is generated.

■ Duplex communication

On duplex communication, transmission and reception can be executed independently at the same time. The frame mode and parity bit of the data used on transmission/reception should have the same polarity.

■ Transfer Bit Count Setup

The transfer bit count is automatically set when the frame mode is specified by the SCnFM1 to 0 flags of SCnMD2 register. If the SCnCMD flag of SCnMD1 register is set to "1" and UART communication is selected, the setting of the synchronous serial transfer bit count selection flags SCnLNG2 to 0 of SCnMD0 register is no longer valid.

■ Data Input Pin Setup

Communication mode can be selected from 2-wire [data output pin (TXDn pin), data input pin (RXDn pin)] mode and 1-wire (data I/O pin TXDn) mode. Set communication mode by the SCnIOM flag of SC0MD1 register. RXDn pin can only be used for serial data input. TXDn pin can be used for serial data input or output. If "data input from TXDn pin" is selected, the communication mode is 1-wire. Transmission and reception can be switched by the direction control of TXDn pin with I/O selection of the port direction control register. At this time, RXDn pin can be used as a general-purpose port.

■ Reception Buffer Empty Flag

When Serial Interface n reception complete interrupt SCnRIRQ is generated, data is automatically stored from internal shift register to RXBUF_n. If data is stored in shift register RXBUF_n, the reception buffer empty flag SCnREMP of SCnSTR register is set to "1" indicating that the reception data is waiting to be read.

SCnREMP is cleared to "0" by reading the RXBUF_n data.

■ Reception BUSY Flag

When a start condition is recognized, the SCnRBSY flag of SCnSTR register is set to "1". It is cleared to "0" after Serial Interface n reception complete interrupt SCnRIRQ is generated. If the SCnSBIS flag is set to "0" during reception, the SCnRBSY flag is cleared to "0".

■ Transmission BUSY Flag

When data is set in TXBUF_n, the SCnTBSY flag of SCnSTR register is set to "1". It is cleared to "0" after Serial Interface n transmission complete interrupt SCnTIRQ is generated. During continuous communication, the set SCnTBSY flag is retained. If the transmission buffer empty flag SCnTEMP is "0" when Serial Interface n transmission complete interrupt SCnTIRQ is generated, the SCnTBSY is cleared to "0". If the SCnSBOS flag is set to "0" during transmission, the SCnTBSY flag is cleared to "0".

■ Frame Mode and Parity Check Setup

The data format of UART communication is shown below.

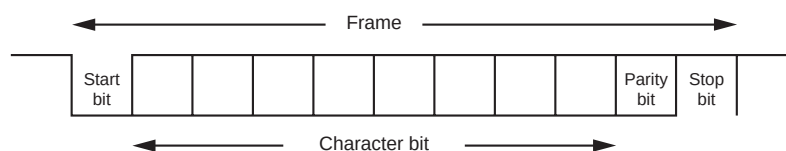


Figure:18.4.1 Transmission/Reception Data Format of UART Serial Interface

The transmission/reception data consists of start bit, character bit, parity bit and stop bit. Table:18.4.1 shows the types of data that can be set.

Table:18.4.1 UART Serial Interface Transmission/Reception Data

Start bit	1 bit
Character bit	7, 8 bits
Parity bit	fixed to "0", fixed to "1", odd, even, none
Stop bit	1, 2 bits

The frame mode is set by the SCnFM1 to 0 flags of SCnMD2 register. Table:18.4.2 shows the types of frame mode that can be set. If the SCnCND flag of SCnMD1 register is set to "1" and UART communication is selected, the transfer bit counts of the SCnLNG2 to 0 flags of SCnMD0 register are no more valid.

Table:18.4.2 UART Serial Interface Frame Mode

SCnMD2 register		Frame mode
SCnFM1	SCnFM0	
0	0	Character bit 7 bits + Stop bit 1 bit
0	1	Character bit 7 bits + Stop bit 2 bits
1	0	Character bit 8 bits + Stop bit 1 bit
1	1	Character bit 8 bits + Stop bit 2 bits

The parity bit is for detecting wrong bits of transmission/reception data. Table:18.4.3 shows the types of the parity bit. The parity bit is set by the SCnNPE and SCnPM1 to 0 flags of SCnMD2 register.

Table:18.4.3 Parity Bit of UART Serial Interface

SCnMD2			Parity bit	Setup
SCnNPE	SCnPM1	SCnPM0		
0	0	0	Fixed at 0	Set parity bit to "0"
0	0	1	Fixed at 1	Set parity bit to "1"
0	1	0	Odd parity	Control the total of "1" of parity bit and character bit to be odd
0	1	1	Even parity	Control the total of "1" of parity bit and character bit to be even
1	-	-	None	Do not add parity bit

■ Break Status Transmission Control Setup

The SCnBRKE flag of SCnMD2 register generates the break status. If SCnBRKE is set to "1" to select the break transmission and all bits from start bit to stop bit are transferred "0".

■ Reception Error

At reception, there are 3 types of errors: overrun error, parity error and framing error. The reception error can be determined by checking the SCnORE, SCnPEK and SCnFEF flags of SCnSTR register. If one of these flags has an error, the SCnERE flag of SCnSTR register is set to "1". The SCnPEK and SCnFEF flags of the reception error flags are renewed during generation of Serial Interface n reception complete interrupt SCnRIRQ. The SCnORE flag is cleared simultaneously when the next SCnRIRQ is generated after the RXBUF_n data is read. The reception error determination should be operated before the next communication is completed. Those error flags have no effects on communication operation. Table:18.4.4 shows reception error factors of reception errors.

Table:18.4.4 Reception Error Factors of UART Serial Interface

Flag	Reception error		
SCnORE	Overrun error	Next data is received before reading the reception buffer	
SCnPEK	Parity error	At fixed to 0	when parity bit is "1"
		At fixed to 1	When parity bit is "0"
		Odd parity	When the total of "1" of parity bit and character bit is even
		Even parity	When the total of "1" of parity bit and character bit is odd
SCnFEF	Framing error	Stop bit is not detected	

■ Determination of Break Status Reception

Reception at break status can be determined by the SCnBRKF flag. If all received data from the start bit and the stop bit are "0", the SCnBRKF flag of SCnMD2 register is set indicating that the break status is generated. The SCnBRKF flag is set when Serial Interface n reception complete interrupt SCnRIRQ is generated.

■ Continuous Communication

This serial interface has a continuous transfer function. If data is set in the transmission data buffer TXBUF_n during transmission, the transmission buffer empty flag SCnTEMP is set and continue transmission is operated automatically. There are no communication blanks between current transfer and next transfer. Set the next data to TXBUF_n between the time when the data in the TXBUF_n is loaded to the transmission shift register and before Serial Interface n transmission complete interrupt SCnTIRQ is generated.

■ Clock Setup

A transfer clock is not necessary at UART communication, but the clock setup is necessary for determining the data transmission/reception timing within the serial interface. Select the timer used as a baud rate timer by SCnMD3 register.



Be sure to set SCnSBIS and SCnSBOS flags of SCnMD1 to "0" before changing a clock.

■ Reception Bit Count and First Transfer Bit

In reception, when transfer bit counts = 7 bits, the data storing method for the reception data buffer RXBUF_n differs depending on the first transfer bit specification. At MSB first, data are stored in the upper bits of RXBUF_n. When transfer bit counts = 7 bits, as shown in Figure:18.4.2, with "G" to be the first bit to transfer and "A" to be the last bit to transfer, data "A" to "G" are stored in bp7 to bp1 of RXBUF_n. At LSB first, data are stored in the lower bits of RXBUF_n. When transfer bit counts = 7 bits, as shown in Figure:18.4.3, with "G" to be the first bit to transfer and "A" to be the last bit to transfer, data "A" to "G" are stored in bp0 to bp6 of RXBUF_n.



Figure:18.4.2 Reception Bit Count and First Transfer Bit (At MSB First)



Figure:18.4.3 Reception Bit Count and First Transfer Bit (At LSB First)

■ Transfer Speed Setup

The transfer speed can be set using a baud rate timer. The setup example for the transfer speed is shown below.

Table:18.4.5 UART Serial Interface Transfer Speed

	Setup	Register	Page
Serial Interface 0, 1, 2 and 3	Serial Interface 0, 1, 2 and 3 clock source (Timer 0 to 2 or A output)	SC0MD3	XVIII-30
		SC1MD3	XVIII-30
		SC2MD3	XVIII-30
		SC3MD3	XVIII-30
	Timer 0 clock source	TM0MD	IX-25
	Timer 1 clock source	TM1MD	IX-27
	Timer 2 clock source	TM2MD	IX-28
	Timer A clock source	TMAMD	X-6
	Timer 0 compare register	TM0OC	IX-20
	Timer 1 compare register	TM1OC	IX-20
	Timer 2 compare register	TM2OC	IX-20
	Timer 3 compare register	TM3OC	IX-20
	Timer 4 compare register	TM4OC	IX-20
	Timer A compare register	TMAOC	X-5

Timer compare register is set as follows;

$$\text{overflow cycle} = (\text{set value of compare register} + 1) \times \text{timer clock cycle}$$

$$\text{baud rate} = 1 / (\text{overflow cycle} \times 2 \times 8) \text{ ("8" is divide-by-8 clock source)}$$

therefore,

$$\text{set value of compare register} = \text{timer clock frequency} / (\text{baud rate} \times 2 \times 8) - 1$$

For example, when setting the baud rate to 300 bps at clock source $f_s/4$ ($f_{pll-div} = 8 \text{ MHz}$, $f_s = f_{pll-div}/2$), the set value should be as follows:

$$\begin{aligned} \text{set value of compare register} &= (8 \times 10^6 / 2 / 4) / (300 \times 2 \times 8) - 1 \\ &= 207 \\ &= 0xCF \end{aligned}$$

The following pages show the timer clock source at the standard transfer rate and the set value of compare register when $f_s = f_{pll-div}/2$.



Transfer rate should be selected under 300 kbps.



When the SCnCMD flag of SCnMD1 register is set to "1" to select "Duplex UART", the transfer clock is divided regardless of the SCnCKM flag.

Table:18.4.6 Setup Value of UART Serial Interface Transfer Speed (decimal) Clock Source Divided by 8

Transfer speed (bit/s)											
fppll-div (MHz)	Clock source (timer)	300		960		1200		2400		4800	
		Set value	Calculated value	Set value	Calculated value	Set value	Calculated value	Set value	Calculated value	Set value	Calculated value
4.00	fppll-div	-	-	-	-	207	1202	103	2404	51	4808
	fppll-div/4	207	300	64	962	51	1202	25	2404	12	4808
	fppll-div/16	51	300	-	-	12	1202	-	-	-	-
	fppll-div/32	25	300	-	-	-	-	-	-	-	-
	fppll-div/64	12	300	-	-	-	-	-	-	-	-
	fs/2	207	300	64	962	51	1202	25	2404	12	4808
4.19	fppll-div	-	-	-	-	217	1201	108	2403	54	4761
	fppll-div/4	217	300	67	963	-	-	-	-	-	-
	fppll-div/16	-	-	16	963	-	-	6	2338	-	-
	fppll-div/32	-	-	-	-	-	-	-	-	-	-
	fppll-div/64	-	-	-	-	-	-	-	-	-	-
	fs/2	217	300	67	963	-	-	-	-	-	-
8.00	fppll-div	-	-	-	-	-	-	207	2404	103	4808
	fppll-div/4	-	-	129	962	103	1202	51	2404	25	4808
	fppll-div/16	103	300	-	-	25	1202	12	2404	-	-
	fppll-div/32	51	300	-	-	12	1202	-	-	-	-
	fppll-div/64	25	300	-	-	-	-	-	-	-	-
	fs/2	-	-	129	962	103	1202	51	2404	25	4808
8.38	fppll-div	-	-	-	-	-	-	217	2403	108	4805
	fppll-div/4	-	-	135	963	108	1201	-	-	-	-
	fppll-div/16	108	300	33	963	-	-	13	2338	-	-
	fppll-div/32	-	-	16	963	-	-	16	2338	-	-
	fppll-div/64	-	-	-	-	-	-	-	-	-	-
	fs/2	-	-	135	963	108	1201	-	-	-	-
12.00	fppll-div	-	-	-	-	-	-	-	-	155	4808
	fppll-div/4	-	-	194	962	155	1202	77	2404	38	4808
	fppll-div/16	155	300	-	-	38	1202	-	-	-	-
	fppll-div/32	77	300	-	-	-	-	-	-	-	-
	fppll-div/64	38	300	-	-	-	-	-	-	-	-
	fs/2	-	-	194	962	155	1202	77	2404	38	4808
16.00	fppll-div	-	-	-	-	-	-	-	-	207	4808
	fppll-div/4	-	-	-	-	207	1202	103	2404	51	4808
	fppll-div/16	207	300	64	962	51	1202	25	2404	12	4808
	fppll-div/32	103	300	-	-	25	1202	12	2404	-	-
	fppll-div/64	51	300	-	-	12	1202	-	-	-	-
	fs/2	-	-	-	-	207	1202	103	2404	51	4808
20.00	fppll-div	-	-	-	-	-	-	-	-	-	-
	fppll-div/4	-	-	-	-	-	-	129	2404	64	4808
	fppll-div/16	-	-	-	-	64	1202	-	-	-	-
	fppll-div/32	129	300	-	-	-	-	-	-	-	-
	fppll-div/64	64	300	-	-	-	-	-	-	-	-
	fs/2	-	-	-	-	-	-	129	2404	64	4808
20.00	fs/4	-	-	162	959	129	1202	64	2404	-	-

Table:18.4.7 Setup Value of UART Serial Interface Transfer Speed (decimal) Clock Source Divided by 8

Transfer speed (bit/s)											
fppll-div (MHz)	Clock source (timer)	9600		19200		28800		31250		38400	
		Set value	Calculated value	Set value	Calculated value	Set value	Calculated value	Set value	Calculated value	Set value	Calculated value
4.00	fppll-div	25	9615	12	19231	-	-	7	31250	-	-
	fppll-div/4	-	-	-	-	-	-	1	31250	-	-
	fppll-div/16	-	-	-	-	-	-	-	-	-	-
	fppll-div/32	-	-	-	-	-	-	-	-	-	-
	fppll-div/64	-	-	-	-	-	-	-	-	-	-
	fs/2	-	-	-	-	-	-	1	31250	-	-
	fs/4	-	-	-	-	-	-	-	-	-	-
4.19	fppll-div	26	9699	-	-	-	-	-	-	-	-
	fppll-div/4	-	-	-	-	-	-	-	-	-	-
	fppll-div/16	-	-	-	-	-	-	-	-	-	-
	fppll-div/32	-	-	-	-	-	-	-	-	-	-
	fppll-div/64	-	-	-	-	-	-	-	-	-	-
	fs/2	-	-	-	-	-	-	-	-	-	-
	fs/4	-	-	-	-	-	-	-	-	-	-
8.00	fppll-div	51	9615	25	19231	-	-	15	31250	12	38462
	fppll-div/4	12	9615	-	-	-	-	3	31250	-	-
	fppll-div/16	-	-	-	-	-	-	-	-	-	-
	fppll-div/32	-	-	-	-	-	-	-	-	-	-
	fppll-div/64	-	-	-	-	-	-	-	-	-	-
	fs/2	12	9615	-	-	-	-	3	31250	-	-
	fs/4	-	-	-	-	-	-	1	31250	-	-
8.38	fppll-div	54	9523	26	19398	-	-	-	-	-	-
	fppll-div/4	-	-	-	-	-	-	-	-	-	-
	fppll-div/16	-	-	-	-	-	-	-	-	-	-
	fppll-div/32	-	-	-	-	-	-	-	-	-	-
	fppll-div/64	-	-	-	-	-	-	-	-	-	-
	fs/2	-	-	-	-	-	-	-	-	-	-
	fs/4	-	-	-	-	-	-	-	-	-	-
12.00	fppll-div	77	9615	38	19231	25	28846	23	31250	-	-
	fppll-div/4	-	-	-	-	-	-	5	31250	-	-
	fppll-div/16	-	-	-	-	-	-	-	-	-	-
	fppll-div/32	-	-	-	-	-	-	-	-	-	-
	fppll-div/64	-	-	-	-	-	-	-	-	-	-
	fs/2	-	-	-	-	-	-	5	31250	-	-
	fs/4	-	-	-	-	-	-	2	31250	-	-
16.00	fppll-div	103	9615	51	19231	-	-	31	31250	25	38462
	fppll-div/4	25	9615	12	19231	-	-	7	31250	-	-
	fppll-div/16	-	-	-	-	-	-	-	-	-	-
	fppll-div/32	-	-	-	-	-	-	-	-	-	-
	fppll-div/64	-	-	-	-	-	-	-	-	-	-
	fs/2	25	9615	-	-	-	-	7	31250	-	-
	fs/4	12	9615	-	-	-	-	2	31250	-	-
20.00	fppll-div	129	9615	64	19231	-	-	39	31250	-	-
	fppll-div/4	-	-	-	-	-	-	9	31250	-	-
	fppll-div/16	-	-	-	-	-	-	-	-	-	-
	fppll-div/32	-	-	-	-	-	-	-	-	-	-
	fppll-div/64	-	-	-	-	-	-	-	-	-	-
	fs/2	-	-	-	-	-	-	9	31250	-	-
	fs/4	-	-	-	-	-	-	4	31250	-	-

Table:18.4.8 Setup Value of UART Serial Interface Transfer Speed (decimal) Clock Source Divided by 16

		Transfer speed (bit/s)									
fppll-div (MHz)	Clock source (timer)	300		960		1200		2400		4800	
		Set value	Calculated value	Set value	Calculated value	Set value	Calculated value	Set value	Calculated value	Set value	Calculated value
4.00	fppll-div	-	-	129	962	103	1202	51	2404	25	4808
	fppll-div/4	104	297	-	-	25	1202	12	2404	-	-
	fppll-div/16	25	300	-	-	-	-	-	-	-	-
	fppll-div/32	12	300	-	-	-	-	-	-	-	-
	fppll-div/64	-	-	-	-	-	-	-	-	-	-
	fs/2	103	300	-	-	25	1202	12	2404	-	-
	fs/4	51	297	-	-	12	1202	-	-	-	-
	fs/8	25	300	-	-	-	-	-	-	-	-
4.19	fppll-div	-	-	135	963	108	1201	-	-	-	-
	fppll-div/4	109	297	33	963	-	-	-	-	-	-
	fppll-div/16	26	303	-	-	-	-	-	-	-	-
	fppll-div/32	-	-	-	-	-	-	-	-	-	-
	fppll-div/64	-	-	-	-	-	-	-	-	-	-
	fs/2	109	297	33	963	-	-	-	-	-	-
	fs/4	54	297	16	963	-	-	6	2338	-	-
	fs/8	26	303	-	-	-	-	-	-	-	-
8.00	fppll-div	-	-	-	-	-	-	103	2404	51	4808
	fppll-div/4	208	300	64	962	51	1202	25	2404	12	4808
	fppll-div/16	51	300	-	-	12	1202	-	-	-	-
	fppll-div/32	25	300	-	-	-	-	-	-	-	-
	fppll-div/64	12	300	-	-	-	-	-	-	-	-
	fs/2	208	300	64	962	51	1202	25	2404	12	4808
	fs/4	103	300	-	-	25	1202	12	2404	-	-
	fs/8	51	300	-	-	12	1202	-	-	-	-
8.38	fppll-div	-	-	-	-	217	1202	108	2403	-	-
	fppll-div/4	218	300	67	963	-	-	27	2338	-	-
	fppll-div/16	55	297	16	963	-	-	6	2338	-	-
	fppll-div/32	27	303	-	-	-	-	-	-	-	-
	fppll-div/64	-	-	-	-	-	-	-	-	-	-
	fs/2	-	-	67	963	-	-	27	2338	-	-
	fs/4	109	300	33	963	-	-	13	2338	-	-
	fs/8	55	297	16	963	-	-	6	2338	-	-
12.00	fppll-div	-	-	-	-	-	-	155	2404	77	4808
	fppll-div/4	-	-	-	-	77	1202	38	2404	-	-
	fppll-div/16	77	300	-	-	-	-	-	-	-	-
	fppll-div/32	38	300	-	-	-	-	-	-	-	-
	fppll-div/64	-	-	-	-	-	-	-	-	-	-
	fs/2	-	-	-	-	77	1202	38	2404	-	-
	fs/4	154	302	-	-	38	1202	-	-	-	-
	fs/8	77	300	-	-	-	-	-	-	-	-
16.00	fppll-div	-	-	-	-	-	-	207	2404	103	4808
	fppll-div/4	-	-	-	-	103	1202	51	2404	25	4808
	fppll-div/16	103	300	-	-	25	1202	12	2404	-	-
	fppll-div/32	51	300	-	-	12	1202	-	-	-	-
	fppll-div/64	25	300	-	-	-	-	-	-	-	-
	fs/2	-	-	-	-	103	1202	51	2404	25	4808
	fs/4	208	300	64	962	51	1202	25	2404	12	4808
	fs/8	103	300	-	-	25	1202	12	2404	-	-
20.00	fppll-div	-	-	-	-	-	-	-	-	-	-
	fppll-div/4	-	-	162	959	129	1201	64	2403	-	-
	fppll-div/16	129	300	-	-	-	-	-	-	-	-
	fppll-div/32	64	300	-	-	-	-	-	-	-	-
	fppll-div/64	-	-	-	-	-	-	-	-	-	-
	fs/2	-	-	162	959	129	1201	64	2403	-	-
	fs/4	129	300	80	965	64	1201	-	-	-	-
	fs/8	129	300	-	-	-	-	-	-	-	-

Table:18.4.9 Setup Value of UART Serial Interface Transfer Speed (decimal) Clock Source Divided by 16

		Transfer speed (bit/s)									
fppll-div (MHz)	Clock source (timer)	300		960		1200		2400		4800	
		Set value	Calculated value	Set value	Calculated value	Set value	Calculated value	Set value	Calculated value	Set value	Calculated value
4.00	fppll-div	12	9615	-	-	-	-	3	31250	-	-
	fppll-div/4	-	-	-	-	-	-	-	-	-	-
	fppll-div/16	-	-	-	-	-	-	-	-	-	-
	fppll-div/32	-	-	-	-	-	-	-	-	-	-
	fppll-div/64	-	-	-	-	-	-	-	-	-	-
	fs/2	-	-	-	-	-	-	-	-	-	-
	fs/4	-	-	-	-	-	-	-	-	-	-
4.19	fppll-div	-	-	-	-	-	-	-	-	-	-
	fppll-div/4	-	-	-	-	-	-	-	-	-	-
	fppll-div/16	-	-	-	-	-	-	-	-	-	-
	fppll-div/32	-	-	-	-	-	-	-	-	-	-
	fppll-div/64	-	-	-	-	-	-	-	-	-	-
	fs/2	-	-	-	-	-	-	-	-	-	-
	fs/4	-	-	-	-	-	-	-	-	-	-
8.00	fppll-div	25	9615	12	19231	-	-	7	31250	-	-
	fppll-div/4	-	-	-	-	-	-	1	31250	-	-
	fppll-div/16	-	-	-	-	-	-	-	-	-	-
	fppll-div/32	-	-	-	-	-	-	-	-	-	-
	fppll-div/64	-	-	-	-	-	-	-	-	-	-
	fs/2	-	-	-	-	-	-	1	31250	-	-
	fs/4	-	-	-	-	-	-	-	-	-	-
8.38	fppll-div	26	9699	-	-	-	-	-	-	-	-
	fppll-div/4	-	-	-	-	-	-	-	-	-	-
	fppll-div/16	-	-	-	-	-	-	-	-	-	-
	fppll-div/32	-	-	-	-	-	-	-	-	-	-
	fppll-div/64	-	-	-	-	-	-	-	-	-	-
	fs/2	-	-	-	-	-	-	-	-	-	-
	fs/4	-	-	-	-	-	-	-	-	-	-
12.00	fppll-div	38	9615	-	-	12	28847	11	31251	-	-
	fppll-div/4	-	-	-	-	-	-	2	31251	-	-
	fppll-div/16	-	-	-	-	-	-	-	-	-	-
	fppll-div/32	-	-	-	-	-	-	-	-	-	-
	fppll-div/64	-	-	-	-	-	-	-	-	-	-
	fs/2	-	-	-	-	-	-	2	31250	-	-
	fs/4	-	-	-	-	-	-	-	-	-	-
16.00	fppll-div	51	9615	25	19231	-	-	11	31250	12	38462
	fppll-div/4	12	9615	-	-	-	-	3	31250	-	-
	fppll-div/16	-	-	-	-	-	-	-	-	-	-
	fppll-div/32	-	-	-	-	-	-	-	-	-	-
	fppll-div/64	-	-	-	-	-	-	-	-	-	-
	fs/2	12	9615	-	-	-	-	3	31250	-	-
	fs/4	-	-	-	-	-	-	1	31250	-	-
20.00	fppll-div	64	9615	-	-	-	-	19	31250	-	-
	fppll-div/4	-	-	-	-	-	-	4	31250	-	-
	fppll-div/16	-	-	-	-	-	-	-	-	-	-
	fppll-div/32	-	-	-	-	-	-	-	-	-	-
	fppll-div/64	-	-	-	-	-	-	-	-	-	-
	fs/2	-	-	-	-	-	-	4	31250	-	-
	fs/4	-	-	-	-	-	-	-	-	-	-
	fs/8	-	-	-	-	-	-	-	-	-	-

The items shown below are the same as clock synchronous serial interface. Refer to the following pages.

■ First Transfer Bit Setup

Refer to: XVIII-43

■ Transmission Data Buffer

Refer to: XVIII-43

■ Reception Data Buffer

Refer to: XVIII-44

■ Transmit Bit Count and First Transfer Bit

Refer to: XVIII-44

■ Transmission Buffer Empty Flag

Refer to: XVIII-48

■ Emergency Reset

Refer to: XVIII-48

18.4.2 Timing

■ Transmission Timing

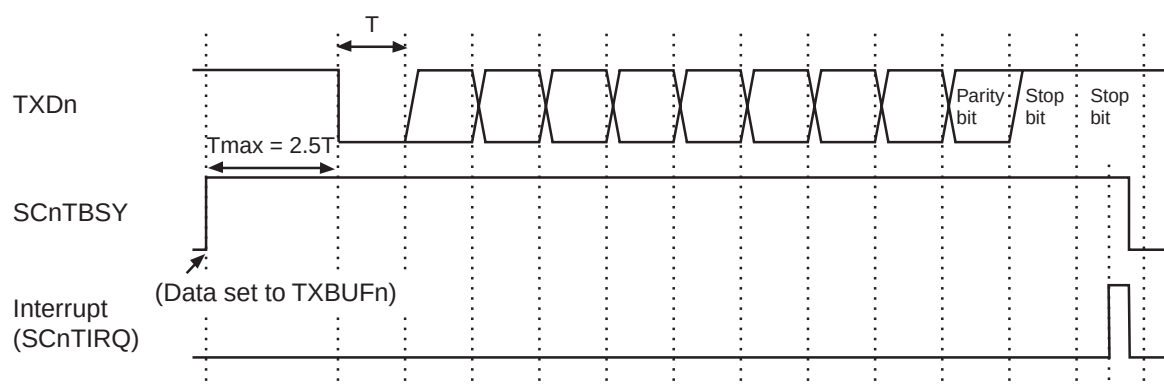


Figure:18.4.4 Transmission Timing (Parity Bit is Enabled)

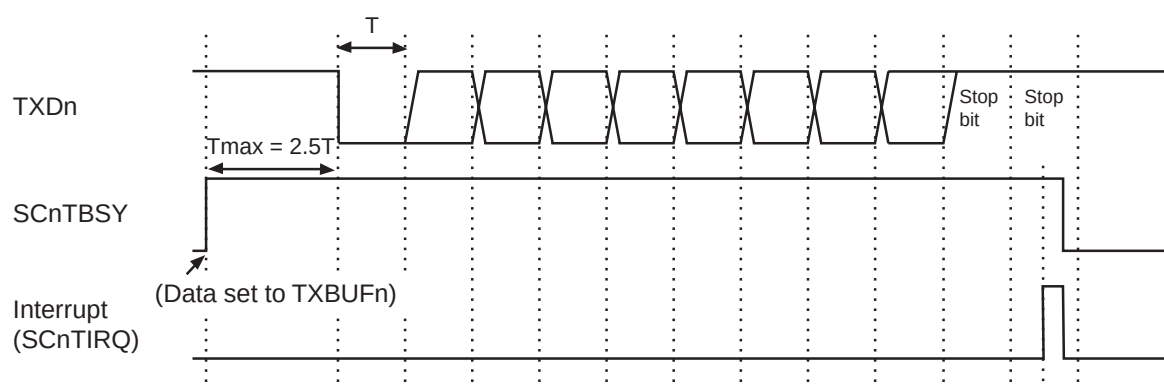


Figure:18.4.5 Transmission Timing (Parity Bit is Disabled)

■ Reception Timing

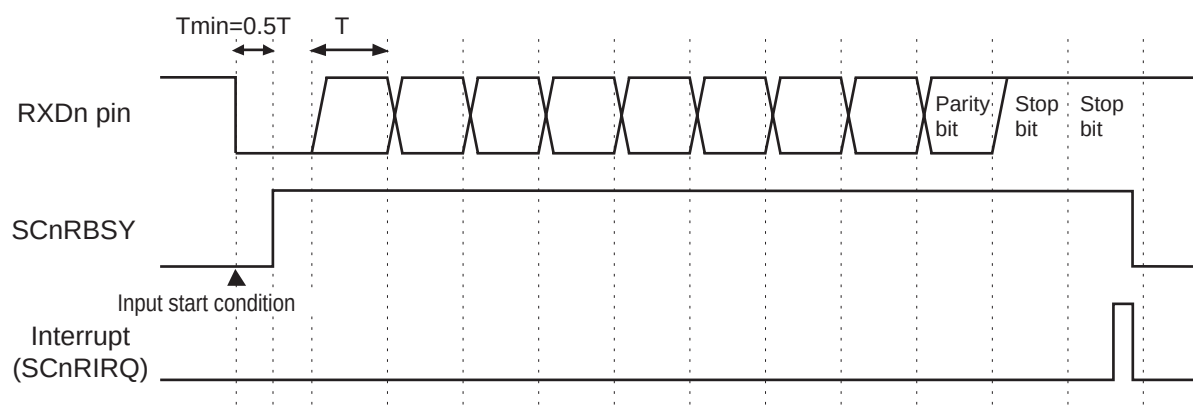


Figure:18.4.6 Reception Timing (Parity Bit is Enabled)

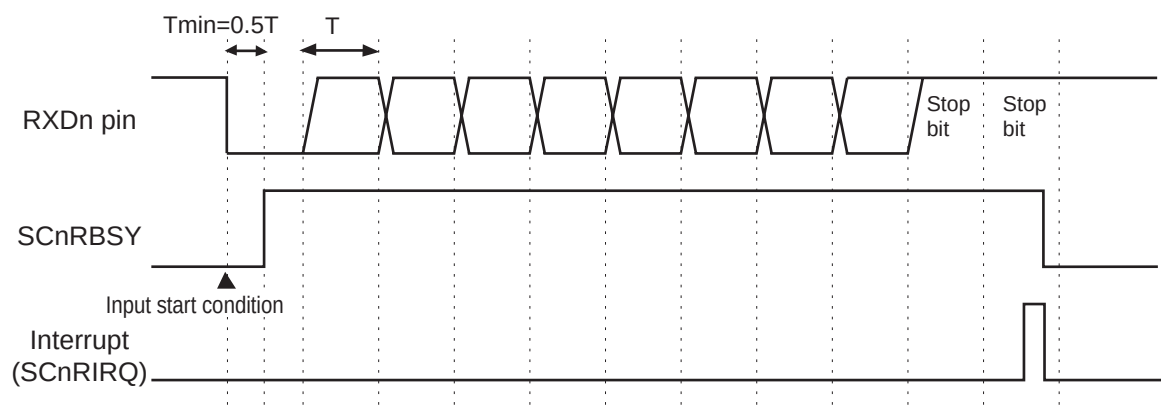


Figure:18.4.7 Reception Timing (Parity Bit is Disabled)

18.4.3 Pin Setup

■ UART Serial Interface 0 Pin Setup

Port	Wire system	Type	Pin	Pin setup (flag setup)						
				SC0SEL register	PnDIR register	PnODC register	PnPLUD register	SC0MD1 register		
				Serial 0 I/O pin switching	I/O mode selection	Nch open-drain output selection Arbitrary setting	Pull-up/ pull-down resistor selection Arbitrary setting	Serial data input selection	Serial input control selection	SBO0 (TXD0) pin function selection
				0: P02 to P03 1: P43 to P44	0: input mode 1: output mode	0: push/pull 1: Nch open-drain	0: not added 1: added	0: data input from RXD0 1: data input from TXD0	0: "1" input 1: serial input	0: port 1: serial data output
				OSL0	PnDIRm	PnODCm	PnPLUDm	SC0IOM	SC0SBIS	SC0SBOS
Port 0	2-wire	Transmission only	P03/TXD0A	0	P0DIR3:1	P0ODC3	P0PLUD3	0	0	1
			-		-	-	-			
		Reception only	-	0	-	-	-	0	1	0
			P02/RXD0A		P0DIR2:0	-	-			
	1-wire	Transmission/reception	P03/TXD0A	0	P0DIR3:1	P0ODC3	P0PLUD3	0	1	1
			P02/RXD0A		P0DIR2:0	-	-			
		Transmission only	P03/TXD0A	0	P0DIR3:1	P0ODC3	P0PLUD3	0	0	1
			-		-	-	-			
Reception only	P03/TXD0A	0	P0DIR3:1	P0ODC3	P0PLUD3	1	1	0		
	-		-	-	-					
Port 4	2-wire	Transmission only	P43/TXD0B	1	P4DIR3:1	P4ODC3	P4PLUD3	0	0	1
			-		-	-	-			
		Reception only	-	1	-	-	-	0	1	0
			P44/RXD0B		P4DIR4:0	-	-			
		Transmission/reception	P43/TXD0B	1	P4DIR3:1	P4ODC3	P4PLUD3	0	1	1
			P44/RXD0B		P4DIR4:0	-	-			
	1-wire	Transmission only	P43/TXD0B	1	P4DIR3:1	P4ODC3	P4PLUD3	0	0	1
			-		-	-	-			
		Reception only	P43/TXD0B	1	P4DIR3:1	P4ODC3	P4PLUD3	1	1	0
			-		-	-	-			

■ UART Serial Interface 1 Pin Setup

Port	Wire system	Type	Pin	Pin setup (flag setup)						
				SC1SEL register	PnDIR register	PnODC register	PnPLUD register	SC1MD1 register		
				Serial 1 I/O pin switching	I/O mode selection	Nch open-drain output selection Arbitrary setting	Pull-up/ pull-down resistor selection Arbitrary setting	Serial data input selection	Serial input control selection	SBO1 (TXD1) pin function selection
				0: P50 to P51 1: P75 to P76	0: input mode 1: output mode	0: push/pull 1: Nch open-drain	0: not added 1: added	0: data input from RXD1 1: data input from TXD1	0: "1" input 1: serial input	0: port 1: serial data output
				OSL1	PnDIRm	PnODCm	PnPLUDm	SC1IOM	SC1SBIS	SC1SBOS
Port 5	2-wire	Transmission only	P50/TXD1A	0	P5DIR0:1	P5ODC0	P5PLUD0	0	0	1
			-		-	-	-			
		Reception only	P51/RXD1A	0	P5DIR1:0	-	-	0	1	0
			-		P5DIR0:1	P5ODC0	P5PLUD0			
	1-wire	Transmission only	P50/TXD1A	0	P5DIR0:1	P5ODC0	P5PLUD0	0	0	1
			-		-	-	-			
		Reception only	P50/TXD1A	0	P5DIR0:1	P5ODC0	P5PLUD0	1	1	0
			-		-	-	-			
Port 7	2-wire	Transmission only	P75/TXD1B	1	P7DIR5:1	P7ODC5	P7PLUD5	0	0	1
			-		-	-	-			
		Reception only	P76/RXD1B	1	P7DIR6:0	-	-	0	1	0
			-		P7DIR5:1	P7ODC5	P7PLUD5			
	1-wire	Transmission only	P75/TXD1B	1	P7DIR5:1	P7ODC5	P7PLUD5	0	0	1
			-		-	-	-			
		Reception only	P75/TXD1B	1	P7DIR5:1	P7ODC5	P7PLUD5	1	1	0
			-		-	-	-			

■ UART Serial Interface 2 Pin Setup

Port	Wire system	Type	Pin	Pin setup (flag setup)						
				SC2SEL register	PnDIR register	PnODC register	PnPLU register	SC2MD1 register		
				Serial 2 I/O pin switching	I/O mode selection	Nch open-drain output selection	Pull-up resistor selection	Serial data input selection	Serial input control selection	SBO2 (TXD2) pin function selection
						Arbitrary setting	Arbitrary setting			
				0: P65 to P66 1: P30 to P31	0: input mode 1: output mode	0: push/pull 1: Nch open-drain	0: not added 1: added	0: data input from RXD2 1: data input from TXD2	0: "1" input 1: serial input	0: port 1: serial data output
				OSL2	PnDIRm	PnODCm	PnPLUm	SC2IOM	SC2SBIS	SC2SBOS
Port 6	2-wire	Transmission only	P65/TXD2A	P6DIR5:1	P6DIR5:1	P6ODC5	P6PLU5	0	0	1
		-	-	-	-	-				
		Reception only	-	-	-	-	-	0	1	0
		P66/RXD2A	P6DIR6:0	P6DIR6:0	-	-				
	Transmission/Reception	P65/TXD2A	P6DIR5:1	P6DIR5:1	P6ODC5	P6PLU5	0	1	1	
	P66/RXD2A	P6DIR6:0	P6DIR6:0	-	-					
	1-wire	Transmission only	P65/TXD2A	P6DIR5:1	P6DIR5:1	P6ODC5	P6PLU5	0	0	1
		-	-	-	-	-				
Reception only		P65/TXD2A	P6DIR5:1	P6DIR5:1	P6ODC5	P6PLU5	1	1	0	
-		-	-	-	-					
Port 3	2-wire	Transmission only	P30/TXD2B	1	P3DIR0:1	P3ODC0	P3PLUD0	0	0	1
		-	-		-	-				
		Reception only	-	1	-	-	-	0	1	0
		P31/RXD2B	P3DIR1:0		-	-				
	Transmission/Reception	P30/TXD2B	1	P3DIR0:1	P3ODC0	P3PLUD0	0	1	1	
	P31/RXD2B	P3DIR1:0		-	-					
	1-wire	Transmission only	P30/TXD2B	1	P3DIR0:1	P3ODC0	P3PLUD0	0	0	1
		-	-		-	-				
Reception only		P30/TXD2B	1	P3DIR0:1	P3ODC0	P3PLUD0	1	1	0	
-		-		-	-					

■ UART Serial Interface 3 Pin Setup (MN101EF76K/MN101EF56K)

Port	Wire system	Type	Pin	Pin setup (flag setup)						
				SC3SEL register	PnDIR register	PnODC register	PnPLUD register	SC3MD1 register		
				Serial 3 I/O pin switching	I/O mode selection	Nch open-drain output selection Arbitrary setting	Pull-up/ pull-down resistor selection Arbitrary setting	Serial data input selection	Serial input control selection	SBO3 (TXD3) pin function selection
				0: P05 to P06 1: P40 to P41	0: input mode 1: output mode	0: push/pull 1: Nch open-drain	0: not added 1: added	0: data input from RXD3 1: data input from TXD3	0: "1" input 1: serial input	0: port 1: serial data output
				OSL3	PnDIRm	PnODCm	PnPLUDm	SC3IOM	SC3SBIS	SC3SBOS
Port 0	2-wire	Transmission only	P05/TXD3A -	0	P0DIR5:1 -	P0ODC5 -	P0PLUD5 -	0	0	1
		Reception only	- P06/RXD3A	0	- P0DIR6:0	- -	- -	0	1	0
		Transmission/ Reception	P05/TXD3A P06/RXD3A	0	P0DIR5:1 P0DIR6:0	P0ODC5 -	P0PLUD5 -	0	1	1
			Transmission only	P05/TXD3A -	0	P0DIR5:1 -	P0ODC5 -	P0PLUD5 -	0	0
	1-wire	Reception only	P05/TXD3A -	0	P0DIR5:0 -	- -	- -	1	1	0
Port 4	2-wire	Transmission only	P40/TXD3B -	1	P4DIR0:1 -	P4ODC0 -	P4PLUD0 -	0	0	1
		Reception only	- P41/RXD3B	1	- P4DIR1:0	- -	- -	0	1	0
		Transmission/ Reception	P40/TXD3B P41/RXD3B	1	P4DIR0:1 P4DIR1:0	P4ODC0 -	P4PLUD0 -	0	1	1
			Transmission only	P40/TXD3B -	1	P4DIR0:1 -	P4ODC0 -	P4PLUD0 -	0	0
	1-wire	Reception only	P40/TXD3B -	1	P4DIR0:1 -	P4ODC0 -	P4PLUD0 -	1	1	0

18.4.4 Setup Example

■ Transmission/Reception Setup

The setup example at UART transmission/reception using Serial Interface 1 is shown below. Table:18.4.10 shows the conditions at transmission/reception. The basic procedures are the same in Serial Interface 0, 2 and 3. Pin settings (2) and (3) differ in each serial interface.

Table:18.4.10 UART Interface Transmission Reception Setup Condition

Setup item	Set to
Serial data input selection	RXD1
Frame mode specification	8 bits + 2 stop bits
First transfer bit	MSB
Clock source	Timer A
TXD1/RXD1 pin style	Nch open-drain
Pull-up resistor of TXD1 pin	Added
Parity bit add/check	"0" added/checked
Serial 1 transmission complete interrupt	Enabled
Serial 1 reception complete interrupt	Enabled

An example setup procedure, with a description of each step is shown below

Setup Procedure	Description
(1) Select the clock source SC1SEL (0x03F20) bp2 to 0: SC1SEL2 to 0 =111 SC1MD3 (0x03F20) bp2 to 0: SC1PSC2 to 0 =111	(1) Set the SC1SEL2 to 0 flags of SC1SEL register and the SC1PSC2 to 0 flags of SC1MD3 register to "111" to select Timer A output as a clock source.
(2) Control the pin style [set the pin corresponding to each serial interface] P5ODC(0x03EF5) bp0: P5ODC0 =1 bp1: P5ODC1 =1 P5PLUD(0x03FA5) bp1: P5PLUD1 =1	(2) Set the P5ODC1 to 0 flags of P5ODC register to "11" to select Nch open-drain as styles of TXD0 pin and RXD0 pin. Set the P5PLUD1 flag of P5PLUD register to "1" to enable the pull-up resistor. (Set the pin corresponding to each serial interface.)
(3) Control the pin direction [Set the pin corresponding to each serial interface] P5DIR(0x03F95) bp1 to 0: P5DIR5 to 0 =01	(3) Set the P5DIR1 to 0 flags of Port 5 direction control register (P5DIR) to "01" to set P51 to input mode (Set the pin corresponding to each serial interface.).
(4) Set SC1MD0 register Select the start condition SC1MD0 (0x03F21) bp3: SC1STE =1 Select the transfer first bit SC1MD0 (0x03F21) bp4: SC1DIR =0	(4) Set the SC1STE flag of SC1MD0 register to "1" to enable start condition. Set the SC1DIR flag of SC1MD0 register to "0" to select MSB as the transfer first bit.

Setup Procedure	Description
(5) Set SC1MD2 register Control the output data SC1MD2 (0x03F23) bp0: SC1BRKE =0 Select the added parity bit SC1MD2 (0x03F23) bp3: SC1NPE =0 bp5 to 4: SC1PM1 to 0 =00 Specify the frame mode SC1MD2 (0x03F23) bp7 to 6: SC1FM1 to 0 =11	(5) Set the SC1BRKE flag of SC1MD2 register to "0" to select serial data transmission. Set the SC1PM1 to 0 flags of SC1MD2 register to "00" to select 0 parity, and set the SC1NPE flag to "0" to enable add parity bit. Set the SC1FM1 to 0 flags of SC1MD2 register to "11" to select 8 bits + 2 stop bits at the frame mode.
(6) Set SC1MD1 register Select the communication type SC1MD1 (0x03F22) bp0: SC1CMD =1 Select the clock dividing SC1MD1 (0x03F22) bp3: SC1CKM =1 bp2: SC1MST =1 bp1: SC1DIV =0 Control the pin function SC1MD1 (0x03F22) bp4: SC1SBOS =1 bp5: SC1SBIS =1 bp7: SC1IOM =0	(6) Set the SC1CMD flag of SC1MD1 register to "1" to select the duplex UART. Set the SC1CKM flag of SC1MD1 register to "1" to select "Divided" for transfer clock division selection. Set the SC1DIV flag of SC1MD1 register to "0" to select "Divide by 8" for transfer clock division value selection. The SC1MST flag should be always set to "1" to select a clock master. Set the SC1SBOS and SC1SBIS flags of SC1MD1 register to "1" to set TXD1 pin to serial data output and RXD1 pin to serial data input.
(7) Set the baud rate timer	(7) Set the baud rate by TMAMD register and TMAOC register. Set the TMAEN flag to "1" to operate Timer A. [Chapter 10 10.4.1 Operation]
(8) Enable the interrupt PSW bp6: MIE =0 SC1RICR(0x03FF7) bp1: SC1RIE =1 SC1TICR(0x03FF6) bp1: SC1TIE =1 PSW bp6: MIE =1	(8) Clear the MIE flag of PSW to disable all maskable interrupts. Set the SC1RIE flag of SC1RICR register to "1" and also set the SC1TIE flag of SC1TICR register to "1" to enable the interrupt request. If any of the interrupt request is already set, clear the request flag. Set the MIE flag of PSW to "1" to enable maskable interrupt.
(9) Start the serial transmission The transmission data → TXBUF1 (0x03F27) The reception data → input to RXD1 pin	(9) Transmission is started by setting transmission data to the serial transmission data buffer (TXBUF1). When the transmission is completed, Serial 1 transmission complete interrupt (SC1TIRQ) is generated. If a start bit is detected, the received data is loaded into the RXBUF1; and, Serial 1 reception complete interrupt (SC1RIRQ) is generated.

* Each setup in (4), (5) and (6) can be set at the same time.



For 1-wire communication, input and output serial data from TXDn pin. Use direction control register of port to switch between input and output. In reception, be sure to set SCnSBIS of SCnMD1 register to "1" to select "serial data input". RXDn pin can be used as a general purpose port.



This serial interface has an emergency reset function. If communication needs to be stopped by this function, set SCnSBOS and SCnSBIS of SCnMD1 register to "0".



Set each flag in accordance with the procedures indicated in the table above. Be sure to start communication after the settings of all control registers (refer to Table:18.2.1 but exclude TXBUF_n and RXBUF_n) are completed.



When setting this serial interface communication mode to "UART", set Serial Interface n mode register 1 (SCnMD1) while the serial data input pin is at "High".



When the SCnCMD flag of SCnMD1 register is set to "1" to select "Duplex UART", the transfer clock is divided regardless of the SCnCKM flag.

18.5 Multi Master IIC Interface

18.5.1 Multi Master IIC Interface

Multi master IIC serial communication is available with Serial Interface 4. This IIC interface communicates by complying with the data transfer format of Philips IIC-BUS. Table:18.1.6 shows IIC serial interface functions.

■ Data I/O Pin Setup

Use SDA4 pin for data input/output, which is commonly used with SBO4 pin. Set the SC4IOM flag of SC4MD1 register to "1" to input the serial data from SDA4 pin. SBI4 pin can be used as a general-purpose port; make sure to set the SC4SBIS flag of SC4MD1 register to "1" to set "serial data input".



Be sure to set the SC4SBIS flag of SC4MD1 register to "serial data input" regardless of transmission/reception in order to detect the start condition and ACK bit reception.



Nch open-drain should be used for pin format because the bus is switched between use and open by hardware during communication. Even in reception, select "output" for direction control of SDA pin.

■ Input Edge/Output Edge Setup

In IIC communication, data is always received at the falling edge of the clock regardless of the SC4CE1 value.

■ Master/Slave Selection

Multi master IIC function (clock master/clock slave) or slave-dedicated IIC function (clock slave) can be selected with the SC4MST of SC4MD1 register. When a start condition from another master or arbitration lost is detected if using clock master, clock slave operation will be selected.

■ Slave Address Setup

In this serial interface, 7-bit or 10-bit slave address can be set. To change slave address for 7 bits, set slave address to the upper 7 bits of SC4AD0 register (SC4ADR7 to SC4ADR1) after setting the SC4ADM flag of SC4MD3 register to "0" to set the 7-bit address mode. To change slave address for 10 bits, set the upper 2 bits of slave address to the lower 10 bits of SC4AD1 register (SC4ADR9 and SC4ADR8), and set the lower 8 bits of slave address to SC4AD0 register after setting the SC4ADM flag of SC4MD3 register to "1" to set the 10-bit address mode.

■ Transfer Format

Two formats are available for IIC bus transfer; addressing format and free data format. In addressing format, 1 to 2 byte address data which consists of the slave address (7 bits/10 bits) and R/W bit (1 bit) is transmitted after the start condition, and transmission/reception is executed. In free data format, data is transmitted immediately after the start condition. Regarding the free data format, this LSI only supports IIC master communication. The following presents the communication sequences. The shaded regions of Figure:18.5.1, Figure:18.5.3 and Figure:18.5.3 are the data transmitted from other IIC.

■ Addressing Format

- 7-bit address

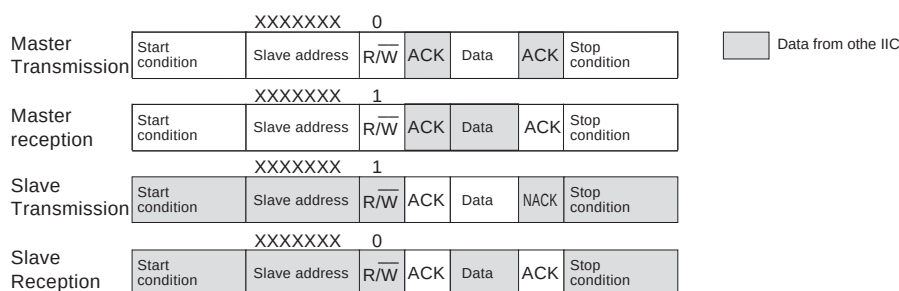


Figure:18.5.1 Communication Sequence in 7-bit Address Mode

- 10-bit address

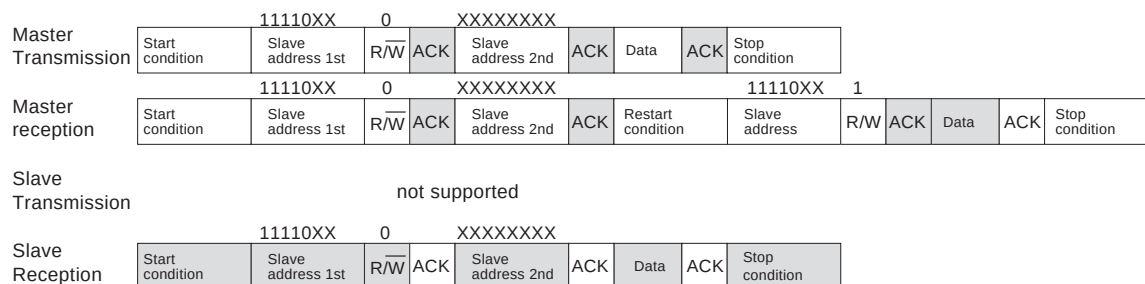


Figure:18.5.2 Communication Sequence in 10-bit Address Mode

- Free data format

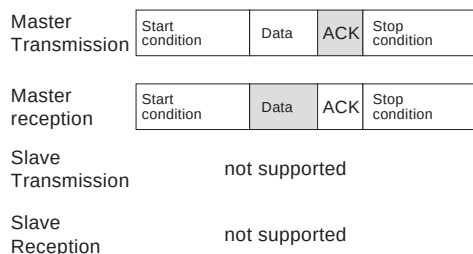


Figure:18.5.3 Communication Sequence in Free Data Format

■ Clock Setup

The clock source is selected from the dedicated prescaler and timer output (Timer 0 to 4 or A) by SC4MD2 register. The dedicated prescaler starts to operate by selecting "prescaler count enable" by the SC4MD2.

The following table shows the clock source selection condition.

Table:18.5.1 Clock Source Selection Condition

Communication condition	Clock source selection condition
Master communication only (Only this IIC is set as the master)	Standard mode : 800 kHz or less High-speed mode : 3.2 MHz or less
Master/slave communication	Standard mode : 500 kHz or more, 800 kHz or less High-speed mode : 3.2 MHz
Slave communication only (SC4MST=0)	Standard mode : 500 kHz or more High-speed mode : 3.4 MHz or more

In master communication, the transfer clock is obtained by dividing the clock source by 8. Duty is "High:Low=1:1" in standard mode and "High:Low=3:5" in high-speed mode. In master communication, set the clock source so that the transfer clock is not over 100 kHz in standard mode and 400 kHz in high-speed mode. Select the transfer rate so that its value is 0.8 times or over of other master.

In slave communication, Set the clock source to 500 kHz or over in standard mode and 3.4 MHz or over in high-speed mode. However, when the transfer rate and duty of the master are identified, any clock source which satisfies the following conditions can be selected.

Table:18.5.2 Transfer Rate Duty and Clock Source Selection Condition

Transfer rate duty	Slave clock source selection condition
1:1	Transfer rate $\times$ 4 or more
1:2/2:1	Transfer rate $\times$ 6 or more
1:3/3:1	Transfer rate $\times$ 8 or more

Table:18.5.3 IIC Interface Clock Source

	Multi master IIC
Clock source (Internal clock)	fpll-div/2
	fpll-div/4
	fpll-div/16
	fpll-div/32
	fs/2
	fs/4
	Timer output (Timer 0 to 4 or A)



In IIC master communication, transfer rate is obtained by dividing the clock source by 8. In master communication, set the clock source with SC4MD2 register so that the transfer rate is not over 100 kHz in the standard mode and 400 kHz in the high-speed mode. Select the transfer rate which is more than 0.8 times of the other master.



In slave communication, set the clock source to 500 kHz or more in the standard mode, 3.4 MHz or more in the high-speed mode, or set to the value which is 6 times or larger than the transfer rate.



When switching the clock setup, always set the SC4SBIS flag and SC4SBOS flag of SC4MD1 register to "0" in advance.

■ Activation Factor for Communication

(Master communication)

Set data (at transmission) or dummy data (at reception) in the transmission buffer (TXBUF4). Regardless of transmission or reception, a start condition and transfer clock are generated to start communication.

(Slave communication)

Reception starts after detecting a start condition. When the received address matches the address set by address setting register 0 and 1, or a general call is detected, the slave address comparison flag is set to start slave communication.

■ Interrupt

This serial interface has two types of interrupt, a Serial 4 interrupt "SC4IRQ" and a Serial 4 stop condition detection interrupt "SC4STPCIRQ". Table:18.5.4 shows the interrupt generation factors.

Table:18.5.4 IIC Communication Interrupt Generation Factor

Interrupt	Interrupt generation factor
SC4IRQ (Serial 4 interrupt)	Master communication completion (after 1 byte data + ACK) Slave address match (after ACK) Slave communication completion (after 1 byte data + ACK) Slave communication completion (communication data instability detection)
SC4STPCIRQ (Serial 4 stop condition detection interrupt)	Detection of stop condition generated by other master

However, in 10-bit address mode communication, even if the first 2 bits addresses transmitted from master match with SC4ADR9 to 8 flags of SC4AD1 register, Serial 4 communication completion interrupt (SC4IRQ) is not generated. Only when the next lower 8 bits addresses match with SC4ADR7 to 0 flags of SC4AD0 register, SC4IRQ is generated

■ Start Condition Setup

In master communication, a start condition is always generated at the first communication regardless of the SC4STE flag value of SC4MD0 register. Select whether the start condition is enabled or disabled after the second byte communication by the SC4STE flag of SC4MD0 register.

Enable/disable of the start condition can be determined in each communication by setting the enable/disable before each communication data setup.

Start Condition Setup Example

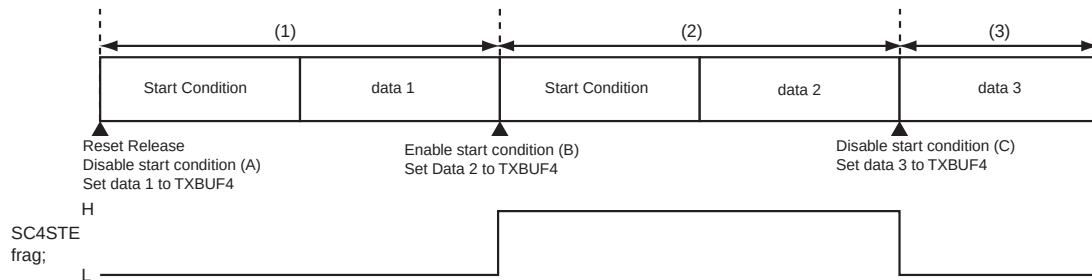


Figure:18.5.4 Start Condition Setup Example

- (1) At the first communication, start condition is added regardless of SC4STE value.
- (2) Start condition is added by setting start condition to "enable" at (B).
- (3) Start condition is not added by setting start condition to "disable" at (C).

■ Stop Condition Generation

Stop condition is formed if the data bus (SDA) changes from "Low" to "High" when the clock bus (SCL) is "High". Writing "1" in the SC4STPC flag of SC4MD3 register by program starts the stop condition output. When the stop condition is generated, the SC4STPC flag is automatically cleared.

The stop condition should be requested only when this IIC occupies the bus as master. If the stop condition is generated by this IIC, Serial 4 stop condition detection interrupt is not generated.

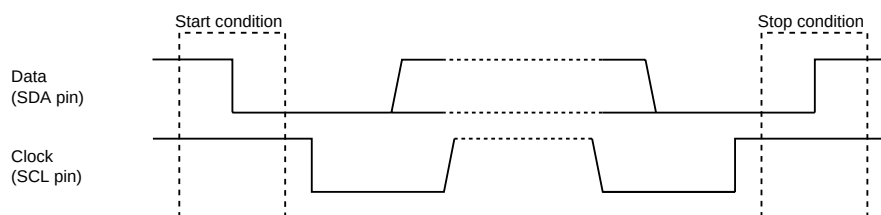


Figure:18.5.5 Start Condition and Stop Condition

■ Start/Restart Condition Detection

If the data bus (SDA) changes from "High" to "Low" when the clock bus (SCL) is "High", a start condition is detected and the SC4STRT flag and SC4BUSBSY flag of SC4STR1 register are set to "1". The SC4STRT flag is cleared to "0" when data is set in TXBUF4 by the interrupt process immediately after the slave address reception. The SC4STRT flag is also set when a restart condition is detected. If the address transmitted from master does not match with the slave address, the address is automatically cleared by the hardware when an address mismatch is detected.

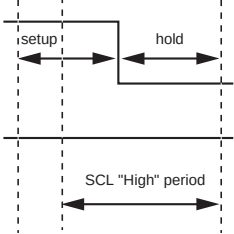
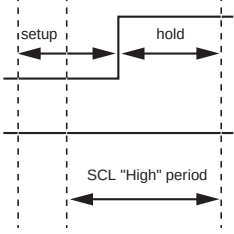
■ Stop Condition Detection

If the data bus (SDA) changes from "Low" to "High" while the clock bus (SCL) is "High", a start condition is detected. SC4STPCIRQ is generated and the SC4BUSBSY flag of SC4STR1 register is cleared.

■ Start Condition/Stop Condition Detection Condition

This IIC detects the start condition and the stop condition on bus lines by the sampling with the internal clock. The detection conditions are as follows.

Table:18.5.5 Start Condition/Stop Condition Detection Condition

Start Condition		SCL "High" period	3-clock source or more
		SDA setup	2-clock source or more
		SDA hold	2-clock source or more
Stop Condition		SCL "High" period	3-clock source or more
		SDA setup	2-clock source or more
		SDA hold	2-clock source or more

■ Communication Data Instability Detection

When a serial circuit detects a changes in the data bus (SDA4 pin) while the clock bus (SCL4 pin) is "High" after a start condition is detected, the communication data is recognized as instability and the SC4DATA_ERR flag of SC4STR1 register is set to "1". Since communication is not properly executed when the SC4DATA_ERR flag is "1", clear the SC4DATA_ERR flag to "0" by a user's program and start the communication again.

The first operation after communication data instability is detected is different between the master communication and the slave communication.

In the case of the master communication, the communication completion interrupt SC4TIRQ is generated after completing 1 byte communication. However, if the arbitration lost is detected, the communication mode is changed from the master to the slave, and the communication may be stopped.

On the other hand, in slave communication, the SC4TIRQ is generated immediately and the communication is completed. When the SC4DEM flag is "1" in slave communication, the circuit can detect start conditions and starts communication even if the SC4DATA_ERR flag is "1".

■ Transmission at Master Communication/Reception Mode Setup and Operation

In master communication, select the transmission or reception mode by the SC4REX flag of SC4MD3 register. The first data always communicates with a start condition regardless of the value set to the SC4STE flag. The start condition is output from this master serial interface.

In master operation, transmission is executed by setting the slave address and R/W bit in the first data after a start condition is generated during the addressing format. In the master reception, check the ACK signal from the slave in the interrupt process after the address data transmission is finished, then switch to the reception mode.

To start a new communication with other devices before the current communication end, regenerate a start condition and transmit the slave address and R/W bit again. In reception, SDA4 line is automatically released and will be in reception wait status. The reception acknowledge (ACK bit) is transmitted after the data storage is completed.

Refer to [Figure:18.5.12 Master Transmission Timing] and [Figure:18.5.13 Master Reception Timing].

■ Slave Communication

This serial interface performs address verification by obtaining the received data automatically after detecting a start condition on IIC bus. SC4IRQ is generated only when the address transmitted from master matches with the assigned slave address. Data transmission and reception are determined by the SC4WRS flag of SC4STR1 register. When SC4WRS = "0", slave reception is selected; when SC4WRS = "1", slave transmission is selected. In slave transmission, the bus line is released by setting the transmission data in TXBUF4 register, and data transmission starts with the clock transmitted from the master. Do not set data to TXBUF4 register because bus line is released automatically when NACK is received. In slave reception, the bus line is released by setting dummy data to TXBUF4 register, and data reception starts with the clock transmitted from the master.

■ Address Compare Flag

When the address transmitted from master matches with the slave address, the address comparison flag SC2ADD_ACC of SC2STR1 is set to "1" and ACK is output automatically. In 10-bit address mode communication, when the first upper 2 bits of addresses transmitted from master match with SC2AD9 to 8 flags of SC2AD1, the SC2ADD_ACC flag is set to "1" and ACK is output automatically. When the next lower 8 bits of addresses transmitted from master match with SC2AD7 to 0 flags of SC2AD0 register, the SC2ADD_ACC flag is held at "1" and ACK bit is output automatically. When the addresses do not match, the SC2ADD_ACC flag is cleared to "0" and also ACK will not be output.

■ General Call Communication outputted

This serial interface supports general call communication. When a general call is detected, the SC4ADD_ACC flag and SC4GCALL flag of SC4STR1 register are set.

■ Reception of Acknowledgement (ACK bit) after Data Transmission

ACK bit can be enabled or disabled by the SC4ACKS flag of SC4MD3 register. If ACK bit is enabled, data (1 to 8 bits) is transmitted and ACK bit is received from the data receiver. The data bus (SDA) is released automatically upon receiving the ACK bit. In master operation, the clock for ACK bit reception is output once, and the ACK bit is stored in the SC4ACK0 flag of SC4MD3 register. There is no shift operation for reception register RXBUF4 by the ACK bit reception clock. When the received ACK bit level is "Low", the reception of the receiver is operated properly. It indicates the reception wait status for the next data.

When the ACK bit level is "High", the receiver may finish the reception process. At master operation, finish the communication by writing "1" to the SC4STPC flag of SC4MD3 register or issue the slave address again by generating a restart condition. At slave operation, it is not necessary to set data to TXBUF4 register because the transmission is finished by releasing the data bus (SDA4) automatically at slave operation. In this case, the slave address compare flag is cleared. To continue the communication, an address match is required again.

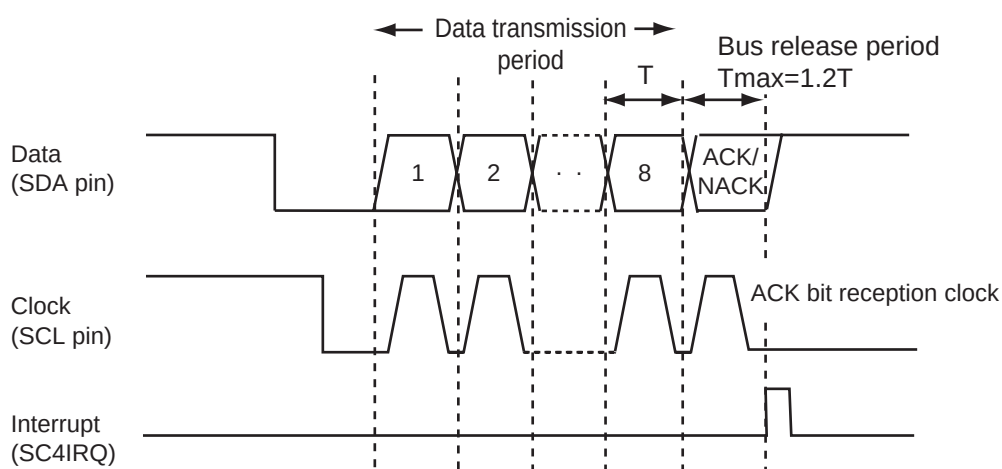


Figure:18.5.6 ACK Bit Reception Timing after 8-bit Data Transmission

■ Transmission of Acknowledgement (ACK bit) after Data Reception

Enable/disable of ACK bit can be selected in the same way as the ACK bit reception. When ACK bit is enabled, ACK bit and clock is output after receiving the data (1 to 8 bit). To continue the reception, output a "Low" level ACK bit. To finish the reception, output a "High" level ACK bit. ACK bit output level can be set by SC4ACK0 flag of SC4MD3 register.

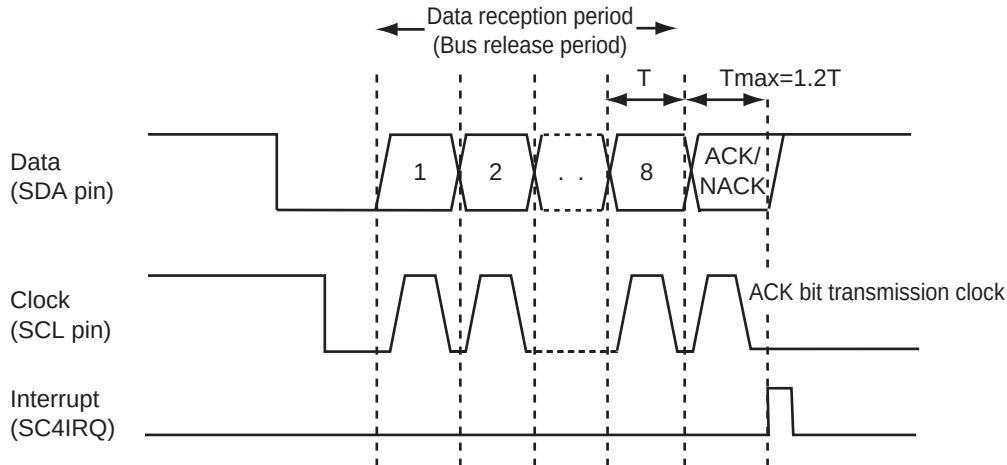


Figure:18.5.7 ACK Bit Transmission Timing after 8-bit Data Reception

■ Arbitration Lost

During master transmission, data bus (SDA) compares every bit of data output from this circuit to detect a race condition with other masters. When the output data does not match with the data bus, it judges as the communication is not permitted (arbitration lost). The SC4ABT_LST flag of SC4STR1 register is set instantly. It releases the data bus and the clock bus and continues slave reception. SC4IRQ is not generated when the arbitration lost is detected. However, SC4IRQ is generated when the slave address is matched after the detection of arbitration lost. Confirm the SC4ABT_LST flag at next interrupt generation timing (SC4STPCIRQ, SC4IRQ). When SC4ABT_LST flag is "1", execute communication again after the release of IIC bus because the master transmission is not formed.

In this case, clear the SC4ABT_LST flag with program.

■ Busy Flag

This serial interface contains 2 types of busy flags (SC4BUSBSY, SC4IICBSY) in SC4STR1 register.

The SC4BUSBSY flag is set to "1" in IIC bus communication. It is automatically set when a start condition is detected on IIC bus and cleared when a stop condition is detected.

The SC4IICBSY flag is "1" when this serial interface is in communication. In master communication, this flag is "1" during data loading, start condition generation, data communication, ACK communication, and stop condition generation. In slave communication, the flag is "1" ACK communication, data communication, and ACK output when the address transmitted from the master matches the slave address. If a restart condition is detected, the SC4IICBSY flag is cleared, and it becomes "1" again at ACK output when the address transmitted from the master matches the slave address. Refer to the timing charts described in [Figure:18.5.12 Master Transmission Timing] for the timing of flag set/clear.

A maximum of one cycle of internal transfer clock is needed before the communication starts after data is set to TXBUF4 register and the flag setting of SC4IICBY is done.

■ Handshake Using Clock Synchronous Mechanism

In master operation, SCL bus-line is monitored by sampling with the clock source. When different signal level between SCL output and SCL bus-line is detected, it is judged that the handshake with clock synchronous mechanism is activated, and the transfer clock is extended. With this operation, the master speed can be matched to the bus clock speed.

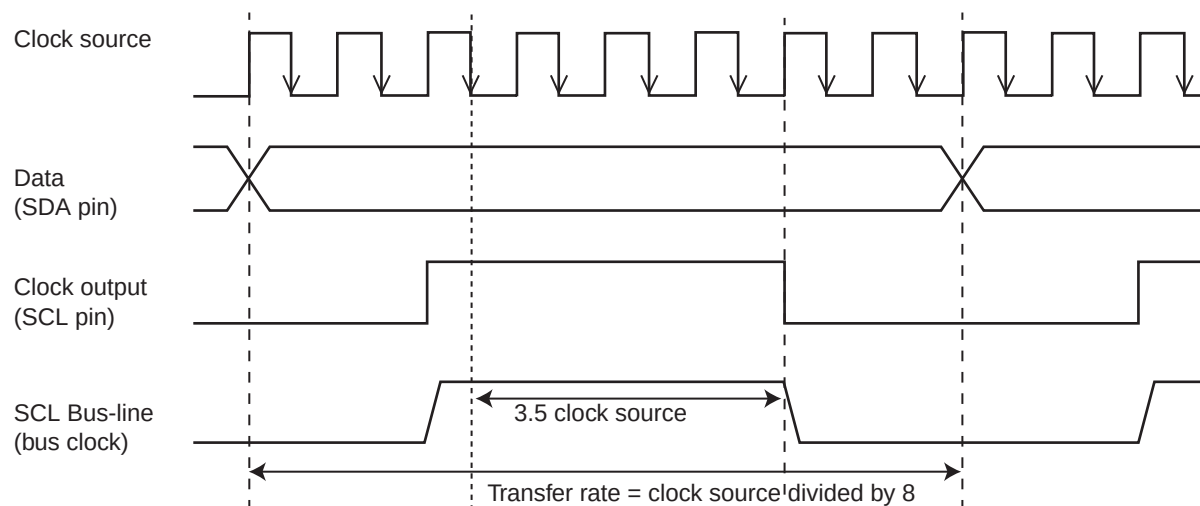


Figure:18.5.8 Without "Low" Period Extension from the Other Clock (Standard Mode)

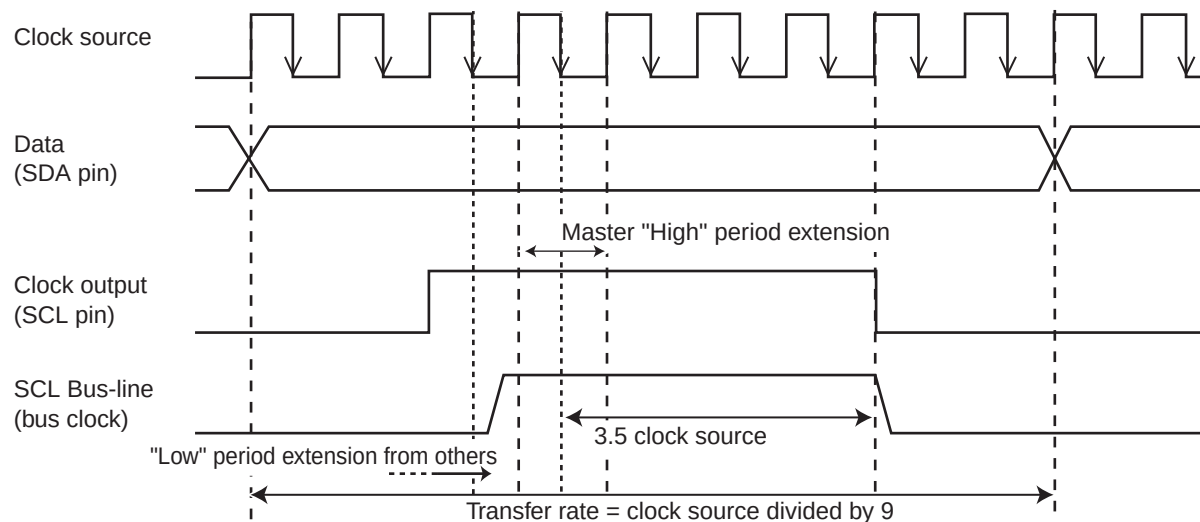


Figure:18.5.9 With "Low" Period Extension from the Other Clock (Standard Mode)

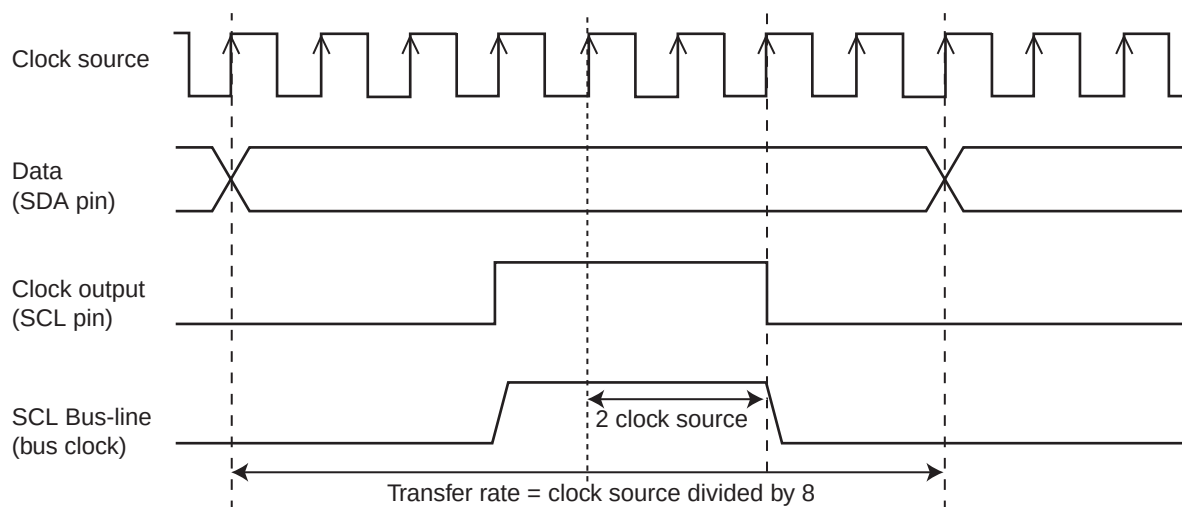


Figure:18.5.10 Without "Low" Period Extension from the Other Clock (High-speed Mode)

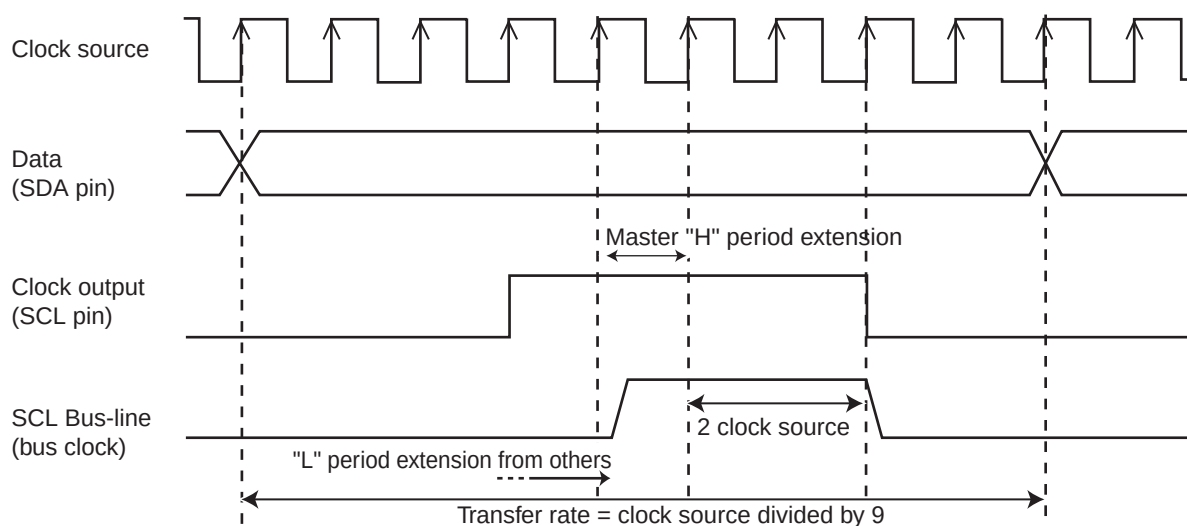


Figure:18.5.11 With "Low" Period Extension from the Other Clock (High-speed Mode))



To obtain the intended transfer rate, design the bus clock so that the rising time of the SCL signal does not exceed 0.5 clock (standard mode) or 1 clock (high-speed mode) of the clock source.

The following items are the same as the clock synchronous serial interface. Refer to the following pages.

■ Automatic Continuous Transfer by ATC1

Refer to: XVIII-45

■ First Transfer Bit Setup

Refer to: XVIII-43

■ Transmission Data Buffer

Refer to: XVIII-43

■ Reception Data Buffer

Refer to: XVIII-44

■ Transmission Bit Count and First Transfer Bit

Refer to: XVIII-44

■ Emergency Reset

Refer to: XVIII-48

■ Master Transmission Timing

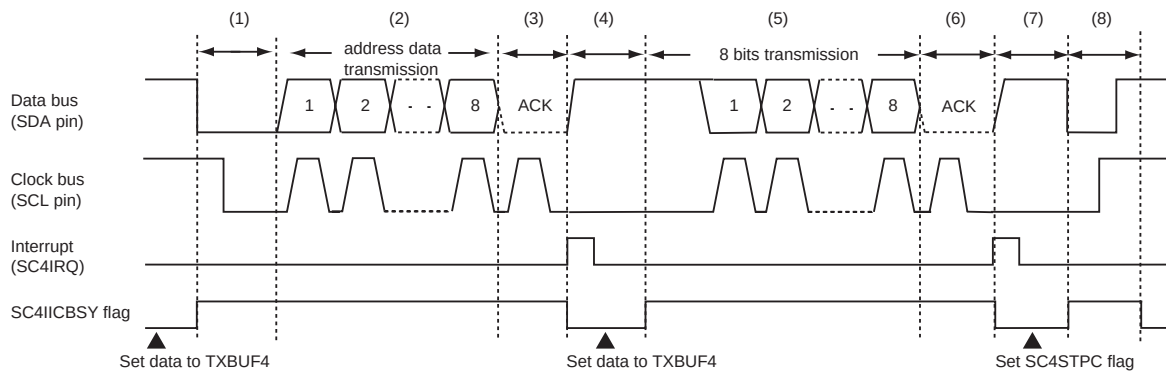


Figure:18.5.12 Master Transmission Timing

(1) Start condition output

- The SC4IICBSY flag is set to "1" after the elapse of 1 to 2 clocks of the clock source when the data is set to TXBUF4.

(2) Address data output

(3) Bus released period, ACK bit reception

(4) Interrupt process

- Communication start: set data to TXBUF4.
- The SC4IICBSY flag is set to "1" after the elapse of 1 to 10 clocks of the clock source when the data is set to TXBUF4.

(5) Transmission data output

(6) Bus released period, ACK bit reception

(7) Interrupt process

- Communication end: set the SC4STPC flag.
- The SC4IICBSY flag is set to "1" after the elapse of 1 to 2 clocks of the clock source when the SC4STPC flag is set.

(8) Stop condition generation

■ Master Reception Timing

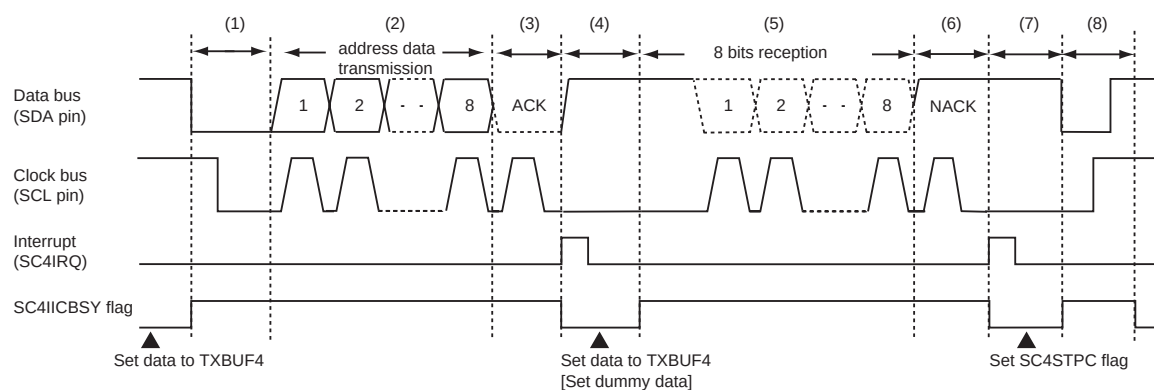


Figure:18.5.13 Master Reception Timing

(1) Start condition output

(2) Address data output

(3) Bus released period, ACK bit reception

(4) Interrupt process

- Reception mode setup: SC4REX = 0 → 1
- Communication start: set dummy data to TXBUF4.
- The SC4IICBSY flag is set to "1" after the elapse of 1 to 10 clocks of the clock source when the dummy data is set to TXBUF4.

(5) Bus released period, ACK bit reception

(6) NACK bit output

(7) Interrupt process

- Communication end: set the SC4STPC flag.
- The SC4IICBSY flag is set to "1" after the elapse of 1 to 2 clocks of the clock source when the SC4STPC flag is set.

(8) Stop condition generation

■ Slave Transmission Timing

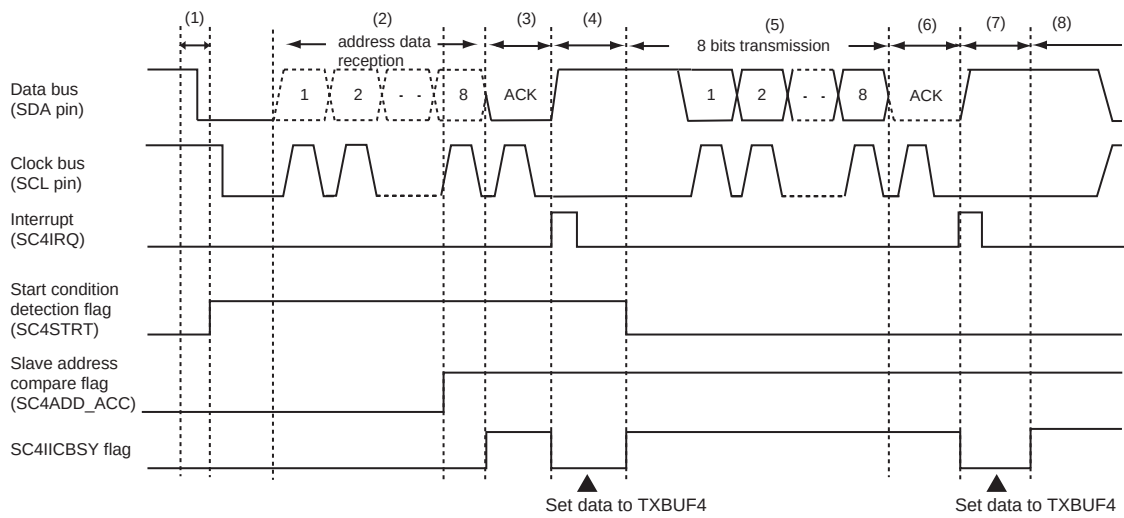


Figure:18.5.14 Slave Transmission Timing

(1) Bus released period, start condition detection

(2) Bus released period, address data reception

(3) ACK bit output

(4) Interrupt process

- Communication start: Set data to TXBUF4.

- The SC4IICBSY flag is set to "1" after the elapse of 1 to 10 clocks of the clock source when the data is set to TXBUF4.

(5) Transmission data output

(6) Bus released period, ACK bit reception

(7) Interrupt process

- Communication start: Set data to TXBUF4.

- The SC4IICBSY flag is set to "1" after the elapse of 1 to 10 clocks of the clock source when the data is set to TXBUF4.

(8) Transmission data output

■ Slave Transmission Timing (NACK Reception)

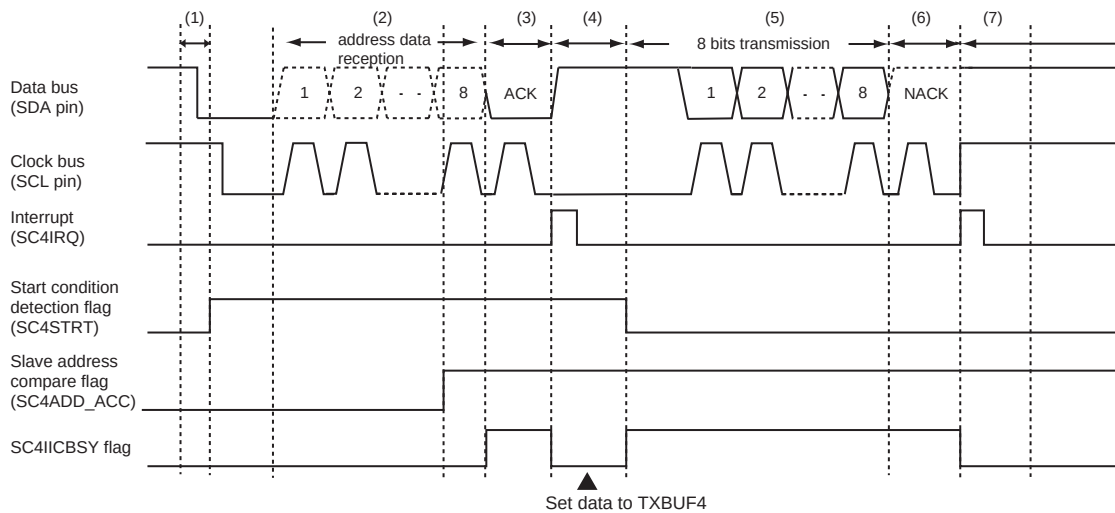


Figure:18.5.15 Slave Transmission Timing (NACK Reception)

(1) Bus released period, start condition detection

(2) Bus released period, address data reception

(3) ACK bit output

(4) Interrupt process

- Communication start: Set data to TXBUF4.

- The SC4IICBSY flag is set to "1" after the elapse of 1 to 10 clocks of the clock source when the data is set to TXBUF4.

(5) Transmission data output

(6) Bus released period, NACK bit reception

(7) Bus released period

■ Slave Reception Timing (Stop Condition Detection)

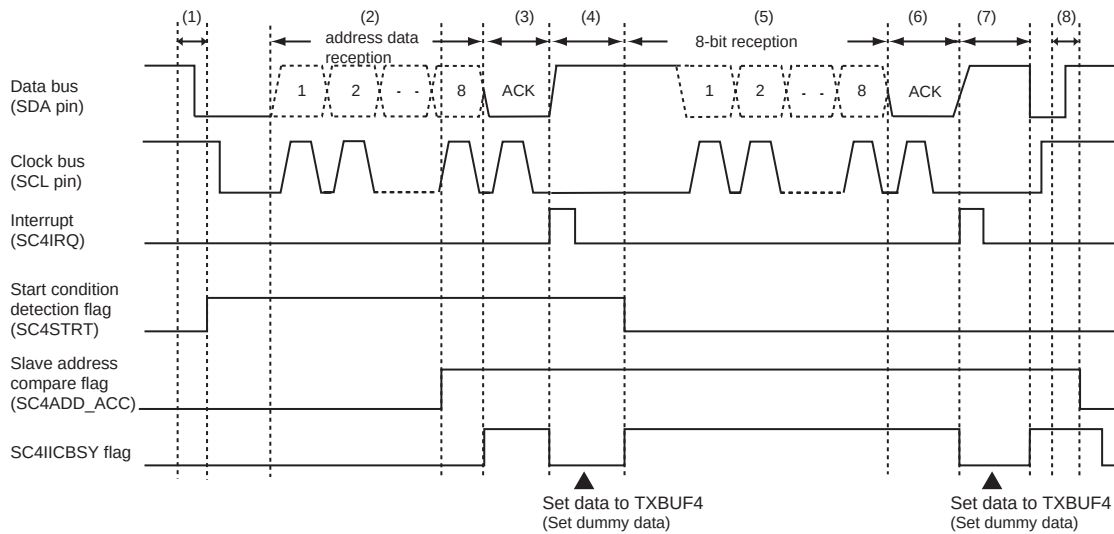


Figure:18.5.16 Slave Reception Timing (Stop Condition Detection)

(1) Bus released period, start condition detection

(2) Bus released period, address data reception

(3) ACK bit output

(4) Interrupt process

- Communication start: Set dummy data to TXBUF4.
- The SC4IICBSY flag is set to "1" after the elapse of 1 to 10 clocks of the clock source when the dummy data is set to TXBUF4.

(5) Bus released period, data reception

(6) ACK bit output

(7) Interrupt process

- Communication start: Set dummy data to TXBUF4.
- The SC4IICBSY flag is set to "1" after the elapse of 1 to 10 clocks of the clock source when the dummy data is set to TXBUF4.

(8) Stop condition detection

■ Slave Reception Timing (Restart Condition Detection)

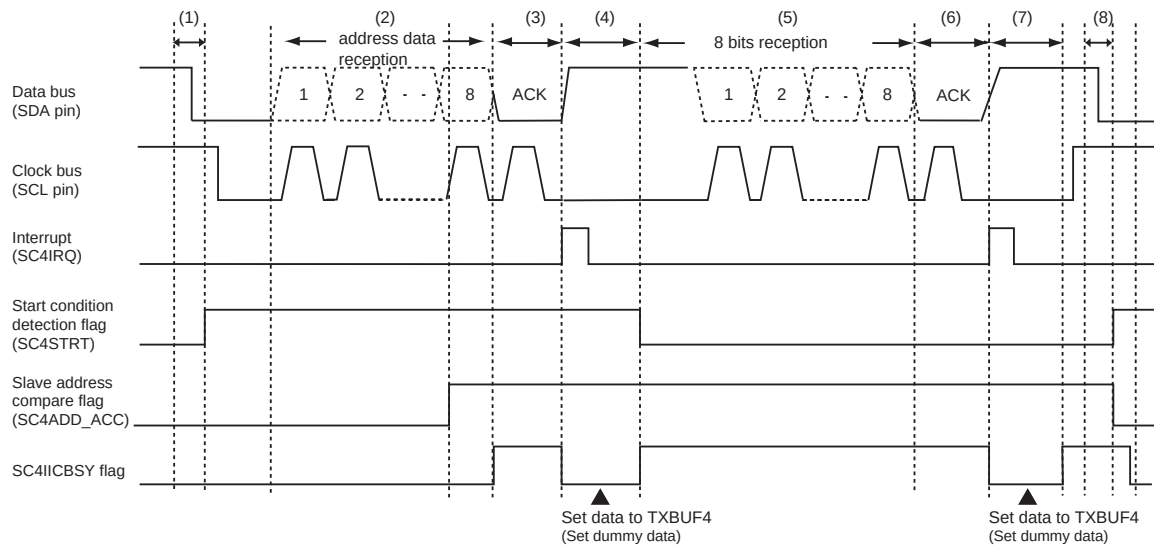


Figure:18.5.17 Slave Reception Timing (Restart Condition Detection)

(1) Bus released period, start condition detection

(2) Bus released period, address data reception

(3) ACK bit output

(4) Interrupt process

- Communication start: Set dummy data to TXBUF4.
- The SC4IICBSY flag is set to "1" after the elapse of 1 to 10 clocks of the clock source when the dummy data is set to TXBUF4.

(5) Bus released period, data reception

(6) ACK bit output

(7) Interrupt process

- Communication start: Set dummy data to TXBUF4.
- The SC4IICBSY flag is set to "1" after the elapse of 1 to 10 clocks of the clock source when the dummy data is set to TXBUF4.

(8) Restart condition detection

* Serial data input selection

18.5.2 PIn Setup

■ IIC (Multi Master) Serial Interface 4 Pin Setup

Port	Type	Pin	Pin setup (flag setup)							
			SC4SEL register	PnDIR register	PnODC register	PnPLUD register	SC4MD1 register			
			Serial 4 I/O pin switching	I/O mode selection	Nch open-drain output selection Arbitrary setting	Pull-up/ pull-down resistor selection Aribitary setting	Serial data input selection	SBT4 pin function selection	Serial input control selection	SBO4 (RXD4) pin function selection
			0: P71 to P72 1: P33 to P35	0: input mode 1: output mode	0: push/pull 1: Nch open-drain	0: not added 1: added	0: data input from SBI4 1: data input from SBO4	0: port 1: transfer clock I/O	0: "1" input 1: serial input	0: port 1: serial data output
			OSL4	PnDIRm	PnODCm	PnPLUDm	SC4IOM	SC4SBTS	SC4SBIS	SC4SBOS
Port 7	Transmission only	P71/SDA4A	0	P7DIR1:1	P7ODC1:1	P7PLUD1:1	1	1	1	1
		P72/SCL4A		P7DIR2:1	P7ODC2:1	P7PLUD2:1				
	Reception only	P71/SDA4A	0	P7DIR1:1	P7ODC1:1	P7PLUD1:1	1	1	1	1
		P72/SCL4A		P7DIR2:1	P7ODC2:1	P7PLUD2:1				
Port 3	Transmission only	P33/SDA4B	1	P3DIR3:1	P3ODC3:1	P3PLU3:1	1	1	1	1
		P34/SCL4B		P3DIR4:1	P3ODC4:1	P3PLU4:1				
	Reception only	P33/SDA4B	1	P3DIR3:1	P3ODC3:1	P3PLU3:1	1	1	1	1
		P34/SCL4B		P3DIR4:1	P3ODC4:1	P3PLU4:1				

18.5.3 Setup Example

■ Master Transmission Setup Example

The following describes the setup example for multiple data transmission to all the devices on IIC bus using Serial Interface 4 multi master IIC interface function. Table:18.5.6 shows communication conditions.

Table:18.5.6 Setup Conditions of Multi Master IIC Communication

Setup item	Set to
Serial data input selection	SDA4
Transfer bit count	8 bits
Start condition	Enabled (after 2nd communication: disabled)
First transfer bit	MSB
ACK bit	Enabled
IIC communication mode	Standard mode
Clock source	fpll-div/32
SCL4/SDA pin type	Nch open-drain
SCL4 pin pull-up resistor	Added
SDA4 pin pull-up resistor	Added
Master/Slave setting	Master

An example setup procedure is shown below.

Setup Procedure	Description
(1) Select the prescaler operation SC4MD2(0x03F52) bp3: SC4PSCE =1 (2) Select the clock source SC4MD2(0x03F52) bp2 to 0: SC4PSC2 to 0 =011 (3) Control the pin type P7ODC(0x03EF7) bp1: P7ODC1 =1 bp2: P7ODC2 =1 P7PLUD(0x03EA0) bp1: P7PLUD1 =1 bp2: P7PLUD2 =1 (4) Control the pin direction P7DIR(0x03E97) bp1: P7DIR1 =1 bp2: P7DIR2 =1	(1) Set the SC4PSCE flag of SC4MD2 register to "1" to select "Enable count" for prescaler count control. (2) Select the clock source by SC4MD2 register. Set SC4PSC2 to 0 flags to "011" to select fpll-div/32. (3) Set the P7ODC1 and P7ODC2 flags of P7ODC register to "1,1" to select Nch open-drain for SDA4/SCL4. Set the P7PLUD2 and P7PLUD1 flags of P7PLUD register to "1,1" to add pull-up resistor. (4) Set the P7DIR1 and P7DIR2 flags of Port 7 direction control register (P7DIR) to "1,1" to set P71 and P72 to output mode.

Setup Procedure	Description
(5) Set SC4MD3 register Set ACK bit SC4MD3(0x03F53) bp0: SC4ACK0 =X bp1: SC4ACKS =1 Set the communication mode. SC4MD3(0x03F53) bp4: SC4TMD =0 Select the communication type. SC4MD3(0x03F53) bp2: SC4CMD =1 Select Transmission/Reception Select transmission/reception mode. SC4MD3(0x03F53) bp3: SC4REX =0 (6) Initialize the monitor flag SC4STR1(0x03F57) bp0: SC4DATA_ERR =0 (7) Set SC4MD0 register Select the transfer bit count SC4MD0(0x03F50) bp2 to 0: SC4LNG2 to 0 =111 Select the start condition SC4MD0(0x03F50) bp3: SC4STE =0 Select the first bit to be transferred SC4MD0(0x03F50) bp4: SC4DIR =0 (8) Set SC4MD1 register Select the transfer clock SC4MD1(0x03F51) bp2: SC4MST =1 Control the pin function SC4MD1(0x03F51) bp4: SC4SBOS =1 bp5: SC4SBIS =1 bp6: SC4SBTS =1 bp7: SC4IOM =1 (9) Set the interrupt level PSW bp6: MIE =0 SC4ICR(0x03FFC) bp7 to 6: SC4LV1 to 0 =10 SC4STPCICR(0x03FFD) bp7 to 6: SC4STPCLV1 to 0 =10	(5) Set the SC4ACKS flag of Serial Interface 4 mode register 3 (SC4MD3) to "1" to select "ACK bit enabled". ACK bit is received at transmission and the level setting of ACK bit with the SC4ACK0 flag is not needed. Set the SC4TMD flag of SC4MD3 register to "0" to set standard mode. Set the SC4CMD flag of SC4MD3 register to "1" to select IIC. Set the SC4REX flag of SC4MD3 register to "0" to select transmission mode. The setting to Serial Interface 4 mode register 3 (SC4MD3) should be executed with one-time MOV instruction in accordance with the setting of each bit. Do not set with BSET/BCLR. Also, do not read out the value of SC4MD3 and write to the specified bit. SC4ACK0 flag will be destroyed. (6) Set the SC4DATA_ERR flag of Serial Interface 4 status register 1 (SC4STR1) to "0" to initialize the communication error detection flag. (7) Set the SC4LNG2 to 0 flags of Serial Interface 4 mode register 0 (SC4MD0) to "111" to set the transfer bit count to 8 bits. Set the SC4STE flag of SC4MD0 register to "0" to disable the stand condition (start condition is not added after second communication). Set the SC4DIR flag of SC4MD0 register to "0" to set the first transfer bit to MSB. (8) Set the SC4MST flag of SC4MD1 register to "1" to select the clock master (internal clock). Set the SC4SBOS, SC4SBIS and SC4SBTS flags of SC4MD1 register to "1" to set SDA4 (SBO4) pin to serial data output, SBI4 pin to serial data input, and SCL4 (SBT4) pin to serial clock I/O. Set the SC4IOM flag to "1" to set SDA4 (SBO4) pin to serial data input. (9) Set the MIE flag of PSW to "1" to enable maskable interrupts. Set the interrupt level by the SC4LV1 to 0 flags of Serial 4 interrupt control register (SC4ICR) and the SC4STPCLV1 to 0 flags of Serial 4 stop condition detection interrupt control register (SC4STPCICR).

Setup Procedure	Description
(10) Enable the interrupt SC4ICR(0x03FFC) bp1: SC4IE =1 bp0: SC4IR =0 SC4STPCICR(0x03FFD) bp1: SC4STPCIE =1 bp0: SC4STPCIR =0 PSW bp6: MIE =1 (11) <Transmission starts> Start the serial transmission. Confirm that SCL4 is "High" Transmission data→TXBUF4(0x03F59) (12) <Transmission ends> <Setup for the next data transmission> Determine the monitor flag SC4STR1(0x03F57) bp0: SC4DATA_ERR (13) Determine the ACK bit level SC4MD3(0x03F53) bp0: SC4ACK0 (14) Set SC4MD0 register Select the transfer bit count SC4MD0(0x03F50) bp2 to 0: SC4LNG2 to 0 (15) <The next data transmission starts> Serial transmission starts [→(14)] (16) <Transmission ends> <End process of IIC communication> Set the SC4STPC flag SC4MD3(0x03F53) bp5: SC4STPC =1	(10) Set the SC4IE flag of SC4ICR register and the SC4STPCIE flag of SC4STPCICR register to "1" to enable interrupts. When the interrupt request flags (SC4IR flag of SC4ICR register and SC4STPCIR flag of SC4STPCICR register) have already been set, clear SC4IR and SC4STPCIR before enabling interrupts. [Chapter 7 7.1.4 Maskable Interrupt Control Register Setup] Set the MIE flag of PSW to "1" to enable maskable interrupts. (11) Set the transmission data in the transmission buffer TXBUF4. A transfer clock is generated and transmission starts. After data is transmitted, if ACK bit is received, Serial 4 communication completion interrupt SC4IRQ is generated. (12) Check the SC4DATA_ERR flag of Serial Interface 4 status register 1 (SC4STR1). When the previous transmission is normally completed, SC4DATA_ERR = "0". If SC4DATA_ERR = "1", reexecute the communication. (13) Check the ACK bit level received by the SC4ACK0 flag of Serial Interface 4 mode register 3 (SC4MD3). When SC4ACK0=0, transmission continues. When SC4ACK0=1, the slave side may not receive data. In this case, finish the communication. (14) When changing the transfer bit count, set the transfer bit count by the SC4LNG2 to 0 flags of Serial Interface 4 mode register 0 (SC4MD0). (15) Set the transmission data in TXBUF4 to start transmission.[→(14)] (16) Set the SC4STPC flag of Serial Interface 4 mode register 3 (SC4MD3) to "1". A stop condition is generated automatically and communication ends.

* (1) and (2) can be set at once.

* Each setup in (7) and (8) can be set at once.

* (9) to (10) can be set at once.



Set each flag in accordance with the order of the setup procedure. Activate the communication after all the control registers (Table:18.2.1, except TXBUF4) are set.

■ Slave Transmission Setup

The following describes the setup examples for slave transmission processing using Serial Interface 4 IIC function. Table:18.5.7 shows the conditions for transmission processing.

Table:18.5.7 Setup Conditions of Slave IIC Communication

Item	Set to
Serial data input selection	SDA4
Transfer bit count	8 bits
First transfer bit	MSB
ACK bit	Enabled
IIC communication mode	Standard mode
Clock source	fpll-div/32
SCL4/SDA4 pin type	Nch open-drain
SCL4 pin pull-up resistor	Added
SDA4 pin pull-up resistor	Added
Address mode	7 bits
Slave address	0110011
Master/slave setup	Master (multi-master)

An example setup procedure is shown below.

Setup Procedure	Description
(1) Select the prescaler operation SC4MD2(0x03F52) bp3: SC4PSCE =1 (2) Select the clock source SC4MD2(0x03F52) bp2 to 0: SC4PSC2 to 0 =011 (3) Control the pin type P7ODC(0x03EF7) bp1: P7ODC1 =1 bp2: P7ODC2 =1 P7PLUD(0x03EA7) bp1: P7PLUD1 =1 bp2: P7PLUD2 =1 (4) Control the pin direction P7DIR(0x03E97) bp1: P7DIR1 =1 bp2: P7DIR2 =1	(1) Set the SC4PSCE flag of SC4MD2 register to "1" to select "Enable count" for prescaler count control. (2) Select clock source by SC4MD2 register. Set bp2 to 0 to "011" to select fpll-div/32. (3) Set the P7ODC1, P7ODC2 flags of P7ODC register to "1,1" to select Nch open-drain for SDA4/SCL4. Set the P7PLUD1, P7PLUD2 flags of P7PLUD register to "1,1" to add pull-up resistor. (4) Set the P7DIR1, P7DIR2 flags of Port 7 direction control register (P7DIR) to "1,1" to set P71 and P72 to output mode.

Setup Procedure	Description
(5) Set the ACK bit SC4MD3(0x03F53) bp0: SC4ACK0 =X bp1: SC4ACKS =1 Set the communication mode SC4MD3(0x03F53) bp4: SC4TMD =0 Select the communication type SC4MD3(0x03F53) bp2: SC4CMD =1	(5) Set the SC4ACKS flag of Serial Interface 4 mode register 3 (SC4MD3) to "1" to select "ACK bit enabled". ACK bit is received at transmission and the level setting of ACK bit with the SC4ACKO flag is not needed. Set the SC4TMD flag of SC4MD3 register to "0" to set standard mode. The setting to Serial Interface 4 mode register 3 (SC4MD3) should be executed with one-time MOV instruction in accordance with the setting of each bit. Do not set with BSET/BCLR. Also, do not read out the value of SC4MD3 and write to the specified bit. SC4ACKO flag will be destroyed.
(6) Initialize the monitor flag SC4STR1(0x03F57) bp0: SC4DATA_ERR =0	(6) Set the SC4DATA_ERR flag of Serial Interface 4 status register 1 (SC4STR1) to "0" to initialize the communication error detection flag.
(7) Set SC4MD0 register Select transfer bit count SC4MD0(0x03F50) bp2 to 0: SC4LNG2 to 0 =111 Select the first transfer bit SC4MD0(0x03F50) bp4: SC4DIR =0	(7) Set the SC4LNG2 to 0 flags of Serial Interface 4 mode register 0 (SC4MD0) to "111" to set the transfer bit count to 8 bits. Set the SC4DIR flag of SC4MD0 register to "0" to set the first transfer bit to MSB.
(8) Set SC4MD1 register Select the transfer clock SC4MD1(0x03F51) bp2: SC4MST =1 Control the pin function SC4MD1(0x03F51) bp4: SC4SBOS =1 bp5: SC4SBIS =1 bp6: SC4SBTS =1 bp7: SC4IOM =1	(8) Set the SC4MST flag of SC4MD1 register to "1" to select the clock master (internal clock). Set the SC4SBOS, SC4SBIS, and SC4SBTS flags of SC4MD1 register to "1" to set SDA4 (SBO4) pin to serial data output, SBI4 pin to serial data input, and SCL4 (SBT4) pin to serial clock I/O. Set the SC4IOM flag to "1" to set SDA4 (SBO4) pin to serial data input.
(9) Set the slave address SC4AD0(0x03F54) bp7 to 1: SC4ADR7 to 1 = 0110011	(9) Set the slave address in the upper 7 bits (SC4ADR7 to 1) of SC4AD0 register.
(10) Set the interrupt level PSW bp6: MIE =0 SC4ICR(0x03FFC) bp7 to 6: SC4LV1 to 0 =10 SC4STPCICR(0x03FFD) bp7 to 6: SC4STPCLV1 to 0 =10	(10) Set the MIE flag of PSW to "1" to enable maskable interrupts. Set the interrupt level by the SC4LV1 to 0 flags of Serial 4 interrupt control register (SC4ICR) and the SC4STPCLV1 to 0 flags of Serial 4 stop condition detection interrupt control register (SC4STPCICR).

Setup Procedure	Description
(11) Enable the interrupt SC4ICR(0x03FFC) bp1: SC4IE =1 bp0: SC4IR =0 SC4STPCICR(0x03FFD) bp1: SC4STPCIE =1 bp0: SC4STPCIR =0 PSW bp6: MIE =1 (12) Start IIC communication. (13) Check the data transmission/reception SC4STR1(0x03F57) bp7: SC4WRS = 0 (14) Set the transmission data TXBUF4(0x03F59) bp7 to 0: TXBUF47 to 0 = 0x55	(11) Set the SC4IE flag of SC4ICR register and the SC4STPCIE flag of SC4STPCICR register to "1" to enable interrupts. When the interrupt request flags (SC4IR flag of SC4ICR register and SC4STPCIR flag of SC4STPCICR register) have been already set, clear SC4IR and SC4STPCIR before enabling interrupts. [Chapter 7 7.1.4 Maskable Interrupt Control Register Setup] Set the MIE flag of PSW to "1" to enable maskable interrupts. (12) The master on the IIC bus starts communication. (13) Serial 4 communication completion interrupt (SC4IRQ) is generated when the address transmitted from the master matches the slave address set in SC4AD0 register. In the interrupt handling routine, the communication is recognized as the slave transmission by verifying that the SC4WRS flag of SC4STR1 register is set to "0". (14) Set the data in TXBUF4 register.

* (1) and (2) can be set at once.

* Each setup in (7) and (8) can be set at once.

* (9) to (10) can be set at once.



Set each flag in accordance with the order of the setup procedure. Activate the communication after all the control registers (Table:18.2.1, except TXBUF4) are set.

Chapter 19 Hardware LIN

19.1 Overview

This LSI includes 1 type of Hardware LIN function to enable LIN communication in cooperation with 8-bit timer function (Timer 0 and 1), 16-bit timer function (Timer 7) and UART function of serial interface function (Serial Interface 0).

LIN is an abbreviation of Local Interconnect Network, and serial communication protocol to reduce costs of in-vehicle network. LIN protocol message frame consists of header frame and response frame.

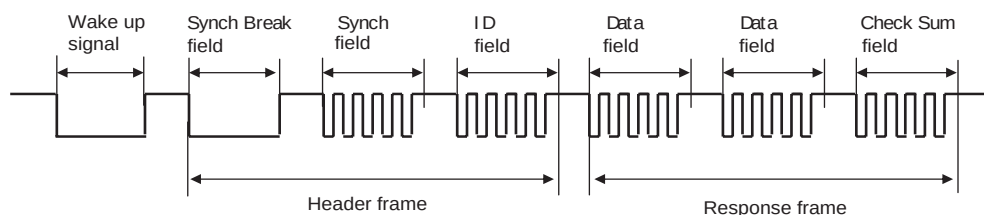


Figure:19.1.1 LIN Message Frame

Table:19.1.1 LIN Communication

	In master	In slave
Wake Up signal	0x80 transmission by Serial Interface 0 UART communication	TXD0 rising edge reception LINIRQ
Synch Break field	12 to 19-bit Low data transmission	Pulse width measurement with Timer 0 (Timer 0 and 1 cascade is possible)
Synch field	0x55 transmission with Serial Interface 0 UART communication	Pulse width measurement with Timer 0 (Timer 0 and 1 cascade is possible)
ID field	Generated by Serial Interface 0 UART function	Reception by Serial Interface 0 UART function
Data field	Generated by Serial Interface 0 UART function	
Check Sum field	Generated by Serial Interface 0 UART function	Reception data measurement with software
Bus collision detection	Detection is available with software every 1 Byte	-
Field interval measurement	Transfer clock number measurement is possible with Timer 7	



Hardware LIN function can be used only in NORMAL mode.
In SLOW and STANDBY mode, this function is not available.

19.1.1 Hardware LIN Function

Table:19.1.2 shows hardware LIN function

Table:19.1.2 Hardware LIN Function

	Hardware LIN	
Interrupt	SC0TIRQ (at master mode) LINIRQ (at slave mode)	
Used pin	TXD0A (P03)	TXD0B (P43)
Master mode	Synch Break is generated	
Slave mode	Synch Break reception (in cooperation with Timer 0 and 1)	
	Synch Field reception (in cooperation with Timer 0 and 1)	

19.1.2 Block Diagram

■ Hardware LIN Block Diagram

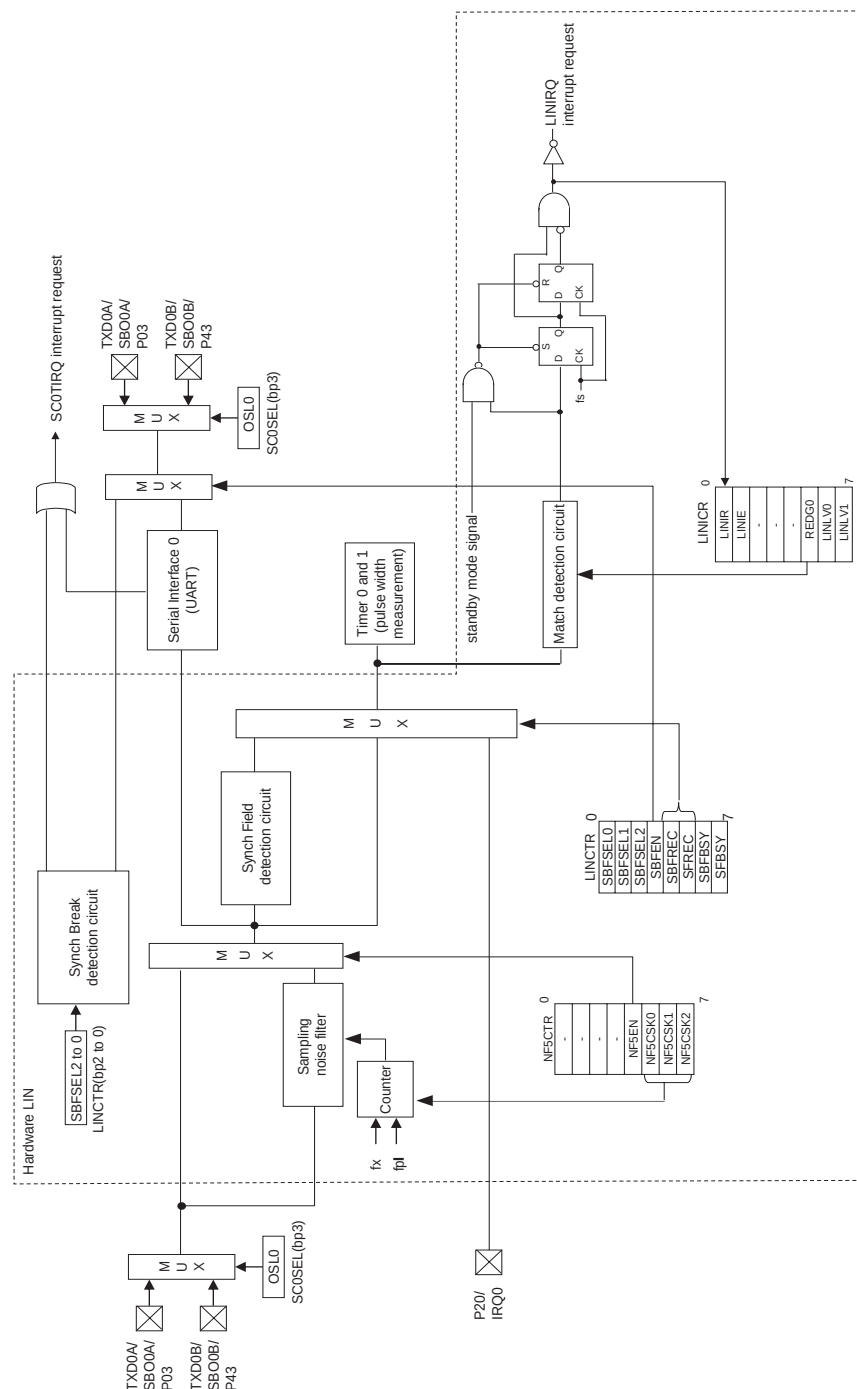


Figure:19.1.2 Hardware LIN Block Diagram

19.2 Registers

19.2.1 LIN Control Register

Hardware LIN function is controlled by LIN control register (LINCTR).

■ LIN Control Register (LINCTR: 0x03F18)

bp	7	6	5	4	3	2	1	0
Flag	SFBSY	SBFBSY	SFREC	SBFREC	SBFEN	SBFSEL2	SBFSEL1	SBFSEL0
At reset	0	0	0	0	0	0	0	0
Access	R	R	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7	SFBSY	Synch field (SF) reception busy flag 0: Except during SF reception 1: During SF reception
6	SBFBSY	Synch Break field (SBF) reception busy flag 0: Except during SBF reception 1: During SBF reception
5	SFREC	Synch field (SF) reception permission 0: Normal operation 1: SF reception
4	SBFREC	Synch Break field (SBF) reception permission 0: Normal operation 1: SBF reception
3	SBFEN	Synch Break field (SBF) transmission trigger/monitor 0: - 1: During SBF transmission
2 to 0	SBFSEL2 SBFSEL1 SBFSEL0	Synch Break field (SBF) transmission width selection 000: 12-bit 001: 13-bit 010: 14-bit 011: 15-bit 100: 16-bit 101: 17-bit 110: 18-bit 111: 19-bit



Do not set SFREC flag and SBFREC flag of LIN control register (LINCTR) to "1" at the same time.



SBFEN flag of LIN control register (LINCTR) is cleared by hardware automatically when SBF transmission is completed.

19.2.2 LIN Noise Filter

LIN noise filter 5 control register selects whether the noise rejection function is included or not, and also selects sampling cycle of the noise rejection function.

■ LIN Noise Filter 5 Control Register (NF5CTR: 0x03FD6)

This register samplings a signal input to TXD0 pin by the sampling time. If the same level is continued in 3 times, these levels are transferred to inside LSI. If the same level is not continued in 3 times, the previous level is transferred. It means, a signal that has more than "sampling time $\times$ 3 sampling clocks" width can be passed the noise filter. A signal that has under this width is rejected as a noise.

bp	7	6	5	4	3	2	1	0
Flag	NF5SCK2	NF5SCK1	NF5SCK0	NF5EN1	-	-	-	-
At reset	0	0	0	0	-	-	-	-
Access	R/W	R/W	R/W	R/W	-	-	-	-

bp	Flag	Description
7 to 5	NF5SCK2 NF5SCK1 NF5SCK0	TXD0 noise sampling frequency selection 000: fpll 001: fpll/2 ⁵ 010: fpll/2 ⁶ 011: fpll/2 ⁷ 100: fpll/2 ⁸ 101: fpll/2 ⁹ 110: fpll/2 ¹⁰ 111: fx
4	NF5EN1	0: Noise filter OFF 1: Noise filter ON
3 to 0	-	-

19.3 Operation

19.3.1 Operation in the Master Mode

Figure:19.3.1 shows basic operation of the header field transmission in LIN communication master mode.

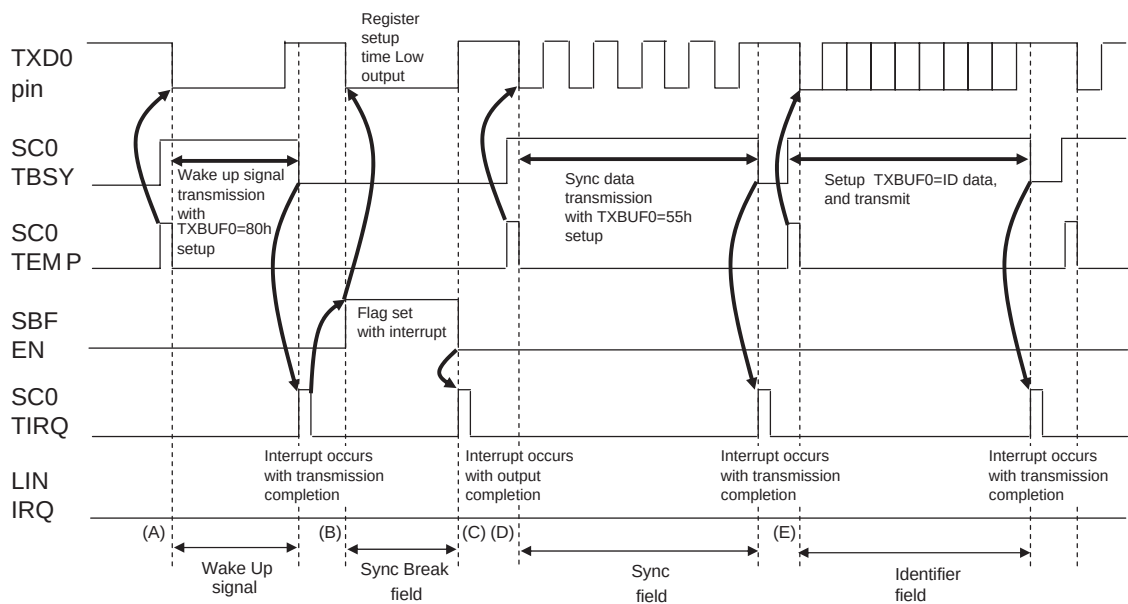


Figure:19.3.1 Basic Operation of Header Field Transmission

- (A) Transmission Wake Up signal
Set "0x80" to transmission data buffer (TXBUF0) of the Serial Interface 0.
- (B) Transmission Synch Break field
By setting SBFEN flag of LIN control register (LINCTR), "Low" level is output from TXD0 pin that is set by SBFSEL2 to 0 flags in advance.
- (C) Transmission completion Synch Break field
SBFEN flag is cleared by hardware automatically by generation of the Serial Interface 0 transmission completion interrupt (SC0IRQ).
- (D) Transmission Synch field
Set "0x55" (no parity, LSB first) to transmission data buffer (TXBUF0) of the Serial Interface 0, and transmit Synch data.
- (E) Transmission ID field
Set ID data to transmission data buffer (TXBUF0) of the Serial Interface 0, and transmit this data.

19.3.2 Operation in Slave Mode

Figure:19.3.2 shows basic operation of header field reception in LIN communication slave mode.

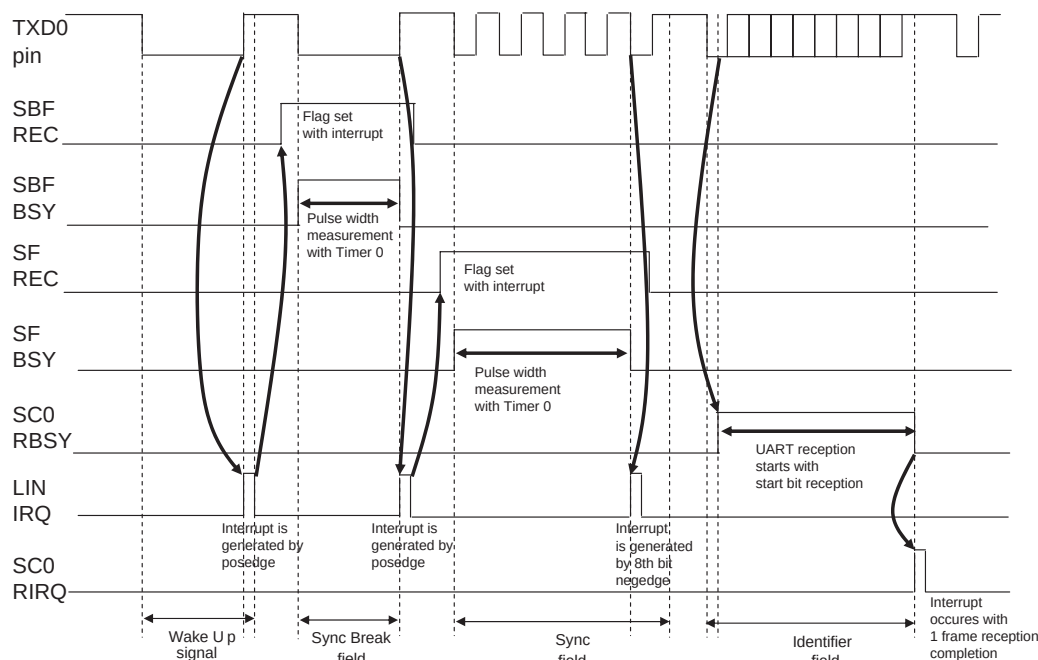


Figure:19.3.2 Basic Operation of Header Field Reception

(A) Enable Wake Up signal reception

The pulse width input signal of the 8-bit timer (Timer 0) is set to TXD0 pin input by setting SBFREC flag of LIN control register (LINCTR). "Low" level width input from TXD0 pin is measured by the pulse width measurement function of Timer 0.

(B) Reception completion Wake Up signal

The LIN interrupt (LINIRQ) is generated by the rising edge of TXD0 pin.

(C) Reception start Synch Break field

As in Wake Up signal reception, "Low" level width input from TXD0 pin is measured by the pulse width measurement function of Timer 0.

(D) Reception completion Synch Break Field

The LIN interrupt (LINIRQ) is generated by the rising edge of TXD0 pin. Clear SBFREC flag in the interrupt processing. And set SFREC flag of LIN control register (LINCTR) for next Synch Field reception.

(E) Reception start Synch field

By setting SFREC flag to "1", the inverted signal of SFBSY that is generated by sampling a signal input from TXD0 pin is input to the pulse width input signal of Timer 0. "Low" level width of this inverted signal is measured by the pulse width measurement function of Timer 0.

(F) Reception completion Synch Field

The LIN interrupt (LINIRQ) is generated by the fifth rising edge of TXD0 pin. Clear SFREC flag in the interrupt processing.

(G) Transmission ID Field

Receive ID field data by UART function of Serial Interface 0.

Chapter 20 A/D Converter

20.1 Overview

This LSI has an A/D converter with 10 bits resolutions. It contains a built-in sample hold circuit. Channel 0 to 23 (AN0 to AN23) of analog input can be switched by software. When A/D converter is stopped, the power consumption can be reduced by turning the built-in resistor ladder OFF. A/D converter is activated by setting a register and an external interrupt.

Table:20.1.1 A/D Converter Pin Functions

Table remarks -: Without function

Special Function	MN101EF76K/MN101EF56K	MN101EF57G
AN0 to AN7	PA0 to PA7	PA0 to PA7
AN8 to AN11	PB0 to PB3	PB0 to PB3
AN12	PB4	P94
AN13	PB5	P93
AN14	PB6	P92
AN15	PB7	P33
AN16 to AN19	P95 to P92	-
AN20 to AN23	P10 to P13	-

20.1.1 Functions

Table:20.1.2 shows the A/D converter functions.

Table:20.1.2 A/D Converter Functions

Special Function	MN101EF76K/MN101EF56K	MN101EF57G
A/D Input Pins	24 pins	16 pins
Pin Name	AN23 to AN0	AN15 to AN0
Interrupt	ADIRQ	
Resolution	10 bits	
Conversion Time (Minimum)	12.93 μ s (T_{AD} = as 800 ns)	
Input Range	V_{SS} to V_{REF+}	
Power Consumption	Built-in resistor ladder (ON/OFF)	



This function can not be used in STOP/HALT mode.



Do not execute mode switching as follows;

- Normal operation state (NORMAL) to Low-speed operation state (SLOW)
- Low-speed operation state (SLOW) to Idle state (IDLE) to Normal operation state (NORMAL)

If the mode switching is executed, we can not guarantee the result of A/D conversion.



To realize a low power consumption, we recommend that the built-in resistor ladder is turned OFF while A/D conversion is not executed.

20.1.2 Block Diagram

■ A/D Converter Block Diagram (MN101EF76K/MN101EF56K)

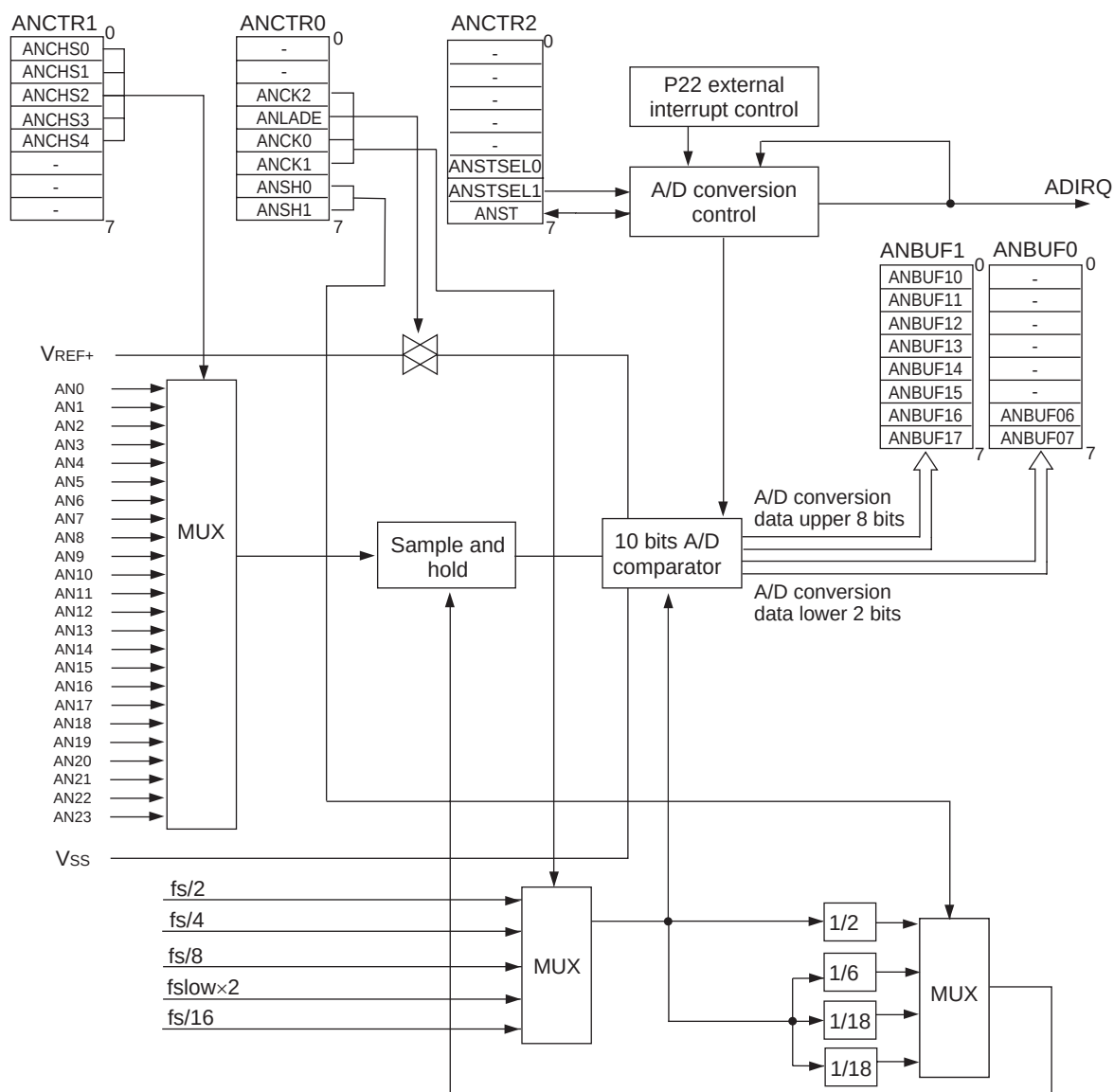


Figure:20.1.1 A/D Converter Block Diagram (MN101EF76K/MN101EF56K)

■ A/D Converter Block Diagram (MN101EF57G)

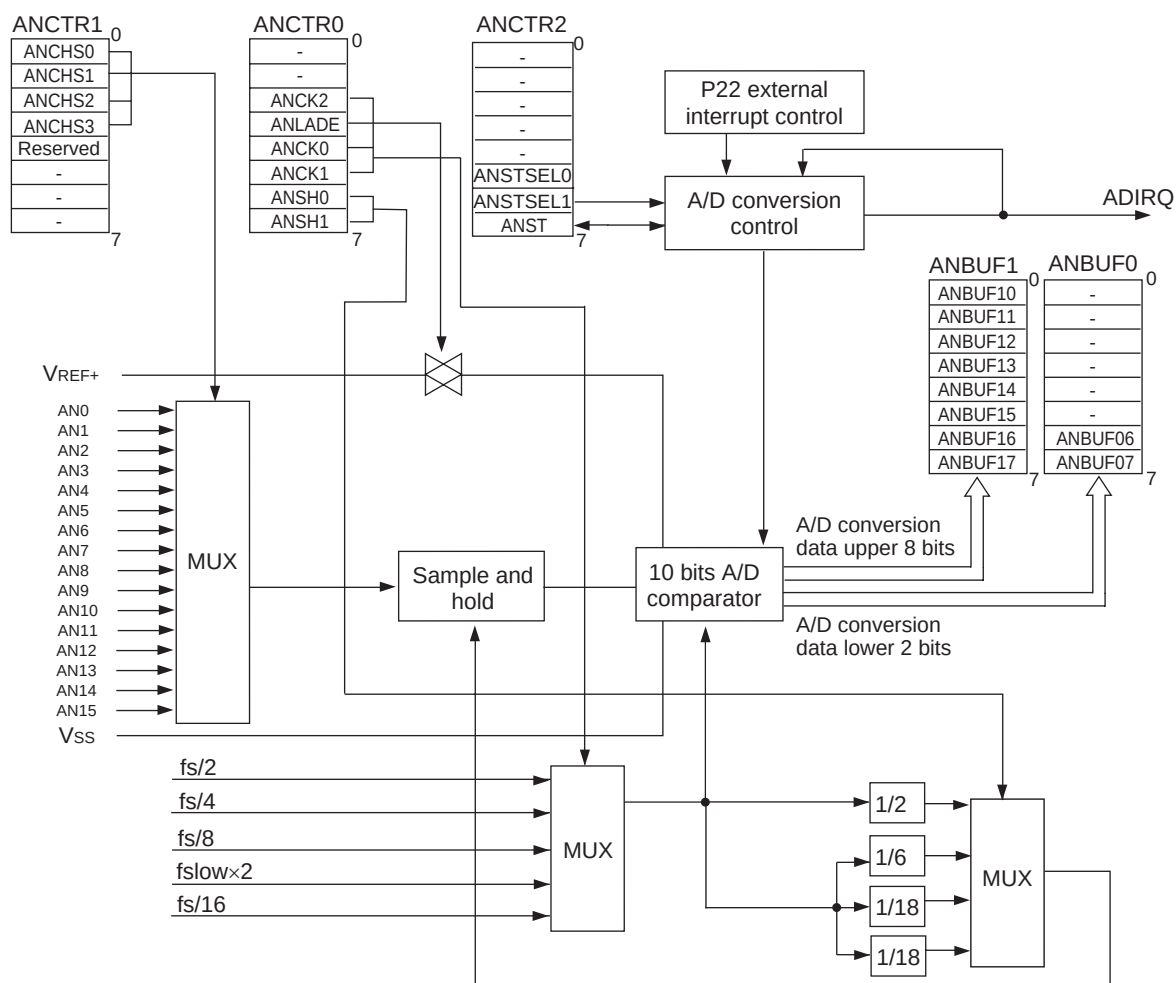


Figure:20.1.2 A/D Converter Block Diagram (MN101EF57G)

20.2 Control Registers

A/D converter consists of the control registers (ANCTR_n) and the data storage buffers (ANBUF_n).

20.2.1 Registers

Table:20.2.1 shows the registers used to control A/D converter.

Table:20.2.1 A/D Converter Control Registers

Table remarks ✓: With function -: Without function

Register	Address	R/W	Function	Page	MN101 EF76K	MN101 EF57G	MN101 EF56K
ANCTR0	0x03FC5	R/W	A/D converter control register 0	XX-7	✓	✓	✓
ANCTR1	0x03FC6	R/W	A/D converter control register 1	XX-9	✓	✓	✓
ANCTR2	0x03FC7	R/W	A/D converter control register 2	XX-10	✓	✓	✓
ANBUF0	0x03FC8	R	A/D converter data storage buffer 0	XX-11	✓	✓	✓
ANBUF1	0x03FC9	R	A/D converter data storage buffer 1	XX-11	✓	✓	✓
PERILR	0x03FFE	R/W	Peripheral functions group interrupt level setting register	VII-37	✓	✓	✓
IRQEXPEN	0x03F4E	R/W	Peripheral functions group interrupt input enable register	VII-38	✓	✓	✓
IRQEXPDT	0x03F4F	R/W	Peripheral functions group interrupt factor holding register	VII-39	✓	✓	✓
EDGDT	0x03F1E	R/W	Both edges interrupt control register	VII-56	✓	✓	✓
PAIMD	0x03ECA	R/W	Port A input mode register	VIII-194	✓	✓	✓
PAPLU	0x03EAA	R/W	Port A pull-up resistor control register	VIII-192	✓	✓	✓
PBIMD	0x03ECB	R/W	Port B input mode register	VIII-215	✓	✓	✓
PBPLU	0x03EAB	R/W	Port B pull-up resistor control register	VIII-214	✓	✓	✓
P9IMD	0x03EC9	R/W	Port 9 input mode register	VIII-179	✓	✓	✓
P9PLU	0x03EA9	R/W	Port 9 pull-up resistor control register	VIII-178	✓	✓	✓
P3IMD	0x03EC3	R/W	Port 3 input mode register	VIII-58	✓	✓	✓
P3PLU	0x03EA3	R/W	Port 3 pull-up resistor control register	VIII-58	✓	✓	✓
P1IMD	0x03EC1	R/W	Port 1 input mode register	VIII-35	✓	-	✓
P1PLUD	0x03EA1	R/W	Port 1 pull-up/pull-down resistor control register	VIII-36	✓	-	✓

R/W: Readable/Writable

R: Read only

20.2.2 Control Registers

■ A/D Converter Control Register 0 (ANCTR0: 0x03FC5)

bp	7	6	5	4	3	2	1	0
Flag	ANSH1	ANSH0	ANCK1	ANCK0	ANLADE	ANCK2	-	-
Reset	0	0	0	0	0	0	-	-
Access	R/W	R/W	R/W	R/W	R/W	R/W	-	-

bp	Flag	Description
7 to 6	ANSH1 ANSH0	Sample and hold time 00: $T_{AD} \times 2$ 01: $T_{AD} \times 6$ 10: $T_{AD} \times 18$ 11: $T_{AD} \times 18$
2, 5 to 4	ANCK2 ANCK1 ANCK0	A/D conversion clock ($f_{tad}=1/T_{AD}$) 000: $f_s/2$ 001: $f_s/4$ 010: $f_s/8$ 011: $f_{slow} \times 2$ 1XX: $f_s/16$ 111: Setting prohibited * as $800\text{ ns} \leq T_{AD} \leq 15.26\text{ }\mu\text{s}$
3	ANLADE	A/D resistor ladder control 0: A/D resistor ladder OFF 1: A/D resistor ladder ON
1 to 0	-	-

■ A/D Converter Control Register 1 (ANCTR1: 0x03FC6)

<MN101EF76K/MN101EF56K>

bp	7	6	5	4	3	2	1	0
Flag	-	-	-	ANCHS4	ANCHS3	ANCHS2	ANCHS1	ANCHS0
Reset	-	-	-	0	0	0	0	0
Access	-	-	-	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7 to 5	-	-
4 to 0	ANCHS4 ANCHS3 ANCHS2 ANCHS1 ANCHS0	Analog input channel 00000: AN0 00001: AN1 00010: AN2 00011: AN3 00100: AN4 00101: AN5 00110: AN6 00111: AN7 01000: AN8 01001: AN9 01010: AN10 01011: AN11 01100: AN12 01101: AN13 01110: AN14 01111: AN15 10000: AN16 10001: AN17 10010: AN18 10011: AN19 10100: AN20 10101: AN21 10110: AN22 10111: AN23 Others: Setting prohibited

■ A/D Converter Control Register 1 (ANCTR1: 0x03FC6)

<MN101EF57G>

bp	7	6	5	4	3	2	1	0
Flag	-	-	-	Reserved	ANCHS3	ANCHS2	ANCHS1	ANCHS0
Reset	-	-	-	0	0	0	0	0
Access	-	-	-	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7 to 5	-	-
4	Reserved	Always set to "0".
3 to 0	ANCHS3 ANCHS2 ANCHS1 ANCHS0	Analog input channel 0000: AN0 0001: AN1 0010: AN2 0011: AN3 0100: AN4 0101: AN5 0110: AN6 0111: AN7 1000: AN8 1001: AN9 1010: AN10 1011: AN11 1100: AN12 1101: AN13 1110: AN14 1111: AN15

■ A/D Converter Control Register 2 (ANCTR2: 0x03FC7)

bp	7	6	5	4	3	2	1	0
Flag	ANST	ANST SEL1	ANST SEL0	-	-	-	-	-
Reset	0	0	0	-	-	-	-	-
Access	R/W	R/W	R/W	-	-	-	-	-

bp	Flag	Description
7	ANST	A/D conversion status 0: Finish, Stop 1: Start, Converting
6 to 5	ANSTSEL1 ANSTSEL0	A/D conversion starting factor selection 00: Set ANST to "1" 01: Set ANST to "1" 10: Set P22 external interrupt, or ANST to "1" 11: Set A/D conversion interrupt, or ANST to "1"
4 to 0	-	-

20.2.3 Data Buffers

■ A/D Converter Data Storage Buffer 0 (ANBUF0: 0x03FC8)

The lower 2 bits results from A/D conversion are stored to this register.

bp	7	6	5	4	3	2	1	0
Flag	ANBUF07	ANBUF06	-	-	-	-	-	-
Reset	X	X	-	-	-	-	-	-
Access	R	R	-	-	-	-	-	-

■ A/D Converter Data Storage Buffer 1 (ANBUF1: 0x03FC9)

The upper 8 bits results from A/D conversion are stored to this register.

bp	7	6	5	4	3	2	1	0
Flag	ANBUF17	ANBUF16	ANBUF15	ANBUF14	ANBUF13	ANBUF12	ANBUF11	ANBUF10
Reset	X	X	X	X	X	X	X	X
Access	R	R	R	R	R	R	R	R



Do not execute the word access to ANBUF1 register.

20.3 Operation

Here is a description of A/D converter circuit setup procedure.

1. Set the analog pins.

Set the analog input pin, set in (2), to "special function pin" by Port A, B, 9, 3, 1 input mode register (PAIMD, PBIMD, P9IMD, P3IMD, P1IMD).

* Be sure to set port input mode register after applying analog voltage to pins.

2. Select the analog input pin.

Select the analog input pin from AN23 to AN0 by the ANCHS4 to 0 flag of A/D converter control register 1 (ANCTR1).

3. Select the A/D conversion clock.

Select the A/D conversion clock by the ANCK2 to ANCK0 flags of A/D converter control register 0 (ANCTR0). Set the converter clock (T_{AD}) so as not to exceed 800 ns depending on the resonator to be used.

4. Set the sample hold time.

Set the sample hold time by the ANSH1 and ANSH0 flags of A/D converter control register 0 (ANCTR0). The sample hold time should be based on analog input impedance.

* The procedures of (2) to (4) are in random order. The procedures of (3) and (4) can be operated simultaneously.

5. Set the A/D resistor ladder.

Set the ANLADE flag of A/D converter control register 0 (ANCTR0) to "1" to apply current to the resistor ladder so that A/D converter will be in standby condition.

6. Select the A/D converter activation factor, then start A/D conversion.

Set the ANST flag of A/D converter control register 2 (ANCTR2) to "1" to enable the A/D converter or set ANSTSEL1, ANSTSEL0 flags of A/D converter control register 2 (ANCTR2) to "10" to enable the A/D converter by external trigger factor.

7. A/D conversion.

A/D conversion is compared and determined sequentially by MSB after the sampling in the sample hold time to be set (4).

8. Complete the A/D conversion.

After the A/D conversion is completed, the result of the conversion is stored in the A/D conversion data store buffer (ANBUF0 and ANBUF1), the A/D conversion interrupt is generated and the ANST flag is cleared to "0".



Set the ANLADE flag to "1" then start A/D conversion after waiting for 12 conversion clock.



When A/D converter is started again after setting the ANST flag to "0" and A/D converter was stopped by force during A/D conversion, start A/D converter after waiting for an equivalent time of (2 system clock) + (2 converter clock) or longer.



When A/D conversion is programmed to start with an external interrupt by setting ANSTSEL1 to 0 flags of ANCTR2 register and A/D conversion starts:
if the conversion is stopped forcibly by setting ANST flag to "0", set the ANSTSEL1 to 0 flags to "00" before setting ANST flag.



If the flags of ANCTR0, ANCTR1 are changed during A/D conversion, the operation and the result of A/D conversion cannot be guaranteed.
Set the ANLADE flag of A/D converter control register 0 (ANCTR0) to "0" to disable the operation before changing the flags.

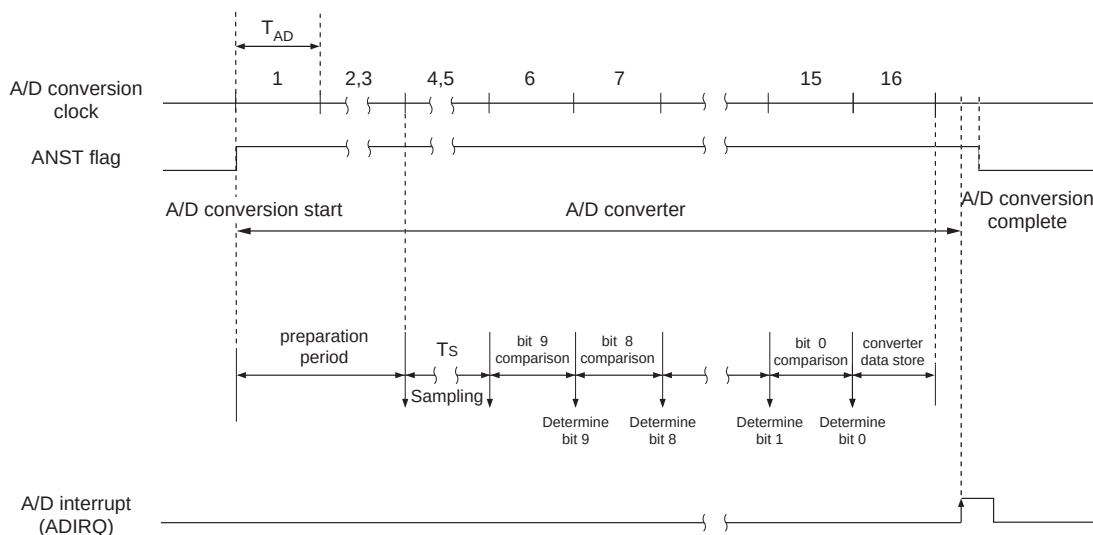


Figure:20.3.1 Operation of A/D conversion (sample hold time at $T_{AD} \times 2$)



Before reading out the value of the A/D conversion, A/D conversion should be done several times to prevent noise error by confirming the match of level by program, or by using the average value.

20.3.1 Setup

■ Input Pins of A/D Conversion Setup

Input pins for A/D converter is selected by the ANCH4 to 0 flags of ANCTR1 register.

■ A/D Conversion Clock Setup

The A/D conversion clock is set by the ANCK2 to ANCK0 flags of ANCTR0 register. Set the A/D conversion cycle (T_{AD}) between 800 ns and 15.26 μ s. Table:20.3.1 shows the machine clock (fpll, fslow, fs) and the A/D conversion cycle (T_{AD}). (calculated as $fs = fpll/2$, $fslow/4$)

Table:20.3.1 A/D Conversion Clock and A/D Conversion Cycle

ANCK2	ANCK1	ANCK0	A/D conversion clock	A/D conversion cycle (T_{AD})		
				High-speed operation		Low-speed operation
				fpll=10 MHz	fpll=4 MHz	fslow=32.768 kHz
0	0	0	fs/2	400 ns (setting prohibited)	477.33 ns (setting prohibited)	244.14 μ s (setting prohibited)
		1	fs/4	800 ns	954.65 ns	488.28 μ s (setting prohibited)
	1	0	fs/8	1.6 μ s	1.91 μ s	976.56 μ s (setting prohibited)
		1	fslow $\times$ 2	15.26 μ s	15.26 μ s	15.26 μ s
1	X	X	fs/16	3.2 μ s	3.82 μ s	1953.12 μ s (setting prohibited)

■ A/D Conversion Sample hold Time (T_S) Setup

The sample hold time of A/D conversion is set with the ANSH1 to 0 flags of ANCTR0 register. The sample hold time of A/D conversion depends on the external circuit, so set the right value based on the analog input impedance.

Table:20.3.2 Sample Hold Time of A/D Conversion and A/D Conversion Time

ANSH1	ANSH0	Sample hold time (T_S)	A/D conversion time [μ s]				
			High-speed operation				Low-speed operation
			$T_{AD}=1.6 \mu$ s (fs=5 MHz)	$T_{AD}=954.65$ ns (fs=4.19 MHz)	$T_{AD}=1.91 \mu$ s (fs=4.19 MHz)	$T_{AD}=15.26 \mu$ s (fs=4.19 MHz)	$T_{AD}=15.26 \mu$ s (fs=8.192 kHz)
0	0	$T_{AD} \times 2$	26.1	15.87	31.16	244.88	610.37
	1	$T_{AD} \times 6$	32.5	19.69	38.8	305.92	671.41
1	0	$T_{AD} \times 18$	51.7	31.15	61.72	488.92	854.53
	1	$T_{AD} \times 18$	51.7	31.15	61.72	488.92	854.53

* Calculated as $fs=fpll/2$, $fslow/4$.

Table:20.3.3 The Calculation method of A/D conversion time

ANCK2	ANCK1	ANCK0	A/D conversion clock	The method of Calculation of A/D conversion time
0	0	0	fs/2	$T_S + 14 \times T_{AD} + 2.5 / fs$
		1	fs/4	
	1	0	fs/8	
		1	fslow $\times$ 2	$T_S + 14 \times T_{AD} + 3 / fs$
1	X	X	fs/16	$T_S + 14 \times T_{AD} + 2.5 / fs$



A/D conversion time indicated in Table:20.3.3 may be shorten up to one cycle time of A/D conversion cycle depending on phase differences between system clock and A/D conversion clock.

■ Built-in Resistor Ladder Control

The ANLADE flag of ANCTR0 register is set to "1" to deliver current to the resistor ladder for A/D conversion. When A/D conversion is stopped, the ANLADE flag of ANCTR0 register is set to "0" to save the power consumption.

■ A/D Conversion Starting Factor Setup

A/D conversion starting factor is set by the ANSTSEL1 to 0 flags of ANCTR2 register. The ANSTSEL1 to 0 flags are set to select the starting factor of an external interrupt or A/D conversion interrupt. In addition, the ANST flag of ANCTR2 register can still possible to be used to start the A/D conversion by setting it to "1".



When External Interrupt 2 is selected as the A/D conversion starting factor, the valid edge should be assigned by REDG2 flag of External Interrupt 2 control register (IRQ2ICR) and EDGSEL1 flag of both edge interrupt control register (EDGDT).



The interrupt valid edge need to be assigned before selecting the interrupt factor for A/D conversion starting factor.

■ A/D Conversion Starting Setup

The A/D conversion start is set with the ANST flag of ANCTR2 register. The A/D conversion is started by setting the ANST flag of ANCTR2 register to "1". When the A/D conversion is started by External Interrupt 2 factor, the ANST flag of ANCTR2 register is automatically set to "1" after External Interrupt 2 is generated and the A/D conversion is started. The ANST flag of ANCTR2 register is cleared to "0" automatically after the conversion data is stored.

20.3.2 Setup Example

■ Example of A/D Conversion Setup by Registers

A/D conversion is started by setting registers. The analog input pins are set to AN0, the conversion clock is set to $f_s/4$, and the sampling hold time is set to $T_{AD} \times 2$. Then, A/D conversion complete interrupt is generated.

An example setup procedure, with a description of each step is shown below.

Setup Procedure	Description
(1) Set the analog input pin PAIMD(0x03ECA) bp0: PAIMD0 =1 PAPLU(0x03EAA) bp0: PAPLU0 =0	(1) Set the analog input pin to be set (2) to the special function pin by Port A input mode register (PAIMD) and set to the no pull-up resistor by Port A pull-up resistor control register (PAPLU).
(2) Select the analog input pin. ANCTR1(0x03FC6) bp4 to 0: ANCHS4 to 0 =000	(2) Select the analog input pin from AN0 to AN23 by setting the ANCHAS4 to 0 flags of A/D converter control register 1 (ANCTR1).
(3) Select the A/D conversion clock ANCTR0(0x03FC5) bp2, 5 to 4: ANCK2 to 0 =001	(3) Select the A/D conversion clock by the ANCK2, ANCK1, ANCK0 flags of A/D converter control register 0 (ANCTR0).
(4) Set the sample and hold time ANCTR0(0x03FC5) bp7 to 6: ANSH1 to 0 =00	(4) Set the sample and hold time by the ANSH1, ANSH0 flags of A/D converter control register 0 (ANCTR0).
(5) Clear the peripheral function group interrupt request flag IRQEXPDT(0x03F4F) bp5: IRQEXPDT5	(5) Clear the peripheral function group interrupt request flag by writing the value read from peripheral function group interrupt factor retention register (IRQEXPDT). Refer to [Chapter 7 7.1.5 Peripheral Function Group Interrupt Control Register Setup].
(6) Set the interrupt level PERIILR(0x03FEF) bp7 to 6: PERIILV1 to 0 =00	(6) Set the interrupt level by the PERIILV1 to 0 flags of peripheral function group interrupt level setting register (PERIILR).
(7) Set the peripheral function group interrupt input IRQEXPEN(0x03F4E) bp5: IRQEXPEN5	(7) Enable the peripheral function group interrupt by setting the IRQEXPEN6 to 0 flags of peripheral function group interrupt input enable register (IRQEXPEN).
(8) Set the A/D resistor ladder ANCTR0(0x03FC5) bp3: ANLADE =1	(8) Set the ANLADE flag of A/D converter control register 0 (ANCTR0) to "1" to deliver current to the resistor ladder for the A/D conversion.
(9) A/D conversion starting factor select ANCTR2(0x03FC7) bp6 to 5: ANSTSEL1 to 0 =00	(9) Clear the ANSTSEL1 to 0 flags of A/D converter control register 2 (ANCTR2) to "00" to set A/D converter starting factor to ANST flag of A/D converter control register 2 (ANCTR2).
(10) Start A/D converter operation ANCTR2(0x03FC7) bp7: ANST =1	(10) Set the ANST flag of A/D converter control register 2 (ANCTR2) to "1" to start the A/D conversion.

* The procedures (3) to (4) above can be set at the same time.



When the conversion is restarted by changing the setting after the A/D conversion is completed, set the ANLADE flag of A/D converter control register 0 (ANCTR0) to "0" to disable the A/D resistor ladder before changing the setting.

Note that operation is not guaranteed when the procedures above are not properly conducted.

■ Example of A/D Conversion Setup by External interrupt

A/D conversion is started by the external interrupt. The analog input pins are set to AN0, the conversion clock is set to $f_s/4$, and the sampling hold time is set to $T_{AD} \times 2$. Then, A/D conversion complete interrupt is generated. An example setup procedure, with a description of each step is shown below.

Setup Procedure	Description
(1) Set the analog input pin PAIMD(0x03ECA) bp0: PAIMD0 =1 PAPLU(0x03EAA) bp0: PAPLU0 =0	(1) Set the analog input pin to the special function pin by Port A input mode register (PAIMD) and set to the no pull-up resistor by Port A pull-up resistor control register (PAPLU).
(2) Select the analog input pin. ANCTR1(0x03FC6) bp4 to 0: ANCHS4 to 0 =0000	(2) Select the analog input pin from AN0 to AN23 by setting the ANCHAS4 to 0 flags of A/D converter control register 1 (ANCTR1).
(3) Select the A/D conversion clock ANCTR0(0x03FC5) bp2, 5 to 4: ANCK2 to 0 =001	(3) Select the A/D conversion clock by the ANCK2, ANCK1, ANCK0 flags of A/D converter control register 0 (ANCTR0).
(4) Set the sample and hold time ANCTR0(0x03FC5) bp7 to 6: ANSH1 to 0 =00	(4) Set the sample and hold time by the ANSH1, ANSH0 flags of A/D converter control register 0 (ANCTR0).
(5) Clear the peripheral function group interrupt request flag IRQEXPDT(0x03F4F) bp5: IRQEXPDT5	(5) Clear the peripheral function group interrupt request flag by writing the value read from peripheral function group interrupt factor retention register (IRQEXPDT). Refer to [Chapter 7 7.1.5 Peripheral Function Group Interrupt Control Register Setup]
(6) Set the interrupt level PERILLR(0x03FEF) bp7 to 6: PERILLV1 to 0 =00	(6) Set the interrupt level by the PERILLV1 to 0 flags of peripheral function group interrupt level setting register (PERILLR).
(7) Set the peripheral function group interrupt input IRQEXPEN(0x03F4E) bp5: IRQEXPEN5	(7) Enable the peripheral function group interrupt by setting the IRQEXPEN6 to 0 flags of peripheral function group interrupt input enable register (IRQEXPEN).
(8) Set the A/D resistor ladder ANCTR0(0x03FC5) bp3: ANLADE =1	(8) Set the ANLADE flag of A/D converter control register 0 (ANCTR0) to "1" to deliver current to the resistor ladder for the A/D conversion.
(9) A/D conversion starting factor select ANCTR2(0x03FC7) bp6 to 5: ANSTSEL1 to 0 =10	(9) Set the ANSTSEL1 to 0 flags of A/D converter control register 2 (ANCTR2) to "10" to set A/D converter starting factor to the external interrupt and ANST flag of A/D converter control register 2 (ANCTR2).
(10) Start A/D converter operation ANCTR2(0x03FC7) bp7: ANST =1	(10) When the external interrupt is generated, ANST flag of A/D converter control register 2 (ANCTR2) is set to "1" automatically, then start A/D conversion. If the external interrupt is not generated, the A/D conversion can start by setting the ANST flag of A/D converter control register 2 (ANCTR2) to "1".



When the conversion is restarted by changing the setting after the A/D conversion is completed, set the ANLADE flag of A/D converter control register 0 (ANCTR0) to "0" to disable the A/D resistor ladder before changing the setting.

Note that operation is not guaranteed when the procedures above are not properly conducted.



Input an external interrupt, which has a longer pulse than the system clock cycle.

If the pulse is shorter than the system clock period, the A/D conversion may not be started.

20.3.3 Cautions

As A/D conversion could be easily damaged by noise, sufficient measures need to be taken to prevent noise.

■ Anti-noise measures

Connect capacitors to pins (analog input pins), which is positioned close to VSS pin of microcontroller to avoid noise.

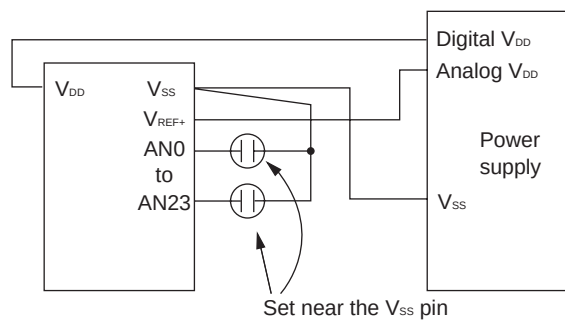


Figure:20.3.2 A/D Converter Recommended Example 1

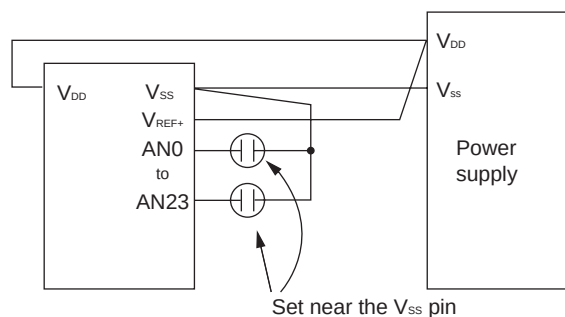


Figure:20.3.3 A/D Converter Recommended Example 2



Be sure to conduct the following procedures to ensure the high precision of A/D conversion.

1. The input impedance R of A/D input pin should be under $500\text{ k}\Omega$, and the external capacitor C (between 1000 pF and $1\text{ }\mu\text{F}$) should be connected to it.
2. The A/D conversion frequency should be set based on the RC time constant.
3. At the A/D conversion, if the output level of microcontroller is changed, or the additional peripheral circuit is switched to ON/OFF, the A/D converter may operate incorrectly, as the analog input pins and power pins cannot be fixed. At the set checking, confirm the waveform of analog input pins.

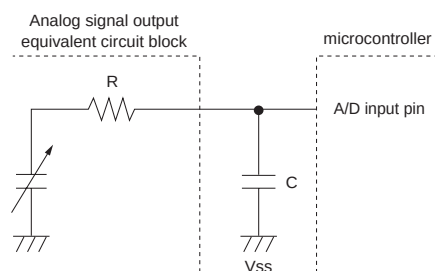


Figure:20.3.4 Recommended Circuit



This LSI contains a sample hold capacitor (C) (approximately. 10 pF).

Set the sample hold time based on the capacitor (C) and the time constant of impedance R of the external analog signal output circuit.

It is recommended to select the sample hold time to be 3 times of the RC time constant or more.

At A/D conversion cycle (T_{AD}) = 800 ns , $1.6\text{ }\mu\text{s}$, $4.8\text{ }\mu\text{s}$, and $14.4\text{ }\mu\text{s}$ can be selected as a sample hold time. Each impedance value of the external analog signal output circuit should be lesser than the following values.

Table:20.3.4 Impedance of analog signal output circuit

Sample hold time ($T_{AD} = 800\text{ ns}$)	Impedance of analog signal output circuit
$T_{AD} \times 2 = 1.6\text{ }\mu\text{s}$	$R < 1.6\text{ }\mu\text{s} / 10\text{ pF} / 3 = 53\text{ k}\Omega$
$T_{AD} \times 6 = 4.8\text{ }\mu\text{s}$	$R < 4.8\text{ }\mu\text{s} / 10\text{ pF} / 3 = 160\text{ k}\Omega$
$T_{AD} \times 18 = 14.4\text{ }\mu\text{s}$	$R < 14.4\text{ }\mu\text{s} / 10\text{ pF} / 3 = 480\text{ k}\Omega$

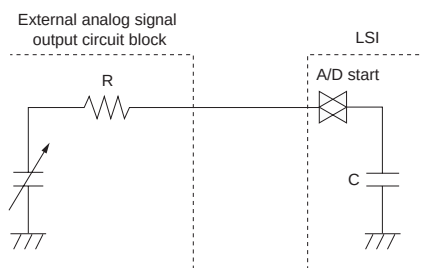


Figure:20.3.5 Circuit Example

21.1 Overview

This LSI has a D/A converter with 8 bits resolution. It contains 4 output channels and 8-bit data registers for each channel. In D/A conversion mode, there are 2 modes: channel fixed conversion mode, 2-channel conversion mode and 4-channel conversion mode. At 2-channel conversion mode and 4-channel conversion mode, the time for switching channels can be programmed by the software. When the D/A converter is not used, the internal resistor ladder can be set to OFF to save the power consumption.

Table:21.1.1 D/A Converter Pin Functions

Table remarks -: Without function

Special Function	MN101EF76K/MN101EF56K	MN101EF57G
DA_A	P95	P07
DA_B	P94	P94
DA_C	P61	-
DA_D	P07	-

21.1.1 Functions

Table:21.1.2 shows the D/A converter functions.

Table:21.1.2 D/A Converter Functions

Special Function	MN101EF76K/MN101EF56K	MN101EF57G
Resolution	8 bits	
Pin Name	DA_A DA_B DA_C DA_D	DA_A DA_B
Power Consumption	Internal resistor ladder ON/OFF	



This function can not be used in STOP/HALT mode.

21.1.2 Block Diagram

■ D/A Converter Block Diagram (MN101EF76K/MN101EF56K)

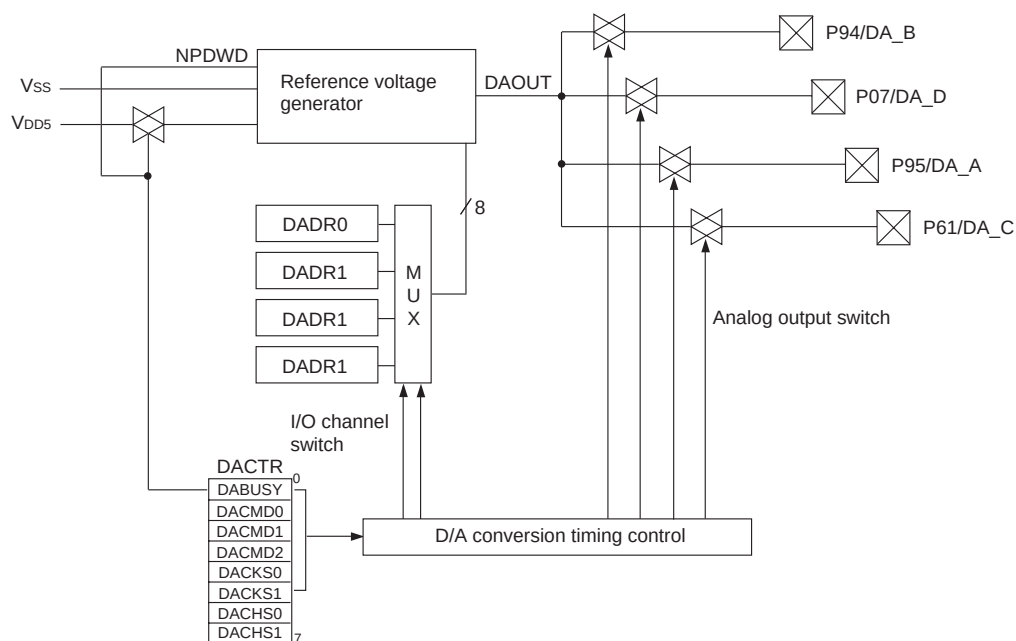


Figure:21.1.1 D/A Converter Block Diagram (MN101EF76K/MN101EF56K)

■ D/A Converter Block Diagram (MN101EF57G)

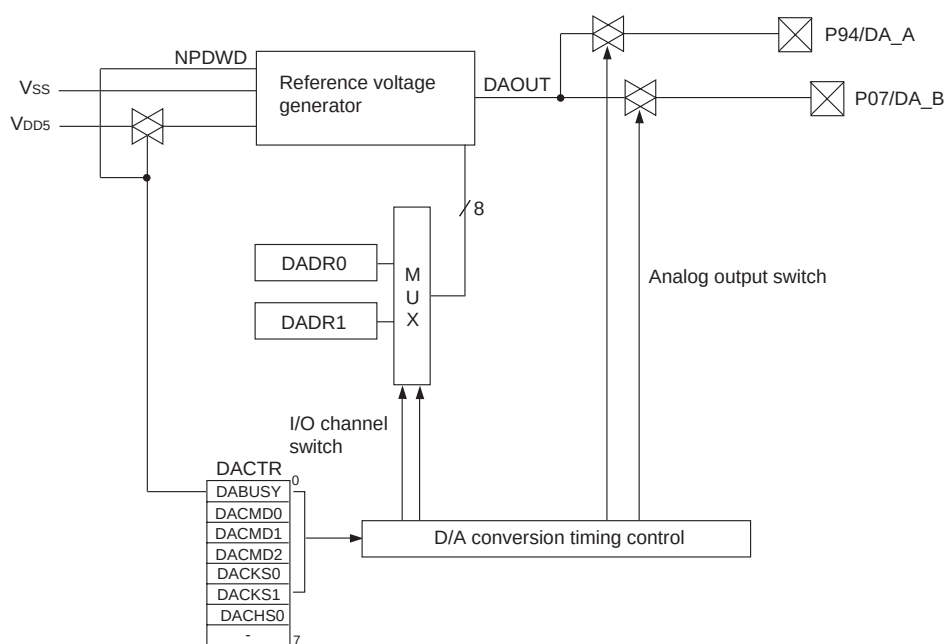


Figure:21.1.2 D/A Converter Block Diagram (MN101EF57G)

21.2 D/A Converter Control Registers

21.2.1 D/A Converter Control Registers

Table:21.2.1 shows the registers used to control D/A converter.

Table:21.2.1 D/A Converter Control Registers

Register	Address	R/W	Function	Page
DACTR	0x03FCB	R/W	D/A converter control register	XXI-5 to XXI-6
DADR0	0x03FCC	R/W	D/A converter input data register 0	XXI-7
DADR1	0x03FCD	R/W	D/A converter input data register 1	XXI-7
DADR2	0x03FCE	R/W	D/A converter input data register 2	XXI-7
DADR3	0x03FCF	R/W	D/A converter input data register 3	XXI-7
P0DIR	0x03E90	R/W	Port 0 direction control register	VIII-9
P0PLUD	0x03EA0	R/W	Port 0 pull-up/pull-down resistor control register	VIII-9
SELUD	0x03EAF	R/W	Pull-up/down resistor selection register	VIII-13
P9DIR	0x03E99	R/W	Port 9 direction control register	VIII-177
P9PLU	0x03EA9	R/W	Port 9 pull-up resistor control register	VIII-178
P6DIR	0x03E96	R/W	Port 6 direction control register	VIII-113
P6PLU	0x03EA6	R/W	Port 6 pull-up resistor control register	VIII-114

R/W: Readable/Writable

21.2.2 D/A Converter Control Register (DACTR)

This is the 8-bit readable/writable register that controls the D/A converter.

■ D/A Converter Control Register (DACTR: 0x03FCB)

<MN101EF76K/MN101EF56K>

bp	7	6	5	4	3	2	1	0
Flag	DACHS1	DACHS0	DACKS1	DACKS0	DACMD2	DACMD1	DACMD0	DABUSY
At reset	X	X	X	X	X	X	X	0
Access	R	R	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7 to 6	DACHS1 DACHS0	Monitor flag of DA conversion 00: During channel 0 conversion 01: During channel 1 conversion 10: During channel 2 conversion 11: During channel 3 conversion
5 to 4	DACKS1 DACKS0	DA scan clock 00: fs/16 01: fs/32 10: fs/64 11: fs/128
3 to 1	DACMD2 DACMD1 DACMD0	DA conversion mode selection 000: Channel 0 fixed conversion 001: Channel 1 fixed conversion 010: Channel 2 fixed conversion 011: Channel 3 fixed conversion 1X0: 2-channel scan conversion (Channel 0 and 1) 1X1: 4-channel scan conversion (Channel 0, 1, 2 and 3)
0	DABUSY	D/A conversion enable flag 0: Disable (resistor ladder OFF) 1: Enable

■ D/A Converter Control Register (DACTR: 0x03FCB)

<MN101EF57G>

bp	7	6	5	4	3	2	1	0
Flag	-	DACHS0	DACKS1	DACKS0	DACMD2	DACMD1	DACMD0	DABUSY
At reset	-	X	X	X	X	X	X	0
Access	-	R	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7	-	-
6	DACHS0	Monitor flag of DA conversion 0: During channel 0 conversion 1: During channel 1 conversion
5 to 4	DACKS1 DACKS0	DA scan clock 00: fs/16 01: fs/32 10: fs/64 11: fs/128
3 to 1	DACMD2 DACMD1 DACMD0	DA conversion mode selection 000: Channel 0 fixed conversion 001: Channel 1 fixed conversion 010: Setting prohibited 011: Setting prohibited 1X0: 2-channel scan conversion (Channel 0 and 1) 1X1: Setting prohibited
0	DABUSY	D/A conversion enable flag 0: Disable (resistor ladder OFF) 1: Enable

21.2.3 D/A Converter Input Data Register

This readable/writable register stores the D/A conversion data.

■ D/A Converter Input Data Register 0 (DADR0: 0x03FCC)

This register stores the D/A converter data (for channel DA0).

bp	7	6	5	4	3	2	1	0
Flag	DA0BUF7	DA0BUF6	DA0BUF5	DA0BUF4	DA0BUF3	DA0BUF2	DA0BUF1	DA0BUF0
At reset	X	X	X	X	X	X	X	X
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

■ D/A Converter Input Data Register 1 (DADR1: 0x03FCD)

This register stores the D/A converter data (for channel DA1).

bp	7	6	5	4	3	2	1	0
Flag	DA1BUF7	DA1BUF6	DA1BUF5	DA1BUF4	DA1BUF3	DA1BUF2	DA1BUF1	DA1BUF0
At reset	X	X	X	X	X	X	X	X
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

■ D/A Converter Input Data Register 2 (DADR2: 0x03FCE)

This register stores the D/A converter data (for channel DA2).

<MN101EF76K/MN101EF56K>

bp	7	6	5	4	3	2	1	0
Flag	DA2BUF7	DA2BUF6	DA2BUF5	DA2BUF4	DA2BUF3	DA2BUF2	DA2BUF1	DA2BUF0
At reset	X	X	X	X	X	X	X	X
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

■ D/A Converter Input Data Register 3 (DADR3: 0x03FCF)

This register stores the D/A converter data (for channel DA3).

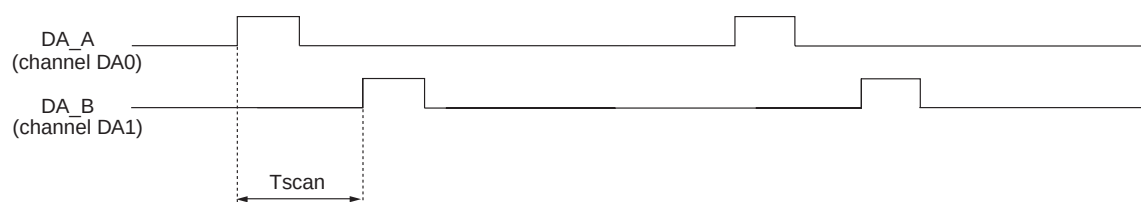
<MN101EF76K/MN101EF56K>

bp	7	6	5	4	3	2	1	0
Flag	DA3BUF7	DA3BUF6	DA3BUF5	DA3BUF4	DA3BUF3	DA3BUF2	DA3BUF1	DA3BUF0
At reset	X	X	X	X	X	X	X	X
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

21.3 Operation

Procedures of D/A converter operation is as follows;

1. Select the D/A conversion mode by the DACMD2 to DACMD0 flags of D/A converter control register (DACTR).
2. Set the switching time of analog output channel by the DACKS1 to DACKS0 flag of DACTR register. When the fixed conversion mode is selected in (1), this setup is not available.
3. Set the DABUSY flag of DACTR register to "1" to send a resistor ladder current to start the D/A conversion.
4. The D/A conversion is done to the data that is set to DADR3 to DADR0 register, and its result output to DA_A to DA_D in accordance with the setup in (1), (2). If the scan conversion is selected, the pins are high impedance while D/A output is not done.
5. When the D/A conversion is stopped, set the DABUSY flag to "0".



* Tscan is the time set by DACKS1 and DACKS0

* For 1/2 period of Tscan, pin is high-impedance not to output inadequately

Figure:21.3.1 4 Channel Scan Conversion Mode Timing



To select D/A scan clock, set more than the settling time.

21.3.1 Setup Example

■ Setup Example by DABUSY Flag

An example of setup procedures for using DA0 as a conversion channel, with a description of each step is shown below.

Setup Procedure	Description
(1) Set Port 9 P9DIR(0x03E99) bp4: P9DIR4 =0 P9PLU9(0x03EA9) bp4: P9PLU4 =0	(1) Set the analog output pin to "input mode" by Port 9 direction control register (P9DIR). And set to the "no pull-up resistor" by Port 9 pull-up/down register (P9PLU).
(2) Set the D/A conversion data DADR0(0x03FCC)	(2) Set the D/A conversion data by D/A converter input data register 0 (DADR0).
(3) Start the D/A conversion DACTR(0x03FCB) bp0: DABUSY =1	(3) Set the DABUSY flag of D/A converter control register (DACTR) to "1" to start the D/A conversion. The result obtained from the conversion is output to DA_A pin.
(4) Stop the D/A conversion DACTR(0x03FCB) bp0: DABUSY =0	(4) Reset the DABUSY flag of D/A converter control register (DACTR) to "0" to stop the D/A conversion.

22.1 Overview

MN101EF76K and MN101EF56K have an LCD driver circuit using 55 segment pins and 4 common pins.
MN101EF57G has an LCD driver circuit using 41 segment pins and 4 common pins.
The LCD driver circuit consists of segment output latch, LCD control registers, prescaler, timing control circuit, multiplexer, segment drivers, common drivers and voltage dividing resistors.

22.1.1 Functions

Table:22.1.1 lists the LCD driver circuit functions.

Table:22.1.1 LCD Functions

Function	MN101EF76K/MN101EF56K	MN101EF57G
Duty	Static 1/2 duty 1/3 duty 1/4 duty	
Segment Output Pins	SEG0 to SEG54	SEG0 to SEG40
Common Output Pins	COM0 to COM3	
LCD Power Supply	V _{LC1} to V _{LC3}	
LCD Voltage Dividing Resistor	V _{LC1} input voltage is divided into 2/3, 1/3. High-resistance and low-resistance are selectable.	
Clock Source (LCDCLK)	fpll/2 ¹¹ fpll/2 ¹² fpll/2 ¹³ fpll/2 ¹⁴ fpll/2 ¹⁵ fpll/2 ¹⁶ fpll/2 ¹⁷ fpll/2 ¹⁸ fslow/2 ⁶ fslow/2 ⁷ fslow/2 ⁸ fslow/2 ⁹ Timer 0 to Timer 4	
fpll: Machine clock (Oscillation for high-speed operation) fslow: Peripheral function clock (Oscillation for low-speed operation) LCDCLK: LCD clock source (selected by LCDCK0 to LCDCK3)		



Use LCD panel with driving voltage (V_{LCD}): $V_{LCD} \leq V_{DD5} \leq 5.5$ V.

22.1.2 LCD Operation in STANDBY Mode

Some LCD operations may be restricted in STANDBY mode.

Table:22.1.2 shows the LCD operation capabilities in STANDBY mode.

Table:22.1.2 LCD Operation in STANDBY Mode

CPU Mode		LCD Clock	
		fpll-based	fslow-based
CPU OPERATION Mode	NORMAL	√	√
	SLOW	-	√
STANDBY Mode	HALT0	Δ	Δ
	HALT1	-	Δ
	STOP	-	-
√: LCD operation is available. Δ: Holding display is available. -: LCD operation is not available.			



For change to CPU mode in which LCD operation is not available, change the CPU mode after turning the LCD off and switch from segment output to ports.

22.1.3 Maximum Pixels

Table:22.1.3 shows the maximum pixels.

Table:22.1.3 Maximum Pixels

Duty	Maximum Pixels (Segment x Common)	8-segment type LCD panel	Common Pins	Bits used by segment output latch
Static	41 (41 x 1)	5 digits	COM0	Bit 0, Bit 4
1/2	82 (32 x 2)	10 digits	COM0 to COM1	Bit 0 to Bit 1, Bit 4 to Bit 5
1/3	123 (32 x 3)	15 digits	COM0 to COM2	Bit 0 to Bit 2, Bit 4 to Bit 6
1/4	164 (32 x 4)	20 digits	COM0 to COM3	Bit 0 to Bit 3, Bit 4 to Bit 7

22.1.4 Switching I/O Ports and LCD Segment/Common Pins

LCD output control register 1 to 7 (LCCTR1 to LCCTR7) switch the general ports to LCD segment pins. Refer to [Chapter 22 22.2 Control Registers].

LCD output control register 0 (LCCTR0) switches the general ports to LCD common pins or VLC pins. Refer to [Chapter 22 22.2 Control Registers].

Segment pins, common pins and VLC pins can be switched by a unit of 1 bit.

22.1.5 Block Diagram

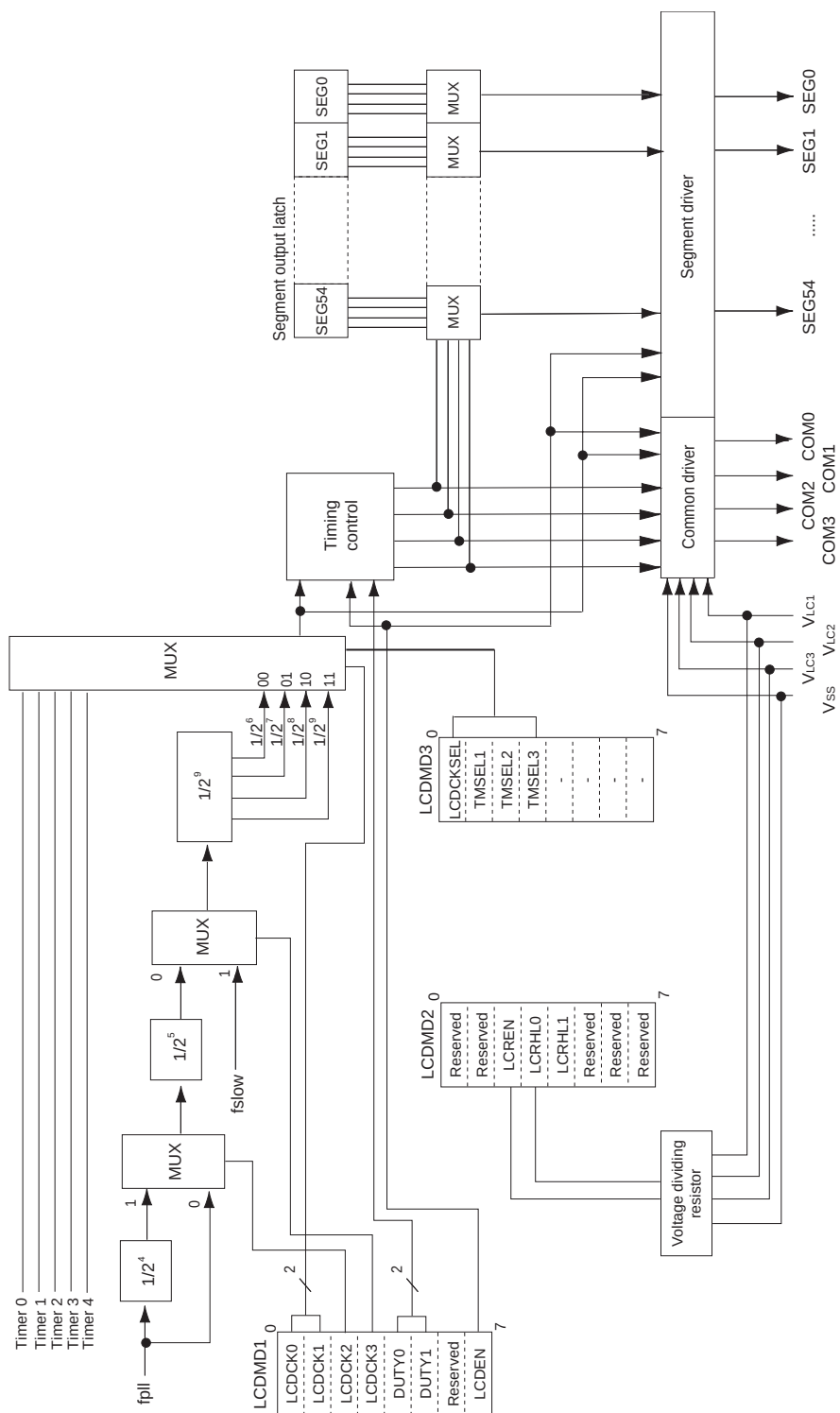


Figure:22.1.1 LCD Driver Circuit Block Diagram

22.2 Control Registers

The LCD is controlled by LCD mode control registers 1 to 3 (LCDMD1 to LCDMD3) and LCD output control registers 0 to 7 (LCCTR0 to 7).

The display data are stored in the segment output latch.

22.2.1 Registers

Table:22.2.1 shows the registers to control LCD.

Table:22.2.1 List of LCD Control Registers

Table remarks ✓: With function -: Without function

Register	Address	R/W	Function	Page	MN101 EF76K	MN101 EF57G	MN101 EF56K
LCDMD1	0x03E40	R/W	LCD mode control register 1	XXII-7	✓	✓	✓
LCDMD2	0x03E41	R/W	LCD mode control register 2	XXII-8	✓	✓	✓
LCDMD3	0x03E42	R/W	LCD mode control register 3	XXII-9	✓	✓	✓
LCCTR0	0x03E43	R/W	LCD output control register 0	XXII-10	✓	✓	✓
LCCTR1	0x03E44	R/W	LCD output control register 1	XXII-11	✓	✓	✓
LCCTR2	0x03E45	R/W	LCD output control register 2	XXII-12	✓	✓	✓
LCCTR3	0x03E46	R/W	LCD output control register 3	XXII-13	✓	✓	✓
LCCTR4	0x03E47	R/W	LCD output control register 4	XXII-14	✓	✓	✓
LCCTR5	0x03E48	R/W	LCD output control register 5	XXII-15	✓	✓	✓
LCCTR6	0x03E49	R/W	LCD output control register 6	XXII-16	✓	✓	✓
LCCTR7	0x03E4A	R/W	LCD output control register 7	XXII-17	✓	-	✓

R/W: Readable/Writable

* Addresses 0x03E20 to 0x03E3B are assigned to the segment output latch.

[Chapter 22 22.2.13 Segment Output Latch]

22.2.2 LCD Mode Control Register 1 (LCDMD1)

LCD mode control register 1 (LCDMD1) controls the LCD clock, the LCD driver circuit ON/OFF and specifies the display duty.

■ LCD Mode Control Register 1 (LCDMD1: 0x03E40)

bp	7	6	5	4	3	2	1	0
Flag	LCDEN	Reserved	DUTY1	DUTY0	LCDCK3	LCDCK2	LCDCK1	LCDCK0
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7	LCDEN	LCD driver circuit start flag 0: Stop 1: Start
6	Reserved	Always set to "0".
5 to 4	DUTY1 DUTY0	LCD display duty selection 00: 1/4 duty 01: 1/3 duty 10: 2/3 duty 11: Static
3 to 0	LCDCK3 LCDCK2 LCDCK1 LCDCK0	LCD clock source selection 0000: $f_{pll}/2^{11}$ 0001: $f_{pll}/2^{12}$ 0010: $f_{pll}/2^{13}$ 0011: $f_{pll}/2^{14}$ 0100: $f_{pll}/2^{15}$ 0101: $f_{pll}/2^{16}$ 0110: $f_{pll}/2^{17}$ 0111: $f_{pll}/2^{18}$ 1X00: $f_{slow}/2^6$ 1X01: $f_{slow}/2^7$ 1X10: $f_{slow}/2^8$ 1X11: $f_{slow}/2^9$

22.2.3 LCD Mode Control Register 2 (LCDMD2)

LCD mode control register 2 (LCDMD2) controls the internal voltage dividing resistor ON/OFF and selecting the type of the internal voltage dividing resistor.

■ LCD Mode Control Register 2 (LCDMD2: 0x03E41)

bp	7	6	5	4	3	2	1	0
Flag	Reserved	Reserved	Reserved	LCRHL1	LCRHL0	LCREN	Reserved	Reserved
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7 to 5	Reserved	Always set to "0".
4 to 3	LCRHL1 LCRHL0	Selection of internal voltage dividing resistor types 00: 10 k Ω in between each V_{LC} 01: 220 k Ω in between each V_{LC} 10: 20 k Ω in between each V_{LC} 11: 100 k Ω in between each V_{LC}
2	LCREN	Selection of internal voltage dividing resistor connection 0: Unconnected 1: Connected
1 to 0	Reserved	Always set to "0".

22.2.4 LCD Mode Control Register 3 (LCDMD3)

LCD mode control register 3 (LCDMD3) controls selecting of the original oscillation dividing or the timers.

■ LCD Mode Control Register 3 (LCDMD3: 0x03E42)

bp	7	6	5	4	3	2	1	0
Flag	-	-	-	-	TMSEL3	TMSEL2	TMSEL1	LCDCKSEL
At reset	-	-	-	-	0	0	0	0
Access	-	-	-	-	R/W	R/W	R/W	R/W

bp	Flag	Description
7 to 4	-	-
3 to 1	TMSEL3 TMSEL2 TMSEL1	Timers 0 to 4 input selection 000: Timer 0 001: Timer 1 010: Timer 2 011: Timer 3 100: Timer 4 101: Setting prohibited 11X: Setting prohibited
0	LCDCKSEL	Selection between the original oscillation dividing and the selected timer 0: Original oscillation dividing 1: Selected timer



To use Timer n (n= 0 to 4) as the LCD clock source, set the LCDEN flag of LCDMD1 register to “1” to start the LCD driver circuit, before starting the count of Timer n by the TMnEN flag of TMnMD register.

To stop the LCD driver circuit, set the LCDEN flag to “0” after stopping the count of Timer n by the TMnEN flag.

22.2.5 LCD Output Control Register 0 (LCCTR0)

The register controls switching between the I/O port (P81 to P84) and the common outputs (COM0 to COM3), and between the I/O port (P85 to P87) and the VLC pins (VLC3 to VLC1). On reset, these ports are set to input mode.

■ LCD Output Control Register 0 (LCCTR0: 0x03E43)

bp	7	6	5	4	3	2	1	0
Flag	COMSL3	COMSL2	COMSL1	COMSL0	Reserved	VLC3SL	VLC2SL	VLC1SL
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7	COMSL3	COM3/P84 selection 0: P84 1: COM3
6	COMSL2	COM2/P83 selection 0: P83 1: COM2
5	COMSL1	COM1/P82 selection 0: P82 1: COM1
4	COMSL0	COM0/P81 selection 0: P81 1: COM0
3	Reserved	Always set to "0".
2	VLC3SL	VLC3/P85 selection 0: P85 1: VLC3
1	VLC2SL	VLC2/P86 selection 0: P86 1: VLC2
0	VLC1SL	VLC1/P87 selection 0: P87 1: VLC1



P87 (VLC1), P86 (VLC2) and P85 (VLC3) can be used as a port when using the internal voltage dividing resistor, but LCD display may not have brightness depending on panels used. This is because the stabilizing capacitor is not be connected to the external of LSI. If LCD display does not have the sufficient brightness, connect to a stabilizing capacitor or use the external voltage dividing resistor.

22.2.6 LCD Output Control Register 1 (LCCTR1)

The register controls switching between the I/O port (P80 and P77 to P71) and the segment outputs (SEG0 to SEG7). On reset, these ports are set to input mode.

■ LCD Output Control Register 1 (LCCTR1: 0x03E44)

bp	7	6	5	4	3	2	1	0
Flag	LC1SL7	LC1SL6	LC1SL5	LC1SL4	LC1SL3	LC1SL2	LC1SL1	LC1SL0
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7	LC1SL7	SEG7/P71 selection 0: P71 1: SEG7
6	LC1SL6	SEG6/P72 selection 0: P72 1: SEG6
5	LC1SL5	SEG5/P73 selection 0: P73 1: SEG5
4	LC1SL4	SEG4/P74 selection 0: P74 1: SEG4
3	LC1SL3	SEG3/P75 selection 0: P75 1: SEG3
2	LC1SL2	SEG2/P76 selection 0: P76 1: SEG2
1	LC1SL1	SEG1/P77 selection 0: P77 1: SEG1
0	LC1SL0	SEG0/P80 selection 0: P80 1: SEG0

22.2.7 LCD Output Control Register 2 (LCCTR2)

The register controls switching between the I/O port and the segment outputs (SEG8 to SEG15).
On reset, these ports are set to input mode.

■ LCD Output Control Register 2 (LCCTR2: 0x03E45)

bp	7	6	5	4	3	2	1	0
Flag	LC2SL7	LC2SL6	LC2SL5	LC2SL4	LC2SL3	LC2SL2	LC2SL1	LC2SL0
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description	
		MN101EF76K/MN101EF56K	MN101EF57G
7	LC2SL7	SEG15/P61 selection 0: P61 1: SEG15	SEG15/P50 selection 0: P50 1: SEG15
6	LC2SL6	SEG14/P62 selection 0: P62 1: SEG14	
5	LC2SL5	SEG13/P63 selection 0: P63 1: SEG13	
4	LC2SL4	SEG12/P64 selection 0: P64 1: SEG12	
3	LC2SL3	SEG11/P65 selection 0: P65 1: SEG11	
2	LC2SL2	SEG10/P66 selection 0: P66 1: SEG10	
1	LC2SL1	SEG9/P67 selection 0: P67 1: SEG9	
0	LC2SL0	SEG8/P70 selection 0: P70 1: SEG8	

22.2.8 LCD Output Control Register 3 (LCCTR3)

The register controls switching between the I/O port and the segment outputs (SEG16 to SEG23).
On reset, these ports are set to input mode.

■ LCD Output Control Register 3 (LCCTR3: 0x03E46)

bp	7	6	5	4	3	2	1	0
Flag	LC3SL7	LC3SL6	LC3SL5	LC3SL4	LC3SL3	LC3SL2	LC3SL1	LC3SL0
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description	
		MN101EF76K/MN101EF56K	MN101EF57G
7	LC3SL7	SEG23/P57 selection 0: P57 1: SEG23	SEG23/P47 selection 0: P47 1: SEG23
6	LC3SL6	SEG22/P56 selection 0: P56 1: SEG22	SEG22/P57 selection 0: P57 1: SEG22
5	LC3SL5	SEG21/P55 selection 0: P55 1: SEG21	SEG21/P56 selection 0: P56 1: SEG21
4	LC3SL4	SEG20/P54 selection 0: P54 1: SEG20	SEG20/P55 selection 0: P55 1: SEG20
3	LC3SL3	SEG19/P53 selection 0: P53 1: SEG19	SEG19/P54 selection 0: P54 1: SEG19
2	LC3SL2	SEG18/P52 selection 0: P52 1: SEG18	SEG18/P53 selection 0: P53 1: SEG18
1	LC3SL1	SEG17/P51 selection 0: P51 1: SEG17	SEG17/P52 selection 0: P52 1: SEG17
0	LC3SL0	SEG16/P50 selection 0: P50 1: SEG16	SEG16/P51 selection 0: P51 1: SEG16

22.2.9 LCD Output Control Register 4 (LCCTR4)

The register controls switching between the I/O port and the segment outputs (SEG24 to SEG31).
On reset, these ports are set to input mode.

■ LCD Output Control Register 4 (LCCTR4: 0x03E47)

bp	7	6	5	4	3	2	1	0
Flag	LC4SL7	LC4SL6	LC4SL5	LC4SL4	LC4SL3	LC4SL2	LC4SL1	LC4SL0
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description	
		MN101EF76K/MN101EF56K	MN101EF57G
7	LC4SL7	SEG31/P40 selection 0: P40 1: SEG31	SEG31/P92 selection 0: P92 1: SEG31
6	LC4SL6	SEG30/P41 selection 0: P41 1: SEG30	SEG30/P33 selection 0: P33 1: SEG30
5	LC4SL5	SEG29/P42 selection 0: P42 1: SEG29	SEG29/P34 selection 0: P34 1: SEG29
4	LC4SL4	SEG28/P43 selection 0: P43 1: SEG28	SEG28/P35 selection 0: P35 1: SEG28
3	LC4SL3	SEG27/P44 selection 0: P44 1: SEG27	SEG27/P43 selection 0: P43 1: SEG27
2	LC4SL2	SEG26/P45 selection 0: P45 1: SEG26	SEG26/P44 selection 0: P44 1: SEG26
1	LC4SL1	SEG25/P46 selection 0: P46 1: SEG25	SEG25/P45 selection 0: P45 1: SEG25
0	LC4SL0	SEG24/P47 selection 0: P47 1: SEG24	SEG24/P46 selection 0: P46 1: SEG24

22.2.10 LCD Output Control Register 5 (LCCTR5)

The register controls switching between the I/O port and the segment outputs (SEG32 to SEG39).
On reset, these ports are set to input mode.

■ LCD Output Control Register 5 (LCCTR5: 0x03E48)

bp	7	6	5	4	3	2	1	0
Flag	LC5SL7	LC5SL6	LC5SL5	LC5SL4	LC5SL3	LC5SL2	LC5SL1	LC5SL0
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description	
		MN101EF76K/MN101EF56K	MN101EF57G
7	LC5SL7	SEG39/P16 selection 0: P16 1: SEG39	SEG39/PA1 selection 0: PA1 1: SEG39
6	LC5SL6	SEG38/P30 selection 0: P30 1: SEG38	SEG38/PA0 selection 0: PA0 1: SEG38
5	LC5SL5	SEG37/P31 selection 0: P31 1: SEG37	SEG37/PB0 selection 0: PB0 1: SEG37
4	LC5SL4	SEG36/P32 selection 0: P32 1: SEG36	SEG36/PB1 selection 0: PB1 1: SEG36
3	LC5SL3	SEG35/P33 selection 0: P33 1: SEG35	SEG35/PB2 selection 0: PB2 1: SEG35
2	LC5SL2	SEG34/P34 selection 0: P34 1: SEG34	SEG34/PB3 selection 0: PB3 1: SEG34
1	LC5SL1	SEG33/P35 selection 0: P35 1: SEG33	SEG33/P94 selection 0: P94 1: SEG33
0	LC5SL0	SEG32/P36 selection 0: P36 1: SEG32	SEG32/P93 selection 0: P93 1: SEG32

22.2.11 LCD Output Control Register 6 (LCCTR6)

The register controls switching between the I/O port and the segment outputs (SEG40 to SEG47).
On reset, these ports are set to input mode.

■ LCD Output Control Register 6 (LCCTR6: 0x03E49)

bp	7	6	5	4	3	2	1	0
Flag	LC6SL7	LC6SL6	LC6SL5	LC6SL4	LC6SL3	LC6SL2	LC6SL1	LC6SL0
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description	
		MN101EF76K/MN101EF56K	MN101EF57G
7	LC6SL7	SEG47/P93 selection 0: P93 1: SEG47	-
6	LC6SL6	SEG46/P30 selection 0: P30 1: SEG46	-
5	LC6SL5	SEG45/P10 selection 0: P10 1: SEG45	-
4	LC6SL4	SEG44/P11 selection 0: P11 1: SEG44	-
3	LC6SL3	SEG43/P12 selection 0: P12 1: SEG43	-
2	LC6SL2	SEG42/P13 selection 0: P13 1: SEG42	-
1	LC6SL1	SEG41/P14 selection 0: P14 1: SEG41	-
0	LC6SL0	SEG40/P15 selection 0: P15 1: SEG40	SEG40/PA2 selection 0: PA2 1: SEG40

22.2.12 LCD Output Control Register 7 (LCCTR7)

The register controls switching between the I/O port (P94 to P95 and PB7 to PB3) and the segment outputs (SEG48 to SEG54). On reset, these ports are set to input mode.

■ LCD Output Control Register 7 (LCCTR7: 0x03E4A)

bp	7	6	5	4	3	2	1	0
Flag	-	LC7SL6	LC7SL5	LC7SL4	LC7SL3	LC7SL2	LC7SL1	LC7SL0
At reset	-	0	0	0	0	0	0	0
Access	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description	
		MN101EF76K/MN101EF56K	MN101EF57G
7	-	-	-
6	LC7SL6	SEG54/PB3 selection 0: PB3 1: SEG46	-
5	LC7SL5	SEG53/PB4 selection 0: PB4 1: SEG45	-
4	LC7SL4	SEG52/PB5 selection 0: PB5 1: SEG44	-
3	LC7SL3	SEG51/PB6 selection 0: PB6 1: SEG43	-
2	LC7SL2	SEG50/PB7 selection 0: PB7 1: SEG42	-
1	LC7SL1	SEG49/P95 selection 0: P95 1: SEG41	-
0	LC7SL0	SEG48/P94 selection 0: P94 1: SEG40	-

22.2.13 Segment Output Latch

Segment output latch stores the data from segment output pins, and can be read and write like a RAM.

Segment output latch address:

0x03E20 to 0x03E3B (MN101EF76K/MN101EF56K)

0x03E20 to 0x03E34 (MN101EF57G)

On reset, segment output latch data is undefined.

4-bit latch is assigned for 1 segment. Bit 0 and Bit 4 are read in synchronization with COM0 timing.

Bit 1 and Bit 5 are read in synchronization with COM1 timing. Bit 2 and Bit 6 are read in synchronization with COM2 timing. Bit 3 and Bit 7 are read in synchronization with COM3 timing.

Selected voltage is output when the bit is "1" and non-selected voltage is output when the bit is "0" from segment output pin.

Figure:22.2.1 and Figure:22.2.2 show the corresponding of segment output latch and segment/common pins.

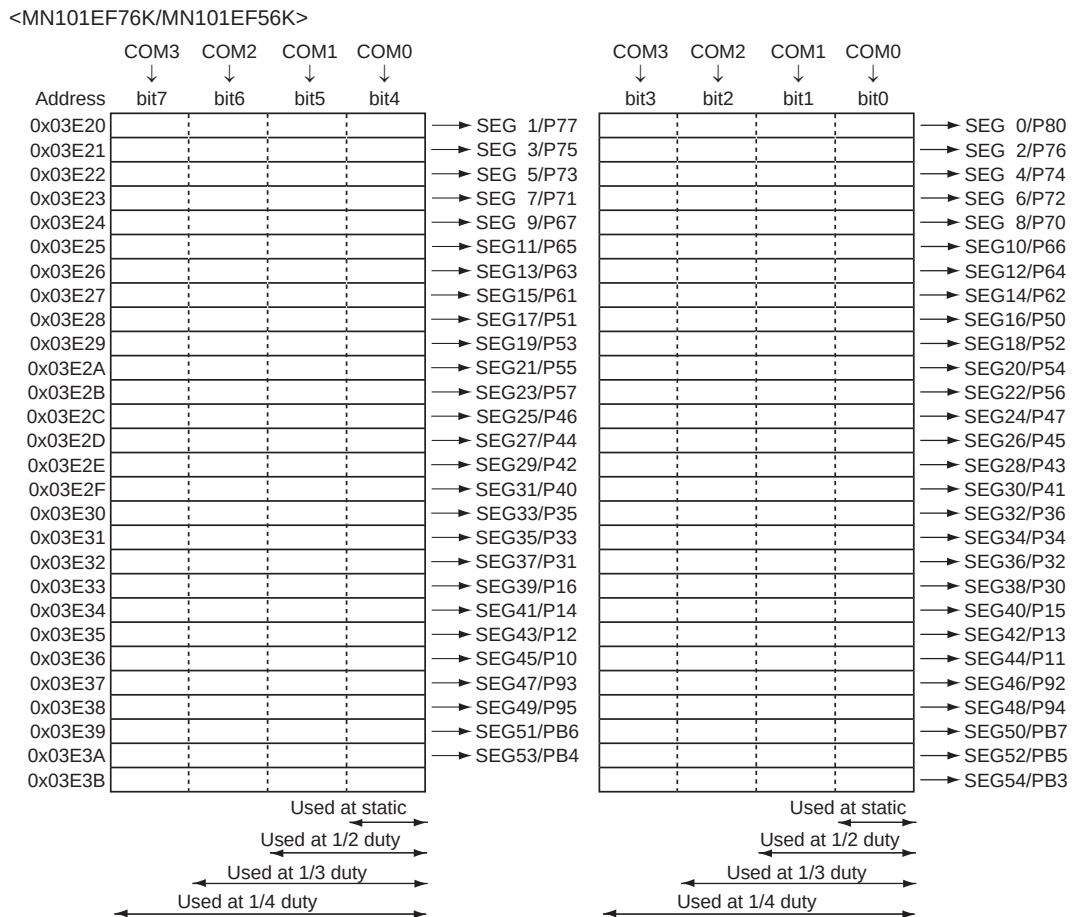


Figure:22.2.1 The Corresponding of Segment Output Latch and Segment/Common Pins (MN101EF76K/MN101EF56K)

<MN101EF57G>

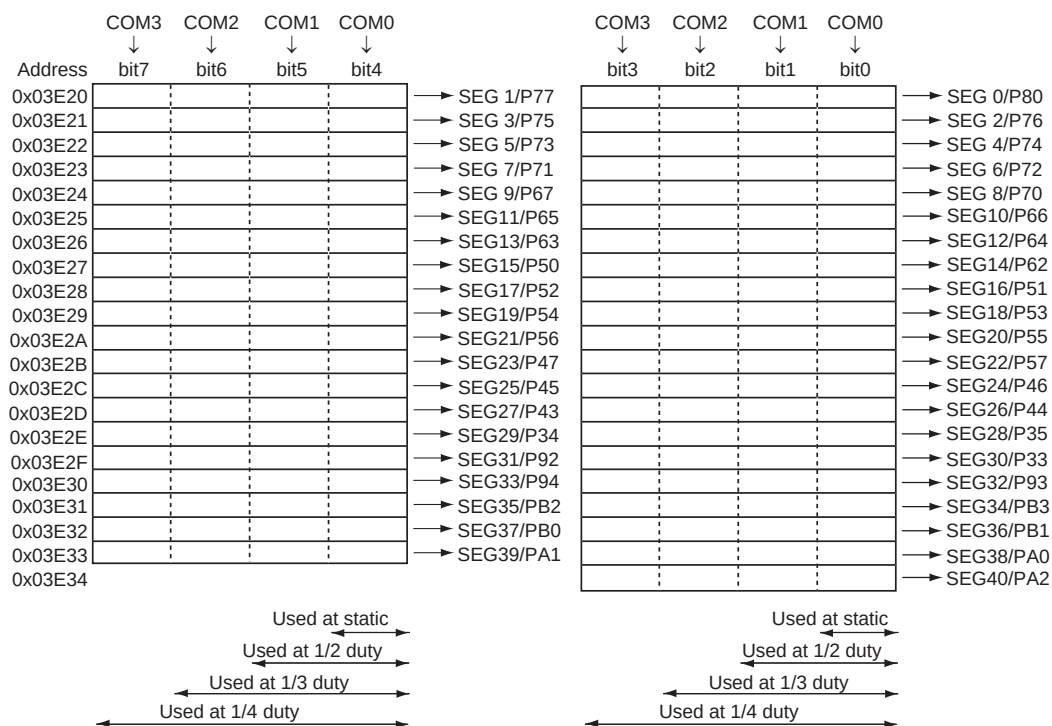


Figure:22.2.2 The Corresponding of Segment Output Latch and Segment/Common Pins (MN101EF57G)

22.3 Operation

22.3.1 LCD Driver Circuit Operation

The LCD driver uses segment output pins (MN101EF76K/MN101EF56K: SEG0 to SEG54, MN101EF56G: SEG0 to SEG40) and common output pins (COM0 to COM3) to drive the LCD with static display and dynamic display (1/2 duty 1/2 bias, 1/3 duty 1/3 bias, 1/4 duty 1/3 bias).

■ LCD driver circuit operation

The LCD driver circuit generates the required timing signals for the controlling of 1/2 duty, 1/3 duty, 1/4 duty and static using the timing control circuit based on the LCD clock (divided by the prescaler) and supplies the signal to the common driver and the multiplexer.

The common driver outputs a common signal needed by the display with the LCD power supply voltage. At LCD OFF, V_{ss} is output and the potential difference between the LCD electrodes are 0 V.

Multiplexer selects the segment output latch data based on a signal from the timing control circuit and supplies it to the segment driver. The segment driver outputs a segment signal by converting the segment output latch into the LCD driving signal by following the supply voltage supplied to the LCD power supply.
At LCD OFF, V_{ss} is output and a potential difference between in LCD electrode becomes 0 V.



On reset, the common pins and the segment pins are in a high-impedance state.
So, display may cause a flicker when the external reset input is long.



LCD can not be driven in STOP mode as the main clock supply is stopped.
Stop the LCD drive circuit by setting LCDEN to "0" (LCD OFF) before transmitting to STOP mode.

22.3.2 LCD Drive Voltage Supply

Power supply pins for LCD drive are V_{LC1} , V_{LC2} and V_{LC3} pins. This LSI includes the voltage dividing resistor which divides the LCD drive voltage. The voltage supplying for LCD driver can be selected from the following two types.

One is to supply the voltage directly to each V_{LC1} , V_{LC2} and V_{LC3} pin from external (using the external voltage dividing resistors). The other is to supply the voltage to V_{LC1} pin from external but using the internal voltage dividing resistors.

The LCD driver power supply and the V_{DD5} power supply are separated, therefore, the LCD panel drive voltage V_{LCD} can be used in the voltage lower than V_{DD5} of microcontroller. (It can be used in $V_{LCD} \leq V_{DD5} \leq 5.5 \text{ V}$.)

The LCD drive voltage from the power supply pin for LCD drive (V_{LC1} , V_{LC2} , V_{LC3}) is supplied to the segment driver and the common driver. The voltage is then converted to the LCD output based on the LCD clock signal and timing control signal.

Table:22.3.1 LCD Drive Voltage Supply Method (when using 1/3 bias)

Power Supply Method		Reference voltage supply pins	Description
1	Direct supply of drive voltage	V_{LC1} V_{LC2} V_{LC3}	A voltage is supplied directly to V_{LC1} , V_{LC2} and V_{LC3} pins from external.
2	Use of external voltage dividing resistor	V_{LC1} V_{LC2} V_{LC3}	A reference voltage is supplied to V_{LC1} pin from external, V_{LC2} and V_{LC3} voltage potentials are generated by the divided external resistor, and the voltage is supplied to each pin.
3	Use of internal voltage dividing resistor	V_{LC1}	A reference voltage is supplied to V_{LC1} pin from external and V_{LC2} and V_{LC3} voltages are generated by the internal resistor.

■ How to Supply the Voltage Using External Voltage Dividing Resistor

Supply the voltage shown in Table:22.3.2.

Table:22.3.2 LCD Power Supply

	static	1/2 bias	1/3 bias
V_{LC1}	$V_{LCD} + V_{SS}$	$V_{LCD} + V_{SS}$	$V_{LCD} + V_{SS}$
V_{LC2}		$1/2 V_{LCD} + V_{SS}$	$2/3 V_{LCD} + V_{SS}$
V_{LC3}	V_{SS}		$1/3 V_{LCD} + V_{SS}$

V_{LCD} : LCD panel drive voltage (supplied voltage maximum value to LCD panel)

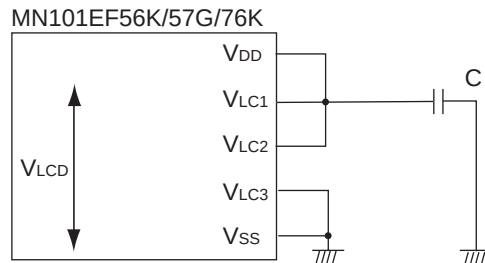


Use LCD panel with driving voltage (V_{LCD}): $V_{LCD} \leq V_{DD5} \leq 5.5 \text{ V}$.

Figure:22.3.4 shows the connection examples of the LCD power supply.

"C = 0.1 μ F" is recommended for the stabilizing capacitor C for the LCD power supply.

(a) Static ($V_{DD5} = V_{LCD}$)



(b) 1/3 duty 1/3 bias, 1/4 duty 1/3 bias ($V_{DD5} = V_{LCD}$)

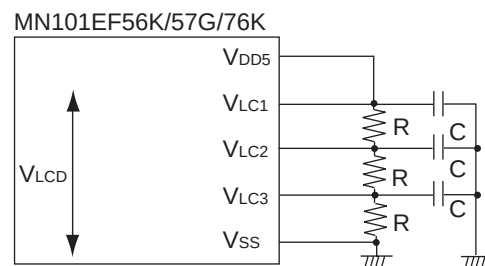


Figure:22.3.1 Examples of Connecting LCD Power Supply (using External Voltage Dividing Resistor)



1. If LCD power supply is connected as indicated in Figure:22.3.1, electric current is always applied to dividing resistors. To stop the current, connect the power supply according to the procedures specified below. ($V_{LC1}=V_{DD5}$)

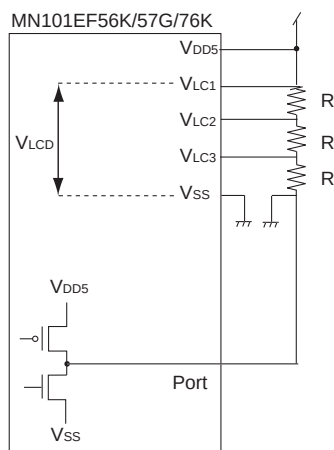


Figure:22.3.2 Connection Example for LCD Power Supply

- (1) By outputting VSS from the port, a potential difference is created across the dividing resistor R. The divided potential is then input to V_{LC1} to V_{LC3} when LCD is operating.
 - (2) V_{DD5} is output from the port. The potential different across the dividing resistor R is equal, thus reduce the current flowing into the dividing resistor.
2. Power supply V_{LC1} to V_{LC3} for LCD are given like following Figure:22.3.3.

A specific value of V_{LCD} is depending on LCD used. Refer to LCD specification for this value.

$$V_{LC1} = V_{LCD} + V_{SS}$$

$$V_{LC2} = \frac{2}{3} V_{LCD} + V_{SS}$$

$$V_{LC3} = \frac{1}{3} V_{LCD} + V_{SS}$$

The potential between V_{DD5} and V_{SS} are divided by the dividing resistors. Usually, the resistor used are ranging from 10th k Ω to 100th k Ω . "C" in Figure:22.3.3 is a bypass capacitor (approximately 0.1 μ F) which decreases the power supply impedance.

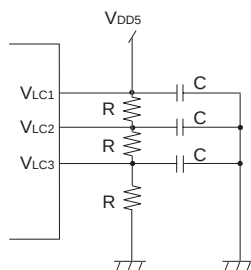


Figure:22.3.3 Supplies of V_{LC1} to V_{LC3}

■ Voltage Supply Method Using Internal Voltage Dividing Resistor

Supply the voltage shown in Figure:22.3.3.

Table:22.3.3 At Use of LCD Internal Voltage Dividing Resistor

	At Static	At 1/2 bias	At 1/3 bias
V _{LC1}	Unusable	V _{LCD}	V _{LCD}
V _{LC2}		V _{LC2} and V _{LC3} short-circuit (1/2 V _{LCD} is output)	(2/3 V _{LCD} is output)
V _{LC3}			(1/3 V _{LCD} is output)

⚠ Adequate brightness in display may not be achieved because the voltages at V_{LC1}, V_{LC2} and V_{LC3} may drop in some LCD panels when the internal voltage dividing resistor is used. For this case, use the external voltage dividing resistor.

⚠ P87 (V_{LC1}), P86 (V_{LC2}) and P85 (V_{LC3}) can be used as a port when using the internal voltage dividing resistor is used, but LCD display may not have brightness depending on panels used. Because the stabilization capacitor can not be connected to outside. If LCD display does not have sufficient brightness, connect the stabilizing capacitor or use the external voltage dividing resistor.



Figure:22.3.4 shows configuration of the internal voltage dividing resistor. When the internal voltage dividing resistor is selected, about 10 kΩ, 20 kΩ, 100 kΩ, 220 kΩ resistances are connected between each V_{LC1} and V_{LC2}, V_{LC2} and V_{LC3}, V_{LC3} and V_{SS}.

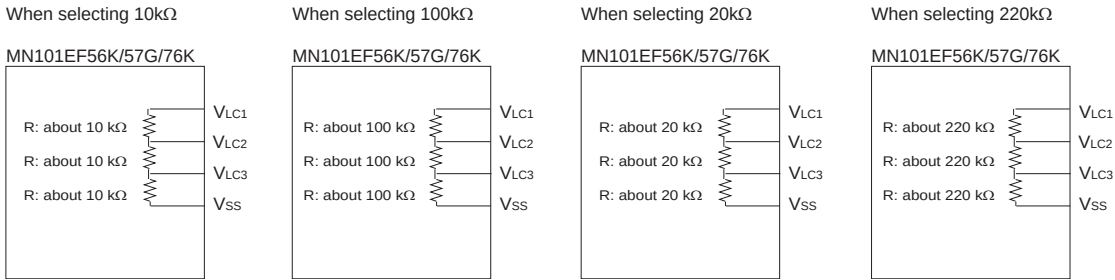


Figure:22.3.4 Configuration and Resistor Value of Internal Voltage Dividing Resistor

22.3.3 LCD Frame Period Setup

■ LCD Frame Period Setup

LCD clock is supplied by dividing "fpll" and "fslow" through a prescaler.

Bit 0 to 3 of LCDMD1 register determine LCD clock and frame period respectively.

Table:22.3.4 shows typical input frequency and its corresponding LCD clocks and frame periods.

Table:22.3.4 Input Frequency and LCD Clock

		Input clock									
Input clock	Duty	10 MHz		8 MHz		4 MHz		2 MHz		32.768 kHz	
LCDCK [3:0]	LCDTY [1:0]	LCD clock	Frame	LCD clock	Frame	LCD clock	Frame	LCD clock	Frame	LCD clock	Frame
0000 (fpll/2 ¹¹)	00 (1/4 duty)	4883 Hz	1221 Hz	3906 Hz	977 Hz	1953 Hz	488 Hz	977 Hz	244 Hz	-	
	01 (1/3 duty)		1628 Hz								
	10 (1/2 duty)		2441 Hz								
	11 (static)		4883 Hz								
0001 (fpll/2 ¹²)	00 (1/4 duty)	2441 Hz	610 Hz	1953 Hz	488 Hz	977 Hz	244 Hz	488 Hz	122 Hz	-	
	01 (1/3 duty)		814 Hz								
	10 (1/2 duty)		1221 Hz								
	11 (static)		2441 Hz								
0010 (fpll/2 ¹³)	00 (1/4 duty)	1221 Hz	305 Hz	977 Hz	244 Hz	488 Hz	122 Hz	244 Hz	61 Hz	-	
	01 (1/3 duty)		407 Hz								
	10 (1/2 duty)		610 Hz								
	11 (static)		1221 Hz								
0011 (fpll/2 ¹⁴)	00 (1/4 duty)	610 Hz	153 Hz	488 Hz	122 Hz	244 Hz	61 Hz	122 Hz	31 Hz	-	
	01 (1/3 duty)		203 Hz								
	10 (1/2 duty)		305 Hz								
	11 (static)		610 Hz								
0100 (fpll/2 ¹⁵)	00 (1/4 duty)	305 Hz	76 Hz	244 Hz	61 Hz	122 Hz	31 Hz	61 Hz	15 Hz	-	
	01 (1/3 duty)		102 Hz								
	10 (1/2 duty)		153 Hz								
	11 (static)		305 Hz								
0101 (fpll/2 ¹⁶)	00 (1/4 duty)	153 Hz	38 Hz	122 Hz	31 Hz	61 Hz	15 Hz	31 Hz	8 Hz	-	
	01 (1/3 duty)		51 Hz								
	10 (1/2 duty)		76 Hz								
	11 (static)		153 Hz								
0110 (fpll/2 ¹⁷)	00 (1/4 duty)	76 Hz	19 Hz	61 Hz	31 Hz	31 Hz	8 Hz	15 Hz	4 Hz	-	
	01 (1/3 duty)		25 Hz								
	10 (1/2 duty)		38 Hz								
	11 (static)		76 Hz								
0111 (fpll/2 ¹⁸)	00 (1/4 duty)	38 Hz	10 Hz	31 Hz	8 Hz	15 Hz	4 Hz	8 Hz	2 Hz	-	
	01 (1/3 duty)		13 Hz								
	10 (1/2 duty)		19 Hz								
	11 (static)		38 Hz								
1X00 (fslow/2 ⁶)	00 (1/4 duty)	-	-	-	-	-	-	-	-	512 Hz	128 Hz
	01 (1/3 duty)										171 Hz
	10 (1/2 duty)										256 Hz
	11 (static)										512 Hz
1X01 (fslow/2 ⁷)	00 (1/4 duty)	-	-	-	-	-	-	-	-	256 Hz	64 Hz
	01 (1/3 duty)										85 Hz
	10 (1/2 duty)										128 Hz
	11 (static)										256 Hz
1X10 (fslow/2 ⁸)	00 (1/4 duty)	-	-	-	-	-	-	-	-	128 Hz	32 Hz
	01 (1/3 duty)										43 Hz
	10 (1/2 duty)										64 Hz
	11 (static)										128 Hz
1X11 (fslow/2 ⁹)	00 (1/4 duty)	-	-	-	-	-	-	-	-	64 Hz	16 Hz
	01 (1/3 duty)										21 Hz
	10 (1/2 duty)										32 Hz
	11 (static)										64 Hz

■ Internal Voltage Dividing Resistor Setup Example

The following example is to display "23" to Type-8 LCD display panel by using SEG0 to SEG3 and COM0 to COM3 with 1/4 duty and 1/3 bias display mode, and using the internal voltage dividing resistor. Refer to Figure:22.3.2 for LCD power supply connection. And refer to [Chapter 22 22.4 LCD Display Examples] for the connection to LCD panels.

Setup Procedure	Description
(1) Select the internal voltage dividing resistor type LCDMD2 (0x03E41) bp3: LCRHL0 = 1 bp4: LCRHL1 = 1	(1) Set the LCRHL1 to 0 flags of LCD mode control register 2 (LCDMD2) to "11" to set the internal voltage dividing resistor to 100 k Ω .
(2) Select the internal voltage dividing resistor connection LCDMD2 (0x03E41) bp2: LCREN = 1	(2) Set the LCREN flag of LCD mode control register 2 (LCDMD2) to "1" to connect the internal voltage dividing resistor between V_{LC1} and V_{LC2} , V_{LC2} and V_{LC3} , V_{LC3} and V_{SS} .
(3) Pins setting LCCTR0 (0x03E43) bp0: VLC1SL = 1 LCCTR0 (0x03E43) bp7 to 4: COMSL3 to 0 = 1111 LCCTR1 (0x03E44) bp3 to 0: LC1SL3 to 0 = 1111	(3) Set the VLC1SL flag of LCD output control register 0 (LCCTR0) to "1" to set VLC1 pin. Set the COMSL3 to 0 flags of LCD output control register 0 (LCCTR0) to "1111" to set COM3 to 0 pins. Set the LC1SL3 to 0 flags of LCD output control register 1 (LCCTR1) to "1111" to set SEG3 to 0 pins.
(4) Select the LCD clock source LCDMD1 (0x03E40) bp3 to 0: LCDCK3 to 0 = 0111	(4) Select fpll/2 ¹⁸ as the LCD clock source by the LCDCK 3 to 0 flags of LCD mode control register 1 (LCDMD1).
(5) Select the LCD display duty LCDMD1 (0x03E40) bp5 to 4: DUTY1 to 0 = 00	(5) Set the DUTY1 to 0 flags of LCD mode control register 1 (LCDMD1) to "00" to set the display duty to 1/4.
(6) Set the LCD panel display data 0x03E20 = 0x5E 0x03E21 = 0x7C	(6) The display data is set to the address of the segment output latch (0x03E20 and 0x03E21). Refer to [Chapter 22 22.4 LCD Display Examples].
(7) Start the LCD driving circuit LCDMD1 (0x03E40) bp7: LCDEN = 1	(7) Set the LCDEN flag of LCD mode control register 1 (LCDMD1) to "1" to start the LCD driving circuit.



When using the internal voltage dividing resistor, adequate brightness in LCD display may not be achieved as the voltages at V_{LC1} , V_{LC2} , V_{LC3} may drop in some LCD panels due to heavy load.

In this case, set the selection bits of the internal voltage dividing resistor type of LCD mode control register 2 (LCDMD2) to "0" to select "low-resistance".

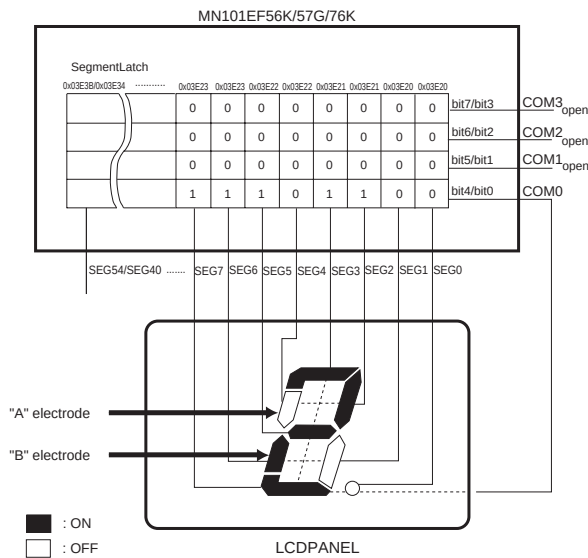
The power supply performance for V_{LC2} and V_{LC3} can be improved by setting "low-resistance".

22.4 LCD Display Examples

22.4.1 LCD Display Example (static)

The connection examples, display examples and driving waveforms of LCD panels in static, 1/2 duty, 1/3 duty and 1/4 duty are shown in [Chapter 22 22.4.1 LCD Display Example (static)] to [Chapter 22 22.4.8 LCD Operation Setup (1/4 duty)].

■ Static



		LCD ON		LCD OFF
		COM=S SEG=S	COM=S SEG=N	
LCD clock				Undefined
	Data			Undefined
COM	V _{Lc1}			
	V _{SS}			
SEG	V _{Lc1}			
	V _{SS}			
COM-SEG	V _{Lcd}			
	0			
	-V _{Lcd}			
		ON	OFF	OFF

S: selected voltage N: non-selected voltage V_{Lcd}: LCD driving voltage
COM (COM0) always outputs the selected voltage in static.

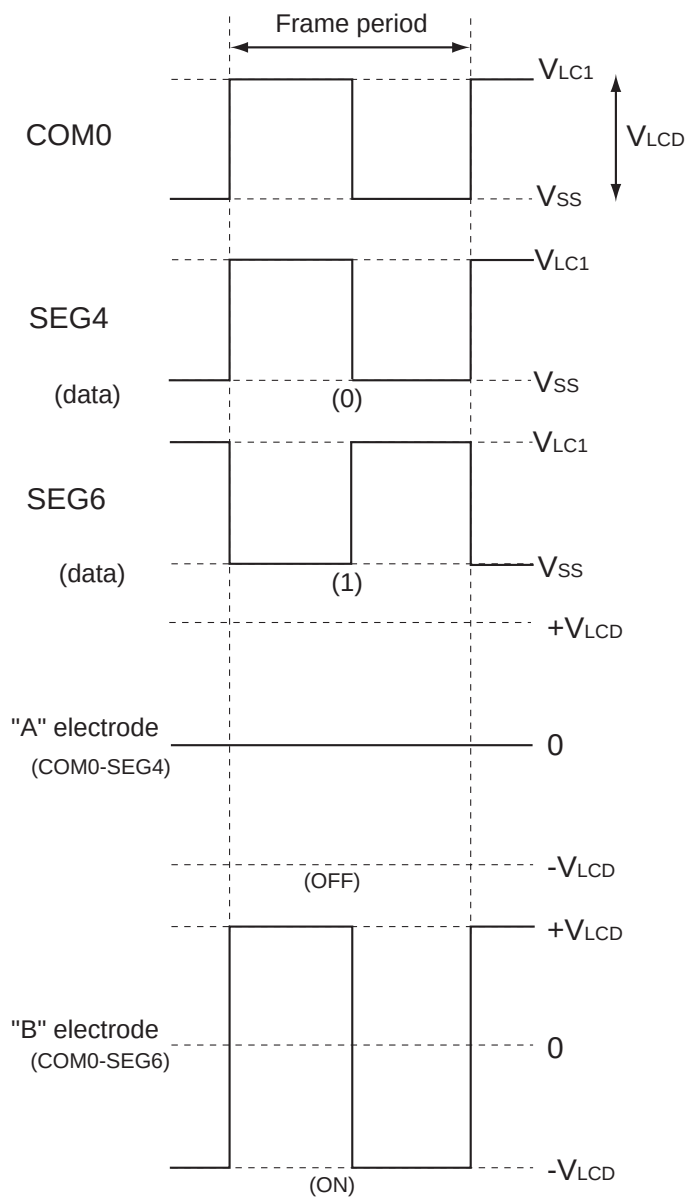


Figure:22.4.1 LCD Display Example in Static

22.4.2 LCD Operation Setup (static)

■ LCD Operation Setup Example (static)

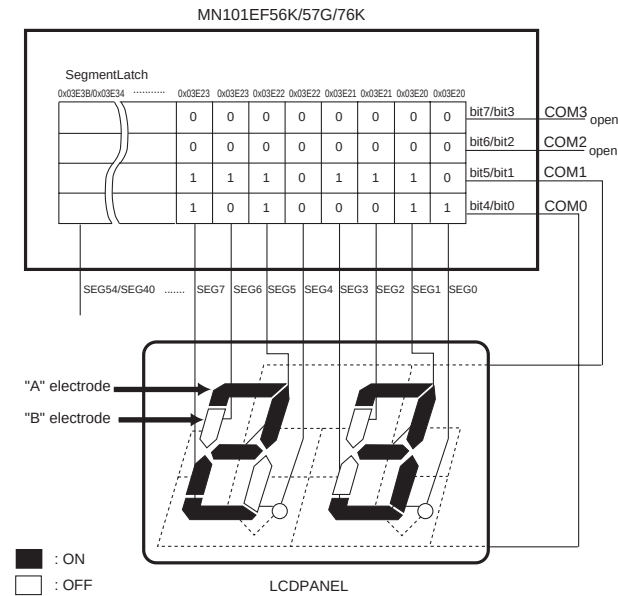
The following example is to display "2" to Type-8 LCD display panel of 1-digit by using the segment signals (SEG0 to SEG7) and the common signal (COM0) with static, and using the external voltage dividing resistor. Refer to [Chapter 22 22.4.1 LCD Display Example (static)].

The oscillation clock is $f_{pll} = 4 \text{ MHz}$, LCD clock is $f_{pll}/2^{15} = 122 \text{ Hz}$ and the frame period is 122 Hz. The setting procedure and description are shown below.

Setup Procedure	Description
(1) Stop LCD operation LCDMD1(0x03E40) bp7: LCDEN = 0	(1) Set the LCDEN flag of LCD mode control register 1 (LCDMD1) to "0" to stop LCD operation.
(2) iSet display duty LCDMD1(0x03E40) bp5 to 4: LCDDTY1 to 0 = 11	(2) Set the LCDTY1 to 0 flags of LCD mode control register1 (LCDMD1) to "11" to set the mode to static driving.
(3) Set LCD clock LCDMD1(0x03E40) bp3 to 0: LCDCK3 to 0 = 0100	(3) Set the source clock of LCD clock to $f_{pll}/2^{15}$ by the LCDCK3 to 0 flags of LCD mode control register 1 (LCDMD1).
(4) Set VLC pin/port shared pin Set segment output pin/port shared pin Set common output pin/port shared pin LCCTR0(0x03E43) bp2 to 0: VLC3 to 1SL = 111 LCCTR0(0x03E43) bp4: COMSL0 = 1 LCCTR1(0x03E44) bp7 to 0: LC1SL7 to 0 = 11111111	(4) Select VLC1 to 3, SEG0 to 7 and COM0 by LCD output control register 0 (LCCTR0) and LCD output control register 1 (LCCTR1).
(5) Set display data Segment output latch SEG1 to 0 (0x03E20) = 0x00 Segment output latch SEG3 to 2 (0x03E21) = 0x11 Segment output latch SEG5 to 4 (0x03E22) = 0x10 Segment output latch SEG7 to 6 (0x03E23) = 0x11	(5) The display data is set to "2" by the segment output latch SEG0 to 7 (0x03E20 to 0x03E23). Refer to [Chapter 22 22.4.1 LCD Display Example (static)]
(6) Start LCD operation LCDMD1(0x03E40) bp7: LCDEN = 1	(6) Set the LCDEN flag of LCD mode control register (LCDMD1) to "1" to start LCD operation.

22.4.3 LCD Display Example (1/2 duty)

■ 1/2 duty



	LCD ON				LCD OFF
	COM=S SEG=S	COM=N SEG=S	COM=S SEG=N	COM=N SEG=N	
LCD clock					Undefined
Data	"1"		"0"		Undefined
COM					
SEG					
COM-SEG					
	ON	OFF	OFF	OFF	OFF

S: selected voltage N: non-selected voltage V_{LCD}: LCD driving voltage

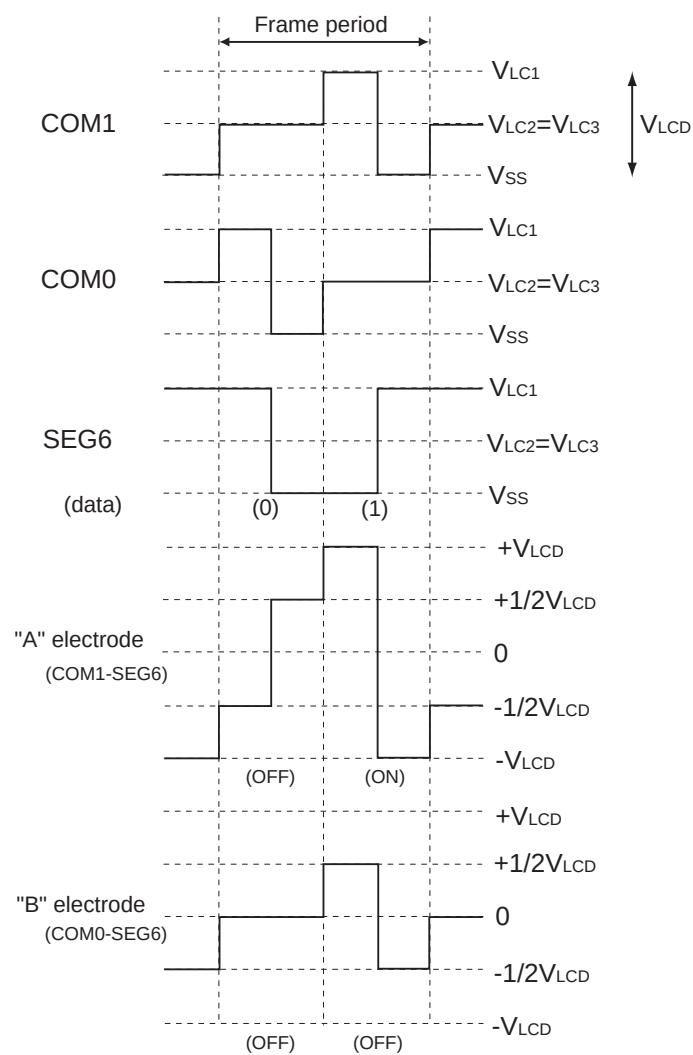


Figure:22.4.2 LCD Display Example in 1/2 duty

22.4.4 LCD Operation Setup (1/2 duty)

■ LCD Operation Setup Example (1/2 duty)

The following example is to display "23" to Type-8 LCD display panel of 2-digit by using the segment signals (SEG0 to SEG7) and the common signals (COM0 to COM1) with 1/2 duty and 1/2 bias, and using the external voltage dividing resistor.

Refer to [Chapter 22 22.4.3 LCD Display Example (1/2 duty)].

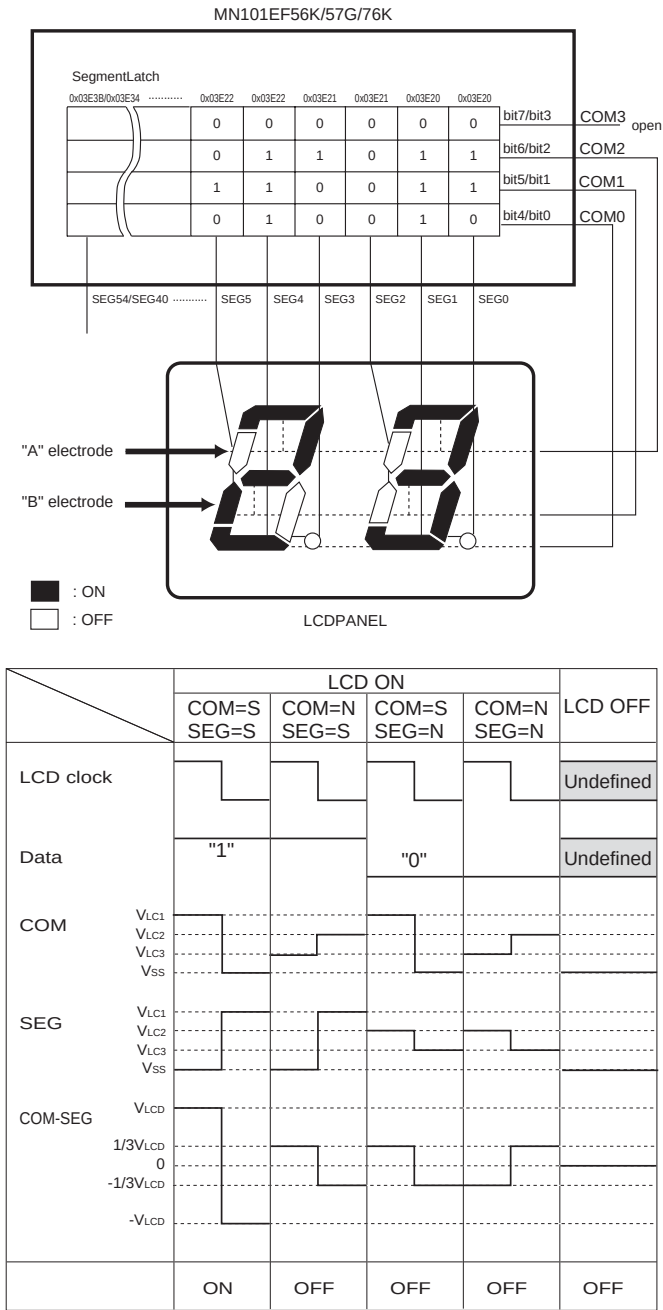
The oscillation clock is $f_{pll} = 4 \text{ MHz}$, LCD clock is $f_{pll}/2^{15} = 122 \text{ Hz}$ and the frame period is 61 Hz.

The setting procedure and description are shown below.

Setup Procedure	Description
(1) Stop LCD operation LCDMD1(0x03E40) bp7: LCDEN = 0	(1) Set the LCDEN flag of LCD mode control register 1 (LCDMD1) to "0" to stop LCD operation.
(2) iSet display duty LCDMD1(0x03E40) bp5 to 4: LCDDTY1 to 0 = 10	(2) Set the DUTY1 to 0 flags of LCD mode control register 1 (LCDMD1) to "10" to set the mode to 1/2 duty driving.
(3) Set LCD clock LCDMD1(0x03E40) bp3 to 0: LCDCK3 to 0 = 0100	(3) Set the source clock of LCD clock to $f_{pll}/2^{15}$ by the LCDCK3 to 0 flags of LCD mode control register 1 (LCDMD1).
(4) Set VLC pin/port shared pin Set segment output pin/port shared pin Set common output pin/port shared pin LCCTR0(0x03E43) bp2 to 0: VLC3SL to VLC1SL = 111 LCCTR0(0x03E43) bp5 to 4: COMSL1 to 0 = 11 LCCTR1(0x03E44) bp7 to 0: LC1SL7 to 0 = 11111111	(4) Select VLC1 to 3, SEG0 to 7 and COM1 to 0 by LCD output control register 0 (LCCTR0) and LCD output control register 1 (LCCTR1).
(5) Set display data Segment output latch SEG1 to 0 (0x03E20) = 0x31 Segment output latch SEG3 to 2 (0x03E21) = 0x22 Segment output latch SEG5 to 4 (0x03E22) = 0x30 Segment output latch SEG7 to 6 (0x03E23) = 0x32	(5) The display data is set to "23" by the segment output latch SEG0 to 7 (0x03E20 to 0x03E23). Refer to [Chapter 22 22.4.3 LCD Display Example (1/2 duty)].
(6) Start LCD operation LCDMD1(0x03E40) bp7: LCDEN = 1	(6) Set the LCDEN flag of LCD mode control register 1 (LCDMD1) to "1" to start LCD operation.

22.4.5 LCD Display Example (1/3 duty)

■ 1/3 duty



S: selected voltage N: non-selected voltage V_{LCD}: LCD driving voltage

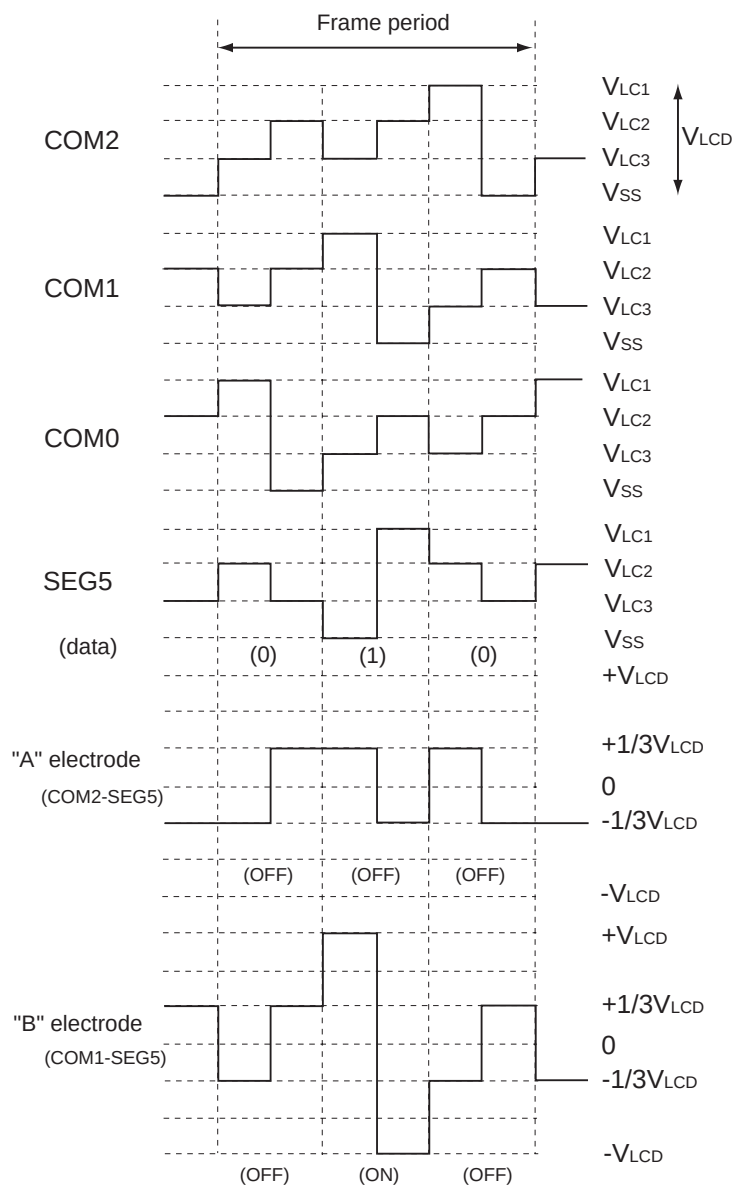


Figure:22.4.3 LCD Display Example in 1/3 duty

22.4.6 LCD Operation Setup (1/3 duty)

■ LCD Operation Setup Example (1/3 duty)

The following example is to display "23" to Type-8 LCD display panel of 2-digits by using the segment signals (SEG0 to SEG5) and the common signals (COM0 to COM2) with 1/3 duty and 1/3 bias, and using the external voltage dividing resistor.

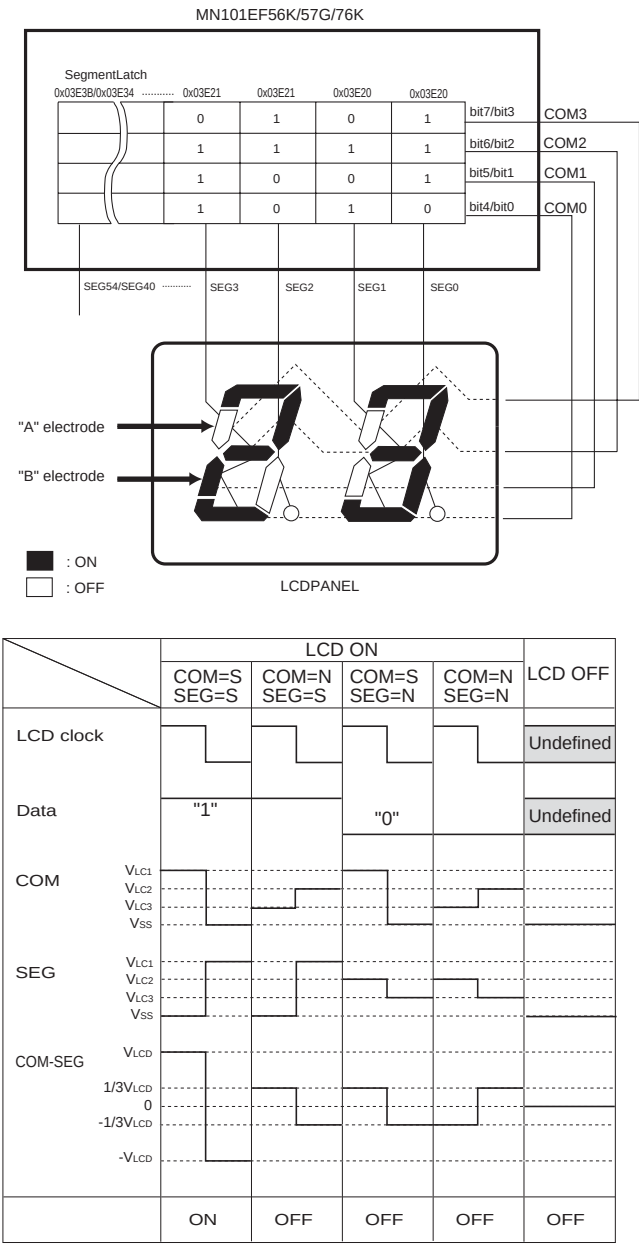
Refer to [22.4.5 LCD Display Examples (1/3 duty)].

The oscillation clock is $f_{pll} = 4 \text{ MHz}$, LCD clock is $f_{pll}/2^{15} = 122 \text{ Hz}$ and the frame period is 41 Hz.
The setting procedure and description are shown below.

Setup Procedure	Description
(1) Stop LCD operation LCDMD1(0x03E40) bp7: LCDEN = 0	(1) Set the LCDEN flag of LCD mode control register 1 (LCDMD1) to "0" to stop LCD operation.
(2) iSet display duty LCDMD1(0x03E40) bp5 to 4: DUTY1 to 0 = 01	(2) Set the DUTY1 to 0 flags of LCD mode control register 1 (LCDMD1) to "01" to set the mode to 1/3 duty driving.
(3) Set LCD clock LCDMD1(0x03E40) bp3 to 0: LCDCK3 to 0 = 0100	(3) Set the source clock of LCD clock to $f_{pll}/2^{15}$ by the LCDCK3 to 0 flags of LCD mode control register 1 (LCDMD1).
(4) Set VLC pin/port shared pin Set segment output pin/port shared pin Set common output pin/port shared pin LCCTR0(0x03E43) bp2 to 0: VLC3 to 1SL = 111 LCCTR0(0x03E43) bp6 to 4: COMSL2 to 0 = 111 LCCTR1(0x03E44) bp5 to 0: LC1SL5 to 0 = 111111	(4) Select VLC1 to 3, SEG0 to 5 and COM2 to 0 by LCD output control register 0 (LCCTR0) and LCD output control register 1 (LCCTR1).
(5) Set display data Segment output latch SEG1 to 0 (0x03E20) = 0x076 Segment output latch SEG3 to 2 (0x03E21) = 0x040 Segment output latch SEG5 to 4 (0x03E22) = 0x027	(5) The display data is set to "23" by the segment output latch SEG0 to 7 (0x03E20 to 0x03E22). Refer to [Chapter 22 22.4.5 LCD Display Example (1/3 duty)].
(6) Start LCD operation LCDMD1(0x03E40) bp7: LCDEN = 1	(6) Set the LCDEN flag of LCD mode control register (LCDMD1) to "1" to start LCD operation.

22.4.7 LCD Display Example (1/4 duty)

■ 1/4 duty



S: selected voltage N: non-selected voltage V_{LCD}: LCD driving voltage

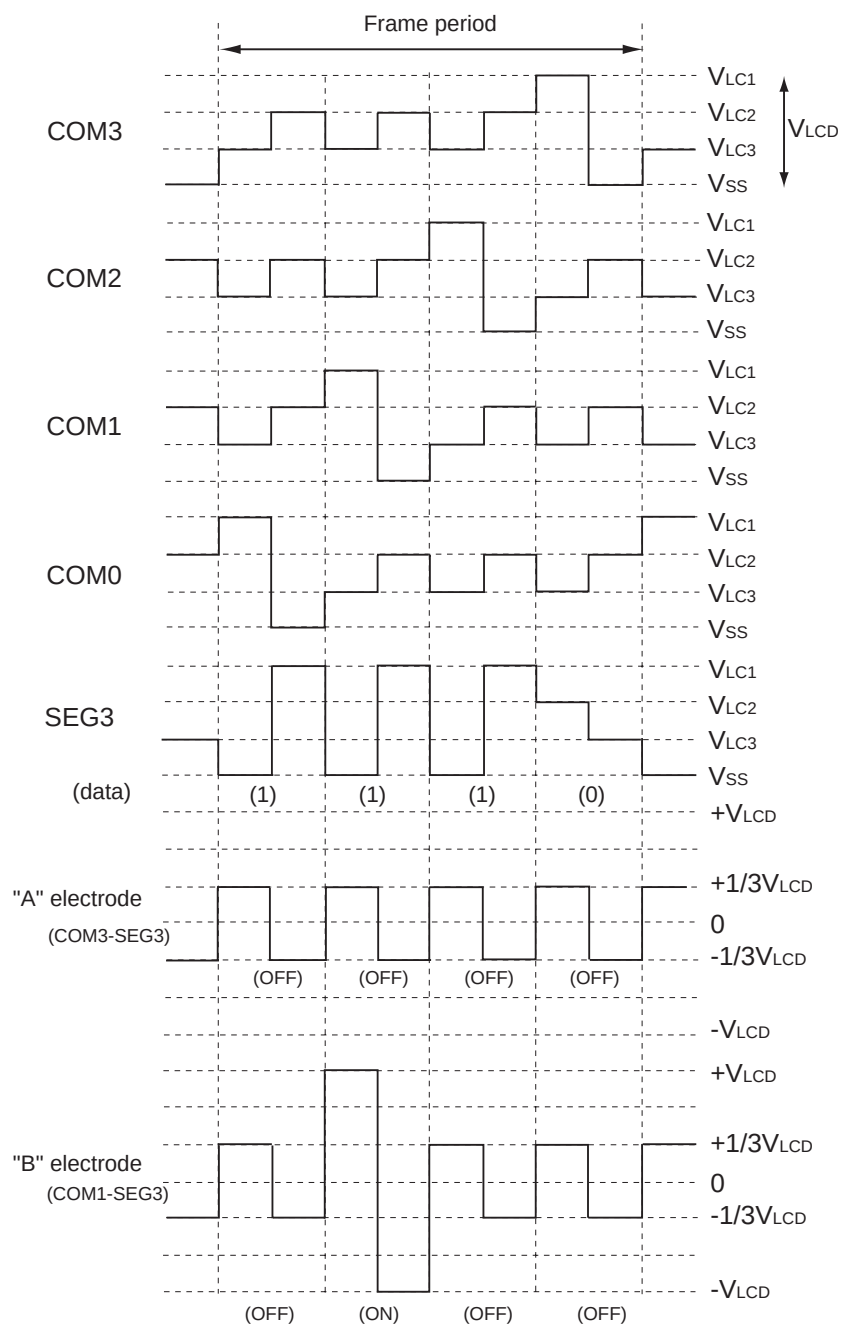


Figure:22.4.4 LCD Display Example in 1/4 duty

22.4.8 LCD Operation Setup (1/4 duty)

■ LCD Operation Setup Example (1/4 duty)

The following example is to display "23" to Type-8 LCD display panel of 2-digits by using the segment signals (SEG0 to SEG4) and the common signals (COM0 to COM3) with 1/4 duty and 1/3 bias, and using the external voltage dividing resistor.

Refer to [Chapter 22 22.4.7 LCD Display Example (1/4 duty)].

The oscillation clock is $f_{pll} = 4 \text{ MHz}$, LCD clock is $f_{pll}/2^{15} = 122 \text{ Hz}$ and the frame period is 31 Hz.

The setting procedure and description are shown below.

Setup Procedure	Description
(1) Stop LCD operation LCDMD1(0x03E40) bp7: LCDEN = 0	(1) Set the LCDEN flag of LCD mode control register 1 (LCDMD1) to "0" to stop LCD operation.
(2) iSet display duty LCDMD1(0x03E40) bp5 to 4: DUTY1 to 0 = 00	(2) Set the DUTY1 to 0 flags of LCD mode control register 1 (LCDMD1) to "00" to set the mode to 1/4 duty driving.
(3) Set LCD clock LCDMD1(0x03E40) bp3 to 0: LCDCK3 to 0 = 0100	(3) Set the source clock of LCD clock to $f_{pll}/2^{15}$ by the LCDCK3 to 0 flags of LCD mode control register 1 (LCDMD1).
(4) Set VLC pin/port shared pin Set segment output pin/port shared pin Set common output pin/port shared pin LCCTR0(0x03E43) bp2 to 0: VLC3 to 1SL = 111 LCCTR0(0x03E43) bp7 to 4: COMSL3 to 0 = 1111 LCCTR1(0x03E44) bp3 to 0: LC1SL1 to 0 = 1111	(4) Select VLC1 to 3, SEG0 to 3 and COM3 to 0 by LCD output control register 0 (LCCTR0) and LCD output control register 1 (LCCTR1).
(5) Set display data Segment output latch SEG1 to 0 (0x03E20) = 0x05E Segment output latch SEG3 to 2 (0x03E21) = 0x07C	(5) The display data is set to "23" by SEG0 to 4 (0x03E20 to 0x03E21). Refer to [Chapter 22 22.4.7 LCD Display Example (1/4 duty)].
(6) Start LCD operation LCDMD1(0x03E40) bp7: LCDEN = 1	(6) Set the LCDEN flag of LCD mode control register 1 (LCDMD1) to "1" to start LCD operation.

23.1 Automatic Transfer Controller

23.1.1 Overview

This LSI contains an automatic transfer controller that uses direct memory access (DMA) to transfer the contents of the whole memory space (1 MB) using the hardware. This function block is called ATC1.

ATC1 is activated by an interrupt or a flag set by the software. Once this occurs, even if it is in the middle of executing an instruction, the microcontroller waits for a time when it can release the bus, stops normal operation, and transfers bus control to ATC1. ATC1 automatically transfer the data using the bus that is released by CPU.

The software sets the activation factor in ATC1 control register 1 (AT1CNT1), then data transfer begins when the AT1ACT flag in ATC1 control register 0 (AT1CNT0) is set to "1". AT1ACT flag is automatically cleared to "0" when ATC1 is activated.

Transfer data counter (AT1TRC) determines the number of transfers that ATC1 makes, up to a maximum of 255 times. There are also 16 transfer modes, set in AT1CNT0 register.



The interrupt enable flag (xxxIE) for interrupt as a trigger factor needs not to be set. This is because the automatic data transfer occurs in the hardware without going through an interrupt service routine. If the interrupt enable flag (xxxIE) is set for the type of interrupt ATC1, the hardware processing of the regular interrupt is operated after the automatic transfer ends.



The order of an interrupt acceptance may be changed by software when setting each interrupt is set as a trigger factor of ATC1, a trigger factor interrupt and an interrupt level of ATC1 interrupt (ATC1IRQ) are in same level.
To prevent this, set each interrupt level differently.



ATC1 can not be used in STANDBY mode.
ATC1 starts the data transfer after recovering CPU OPERATION mode when ATC1 activation factor is generated in STANDBY mode.

23.1.2 Functions

Table:23.1.1 provides a list of the ATC1 trigger factors and transfer modes.

■ ATC1 Trigger Factors

Table:23.1.1 ATC1 Trigger Factors

Trigger Factors	External Interrupt 0
	External Interrupt 1
	External Interrupt 2
	External Interrupt 3
	Timer 0 interrupt, External Interrupt 4
	Timer 1 interrupt, LIN interrupt
	Timer 2 interrupt, Timer 4 interrupt
	Timer 3 interrupt, Timer 8 interrupt
	Timer 7 interrupt, Timer 8 capture trigger
	Timer 7 capture trigger, Serial Interface 0 transmission complete interrupt
	Serial Interface 0 transmission buffer empty, Serial Interface 1 transmission complete interrupt
	Serial Interface 1 transmission buffer empty, Serial Interface 2 transmission complete interrupt
	Serial Interface 2 transmission buffer empty
	Serial Interface 4 transmission complete interrupt
	A/D converter interrupt, Serial Interface 4 transmission buffer empty
	Software interrupt, Peripheral function group interrupt

■ Transfer Modes

Table:23.1.2 Transfer Modes

Transfer Mode	Transfer Direction (*)			Pointer Increment Control		Transfer Operation
	cycle	Source Address	→ Destination Address	AT1MAP0	AT1MAP1	
Transfer mode 0		AT1MAP0	→ AT1MAP1 (I/O area)	-	-	1-byte data transfer
Transfer mode 1		AT1MAP1 (I/O area)	→ AT1MAP0	-	-	1-byte data transfer
Transfer mode 2		AT1MAP0	→ AT1MAP1 (I/O area)	AT1MAP0+1	-	1-byte data transfer
Transfer mode 3		AT1MAP1 (I/O area)	→ AT1MAP0	AT1MAP0+1	-	1-byte data transfer
Transfer mode 4	1st	AT1MAP0	→ AT1MAP1 (I/O area: even ADR)	AT1MAP0+1	-	1-word data transfer (An even address must be set in AT1MAP1)
	2nd	AT1MAP0 [=AT1MAP0+1]	→ AT1MAP1 (I/O area: odd ADR)	AT1MAP0+1	-	
Transfer mode 5	1st	AT1MAP1 (I/O area: even ADR)	→ AT1MAP0	AT1MAP0+1	-	1-word data transfer (An even address must be set in AT1MAP1)
	2nd	AT1MAP1 (I/O area: odd ADR)	→ AT1MAP0 [=AT1MAP0+1]	AT1MAP0+1	-	
Transfer mode 6	1st	AT1MAP1 (I/O area)	→ AT1MAP0	AT1MAP0+1	-	Two 1-byte data transfers
	2nd	AT1MAP0 [=AT1MAP0+1]	→ AT1MAP1 (I/O area)	AT1MAP0+1	-	
Transfer mode 7	1st	AT1MAP1 (I/O area)	→ AT1MAP0	AT1MAP0+1	-	Two 1-byte data transfers
	2nd	AT1MAP0 [=AT1MAP0+1]	→ AT1MAP1 (I/O area)	-	-	
Transfer mode 8	1st	AT1MAP1 (I/O area: even ADR)	→ AT1MAP0	AT1MAP0+1	-	Two 1-byte data transfers (An even address must be set in AT1MAP1)
	2nd	AT1MAP0 [=AT1MAP0+1]	→ AT1MAP1 (I/O area: odd ADR)	AT1MAP0+1	-	
Transfer mode 9	1st	AT1MAP1 (I/O area: even ADR)	→ AT1MAP0	AT1MAP0+1	-	Two 1-byte data transfers (An even address must be set in AT1MAP1)
	2nd	AT1MAP0 [=AT1MAP0+1]	→ AT1MAP1 (I/O area: odd ADR)	-	-	
Transfer mode A		AT1MAP0	→ AT1MAP1	-	-	1-byte data transfer (whole memory area)
Transfer mode B		AT1MAP1	→ AT1MAP0	-	-	1-byte data transfer (whole memory area)
Transfer mode C		AT1MAP0	→ AT1MAP1	AT1MAP0+1	AT1MAP1+1	1-byte data transfer (whole memory area)
Transfer mode D		AT1MAP1	→ AT1MAP0	AT1MAP0+1	AT1MAP1+1	1-byte data transfer (whole memory area)
Transfer mode E		AT1MAP0	→ AT1MAP1	AT1MAP0+1	AT1MAP1+1	Burst transfer (continues until AT1TCR=0)
Transfer mode F		AT1MAP1	→ AT1MAP0	AT1MAP0+1	AT1MAP1+1	Burst transfer (continues until AT1TCR=0)

(*) When a memory pointer points to the I/O space, only the lower 8 bits of the pointer are valid.



Change the ATC1 activation factor and the transfer mode while ATC1 transfer is disabled (AT1EN flag of AT1CNT0 register is "0").

23.1.3 Block Diagram

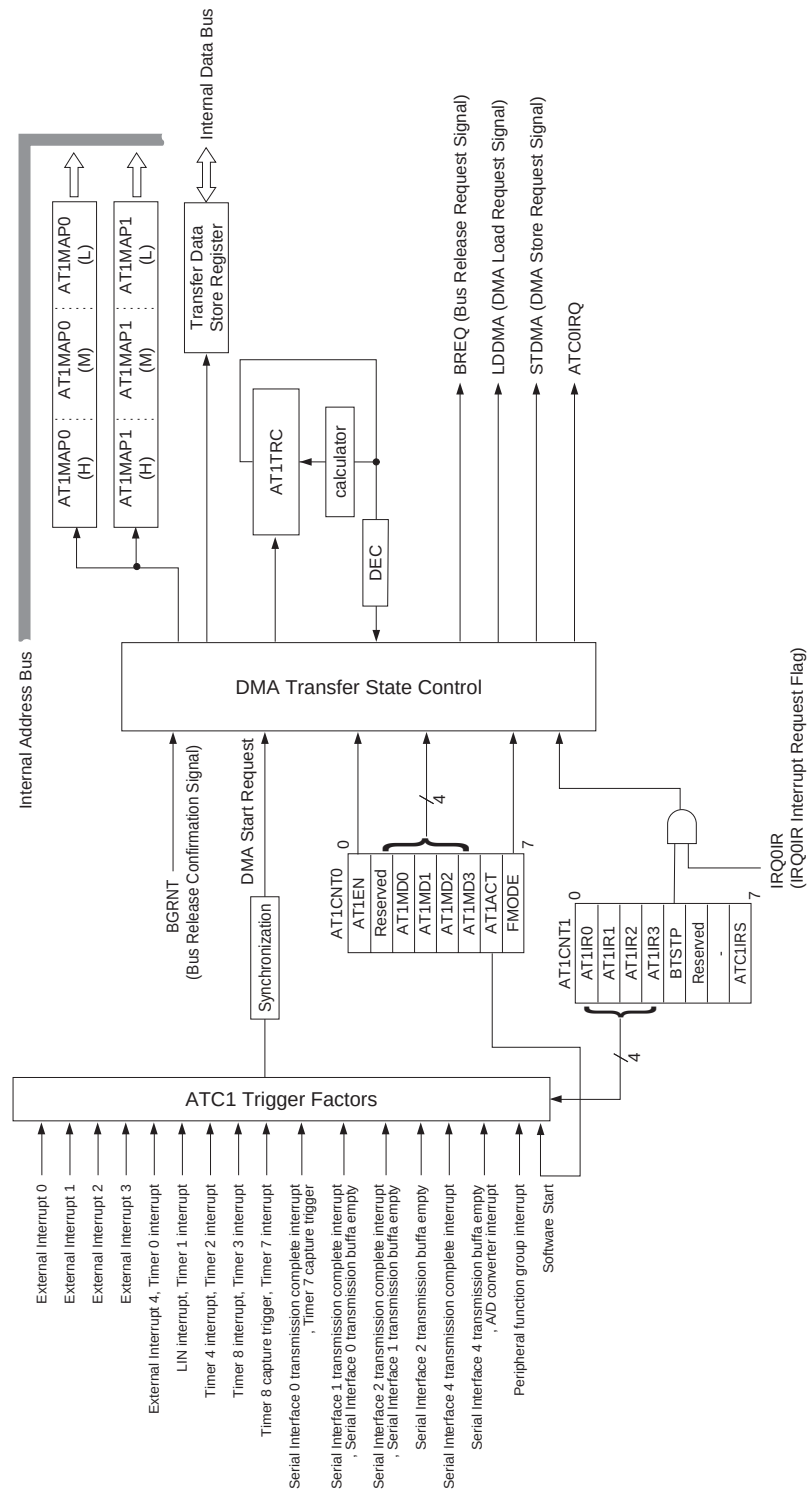


Figure:23.1.1 ATC1 Block Diagram

23.2 Control Registers

23.2.1 Registers

Table:23.2.1 shows the registers used to control ATC1.

Table:23.2.1 ATC1 Control Registers

	Register	Address	R/W	Function	Page
ATC1	AT1CNT0	0X03E60	R/W	ATC1 control register 0	XXIII-7
	AT1CNT1	0X03E61	R/W	ATC1 control register 1	XXIII-8
	AT1TRC	0X03E62	R/W	ATC1 transfer data counter	XXIII-9
	AT1MAP0L	0X03E63	R/W	ATC1 memory pointer 0 (lower 8 bits)	XXIII-9
	AT1MAP0M	0X03E64	R/W	ATC1 memory pointer 0 (middle 8 bits)	XXIII-9
	AT1MAP0H	0X03E65	R/W	ATC1 memory pointer 0 (upper 4 bits)	XXIII-9
	AT1MAP1L	0X03E66	R/W	ATC1 memory pointer 1 (lower 8 bits)	XXIII-10
	AT1MAP1M	0X03E67	R/W	ATC1 memory pointer 1 (middle 8 bits)	XXIII-10
	AT1MAP1H	0X03E68	R/W	ATC1 memory pointer 1 (upper 4 bits)	XXIII-10

R/W: Readable/Writable

■ ATC1 Control Register 0 (AT1CNT0: 0x03E60)

bp	7	6	5	4	3	2	1	0
Flag	FM0DE	AT1ACT	AT1MD3	AT1MD2	AT1MD1	AT1MD0	Reserved	AT1EN
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7	FM0DE	Increment control flag for memory pointer 0 0: Increment depending on transfer mode 1: Disable incrementing of memory pointer 0
6	AT1ACT	ATC1 software activation flag 0: Do not activate ATC1 1: Activate ATC1
5 to 2	AT1MD3 AT1MD2 AT1MD1 AT1MD0	ATC1 data transfer mode 0000: Transfer mode 0 0001: Transfer mode 1 0010: Transfer mode 2 0011: Transfer mode 3 0100: Transfer mode 4 0101: Transfer mode 5 0110: Transfer mode 6 0111: Transfer mode 7 1000: Transfer mode 8 1001: Transfer mode 9 1010: Transfer mode A 1011: Transfer mode B 1100: Transfer mode C 1101: Transfer mode D 1110: Transfer mode E 1111: Transfer mode F
1	Reserved	Always set to "0".
0	AT1EN	ATC1 transfer enable flag 0: ATC1 transfer disable 1: ATC1 transfer enable



AT1ACT flag of ATC1 control register 0 (AT1CNT0) is cleared by hardware automatically when the transfer completes.

■ ATC1 Control Register 1 (AT1CNT1: 0x03E61)

bp	7	6	5	4	3	2	1	0
Flag	AT1IRS	-	Reserved	BTSTP	AT1IR3	AT1IR2	AT1IR1	AT1IR0
At reset	0	-	0	0	0	0	0	0
Access	R/W	-	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7	AT1IRS	Activation factor selection flag 0: Activation factor 1 1: Activation factor 2
6	-	-
5	Reserved	Always set to "0".
4	BTSTP	Burst transfer stop enable 0: Burst transfer stop disable 1: Burst transfer stop enable (Transfer stops when External Interrupt 0 occurs.)
3 to 0	AT1IR3 AT1IR2 AT1IR1 AT1IR0	ATC1 trigger factor setting flag (AT1IRS=0) 0000: External Interrupt 0 0001: External Interrupt 1 0010: External Interrupt 2 0011: External Interrupt 3 0100: Timer 0 interrupt 0101: Timer 1 interrupt 0110: Timer 2 interrupt 0111: Timer 3 interrupt 1000: Timer 7 interrupt 1001: Timer 7 capture trigger 1010: Serial 0 transmission buffer empty 1011: Serial 1 transmission buffer empty 1100: Serial 2 transmission buffer empty 1101: Setting prohibited 1110: A/D converter interrupt 1111: Software interrupt (AT1IRS=1) 0000: Setting prohibited 0001: Setting prohibited 0010: Setting prohibited 0011: Setting prohibited 0100: External Interrupt 4 0101: LIN interrupt 0110: Timer 4 interrupt 0111: Timer 8 interrupt 1000: Timer 8 capture trigger 1001: Serial 0 transmission complete interrupt 1010: Serial 1 transmission complete interrupt 1011: Serial 2 transmission complete interrupt 1100: Setting prohibited 1101: Serial 4 transmission complete interrupt 1110: Serial 4 buffer empty 1111: Peripheral function group interrupt



Do not select External Interrupt 0 for ATC1 trigger factor when the burst transfer stop is enabled.



Bp5 of ATC1 control register 1 (AT1CNT1) may be set by hardware automatically. If automatic setting is operated by hardware, the data automatic transfer function operates normally.

■ ATC1 Transfer Counter (AT1TRC1: 0x03E62)

bp	7	6	5	4	3	2	1	0
Flag	AT1TRC7	AT1TRC6	AT1TRC5	AT1TRC4	AT1TRC3	AT1TRC2	AT1TRC1	AT1TRC0
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7 to 0	AT1TRC7 to AT1TRC0	ATC1 Transfer Data Count Setting · For transfer modes 0 to D, set this register to the number of ATC activations. · For transfer modes E and F, set this register to number of burst transfers.

■ ATC1 Memory Pointer 0: Lower 8 bits (AT1MAP0L: 0x03E63)

bp	7	6	5	4	3	2	1	0
Flag	AT1MA P0L7	AT1MA P0L6	AT1MA P0L5	AT1MA P0L4	AT1MA P0L3	AT1MA P0L2	AT1MA P0L1	AT1MA P0L0
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

■ ATC1 Memory Pointer 0: Middle 8 bits (AT1MAP0M: 0x03E64)

bp	7	6	5	4	3	2	1	0
Flag	AT1MA P0M7	AT1MA P0M6	AT1MA P0M5	AT1MA P0M4	AT1MA P0M3	AT1MA P0M2	AT1MA P0M1	AT1MA P0M0
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

■ ATC1 Memory Pointer 0: Upper 4 bits (AT1MAP0H: 0x03E65)

bp	7	6	5	4	3	2	1	0
Flag	-	-		-	AT1MA P0H3	AT1MA P0H2	AT1MA P0H1	AT1MA P0H0
At reset	-	-	-	-	0	0	0	0
Access	-	-	-	-	R/W	R/W	R/W	R/W

■ ATC1 Memory Pointer 1: Lower 8 bits (AT1MAP1L: 0x03E66)

bp	7	6	5	4	3	2	1	0
Flag	AT1MA P1L7	AT1MA P1L6	AT1MA P1L5	AT1MA P1L4	AT1MA P1L3	AT1MA P1L2	AT1MA P1L1	AT1MA P1L0
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

■ ATC1 Memory Pointer 1: Middle 8 bits (AT1MAP1M: 0x03E67)

bp	7	6	5	4	3	2	1	0
Flag	AT1MA P1M7	AT1MA P1M6	AT1MA P1M5	AT1MA P1M4	AT1MA P1M3	AT1MA P1M2	AT1MA P1M1	AT1MA P1M0
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

■ ATC1 Memory Pointer 1: Upper 4 bits (AT1MAP1H: 0x03E68)

bp	7	6	5	4	3	2	1	0
Flag	-	-		-	AT1MA P1H3	AT1MA P1H2	AT1MA P1H1	AT1MA P1H0
At reset	-	-	-	-	0	0	0	0
Access	-	-	-	-	R/W	R/W	R/W	R/W

23.3 Operation

23.3.1 Basic Operations and Timing

ATC1 is a DMA block that enables the hardware to transfer the whole memory space (1 MB). This section provides a description of and timing for the basic ATC1 operations.

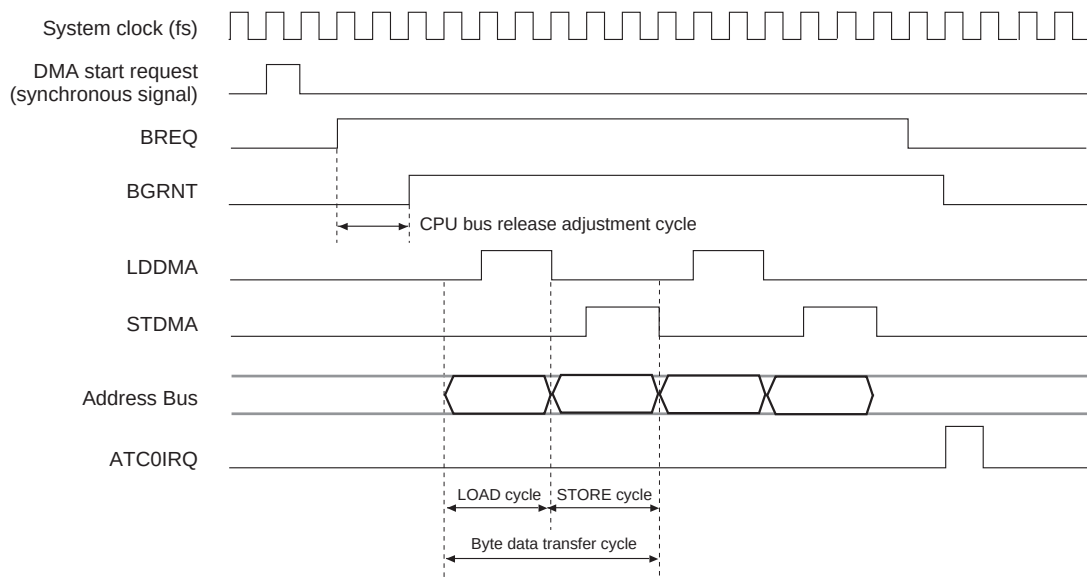


Figure:23.3.1 ATC1 Operation Timing

■ ATC1 Activation and Internal Bus Acquisition

ATC1 activates either when the selected interrupt factor occurs or when the software sets the activation flag. Set the ATC1 trigger factor in ATC1 control register 1 (AT1CNT1). When ATC1 starts, the ATC1 controller asserts the BREQ signal, which requests CPU to release the bus. When the core receives the BREQ signal, it stops all normal executions, even if it is in the middle of executing an instruction, and releases the bus at the next available timing. The core takes a minimum of two cycles from the time it receives the BREQ signal until it actually releases the bus. After it releases the internal bus, the core returns the bus granted signal, BGRNT, to ATC1. ATC1 can then begin using the bus to transfer data.



When an external interrupt is selected as an ATC1 trigger factor, specify the activation valid edge by the REDGn flag of external interrupt control register and the EDGSELn flag of both edges interrupt control register (EDGDT). [Chapter 7 7.3 External Interrupts]



The pulse input which is shorter than the system clock cycle may sometimes be ignored as the external interrupt input is sampled by the system clock.



Set the valid edge for external interrupts before ATC1 activates.



When the software activation is selected as an activation factor of ATC1, maximum four instructions are needed from setting of the software activation flag (AT1ACT) until ATC1 transfer activation.

And maximum three instructions are needed until ATC1 transfer complete interrupt (ATC1IRQ) generation.

■ Data Transfer

The basic ATC1 operation cycle is the "byte-data transfer cycle", in which ATC1 transfers a single byte of data. This operation consists of two instruction cycles, a load and a store cycle. In the load cycle, ATC1 reads the data from the source address of the source memory, and in the store cycle, ATC1 stores the read data to the destination address of the destination memory. ATC1 transfers word-length data or a multi-byte stream of data by repeating the byte-data transfer cycle as many times as necessary.

■ Transfer End

Once it has transferred all the data, ATC1 generates an interrupt (ATC1IRQ) and stop the automatic transfer. In this way, the ATC1 block bypasses the software and automatically transfers data in a continuous DMA operation.



In both the load and store cycles, the read and write access occurs to the memory exactly as it does in a normal instruction execution. This means that the access timing is different depending on the memory space. Also, the wait settings for I/O apply. The following is the access timing for each memory space, assuming no-wait situation.

- Internal ROM/RAM space: 2 cycles
- I/O space (special registers): 3 cycles

LOAD cycle and STORE cycle are set as follows;
An access timing corresponding to each memory space + 1 cycle



In Figure:23.3.1 ATC1 Operation Timing, the time from the rising of DMA activation request signal to the starting of LOAD cycle depends on the state of CPU, but it takes minimum of 6 cycles.

23.3.2 Memory Address Setting

■ Setting of Transfer Addresses to Memory Pointers

The address of the memory space for an automatic data transfer (ATC1) should be set in the both of memory pointer 0 (AT1MAP0) and memory pointer 1 (AT1MAP1). In each transfer mode, one of those pointer is the source address, and another is the destination address.

■ Memory Pointer 0 Functions

Memory pointer 0 is consists of three 8-bit registers, AT1MAP0H, AT1MAP0M and AT1MAP0L. AT1MAP0H holds upper 4 bits of the 20-bit address, AT1MAP0M contains the middle 8 bits, and AT1MAP0L contains lower 8 bits. The 20-bit address set in memory pointer 0 points to a specific address in the total memory space of 1 MB. Memory pointer 0 also contains a computational function that enables it to increment the address based on the transfer state. You can disable this function for all transfer modes by setting the FMODE bit of ATC1 control register 0 (AT1CNT0) to "1".

■ Memory Pointer 1 Functions

Memory pointer 1 is consists of three 8-bit registers, AT1MAP1H, AT1MAP1M and AT1MAP1L. AT1MAP1H holds upper 4 bits of the 20-bit address, AT1MAP1M contains the middle 8 bits, and AT1MAP1L contains lower 8 bits. Depending on the transfer mode, either all 20 bits are valid, or only the least significant 8 bits (in AT1MAP1L) are valid. When only the 8 bits in AT1MAP1L are valid, the value 0x03F is assigned to the 12 bits in AT1MAP1H and AT1MAP1M, and the pointer points to the I/O space (special registers). Memory pointer 1 also contains a computational function that enables it to increment the address based on the transfer state.



Set the memory address while ATC1 transfer is disabled (AT1EN flag of AT1CNT0 register is "0").

23.3.3 Data Transfer Count Setting

■ Transfer Data Counter (AT1TRC) Function

You can preset the data transfer count by ATC1. Set the value in the ATC1 transfer counter (AT1TRC). The counter decrements every time when ATC1 transfers one byte of data.

The value in the transfer data counter is 0x00 at reset. Set the data transfer counts before activating ATC1. Note that ATC1 cannot be activated if the transfer data counter is set to 0x00.

■ Data Transfer Operations Using the Transfer Data Counter (AT1TRC)

There are two main types of ATC1 data transfers, standard and burst transfers. Refer to [Chapter 23 23.3.4 Data Transfer Modes Setting]. The transfer counter operates differently depending on the transfer type.

1. Standard transfers [Transfer modes 0 to D]

In standard transfers, the transfer counter decrements every time when ATC1 is activated. When the counter reaches 0x00 after a data transfer, ATC1 generates an interrupt (ATC1IRQ). This means that for standard transfers, the program must set the counter to the number of times ATC1 needs to be activated.

2. Burst transfers [Transfer modes E to F]

In burst transfers, one activation of ATC1 continuously transfers multiple bytes of data. In this case, the program must set the counter to the number of data bytes contained in the burst transfer. When the burst transfer starts, the transfer counter decrements every time when one byte of data is transferred. When the counter reaches 0x00, ATC1 generates an interrupt (ATC1IRQ). It is also possible to force ATC1 to shut down during the burst transfer using External Interrupt 0 by setting the BTSTP flag of ATC1 control register 1 (AT1CNT1) to "1". Refer to [Chapter 23 23.3.4 Data Transfer Modes Setting].

■ Transfer Data Counter (AT1TRC)

Transfer data counter can be set to a maximum 255 transfers (for standard transfers) or 255 bytes (for burst transfers). Note that setting the counter to 0x00 disables transfers.



Set the data transfer count while ATC1 transfer is disabled (AT1EN flag of AT1CNT0 register is "0").

23.3.4 Data Transfer Modes Setting

■ Data Transfer Modes

There are two types of ATC1 transfers, standard and burst, and sixteen transfer modes. Set the transfer mode in ATC1 control register 0 (AT1CNT0). [Table:23.1.2 Transfer Modes]

■ Standard and Burst Transfers

The ATC1 transfer modes are divided into standard transfer modes and burst transfer modes. There are fourteen standard modes, 0 to D, and two burst modes, E and F.

In standard modes, the operation specified for that mode executes every time when ATC1 is activated. When the transfer ends, the value set in transfer counter (AT1TRC) decrements and bus control returns to CPU. This operation repeats until the transfer counter reaches 0x00. When this happens, ATC1 completes the final data transfer, then generates an interrupt (ATC1IRQ).

For instance, if the initial transfer counter value is 0x05, and the ATC1 activation factor is set to a timer 0 interrupt, ATC1 is activated every time when an interrupt request of timer 0 interrupt is generated and the automatic transfer begins. After fifth data transfers (activated by generation of 5th interrupt request) is completed, the transfer counter value becomes 0x00, an ATC1 interrupt occurs, and the operation ends. Timer 0 overflows occurring after this point do not activate ATC1. For standard transfers, the program must set the transfer counter to the number of ATC1 activations required.

In burst modes, once ATC1 is activated, it transfers in one operation the number of bytes set in the transfer counter (AT1TRC). After the burst transfer begins, the transfer counter decrements every time when ATC1 transfers one byte of data. When the counter reaches 0x00, ATC1 generates an interrupt (ATC1IRQ) and the burst transfer ends. For burst transfers, the program must set the transfer counter to the number of data bytes in the burst transfer.

External Interrupt 0 can also be used to shut down ATC1 during a burst transfer. To enable this function, set the burst transfer stop enable bit (BTSTP) in ATC1 control register 1 (AT1CNT1) to 1. When BTSTP = 1, ATC1 data transfers stop when the interrupt request flag of External Interrupt 0 (IRQ0IR flag in IRQ0ICR register) is set. In an emergency shutdown, the transfer counter and memory pointer save the values prior to the shutdown. When the interrupt service routine ends, a new activation factor restarts ATC1, and the burst transfer begins transferring data from the point at which it stopped.



Do not select External Interrupt 0 for ATC1 trigger factor when the burst transfer stop is enabled.



Set the data transfer mode while ATC1 transfer is disabled (AT1EN flag of AT1CNT0 register is "0").

23.3.5 Transfer Mode 0

In transfer mode 0, ATC1 automatically transfers one byte of data from any memory space to the I/O space (special registers: 0x03F00 to 0x03FFF) every time an ATC1 activation request occurs.

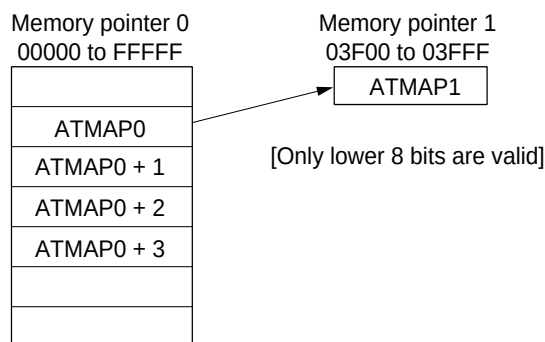


Figure:23.3.2 Transfer Mode 0

Set the source address in 20-bit memory pointer 0 (AT1MAP0H, M, L), and set the destination I/O address in lower 8 bits of memory pointer 1 (AT1MAP1L). You do not have to set the upper 12 bits of the I/O space address (0x03F) in AT1MAP1H and AT1MAP1M.

Transfer mode 0 does not have an increment function for the memory pointers and executes data transfer for a fixed address.

Set the data transfer count for ATC1 in transfer data counter (AT1TRC). Up to 255 transfers can be set. The counter decrements every time an ATC1 is activated. When it reaches 0x00, an interrupt (ATC1IRQ) occurs and the automatic transfer ends.

23.3.6 Transfer Mode 1

In transfer mode 1, ATC1 automatically transfers one byte of data from the I/O space (special registers: 0x03F00 to 0x03FFF) to any memory space every time an ATC1 activation request occurs.

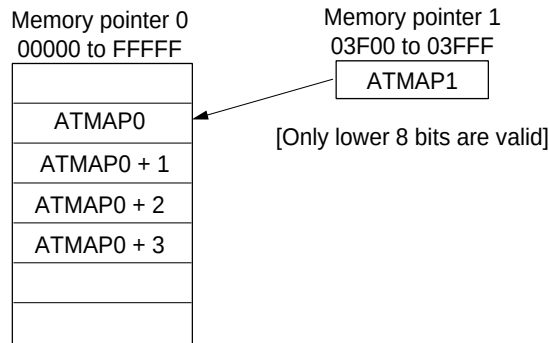


Figure:23.3.3 Transfer Mode 1

Set the source I/O address in lower 8 bits of memory pointer 1 (`AT1MAP1L`), and set the destination address in 20-bit memory pointer 0 (`AT1MAP0H`, `M`, `L`). You do not have to set the upper 12 bits of the I/O space address (0x03F) in `AT1MAP1H` and `AT1MAP1M`.

Transfer mode 1 does not have an increment function for the memory pointers and executes data transfer for a fixed address.

Set the data transfer count for ATC1 in transfer data counter (`AT1TRC`). Up to 255 transfers can be set. The counter decrements every time an ATC1 is activated. When it reaches 0x00, an interrupt (`ATC1IRQ`) occurs and the automatic transfer ends.

23.3.7 Transfer Mode 2

In transfer mode 2, ATC1 automatically transfers one byte of data from any memory space to the I/O space (special registers: 0x03F00 to 0x03FFF) every time an ATC1 activation request occurs.

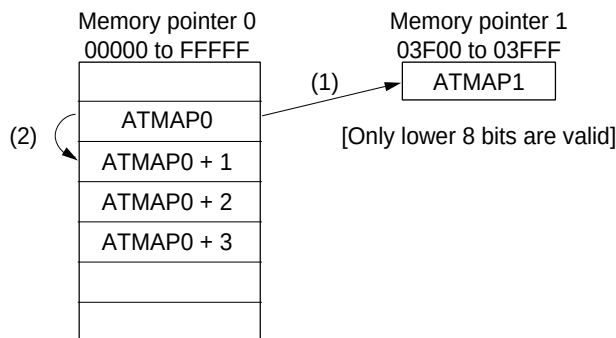


Figure:23.3.4 Transfer Mode 2

Set the source address in 20-bit memory pointer 0 (AT1MAP0H, M, L), and set the destination I/O address in lower 8 bits of memory pointer 1 (AT1MAP1L). You do not have to set the upper 12 bits of the I/O space address (0x03F) in AT1MAP1H and AT1MAP1M.

In transfer mode 2, the value in memory pointer 0 increments every time a byte-length data transfer ends. As a result, the source address for the next transfer is one address higher than that for the previous transfer.

Set the data transfer count for ATC1 in transfer data counter (AT1TRC). Up to 255 transfers can be set. The counter decrements every time an ATC1 is activated. When it reaches 0x00, an interrupt (ATC1IRQ) occurs and the automatic transfer ends.

23.3.8 Transfer Mode 3

In transfer mode 3, ATC1 automatically transfers one byte of data from the I/O space (special registers: 0x03F00 to 0x03FFF) to any memory space every time an ATC1 activation request occurs.

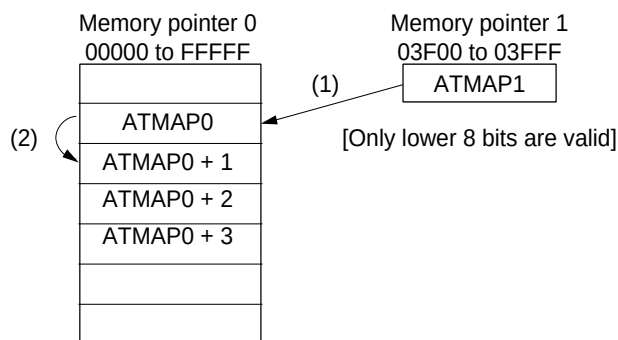


Figure:23.3.5 Transfer Mode 3

Set the source I/O address in lower 8 bits of memory pointer 1 (AT1MAP1L), and set the destination address in 20-bit memory pointer 0 (AT1MAP0H, M, L). You do not have to set the upper 12 bits of the I/O space address (0x3F) in AT1MAP1H and AT1MAP1M.

In transfer mode 3, the value in memory pointer 0 increments every time a byte-length data transfer ends. As a result, the destination address for the next transfer is one address higher than that for the previous transfer.

Set the data transfer count for ATC1 in transfer data counter (AT1TRC). Up to 255 transfers can be set. The counter decrements every time an ATC1 is activated. When it reaches 0x00, an interrupt (ATC1IRQ) occurs and the automatic transfer ends.

23.3.9 Transfer Mode 4

In transfer mode 4, ATC1 automatically transfers two bytes (one word) of data from any memory space to the I/O space (special registers: 0x03F00 to 0x03FFF) every time an ATC1 activation request occurs.

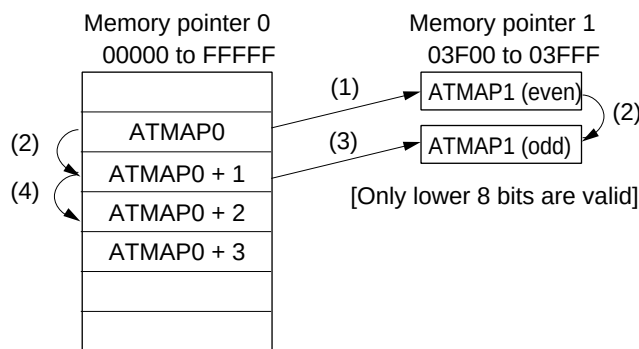


Figure:23.3.6 Transfer Mode 4

Set the source address in 20-bit memory pointer 0 (AT1MAP0H, M, L), and set the destination I/O address in the lower 8 bits of memory pointer 1 (AT1MAP1L). You do not have to set the upper 12 bits of the I/O space address (0x03F) in AT1MAP1H and AT1MAP1M.



Always set an even address as the destination I/O address in memory pointer 1. When ATC1 transfers one word to the I/O space, ATC1 can transfer the even address set in memory pointer 1 and the consecutive odd address.

In transfer mode 4, ATC1 executes a data byte transfer twice to send one data word every time when activated. The value in memory pointer 0 increments every time a byte-length data transfer ends. As a result, the source address for the next ATC1 operation is two addresses higher than that for the previous operation.

In this word-length transfer, ATC1 transfers the first data byte to an even address in the I/O space and the second data byte to an odd address in the I/O space.

Set the data transfer data count for ATC1 in transfer data counter (AT1TRC). Up to 255 transfers can be set. The counter decrements every time ATC1 is activated (after each word transfer). When it reaches "0x00", an interrupt (ATC1IRQ) occurs and the automatic transfer ends.

23.3.10 Transfer Mode 5

In transfer mode 5, ATC1 automatically transfers two bytes (one word) of data from the I/O space (special registers: 0x03F00 to 0x03FFF) to any memory space every time an ATC1 activation request occurs.

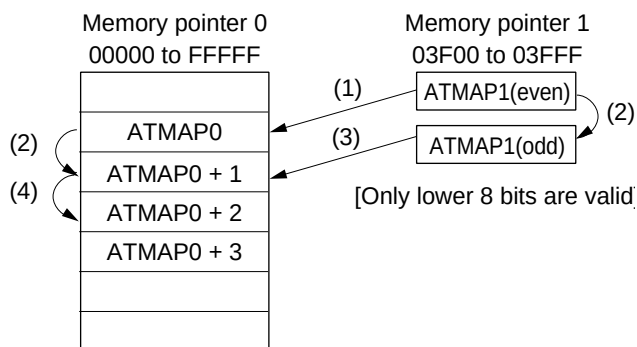


Figure:23.3.7 Transfer Mode 5

Set the source I/O address in lower 8 bits of memory pointer 1 (AT1MAP1L), and set the destination address in 20-bit memory pointer 0 (AT1MAP0H, M, L). You do not have to set the upper 12 bits of the I/O space address (0x03F) in AT1MAP1H and AT1MAP1M.



Always set an even address as the source I/O address in memory pointer 1. When ATC1 transfers one word from the I/O space, ATC1 can transfer the even address set in memory pointer 1 and the consecutive odd address.

In transfer mode 5, ATC1 executes a data byte transfer twice to send one data word every time when activated. The value in memory pointer 0 increments by one each time a byte-length data transfer ends. As a result, the destination address for the next ATC1 operation is two addresses higher than that for the previous operation.

In this word-length transfer, ATC1 transfers the first data byte from an even address in the I/O space and the second data byte from an odd address in the I/O space.

Set the data transfer count for ATC1 in transfer data counter (AT1TRC). Up to 255 transfers can be set. The counter decrements every time an ATC1 is activated (after each word transfer). When it reaches "0x00", an interrupt (ATC1IRQ) occurs and the automatic transfer ends.

23.3.11 Transfer Mode 6

In transfer mode 6, ATC1 automatically transfers one byte of data two times every time an ATC1 activation request occurs.

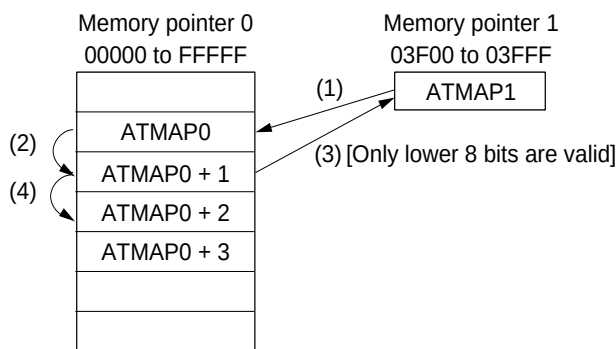


Figure:23.3.8 Transfer Mode 6

In this mode the transfer direction indicated by memory pointers 0 and 1 reverses for the second data byte transfer.

In the first data byte transfer, the I/O space address (0x03F00 to 0x03FFF) in memory pointer 1 is the source address, and the address in memory pointer 0, for any memory space, is the destination address. When the first data byte transfer ends, the address in memory pointer 0 increments by one.

In the second data byte transfer, the incremented address in memory pointer 0 becomes the source address, and the I/O space address (0x03F00 to 0x03FFF) in memory pointer 1 becomes the destination address. The address in memory pointer 0 increments again after the second data byte transfer ends.

Set the I/O address in lower 8 bits of memory pointer 1 (AT1MAP1L). You do not have to set the upper 12 bits of the I/O space address (0x03F) in AT1MAP1H, AT1MAP1M.

In transfer mode 6, ATC1 executes a data byte transfer twice every time when activated. The value in memory pointer 0 increments by one every time a byte-length data transfer ends. As a result, the source address for the next ATC1 operation is two addresses higher than that for the previous operation.

Set the data transfer count for ATC1 in transfer data counter (AT1TRC). Up to 255 transfers can be set. The counter decrements every time an ATC1 is activated (after one byte of data is transferred twice). When it reaches "0x00", an interrupt (ATC1IRQ) occurs and the automatic transfer ends.

23.3.12 Transfer Mode 7

In transfer mode 7, ATC1 automatically transfers one byte of data two times every time an ATC1 activation request occurs.

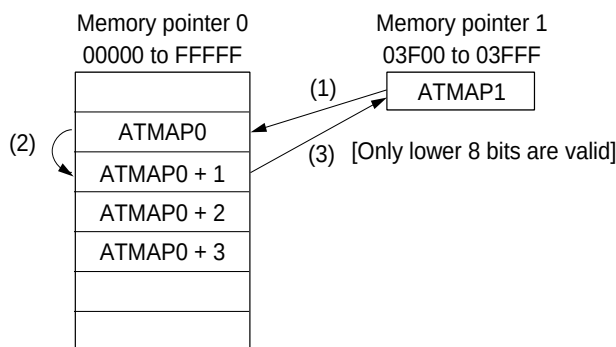


Figure:23.3.9 Transfer Mode 7

In this mode the transfer direction indicated by memory pointers 0 and 1 reverses for the second data byte transfer.

In the first data byte transfer, the I/O space address (0x03F00 to 0x03FFF) in memory pointer 1 is the source address, and the address in memory pointer 0, for any memory space, is the destination address. When the first data byte transfer ends, the address in memory pointer 0 increments by one.

In the second data byte transfer, the incremented address in memory pointer 0 becomes the source address, and the I/O space address (0x03F00 to 0x03FFF) in memory pointer 1 becomes the destination address. Unlike in the case of transfer mode 6, the address in memory pointer 0 remains unchanged after the second data byte transfer ends.

Set the I/O address in lower 8 bits of memory pointer 1 (AT1MAP1L). You do not have to set the upper 12 bits of the I/O space address (0x03F) in AT1MAP1H, AT1MAP1M.

In transfer mode 7, ATC1 executes a data byte transfer twice every time when activated. However, the value in memory pointer 0 increments by one only after the first transfer ends. As a result, the source address for the next ATC1 operation is one address higher than that for the previous operation.

Set the data transfer count for ATC1 in transfer data counter (AT1TRC). Up to 255 transfers can be set. The counter decrements every time an ATC1 is activated (after one byte of data has been transferred twice). When it reaches "0x00", an interrupt (ATC1IRQ) occurs and the automatic transfer ends.

23.3.13 Transfer Mode 8

In transfer mode 8, ATC1 automatically transfers one byte of data two times every time an ATC1 activation request occurs.

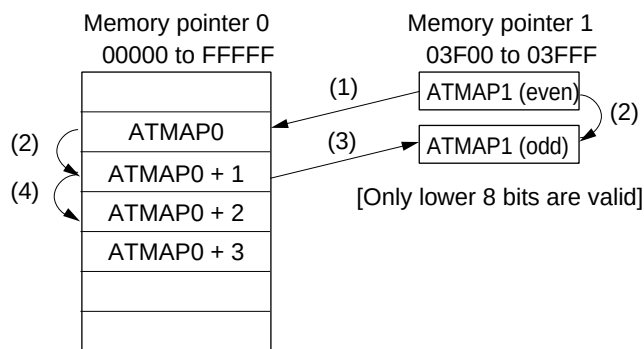


Figure:23.3.10 Transfer Mode 8

In this mode the transfer direction indicated by memory pointers 0 and 1 reverses for the second data byte transfer.

In the first data byte transfer, the I/O space address (0x03F00 to 0x03FFF) in memory pointer 1 is the source address, and the address in memory pointer 0, for any memory space, is the destination address. When the first data byte transfer ends, the address in memory pointer 0 increments by one.

In the second data byte transfer, the incremented address in memory pointer 0 becomes the source address, and the I/O space address (0x03F00 to 0x03FFF) in memory pointer 1 becomes the destination address. When the second data byte transfer ends, the address in memory pointer 0 increments again.

Set an even I/O address in lower 8 bits of memory pointer 1 (AT1MAP1L). You do not have to set the upper 12 bits of the I/O space address (0x03F) in AT1MAP1H, AT1MAP1M.



Always set an even I/O address in memory pointer 1. In this double transfer of a data byte from and to the I/O space, ATC1 targets the even I/O address set in memory pointer 1 and the consecutive odd address. In this mode, the first data byte transfer accesses an even I/O address and the second data byte transfer accesses an odd I/O address.



Transfer mode 8 can be used to support continuous transmission/ reception for Serial Interface 0, 1, 2, 3 and 4. Set the memory pointer 1 to point to serial reception buffer (RXBUF0, RXBUF1) and select serial interrupts as the ATC1 trigger factor. In this way, every time the serial communication ends, the MCU continuously reads the reception data (first data byte transfer), then writes the transmission data to transmission buffer (TXBUF0, TXBUF1) (second data byte transfer) up to 255 times, entirely through the hardware.



Before execute a continuous serial transaction, store the serial transmission data in the memory space that memory pointer 0 points, the transmission data must fill every other address in the space. Once the serial transaction ends, the received data is stored in empty (skipped) addresses and the transmission and reception data at stored in an alternative pattern.

In transfer mode 8, ATC1 executes a data byte transfer twice each time it is activated. The value in memory pointer 0 increments by one every time a byte-length data transfer ends. As a result, the source address for the next ATC1 operation is two addresses higher than that for the previous operation.

Set the data transfer count for ATC1 in transfer data counter (AT1TRC). Up to 255 transfers can be set. The counter decrements every time an ATC1 is activated (after one byte of data is transferred twice). When it reaches "0x00", an interrupt (ATC1IRQ) occurs and the automatic transfer ends.

23.3.14 Transfer Mode 9

In transfer mode 9, ATC1 automatically transfers one byte of data two times every time an ATC1 activation request occurs.

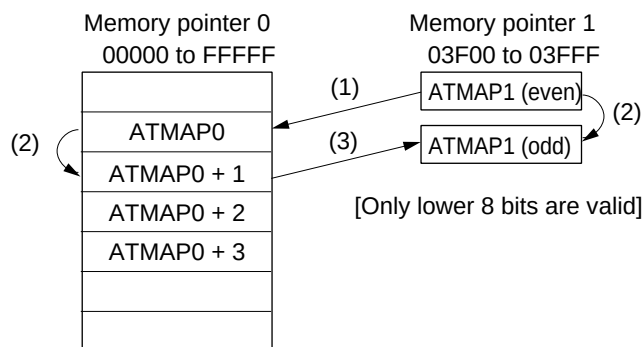


Figure:23.3.11 Transfer Mode 9

In this mode the transfer direction indicated by memory pointers 0 and 1 reverses for the second data byte transfer.

In the first data byte transfer, the I/O space address (0x03F00 to 0x03FFF) in memory pointer 1 is the source address, and the address in memory pointer 0 for any memory space is the destination address. When the first data byte transfer ends, the address in memory pointer 0 increments by one.

In the second data byte transfer, the incremented address in memory pointer 0 becomes the source address, and the I/O space address (0x03F00 to 0x03FFF) in memory pointer 1 becomes the destination address. The address in memory pointer 0 remains unchanged after the second data byte transfer ends.

Set an even I/O address in lower 8 bits of memory pointer 1 (AT1MAP1L). You do not have to set the upper 12 bits of the I/O space address (0x03F) in AT1MAP1H, AT1MAP1M.



Always set an even I/O address in memory pointer 1. In this double transfer of a data byte from and to the I/O space, ATC1 targets the even I/O address set in memory pointer 1 and the consecutive odd address. In this mode, the first data byte transfer accesses an even I/O address and the second data byte transfer accesses an odd I/O address.



Transfer mode 9 can be used to support continuous transmission/ reception for Serial Interface 0, 1 and 2. Set the memory pointer 1 to point to serial reception buffer (RXBUF0, RXBUF1) and select serial interrupts as the ATC1 trigger factor. In this way, every time a serial communication ends, the MCU continuously reads the reception data (first data byte transfer), then writes the transmission data to transmission buffer (TXBUF0, TXBUF1) (second data byte transfer) up to 255 times, entirely through the hardware.



Before execute a continuous serial transaction, store the serial transmission data in the memory space that memory pointer 0 points, once the serial communication ends, the MCU has written to the reception data over the transmission data, so that only reception data remains in the memory.

In transfer mode 9, ATC1 executes a data byte transfer twice every time when activated. However, the value in memory pointer 0 increments by one only after the first transfer ends. As a result, the source address for the next ATC1 operation is one address higher than that for the previous operation.

Set the data transfer count for ATC1 in transfer data counter (AT1TRC). Up to 255 transfers can be set. The counter decrements every time an ATC1 is activated (after one byte of data is transferred twice). When it reaches 0x00, an interrupt (ATC1IRQ) occurs and the automatic transfer ends.

23.3.15 Transfer mode A

In transfer mode A, ATC1 automatically transfers one byte of data from any memory space to any other memory space every time an ATC1 activation request occurs.

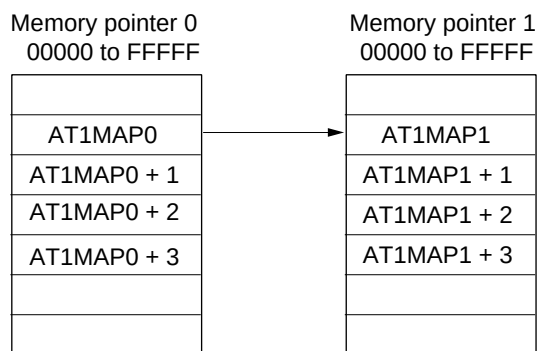


Figure:23.3.12 Transfer Mode A

Set the source address in 20-bit memory pointer 0 (AT1MAP0H, M, L), and set the destination address in 20-bit memory pointer 1 (AT1MAP0H, M, L).

Transfer mode A does not have an increment function for the memory pointers and executes data transfer for a fixed address.

Set the data transfer count for ATC1 in transfer data counter (AT1TRC). Up to 255 transfers can be set. The counter decrements every time an ATC1 is activated. When it reaches "0x00", an interrupt (ATC1IRQ) occurs and the automatic transfer ends.

23.3.16 Transfer Mode B

In transfer mode B, ATC1 automatically transfers one byte of data from any memory space to any other memory space every time an ATC1 activation request occurs.

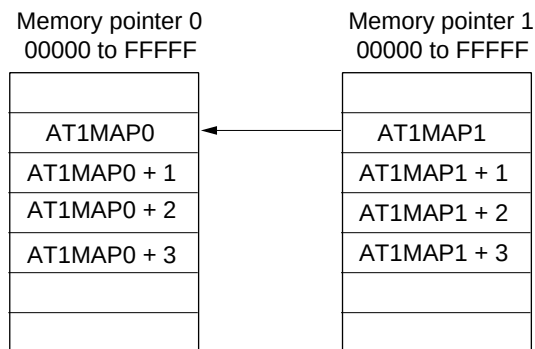


Figure:23.3.13 Transfer Mode B

Set the source address in 20-bit memory pointer 1 (AT1MAP1H, M, L), and set the destination address in 20-bit memory pointer 0 (AT1MAP0H, M, L).

Transfer mode B does not have an increment function for the memory pointers and executes data transfer for a fixed address.

Set the data transfer count for ATC1 in transfer data counter (AT1TRC). Up to 255 transfers can be set. The counter decrements every time an ATC1 is activated. When it reaches "0x00", an interrupt (ATC1IRQ) occurs and the automatic transfer ends.

23.3.17 Transfer Mode C

In transfer mode C, ATC1 automatically transfers one byte of data from any memory space to any other memory space every time an ATC1 activation request occurs.

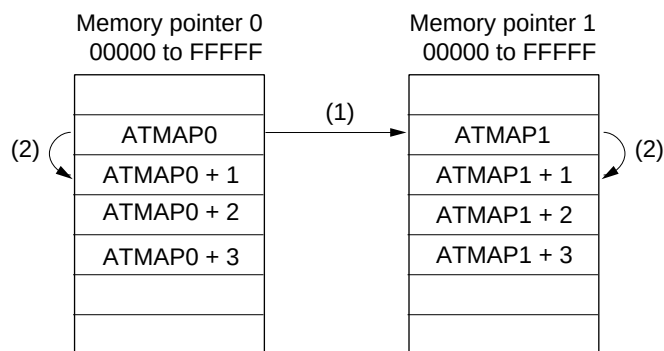


Figure:23.3.14 Transfer Mode C

Set the source address in 20-bit memory pointer 0 (AT1MAP0H, M, L), and set the destination address in 20-bit memory pointer 1 (AT1MAP1H, M, L).

In transfer mode C, the values in memory pointers 0 and 1 increment every time a byte-length data transfer ends. As a result, the source and destination addresses for the next transfer are one address higher than those for the original transfer.

Set the data transfer count for ATC1 in transfer data counter (AT1TRC). Up to 255 transfers can be set. The counter decrements every time an ATC1 is activated. When it reaches "0x00", an interrupt (ATC1IRQ) occurs and the automatic transfer ends.

23.3.18 Transfer Mode D

In transfer mode D, ATC1 automatically transfers one byte of data from any memory space to any other memory space every time an ATC1 activation request occurs.

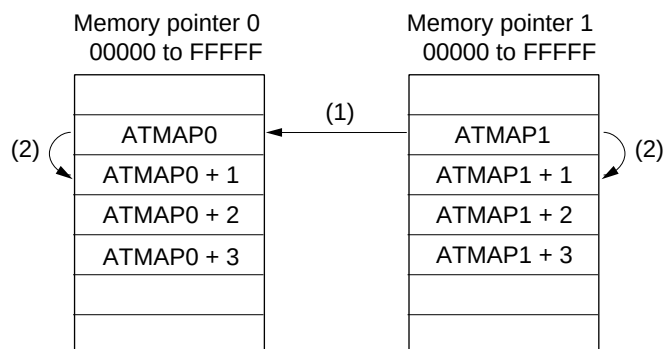


Figure:23.3.15 Transfer Mode D

Set the source address in 20-bit memory pointer 1 (AT1MAP1H, M, L), and set the destination address in 20-bit memory pointer 0 (AT1MAP0H, M, L).

In transfer mode D, the values in memory pointers 0 and 1 increment every time a byte-length data transfer ends. As a result, the source and destination addresses for the next transfer are one address higher than those for the original transfer.

Set the data transfer count for ATC1 in transfer data counter (AT1TRC). Up to 255 transfers can be set. The counter decrements every time an ATC1 is activated. When it reaches "0x00", an interrupt (ATC1IRQ) occurs and the automatic transfer ends.

23.3.19 Transfer Mode E

Transfer mode E is a burst mode. In this mode, when ATC1 is activated, it automatically transfers the number of data bytes set in the transfer data counter (AT1TRC) in one continuous operation.

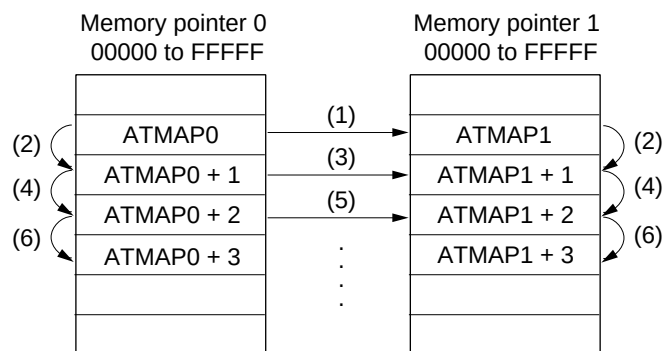


Figure:23.3.16 Transfer Mode E

Set the source address in 20-bit memory pointer 0 (AT1MAP0H, M, L), and set the destination address in 20-bit memory pointer 1 (AT1MAP1H, M, L). Once ATC1 is activated, memory pointers 0 and 1 increment every time a byte-length data transfer ends.

For burst transfers, set the number of data bytes to be transferred in the transfer data counter (AT1TRC). Up to 255 transfers can be set. Once the burst transfer starts, the counter decrements every time an ATC1 transfers one byte of data. When it reaches 0x00, an interrupt (ATC1IRQ) occurs and the burst transfer ends.

You can shut down ATC1 during burst transfers using External Interrupt 0. You can enable or disable ATC1 shutdown with the burst transfer stop enable flag (BSTP) of ATC1 control register 1 (AT1CNT1).

When BTSTP=1 and the interrupt request flag for External Interrupt 0 (the IRQ0IR flag in IRQ0ICR register) is set, the ATC1 data transfer shuts down immediately at 1 byte transfer complete timing. During this shutdown, the transfer counter and the memory pointers save the values they contained prior to the shutdown. When the interrupt service routine ends and a new ATC1 trigger factor occurs, the burst transfer restarts from the point at which it stopped.



Do not select External Interrupt 0 for ATC1 trigger factor when the burst transfer stop is enabled.

23.3.20 Transfer Mode F

Transfer mode F is a burst mode. In this mode, when ATC1 is activated, it automatically transfers the number of data bytes set in the transfer data counter (AT1TRC) in one continuous operation.

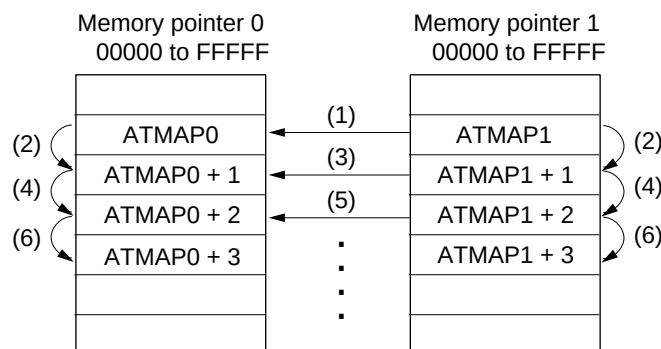


Figure:23.3.17 Transfer Mode F

Set the source address in 20-bit memory pointer 1 (AT1MAP1H, M, L), and set the destination address in 20-bit memory pointer 0 (AT1MAP0H, M, L). Once ATC1 is activated, memory pointers 0 and 1 increment every time a byte-length data transfer ends.

For burst transfers, set the number of data bytes to be transferred in the transfer data counter (AT1TRC). Up to 255 transfers can be set. Once the burst transfer starts, the counter decrements every time ATC1 transfers one byte of data. When it reaches 0x00, an interrupt (ATC1IRQ) occurs and the burst transfer ends.

You can shut down ATC1 during burst transfers using External Interrupt 0. You can enable or disable ATC1 shutdown with the burst transfer stop enable flag (BSTP) of ATC1 control register 1 (AT1CNT1).

When BTSTP=1 and the interrupt request flag for External Interrupt 0 (the IRQ0IR flag in IRQ0ICR register) is set, the ATC1 data transfer shuts down immediately at 1 byte transfer complete timing. During this shutdown, the transfer counter and the memory pointers save the values they contained prior to the shutdown. When the interrupt service routine ends and a new ATC1 trigger factor occurs, the burst transfer restarts from the point at which it stopped.



Do not select External Interrupt 0 for ATC1 trigger factor when the burst transfer stop is enabled.

23.4 Setup Example

An example setup procedure, with a description of each step is as follows ;

Setup Procedure	Description
(1) Disable the data automatic transfer AT1CNT0 (0x03E60) bp0: AT1EN = 0 (2) Set the data transfer mode AT1CNT0 (0x03E60) bp7: FMODE bp6: AT1ACT = 0 bp5 to 2: AT1MD3 to 0 bp0: AT1EN = 0 (3) Set memory pointer 0 AT1MAP0L (0x03E63) AT1MAP0M (0x03E64) AT1MAP0H (0x03E65) (4) Set memory pointer 1 AT1MAP1L (0x03E66) AT1MAP1M (0x03E67) AT1MAP1H (0x03E68) (5) Set the transfer data counter AT1TRC (0x03E62) (6) Select the ATC1 activation factor AT1CNT1 (0x03E61) bp4: BTSTP bp3 to 0: AT1IR3 to 0 (7) Enable ATC operation AT1CNT0 (0x03E60) bp0: AT1EN = 1	(1) Set the AT1EN flag of AT1CNT0 register to "0" to disable ATC1 data automatic transfer. (2) Select the data transfer mode with the AT1MD flag in AT1CNT0 register. No matter which mode you select, setting the FMODE flag disables the increment function in memory pointer 0. Normally set this flag to 0. Note that you must set the ATC1 enable flag, AT1EN, to 0 at this step. Only enable ATC1 after setting all the other registers. (3) Set the source or destination address in AT1MAP0 registers depending on the transfer mode you select. (4) Set the source or destination address in AT1MAP1 registers depending on the transfer mode you select. (5) Set the ATC1 data transfer count in AT1TRC register. (6) Select the ATC1 activation factor with the AT1IR flag in AT1CNT1 register. If you select a burst-type transfer mode, then you must also enable or disable ATC1 shutdown at this step, by setting the BTSTP. (7) Enable ATC1 data transfers with the AT1EN flag in AT1CNT0 register.



To activate ATC1 in the software, first complete steps (1) to (6), then set the AT1ACT flag in AT1CNT0 register. After the AT1ACT flag is set, ATC1 is started and data transfer is started. The hardware automatically clears AT1ACT flag when ATC1 is activated. In standard transfer mode, set a program that sets flags as many as necessary for the data transfer.



Set the data automatic transfer while ATC1 transfer is disabled (AT1EN flag of AT1CNT0 register is "0").

24.1 Overview

MN101EF76K and MN101EF56K include a 256 KB flash memory as an internal instruction memory.
MN101EF57G includes a 128 KB flash memory as an internal instruction memory.

The flash memory is divided to the areas called "sector" as indicated in Figure:24.1.1. This memory is writable in bytes and erasable in sectors.

For rewriting of the flash memory, it can be rewritten not only by a programmer but also by user program.

To rewrite the flash memory by user program, perform a subroutine call of command library stored in the dedicated area from user program in advance.

	MN101EF76K MN101EF56K		MN101EF57G
0x04000	Sector 0 32 Kbyte		Sector 0 32 Kbyte
0x0C000	Sector 1 32 Kbyte		Sector 1 32 Kbyte
0x14000	Sector 2 32 Kbyte		Sector 2 32 Kbyte
0x1C000	Sector 3 20 Kbyte		Sector 3 20 Kbyte
0x21000	Sector 4 4 Kbyte		Sector 4 4 Kbyte
0x22000	Sector 5 4 Kbyte		Sector 5 4 Kbyte
0x23000	Sector 6 4 Kbyte	0x23FFF	Sector 6 4 Kbyte
0x24000	Sector 7 32 Kbyte		
0x2C000	Sector 8 32 Kbyte		
0x34000	Sector 9 32 Kbyte		
0x3C000	Sector 10 20 Kbyte		
0x41000	Sector 11 4 Kbyte		
0x42000	Sector 12 4 Kbyte		
0x43000	Sector 13 4 Kbyte		
0x43FFF			

Figure:24.1.1 Sector Structure

24.1.1 Outline Specification

Table:24.1.1 shows the outline specification of the flash memory.

Table:24.1.1 Outline Specification of Flash Memory

Function	Description
Capacity	MN101EF76K: 256 KB MN101EF57G: 128 KB MN101EF56K: 256 KB
Number of times of rewriting in each sector	Max. 1000 times
Voltage for rewriting	VDD5: 2.7 to 5.5 V
Voltage for reading	VDD5: 1.8 to 5.5 V (maximum 8 MHz) 2.7 to 5.5 V (maximum 20 MHz)
Data retention period	10 years
Flash memory rewriting method	- Parallel programmer rewriting - Serial programmer rewriting - User mode microcontroller rewriting - Boot mode microcontroller rewriting
Erasable unit	Sector
Programmable unit	Byte (Up to 64 byte of continuous writing is possible)
Data protection function	- Protect function - Security function



The number of times of rewriting is counted by the number of time a sector is erased. It is controlled on sector basis.

For example, if writing 1 byte of data in any sector for hundred of times and then erasing the sector, a single rewriting is counted. Also, the number of times of rewriting in another sector, in which erasing is not performed, is not counted.

Overwriting data is disabled. To rewrite data, write the data after erasing sectors.

24.1.2 Flash Memory Rewriting Method

This LSI uses two rewriting methods for writing and erasing in flash memory: rewriting by a programmer (Programmer rewriting method) and rewriting by user program (Microcontroller rewriting method).

For a programmer rewriting method, parallel programmer rewriting method using parallel programmer and serial programmer writing method using serial programmer are provided.

For a microcontroller rewriting method, user mode microcontroller rewriting method is provided. In this method, the LSI is activated from user program and rewriting is performed during user program execution.

■ Parallel Programmer Rewriting Method

This method is generally used as a rewriting method of flash memory. Parallel programmer and a dedicated writer adapter are used to rewrite.

In this method, a single LSI before implemented on board can be rewritten.

By gang programmer, more than one LSI can be rewritten simultaneously. It is suitable for mass production of the LSI developed completely.

■ Serial Programmer Rewriting Method

This method is performed by using serial programmer.

In this method, the LSI implemented on board can be rewritten, and it is suitable for the usage of program development and parameter rewriting of individual equipment. Also, rewriting the LSI implemented on board allows more time to develop software than using parallel programmer rewriting method.

■ User Mode Microcontroller Rewriting Method

This method is performed by using serial programmer.

In this method, rewriting to any interface can be performed during user program execution after the LSI is implemented on board. It is suitable for program update during program execution and alternative EEPROM to retain data.

■ BOOT Mode Microcontroller Rewriting Method

This method is used to rewrite user program, and activated by the rewrite-only program written in the flash memory in advance.

In this method, rewriting to any interface can be performed when the LSI is implemented on board.

As the rewriting is performed without the need of a programmer, thus it is still possible to rewrite the flash memory even if the user program is destroyed.

Table:24.1.2 shows the outline of flash memory rewriting method.

Table:24.1.2 Outline of Flash Memory Rewriting Method

Rewriting method	Parallel programmer rewriting method	Serial programmer rewriting method	User mode microcontroller rewriting method	Boot mode microcontroller rewriting method
Description	Rewriting by parallel programmer and rewriting adapter.	Rewriting by serial programmer	Rewriting after activation from user program execution.	Rewriting after activation from rewriting program execution.
Rewritable area	All sectors	All sectors	All sectors	All sectors
Required equipment	- Parallel programmer - rewriting adapter	- Serial programmer	-	-
Features	- Rewritable before board implementation - Rewriting program is not required. - Multiple LSIs are rewritable simultaneously (When using gang programmer). - Prior writing is not required.	- Rewritable after board implementation - Rewriting program such as serial communication control is not required. - Prior writing is not required.	- Rewritable after board implementation - Programmer is not required. - Rewritable in state of user program execution. - Rewriting program is required.	- Rewritable after board implementation - Programmer is not required. - The last sector is used as activating area. - Rewriting program is required.

Table:24.1.3 and Table:24.1.4 show the memory map of the internal flash memory.

When the BOOT area is used, the memory map changes depending on the setting of rewriting enable register (FBEWER).

Table:24.1.3 Memory Map of Internal Flash Memory (MN101EF76K/MN101EF56K: 256 KB)

Sector	Sector size	When BOOT area is not in used	When BOOT area is in used	
		MAIN area	MAIN area (FBEWER = 0x4B)	BOOT area (FBEWER ≠ 0x4B)
Sector 0	32 KB	0x04000 to 0x0BFFF	0x04000 to 0x0BFFF	(Accessing Prohibited)
Sector 1	32 KB	0x0C000 to 0x13FFF	0x0C000 to 0x13FFF	
Sector 2	32 KB	0x14000 to 0x1BFFF	0x14000 to 0x1BFFF	
Sector 3	20 KB	0x1C000 to 0x20FFF	0x1C000 to 0x20FFF	
Sector 4	4 KB	0x21000 to 0x21FFF	0x21000 to 0x21FFF	
Sector 5	4 KB	0x22000 to 0x22FFF	0x22000 to 0x22FFF	
Sector 6	4 KB	0x23000 to 0x23FFF	0x23000 to 0x23FFF	
Sector 7	32 KB	0x24000 to 0x2BFFF	0x24000 to 0x2BFFF	
Sector 8	32 KB	0x2C000 to 0x33FFF	0x2C000 to 0x33FFF	
Sector 9	32 KB	0x34000 to 0x3BFFF	0x34000 to 0x3BFFF	
Sector 10	20 KB	0x3C000 to 0x40FFF	0x3C000 to 0x40FFF	
Sector 11	4 KB	0x41000 to 0x41FFF	0x41000 to 0x41FFF	
Sector 12	4 KB	0x42000 to 0x42FFF	0x42000 to 0x42FFF	
Sector 13 (also used as BOOT area)	4 KB	0x43000 to 0x43FFF	0x43000 to 0x43FFF	0x04000 to 0x04FFF

Table:24.1.4 Memory Map of Internal Flash Memory (MN101EF57G: 32 KB)

Sector	Sector size	When BOOT area is not in used	When BOOT area is in used	
		MAIN area	MAIN area (FBEWER = 0x4B)	BOOT area (FBEWER ≠ 0x4B)
Sector 0	32KB	0x04000 to 0x0BFFF	0x04000 to 0x0BFFF	(Accessing Prohibited)
Sector 1	32KB	0x0C000 to 0x13FFF	0x0C000 to 0x13FFF	
Sector 2	32KB	0x14000 to 0x1BFFF	0x14000 to 0x1BFFF	
Sector 3	20KB	0x1C000 to 0x20FFF	0x1C000 to 0x20FFF	
Sector 4	4KB	0x21000 to 0x21FFF	0x21000 to 0x21FFF	
Sector 5	4KB	0x22000 to 0x22FFF	0x22000 to 0x22FFF	
Sector 6 (also used as BOOT area)	4KB	0x23000 to 0x23FFF	0x23000 to 0x23FFF	0x04000 to 0x04FFF

24.1.3 Flash Memory Area

Flash memory is divided into two areas: MAIN area reserved area.

MAIN area stores user program, and it is rewritable.

BOOT area activates BOOT mode microcontroller rewriting method, and it is rewritable.

Reserved area stores the command library which is used for flash memory rewriting, and it is not rewritable. The capacity of the reserved area is not included in the flash memory capacity.

24.1.4 Protection Function

Protect function can disable writing/erasing to prevent writing/erasing errors for the flash memory in which data has been written. Protection can be set in sectors. The protection cannot be released once it is set.



Note that once protection is set, writing and erasing cannot be performed.

24.1.5 Security Function

Security function prevents alteration or leakage of the program using programmer or debugger.

Writing, reading and erasing for all the area of the flash memory can be disabled by security function. To set or release the security, a key code (128 bit) is required. The key code cannot be changed once it is set.



When enabling the security function, collectively erasing cannot be executed. Note that writing or erasing cannot be performed if the key code is lost.

24.2 Parallel Programmer Rewriting

Parallel programmer rewriting is used to rewrite the flash memory with a rewriting adopter and parallel programmer corresponding to this LSI. Rewriting of the flash memory is possible for the LSI before implemented on board.



To rewrite entire area, batch rewriting of MAIN area and BOOT area can be performed. The following description shows how to create a data file using debugger or PX-FW2 programmer for the batch rewriting of MAIN area and BOOT area. For using the other programmer, it is recommended to consult the programmer manufacturer.

- (a) Using PanaXSeries debugger DebugFactory Builder (DebugFactory Builder)
1. Prepare data files for MAIN area (from 0x04000) and BOOT area (from 0x04000) in executable format (EX format file).
 2. Load the data file for BOOT area using L command of DebugFactory Builder.
>l boot.ex
 3. Transfer the data for BOOT area from 0x04000 to 0x43000[0x23000] using M command of DebugFactory Builder.
>m 4000, 4fff, 43000[23000]
 4. Load the data file for MAIN area using L command of DebugFactory Builder.
>l main.ex
 5. Write the contents of the memory using WR command of DebugFactory Builder.
>wr main-boot. s, 4000, 43fff[23fff]

* [] shows for MN101EF75G

- (b) Using PX-FW2 programmer
1. Prepare data files for MAIN area (from 0x04000) and BOOT area (from 0x04000) in programmable format.
In addition, prepare this LSI for writing data actually.
 2. Write the data file for BOOT area in BOOT area.
 3. Write the data file for MAIN area in MAIN area.
 4. Read the data of the entire area, and write it into the file.

For the memory map at parallel programmer rewriting, refer to Figure:24.2.1.

	MN101EF76K/MN101EF56K	MN101EF57G
Rewriting area	MAIN area/BOOT area *1 256 KB of data from the address of 0x04000 is written in all sectors	MAIN area/BOOT area *1 128 KB of data from the address of 0x04000 is written in all sectors
Memory map	Address	Address
	0x04000	0x04000
	Sector 0 32 Kbyte	Sector 0 32 Kbyte
	0x0BFFF 0x0C000	0x0BFFF 0x0C000
	Sector 1 32 Kbyte	Sector 1 32 Kbyte
	0x13FFF 0x14000	0x13FFF 0x14000
	Sector 2 32 Kbyte	Sector 2 32 Kbyte
	0x1BFFF 0x1C000	0x1BFFF 0x1C000
	Sector 3 20 Kbyte	Sector 3 20 Kbyte
	0x20FFF 0x21000	0x20FFF 0x21000
	Sector 4 4 Kbyte	Sector 4 4 Kbyte
	0x21FFF 0x22000	0x21FFF 0x22000
	Sector 5 4 Kbyte	Sector 5 4 Kbyte
	0x22FFF 0x23000	0x22FFF 0x23000
	Sector 6 4 Kbyte	Sector 6 4 Kbyte
	0x23FFF 0x24000	0x23FFF

*1 When using BOOT Mode Microcontroller Rewriting Method

Figure:24.2.1 Parallel Programmer Rewriting Memory Map

24.2.1 Equipment Configuration in Parallel Programmer Rewriting

Figure:24.2.2 shows the equipment configuration in parallel programmer rewriting. Perform rewriting by attaching a rewriting adapter dedicated for this LSI to the parallel programmer corresponding to this LSI. For the parallel programmer, the following two programmers are provided: programmer which can rewrite without host computer, and programmer for gang programming to rewrite multiple LSIs collectively. For the procedure of rewriting, refer to the parallel programmer instruction manual.

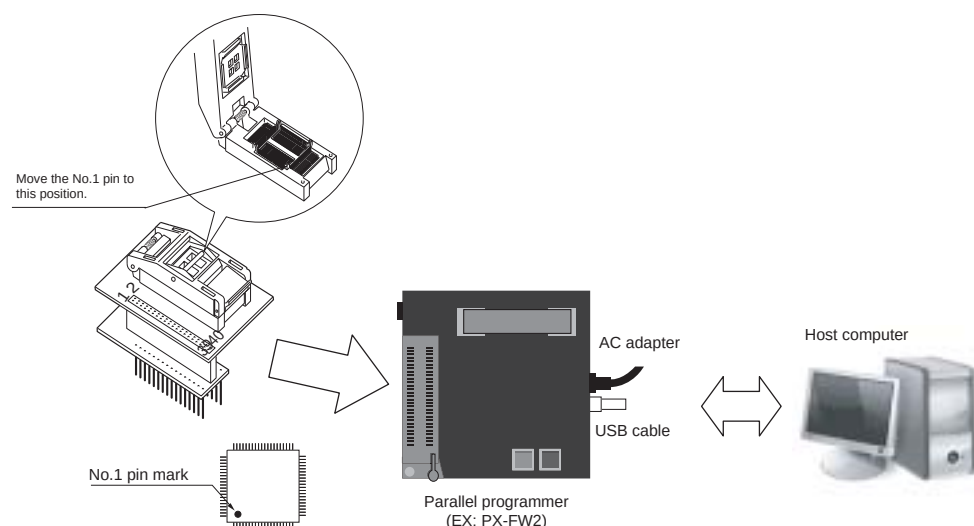


Figure:24.2.2 Equipment Configuration of Parallel Programmer Rewriting

Table:24.2.1 shows the corresponding table between rewriting adapter and parallel programmer. Our website [Semiconductor Technical Support System] shows the most up-to-date information on the rewriting adapter and parallel programmer. For our website URL, refer to the end of this manual.

Table:24.2.1 Corresponding Table between rewriting adapter and Parallel Programmer

Product	Package	Rewriting adapter	Parallel programmer	
			Manufacturer	Model
MN101EF76K	LQFP128-P-1818C	TRF009-101EF76K128	Panasonic Corporation	PX-FW2
MN101EF57G	LQFP080-P-1414A	TRF009-101EF57G/D80	Panasonic Corporation	PX-FW2
			Flash Support Group, Inc.	AF9723

24.3 Serial Programmer Rewriting

Serial programmer rewriting is used to rewrite the flash memory with the serial programmer corresponding to this LSI. The flash memory can be rewritten during implementation of the LSI on the circuit board. It is not required to write serial communication control program in the flash memory in advance.



To rewrite entire area, batch rewriting of MAIN area and BOOT area can be performed. The following description shows how to create a data file using debugger or programmer for the batch rewriting of MAIN area and BOOT area.

(a) Using DebugFactory Builder

1. Prepare data files for MAIN area (from 0x04000) and BOOT area (from 0x04000) in executable format (EX format file).
2. Load the data file for BOOT area using L command of DebugFactory Builder.
>l boot.ex
3. Transfer the data for BOOT area from 0x04000 to 0x43000[0x23000] using M command of DebugFactory Builder.
>m 4000, 4fff, 43000[23000]
4. Load the data file for MAIN area using L command of DebugFactory Builder.
>l main.ex
5. Write the contents of the memory using WR command of DebugFactory Builder.
>wr main-boot. s, 4000, 43fff[23fff]

* [] shows for MN101EF75G

(b) Using PX-FW2 programmer

1. Prepare data files for MAIN area (from 0x04000) and BOOT area (from 0x04000) in programmable format.
In addition, prepare this LSI for actual data writing.
2. Write the data file for BOOT area in BOOT area.
3. Write the data file for MAIN area in MAIN area.
4. Read the data of the entire area, and write it into the file.

For the memory map at serial programmer rewriting, refer to Figure:24.3.1.

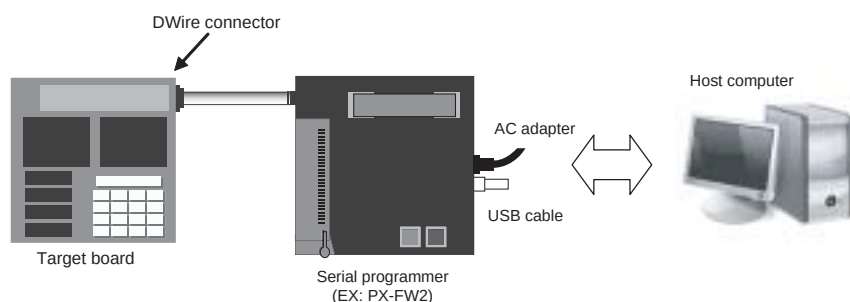
	MN101EF76K/MN101EF56K	MN101EF57G																																																		
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Memory map	<table><tr><td>Address</td><td></td></tr><tr><td>0x04000</td><td>Sector 0 32 Kbyte</td></tr><tr><td>0x0BFFF 0x0C000</td><td>Sector 1 32 Kbyte</td></tr><tr><td>0x13FFF 0x14000</td><td>Sector 2 32 Kbyte</td></tr><tr><td>0x1BFFF 0x1C000</td><td>Sector 3 20 Kbyte</td></tr><tr><td>0x20FFF 0x21000</td><td>Sector 4 4 Kbyte</td></tr><tr><td>0x21FFF 0x22000</td><td>Sector 5 4 Kbyte</td></tr><tr><td>0x22FFF 0x23000</td><td>Sector 6 4 Kbyte</td></tr><tr><td>0x23FFF 0x24000</td><td>Sector 7 32 Kbyte</td></tr><tr><td>0x2BFFF 0x2C000</td><td>Sector 8 32 Kbyte</td></tr><tr><td>0x33FFF 0x34000</td><td>Sector 9 32 Kbyte</td></tr><tr><td>0x3BFFF 0x3C000</td><td>Sector 10 20 Kbyte</td></tr><tr><td>0x40FFF 0x41000</td><td>Sector 11 4 Kbyte</td></tr><tr><td>0x41FFF 0x42000</td><td>Sector 12 4 Kbyte</td></tr><tr><td>0x42FFF 0x43000</td><td>Sector 13 4 Kbyte</td></tr><tr><td>0x43FFF</td><td></td></tr></table>	Address		0x04000	Sector 0 32 Kbyte	0x0BFFF 0x0C000	Sector 1 32 Kbyte	0x13FFF 0x14000	Sector 2 32 Kbyte	0x1BFFF 0x1C000	Sector 3 20 Kbyte	0x20FFF 0x21000	Sector 4 4 Kbyte	0x21FFF 0x22000	Sector 5 4 Kbyte	0x22FFF 0x23000	Sector 6 4 Kbyte	0x23FFF 0x24000	Sector 7 32 Kbyte	0x2BFFF 0x2C000	Sector 8 32 Kbyte	0x33FFF 0x34000	Sector 9 32 Kbyte	0x3BFFF 0x3C000	Sector 10 20 Kbyte	0x40FFF 0x41000	Sector 11 4 Kbyte	0x41FFF 0x42000	Sector 12 4 Kbyte	0x42FFF 0x43000	Sector 13 4 Kbyte	0x43FFF		<table><tr><td>Address</td><td></td></tr><tr><td>0x04000</td><td>Sector 0 32 Kbyte</td></tr><tr><td>0x0BFFF 0x0C000</td><td>Sector 1 32 Kbyte</td></tr><tr><td>0x13FFF 0x14000</td><td>Sector 2 32 Kbyte</td></tr><tr><td>0x1BFFF 0x1C000</td><td>Sector 3 20 Kbyte</td></tr><tr><td>0x20FFF 0x21000</td><td>Sector 4 4 Kbyte</td></tr><tr><td>0x21FFF 0x22000</td><td>Sector 5 4 Kbyte</td></tr><tr><td>0x22FFF 0x23000</td><td>Sector 6 4 Kbyte</td></tr><tr><td>0x23FFF</td><td></td></tr></table>	Address		0x04000	Sector 0 32 Kbyte	0x0BFFF 0x0C000	Sector 1 32 Kbyte	0x13FFF 0x14000	Sector 2 32 Kbyte	0x1BFFF 0x1C000	Sector 3 20 Kbyte	0x20FFF 0x21000	Sector 4 4 Kbyte	0x21FFF 0x22000	Sector 5 4 Kbyte	0x22FFF 0x23000	Sector 6 4 Kbyte	0x23FFF	
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0x22FFF 0x23000	Sector 6 4 Kbyte																																																			
0x23FFF																																																				

*1 When using BOOT Mode Microcontroller Rewriting Method

Figure:24.3.1 Serial Programmer Rewriting Memory Map

24.3.1 Equipment Configuration in Serial Programmer Rewriting

Figure:24.3.2 shows the equipment configuration in serial programmer rewriting. Install a connector to connect the serial programmer to this LSI on the circuit board. Connect the serial programmer via this connector. For the serial programmer, the programmer which can rewrite without host computer is provided. For procedure of rewriting, refer to the serial programmer instruction manual.



*1: DWire: Dedicated Serial I/F for debugging (Dedicated for Panasonic microcontroller)

Figure:24.3.2 Equipment Configuration of Serial Programmer Rewriting

Table:24.3.1 shows the corresponding table between rewriting adapter and serial programmer. Our website [Semiconductor Technical Support System] shows the most up-to-date information on rewriting adapter and serial programmer. For our website URL, refer to the end of this manual.

Table:24.3.1 Corresponding Table between rewriting adapter and Serial Programmer

Product	Serial programmer	
	Manufacturer	Model
MN101EF76K	Panasonic Corporation	PX-FW2
MN101EF57G	Panasonic Corporation	PX-FW2
	Yokogawa Electric Corporation	MegaNETIMPRESS (AF420/AF320)
	OBJECT Co., Ltd.	OBP1120/OBP1122
	Flash Support Group, Inc.	AF9101

24.3.2 Pin Configuration in Serial Programmer Rewriting

Table:24.3.2 shows the pin and signal functions in serial programmer rewriting, and Figure:24.3.3 shows the pin connecting diagram.

Table:24.3.2 Pin and Signal Functions

Pin	Function	I/O	Remarks
VDD5	Power supply	-	Voltage in writing is 2.7 to 5.5 V
VSS	GND	-	
NRST	Reset pin	I	
P00/OCD_DATA	Data I/O pin	I/O	
P01/OCD_CLK	Clock input pin 1	I	
DMOD	Mode switch input pin	I	
MMOD	Memory mode switch input pin	I	Fix to "0" in serial programmer rewriting
RON	Regulator control pin	I	Fix to V _{DD5} level

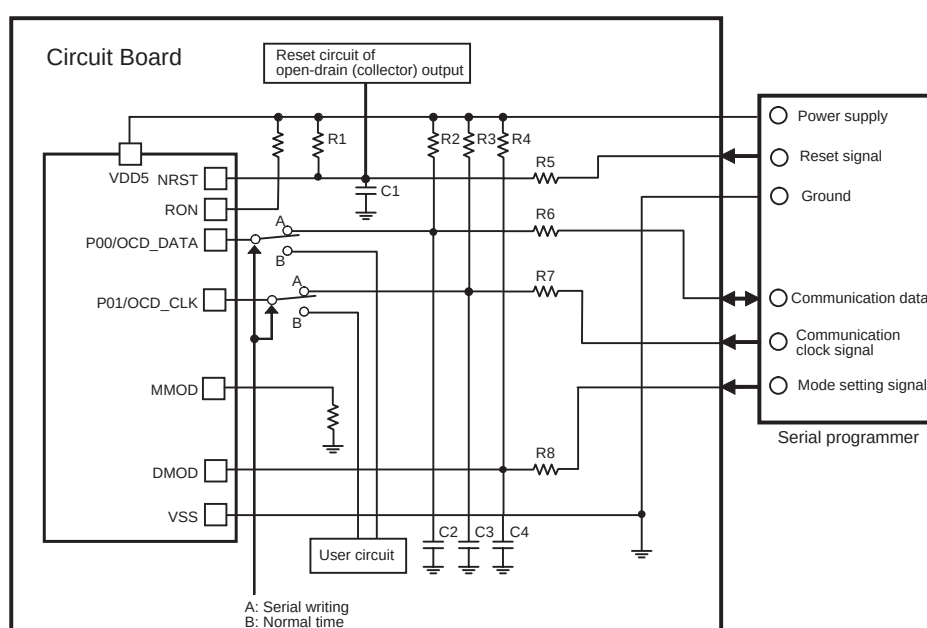


Figure:24.3.3 Pin Connecting Diagram in Serial Programmer Rewriting *1

*1: The pin connecting diagram above is an example. The pin to control and the external circuit differ depending on programmer.

For details, refer to the manual of programmer.

24.4 Component Value Calculations

This section presents the calculations for each of the components used for connection with the PX-FW2.

24.4.1 Component Values

The values of the components used in Figure:24.3.3 are shown below.

- The value of Resistor R1 must be greater than R_{upRst} .
- The values of Resistor R2,R3 and R4 must be greater than R_{upMin} and greater than 1 k Ω .
- The value of Resistor R5 must be less than 1/10 that of R1 and less than R_{sMax} .
- The value of Resistor R6 must be less than 1/10 that of R2 and less than R_{sMax} .
- The value of Resistor R7 must be less than 1/10 that of R3 and less than R_{sMax} .
- The value of Resistor R8 must be less than 1/10 that of R4 and less than R_{sMax} .
- C1 must be less than C_{rst} and less than 100 μ F.
- C2, C3 and C4 must be under 50 pF.
- Except for the V_{DD5} and V_{SS} lines, the line length of the signal from the connector to the microcomputer must be less than 50 cm.



This document is written assuming that the clock and data outputs used from PX-FW2 are the push-pull outputs. The required component values will differ from those shown here if PX-FW2 open-drain outputs are used.

24.4.2 Reset Signal Capacitor (C1) Maximum Value Calculation

After applying a reset to the target microcomputer, the PX-FW2 waits for a fixed period before initiating a communication. Determine the maximum value, C_{rst} , of the reset signal capacitor according to Formula (1) so that the sum of the time constant and the oscillator stabilization time (T_{wait}) is under 250 ms.

$$C_{rst} = \frac{250 \times 10^{-3} - T_{wait}}{R1} \quad \dots \text{Formula (1)}$$

24.4.3 Pull-up Resistor (R1) Minimum Value Calculation

The maximum output current from the PX-FW2 is 12 mA. Since this value is the maximum load current available for outputting a low level, R_{upRst} can be determined according to Formula (2).

$$R_{upRst} = \frac{\text{Operating supply voltage (V}_{DD5}\text{)}}{12 \text{ mA}} - R_{sRst} \quad \dots \text{Formula (2)}$$

24.4.4 Relationship Between R_{upRst} and R_{sRst}

If a resistor with a large value is inserted in series with the reset pin, the pull-up resistor (R_{upRst}) and the series resistor (R_{sRst}) must meet the condition in Formula (3) to ensure that the signal level falls all the way to the low level.

$$R_{upRst} \times \frac{1}{10} \geq R_{sRst} \quad \dots \text{Formula (3)}$$

24.4.5 Pull-up Resistor (R2, R3 and R4) Minimum Value Calculations

Find the maximum output current, I_{OL} , for the pins used for the communication from [Chapter 1 1.5.3 DC Characteristics]. Since the value is the maximum load current available to output a low level, R_{upMin} can be determined according to Formula (4).

$$R_{upMin} = \frac{\text{Operating supply voltage (V}_{DD5}\text{)}}{\text{Pin maximum output current (I}_{OL}\text{)}} \quad \dots \text{Formula (4)}$$

24.4.6 Communication Pin Series Resistor (R6, R7, R8) Maximum Value Calculations

If series resistors are inserted in the communication pin lines, the signal transmission speed will slow down due to the influence of the load capacitors (C2, C3). To ensure a reliable communication, the time for the signal voltage to change by 63% of the supply voltage (i.e. the time constant) must be held to under 1/8 of the communication period.

When load capacity is 50 pF and the communication frequency is 1 MHz, the maximum resistor value (R_{sMax}) for stable communication will be 2.5 k Ω by calculations according to Formula (5).

$$R_{sMax} = \frac{1}{8 \times \text{Communication frequency (f)} \times \text{Load capacitance (C)}} \quad \dots \text{Formula (5)}$$

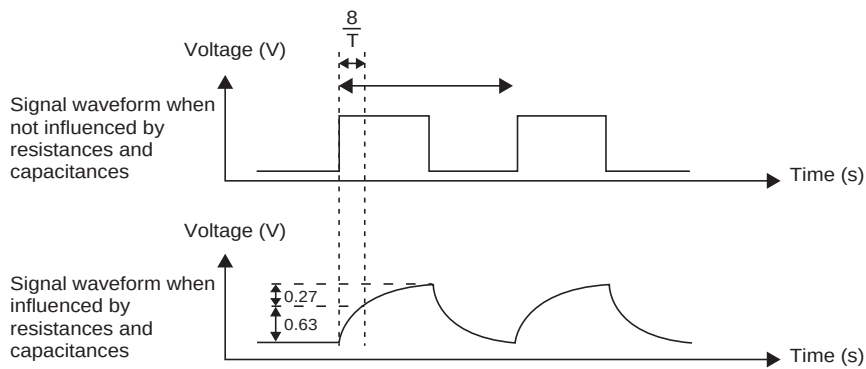


Figure:24.4.1 Relationship Between Communication Frequency, R and C



When using on-board debugger, set the resistance value of 100 Ω or lower, and connect it in series with a communication terminal.

24.4.7 Relationship Between R_{upMin} and R_{sMax}

It is possible to insert resistors with large values in series with the communication lines if the communication speed is lowered. However, even in that case, the pull-up resistor (R_{upMin}) and the series resistor (R_{sMax}) must meet the condition in Formula (6) to ensure that the signals fall to the low level.

$$R_{upMin} \times \frac{1}{10} \geq R_{sMax} \quad \dots \text{Formula (6)}$$

24.5 User Mode Microcontroller Rewriting

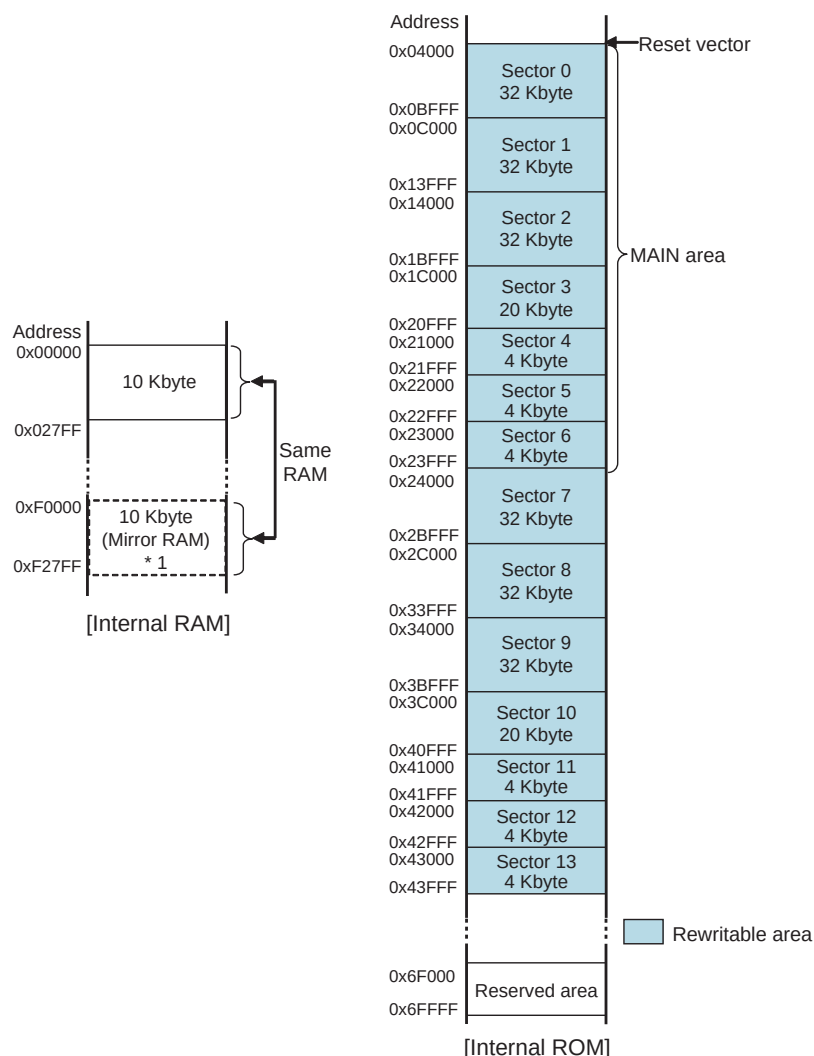
User mode microcontroller rewriting is used to rewrite the internal flash memory by the rewriting program included in the user program in advance. On any interface, rewriting during user program execution after implemented on board is possible.

It is suitable for a program which update during user program execution, or for alteration of EEPROM which retains data.

In this rewriting method, the entire area can be rewritten.

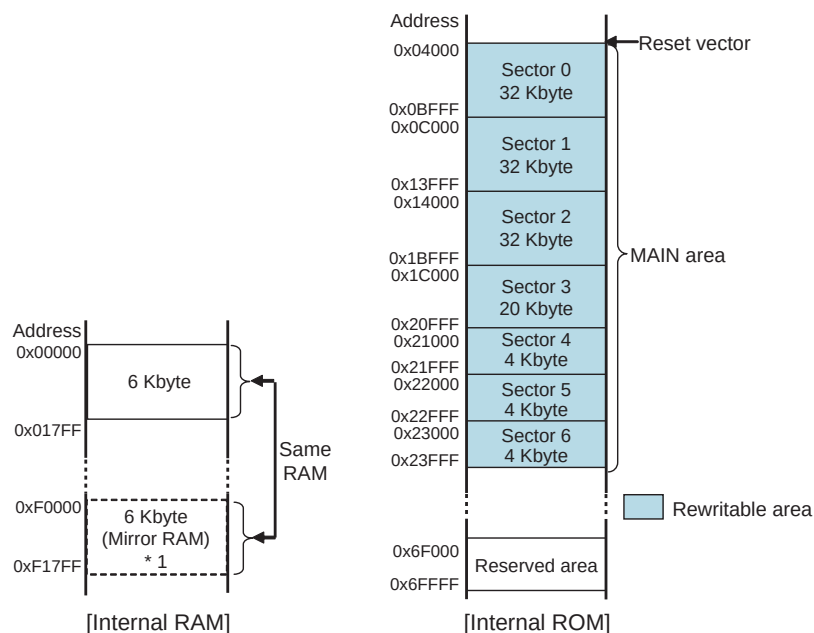
To set the user mode microcontroller rewriting method, release reset during the activation mode setting pin MMOD = "Low".

Figure:24.5.1 and Figure:24.5.2 show the memory map in the user mode microcontroller rewriting.



*1: Mirror RAM: Refer to [Chapter 2 2. 2 Memory Space].

Figure:24.5.1 Memory Map in User Mode Microcontroller Rewriting (MN101EF76K/MN101EF56K)



*1: Mirror RAM: Refer to [Chapter 2 2. 2 Memory Space].

Figure:24.5.2 Memory Map in User Mode Microcontroller Rewriting (MN101EF57G)

When reset is released while the activation mode setting pin MMOD = "Low", Sector 0 to 13 [Sector 0 to 6] are addressed in 0x04000 to 0x43FFF [0x23FFF], and activated from 0x04000. When 0x4B is set in the rewriting enable register (FBWER), reserved area is addressed in 0x6F000 to enable command library, and Sector 0 to 13 [Sector 0 to 6] can be rewritten.

Flash memory rewriting is performed from user program. Therefore, when the program is accidentally destroyed, the LSI could no longer execute a proper operation. At that time, use the serial programmer rewriting method to rewrite the program correctly.

* [] shows for MN101EF57G.

24.5.1 Rewriting by Command Library

The flash memory can be easily rewritten by a subroutine call of command library inside the program. The command library is stored in the reserved area of the flash memory.

If the command library is executed by the subroutine call, the program required for rewriting is automatically transferred from the reserved area of the flash memory to the internal RAM. If the rewriting of the flash memory is controlled by the program in the internal RAM after the program is transferred and the rewriting is completed, it returns from the command library. The subroutine call of the command library is executed from the program on the flash memory. But if the sector, where the program to call the command library exists, is erased, the returning address from the command library is erased and the microcontroller does not operate. When rewriting the sector where the rewriting program exists, the rewriting program needs to be allocated in the internal RAM and the command library is called up from the internal RAM.

Refer to [Chapter 24 24.5.3 Rewriting Procedure-2] for details.

24.5.2 Rewriting Procedure-1

Figure:24.5.3 shows a sample of the rewriting procedure suitable for rewriting a part of the flash memory. This procedure is not used for the rewriting of the sector where the rewriting program stores. Please refer to [Chapter 24 24.5.3 Rewriting Procedure-2] when rewriting the sector including the rewriting program.

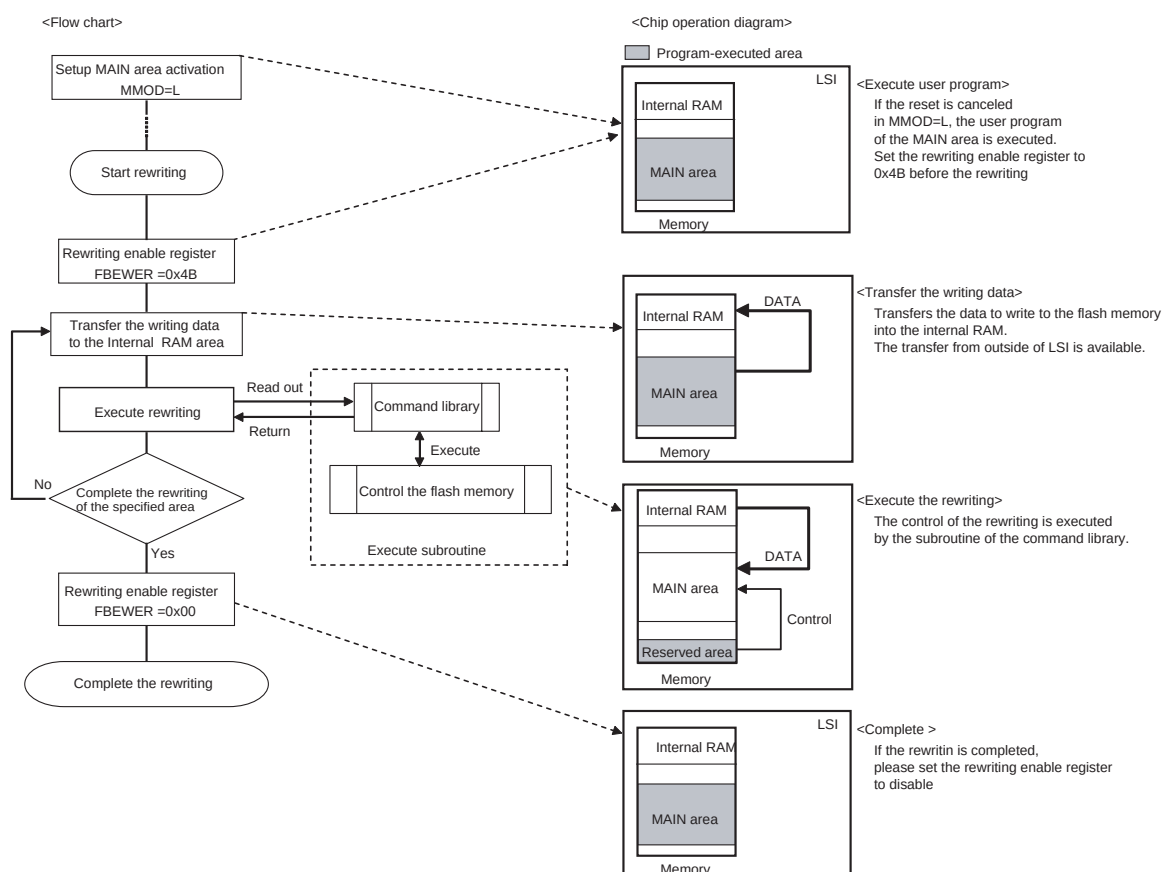


Figure:24.5.3 Rewriting Procedure-1 in User Mode Microcontroller Rewriting

24.5.3 Rewriting Procedure-2

In order to update the sector including the rewriting program, the rewriting program must be executed by the internal RAM. Figure:24.5.4 shows a sample procedure to execute the rewriting program by the internal RAM to rewrite the flash memory.

The following procedure shows a sample of whole flash memory rewriting, but it is possible to specify a certain sectors and rewrite them.

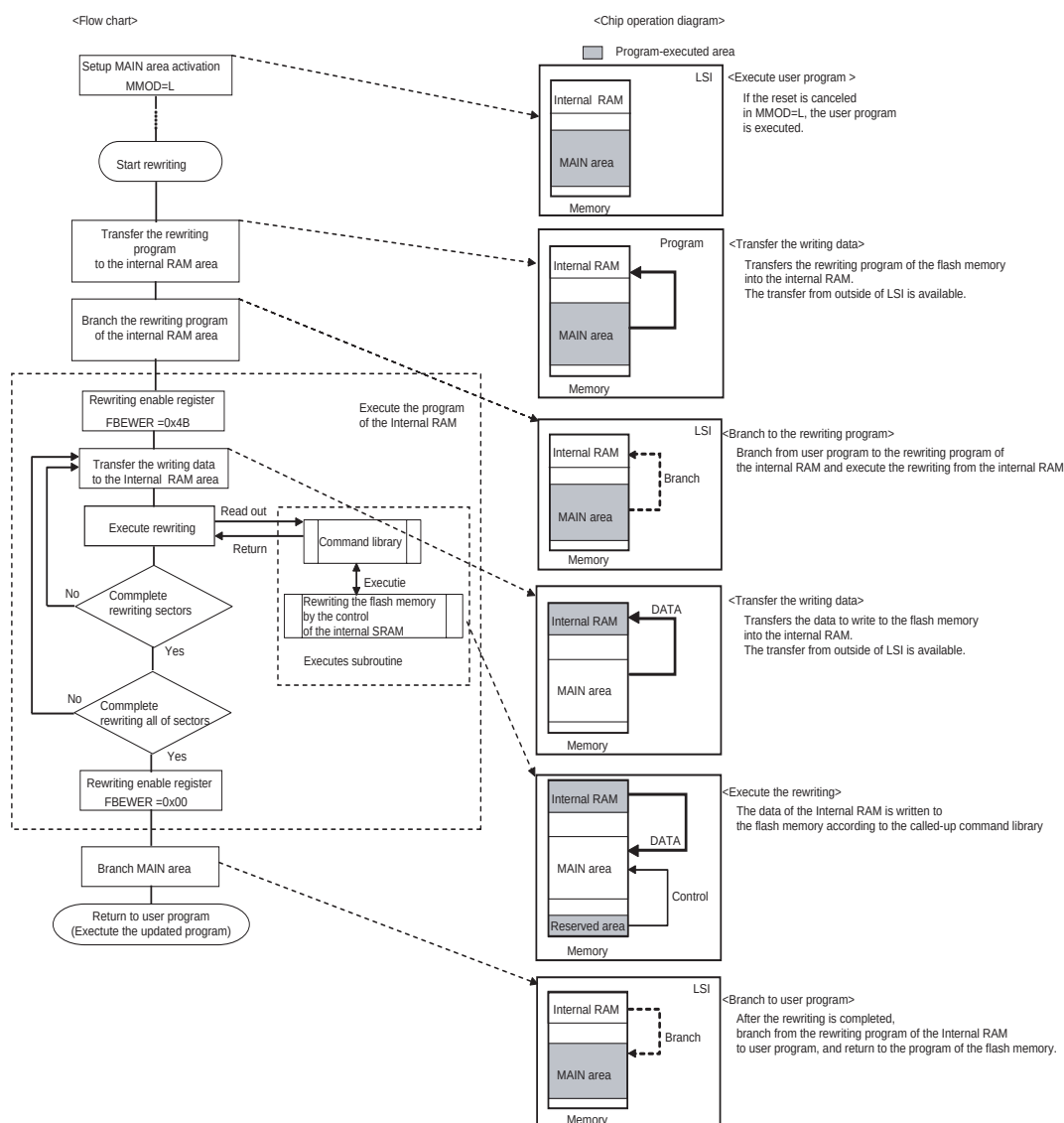


Figure:24.5.4 Rewriting Procedure-2 in User Mode Microcontroller Rewriting

24.6 BOOT Mode Microcontroller Rewriting

BOOT mode microcontroller rewriting is used to rewrite the internal flash memory by the rewrite-only program written in BOOT area in advance. In this rewriting method, rewriting is performed by the rewrite-only program independent of user program. As a result, user program can be updated without programmer even of the user program is destroyed accidentally.

In this rewriting method, the entire area can be rewritten. However, if BOOT area is rewritten accidentally, rewriting with this method cannot be performed.

The LSI activates by releasing reset when the activation mode setting pin MMOD = "High".

24.6.1 Rewriting Procedure

It is required to write the program for rewriting in advance by using serial programmer rewriting method or parallel programmer rewriting method.

Figure:24.6.1 and Figure:24.6.2 show the memory map in BOOT mode microcontroller rewriting.

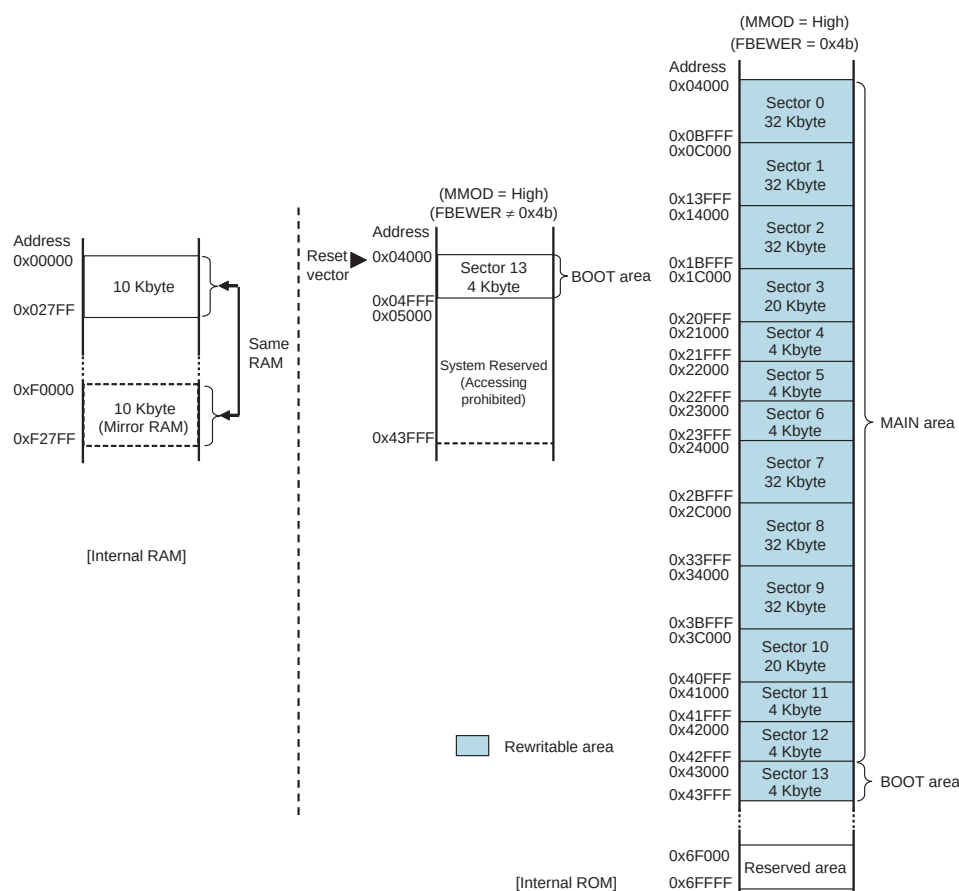


Figure:24.6.1 Memory Map in BOOT Mode Microcontroller Rewriting (MN101EF76K/MN101EF56K)

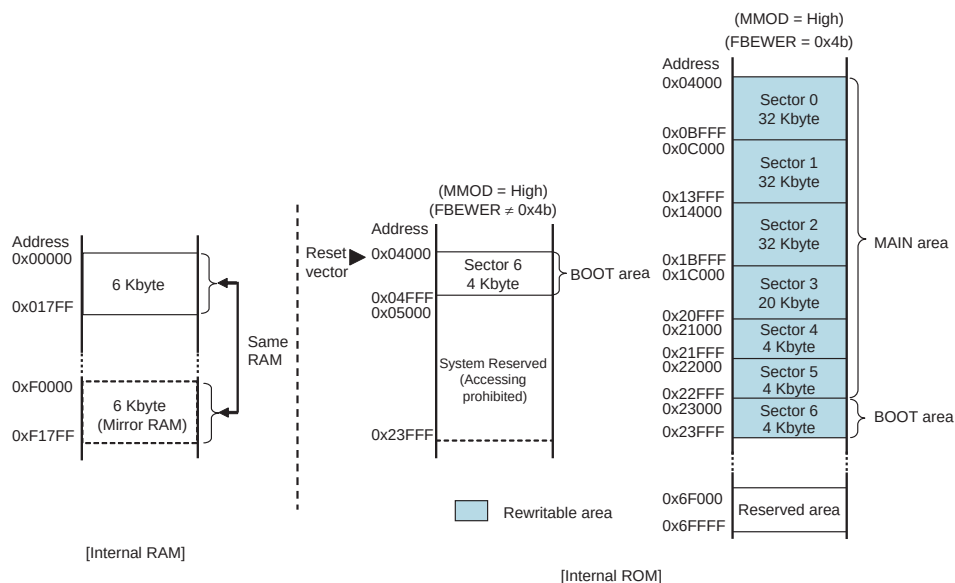


Figure:24.6.2 Memory Map in BOOT Mode Microcontroller Rewriting (MN101EF57G)

When releasing reset when the activation mode setting pin $MMOD = \text{"High"}$, BOOT area is addressed to 0x04000 and the LSI activates from BOOT area.

If the rewriting enable register (FBEWER) is set to 0x4B, the memory map is changed. MAIN area (Sector 0 to 12[Sector 0 to 5]) is addressed to 0x04000 and BOOT area (Sector 13[Sector 6]) is addressed to 0x43000[0x23000]. In addition, the reserved area is addressed to 0x6F000 and the command library becomes available to enable the rewriting of MAIN area (Sector 0 to 12[Sector 0 to 5]) and BOOT area (Sector 13[Sector 6]).

If the program is executed from BOOT area when the memory map is changed, programming and reading of data will be disabled. The setting routine of the rewriting enable register must be allocated in the internal RAM and executed in the internal RAM.

After rewriting, the activation mode setting pin $MMOD$ is set to "Low" to execute reset, the rewritten user program which is activated from MAIN area will be executed.

Figure:24.6.3 shows a sample procedure of rewriting by BOOT activation.

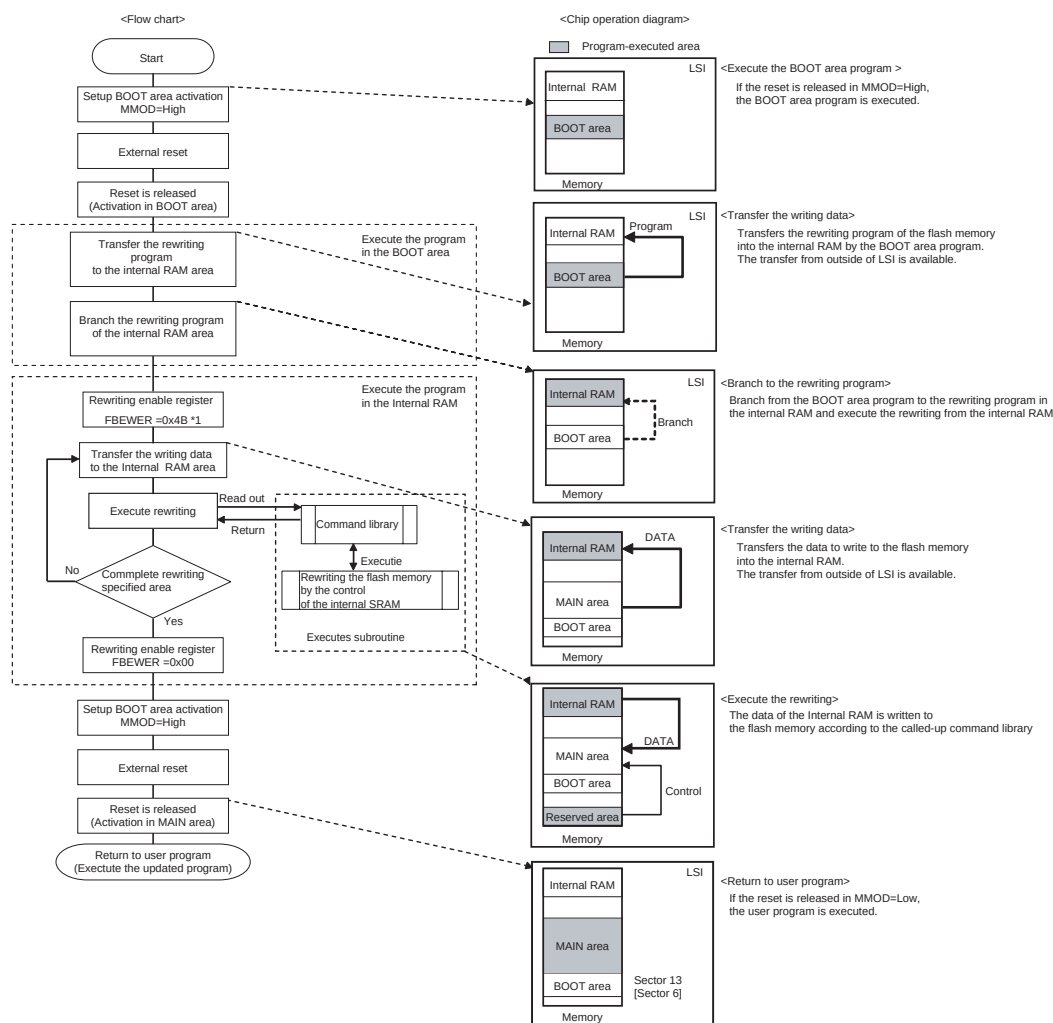


Figure:24.6.3 Rewriting Procedure in BOOT Mode Microcontroller Rewriting

*1 If the rewriting enable register (FBEWER) is set to 0x4B in BOOT area, the memory map is changed to disable programming and reading of data. The setting routine of rewriting enable register must be allocated in the internal RAM and executed on the internal RAM.
(In user mode microcontroller rewriting method, this constraint is not available.)

24.6.2 Protection of BOOT Area

If BOOT area (Sector 13[Sector 6]) is rewritten accidentally, rewriting with BOOT mode microcontroller rewriting method will not be performed. To use Sector 13[Sector 6] as BOOT area, it is recommended to protect Sector 13[Sector 6] in advance.

* [] shows for MN101EF57G.

24.7 Appendix

24.7.1 Rewriting Control Register

Table:24.7.1 shows the rewriting control register.

Table:24.7.1 Rewriting Control Register

Rewriting Control Register	
Registers	FBEWER
Address	0x3FBD
R/W	R/W
Initial value	0x00
Function	Setup write/erase enable of the flash memory
Setup	FBEWER=0x4B: Enable write/erase enable of the flash memory Command library is available. FBEWER≠0x4B: Disable write/erase enable of the flash memory When using BOOT mode microcontroller rewriting method, BOOT area is addressed at 0x04000.

24.7.2 Mode Setting

Table:24.7.2 shows the pin setup in microcontroller rewriting mode.

When the reset is released, the status of MMOD pin determines the microcontroller rewriting mode. Microcontroller rewriting mode can be activated from MAIN area or BOOT area.

Table:24.7.2 Pin Setup in Microcontroller Rewriting Mode

Activation Area	Rewriting Area	Pin Setup	
		NRST	MMOD
MAIN area	MAIN area	Low → High	Low
BOOT area	MAIN area + BOOT area		High

24.7.3 Procedure

■ Whole Chip Rewriting Procedure

Figure:24.7.1 shows rewriting procedures for the whole chip.

First of all, erase all of sectors by the control command. The erase is executed by the sector unit and multiple sectors can be specified.

Next, execute a writing by the control command. At that time, up to 64-byte consecutive writing is possible. But writing crossing over the sectors is not possible.

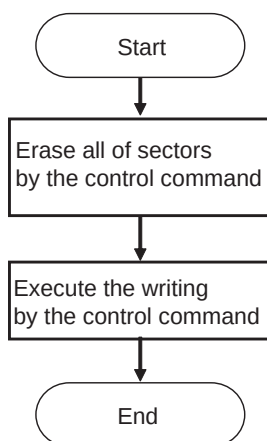


Figure:24.7.1 Whole Chip Rewriting Procedure

■ Sector Unit Rewriting Procedure

Figure:24.7.2 shows rewriting procedures by sector.

First, erase sectors to be rewritten using the control command.

Next, execute writing by a control command.

To rewrite multiple sectors, multiple sectors can be specified by a control command. Writing crossing over sectors cannot be executed.

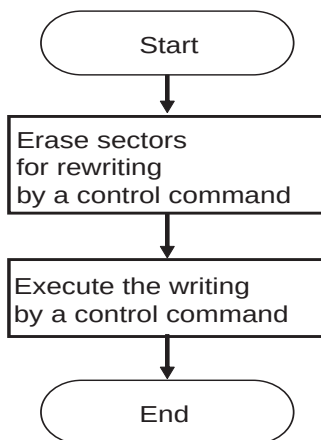


Figure:24.7.2 Sector Unit Rewriting Procedure

■ Partial Sectors Rewriting Procedure

Figure:24.7.3 shows partial sector rewriting procedure.

The erase can be executed only by the sector unit. If rewriting partial sector, back-up all the data in the sector to be rewritten before the erase, and execute the writing of the revised data after the erase. But if all the data in the sectors to be rewritten can be transferred from an external device, the data back-up is not necessary. In this case, please erase the target sectors and transfer the data from the external device for writing.

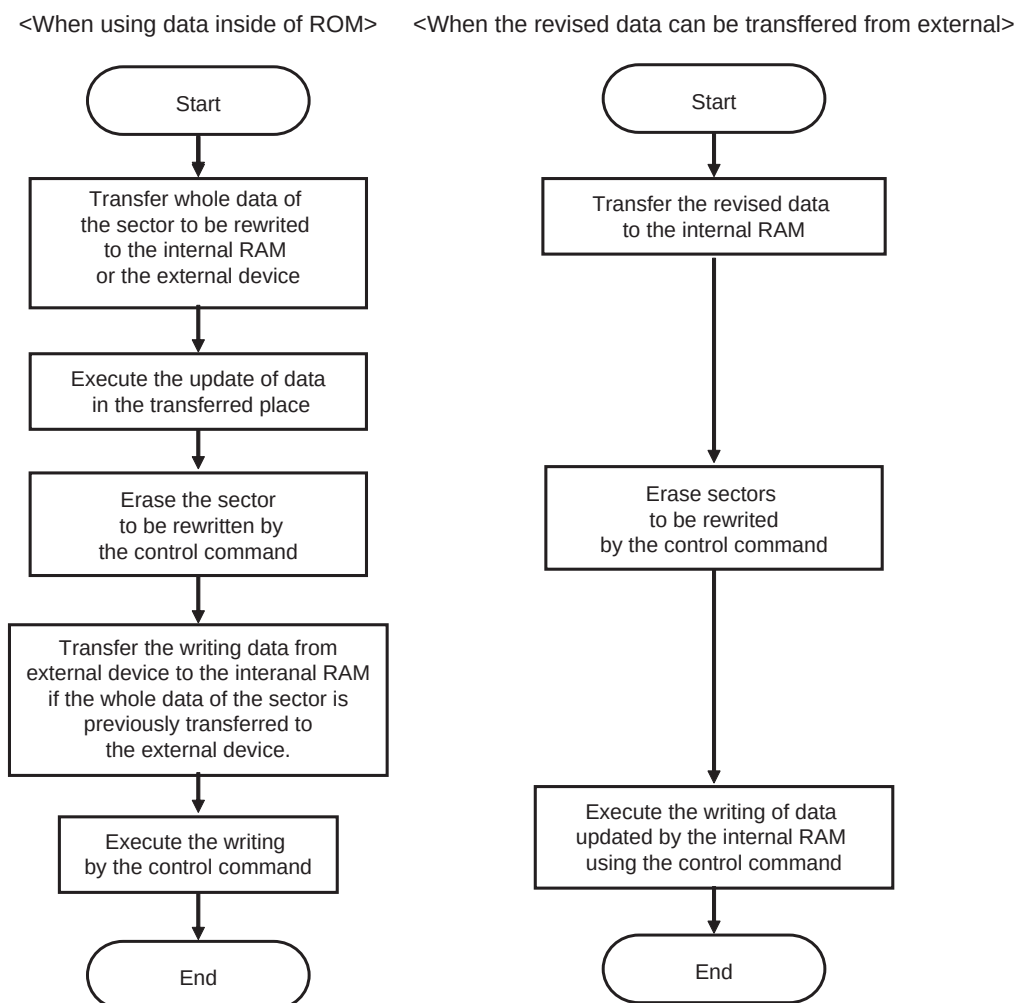


Figure:24.7.3 Partial Sectors Rewriting Procedure

24.7.4 Precautions

The flash memory rewriting is executed by the subroutine call at the initial address of the command library by using the JSR instruction. In rewriting, please pay attention to the following.

- If the command library is executed by the subroutine call, the rewriting enable register (FBEWER) must be set to "0x4B". When the rewriting is completed, other value except "0x4B" must be set to the rewriting enable register (FBEWER) in order to avoid error in writing.
And the subroutine call of the command library is prohibited when the rewriting enable register (FBEWER) is not set to "0x4B". As the reserved area where the command library is stored is not allocated, a non-maskable interrupt is generated by a undefined instruction.
- Please execute subroutine call to the command library, after the initial address of the structure which stores the rewriting parameter is stored to address register A0. The executed result of the command library is stored in data register D0.
The command library is a common specification that the argument is stored in address register A0 and the return value is stored in data register D0, and can be called up from C program.
Please refer to [Chapter 24 24.7.5 Command Library] and the attached papers for the overview of the command library and for specification, respectively.
- The command library needs up to 623 byte as the work area except the structure which stores the rewriting parameter. The work area can be set by ram_adr parameter. The structure and work area must be on the internal RAM. The required work area is different by the command library.
- Part of the command library can accept interrupts during command library execution. During this period, the interrupt vector base is automatically changed to 0x00100. Refer to the attached paper for the interrupts during command library execution.
- To execute rewriting flash memory using command library, it is necessary to set watchdog error detect cycle for watchdog timers as indicated below.

Watchdog timer 1: Watchdog error detect cycle = "2²⁰ of system clock cycle"

Watchdog timer 2: Watchdog error detect cycle ≥ "frcs/2¹¹"

- When calling command library, it is required to set CPU OPERATION mode to NORMAL mode in advance.

24.7.5 Command Library

Following are the functions as command library for flash memory rewriting.
The flash memory rewriting is executed by the subroutine call at the initial address of the command library by using the JSR instruction.

Table:24.7.3 Command library

Command library	Address	Argument structure (Byte)	Typical return value	Function
Program_Lib	0x6F89E	ram_adr (2) dst_adr (2) dst_bank (1) bc (1) src_adr (2)	0x00: Normal end 0x71: Writing error 0x72: 1over0 error 0x73: Security error 0x74: Protect error 0x75: Address violation error 0x76: Double activation error 0x77: Double activation error 0x7A: High-speed interrupt timeover error	Write bc+1 bytes of data stored in src_adr to dst_bank,dst_adr
Erase_Lib	0x6F8A8	ram_adr (2) sector_number (1)	0x00: Normal end 0x71: Erase error 0x73: Security error 0x74: Protect error 0x75: Address violation error 0x76: Double activation error 0x77: Double activation error 0x78: Erase error 0x7A: High-speed interrupt timeover error	Erase the sector specified by sector_number
Sector_Protect_Lib	0x6F88A	ram_adr (2) sector_number (1)	0x00: Normal end 0x71: Protect setup error 0x72: Reset error 0x73: Security error 0x74: Already protected 0x75: Address violation error 0x76: Double activation error 0x77: Double activation error 0x7A: High-speed interrupt timeover error	Protect setup the sector specified by sector_number
Security_Key_Program_Lib	0x6F88F	ram_adr (2) key_adr (2)	0x00: Normal end 0x71: KEY setup error 0x72: Reset error 0x74: Already secured 0x76: Double activation error 0x77: Double activation error 0x7A: High-speed interrupt timeover error	Set up Security Key cord (128-bit) stored in key_adr
Security_Key_Check_Lib	0x6F894	ram_adr (2) key_adr (2)	0x00: Authentication accord 0x76: Double activation error 0x77: Double activation error 0x7F: Authentication disaccord 0x80: No key setup	Authenticate Security Key cord (128-bit) stored in key_adr
Sector_Blank_Check_Lib	0x6F8AD	ram_adr (2) sector_number (1)	0x00: Blank 0x75: Address violation error 0x76: Double activation error 0x77: Double activation error 0x7F: Not Blank	Confirm the sector specified by sector_number is set to Blank (All 0xFF)
Sector_Sum_Verify_Lib	0x6F8B2	ram_adr (2) sector_number (1) sum_data (4)	0x00: Verify Pass 0x75: Address violation error 0x7F: Verify Fail	Verify the sector specified by sector_number by Check Sum value of 4-byte data.
Read_Status_Lib	0x6F8B7	ram_adr (2)	None	Store Security/Protect state to the address stored by ram_adr

Command library	Address	Argument structure (Byte)	Typical return value	Function
Program_1ms_Lib	0x6F899	ram_adr (2) dst_adr (2) dst_bank (1) bc (1) src_adr (2)	0x00: Normal end 0x71: Writing error 0x72: 1over0 error 0x73: Security error 0x74: Protect error 0x75: Address volation error 0x76: Double activation error 0x77: Double activation error 0x7A: High-speed interrupt timeover error 0x80: Suspend	Write bc+1 bytes of data stored in src_adr to dst_bank, dst_adr
Erase_1ms_Lib	0x6F8A3	ram_adr (2) sector_number (1)	0x00: Normal end 0x71: Erase error 0x73: Security error 0x74: Protect error 0x75: Address volation error 0x76: Double activation error 0x77: Double activation error 0x78: Erase error 0x7A: High-speed interrupt timeover error 0x80: Suspend	Erase the sector specified by sector_number

24.8 ROM Programming Service

Panasonic Corporation provides ROM programming service.

This LSI can be produced in which the arbitrary data has been written in advance.

The protect information to prevent writing/erasing errors and the security key code to prevent alteration or leakage of the program can be set.

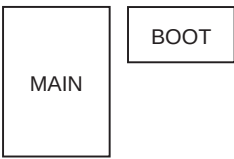
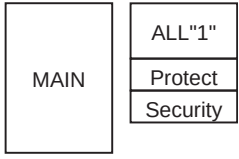
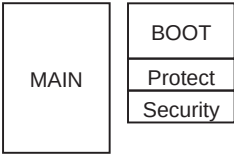
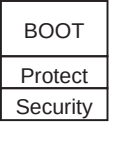
Request it from our ROM order service.

24.8.1 ROM Data Configuration

For your ROM structure, select one among six configurations of ROM data according to your usage.

Table:24.8.1 shows ROM data configuration.

Table:24.8.1 ROM Data Configuration

		BOOT Mode Microcontroller Rewriting Method		
		Unused	Used	
Protect/ Security Function	Unused	Configuration 1	Configuration 2	Configuration 3
				
	Used	Configuration 4	Configuration 5	Configuration 6
				

MAIN: Data for MAIN area

BOOT: Data for BOOT area

ALL"1": 0xFF padding data

Protect: Protect information

Security: Key code for security

Configuration 1

BOOT mode microcontroller rewriting method: Unused

Protect / Security function : Unused

Prepare only the data file for MAIN area.

MN101EF76K/MN101EF56K:

The size of the data file for MAIN area should be adjusted to less than 256 KB.

MN101EF57G:

The size of the data file for MAIN area should be adjusted to less than 128 KB.

Configuration 2, 3

BOOT mode microcontroller rewriting method: Used

Protect / Security function : Unused

Prepare the data file for MAIN area and the data file for BOOT area. (Configuration 2)

MN101EF76K/MN101EF56K:

Sector 13 is used as BOOT area.

The size of the data file for MAIN area should be adjusted to less than 252 KB.

MN101EF57G:

Sector 6 is used as BOOT area.

The size of the data file for MAIN area should be adjusted to less than 124 KB.

The size of the data file for BOOT area should be adjusted to less than 4 KB.

Prepare only the data file for BOOT area. (Configuration 3)

Configuration 4

BOOT mode microcontroller rewriting method: Unused

Protect / Security function : Used

Prepare the data file for MAIN area and the data file for protect / security.

MN101EF76K/MN101EF56K:

The size of the data file for MAIN area should be adjusted to less than 256 KB.

MN101EF57G:

The size of the data file for MAIN area should be adjusted to less than 128 KB.

For the data file for protect/security, refer to [Chapter 24 24.8.2 File for Protect / Security].

Configuration 5 and 6

BOOT mode microcontroller rewriting method: Used

Protect / Security function : Used

Prepare the data file for MAIN area and the data file for BOOT area/ protect/ security. (Configuration 5)

MN101EF76K/MN101EF56K:

Sector 13 is used as BOOT area.

The size of the data file for MAIN area should be adjusted to less than 252 KB.

MN101EF57G:

Sector 6 is used as BOOT area.

The size of the data file for MAIN area should be adjusted to less than 124 KB.

For the data file for BOOT area/ protect/ security, refer to [Chapter 24 24.8.2 File for Protect / Security].

Prepare only the data file for BOOT area/ protect/ security. (Configuration 6).

24.8.2 File for Protect / Security

When using protect / security function, prepare the data for BOOT area and the data for protect / security with one file. Figure:24.8.1 shows the configuration of the file for protect / security. Table:24.8.2 and Table:24.8.3 show the protect information.

Address		Data Size
0x04000	ALL "FF" Data	4 Kbyte
0x05000	ID code (ALL "AA")	16 byte
0x05010	Protect Information	4 byte
0x05014	ALL "FF" Data	12 byte
0x05020	ID code (ALL "55")	16 byte
0x05030	Security Key Code	16 byte

Figure:24.8.1 File for Protect / Security

Table:24.8.2 Protect Information (MN101EF76K/MN101EF56K)

Address	Protect Setup			
0x05010	Bit 0	Sector 0	"0": Enable Protect "1": Disable Protect	
	Bit 1	Sector 1		
	Bit 2	Sector 2		
	Bit 3	Sector 3		
	Bit 4	Sector 4		
	Bit 5	Sector 5		
	Bit 6	Sector 6		
	Bit 7	Sector 7		
0x05011	Bit 0	Sector 8		
	Bit 1	Sector 9		
	Bit 2	Sector 10		
	Bit 3	Sector 11		
	Bit 4	Sector 12		
	Bit 5	Sector 13		
	Bit 6	No use		Always set to "1"
	Bit 7			
0x05012	All bits	No use	Always set to "0xFF"	
0x05013				

Table:24.8.3 Protect Information (MN101EF57G)

Address	Protect Setup		
0x05010	Bit 0	Sector 0	"0": Enable Protect "1": Disable Protect
	Bit 1	Sector 1	
	Bit 2	Sector 2	
	Bit 3	Sector 3	
	Bit 4	Sector 4	
	Bit 5	Sector 5	
	Bit 6	Sector 6	
	Bit 7	No use	Always set to "1"
0x05011	All bits	No use	Always set to "0xFF"
0x05012			
0x05013			

The following description shows precautions for creation of the file for protect / security.

- Compose the file for protect/security of 4 KB + 64 byte.
- When the data for BOOT area is less than 4 KB, perform padding by "0xFF".
- When BOOT mode microcontroller rewriting method is not used, perform padding by "0xFF" to BOOT area.
- When the protect function is not used, set all the data of the protect information to "0xFF".
- When the security function is not used, set all the data of the security key code to "0xFF".
- Even when the protect or the security function is not used, allocate the ID code.

24.8.3 ROM Order Service

For the ROM order service, consult our sales offices.

25.1 Special Function Registers List

25.1.1 Special Function Registers List (MN101EF76K/MN101EF56K)

<MN101EF76K/MN101EF56K> (*: Reserved register in MN101EF56K)

Address	Register	Bit symbol								Page
		bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0	
0x03DF0	CHDRVSEL	-	-	-	-	-	-	DRVSOSC_SEL	DRVOSC_SEL	III-18
		-	-	-	-	-	-	0	0	
		-	-	-	-	-	-	Low-speed/high-speed oscillation performance selection		
0x03E00	PWMMDL	PCRAEN	PCRBEN	INTAEN	INTBEN	DTEN	ORMD	TCEN	WAVEMD	XII-8
		0	0	0	0	0	0	0	0	
PWM mode control register lower 8-bit										
0x03E01	PWMMDH	-	INTCEN	-	Reserved	-	DTSEL	SDSELA	SDSELB	XII-9
		-	0	-	0	-	0	0	0	
		PWM mode control register upper 8-bit								
0x03E02	PWMSSELL	PSELN00	PSEL00	OTLVN02	OTLV02	OTLVN01	OTLV01	OTLVN00	OTLV000	XII-11
		0	0	0	0	0	0	0	0	
PWM output control register lower 8-bit										
0x03E03	PWMSSELH	-	-	-	-	PSELN02	PSEL02	PSELN01	PSEL01	XII-11
		-	-	-	-	0	0	0	0	
		PWM output control register upper 8-bit								
0x03E04	PWMSETL	PMSET07	PMSET06	PMSET05	PMSET04	PMSET03	PMSET02	PMSET01	PMSET00	XII-12
		0	0	0	0	0	0	0	0	
PWM cycle setting register lower 8-bit										
0x03E05	PWMSETH	PMSET0F	PMSET0E	PMSET0D	PMSET0C	PMSET0B	PMSET0A	PMSET09	PMSET08	XII-12
		0	0	0	0	0	0	0	0	
PWM cycle setting register upper 8-bit										
0x03E06	TCMPAL	TCPA07	TCPA06	TCPA05	TCPA04	TCPA03	TCPA02	TCPA01	TCPA00	XII-13
		0	0	0	0	0	0	0	0	
PWM 0-phase comparison setting register lower 8-bit										
0x03E07	TCMPAH	TCPA0F	TCPA0E	TCPA0D	TCPA0C	TCPA0B	TCPA0A	TCPA09	TCPA08	XII-13
		0	0	0	0	0	0	0	0	
PWM 0-phase comparison setting register upper 8-bit										
0x03E08	TCMPBL	TCPB07	TCPB06	TCPB05	TCPB04	TCPB03	TCPB02	TCPB01	TCPB00	XII-13
		0	0	0	0	0	0	0	0	
PWM 1-phase comparison setting register lower 8-bit										
0x03E09	TCMPBH	TCPB0F	TCPB0E	TCPB0D	TCPB0C	TCPB0B	TCPB0A	TCPB09	TCPB08	XII-14
		0	0	0	0	0	0	0	0	
PWM 1-phase comparison setting register upper 8-bit										
0x03E0A	TCMPCL	TCPC07	TCPC06	TCPC05	TCPC04	TCPC03	TCPC02	TCPC01	TCPC00	XII-14
		0	0	0	0	0	0	0	0	
PWM 2-phase comparison setting register lower 8-bit										
0x03E0B	TCMPCH	TCPC0F	TCPC0E	TCPC0D	TCPC0C	TCPC0B	TCPC0A	TCPC09	TCPC08	XII-14
		0	0	0	0	0	0	0	0	
PWM 2-phase comparison setting register upper 8-bit										
0x03E0C	OUTMD	-	-	PXDTNW	PXDTW	PXDTNV	PXDTV	PXDTNU	PXDTU	XII-10
		-	-	0	0	0	0	0	0	
PWM output polarity control register										
0x03E0D	DTMSET	DTST07	DTST06	DTST05	DTST04	DTST03	DTST02	DTST01	DTST00	XII-15
		0	0	0	0	0	0	0	0	
PWM dead time setting register										

<MN101EF76K/MN101EF56K> (*: Reserved register in MN101EF56K)

Address	Register	Bit symbol								Page
		bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0	
0x03E0E	DTMSET1	DTST17	DTST16	DTST15	DTST14	DTST13	DTST12	DTST11	DTST10	XII-15
		0	0	0	0	0	0	0	0	
		PWM dead time setting register 1								
0x03E10	PWMBC1	PWMBC07	PWMBC06	PWMBC05	PWMBC04	PWMBC03	PWMBC02	PWMBC01	PWMBC00	XII-16
		0	0	0	0	0	0	0	0	
		PWM BC value read register lower 8-bit								
0x03E11	PWMBC8	PWMBC15	PWMBC14	PWMBC13	PWMBC12	PWMBC11	PWMBC10	PWMBC09	PWMBC08	XII-16
		0	0	0	0	0	0	0	0	
		PWM BC value read register upper 8-bit								
0x03E12	BCSTR	-	-	-	-	-	-	-	PWMSTR	XII-17
		-	-	-	-	-	-	-	1	
		PWM BC status register								
0x03E13	PWMOFFL	PRTANU1	PRTANU0	PRTAU1	PRTAU0	IRQSEL2	IRQSEL1	IRQSEL0	OUTEN0	XII-18
		0	0	0	0	0	0	0	0	
		PWM pin protection control register lower 8-bit								
0x03E14	PWMOFFH	PRTANW1	PRTANW0	PRTAW1	PRTAW0	PRTANV1	PRTANV0	PRTAV1	PRTAV0	XII-19
		0	0	0	0	0	0	0	0	
		PWM pin protection control register upper 8-bit								
0x03E15	IRQCULL	REGSELEDGE 1	REGSELEDGE 0	REGCULLCM P1	REGCULLCM P0	REGCULLUDF 1	REGCULLUDF 0	REGCULLOVF 1	REGCULLOVF 0	XII-20
		0	0	0	0	0	0	0	0	
		PWM interrupt signal culling selection register								
0x03E16	PWMTMCNT	-	-	-	PSCSEL1	PSCSEL0	PWMCKSEL1	PWMCKSEL0	SEL_PWM_TM	XII-21
		-	-	-	0	0	0	0	0	
		PWM timer operation control register								
0x03E17	RELCTR	RELIRQA	RELIRQB	RELIRQC	RELIRQD	-	MD1CTR	RELMD		XII-22
		0	0	0	0	-	0	0	0	
		1st phase inter- rupt event enable control	2nd phase interrupt event enable control	3rd phase interrupt event enable control	4th phase inter- rupt event enable control	-	PWM output order control in 1-phase mode	RELATIONAL PWM mode		
0x03E18	PWMODR	-	-	-	-	-	-	-	ORDER	XII-23
		-	-	-	-	-	-	-	0	
		-	-	-	-	-	-	-	4-phase PWM output order	
0x03E19	RELSTAT	-	-	-	-	-	-	STAT1	STAT0	XII-23
		-	-	-	-	-	-	0	0	
		-	-	-	-	-	-	4-phase PWM output order		
0x03E1A	PWMCMP1	PWMCMP17	PWMCMP16	PWMCMP15	PWMCMP14	PWMCMP13	PWMCMP12	PWMCMP11	PWMCMP10	XII-24
		0	0	0	0	0	0	0	0	
		Lower 8-bit of PWM cycle register and PWMn phase comparison register								
0x03E1B	PWMCMP2	PWMCMPH7	PWMCMPH6	PWMCMPH5	PWMCMPH4	PWMCMPH3	PWMCMPH2	PWMCMPH1	PWMCMPH0	XII-24
		0	0	0	0	0	0	0	0	
		Upper 8-bit of PWM cycle register and PWMn phase comparison register								
0x03E20	LC00BUF	LC00BUF7	LC00BUF6	LC00BUF5	LC00BUF4	LC00BUF3	LC00BUF2	LC00BUF1	LC00BUF0	XXII-18
		X	X	X	X	X	X	X	X	
		Segment output latch data								
0x03E21	LC01BUF	LC01BUF7	LC01BUF6	LC01BUF5	LC01BUF4	LC01BUF3	LC01BUF2	LC01BUF1	LC01BUF0	XXII-18
		X	X	X	X	X	X	X	X	
		Segment output latch data								
0x03E22	LC02BUF	LC02BUF7	LC02BUF6	LC02BUF5	LC02BUF4	LC02BUF3	LC02BUF2	LC02BUF1	LC02BUF0	XXII-18
		X	X	X	X	X	X	X	X	
		Segment output latch data								
0x03E23	LC03BUF	LC03BUF7	LC03BUF6	LC03BUF5	LC03BUF4	LC03BUF3	LC03BUF2	LC03BUF1	LC03BUF0	XXII-18
		X	X	X	X	X	X	X	X	
		Segment output latch data								
0x03E24	LC04BUF	LC04BUF7	LC04BUF6	LC04BUF5	LC04BUF4	LC04BUF3	LC04BUF2	LC04BUF1	LC04BUF0	XXII-18
		X	X	X	X	X	X	X	X	
		Segment output latch data								

<MN101EF76K/MN101EF56K> (*: Reserved register in MN101EF56K)

Address	Register	Bit symbol								Page
		bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0	
0x03E25	LC05BUF	LC05BUF7	LC05BUF6	LC05BUF5	LC05BUF4	LC05BUF3	LC05BUF2	LC05BUF1	LC05BUF0	XXII-18
		X	X	X	X	X	X	X	X	
		Segment output latch data								
0x03E26	LC06BUF	LC06BUF7	LC06BUF6	LC06BUF5	LC06BUF4	LC06BUF3	LC06BUF2	LC06BUF1	LC06BUF0	XXII-18
		X	X	X	X	X	X	X	X	
		Segment output latch data								
0x03E27	LC07BUF	LC07BUF7	LC07BUF6	LC07BUF5	LC07BUF4	LC07BUF3	LC07BUF2	LC07BUF1	LC07BUF0	XXII-18
		X	X	X	X	X	X	X	X	
		Segment output latch data								
0x03E28	LC08BUF	LC08BUF7	LC08BUF6	LC08BUF5	LC08BUF4	LC08BUF3	LC08BUF2	LC08BUF1	LC08BUF0	XXII-18
		X	X	X	X	X	X	X	X	
		Segment output latch data								
0x03E29	LC09BUF	LC09BUF7	LC09BUF6	LC09BUF5	LC09BUF4	LC09BUF3	LC09BUF2	LC09BUF1	LC09BUF0	XXII-18
		X	X	X	X	X	X	X	X	
		Segment output latch data								
0x03E2A	LC0ABUF	LC0ABUF7	LC0ABUF6	LC0ABUF5	LC0ABUF4	LC0ABUF3	LC0ABUF2	LC0ABUF1	LC0ABUF0	XXII-18
		X	X	X	X	X	X	X	X	
		Segment output latch data								
0x03E2B	LC0BBUF	LC0BBUF7	LC0BBUF6	LC0BBUF5	LC0BBUF4	LC0BBUF3	LC0BBUF2	LC0BBUF1	LC0BBUF0	XXII-18
		X	X	X	X	X	X	X	X	
		Segment output latch data								
0x03E2C	LC0CBUF	LC0CBUF7	LC0CBUF6	LC0CBUF5	LC0CBUF4	LC0CBUF3	LC0CBUF2	LC0CBUF1	LC0CBUF0	XXII-18
		X	X	X	X	X	X	X	X	
		Segment output latch data								
0x03E2D	LC0DBUF	LC0DBUF7	LC0DBUF6	LC0DBUF5	LC0DBUF4	LC0DBUF3	LC0DBUF2	LC0DBUF1	LC0DBUF0	XXII-18
		X	X	X	X	X	X	X	X	
		Segment output latch data								
0x03E2E	LC0EBUF	LC0EBUF7	LC0EBUF6	LC0EBUF5	LC0EBUF4	LC0EBUF3	LC0EBUF2	LC0EBUF1	LC0EBUF0	XXII-18
		X	X	X	X	X	X	X	X	
		Segment output latch data								
0x03E2F	LC0FBUF	LC0FBUF7	LC0FBUF6	LC0FBUF5	LC0FBUF4	LC0FBUF3	LC0FBUF2	LC0FBUF1	LC0FBUF0	XXII-18
		X	X	X	X	X	X	X	X	
		Segment output latch data								
0x03E30	LC10BUF	LC10BUF7	LC10BUF6	LC10BUF5	LC10BUF4	LC10BUF3	LC10BUF2	LC10BUF1	LC10BUF0	XXII-18
		X	X	X	X	X	X	X	X	
		Segment output latch data								
0x03E31	LC11BUF	LC11BUF7	LC11BUF6	LC11BUF5	LC11BUF4	LC11BUF3	LC11BUF2	LC11BUF1	LC11BUF0	XXII-18
		X	X	X	X	X	X	X	X	
		Segment output latch data								
0x03E32	LC12BUF	LC12BUF7	LC12BUF6	LC12BUF5	LC12BUF4	LC12BUF3	LC12BUF2	LC12BUF1	LC12BUF0	XXII-18
		X	X	X	X	X	X	X	X	
		Segment output latch data								
0x03E33	LC13BUF	LC13BUF7	LC13BUF6	LC13BUF5	LC13BUF4	LC13BUF3	LC13BUF2	LC13BUF1	LC13BUF0	XXII-18
		X	X	X	X	X	X	X	X	
		Segment output latch data								
0x03E34	LC14BUF	LC14BUF7	LC14BUF6	LC14BUF5	LC14BUF4	LC14BUF3	LC14BUF2	LC14BUF1	LC14BUF0	XXII-18
		X	X	X	X	X	X	X	X	
		Segment output latch data								
0x03E35	LC15BUF	LC15BUF7	LC15BUF6	LC15BUF5	LC15BUF4	LC15BUF3	LC15BUF2	LC15BUF1	LC15BUF0	XXII-18
		X	X	X	X	X	X	X	X	
		Segment output latch data								
0x03E36	LC16BUF	LC16BUF7	LC16BUF6	LC16BUF5	LC16BUF4	LC16BUF3	LC16BUF2	LC16BUF1	LC16BUF0	XXII-18
		X	X	X	X	X	X	X	X	
		Segment output latch data								
0x03E37	LC17BUF	LC17BUF7	LC17BUF6	LC17BUF5	LC17BUF4	LC17BUF3	LC17BUF2	LC17BUF1	LC17BUF0	XXII-18
		X	X	X	X	X	X	X	X	
		Segment output latch data								

<MN101EF76K/MN101EF56K> (*: Reserved register in MN101EF56K)

Address	Register	Bit symbol								Page
		bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0	
0x03E38	LC18BUF	LC18BUF7	LC18BUF6	LC18BUF5	LC18BUF4	LC18BUF3	LC18BUF2	LC18BUF1	LC18BUF0	XXII-18
		X	X	X	X	X	X	X	X	
		Segment output latch data								
0x03E39	LC19BUF	LC19BUF7	LC19BUF6	LC19BUF5	LC19BUF4	LC19BUF3	LC19BUF2	LC19BUF1	LC19BUF0	XXII-18
		X	X	X	X	X	X	X	X	
		Segment output latch data								
0x03E3A	LC1ABUF	LC1ABUF7	LC1ABUF6	LC1ABUF5	LC1ABUF4	LC1ABUF3	LC1ABUF2	LC1ABUF1	LC1ABUF0	XXII-18
		X	X	X	X	X	X	X	X	
		Segment output latch data								
0x03E3B	LC1BBUF	-	-	-	-	LC1BBUF3	LC1BBUF2	LC1BBUF1	LC1BBUF0	XXII-18
		-	-	-	-	X	X	X	X	
		-	-	-	-	Segment output latch data				
0x03E40	LCDMD1	LCDEN	Reserved	DUTY1	DUTY0	LCDCCK3	LCDCCK2	LCDCCK1	LCDCCK0	XXII-7
		0	0	0	0	0	0	0	0	
		LCD driving circuit start flag	Always set to "0"	LCD display duty selection		LCD clock source selection				
0x03E41	LCDMD2	Reserved	Reserved	Reserved	LCRHL1	LCRHL0	LCREN	Reserved	Reserved	XXII-8
		0	0	0	0	0	0	0	0	
		Always set to "0"	Always set to "0"	Always set to "0"	Internal dividing resistor type selection		Internal dividing resistor connecting selection	Always set to "0"	Always set to "0"	
0x03E42	LCDMD3	-	-	-	-	TMSEL3	TMSEL2	TMSEL1	LCDCCKSEL	XXII-9
		-	-	-	-	0	0	0	0	
		-	-	-	-	Timer 0 to 4 and A input selection				
0x03E43	LCCTR0	COMSL3	COMSL2	COMSL1	COMSL0	Reserved	VLC3SL	VLC2SL	VLC1SL	XXII-10
		0	0	0	0	0	0	0	0	
		COM3/P84 selection	COM2/P83 selection	COM1/P82 selection	COM0/P81 selection	Always set to "0"	VLC3/P85 selection	VLC2/P86 selection	VLC1/P87 selection	
0x03E44	LCCTR1	LC1SL7	LC1SL6	LC1SL5	LC1SL4	LC1SL3	LC1SL2	LC1SL1	LC1SL0	XXII-11
		0	0	0	0	0	0	0	0	
		SEG7/P71 selection	SEG6/P72 selection	SEG5/P73 selection	SEG4/P74 selection	SEG3/P75 selection	SEG2/P76 selection	SEG1/P77 selection	SEG0/P80 selection	
0x03E45	LCCTR2	LC2SL7	LC2SL6	LC2SL5	LC2SL4	LC2SL3	LC2SL2	LC2SL1	LC2SL0	XXII-12
		0	0	0	0	0	0	0	0	
		SEG15/P61 selection	SEG14/P62 selection	SEG13/P63 selection	SEG12/P64 selection	SEG11/P65 selection	SEG10/P66 selection	SEG9/P67 selection	SEG8/P70 selection	
0x03E46	LCCTR3	LC3SL7	LC3SL6	LC3SL5	LC3SL4	LC3SL3	LC3SL2	LC3SL1	LC3SL0	XXII-13
		0	0	0	0	0	0	0	0	
		SEG23/P57 selection	SEG22/P56 selection	SEG21/P55 selection	SEG20/P54 selection	SEG19/P53 selection	SEG18/P52 selection	SEG17/P51 selection	SEG16/P50 selection	
0x03E47	LCCTR4	LC4SL7	LC4SL6	LC4SL5	LC4SL4	LC4SL3	LC4SL2	LC4SL1	LC4SL0	XXII-14
		0	0	0	0	0	0	0	0	
		SEG31/P40 selection	SEG30/P41 selection	SEG29/P42 selection	SEG28/P43 selection	SEG27/P44 selection	SEG26/P45 selection	SEG25/P46 selection	SEG24/P47 selection	
0x03E48	LCCTR5	LC5SL7	LC5SL6	LC5SL5	LC5SL4	LC5SL3	LC5SL2	LC5SL1	LC5SL0	XXII-15
		0	0	0	0	0	0	0	0	
		SEG39/P16 selection	SEG38/P30 selection	SEG37/P31 selection	SEG36/P32 selection	SEG35/P33 selection	SEG34/P34 selection	SEG33/P35 selection	SEG32/P36 selection	
0x03E49	LCCTR6	LC6SL7	LC6SL6	LC6SL5	LC6SL4	LC6SL3	LC6SL2	LC6SL1	LC6SL0	XXII-16
		0	0	0	0	0	0	0	0	
		SEG47/P93 selection	SEG46/P92 selection	SEG45/P10 selection	SEG44/P11 selection	SEG43/P12 selection	SEG42/P13 selection	SEG41/P14 selection	SEG40/P15 selection	
0x03E4A	LCCTR7	-	LC7SL6	LC7SL5	LC7SL4	LC7SL3	LC7SL2	LC7SL1	LC7SL0	XXII-17
		-	0	0	0	0	0	0	0	
		-	SEG54/PB3 selection	SEG53/PB4 selection	SEG52/PB5 selection	SEG51/PB6 selection	SEG50/PB7 selection	SEG49/P95 selection	SEG48/P94 selection	
0x03E50	PRTKEY	PRTK7	PRTK6	PRTK5	PRTK4	PRTK3	PRTK2	PRTK1	PRTK0	V-4
		0	0	0	0	0	0	0	0	
		Register protect control								

<MN101EF76K/MN101EF56K> (*: Reserved register in MN101EF56K)

Address	Register	Bit symbol								Page
		bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0	
0x03E51	CKTRMD	-	-	-	-	-	-	-	TRMD	IV-10
		-	-	-	-	-	-	-	0	
		-	-	-	-	-	-	-	Clock transition mode	
0x03E52	CKMONEN	-	-	-	-	-	MVCLK	DETIRQ	MONIEN	IV-5
		-	-	-	-	-	0	0	0	
		-	-	-	-	-	Clock switching selection	Detection interrupt selection	Clock monitoring function enable	
0x03E53	CKMONSETU	MNUSET7	MNUSET6	MNUSET5	MNUSET4	MNUSET3	MNUSET2	MNUSET1	MNUSET0	IV-6
		0	0	0	0	0	0	0	0	
0x03E54	CKMONSETL	Clock monitoring frequency upper limit setting value								IV-8
		MNLSET7	MNLSET6	MNLSET5	MNLSET4	MNLSET3	MNLSET2	MNLSET1	MNLSET0	
		0	0	0	0	0	0	0	0	
0x03E55	CKMONSTAT	Clock monitoring frequency lower limit setting value								IV-9
		-	-	-	-	-	PLLERR	XIERR	OSCERR	
		-	-	-	-	-	0/Retained	0/Retained	0/Retained	
0x03E56	OSCLOCK	-	-	-	-	-	-	-	LOCKEN	III-19
		-	-	-	-	-	-	-	0	
		-	-	-	-	-	-	-	Oscillation control register writing control	
0x03E60	AT1CNT0	FM0DE	AT1ACT	AT1MD3	AT1MD2	AT1MD1	AT1MD0	Reserved	AT1EN	XXIII-7
		0	0	0	0	0	0	0	0	
		Increment control for memory pointer 0	ATC1 software activation flag	ATC1 data transfer mode				Always set to "0"	ATC1 transfer enable flag	
0x03E61	AT1CNT1	AT1IRS	-	Reserved	BTSTP	AT1IR3	AT1IR2	AT1IR1	AT1IR0	XXIII-8
		0	-	0	0	0	0	0	0	
		Activation factor selection flag	-	Always set to "0"	Burst transfer stop enable	ATC1 trigger factor setting flag				
0x03E62	AT1TRC	AT1TRC7	AT1TRC6	AT1TRC5	AT1TRC4	AT1TRC3	AT1TRC2	AT1TRC1	AT1TRC0	XXIII-9
		0	0	0	0	0	0	0	0	
		ATC1 Transfer Data Count Setting								
0x03E63	AT1MAP0L	AT1MAP0L7	AT1MAP0L6	AT1MAP0L5	AT1MAP0L4	AT1MAP0L3	AT1MAP0L2	AT1MAP0L1	AT1MAP0L0	XXIII-9
		0	0	0	0	0	0	0	0	
		ATC1 Memory Pointer 0: Lower 8 bits								
0x03E64	AT1MAP0M	AT1MAP0M7	AT1MAP0M6	AT1MAP0M5	AT1MAP0M4	AT1MAP0M3	AT1MAP0M2	AT1MAP0M1	AT1MAP0M0	XXIII-9
		0	0	0	0	0	0	0	0	
		ATC1 Memory Pointer 0: Middle 8 bits								
0x03E65	AT1MAP0H	-	-	-	-	AT1MAP0H3	AT1MAP0H2	AT1MAP0H1	AT1MAP0H0	XXIII-9
		-	-	-	-	0	0	0	0	
		-	-	-	-	ATC1 Memory Pointer 0: Upper 4 bits				
0x03E66	AT1MAP1L	AT1MAP1L7	AT1MAP1L6	AT1MAP1L5	AT1MAP1L4	AT1MAP1L3	AT1MAP1L2	AT1MAP1L1	AT1MAP1L0	XXIII-10
		0	0	0	0	0	0	0	0	
		ATC1 Memory Pointer 1: Lower 8 bits								
0x03E67	AT1MAP1M	AT1MAP1M7	AT1MAP1M6	AT1MAP1M5	AT1MAP1M4	AT1MAP1M3	AT1MAP1M2	AT1MAP1M1	AT1MAP1M0	XXIII-10
		0	0	0	0	0	0	0	0	
		ATC1 Memory Pointer 1: Middle 8 bits								
0x03E68	AT1MAP1H	-	-	-	-	AT1MAP1H3	AT1MAP1H2	AT1MAP1H1	AT1MAP1H0	XXIII-10
		-	-	-	-	0	0	0	0	
		-	-	-	-	ATC1 Memory Pointer 1: Upper 4 bits				
0x03E70	P0OUT	P0OUT7	P0OUT6	P0OUT5	P0OUT4	P0OUT3	P0OUT2	P0OUT1	P0OUT0	VIII-8
		x	x	X	X	X	X	X	X	
		Port 0 output data								

<MN101EF76K/MN101EF56K> (*: Reserved register in MN101EF56K)

Address	Register	Bit symbol								Page
		bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0	
0x03E71	P1OUT	-	P1OUT6	P1OUT5	P1OUT4	P1OUT3	P1OUT2	P1OUT1	P1OUT0	VIII-32
		-	x	X	X	X	X	X	X	
		Port 1 output data								
0x03E72	P2OUT	P2OUT7	P2OUT6	P2OUT5	P2OUT4	P2OUT3	P2OUT2	P2OUT1	P2OUT0	VIII-45
		1	x	x	X	X	X	X	X	
		Port 2 output data								
0x03E73	P3OUT	-	P3OUT6	P3OUT5	P3OUT4	P3OUT3	P3OUT2	P3OUT1	P3OUT0	VIII-55
		-	x	X	X	X	X	X	X	
		Port 3 output data								
0x03E74	P4OUT	P4OUT7	P4OUT6	P4OUT5	P4OUT4	P4OUT3	P4OUT2	P4OUT1	P4OUT0	VIII-72
		X	X	X	X	X	X	X	X	
		Port 4 output data								
0x03E75	P5OUT	P5OUT7	P5OUT6	P5OUT5	P5OUT4	P5OUT3	P5OUT2	P5OUT1	P5OUT0	VIII-93
		X	X	X	X	X	X	X	X	
		Port 5 output data								
0x03E76	P6OUT	P6OUT7	P6OUT6	P6OUT5	P6OUT4	P6OUT3	P6OUT2	P6OUT1	-	VIII-111
		X	X	X	X	X	X	X	-	
		Port 6 output data								
0x03E77	P7OUT	P7OUT7	P7OUT6	P7OUT5	P7OUT4	P7OUT3	P7OUT2	P7OUT1	P7OUT0	VIII-131
		X	X	X	X	X	X	X	X	
		Port 7 output data								
0x03E78	P8OUT	P8OUT7	P8OUT6	P8OUT5	P8OUT4	P8OUT3	P8OUT2	P8OUT1	P8OUT0	VIII-153
		X	X	X	X	X	X	X	X	
		Port 8 output data								
0x03E79	P9OUT	-	-	P9OUT5	P9OUT4	P9OUT3	P9OUT2	P9OUT1	P9OUT0	VIII-175
		-	-	x	x	x	x	x	x	
		Port 9 output data								
0x03E7A	PAOUT	PAOUT7	PAOUT6	PAOUT5	PAOUT4	PAOUT3	PAOUT2	PAOUT1	PAOUT0	VIII-191
		X	X	X	X	X	X	X	X	
		Port A output data								
0x03E7B	PBOUT	PBOUT7	PBOUT6	PBOUT5	PBOUT4	PBOUT3	PBOUT2	PBOUT1	PBOUT0	VIII-211
		X	X	X	X	x	x	x	x	
		Port B output data								
0x03E7C	PCOUT *	-	PCOUT6	PCOUT5	PCOUT4	PCOUT3	PCOUT2	PCOUT1	PCOUT0	VIII-221
		-	X	X	X	x	x	x	x	
		Port B output data								
0x03E7D	PDOUT *	-	PDOUT6	PDOUT5	PDOUT4	PDOUT3	PDOUT2	PDOUT1	PDOUT0	VIII-228
		-	X	X	X	x	x	x	x	
		Port B output data								
0x03E80	P0IN	P0IN7	P0IN6	P0IN5	P0IN4	P0IN3	P0IN2	P0IN1	P0IN0	VIII-9
		X	X	X	X	X	X	X	X	
		Port 0 input data								
0x03E81	P1IN	-	P1IN6	P1IN5	P1IN4	P1IN3	P1IN2	P1IN1	P1IN0	VIII-33
		-	X	X	X	X	X	X	X	
		Port 1 input data								
0x03E82	P2IN	P2IN7	P2IN6	P2IN5	P2IN4	P2IN3	P2IN2	P2IN1	P2IN0	VIII-45
		X	X	X	X	X	X	X	X	
		Port 2 input data								
0x03E83	P3IN	-	P3IN6	P3IN5	P3IN4	P3IN3	P3IN2	P3IN1	P3IN0	VIII-56
		-	X	X	X	X	X	X	X	
		Port 3 input data								
0x03E84	P4IN	P4IN7	P4IN6	P4IN5	P4IN4	P4IN3	P4IN2	P4IN1	P4IN0	VIII-73
		X	X	X	X	X	X	X	X	
		Port 4 input data								
0x03E85	P5IN	P5IN7	P5IN6	P5IN5	P5IN4	P5IN3	P5IN2	P5IN1	P5IN0	VIII-93
		X	X	X	X	X	X	X	X	
		Port 5 input data								

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Address	Register	Bit symbol								Page
		bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0	
0x03E86	P6IN	P6IN7	P6IN6	P6IN5	P6IN4	P6IN3	P6IN2	P6IN1	-	VIII-112
		X	X	X	X	X	X	X	-	
		Port 6 input data								
0x03E87	P7IN	P7IN7	P7IN6	P7IN5	P7IN4	P7IN3	P7IN2	P7IN1	P7IN0	VIII-131
		X	X	X	X	X	X	X	X	
		Port 7 input data								
0x03E88	P8IN	P8IN7	P8IN6	P8IN5	P8IN4	P8IN3	P8IN2	P8IN1	P8IN0	VIII-153
		X	X	X	X	X	X	X	X	
		Port 8 input data								
0x03E89	P9IN	-	-	P9IN5	P9IN4	P9IN3	P9IN2	P9IN1	P9IN0	VIII-176
		-	-	X	X	X	X	X	X	
		Port 9 input data								
0x03E8A	PAIN	PAIN7	PAIN6	PAIN5	PAIN4	PAIN3	PAIN2	PAIN1	PAIN0	VIII-191
		X	X	X	X	X	X	X	X	
		Port A input data								
0x03E8B	PBIN	-	-	-	-	PBIN3	PBIN2	PBIN1	PBIN0	VIII-212
		-	-	-	-	X	X	X	X	
		Port B input data								
0x03E8C	PCIN *	-	PCIN6	PCIN5	PCIN4	PCIN3	PCIN2	PCIN1	PCIN0	VIII-221
		-	X	X	X	X	X	X	X	
		Port C input data								
0x03E8D	PDIN *	-	PDIN6	PDIN5	PDIN4	PDIN3	PDIN2	PDIN1	PDIN0	VIII-228
		-	X	X	X	X	X	X	X	
		Port D input data								
0x03E8F	SWCNT	-	-	-	-	-	SWCNT_A	-	-	VIII-195 VIII-243
		-	-	-	-	-	0	-	-	
		-	-	-	-	-	Port A Input voltage level selection	-	-	
0x03E90	P0DIR	P0DIR7	P0DIR6	P0DIR5	P0DIR4	P0DIR3	P0DIR2	P0DIR1	P0DIR0	VIII-9
		0	0	0	0	0	0	0	0	
		Port 0 direction control								
0x03E91	P1DIR	-	P1DIR6	P1DIR5	P1DIR4	P1DIR3	P1DIR2	P1DIR1	P1DIR0	VIII-33
		-	0	0	0	0	0	0	0	
		Port 1 direction control								
0x03E92	P2DIR	-	P2DIR6	P2DIR5	P2DIR4	P2DIR3	P2DIR2	P2DIR1	P2DIR0	VIII-46
		-	0	0	0	0	0	0	0	
		Port 2 direction control								
0x03E93	P3DIR	-	P3DIR6	P3DIR5	P3DIR4	P3DIR3	P3DIR2	P3DIR1	P3DIR0	VIII-57
		-	0	0	0	0	0	0	0	
		Port 3 direction control								
0x03E94	P4DIR	P4DIR7	P4DIR6	P4DIR5	P4DIR4	P4DIR3	P4DIR2	P4DIR1	P4DIR0	VIII-74
		0	0	0	0	0	0	0	0	
		Port 4 direction control								
0x03E95	P5DIR	P5DIR7	P5DIR6	P5DIR5	P5DIR4	P5DIR3	P5DIR2	P5DIR1	P5DIR0	VIII-94
		0	0	0	0	0	0	0	0	
		Port 5 direction control								
0x03E96	P6DIR	P6DIR7	P6DIR6	P6DIR5	P6DIR4	P6DIR3	P6DIR2	P6DIR1	-	VIII-113
		0	0	0	0	0	0	0	-	
		Port 6 direction control								
0x03E97	P7DIR	P7DIR7	P7DIR6	P7DIR5	P7DIR4	P7DIR3	P7DIR2	P7DIR1	P7DIR0	VIII-132
		0	0	0	0	0	0	0	0	
		Port 7 direction control								
0x03E98	P8DIR	P8DIR7	P8DIR6	P8DIR5	P8DIR4	P8DIR3	P8DIR2	P8DIR1	P8DIR0	VIII-154
		0	0	0	0	0	0	0	0	
		Port 8 direction control								

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Address	Register	Bit symbol								Page
		bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0	
0x03E99	P9DIR	-	-	P9DIR5	P9DIR4	P9DIR3	P9DIR2	P9DIR1	P9DIR0	VIII-177
		-	-	0	0	0	0	0	0	
		Port 9 direction control								
0x03E9A	PADIR	PADIR7	PADIR6	PADIR5	PADIR4	PADIR3	PADIR2	PADIR1	PADIR0	VIII-192
		0	0	0	0	0	0	0	0	
		Port A direction control								
0x03E9B	PBDIR	PBDIR7	PBDIR6	PBDIR5	PBDIR4	PBDIR3	PBDIR2	PBDIR1	PBDIR0	VIII-213
		0	0	0	0	0	0	0	0	
		Port B direction control								
0x03E9C	PCDIR *	-	PCDIR6	PCDIR5	PCDIR4	PCDIR3	PCDIR2	PCDIR1	PCDIR0	VIII-222
		-	0	0	0	0	0	0	0	
		Port C direction control								
0x03E9D	PDDIR *	-	PDDIR6	PDDIR5	PDDIR4	PDDIR3	PDDIR2	PDDIR1	PDDIR0	VIII-229
		-	0	0	0	0	0	0	0	
		Port D direction control								
0x03EA0	P0PLUD	P0PLUD7	P0PLUD6	P0PLUD5	P0PLUD4	P0PLUD3	P0PLUD2	P0PLUD1	P0PLUD0	VIII-9
		0	0	0	0	0	0	0	0	
		Port 0 pull-up/pull-down resistor control								
0x03EA1	P1PLUD	P1PLUD7	P1PLUD6	P1PLUD5	P1PLUD4	P1PLUD3	P1PLUD2	P1PLUD1	P1PLUD0	VIII-33
		0	0	0	0	0	0	0	0	
		Port 1 pull-up/pull-down resistor control								
0x03EA2	P2PLU	-	P2PLU6	P2PLU5	P2PLU4	P2PLU3	P2PLU2	P2PLU1	P2PLU0	VIII-46
		-	0	0	0	0	0	0	0	
		Port 2 pull-up resistor control								
0x03EA3	P3PLU	-	P3PLU6	P3PLU5	P3PLU4	P3PLU3	P3PLU2	P3PLU1	P3PLU0	VIII-58
		-	0	0	0	0	0	0	0	
		Port 3 pull-up resistor control								
0x03EA4	P4PLUD	P4PLUD7	P4PLUD6	P4PLUD5	P4PLUD4	P4PLUD3	P4PLUD2	P4PLUD1	P4PLUD0	VIII-75
		0	0	0	0	0	0	0	0	
		Port 4 pull-up/pull-down resistor control								
0x03EA5	P5PLUD	P5PLUD7	P5PLUD6	P5PLUD5	P5PLUD4	P5PLUD3	P5PLUD2	P5PLUD1	P5PLUD0	VIII-94
		0	0	0	0	0	0	0	0	
		Port 5 pull-up/pull-down resistor control								
0x03EA6	P6PLU	P6PLU7	P6PLU6	P6PLU5	P6PLU4	P6PLU3	P6PLU2	P6PLU1	-	VIII-114
		0	0	0	0	0	0	0	-	
		Port 6 pull-up resistor control								
0x03EA7	P7PLUD	P7PLUD7	P7PLUD6	P7PLUD5	P7PLUD4	P7PLUD3	P7PLUD2	P7PLUD1	P7PLUD0	VIII-132
		0	0	0	0	0	0	0	0	
		Port 7 pull-up/pull-down resistor control								
0x03EA8	P8PLU	P8PLU7	P8PLU6	P8PLU5	P8PLU4	P8PLU3	P8PLU2	P8PLU1	P8PLU0	VIII-154
		0	0	0	0	0	0	0	0	
		Port 8 pull-up resistor control								
0x03EA9	P9PLU	-	-	P9PLU5	P9PLU4	P9PLU3	P9PLU2	P9PLU1	P9PLU0	VIII-178
		-	-	0	0	0	0	0	0	
		Port 9 pull-up resistor control								
0x03EAA	PAPLU	PAPLU7	PAPLU6	PAPLU5	PAPLU4	PAPLU3	PAPLU2	PAPLU1	PAPLU0	VIII-192
		0	0	0	0	0	0	0	0	
		Port A pull-up resistor control								
0x03EAB	PBPLU	PBPLU7	PBPLU6	PBPLU5	PBPLU4	PBPLU3	PBPLU2	PBPLU1	PBPLU0	VIII-214
		0	0	0	0	0	0	0	0	
		Port B pull-up resistor control								
0x03EAC	PCPLU *	PCPLU7	PCPLU6	PCPLU5	PCPLU4	PCPLU3	PCPLU2	PCPLU1	PCPLU0	VIII-222
		0	0	0	0	0	0	0	0	
		Port C pull-up resistor control								
0x03EAD	PDPLU *	PDPLU7	PDPLU6	PDPLU5	PDPLU4	PDPLU3	PDPLU2	PDPLU1	PDPLU0	VIII-229
		0	0	0	0	0	0	0	0	
		Port D pull-up resistor control								

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Address	Register	Bit symbol								Page
		bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0	
0x03EAF	SELUD	SELUD7	-	SELUD5	SELUD4	-	-	SELUD1	SELUD0	VIII-13 VIII-36 VIII-77 VIII-96 VIII-134
		0	-	0	0	-	-	0	0	
		Pull-up/Pull-down resistor selection								
0x03EB0	P0OMD1	-	-	-	P0OMD14	P0OMD13	P0OMD12	P0OMD11	-	VIII-10
		-	-	-	0	0	0	0	-	
		-	-	-	Port/TM2IOB/ TM0IOB selec- tion	Port/TM8IOB selection	Port/TM7IOB selection	Port/TM9IOB selection	-	
0x03EB1	P1OMD1	-	P1OMD16	P1OMD15	P1OMD14	P1OMD13	P1OMD12	P1OMD11	P1OMD10	VIII-34
		-	0	0	0	0	0	0	0	
		-	Port/TM8IOC/ NBUZZERB selection	Port/TM7IOC/ BUZZERB selection	Port/TM4IOC selection	Port/TM3IOC selection	Port/TM2IOC selection	Port/TM1IOC selection	Port/TM0IOC selection	
0x03EB5	P5OMD	P5OMD7	P5OMD6	-	-	-	-	-	-	VIII-95 XVII-4
		0	0	-	-	-	-	-	-	
		Port/BUZZERA selection	Port/ NBUZZERA selection	-	-	-	-	-	-	
0x03EB6	P6OMD	-	-	-	P6OMD4	P6OMD3	P6OMD2	-	-	VIII-115
		-	-	-	0	0	0	-	-	
		-	-	-	Port/TM4IOB selection	Port/TM3IOB selection	Port/TM1IOB selection	-	-	
0x03EB8	P8OMD	-	-	P8OMD5	P8OMD4	P8OMD3	P8OMD2	P8OMD1	P8OMD0	VIII-155
		-	-	0	0	0	0	0	0	
		-	-	Port/TM9OD5A selection	Port/TM9OD4A selection	Port/TM9OD3A selection	Port/TM9OD2A selection	Port/TM9OD1A selection	Port/TM9OD0A selection	
0x03EBA	PAOMD1	PAOMD7	PAOMD6	PAOMD5	PAOMD4	PAOMD3	PAOMD2	PAOMD1	PAOMD0	VIII-193
		0	0	0	0	0	0	0	0	
		Port/TM9IOA selection	Port/TM8IOA selection	Port/TM7IOA selection	Port/TM4IOA selection	Port/TM3IOA selection	Port/TM2IOA selection	Port/TM1IOA selection	Port/TM0IOA selection	
0x03EBD	PDOMD *	-	-	PDOMD5	PDOMD4	PDOMD3	PDOMD2	PDOMD1	PDOMD0	VIII-230
		-	-	0	0	0	0	0	0	
		-	-	Port/TM9OD5B selection	Port/TM9OD4B selection	Port/TM9OD3B selection	Port/TM9OD2B selection	Port/TM9OD1B selection	Port/TM9OD0B selection	
0x03EC0	P0OMD2	-	-	-	P0OMD24	-	-	-	-	VIII-10
		-	-	-	0	-	-	-	-	
		-	-	-	Special func- tion selection (TM0IOB/ TM2IOB)	-	-	-	-	
0x03EC1	P1OMD2	-	P1OMD26	P1OMD25	-	-	-	-	-	VIII-35
		-	0	0	-	-	-	-	-	
		-	Special function selection (TM8IOC/ NBUZZERB)	Special function selection (TM7IOC/ BUZZERB)	-	-	-	-	-	
0x03EC9	P9IMD	-	-	P9IMD5	P9IMD4	P9IMD3	P9IMD2	-	-	VIII-179
		-	-	0	0	0	0	-	-	
		Port/analog input selection								
0x03ECA	PAIMD	PAIMD7	PAIMD6	PAIMD5	PAIMD4	PAIMD3	PAIMD2	PAIMD1	PAIMD0	VIII-194
		0	0	0	0	0	0	0	0	
		Port/analog input selection								
0x03ECB	PBIMD	PBIMD7	PBIMD6	PBIMD5	PBIMD4	PBIMD3	PBIMD2	PBIMD1	PBIMD0	VIII-215
		0	0	0	0	0	0	0	0	
		Port/analog input selection								
0x03ED1	P1IMD	-	-	-	-	P1IMD3	P1IMD2	P1IMD1	P1IMD0	VIII-35
		-	-	-	-	0	0	0	0	
		Port/analog input selection								
0x03EDA	PACNT	-	-	PACNT5	PACNT4	PACNT3	PACNT2	PACNT1	PACNT0	VIII-195
		-	-	0	0	0	0	0	0	
		Port A real-time control								

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Address	Register	Bit symbol								Page
		bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0	
0x03EE0	POLED	POLED7	POLED6	POLED5	POLED4	POLED3	POLED2	POLED1	POLED0	VIII-12
		0	0	0	0	0	0	0	0	
0x03EF0	P0ODC	LED output selection								VIII-11
		P0ODC7	-	P0ODC5	P0ODC4	P0ODC3	-	-	-	
		0	-	0	0	0	-	-	-	
0x03EE3	P3ODC	Nch open-drain output selection								VIII-59
		-	-	-	P3ODC4	P3ODC3	P3ODC2	-	P3ODC0	
		-	-	-	0	0	0	-	0	
0x03EE4	P4ODC	Nch open-drain output selection								VIII-76
		-	-	P4ODC5	-	P4ODC3	P4ODC2	-	P4ODC0	
		-	-	0	-	0	0	-	0	
0x03EE5	P5ODC	Nch open-drain output selection								VIII-94
		-	-	-	-	-	P5ODC2	-	P5ODC0	
		-	-	-	-	-	0	-	0	
0x03EE6	P6ODC	Nch open-drain output selection								VIII-115
		P6ODC7	-	P6ODC5	-	-	-	-	-	
		0	-	0	-	-	-	-	-	
0x03EE7	P7ODC	Nch open-drain output selection								VIII-133
		P7ODC7	-	P7ODC5	-	-	P7ODC2	P7ODC1	-	
		0	-	0	-	-	0	0	-	
0x03EF8	P8SYO	Nch open-drain output selection								VIII-156
		P8SYO7	P8SYO6	P8SYO5	P8SYO4	P8SYO3	P8SYO2	P8SYO1	P8SYO0	
		0	0	0	0	0	0	0	0	
0x03EF9	P8SEV	Synchronous output selection								VIII-157
		-	-	-	-	-	-	P8SEV1	P8SEV0	
		-	-	-	-	-	-	0	0	
0x03F00	CPUM	Synchronous output event selection								III-22
		Reserved	OSCSEL1	OSCSEL0	OSCDL	STOP	HALT	OSC1	OSC0	
		0	0	1	0	0	0	0	0	
0x03F01	MEMCTR	Always set to "0"	Division ratio		Internal system clock	Operation mode control				II-38
		IOW1	IOW0	IVBM	Reserved	Reserved	IRWE	Reserved	Reserved	
		1	1	0	0	1	0	1	1	
0x03F02	WDCTR	Wait cycles when accessing special register area		Base address specification for interrupt vector table	Always set to "0"	Always set to "1"	Software write setting for interrupt request flag	Always set to "1"		XV-5
		-	-	Reserved	Reserved	Reserved	WDTS1	WDTS0	WDEN	
		-	-	0	0	0	1	1	0	
0x03F03	DLYCTR	-	-	Always set to "0"	Always set to "0"	Always set to "0"	Watchdog error detect cycle		Watchdog timer enable	XV-6
		BUZOE	BUZS2	BUZS1	BUZS0	DLYS1	DLYS0	-	-	
		0	0	0	0	0	1	-	-	
0x03F05	WD2CTR	Buzzer output enable	Buzzer output frequency			Oscillation stabilization wait cycle		-	-	XVI-5
		Reserved	Reserved	-	-	WDTS23	WDTS22	WDTS21	WDTS20	
		0	0	-	-	1	0	1	1	
0x03F06	HANDSHAKE	Always set to "0"	Always set to "0"	-	-	Watchdog timer 2 cycle setup				III-24
		-	-	-	-	-	ROMHND	-	-	
		-	-	-	-	-	1	-	-	
0x03F07	AUCTR	-	-	-	-	-	Internal ROM access method switching	-	-	II-41
		-	-	-	-	-	AUDIVU	AUMUL	AUMULU	
		-	-	-	-	-	0	0	0	
0x03F07	AUCTR	-	-	-	-	-	Unsigned division execution	Signed multiplication execution	Unsigned multiplication execution	II-41
		-	-	-	-	-	0	0	0	
		-	-	-	-	-	Unsigned division execution	Signed multiplication execution	Unsigned multiplication execution	

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Address	Register	Bit symbol								Page
		bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0	
0x03F0A	SBNKR	-	-	-	-	SBA3	SBA2	SBA1	SBA0	II-21
		-	-	-	-	0	0	0	0	
		-	-	-	-	Bank setting for source address				
0x03F0B	DBNKR	-	-	-	-	DBA3	DBA2	DBA1	DBA0	II-22
		-	-	-	-	0	0	0	0	
		-	-	-	-	Bank setting for destination address				
0x03F10	SC0SEL	SBO0SEL	SC0BRP2	SC0BRP1	SC0BRP0	OSL0	SC0SEL2	SC0SEL1	SC0SEL0	XVIII-19
		0	0	0	0	0	0	0	0	
		UART reverse output selection	Timer clock output selection			Serial output port selection	Timer selection			
0x03F11	SC0MD0	SC0CE1	SC0SSC	Reserved	SC0DIR	SC0STE	SC0LNG2	SC0LNG1	SC0LNG0	XVIII-26
		0	0	0	0	0	1	1	1	
		Communication data I/O edge	Internal operating clock source	Always set to "0"	Transfer bit specify	Start condition selection	Number of synchronous serial transfer bit			
0x03F12	SC0MD1	SC0IOM	SC0SBTS	SC0SBIS	SC0SBOS	SC0CKM	SC0MST	SC0DIV	SC0CMD	XVIII-28
		0	0	0	0	0	0	0	0	
		Serial data input pin selection	SBT function selection	Serial input control	SBO (TXD) function selection	Transfer clock dividing selection	Clock master/slave selection	Transfer clock dividing value selection	Clock synchronous/UART selection	
0x03F13	SC0MD2	SC0FM1	SC0FM0	SC0PM1	SC0PM0	SC0NPE	Reserved	SC0BRKF	SC0BRKE	XVIII-29
		0	0	0	0	0	0	0	0	
		Frame mode specify		Additional bit specify		Parity enable	Always set to "0"	Break status reception monitor	Break status transmission control	
0x03F14	SC0MD3	SC0FDC1	SC0FDC0	-	-	SC0PSCE	SC0PSC2	SC0PSC1	SC0PSC0	XVIII-30
		0	0	-	-	0	0	0	0	
		Output selection after SBO final data transmit		-	-	Prescaler count control	Clock selection			
0x03F15	SC0STR	SC0TBSY	SC0RBSY	SC0TEMP	SC0REMP	SC0FEF	SC0PEK	SC0ORE	SC0ERE	XVIII-31
		0	0	0	0	0	0	0	0	
		Serial bus transmission status	Serial bus reception status	Transmission buffer empty flag	Reception buffer empty flag	Frame error detection	Parity error detection	Overrun error detection	Error monitor flag	
0x03F16	RXBUF0	RXBUF07	RXBUF06	RXBUF05	RXBUF04	RXBUF03	RXBUF02	RXBUF01	RXBUF00	XVIII-25
		x	x	x	x	x	x	x	x	
		Serial interface 0 reception data buffer								
0x03F17	TXBUF0	TXBUF07	TXBUF06	TXBUF05	TXBUF04	TXBUF03	TXBUF02	TXBUF01	TXBUF00	XVIII-25
		x	x	x	x	x	x	x	x	
		Serial interface 0 transmission data buffer								
0x03F18	LINCTR	SFBSY	SBFBSY	SFREC	SBFREC	SBFEN	SBFSEL2	SBFSEL1	SBFSEL0	XIX-5
		0	0	0	0	0	0	0	0	
		SF reception busy flag	SBF reception busy flag	SF reception enable	SBF reception enable	SBF transmission trigger/monitor	SBF transmission width selection			
0x03F19	RSTFACT	-	-	-	-	-	-	-	RSTMON	II-54
		-	-	-	-	-	-	-	0	
		-	-	-	-	-	-	-	Reset factor determination	
0x03F1A	OSCCNT	HOSCDIVSEL2	HOSCDIVSEL1	HOSCDIVSEL0	HOSCDIVS-TOP	-	-	HOSCSEL	HOSCCNT	III-10
		0	0	0	0	-	-	0	0	
		Division setting of high-speed oscillation for peripheral functions			High-speed oscillation for peripheral functions control	-	-	Operation clock selection	P25/P26 function selection	
0x03F1B	RCCNT	Reserved	-	-	-	-	-	RCFSEL	RCON	III-12
		1	-	-	-	-	-	0	0	
		Always set to "1"	-	-	-	-	-	Internal high-speed oscillation frequency selection	Internal high-speed oscillation control	

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Address	Register	Bit symbol								Page
		bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0	
0x03F1C	OSCSCNT	-	-	-	-	-	-	SOSCCSEL	SOSCCNT	III-14
		-	-	-	-	-	-	0	0	
		-	-	-	-	-	-	Operation clock selection	P90/P91 function selection	
0x03F1D	RCSCNT	-	-	-	-	-	-	-	RCON	III-16
		-	-	-	-	-	-	-	0	
		-	-	-	-	-	-	-	Internal low-speed oscillation control	
0x03F1E	EDGDT	EDGSEL7	-	EDGSEL5	-	EDGSEL3	EDGSEL2	-	EDGSEL0	VII-56
		0	-	0	-	0	0	-	0	
		IRQ4 both edges operation selection	-	LNIRQ both edges operation selection	-	IRQ3 both edges operation selection	IRQ2 both edges operation selection	-	Key interrupt selection	
0x03F1F	PLLCNT	PLLCK3	PLLCK2	PLLCK1	PLLCK0	-	-	PLLEN	PLLSTART	III-20
		0	0	0	0	-	-	0	0	
		Multiplication setting				-	-	PLL clock enable	PLL operation control	
0x03F20	SC1SEL	SBO1SEL	SC1BRP2	SC1BRP1	SC1BRP0	OSL1	SC1SEL2	SC1SEL1	SC1SEL0	XVIII-20
		0	0	0	0	0	0	0	0	
		UART reverse output selection	Timer clock output selection			Serial output port selection	Timer selection			
0x03F21	SC1MD0	SC1CE1	SC1SSC	Reserved	SC1DIR	SC1STE	SC1LNG2	SC1LNG1	SC1LNG0	XVIII-26
		0	0	0	0	0	1	1	1	
		Communication data I/O edge	Internal operating clock source	Always set to "0"	Transfer bit specify	Start condition selection	Number of synchronous serial transfer bit			
0x03F22	SC1MD1	SC1IOM	SC1SBTS	SC1SBIS	SC1SBOS	SC1CKM	SC1MST	SC1DIV	SC1CMD	XVIII-28
		0	0	0	0	0	0	0	0	
		Serial data input pin selection	SBT function selection	Serial input control	SBO (TXD) function selection	Transfer clock dividing selection	Clock master/slave selection	Transfer clock dividing value selection	Clock synchronous/UART selection	
0x03F23	SC1MD2	SC1FM1	SC1FM0	SC1PM1	SC1PM0	SC1NPE	Reserved	SC1BRKF	SC1BRKE	XVIII-29
		0	0	0	0	0	0	0	0	
		Frame mode specify		Additional bit specify		Parity enable	Always set to "0"	Break status reception monitor	Break status transmission control	
0x03F24	SC1MD3	SC1FDC1	SC1FDC0	-	-	SC1PSCE	SC1PSC2	SC1PSC1	SC1PSC0	XVIII-30
		0	0	-	-	0	0	0	0	
		Output selection after SBO final data transmit		-	-	Prescaler count control	Clock selection			
0x03F25	SC1STR	SC1TBSY	SC1RBSY	SC1TEMP	SC1REMP	SC1FEF	SC1PEK	SC1ORE	SC1ERE	XVIII-31
		0	0	0	0	0	0	0	0	
		Serial bus transmission status	Serial bus reception status	Transmission buffer empty flag	Reception buffer empty flag	Frame error detection	Parity error detection	Overrun error detection	Error monitor flag	
0x03F26	RXBUF1	RXBUF17	RXBUF16	RXBUF15	RXBUF14	RXBUF13	RXBUF12	RXBUF11	RXBUF10	XVIII-25
		x	x	x	x	x	x	x	x	
		Serial interface 1 reception data buffer								
0x03F27	TXBUF1	TXBUF17	TXBUF16	TXBUF15	TXBUF14	TXBUF13	TXBUF12	TXBUF11	TXBUF10	XVIII-25
		x	x	x	x	x	x	x	x	
		Serial interface 1 transmission data buffer								
0x03F28	SCINTSEL	-	-	-	-	-	-	SC1IGC	SC0IGC	XVIII-33
		-	-	-	-	-	-	0	0	
		-	-	-	-	-	-	Serial 1 reception interrupt generation timing	Serial 0 reception interrupt generation timing	
0x03F2E	ACZCTR	P21IM	-	-	-	P20IM	-	-	-	VII-55
		0	-	-	-	0	-	-	-	
		ACZ1 input enable	-	-	-	ACZ0 input enable	-	-	-	

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Address	Register	Bit symbol								Page
		bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0	
0x03F2F	LVIMD	LVIMOD	LVICTR2	LVICTR1	LVICTR0	-	-	LVIMON	LVION	VI-3
		0	0	0	0	-	-	0	0	
		Power supply voltage detection voltage setting				-	-	Power supply voltage monitor flag	Power supply detection enable	
0x03F30	SC2SEL	SBO2SEL	SC2BRP2	SC2BRP1	SC2BRP0	OSL2	SC2SEL2	SC2SEL1	SC2SEL0	XVIII-21
		0	0	0	0	0	0	0	0	
		UART reverse output selection	Timer clock output selection			Serial output port selection	Timer selection			
0x03F31	SC2MD0	SC2CE1	SC2SSC	Reserved	SC2DIR	SC2STE	SC2LNG2	SC2LNG1	SC2LNG0	XVIII-26
		0	0	0	0	0	1	1	1	
		Communication data I/O edge	Internal operating clock source	Always set to "0"	Transfer bit specify	Start condition selection	Number of synchronous serial transfer bit			
0x03F32	SC2MD1	SC2IOM	SC2SBTS	SC2SBIS	SC2SBOS	SC2CKM	SC2MST	SC2DIV	SC2CMD	XVIII-28
		0	0	0	0	0	0	0	0	
		Serial data input pin selection	SBT function selection	Serial input control	SBO (TXD) function selection	Transfer clock dividing selection	Clock master/slave selection	Transfer clock dividing value selection	Clock synchronous/UART selection	
0x03F33	SC2MD2	SC2FM1	SC2FM0	SC2PM1	SC2PM0	SC2NPE	Reserved	SC2BRKF	SC2BRKE	XVIII-29
		0	0	0	0	0	0	0	0	
		Frame mode specify		Additional bit specify		Parity enable	Always set to "0"	Break status reception monitor	Break status transmission control	
0x03F34	SC2MD3	SC2FDC1	SC2FDC0	-	-	SC2PSCE	SC2PSC2	SC2PSC1	SC2PSC0	XVIII-30
		0	0	-	-	0	0	0	0	
		Output selection after SBO final data transmit		-	-	Prescaler count control	Clock selection			
0x03F35	SC2STR	SC2TBSY	SC2RBSY	SC2TEMP	SC2REMP	SC2FEF	SC2PEK	SC2ORE	SC2ERE	XVIII-31
		0	0	0	0	0	0	0	0	
		Serial bus transmission status	Serial bus reception status	Transmission buffer empty flag	Reception buffer empty flag	Frame error detection	Parity error detection	Overrun error detection	Error monitor flag	
0x03F36	RXBUF2	RXBUF27	RXBUF26	RXBUF25	RXBUF24	RXBUF23	RXBUF22	RXBUF21	RXBUF20	XVIII-25
		x	x	x	x	x	x	x	x	
		Serial interface 2 reception data buffer								
0x03F37	TXBUF2	TXBUF27	TXBUF26	TXBUF25	TXBUF24	TXBUF23	TXBUF22	TXBUF21	TXBUF20	XVIII-25
		x	x	x	x	x	x	x	x	
		Serial interface 2 transmission data buffer								
0x03F3E	KEYT3_1IMD	KEYT3SEL	-	-	-	KEYT3_1EN3	KEYT3_1EN2	KEYT3_1EN1	KEYT3_1EN0	VII-58
		0	-	-	-	0	0	0	0	
		Interrupt source selection	-	-	-	KEY interrupt selection				
0x03F3F	KEYT3_2IMD	-	-	-	-	KEYT3_2EN3	KEYT3_2EN2	KEYT3_2EN1	KEYT3_2EN0	VII-59
		-	-	-	-	0	0	0	0	
		KEY interrupt selection								
0x03F40	SC3SEL	SBO3SEL	SC3BRP2	SC3BRP1	SC3BRP0	OSL3	SC3SEL2	SC3SEL1	SC3SEL0	XVIII-23
		0	0	0	0	0	0	0	0	
		UART reverse output selection	Timer clock output selection			Serial output port selection	Timer selection			
0x03F41	SC3MD0	SC3CE1	SC3SSC	Reserved	SC3DIR	SC3STE	SC3LNG2	SC3LNG1	SC3LNG0	XVIII-26
		0	0	0	0	0	1	1	1	
		Communication data I/O edge	Internal operating clock source	Always set to "0"	Transfer bit specify	Start condition selection	Number of synchronous serial transfer bit			
0x03F42	SC3MD1	SC3IOM	SC3SBTS	SC3SBIS	SC3SBOS	SC3CKM	SC3MST	SC3DIV	SC3CMD	XVIII-28
		0	0	0	0	0	0	0	0	
		Serial data input pin selection	SBT function selection	Serial input control	SBO (TXD) function selection	Transfer clock dividing selection	Clock master/slave selection	Transfer clock dividing value selection	Clock synchronous/UART selection	

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Address	Register	Bit symbol								Page
		bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0	
0x03F43	SC3MD2	SC3FM1	SC3FM0	SC3PM1	SC3PM0	SC3NPE	Reserved	SC3BRKF	SC3BRKE	XVIII-29
		0	0	0	0	0	0	0	0	
		Frame mode specify		Additional bit specify		Parity enable	Always set to "0"	Break status reception monitor	Break status transmission control	
0x03F44	SC3MD3	SC3FDC1	SC3FDC0	-	-	SC3PSCE	SC3PSC2	SC3PSC1	SC3PSC0	XVIII-30
		0	0	-	-	0	0	0	0	
		Output selection after SBO final data transmit		-	-	Prescaler count control	Clock selection			
0x03F45	SC3STR	SC3TBSY	SC3RBSY	SC3TEMP	SC3REMP	SC3FEF	SC3PEK	SC3ORE	SC3ERE	XVIII-31
		0	0	0	0	0	0	0	0	
		Serial bus transmission status	Serial bus reception status	Transmission buffer empty flag	Reception buffer empty flag	Frame error detection	Parity error detection	Overrun error detection	Error monitor flag	
0x03F46	RXBUF3	RXBUF37	RXBUF36	RXBUF35	RXBUF34	RXBUF33	RXBUF32	RXBUF31	RXBUF30	XVIII-25
		x	x	x	x	x	x	x	x	
		Serial interface 3 reception data buffer								
0x03F47	TXBUF3	TXBUF37	TXBUF36	TXBUF35	TXBUF34	TXBUF33	TXBUF32	TXBUF31	TXBUF30	XVIII-25
		x	x	x	x	x	x	x	x	
		Serial interface 3 transmission data buffer								
0x03F4C	WD2MDSEL	WD2MD7	WD2MD6	WD2MD5	WD2MD4	WD2MD3	WD2MD2	WD2MD1	WD2MD0	XVI-6
		1	0	1	0	1	0	1	0	
		WDT2 operation mode setup								
0x03F4D	WD2CLR	WD2CL7	WD2CL6	WD2CL5	WD2CL4	WD2CL3	WD2CL2	WD2CL1	WD2CL0	XVI-8
		0	0	0	0	0	0	0	0	
		Watchdog timer 2 counter clear								
0x03F4E	IRQEXPEN	Reserved	IRQEXPEN6	IRQEXPEN5	IRQEXPEN4	IRQEXPEN3	IRQEXPEN2	IRQEXPEN1	IRQEXPEN0	VII-38
		0	0	0	0	0	0	0	0	
		Always set to "0"	ATC1 interrupt enable	A/D conversion interrupt enable	24H timer alarm interrupt enable	PWM underflow interrupt enable	PWM overflow interrupt enable	TM8 compare 2 match interrupt enable	TM7 compare 2 match interrupt enable	
0x03F4F	IRQEXPDT	Reserved	IRQEXPDT6	IRQEXPDT5	IRQEXPDT4	IRQEXPDT3	IRQEXPDT2	IRQEXPDT1	IRQEXPDT0	VII-39
		0	0	0	0	0	0	0	0	
		Always set to "0"	ATC1 interrupt request	A/D conversion interrupt request	24H timer alarm interrupt request	PWM underflow interrupt request	PWM overflow interrupt request	TM8 compare 2 match interrupt request	TM7 compare 2 match interrupt request	
0x03F50	SC4MD0	SC4CE1	-	SC4DEM	SC4DIR	SC4STE	SC4LNG2	SC4LNG1	SC4LNG0	XVIII-35
		0	-	0	0	0	1	1	1	
		Communication data I/O edge	-	Operation selection after detecting instability of IIC slave communication	Transfer bit specify	Start condition selection	Number of transfer bit			
0x03F51	SC4MD1	SC4IOM	SC4SBTS	SC4SBIS	SC4SBOS	-	SC4MST	-	-	XVIII-36
		0	0	0	0	-	0	-	-	
		Serial data input pin selection	SBT function selection	Serial input control	SB0 (SDA) function selection	-	Clock master/slave selection	-	-	
0x03F52	SC4MD2	SC4FDC1	SC4FDC0	-	-	SC4PSCE	SC4PSC2	SC4PSC1	SC4PSC0	XVIII-37
		0	0	-	-	0	0	0	0	
		Output selection after SBO final data transmit		-	-	Prescaler count control	Clock selection			
0x03F53	SC4MD3	Reserved	SC4ADM	SC4STPC	SC4TMD	SC4REX	SC4CMD	SC4ACKS	SC4ACK0	XVIII-38
		0	0	0	0	0	0	0	0	
		Always set to "0"	Address format selection	Stop condition generation flag in IIC	Communication mode selection in IIC	Transmission/reception selection in IIC master communication	Clock synchronous/IIC selection	ACK bit enable	ACK bit selection in transmission/reception mode	
0x03F54	SC4AD0	SC4ADR7	SC4ADR6	SC4ADR5	SC4ADR4	SC4ADR3	SC4ADR2	SC4ADR1	SC4ADR0	XVIII-39
		0	0	0	0	0	0	0	0	
		Serial interface 4 address setting register 0								

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Address	Register	Bit symbol								Page
		bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0	
0x03F55	SC4AD1	-	-	-	-	-	-	SC4ADR9	SC4ADR8	XVIII-39
		-	-	-	-	-	-	0	0	
		Serial interface 4 address setting register 1								
0x03F56	SC4STR0	SC4BSY	-	SC4TEMP	SC4REMP	-	-	-	SC4ORE	XVIII-40
		0	-	0	0	-	-	-	0	
		Serial bus status in clock synchronous	-	Transmission buffer empty flag	Reception buffer empty flag	-	-	-	Overrun error detection	
0x03F57	SC4STR1	SC4WRS	SC4ABT_LST	SC4ADD_ACC	SC4STRT	SC4BUSBSY	SC4IICBSY	SC4GCALL	SC4DATA_ER R	XVIII-41
		0	0	0	0	0	0	0	0	
		Data transfer determination flag in slave mode	Arbitration lost detection flag	Slave address comparison flag	Start condition detection flag	Bus busy flag	Serial bus status in IIC	General-call detection flag	Communication abnormal detection flag	
0x03F58	RXBUF4	RXBUF47	RXBUF46	RXBUF45	RXBUF44	RXBUF43	RXBUF42	RXBUF41	RXBUF40	XVIII-34
		x	x	x	x	x	x	x	x	
		Serial interface 4 reception data buffer								
0x03F59	TXBUF4	TXBUF47	TXBUF46	TXBUF45	TXBUF44	TXBUF43	TXBUF42	TXBUF41	TXBUF40	XVIII-34
		x	x	x	x	x	x	x	x	
		Serial interface 4 transmission data buffer								
0x03F5A	SC4SEL	Reserved	SC4BRP2	SC4BRP1	SC4BRP0	OSL4	SC4SEL2	SC4SEL1	SC4SEL0	XVIII-24
		0	0	0	0	0	0	0	0	
		Always set to "0"	Timer clock output selection			Serial output port selection	Timer selection			
0x03F5E	STB_EDG	-	-	-	STB_EDG_IR Q4	STB_EDG_IR Q3	STB_EDG_IR Q2	STB_EDG_IR Q1	STB_EDGIRQ 0	VII-60
		-	-	-	0	0	0	0	0	
		Edge Interrupt control in STANDBY mode								
0x03F5F	STB_MSK	-	-	-	STB_MSK_IR Q4	STB_MSK_IR Q3	STB_MSK_IR Q2	STB_MSK_IR Q1	STB_MSK_IR Q0	VII-61
		-	-	-	0	0	0	0	0	
		Interrupt mask control at transition to STANDBY mode								
0x03F60	TM0BC	TM0BC7	TM0BC6	TM0BC5	TM0BC4	TM0BC3	TM0BC2	TM0BC1	TM0BC0	IX-22
		0	0	0	0	0	0	0	0	
		Timer 0 binary counter								
0x03F61	TM1BC	TM1BC7	TM1BC6	TM1BC5	TM1BC4	TM1BC3	TM1BC2	TM1BC1	TM1BC0	IX-22
		0	0	0	0	0	0	0	0	
		Timer 1 binary counter								
0x03F62	TM0OC	TM0OC7	TM0OC6	TM0OC5	TM0OC4	TM0OC3	TM0OC2	TM0OC1	TM0OC0	IX-20
		X	X	X	X	X	X	X	X	
		Timer 0 compare register								
0x03F63	TM1OC	TM1OC7	TM1OC6	TM1OC5	TM1OC4	TM1OC3	TM1OC2	TM1OC1	TM1OC0	IX-20
		X	X	X	X	X	X	X	X	
		Timer 1 compare register								
0x03F64	TM0MD	TM0BUFSEL	TM0POP	TM0MOD	TM0PWM	TM0EN	TM0CK2	TM0CK1	TM0CK0	IX-23
		0	0	0	0	0	0	0	0	
		Compare register buffer selection	Output signal start polarity selection	Pulse width measurement control	Operation mode selection	Count control	Clock source selection			
0x03F65	TM1MD	TM1BUFSEL	Reserved	TM1CAS1	TM1CAS0	TM1EN	TM1CK2	TM1CK1	TM1CK0	IX-26
		0	0	0	0	0	0	0	0	
		Compare register buffer selection	Always set to "0"	Operation mode		Count control	Clock source selection			
0x03F66	CK0MD	-	TM0ADD1	TM0ADD0	TM0ADDEN	TM0PSC2	TM0PSC1	TM0PSC0	TM0BAS	IX-17
		-	0	0	0	0	0	0	0	
		-	Additional pulse location			Additional pulse method PWM control	Clock source selection			
0x03F67	CK1MD	-	-	-	-	TM1PSC2	TM1PSC1	TM1PSC0	TM1BAS	IX-18
		-	-	-	-	0	0	0	0	
		-	-	-	-	Clock source selection				

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Address	Register	Bit symbol								Page
		bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0	
0x03F68	TM2BC	TM2BC7	TM2BC6	TM2BC5	TM2BC4	TM2BC3	TM2BC2	TM2BC1	TM2BC0	IX-22
		0	0	0	0	0	0	0	0	
		Time 2 binary counter								
0x03F69	TM3BC	TM3BC7	TM3BC6	TM3BC5	TM3BC4	TM3BC3	TM3BC2	TM3BC1	TM3BC0	IX-22
		0	0	0	0	0	0	0	0	
		Time 3 binary counter								
0x03F6A	TM2OC	TM2OC7	TM2OC6	TM2OC5	TM2OC4	TM2OC3	TM2OC2	TM2OC1	TM2OC0	IX-20
		X	X	X	X	X	X	X	X	
		Time 2 compare register								
0x03F6B	TM3OC	TM3OC7	TM3OC6	TM3OC5	TM3OC4	TM3OC3	TM3OC2	TM3OC1	TM3OC0	IX-21
		X	X	X	X	X	X	X	X	
		Time 3 compare register								
0x03F6C	TM2MD	TM2BUFSEL	TM2POP	TM2MOD	TM2PWM	TM2EN	TM2CK2	TM2CK1	TM2CK0	IX-28
		0	0	0	0	0	0	0	0	
		Compare register buffer selection	Output signal start polarity selection	Pulse width measurement control	Operation mode selection	Count control	Clock source selection			
0x03F6D	TM3MD	TM3BUFSEL	-	-	TM3CAS	TM3EN	TM3CK2	TM3CK1	TM3CK0	IX-30
		0	-	-	0	0	0	0	0	
		Compare register buffer selection	-	-	Operation mode selection	Count control	Clock source selection			
0x03F6E	CK2MD	-	TM2ADD1	TM2ADD0	TM2ADDEN	TM2PSC2	TM2PSC1	TM2PSC0	TM2BAS	IX-18
		-	0	0	0	0	0	0	0	
		-	Additional pulse location		Additional pulse method PWM control	Clock source selection				
0x03F6F	CK3MD	-	-	-	-	TM3PSC2	TM3PSC1	TM3PSC0	TM3BAS	IX-19
		-	-	-	-	0	0	0	0	
		-	-	-	-	Clock source selection				
0x03F70	TM4BC	TM4BC7	TM4BC6	TM4BC5	TM4BC4	TM4BC3	TM4BC2	TM4BC1	TM4BC0	IX-22
		0	0	0	0	0	0	0	0	
		Time 4 binary counter								
0x03F71	TMABC	TMABC7	TMABC6	TMABC5	TMABC4	TMABC3	TMABC2	TMABC1	TMABC0	X-5
		0	0	0	0	0	0	0	0	
		Time A binary counter								
0x03F72	TM4OC	TM4OC7	TM4OC6	TM4OC5	TM4OC4	TM4OC3	TM4OC2	TM4OC1	TM4OC0	IX-21
		X	X	X	X	X	X	X	X	
		Time 4 compare register								
0x03F73	TMAOC	TMAOC7	TMAOC6	TMAOC5	TMAOC4	TMAOC3	TMAOC2	TMAOC1	TMAOC0	X-5
		X	X	X	X	X	X	X	X	
		Time A compare register								
0x03F74	TM4MD	-	TM4POP	TM4MOD	TM4PWM	TM4EN	TM4CK2	TM4CK1	TM4CK0	IX-31
		-	0	0	0	0	0	0	0	
		-	Output signal start polarity selection	Pulse width measurement control	Operation mode selection	Count control	Clock source selection			
0x03F75	TMAMD1	-	-	-	-	TMAEN	TMACK2	TMACK1	TMACK0	X-6
		-	-	-	-	0	0	0	0	
		-	-	-	-	Count control	Clock source selection			
0x03F76	CK4MD	-	TM4ADD1	TM4ADD0	TM4ADDEN	TM4PSC2	TM4PSC1	TM4PSC0	TM4BAS	IX-19
		-	0	0	0	0	0	0	0	
		-	Additional pulse location		Additional pulse method PWM control	Clock source selection				
0x03F77	TMAMD2	Reserved	PSCEN	-	-	-	-	-	-	X-6
		0	0	-	-	-	-	-	-	
		Always set to "0"	Prescaler operation control	-	-	-	-	-	-	

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Address	Register	Bit symbol								Page
		bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0	
0x03F78	TM6BC	TM6BC7	TM6BC6	TM6BC5	TM6BC4	TM6BC3	TM6BC2	TM6BC1	TM6BC0	XIV-6
		0	0	0	0	0	0	0	0	
		Timer 6 binary counter								
0x03F79	TM6OC	TM6OC7	TM6OC6	TM6OC5	TM6OC4	TM6OC3	TM6OC2	TM6OC1	TM6OC0	XIV-6
		X	X	X	X	X	X	X	X	
		Timer 6 compare register								
0x03F7A	TM6MD	TM6CLRS	TM6IR2	TM6IR1	TM6IR0	TM6CK3	TM6CK2	TM6CK1	TM6CK0	XIV-8
		0	0	0	0	0	0	0	0	
		Binary counter clear selection	Time base timer interrupt cycle selection			Clock source selection			Time base timer clock source selection	
0x03F7B	TBCLR	TBCLR7	TBCLR6	TBCLR5	TBCLR4	TBCLR3	TBCLR2	TBCLR1	TBCLR0	XIV-6
		-	-	-	-	-	-	-	-	
		Time base timer clear control register								
0x03F7C	TM6BEN	-	-	-	-	-	Reserved	TBEN	TM6EN	XIV-7
		-	-	-	-	-	0	0	0	
		-	-	-	-	-	Always set to "0"	Time base timer operation control	Timer6 operation control	
0x03F7E	TM7MD4	Reserved	T7IGBTSFT	T7IGBTCNT	T7ONESHOT	T7NODED	-	T7ICT2	T7CAPCLR	XI-24
		0	0	0	0	0	-	0	0	
		Always set to "0"	IGBT software activation	BC operation at IGBT disable is invalid	One shot pulse selection	Dead time selection	-	Capture trigger selection	BC clear at capture clear	
0x03F7F	TM8MD4	Reserved	Reserved	Reserved	TM8SEL_C	Reserved	TM8SEL_A	T8ICT2	T8CAPCLRA	XI-29
		0	0	0	0	0	0	0	0	
		Always set to "0"	Always set to "0"	Always set to "0"	Timer8/IGBT selection	Always set to "0"	Timer8/IGBT selection	Capture trigger selection	BC clear at capture clear	
0x03F80	TM7BCL	TM7BCL7	TM7BCL6	TM7BCL5	TM7BCL4	TM7BCL3	TM7BCL2	TM7BCL1	TM7BCL0	XI-14
		0	0	0	0	0	0	0	0	
		Timer 7 binary counter lower 8-bit								
0x03F81	TM7BCH	TM7BCH7	TM7BCH6	TM7BCH5	TM7BCH4	TM7BCH3	TM7BCH2	TM7BCH1	TM7BCH0	XI-14
		0	0	0	0	0	0	0	0	
		Timer 7 binary counter upper 8-bit								
0x03F82	TM7OC1L	TM7OC1L7	TM7OC1L6	TM7OC1L5	TM7OC1L4	TM7OC1L3	TM7OC1L2	TM7OC1L1	TM7OC1L0	XI-12
		X	X	X	X	X	X	X	X	
		Timer 7 compare register lower 8-bit								
0x03F83	TM7OC1H	TM7OC1H7	TM7OC1H6	TM7OC1H5	TM7OC1H4	TM7OC1H3	TM7OC1H2	TM7OC1H1	TM7OC1H0	XI-12
		X	X	X	X	X	X	X	X	
		Timer 7 compare register upper 8-bit								
0x03F84	TM7PR1L	TM7PR1L7	TM7PR1L6	TM7PR1L5	TM7PR1L4	TM7PR1L3	TM7PR1L2	TM7PR1L1	TM7PR1L0	XI-13
		X	X	X	X	X	X	X	X	
		Timer 7 preset register 1 lower 8-bit								
0x03F85	TM7PR1H	TM7PR1H7	TM7PR1H6	TM7PR1H5	TM7PR1H4	TM7PR1H3	TM7PR1H2	TM7PR1H1	TM7PR1H0	XI-13
		X	X	X	X	X	X	X	X	
		Timer 7 preset register 1 upper 8-bit								
0x03F86	TM7ICL	TM7ICL7	TM7ICL6	TM7ICL5	TM7ICL4	TM7ICL3	TM7ICL2	TM7ICL1	TM7ICL0	XI-15
		X	X	X	X	X	X	X	X	
		Timer 7 input capture lower 8-bit								
0x03F87	TM7ICH	TM7ICH7	TM7ICH6	TM7ICH5	TM7ICH4	TM7ICH3	TM7ICH2	TM7ICH1	TM7ICH0	XI-15
		X	X	X	X	X	X	X	X	
		Timer 7 input capture upper 8-bit								
0x03F88	TM7MD1	Reserved	T7ICEDG1	TM7CL	TM7EN	TM7PS1	TM7PS0	TM7CK1	TM7CK0	XI-21
		0	0	1	0	0	0	0	0	
		Always set to "0"	Capture trigger edge selection	Timer output reset	Timer count control	Count clock selection		Clock source selection		
0x03F89	TM7MD2	TM7ICEDG0	T7PWMSL	TM7BCR	TM7PWM	TM7IRS1	T7ICEN	T7ICT1	T7ICT0	XI-22
		0	0	0	0	0	0	0	0	
		Capture trigger edge selection	PWM mode selection	Count clear factor selection	Timer output wave selection	Interrupt factor selection	Input capture enable	Capture trigger selection		

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Address	Register	Bit symbol								Page
		bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0	
0x03F8A	TM7OC2L	TM7OC2L7	TM7OC2L6	TM7OC2L5	TM7OC2L4	TM7OC2L3	TM7OC2L2	TM7OC2L1	TM7OC2L0	XI-12
		X	X	X	X	X	X	X	X	
		Timer 7 compare register 2 lower 8-bit								
0x03F8B	TM7OC2H	TM7OC2H7	TM7OC2H6	TM7OC2H5	TM7OC2H4	TM7OC2H3	TM7OC2H2	TM7OC2H1	TM7OC2H0	XI-12
		X	X	X	X	X	X	X	X	
		Timer 7 compare register 2 upper 8-bit								
0x03F8C	TM7PR2L	TM7PR2L7	TM7PR2L6	TM7PR2L5	TM7PR2L4	TM7PR2L3	TM7PR2L2	TM7PR2L1	TM7PR2L0	XI-13
		X	X	X	X	X	X	X	X	
		Timer 7 preset register 2 lower 8-bit								
0x03F8D	TM7PR2H	TM7PR2H7	TM7PR2H6	TM7PR2H5	TM7PR2H4	TM7PR2H3	TM7PR2H2	TM7PR2H1	TM7PR2H0	XI-13
		X	X	X	X	X	X	X	X	
		Timer 7 preset register 2 upper 8-bit								
0x03F8E	TM7DPR1	TM7DPR17	TM7DPR16	TM7DPR15	TM7DPR14	TM7DPR13	TM7DPR12	TM7DPR11	TM7DPR10	XI-17
		0	0	0	0	0	0	0	0	
		Timer 7 dead-time preset register 1								
0x03F8F	TM7DPR2	TM7DPR27	TM7DPR26	TM7DPR25	TM7DPR24	TM7DPR23	TM7DPR22	TM7DPR21	TM7DPR20	XI-17
		0	0	0	0	0	0	0	0	
		Timer 7 dead-time preset register 2								
0x03F90	TM8BCL	TM8BCL7	TM8BCL6	TM8BCL5	TM8BCL4	TM8BCL3	TM8BCL2	TM8BCL1	TM8BCL0	XI-20
		0	0	0	0	0	0	0	0	
		Timer 8 binary counter lower 8-bit								
0x03F91	TM8BCH	TM8BCH7	TM8BCH6	TM8BCH5	TM8BCH4	TM8BCH3	TM8BCH2	TM8BCH1	TM8BCH0	XI-20
		0	0	0	0	0	0	0	0	
		Timer 8 binary counter upper 8-bit								
0x03F92	TM8OC1L	TM8OC1L7	TM8OC1L6	TM8OC1L5	TM8OC1L4	TM8OC1L3	TM8OC1L2	TM8OC1L1	TM8OC1L0	XI-18
		X	X	X	X	X	X	X	X	
		Timer 8 compare register 1 lower 8-bit								
0x03F93	TM8OC1H	TM8OC1H7	TM8OC1H6	TM8OC1H5	TM8OC1H4	TM8OC1H3	TM8OC1H2	TM8OC1H1	TM8OC1H0	XI-18
		X	X	X	X	X	X	X	X	
		Timer 8 compare register 1 upper 8-bit								
0x03F94	TM8PR1L	TM8PR1L7	TM8PR1L6	TM8PR1L5	TM8PR1L4	TM8PR1L3	TM8PR1L2	TM8PR1L1	TM8PR1L0	XI-19
		X	X	X	X	X	X	X	X	
		Timer 8 preset register 1 lower 8-bit								
0x03F95	TM8PR1H	TM8PR1H7	TM8PR1H6	TM8PR1H5	TM8PR1H4	TM8PR1H3	TM8PR1H2	TM8PR1H1	TM8PR1H0	XI-19
		X	X	X	X	X	X	X	X	
		Timer 8 preset register 1 upper 8-bit								
0x03F96	TM8ICL	TM8ICL7	TM8ICL6	TM8ICL5	TM8ICL4	TM8ICL3	TM8ICL2	TM8ICL1	TM8ICL0	XI-20
		X	X	X	X	X	X	X	X	
		Timer 8 input capture register lower 8-bit								
0x03F97	TM8ICH	TM8ICH7	TM8ICH6	TM8ICH5	TM8ICH4	TM8ICH3	TM8ICH2	TM8ICH1	TM8ICH0	XI-20
		X	X	X	X	X	X	X	X	
		Timer 8 input capture register upper 8-bit								
0x03F98	TM8MD1	Reserved	T8ICEDG1	TM8CL	TM8EN	TM8PS1	TM8PS0	TM8CK1	TM8CK0	XI-26
		0	0	1	0	0	0	0	0	
		Always set to "0"	Capture trigger edge selection	Timer output reset	Timer count control	Count clock selection		Clock source selection		
0x03F99	TM8MD2	T8ICERDGO	TM8PWMSL	TM8BCR	TM8PWM	TM8IRS1	T8ICEN	T8ICT1	T8ICT0	XI-27
		0	0	0	0	0	0	0	0	
		Capture trigger edge selection	PWM mode selection	Count clear factor selection	Timer output wave selection	Interrupt factor selection	Input capture enable	Capture trigger selection		
0x03F9A	TM8OC2L	TM8OC2L7	TM8OC2L6	TM8OC2L5	TM8OC2L4	TM8OC2L3	TM8OC2L2	TM8OC2L1	TM8OC2L0	XI-18
		X	X	X	X	X	X	X	X	
		Timer 8 compare register 2 lower 8-bit								
0x03F9B	TM8OC2H	TM8OC2H7	TM8OC2H6	TM8OC2H5	TM8OC2H4	TM8OC2H3	TM8OC2H2	TM8OC2H1	TM8OC2H0	XI-18
		X	X	X	X	X	X	X	X	
		Timer 8 compare register 2 upper 8-bit								

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Address	Register	Bit symbol								Page
		bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0	
0x03F9C	TM8PR2L	TM8PR2L7	TM8PR2L6	TM8PR2L5	TM8PR2L4	TM8PR2L3	TM8PR2L2	TM8PR2L1	TM8PR2L0	XI-19
		X	X	X	X	X	X	X	X	
		Timer 8 preset register 2 lower 8-bit								
0x03F9D	TM8PR2H	TM8PR2H7	TM8PR2H6	TM8PR2H5	TM8PR2H4	TM8PR2H3	TM8PR2H2	TM8PR2H1	TM8PR2H0	XI-19
		X	X	X	X	X	X	X	X	
		Timer 8 preset register 2 upper 8-bit								
0x03F9E	TM7MD3	TM7CKSMP	TM7BUFSEL	TM7CKEDG	TM7IGBTTR	T7IGBTDT	T7IGBTEN	T7IGBT1	T7IGBT0	XI-23
		0	0	0	0	0	0	0	0	
		Capture sampling selection	Buffer selection	TM7IO count edge selection	IGBT trigger level selection	IGBT dead-time insertion timing	IGBT operation enable	IGBT/timer activation factor selection		
0x03F9F	TM8MD3	TM8CKSMP	TM8BUFSEL	-	TM8CKEDG	-	TM8PWMF	TM8PWMO	TM8CAS	XI-28
		0	0	-	0	-	0	0	0	
		Capture sampling selection	Buffer selection	-	TM8IO count edge selection	-	PWM output value selection	PWM output polarity selection	Cascade selection	
0x03FA0	TM7ICL1	TM7ICL17	TM7ICL16	TM7ICL15	TM7ICL14	TM7ICL13	TM7ICL12	TM7ICL11	TM7ICL10	XI-15
		X	X	X	X	X	X	X	X	
		Timer 7 input capture 1 lower 8-bit								
0x03FA1	TM7ICH1	TM7ICH17	TM7ICH16	TM7ICH15	TM7ICH14	TM7ICH13	TM7ICH12	TM7ICH11	TM7ICH10	XI-15
		X	X	X	X	X	X	X	X	
		Timer 7 input capture 1 upper 8-bit								
0x03FA2	TM7ICLD	TM7ICLD7	TM7ICLD6	TM7ICLD5	TM7ICLD4	TM7ICLD3	TM7ICLD2	TM7ICLD1	TM7ICLD0	XI-16
		X	X	X	X	X	X	X	X	
		Timer 7 input capture D lower 8-bit								
0x03FA3	TM7ICHD	TM7ICHD7	TM7ICHD6	TM7ICHD5	TM7ICHD4	TM7ICHD3	TM7ICHD2	TM7ICHD1	TM7ICHD0	XI-16
		X	X	X	X	X	X	X	X	
		Timer 7 input capture D upper 8-bit								
0x03FA4	TM7ICLD1	TM7ICLD17	TM7ICLD16	TM7ICLD15	TM7ICLD14	TM7ICLD13	TM7ICLD12	TM7ICLD11	TM7ICLD10	XI-16
		X	X	X	X	X	X	X	X	
		Timer 7 input capture D1 lower 8-bit								
0x03FA5	TM7ICHD1	TM7ICHD17	TM7ICHD16	TM7ICHD15	TM7ICHD14	TM7ICHD13	TM7ICHD12	TM7ICHD11	TM7ICHD10	XI-16
		X	X	X	X	X	X	X	X	
		Timer 7 input capture D1 upper 8-bit								
0x03FA6	TM7MD5	-	-	-	-	-	-	T7DBUF	T7UD	XI-25
		-	-	-	-	-	-	0	0	
		-	-	-	-	-	-	IC register buffer selection	IC register 1 selection	
0x03FA8	TMRMD1	-	TMRIRSEL2	TMRIRSEL1	TMRIRSEL0	-	-	TMRALMEN	TMREN	XIII-5
		-	0	0	0	-	-	0	0	
		-	Interrupt generation cycle selection			-	-	Alarm function operation enable	24H timer operation enable	
0x03FA9	TMRMD2	Reserved	Reserved	Reserved	Reserved	TMRCK3	TMRCK2	TMRCK1	TMRCK0	XIII-6
		0	0	0	0	0	0	0	0	
		Always set to "0"				24H timer frequency selection		clock specification		
0x03FAA	TMRBCL	-	TMRBC6	TMRBC5	TMRBC4	TMRBC3	TMRBC2	TMRBC1	TMRBC0	XIII-7
		-	X	X	X	X	X	X	X	
		24 H timer binary counter L								
0x03FAB	TMRBCM	-	-	TMRBC12	TMRBC11	TMRBC10	TMRBC9	TMRBC8	TMRBC7	XIII-7
		-	-	X	X	X	X	X	X	
		24 H timer binary counter M								
0x03FAC	TMRBCH	-	-	-	TMRBC17	TMRBC16	TMRBC15	TMRBC14	TMRBC13	XIII-7
		-	-	-	X	X	X	X	X	
		24 H timer binary counter H								
0x03FAD	TMRALL	-	TMRAL6	TMRAL5	TMRAL4	TMRAL3	TMRAL2	TMRAL1	TMRAL0	XIII-9
		-	X	X	X	X	X	X	X	
		24 H timer alarm setting register L								

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Address	Register	Bit symbol								Page
		bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0	
0x03FAE	TMRALM	-	-	TMRAL12	TMRAL11	TMRAL10	TMRAL9	TMRAL8	TMRAL7	XIII-9
		-	-	X	X	X	X	X	X	
		24 H timer alarm setting register M								
0x03FAF	TMRALH	-	-	-	TMRAL17	TMRAL16	TMRAL15	TMRAL14	TMRAL13	XIII-9
		-	-	-	X	X	X	X	X	
		24 H timer alarm setting register H								
0x03FB0	TMCKSEL1	TM6CKSEL	-	-	TM4IOSEL	TM3IOSEL	TM2IOSEL	TM1IOSEL	TM0IOSEL	IX-32 XIV-9
		0	-	-	0	0	0	0	0	
		Selection of Time Base Timer clock dividing ratio	-	-	Timer 4 input selection	Timer 3 input selection	Timer 2 input selection	Timer 1 input selection	Timer 0 input selection	
0x03FB1	TMCKSEL2	-	TM8IOSEL2	TM7IOSEL3	TM7IOSEL2	-	TM9IOSEL	TM8IOSEL	TM7IOSEL	XI-30
		-	0	0	0	-	0	0	0	
		-	Timer 8 input selection 2	Timer 7 input selection 3	Timer 7 input selection 2	-	Timer 9 input selection	Timer 8 input selection	Timer 7 input selection	
0x03FB2	TMINSEL1	TMINSEL17	TMINSEL16	TMINSEL15	TMINSEL14	TMINSEL13	TMINSEL12	TMINSEL11	TMINSEL10	IX-33
		0	0	0	0	0	0	0	0	
		Timer 3 input selection		Timer 2 input selection		Timer 1 input selection		Timer 0 input selection		
0x03EB3	TMINSEL2	TMINSEL27	TMINSEL26	TMINSEL25	TMINSEL24	TMINSEL23	TMINSEL22	TMINSEL21	TMINSEL20	IX-35 XI-31
		0	0	0	0	0	0	0	0	
		Timer 9 input selection		Timer 8 input selection		Timer 7 input selection		Timer 4 input selection		
0x03FBD	FBEWER	BEW7	BEW6	BEW5	BEW4	BEW3	BEW2	BEW1	BEW0	XXIV-25
		0	0	0	0	0	0	0	0	
		On-board rewriting control								
0x03FBF	FEWSPD	Reserved	-	-	-	NSTOP	-	-	Reserved	III-26
		0	-	-	-	0	-	-	1	
		Always set to "0"	-	-	-	Internal flash control	-	-	-	
0x03FC0	RCAPL	RCAPL7	RCAPL6	RCAPL5	RCAPL4	RCAPL3	RCAPL2	RCAPL1	RCAPL0	II-32
		0	0	0	0	0	0	0	0	
		ROM correction address setting register for lower 8bits								
0x03FC1	RCAPM	RCAPM7	RCAPM6	RCAPM5	RCAPM4	RCAPM3	RCAPM2	RCAPM1	RCAPM0	II-32
		0	0	0	0	0	0	0	0	
		ROM correction address setting register for middle 8bits								
0x03FC2	RCAPH	-	-	-	-	RCAPH3	RCAPH2	RCAPH1	RCAPH0	II-32
		-	-	-	-	0	0	0	0	
		ROM correction address setting register for upper 4bits								
0x03FC3	RCPSR	-	-	-	-	Reserved	RCPSR2	RCPSR1	RCPSR0	II-30
		-	-	-	-	0	0	0	0	
		-	-	-	-	Always set to "0"	Specification of ROM correction pointer			
0x03FC4	RCCTR0	Reserved	RC6EN	RC5EN	RC4EN	RC3EN	RC2EN	RC1EN	RC0EN	II-31
		0	0	0	0	0	0	0	0	
		Always set to "0"	ROM correction control							
0x03FC5	ANCTR0	ANSH1	ANSH0	ANCK1	ANCK0	ANLADE	ANCK2	-	-	XX-7
		0	0	0	0	0	0	-	-	
		Sample and hold time		A/D conversion clock		A/D resistor ladder control	A/D conversion clock	-	-	
0x03FC6	ANCTR1	-	-	-	ANCHS4	ANCHS3	ANCHS2	ANCHS1	ANCHS0	XX-8
		-	-	-	0	0	0	0	0	
		-	-	-	Analog input channel selection					
0x03FC7	ANCTR2	ANST	ANSTSEL1	ANSTSEL0	-	-	-	-	-	XX-10
		0	0	0	-	-	-	-	-	
		A/D conversion status	A/D conversion starting factor selection		-	-	-	-	-	
0x03FC8	ANBUF0	ANBUF07	ANBUF06	-	-	-	-	-	-	XX-11
		X	X	-	-	-	-	-	-	
		Conversion data storage buffer 0		-	-	-	-	-	-	

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Address	Register	Bit symbol								Page
		bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0	
0x03FC9	ANBUF1	ANBUF17	ANBUF16	ANBUF15	ANBUF14	ANBUF13	ANBUF12	ANBUF11	ANBUF10	XX-11
		X	X	X	X	X	X	X	X	
		Conversion data storage buffer 1								
0x03FCB	DACTR	DACHS1	DACHS0	DACKS1	DACKS0	DACMD2	DACMD1	DACMD0	DABUSY	XXI-5
		X	X	X	X	X	X	X	0	
		Monitor flag of DA conversion		DA scan clock		DA conversion mode selection			D/A conversion enable	
0x03FCC	DADR0	DA0BUF7	DA0BUF6	DA0BUF5	DA0BUF4	DA0BUF3	DA0BUF2	DA0BUF1	DA0BUF0	XXI-7
		X	X	X	X	X	X	X	X	
		D/A Converter Input Data Register 0								
0x03FCD	DADR1	DA1BUF7	DA1BUF6	DA1BUF5	DA1BUF4	DA1BUF3	DA1BUF2	DA1BUF1	DA1BUF0	XXI-7
		X	X	X	X	X	X	X	X	
		D/A Converter Input Data Register 1								
0x03FCE	DADR2	DA2BUF7	DA2BUF6	DA2BUF5	DA2BUF4	DA2BUF3	DA2BUF2	DA2BUF1	DA2BUF0	XXI-7
		X	X	X	X	X	X	X	X	
		D/A Converter Input Data Register 2								
0x03FCF	DADR3	DA3BUF7	DA3BUF6	DA3BUF5	DA3BUF4	DA3BUF3	DA3BUF2	DA3BUF1	DA3BUF0	XXI-7
		X	X	X	X	X	X	X	X	
		D/A Converter Input Data Register 3								
0x03FD0	IRQCNT	-	-	-	P24EN	P23EN	P22EN	P21EN	P20EN	VII-51
		-	-	-	0	0	0	0	0	
		-	-	-	Pin setting for External Interrupt					
0x03FD1	NF0CTR	NF0SCK2	NF0SCK1	NF0SCK0	NF0EN1	-	-	-	-	VII-54
		0	0	0	0	-	-	-	-	
		IRQ0 noise sampling frequency			Noise filter ON/OFF control	-	-	-	-	
0x03FD2	NF1CTR	NF1SCK2	NF1SCK1	NF1SCK0	NF1EN1	-	-	-	-	VII-54
		0	0	0	0	-	-	-	-	
		IRQ1 noise sampling frequency			Noise filter ON/OFF control	-	-	-	-	
0x03FD3	NF2CTR	NF2SCK2	NF2SCK1	NF2SCK0	NF2EN1	-	-	-	-	VII-54
		0	0	0	0	-	-	-	-	
		IRQ2 noise sampling frequency			Noise filter ON/OFF control	-	-	-	-	
0x03FD4	NF3CTR	NF3SCK2	NF3SCK1	NF3SCK0	NF3EN1	-	-	-	-	VII-54
		0	0	0	0	-	-	-	-	
		IRQ3 noise sampling frequency			Noise filter ON/OFF control	-	-	-	-	
0x03FD5	NF4CTR	NF4SCK2	NF4SCK1	NF4SCK0	NF4EN1	-	-	-	-	VII-54
		0	0	0	0	-	-	-	-	
		IRQ4 noise sampling frequency			Noise filter ON/OFF control	-	-	-	-	
0x03FD6	NF5CTR	NF5SCK2	NF5SCK1	NF5SCK0	NF5EN1	-	-	-	-	XIX-7
		0	0	0	0	-	-	-	-	
		TXD0 noise sampling frequency			Noise filter ON/OFF control	-	-	-	-	
0x03FD7	LVLMD	-	-	EXLVL4	LVLEN4	EXLVL3	LVLEN3	EXLVL2	LVLEN2	VII-57
		-	-	0	0	0	0	0	0	
		-	-	IRQ4 valid input level setting	IRQ4 valid input setting	IRQ3 valid input level setting	IRQ3 valid input setting	IRQ2 valid input level setting	IRQ2 valid input setting	
0x03FD8	IRQINSEL *	-	-	-	IRQINSEL4	IRQINSEL3	IRQINSEL2	IRQINSEL1	IRQINSEL0	VII-53
		-	-	-	0	0	0	0	0	
		External Interrupt pin selection								
0x03FE1	NMICR	-	-	-	-	-	IRQNPG	IRQNWDG	IRQNPRI	VII-23
		-	-	-	-	-	0	0	0	
		-	-	-	-	-	Program interrupt request flag	Watchdog interrupt request flag	Peripheral functions NMI request flag	

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Address	Register	Bit symbol								Page
		bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0	
0x03FE2	IRQ0ICR	IRQ0LV1	IRQ0LV0	REDG0	-	Reserved	-	IRQ0IE	IRQ0IR	VII-24
		0	0	0	-	0	-	0	0	
		Interrupt level specification flag		Interrupt valid edge flag	-	Always set to "0"	-	Interrupt enable flag	Interrupt request flag	
0x03FE3	IRQ1ICR	IRQ1LV1	IRQ1LV0	REDG1	-	Reserved	-	IRQ1IE	IRQ1IR	VII-24
		0	0	0	-	0	-	0	0	
		Interrupt level specification flag		Interrupt valid edge flag	-	Always set to "0"	-	Interrupt enable flag	Interrupt request flag	
0x03FE4	IRQ2ICR	IRQ2LV1	IRQ2LV0	REDG2	-	Reserved	-	IRQ2IE	IRQ2IR	VII-24
		0	0	0	-	0	-	0	0	
		Interrupt level specification flag		Interrupt valid edge flag	-	Always set to "0"	-	Interrupt enable flag	Interrupt request flag	
0x03FE5	IRQ3ICR	IRQ3LV1	IRQ3LV0	REDG3	-	Reserved	-	IRQ3IE	IRQ3IR	VII-24
		0	0	0	-	0	-	0	0	
		Interrupt level specification flag		Interrupt valid edge flag	-	Always set to "0"	-	Interrupt enable flag	Interrupt request flag	
0x03FE6	IRQ4ICR	IRQ4LV1	IRQ4LV0	REDG4	-	Reserved	-	IRQ4IE	IRQ4IR	VII-24
		0	0	0	-	0	-	0	0	
		Interrupt level specification flag		Interrupt valid edge flag	-	Always set to "0"	-	Interrupt enable flag	Interrupt request flag	
0x03FE7	LINICR	LINLV1	LINLV0	LINREDG	-	Reserved	-	LINIE	LINIR	VII-25
		0	0	0	-	0	-	0	0	
		Interrupt level specification flag		Interrupt valid edge flag	-	Always set to "0"	-	Interrupt enable flag	Interrupt request flag	
0x03FE8	LVIIICR	LVILV1	LVILV0	-	-	Reserved	-	LVIIIE	LVIIIR	VII-26
		0	0	-	-	0	-	0	0	
		Interrupt level specification flag		-	-	Always set to "0"	-	Interrupt enable flag	Interrupt request flag	
0x03FE9	TM0ICR	TM0LV1	TM0LV0	-	-	Reserved	-	TM0IE	TM0IR	VII-27
		0	0	-	-	0	-	0	0	
		Interrupt level specification flag		-	-	Always set to "0"	-	Interrupt enable flag	Interrupt request flag	
0x03FEA	TM1ICR	TM1LV1	TM1LV0	-	-	Reserved	-	TM1IE	TM1IR	VII-27
		0	0	-	-	0	-	0	0	
		Interrupt level specification flag		-	-	Always set to "0"	-	Interrupt enable flag	Interrupt request flag	
0x03FEB	TM2ICR	TM2LV1	TM2LV0	-	-	Reserved	-	TM2IE	TM2IR	VII-27
		0	0	-	-	0	-	0	0	
		Interrupt level specification flag		-	-	Always set to "0"	-	Interrupt enable flag	Interrupt request flag	
0x03FEC	TM3ICR	TM3LV1	TM3LV0	-	-	Reserved	-	TM3IE	TM3IR	VII-27
		0	0	-	-	0	-	0	0	
		Interrupt level specification flag		-	-	Always set to "0"	-	Interrupt enable flag	Interrupt request flag	
0x03FED	TM4ICR	TM4LV1	TM4LV0	-	-	Reserved	-	TM4IE	TM4IR	VII-27
		0	0	-	-	0	-	0	0	
		Interrupt level specification flag		-	-	Always set to "0"	-	Interrupt enable flag	Interrupt request flag	
0x03FEE	TMRICR	TMRLV1	TMRLV0	-	-	Reserved	-	TMRIE	TMRIR	VII-28
		0	0	-	-	0	-	0	0	
		Interrupt level specification flag		-	-	Always set to "0"	-	Interrupt enable flag	Interrupt request flag	
0x03FEF	TM6ICR	TM6LV1	TM6LV0	-	-	Reserved	-	TM6IE	TM6IR	VII-27
		0	0	-	-	0	-	0	0	
		Interrupt level specification flag		-	-	Always set to "0"	-	Interrupt enable flag	Interrupt request flag	
0x03FF0	TBICR	TBLV1	TBLV0	-	-	Reserved	-	TBIE	TBIR	VII-29
		0	0	-	-	0	-	0	0	
		Interrupt level specification flag		-	-	Always set to "0"	-	Interrupt enable flag	Interrupt request flag	

<MN101EF76K/MN101EF56K> (*: Reserved register in MN101EF56K)

Address	Register	Bit symbol								Page
		bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0	
0x03FF1	TM7ICR	TM7LV1	TM7LV0	-	-	Reserved	-	TM7IE	TM7IR	VII-30
		0	0	-	-	0	-	0	0	
		Interrupt level specification flag		-	-	Always set to "0"	-	Interrupt enable flag	Interrupt request flag	
0x03FF2	TM8ICR	TM8LV1	TM8LV0	-	-	Reserved	-	TM8IE	TM8IR	VII-31
		0	0	-	-	0	-	0	0	
		Interrupt level specification flag		-	-	Always set to "0"	-	Interrupt enable flag	Interrupt request flag	
0x03FF3	TM9OC2ICR	TM9OC2LV1	TM9OC2LV0	-	-	Reserved	-	TM9OC2IE	TM9OC2IR	VII-32
		0	0	-	-	0	-	0	0	
		Interrupt level specification flag		-	-	Always set to "0"	-	Interrupt enable flag	Interrupt request flag	
0x03FF4	SC0TICR	SC0TLV1	SC0TLV0	-	-	Reserved	-	SC0TIE	SC0TIR	VII-33
		0	0	-	-	0	-	0	0	
		Interrupt level specification flag		-	-	Always set to "0"	-	Interrupt enable flag	Interrupt request flag	
0x03FF5	SC0RICR	SC0RLV1	SC0RLV0	-	-	Reserved	-	SC0RIE	SC0RIR	VII-34
		0	0	-	-	0	-	0	0	
		Interrupt level specification flag		-	-	Always set to "0"	-	Interrupt enable flag	Interrupt request flag	
0x03FF6	SC1TICR	SC1TLV1	SC1TLV0	-	-	Reserved	-	SC1TIE	SC1TIR	VII-33
		0	0	-	-	0	-	0	0	
		Interrupt level specification flag		-	-	Always set to "0"	-	Interrupt enable flag	Interrupt request flag	
0x03FF7	SC1RICR	SC1RLV1	SC1RLV0	-	-	Reserved	-	SC1RIE	SC1RIR	VII-34
		0	0	-	-	0	-	0	0	
		Interrupt level specification flag		-	-	Always set to "0"	-	Interrupt enable flag	Interrupt request flag	
0x03FF8	SC2TICR	SC2TLV1	SC2TLV0	-	-	Reserved	-	SC2TIE	SC2TIR	VII-33
		0	0	-	-	0	-	0	0	
		Interrupt level specification flag		-	-	Always set to "0"	-	Interrupt enable flag	Interrupt request flag	
0x03FF9	SC2RICR	SC2RLV1	SC2RLV0	-	-	Reserved	-	SC2RIE	SC2RIR	VII-34
		0	0	-	-	0	-	0	0	
		Interrupt level specification flag		-	-	Always set to "0"	-	Interrupt enable flag	Interrupt request flag	
0x03FFA	SC3TICR	SC3TLV1	SC3TLV0	-	-	Reserved	-	SC3TIE	SC3TIR	VII-33
		0	0	-	-	0	-	0	0	
		Interrupt level specification flag		-	-	Always set to "0"	-	Interrupt enable flag	Interrupt request flag	
0x03FFB	SC3RICR	SC3RLV1	SC3RLV0	-	-	Reserved	-	SC3RIE	SC3RIR	VII-34
		0	0	-	-	0	-	0	0	
		Interrupt level specification flag		-	-	Always set to "0"	-	Interrupt enable flag	Interrupt request flag	
0x03FFC	SC4ICR	SC4LV1	SC4LV0	-	-	Reserved	-	SC4IE	SC4IR	VII-35
		0	0	-	-	0	-	0	0	
		Interrupt level specification flag		-	-	Always set to "0"	-	Interrupt enable flag	Interrupt request flag	
0x03FFD	SC4STPCICR	SC4STPCLV1	SC4STPCLV0	-	-	Reserved	-	SC4STPCIE	SC4STPCIR	VII-36
		0	0	-	-	0	-	0	0	
		Interrupt level specification flag		-	-	Always set to "0"	-	Interrupt enable flag	Interrupt request flag	
0x03FFE	PERILR	PERILV1	PERILV0	-	-	Reserved	-	-	-	VII-37
		0	0	-	-	0	-	-	-	
		Interrupt level specification flag		-	-	Always set to "0"	-	-	-	

25.1.2 Special Function Registers List (MN101EF57G)

<MN101EF57G>

Address	Register	Bit symbol								Page
		bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0	
0x03DF0	CHDRVSEL	-	-	-	-	-	-	DRVSOSC_SEL	DRVOSC_SEL	III-18
		-	-	-	-	-	-	0	0	
		-	-	-	-	-	-	Low-speed/high-speed oscillation performance selection		
0x03E00	PWMMDL	PCRAEN	PCRBEN	INTAEN	INTBEN	DTEN	ORMD	TCEN	WAVEMD	XII-8
		0	0	0	0	0	0	0	0	
		PWM mode control register lower 8-bit								
0x03E01	PWMMDH	-	INTCEN	-	Reserved	-	DTSEL	SDSELA	SDSELB	XII-9
		-	0	-	0	-	0	0	0	
		PWM mode control register upper 8-bit								
0x03E02	PWMSSELL	PSELN00	PSEL00	OTLVN02	OTLV02	OTLVN01	OTLV01	OTLVN00	OTLV000	XII-11
		0	0	0	0	0	0	0	0	
		PWM output control register lower 8-bit								
0x03E03	PWMSELH	-	-	-	-	PSELN02	PSEL02	PSELN01	PSEL01	XII-11
		-	-	-	-	0	0	0	0	
		PWM output control register upper 8-bit								
0x03E04	PWMSETL	PMSET07	PMSET06	PMSET05	PMSET04	PMSET03	PMSET02	PMSET01	PMSET00	XII-12
		0	0	0	0	0	0	0	0	
		PWM cycle setting register lower 8-bit								
0x03E05	PWMSETH	PMSET0F	PMSET0E	PMSET0D	PMSET0C	PMSET0B	PMSET0A	PMSET09	PMSET08	XII-12
		0	0	0	0	0	0	0	0	
		PWM cycle setting register upper 8-bit								
0x03E06	TCMPAL	TCPA07	TCPA06	TCPA05	TCPA04	TCPA03	TCPA02	TCPA01	TCPA00	XII-13
		0	0	0	0	0	0	0	0	
		PWM 0-phase comparison setting register lower 8-bit								
0x03E07	TCMPAH	TCPA0F	TCPA0E	TCPA0D	TCPA0C	TCPA0B	TCPA0A	TCPA09	TCPA08	XII-13
		0	0	0	0	0	0	0	0	
		PWM 0-phase comparison setting register upper 8-bit								
0x03E08	TCMPBL	TCPB07	TCPB06	TCPB05	TCPB04	TCPB03	TCPB02	TCPB01	TCPB00	XII-13
		0	0	0	0	0	0	0	0	
		PWM 1-phase comparison setting register lower 8-bit								
0x03E09	TCMPBH	TCPB0F	TCPB0E	TCPB0D	TCPB0C	TCPB0B	TCPB0A	TCPB09	TCPB08	XII-14
		0	0	0	0	0	0	0	0	
		PWM 1-phase comparison setting register upper 8-bit								
0x03E0A	TCMPCL	TCPC07	TCPC06	TCPC05	TCPC04	TCPC03	TCPC02	TCPC01	TCPC00	XII-14
		0	0	0	0	0	0	0	0	
		PWM 2-phase comparison setting register lower 8-bit								
0x03E0B	TCMPCH	TCPC0F	TCPC0E	TCPC0D	TCPC0C	TCPC0B	TCPC0A	TCPC09	TCPC08	XII-14
		0	0	0	0	0	0	0	0	
		PWM 2-phase comparison setting register upper 8-bit								
0x03E0C	OUTMD	-	-	PXDTNW	PXDTW	PXDTNV	PXDTV	PXDTNU	PXDTU	XII-10
		-	-	0	0	0	0	0	0	
		PWM output polarity control register								
0x03E0D	DTMSET	DTST07	DTST06	DTST05	DTST04	DTST03	DTST02	DTST01	DTST00	XII-15
		0	0	0	0	0	0	0	0	
		PWM dead time setting register								
0x03E0E	DTMSET1	DTST17	DTST16	DTST15	DTST14	DTST13	DTST12	DTST11	DTST10	XII-15
		0	0	0	0	0	0	0	0	
		PWM dead time setting register 1								
0x03E10	PWMBCL	PWMB07	PWMB06	PWMB05	PWMB04	PWMB03	PWMB02	PWMB01	PWMB00	XII-16
		0	0	0	0	0	0	0	0	
		PWM BC value read register lower 8-bit								

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Address	Register	Bit symbol								Page
		bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0	
0x03E11	PWMBCH	PWMB15	PWMB14	PWMB13	PWMB12	PWMB11	PWMB10	PWMB09	PWMB08	XII-16
		0	0	0	0	0	0	0	0	
		PWM BC value read register upper 8-bit								
0x03E12	BCSTR	-	-	-	-	-	-	-	PWMSTR	XII-17
		-	-	-	-	-	-	-	1	
		PWM BC status register								
0x03E13	PWMOFFL	PRTANU1	PRTANU0	PRTAU1	PRTAU0	IRQSEL2	IRQSEL1	IRQSEL0	OUTEN0	XII-18
		0	0	0	0	0	0	0	0	
		PWM pin protection control register lower 8-bit								
0x03E14	PWMOFFH	PRTANW1	PRTANW0	PRTAW1	PRTAW0	PRTANV1	PRTANV0	PRTAV1	PRTAV0	XII-19
		0	0	0	0	0	0	0	0	
		PWM pin protection control register upper 8-bit								
0x03E15	IRQCULL	REGSELEDGE1	REGSELEDGE0	REGCULLCMP1	REGCULLCMP0	REGCULLUDF1	REGCULLUDF0	REGCULLOVF1	REGCULLOVF0	XII-20
		0	0	0	0	0	0	0	0	
		PWM interrupt signal culling selection register								
0x03E16	PWMTMCNT	-	-	-	PSCSEL1	PSCSEL0	PWMCKSEL1	PWMCKSEL0	SEL_PWM_TM	XII-21
		-	-	-	0	0	0	0	0	
		PWM timer operation control register								
0x03E17	RELCTR	RELIRQA	RELIRQB	RELIRQC	RELIRQD	-	MD1CTR	RELMD		XII-22
		0	0	0	0	-	0	0	0	
		1st phase interrupt event enable control	2nd phase interrupt event enable control	3rd phase interrupt event enable control	4th phase interrupt event enable control	-	PWM output order control in 1-phase mode	RELATIONAL PWM mode		
0x03E18	PWMODR	-	-	-	-	-	-	-	ORDER	XII-23
		-	-	-	-	-	-	-	0	
		-	-	-	-	-	-	-	4-phase PWM output order	
0x03E19	RELSTAT	-	-	-	-	-	-	STAT1	STAT0	XII-23
		-	-	-	-	-	-	0	0	
		-	-	-	-	-	-	4-phase PWM output order		
0x03E1A	PWMCMP1	PWMCMP17	PWMCMP16	PWMCMP15	PWMCMP14	PWMCMP13	PWMCMP12	PWMCMP11	PWMCMP10	XII-24
		0	0	0	0	0	0	0	0	
		Lower 8-bit of PWM cycle register and PWMn phase comparison register								
0x03E1B	PWMCMP2	PWMCMPH7	PWMCMPH6	PWMCMPH5	PWMCMPH4	PWMCMPH3	PWMCMPH2	PWMCMPH1	PWMCMPH0	XII-24
		0	0	0	0	0	0	0	0	
		Upper 8-bit of PWM cycle register and PWMn phase comparison register								
0x03E20	LC00BUF	LC00BUF7	LC00BUF6	LC00BUF5	LC00BUF4	LC00BUF3	LC00BUF2	LC00BUF1	LC00BUF0	XXII-18
		X	X	X	X	X	X	X	X	
		Segment output latch data								
0x03E21	LC01BUF	LC01BUF7	LC01BUF6	LC01BUF5	LC01BUF4	LC01BUF3	LC01BUF2	LC01BUF1	LC01BUF0	XXII-18
		X	X	X	X	X	X	X	X	
		Segment output latch data								
0x03E22	LC02BUF	LC02BUF7	LC02BUF6	LC02BUF5	LC02BUF4	LC02BUF3	LC02BUF2	LC02BUF1	LC02BUF0	XXII-18
		X	X	X	X	X	X	X	X	
		Segment output latch data								
0x03E23	LC03BUF	LC03BUF7	LC03BUF6	LC03BUF5	LC03BUF4	LC03BUF3	LC03BUF2	LC03BUF1	LC03BUF0	XXII-18
		X	X	X	X	X	X	X	X	
		Segment output latch data								
0x03E24	LC04BUF	LC04BUF7	LC04BUF6	LC04BUF5	LC04BUF4	LC04BUF3	LC04BUF2	LC04BUF1	LC04BUF0	XXII-18
		X	X	X	X	X	X	X	X	
		Segment output latch data								
0x03E25	LC05BUF	LC05BUF7	LC05BUF6	LC05BUF5	LC05BUF4	LC05BUF3	LC05BUF2	LC05BUF1	LC05BUF0	XXII-18
		X	X	X	X	X	X	X	X	
		Segment output latch data								
0x03E26	LC06BUF	LC06BUF7	LC06BUF6	LC06BUF5	LC06BUF4	LC06BUF3	LC06BUF2	LC06BUF1	LC06BUF0	XXII-18
		X	X	X	X	X	X	X	X	
		Segment output latch data								

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Address	Register	Bit symbol								Page
		bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0	
0x03E27	LC07BUF	LC07BUF7	LC07BUF6	LC07BUF5	LC07BUF4	LC07BUF3	LC07BUF2	LC07BUF1	LC07BUF0	XXII-18
		X	X	X	X	X	X	X	X	
		Segment output latch data								
0x03E28	LC08BUF	LC08BUF7	LC08BUF6	LC08BUF5	LC08BUF4	LC08BUF3	LC08BUF2	LC08BUF1	LC08BUF0	XXII-18
		X	X	X	X	X	X	X	X	
		Segment output latch data								
0x03E29	LC09BUF	LC09BUF7	LC09BUF6	LC09BUF5	LC09BUF4	LC09BUF3	LC09BUF2	LC09BUF1	LC09BUF0	XXII-18
		X	X	X	X	X	X	X	X	
		Segment output latch data								
0x03E2A	LC0ABUF	LC0ABUF7	LC0ABUF6	LC0ABUF5	LC0ABUF4	LC0ABUF3	LC0ABUF2	LC0ABUF1	LC0ABUF0	XXII-18
		X	X	X	X	X	X	X	X	
		Segment output latch data								
0x03E2B	LC0BBUF	LC0BBUF7	LC0BBUF6	LC0BBUF5	LC0BBUF4	LC0BBUF3	LC0BBUF2	LC0BBUF1	LC0BBUF0	XXII-18
		X	X	X	X	X	X	X	X	
		Segment output latch data								
0x03E2C	LC0CBUF	LC0CBUF7	LC0CBUF6	LC0CBUF5	LC0CBUF4	LC0CBUF3	LC0CBUF2	LC0CBUF1	LC0CBUF0	XXII-18
		X	X	X	X	X	X	X	X	
		Segment output latch data								
0x03E2D	LC0DBUF	LC0DBUF7	LC0DBUF6	LC0DBUF5	LC0DBUF4	LC0DBUF3	LC0DBUF2	LC0DBUF1	LC0DBUF0	XXII-18
		X	X	X	X	X	X	X	X	
		Segment output latch data								
0x03E2E	LC0EBUF	LC0EBUF7	LC0EBUF6	LC0EBUF5	LC0EBUF4	LC0EBUF3	LC0EBUF2	LC0EBUF1	LC0EBUF0	XXII-18
		X	X	X	X	X	X	X	X	
		Segment output latch data								
0x03E2F	LC0FBUF	LC0FBUF7	LC0FBUF6	LC0FBUF5	LC0FBUF4	LC0FBUF3	LC0FBUF2	LC0FBUF1	LC0FBUF0	XXII-18
		X	X	X	X	X	X	X	X	
		Segment output latch data								
0x03E30	LC10BUF	LC10BUF7	LC10BUF6	LC10BUF5	LC10BUF4	LC10BUF3	LC10BUF2	LC10BUF1	LC10BUF0	XXII-18
		X	X	X	X	X	X	X	X	
		Segment output latch data								
0x03E31	LC11BUF	LC11BUF7	LC11BUF6	LC11BUF5	LC11BUF4	LC11BUF3	LC11BUF2	LC11BUF1	LC11BUF0	XXII-18
		X	X	X	X	X	X	X	X	
		Segment output latch data								
0x03E32	LC12BUF	LC12BUF7	LC12BUF6	LC12BUF5	LC12BUF4	LC12BUF3	LC12BUF2	LC12BUF1	LC12BUF0	XXII-18
		X	X	X	X	X	X	X	X	
		Segment output latch data								
0x03E33	LC13BUF	LC13BUF7	LC13BUF6	LC13BUF5	LC13BUF4	LC13BUF3	LC13BUF2	LC13BUF1	LC13BUF0	XXII-18
		X	X	X	X	X	X	X	X	
		Segment output latch data								
0x03E34	LC14BUF	-	-	-	-	LC14BUF3	LC14BUF2	LC14BUF1	LC14BUF0	XXII-18
		-	-	-	-	X	X	X	X	
		Segment output latch data								
0x03E40	LCDMD1	LCDEN	Reserved	DUTY1	DUTY0	LCDCCK3	LCDCCK2	LCDCCK1	LCDCCK0	XXII-7
		0	0	0	0	0	0	0	0	
		LCD driving circuit start flag	Always set to "0"	LCD display duty selection			LCD clock source selection			
0x03E41	LCDMD2	Reserved	Reserved	Reserved	LCRHL1	LCRHL0	LCREN	Reserved	Reserved	XXII-8
		0	0	0	0	0	0	0	0	
		Always set to "0"	Always set to "0"	Always set to "0"	Internal dividing resistor type selection		Internal dividing resistor connecting selection	Always set to "0"	Always set to "0"	
0x03E42	LCDMD3	-	-	-	-	TMSEL3	TMSEL2	TMSEL1	LCDCCKSEL	XXII-9
		-	-	-	-	0	0	0	0	
		-	-	-	-	Timer 0 to 4 and A input selection				

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Address	Register	Bit symbol								Page
		bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0	
0x03E43	LCCTR0	COMSL3	COMSL2	COMSL1	COMSL0	Reserved	VLC3SL	VLC2SL	VLC1SL	XXII-10
		0	0	0	0	0	0	0	0	
		COM3/P84 selection	COM2/P83 selection	COM1/P82 selection	COM0/P81 selection	Always set to "0"	VLC3/P85 selection	VLC2/P86 selection	VLC1/P87 selection	
0x03E44	LCCTR1	LC1SL7	LC1SL6	LC1SL5	LC1SL4	LC1SL3	LC1SL2	LC1SL1	LC1SL0	XXII-11
		0	0	0	0	0	0	0	0	
		SEG7/P71 selection	SEG6/P72 selection	SEG5/P73 selection	SEG4/P74 selection	SEG3/P75 selection	SEG2/P76 selection	SEG1/P77 selection	SEG0/P80 selection	
0x03E45	LCCTR2	LC2SL7	LC2SL6	LC2SL5	LC2SL4	LC2SL3	LC2SL2	LC2SL1	LC2SL0	XXII-12
		0	0	0	0	0	0	0	0	
		SEG15/P50 selection	SEG14/P62 selection	SEG13/P63 selection	SEG12/P64 selection	SEG11/P65 selection	SEG10/P66 selection	SEG9/P67 selection	SEG8/P70 selection	
0x03E46	LCCTR3	LC3SL7	LC3SL6	LC3SL5	LC3SL4	LC3SL3	LC3SL2	LC3SL1	LC3SL0	XXII-13
		0	0	0	0	0	0	0	0	
		SEG23/P47 selection	SEG22/P57 selection	SEG21/P56 selection	SEG20/P55 selection	SEG19/P54 selection	SEG18/P53 selection	SEG17/P52 selection	SEG16/P51 selection	
0x03E47	LCCTR4	LC4SL7	LC4SL6	LC4SL5	LC4SL4	LC4SL3	LC4SL2	LC4SL1	LC4SL0	XXII-14
		0	0	0	0	0	0	0	0	
		SEG31/P92 selection	SEG30/P33 selection	SEG29/P34 selection	SEG28/P35 selection	SEG27/P43 selection	SEG26/P44 selection	SEG25/P45 selection	SEG24/P46 selection	
0x03E48	LCCTR5	LC5SL7	LC5SL6	LC5SL5	LC5SL4	LC5SL3	LC5SL2	LC5SL1	LC5SL0	XXII-15
		0	0	0	0	0	0	0	0	
		SEG39/PA1 selection	SEG38/PA0 selection	SEG37/PB0 selection	SEG36/PB1 selection	SEG35/PB2 selection	SEG34/PB3 selection	SEG33/P94 selection	SEG32/P93 selection	
0x03E49	LCCTR6	LC6SL7	LC6SL6	LC6SL5	LC6SL4	LC6SL3	LC6SL2	LC6SL1	LC6SL0	XXII-16
		0	0	0	0	0	0	0	0	
		Always set to "0"								
0x03E50	PRTKEY	PRTK7	PRTK6	PRTK5	PRTK4	PRTK3	PRTK2	PRTK1	PRTK0	V-4
		0	0	0	0	0	0	0	0	
		Register protect control								
0x03E51	CKTRMD	-	-	-	-	-	-	-	TRMD	IV-10
		-	-	-	-	-	-	-	0	
		-	-	-	-	-	-	-	Clock transition mode	
0x03E52	CKMONEN	-	-	-	-	-	MVCLK	DETIRQ	MONIEN	IV-5
		-	-	-	-	-	0	0	0	
		-	-	-	-	-	Clock switching selection	Detection interrupt selection	Clock monitoring function enable	
0x03E53	CKMONSETU	MNUSET7	MNUSET6	MNUSET5	MNUSET4	MNUSET3	MNUSET2	MNUSET1	MNUSET0	IV-6
		0	0	0	0	0	0	0	0	
		Clock monitoring frequency upper limit setting value								
0x03E54	CKMONSETL	MNLSET7	MNLSET6	MNLSET5	MNLSET4	MNLSET3	MNLSET2	MNLSET1	MNLSET0	IV-8
		0	0	0	0	0	0	0	0	
		Clock monitoring frequency lower limit setting value								
0x03E55	CKMONSTAT	-	-	-	-	-	PLLERR	XIERR	OSCERR	IV-9
		-	-	-	-	-	0/Retained	0/Retained	0/Retained	
		-	-	-	-	-	PLL using flag	External low-speed using flag	External high-speed using flag	
0x03E56	OSLOCK	-	-	-	-	-	-	-	LOCKEN	III-19
		-	-	-	-	-	-	-	0	
		-	-	-	-	-	-	-	Oscillation control register writing control	
0x03E60	AT1CNT0	FM0DE	AT1ACT	AT1MD3	AT1MD2	AT1MD1	AT1MD0	Reserved	AT1EN	XXIII-7
		0	0	0	0	0	0	0	0	
		Increment control for memory pointer 0	ATC1 software activation flag	ATC1 data transfer mode					Always set to "0"	

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Address	Register	Bit symbol								Page
		bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0	
0x03E61	AT1CNT1	AT1IRS	-	Reserved	BTSTP	AT1IR3	AT1IR2	AT1IR1	AT1IR0	XXIII-8
		0	-	0	0	0	0	0	0	
		Activation factor selection flag	-	Always set to "0"	Burst transfer stop enable	ATC1 trigger factor setting flag				
0x03E62	AT1TRC	AT1TRC7	AT1TRC6	AT1TRC5	AT1TRC4	AT1TRC3	AT1TRC2	AT1TRC1	AT1TRC0	XXIII-9
		0	0	0	0	0	0	0	0	
		ATC1 Transfer Data Count Setting								
0x03E63	AT1MAP0L	AT1MAP0L7	AT1MAP0L6	AT1MAP0L5	AT1MAP0L4	AT1MAP0L3	AT1MAP0L2	AT1MAP0L1	AT1MAP0L0	XXIII-9
		0	0	0	0	0	0	0	0	
		ATC1 Memory Pointer 0: Lower 8 bits								
0x03E64	AT1MAP0M	AT1MAP0M7	AT1MAP0M6	AT1MAP0M5	AT1MAP0M4	AT1MAP0M3	AT1MAP0M2	AT1MAP0M1	AT1MAP0M0	XXIII-9
		0	0	0	0	0	0	0	0	
		ATC1 Memory Pointer 0: Middle 8 bits								
0x03E65	AT1MAP0H	-	-	-	-	AT1MAP0H3	AT1MAP0H2	AT1MAP0H1	AT1MAP0H0	XXIII-9
		-	-	-	-	0	0	0	0	
		-	-	-	-	ATC1 Memory Pointer 0: Upper 4 bits				
0x03E66	AT1MAP1L	AT1MAP1L7	AT1MAP1L6	AT1MAP1L5	AT1MAP1L4	AT1MAP1L3	AT1MAP1L2	AT1MAP1L1	AT1MAP1L0	XXIII-10
		0	0	0	0	0	0	0	0	
		ATC1 Memory Pointer 1: Lower 8 bits								
0x03E67	AT1MAP1M	AT1MAP1M7	AT1MAP1M6	AT1MAP1M5	AT1MAP1M4	AT1MAP1M3	AT1MAP1M2	AT1MAP1M1	AT1MAP1M0	XXIII-10
		0	0	0	0	0	0	0	0	
		ATC1 Memory Pointer 1: Middle 8 bits								
0x03E68	AT1MAP1H	-	-	-	-	AT1MAP1H3	AT1MAP1H2	AT1MAP1H1	AT1MAP1H0	XXIII-10
		-	-	-	-	0	0	0	0	
		-	-	-	-	ATC1 Memory Pointer 1: Upper 4 bits				
0x03E70	P0OUT	P0OUT7	P0OUT6	P0OUT5	P0OUT4	P0OUT3	P0OUT2	P0OUT1	P0OUT0	VIII-8
		x	x	X	X	X	X	X	X	
		Port 0 output data								
0x03E72	P2OUT	P2OUT7	P2OUT6	P2OUT5	P2OUT4	P2OUT3	P2OUT2	P2OUT1	P2OUT0	VIII-45
		1	x	x	X	X	X	X	X	
		Port 2 output data								
0x03E73	P3OUT	-	-	P3OUT5	P3OUT4	P3OUT3	-	-	-	VIII-55
		-	-	X	X	X	-	-	-	
		Port 3 output data								
0x03E74	P4OUT	P4OUT7	P4OUT6	P4OUT5	P4OUT4	P4OUT3	-	-	-	VIII-72
		X	X	X	X	X	-	-	-	
		Port 4 output data								
0x03E75	P5OUT	P5OUT7	P5OUT6	P5OUT5	P5OUT4	P5OUT3	P5OUT2	P5OUT1	P5OUT0	VIII-93
		X	X	X	X	X	X	X	X	
		Port 5 output data								
0x03E76	P6OUT	P6OUT7	P6OUT6	P6OUT5	P6OUT4	P6OUT3	P6OUT2	-	-	VIII-111
		X	X	X	X	X	X	-	-	
		Port 6 output data								
0x03E77	P7OUT	P7OUT7	P7OUT6	P7OUT5	P7OUT4	P7OUT3	P7OUT2	P7OUT1	P7OUT0	VIII-131
		X	X	X	X	X	X	X	X	
		Port 7 output data								
0x03E78	P8OUT	P8OUT7	P8OUT6	P8OUT5	P8OUT4	P8OUT3	P8OUT2	P8OUT1	P8OUT0	VIII-153
		X	X	X	X	X	X	X	X	
		Port 8 output data								
0x03E79	P9OUT	-	-	-	P9OUT4	P9OUT3	P9OUT2	P9OUT1	P9OUT0	VIII-175
		-	-	-	x	x	x	x	x	
		Port 9 output data								
0x03E7A	PAOUT	PAOUT7	PAOUT6	PAOUT5	PAOUT4	PAOUT3	PAOUT2	PAOUT1	PAOUT0	VIII-191
		X	X	X	X	X	X	X	X	
		Port A output data								

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Address	Register	Bit symbol								Page
		bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0	
0x03E7B	PBOUT	-	-	-	-	PBOUT3	PBOUT2	PBOUT1	PBOUT0	VIII-211
		-	-	-	-	x	x	x	x	
		Port B output data								
0x03E80	P0IN	P0IN7	P0IN6	P0IN5	P0IN4	P0IN3	P0IN2	P0IN1	P0IN0	VIII-9
		X	X	X	X	X	X	X	X	
		Port 0 input data								
0x03E82	P2IN	P2IN7	P2IN6	P2IN5	P2IN4	P2IN3	P2IN2	P2IN1	P2IN0	VIII-45
		X	X	X	X	X	X	X	X	
		Port 2 input data								
0x03E83	P3IN	-	-	P3IN5	P3IN4	P3IN3	-	-	-	VIII-56
		-	-	X	X	X	-	-	-	
		Port 3 input data								
0x03E84	P4IN	P4IN7	P4IN6	P4IN5	P4IN4	P4IN3	-	-	-	VIII-73
		X	X	X	X	X	-	-	-	
		Port 4 input data								
0x03E85	P5IN	P5IN7	P5IN6	P5IN5	P5IN4	P5IN3	P5IN2	P5IN1	P5IN0	VIII-93
		X	X	X	X	X	X	X	X	
		Port 5 input data								
0x03E86	P6IN	P6IN7	P6IN6	P6IN5	P6IN4	P6IN3	P6IN2	-	-	VIII-112
		X	X	X	X	X	X	-	-	
		Port 6 input data								
0x03E87	P7IN	P7IN7	P7IN6	P7IN5	P7IN4	P7IN3	P7IN2	P7IN1	P7IN0	VIII-131
		X	X	X	X	X	X	X	X	
		Port 7 input data								
0x03E88	P8IN	P8IN7	P8IN6	P8IN5	P8IN4	P8IN3	P8IN2	P8IN1	P8IN0	VIII-153
		X	X	X	X	X	X	X	X	
		Port 8 input data								
0x03E89	P9IN	-	-	-	P9IN4	P9IN3	P9IN2	P9IN1	P9IN0	VIII-176
		-	-	-	X	X	X	X	X	
		Port 9 input data								
0x03E8A	PAIN	PAIN7	PAIN6	PAIN5	PAIN4	PAIN3	PAIN2	PAIN1	PAIN0	VIII-191
		X	X	X	X	X	X	X	X	
		Port A input data								
0x03E8B	PBIN	-	-	-	-	PBIN3	PBIN2	PBIN1	PBIN0	VIII-212
		-	-	-	-	X	X	X	X	
		Port B input data								
0x03E8F	SWCNT	-	-	-	-	-	SWCNT_A	-	-	VIII-195 VIII-243
		-	-	-	-	-	0	-	-	
		-	-	-	-	-	Port A Input voltage level selection	-	-	
0x03E90	P0DIR	P0DIR7	P0DIR6	P0DIR5	P0DIR4	P0DIR3	P0DIR2	P0DIR1	P0DIR0	VIII-9
		0	0	0	0	0	0	0	0	
		Port 0 direction control								
0x03E92	P2DIR	-	P2DIR6	P2DIR5	P2DIR4	P2DIR3	P2DIR2	P2DIR1	P2DIR0	VIII-46
		-	0	0	0	0	0	0	0	
		Port 2 direction control								
0x03E93	P3DIR	-	-	P3DIR5	P3DIR4	P3DIR3	-	-	-	VIII-57
		-	-	0	0	0	-	-	-	
		Port 3 direction control								
0x03E94	P4DIR	P4DIR7	P4DIR6	P4DIR5	P4DIR4	P4DIR3	-	-	-	VIII-74
		0	0	0	0	0	-	-	-	
		Port 4 direction control								
0x03E95	P5DIR	P5DIR7	P5DIR6	P5DIR5	P5DIR4	P5DIR3	P5DIR2	P5DIR1	P5DIR0	VIII-94
		0	0	0	0	0	0	0	0	
		Port 5 direction control								

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Address	Register	Bit symbol								Page
		bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0	
0x03E96	P6DIR	P6DIR7	P6DIR6	P6DIR5	P6DIR4	P6DIR3	P6DIR2	-	-	VIII-113
		0	0	0	0	0	0	-	-	
		Port 6 direction control								
0x03E97	P7DIR	P7DIR7	P7DIR6	P7DIR5	P7DIR4	P7DIR3	P7DIR2	P7DIR1	P7DIR0	VIII-132
		0	0	0	0	0	0	0	0	
		Port 7 direction control								
0x03E98	P8DIR	P8DIR7	P8DIR6	P8DIR5	P8DIR4	P8DIR3	P8DIR2	P8DIR1	P8DIR0	VIII-154
		0	0	0	0	0	0	0	0	
		Port 8 direction control								
0x03E99	P9DIR	-	-	-	P9DIR4	P9DIR3	P9DIR2	P9DIR1	P9DIR0	VIII-177
		-	-	-	0	0	0	0	0	
		Port 9 direction control								
0x03E9A	PADIR	PADIR7	PADIR6	PADIR5	PADIR4	PADIR3	PADIR2	PADIR1	PADIR0	VIII-192
		0	0	0	0	0	0	0	0	
		Port A direction control								
0x03E9B	PBDIR	-	-	-	-	PBDIR3	PBDIR2	PBDIR1	PBDIR0	VIII-213
		-	-	-	-	0	0	0	0	
		Port B direction control								
0x03EA0	P0PLUD	P0PLUD7	P0PLUD6	P0PLUD5	P0PLUD4	P0PLUD3	P0PLUD2	P0PLUD1	P0PLUD0	VIII-9
		0	0	0	0	0	0	0	0	
		Port 0 pull-up/pull-down resistor control								
0x03EA2	P2PLU	-	P2PLU6	P2PLU5	P2PLU4	P2PLU3	P2PLU2	P2PLU1	P2PLU0	VIII-46
		-	0	0	0	0	0	0	0	
		Port 2 pull-up resistor control								
0x03EA3	P3PLU	-	-	P3PLU5	P3PLU4	P3PLU3	-	-	-	VIII-58
		-	-	0	0	0	-	-	-	
		Port 3 pull-up resistor control								
0x03EA4	P4PLUD	P4PLUD7	P4PLUD6	P4PLUD5	P4PLUD4	P4PLUD3	-	-	-	VIII-75
		0	0	0	0	0	-	-	-	
		Port 4 pull-up/pull-down resistor control								
0x03EA5	P5PLUD	P5PLUD7	P5PLUD6	P5PLUD5	P5PLUD4	P5PLUD3	P5PLUD2	P5PLUD1	P5PLUD0	VIII-94
		0	0	0	0	0	0	0	0	
		Port 5 pull-up/pull-down resistor control								
0x03EA6	P6PLU	P6PLU7	P6PLU6	P6PLU5	P6PLU4	P6PLU3	P6PLU2	-	-	VIII-114
		0	0	0	0	0	0	-	-	
		Port 6 pull-up resistor control								
0x03EA7	P7PLUD	P7PLUD7	P7PLUD6	P7PLUD5	P7PLUD4	P7PLUD3	P7PLUD2	P7PLUD1	P7PLUD0	VIII-132
		0	0	0	0	0	0	0	0	
		Port 7 pull-up/pull-down resistor control								
0x03EA8	P8PLU	P8PLU7	P8PLU6	P8PLU5	P8PLU4	P8PLU3	P8PLU2	P8PLU1	P8PLU0	VIII-154
		0	0	0	0	0	0	0	0	
		Port 8 pull-up resistor control								
0x03EA9	P9PLU	-	-	-	P9PLU4	P9PLU3	P9PLU2	P9PLU1	P9PLU0	VIII-178
		-	-	-	0	0	0	0	0	
		Port 9 pull-up resistor control								
0x03EAA	PAPLU	PAPLU7	PAPLU6	PAPLU5	PAPLU4	PAPLU3	PAPLU2	PAPLU1	PAPLU0	VIII-192
		0	0	0	0	0	0	0	0	
		Port A pull-up resistor control								
0x03EAB	PBPLU	-	-	-	-	PBPLU3	PBPLU2	PBPLU1	PBPLU0	VIII-214
		-	-	-	-	0	0	0	0	
		Port B pull-up resistor control								
0x03EAF	SELUD	SELUD7	-	SELUD5	SELUD4	-	-	-	SELUD0	VIII-13 VIII-36 VIII-77 VIII-96 VIII-134
		0	-	0	0	-	-	-	0	
		Pull-up/Pull-down resistor selection								

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Address	Register	Bit symbol								Page
		bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0	
0x03EB0	P0OMD1	-	-	-	P0OMD14	P0OMD13	P0OMD12	P0OMD11	-	VIII-10
		-	-	-	0	0	0	0	-	
		-	-	-	Port/TM2IOB/ TM0IOB selec- tion	Port/TM8IOB selection	Port/TM7IOB selection	Port/TM9IOB selection	-	
0x03EB5	P5OMD1	P5OMD7	P5OMD6	-	-	-	-	-	-	VIII-95
		0	0	-	-	-	-	-	-	
		Port/BUZZERA selection	Port/ NBUZZERA selection	-	-	-	-	-	-	
0x03EB6	P6OMD	-	-	-	P6OMD4	P6OMD3	P6OMD2	-	-	VIII-115
		-	-	-	0	0	0	-	-	
		-	-	-	Port/TM4IOB selection	Port/TM3IOB selection	Port/TM1IOB selection	-	-	
0x03EB8	P8OMD	-	-	P8OMD5	P8OMD4	P8OMD3	P8OMD2	P8OMD1	P8OMD0	VIII-155
		-	-	0	0	0	0	0	0	
		-	-	Port/TM9OD5A selection	Port/TM9OD4A selection	Port/TM9OD3A selection	Port/TM9OD2A selection	Port/TM9OD1A selection	Port/TM9OD0A selection	
0x03EBA	PAOMD1	PAOMD7	PAOMD6	PAOMD5	PAOMD4	PAOMD3	PAOMD2	PAOMD1	PAOMD0	VIII-193
		0	0	0	0	0	0	0	0	
		Port/TM9IOA selection	Port/TM8IOA selection	Port/TM7IOA selection	Port/TM4IOA selection	Port/TM3IOA selection	Port/TM2IOA selection	Port/TM1IOA selection	Port/TM0IOA selection	
0x03EC0	P0OMD2	-	-	-	P0OMD24	-	-	-	-	VIII-10
		-	-	-	0	-	-	-	-	
		-	-	-	Special func- tion selec- tion (TM0IOB/ TM2IOB)	-	-	-	-	
0x03EC3	P3IMD	-	-	-	-	P3IMD3	-	-	-	VIII-58
		-	-	-	-	0	-	-	-	
		-	-	-	-	Port/analog input selection	-	-	-	
0x03EC9	P9IMD	-	-	-	P9IMD4	P9IMD3	P9IMD2	-	-	VIII-179
		-	-	-	0	0	0	-	-	
		Port/analog input selection								
0x03ECA	PAIMD	PAIMD7	PAIMD6	PAIMD5	PAIMD4	PAIMD3	PAIMD2	PAIMD1	PAIMD0	VIII-194
		0	0	0	0	0	0	0	0	
		Port/analog input selection								
0x03ECB	PBIMD	-	-	-	-	PBIMD3	PBIMD2	PBIMD1	PBIMD0	VIII-215
		-	-	-	-	0	0	0	0	
		Port/analog input selection								
0x03EDA	PACNT	-	-	PACNT5	PACNT4	PACNT3	PACNT2	PACNT1	PACNT0	VIII-195
		-	-	0	0	0	0	0	0	
		Port A real-time control								
0x03EE0	P0LED	P0LED7	P0LED6	P0LED5	P0LED4	P0LED3	P0LED2	P0LED1	P0LED0	VIII-12
		0	0	0	0	0	0	0	0	
		LED output selection								
0x03EF0	P0ODC	Reserved	-	Reserved	P0ODC4	P0ODC3	-	-	-	VIII-11
		0	-	0	0	0	-	-	-	
		Always set to "0".	-	Always set to "0".	Nch open-drain output selection		-	-	-	
0x03EE3	P3ODC	-	-	-	P3ODC4	P3ODC3	-	-	-	VIII-59
		-	-	-	0	0	-	-	-	
		Nch open-drain output selection								
0x03EE4	P4ODC	-	-	P4ODC5	-	P4ODC3	-	-	-	VIII-76
		-	-	0	-	0	-	-	-	
		Nch open-drain output selection								
0x03EE5	P5ODC	-	-	-	-	-	P5ODC2	-	P5ODC0	VIII-94
		-	-	-	-	-	0	-	0	
		Nch open-drain output selection								

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Address	Register	Bit symbol								Page
		bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0	
0x03EE6	P6ODC	P6ODC7	-	P6ODC5	-	-	-	-	-	VIII-115
		0	-	0	-	-	-	-	-	
		Nch open-drain output selection								
0x03EE7	P7ODC	P7ODC7	-	P7ODC5	-	-	P7ODC2	P7ODC1	-	VIII-133
		0	-	0	-	-	0	0	-	
		Nch open-drain output selection								
0x03EF8	P8SYO	P8SYO7	P8SYO6	P8SYO5	P8SYO4	P8SYO3	P8SYO2	P8SYO1	P8SYO0	VIII-156
		0	0	0	0	0	0	0	0	
		Synchronous output selection								
0x03EF9	P8SEV	-	-	-	-	-	-	P8SEV1	P8SEV0	VIII-157
		-	-	-	-	-	-	0	0	
		-	-	-	-	-	-	Synchronous output event selection		
0x03F00	CPUM	Reserved	OSCSEL1	OSCSEL0	OSCDBL	STOP	HALT	OSC1	OSC0	III-22
		0	0	1	0	0	0	0	0	
		Always set to "0"	Division ratio		Internal system clock	Operation mode control				
0x03F01	MEMCTR	IOW1	IOW0	IVBM	Reserved	Reserved	IRWE	Reserved	Reserved	II-38
		1	1	0	0	1	0	1	1	
		Wait cycles when accessing special register area		Base address specification for interrupt vector table	Always set to "0"	Always set to "1"	Software write setting for interrupt request flag	Always set to "1"		
0x03F02	WDCTR	-	-	Reserved	Reserved	Reserved	WDTS1	WDTS0	WDEN	XV-5
		-	-	0	0	0	1	1	0	
		-	-	Always set to "0"	Always set to "0"	Always set to "0"	Watchdog error detect cycle		Watchdog timer enable	
0x03F03	DLYCTR	BUZOE	BUZS2	BUZS1	BUZS0	DLYS1	DLYS0	-	-	XV-6
		0	0	0	0	0	1	-	-	
		Buzzer output enable	Buzzer output frequency			Oscillation stabilization wait cycle		-	-	
0x03F05	WD2CTR	Reserved	Reserved	-	-	WDTS23	WDTS22	WDTS21	WDTS20	XVI-5
		0	0	-	-	1	0	1	1	
		Always set to "0"	Always set to "0"	-	-	Watchdog timer 2 cycle setup				
0x03F06	HANDSHAKE	-	-	-	-	-	ROMHND	-	-	III-24
		-	-	-	-	-	1	-	-	
		-	-	-	-	-	Internal ROM access method switching	-	-	
0x03F07	AUCTR	-	-	-	-	-	AUDIVU	AUMUL	AUMULU	II-41
		-	-	-	-	-	0	0	0	
		-	-	-	-	-	Unsigned division execution	Signed multiplication execution	Unsigned multiplication execution	
0x03F0A	SBNKR	-	-	-	-	SBA3	SBA2	SBA1	SBA0	II-21
		-	-	-	-	0	0	0	0	
		-	-	-	-	Bank setting for source address				
0x03F0B	DBNKR	-	-	-	-	DBA3	DBA2	DBA1	DBA0	II-22
		-	-	-	-	0	0	0	0	
		-	-	-	-	Bank setting for destination address				
0x03F10	SC0SEL	SBO0SEL	SC0BRP2	SC0BRP1	SC0BRP0	OSL0	SC0SEL2	SC0SEL1	SC0SEL0	XVIII-19
		0	0	0	0	0	0	0	0	
		UART reverse output selection	Timer clock output selection			Serial output port selection	Timer selection			
0x03F11	SC0MD0	SC0CE1	-	Reserved	SC0DIR	SC0STE	SC0LNG2	SC0LNG1	SC0LNG0	XVIII-27
		0	-	0	0	0	1	1	1	
		Communication data I/O edge	-	Always set to "0"	Transfer bit specify	Start condition selection	Number of synchronous serial transfer bit			

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Address	Register	Bit symbol								Page
		bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0	
0x03F12	SC0MD1	SC0IOM	SC0SBTS	SC0SBIS	SC0SBOS	SC0CKM	SC0MST	SC0DIV	SC0CMD	XVIII-28
		0	0	0	0	0	0	0	0	
		Serial data input pin selection	SBT function selection	Serial input control	SBO (TXD) function selection	Transfer clock dividing selection	Clock master/slave selection	Transfer clock dividing value selection	Clock synchronous/UART selection	
0x03F13	SC0MD2	SC0FM1	SC0FM0	SC0PM1	SC0PM0	SC0NPE	Reserved	SC0BRKF	SC0BRKE	XVIII-29
		0	0	0	0	0	0	0	0	
		Frame mode specify		Additional bit specify		Parity enable	Always set to "0"	Break status reception monitor	Break status transmission control	
0x03F14	SC0MD3	SC0FDC1	SC0FDC0	-	-	SC0PSCE	SC0PSC2	SC0PSC1	SC0PSC0	XVIII-30
		0	0	-	-	0	0	0	0	
		Output selection after SBO final data transmit		-	-	Prescaler count control	Clock selection			
0x03F15	SC0STR	SC0TBSY	SC0RBSY	SC0TEMP	SC0REMP	SC0FEF	SC0PEK	SC0ORE	SC0ERE	XVIII-31
		0	0	0	0	0	0	0	0	
		Serial bus transmission status	Serial bus reception status	Transmission buffer empty flag	Reception buffer empty flag	Frame error detection	Parity error detection	Overrun error detection	Error monitor flag	
0x03F16	RXBUF0	RXBUF07	RXBUF06	RXBUF05	RXBUF04	RXBUF03	RXBUF02	RXBUF01	RXBUF00	XVIII-25
		x	x	x	x	x	x	x	x	
		Serial interface 0 reception data buffer								
0x03F17	TXBUF0	TXBUF07	TXBUF06	TXBUF05	TXBUF04	TXBUF03	TXBUF02	TXBUF01	TXBUF00	XVIII-25
		x	x	x	x	x	x	x	x	
		Serial interface 0 transmission data buffer								
0x03F18	LINCTR	SFBSY	SBFBSY	SFREC	SBFREC	SBFEN	SBFSEL2	SBFSEL1	SBFSEL0	XIX-5
		0	0	0	0	0	0	0	0	
		SF reception busy flag	SBF reception busy flag	SF reception enable	SBF reception enable	SBF transmission trigger/monitor	SBF transmission width selection			
0x03F19	RSTFACT	-	-	-	-	-	-	-	RSTMON	II-54
		-	-	-	-	-	-	-	0	
		-	-	-	-	-	-	-	Reset factor determination	
0x03F1A	OSCCNT	HOSCDIVSEL2	HOSCDIVSEL1	HOSCDIVSEL0	HOSCDIVS-TOP	-	-	HOSCSEL	HOSCCNT	III-10
		0	0	0	0	-	-	0	0	
		Division setting of high-speed oscillation for peripheral functions			High-speed oscillation for peripheral functions control	-	-	Operation clock selection	P25/P26 function selection	
0x03F1B	RCCNT	Reserved	-	-	-	-	-	RCFSEL	RCON	III-12
		1	-	-	-	-	-	0	0	
		Always set to "1"	-	-	-	-	-	Internal high-speed oscillation frequency selection	Internal high-speed oscillation control	
0x03F1C	OSCSCNT	-	-	-	-	-	-	SOSCSEL	SOSCCNT	III-14
		-	-	-	-	-	-	0	0	
		-	-	-	-	-	-	Operation clock selection	P90/P91 function selection	
0x03F1D	RCSCNT	-	-	-	-	-	-	-	RCSON	III-16
		-	-	-	-	-	-	-	0	
		-	-	-	-	-	-	-	Internal low-speed oscillation control	
0x03F1E	EDGDT	EDGSEL7	-	EDGSEL5	-	EDGSEL3	EDGSEL2	-	EDGSEL0	VII-56
		0	-	0	-	0	0	-	0	
		IRQ4 both edges operation selection	-	LINIRQ both edges operation selection	-	IRQ3 both edges operation selection	IRQ2 both edges operation selection	-	Key interrupt selection	
0x03F1F	PLLCNT	PLLCK3	PLLCK2	PLLCK1	PLLCK0	-	-	PLLEN	PLLSTART	III-20
		0	0	0	0	-	-	0	0	
		Multiplication setting				-	-	PLL clock enable	PLL operation control	

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Address	Register	Bit symbol								Page
		bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0	
0x03F20	SC1SEL	SBO1SEL	SC1BRP2	SC1BRP1	SC1BRP0	OSL1	SC1SEL2	SC1SEL1	SC1SEL0	XVIII-20
		0	0	0	0	0	0	0	0	
		UART reverse output selection	Timer clock output selection			Serial output port selection	Timer selection			
0x03F21	SC1MD0	SC1CE1	-	Reserved	SC1DIR	SC1STE	SC1LNG2	SC1LNG1	SC1LNG0	XVIII-27
		0	-	0	0	0	1	1	1	
		Communication data I/O edge	-	Always set to "0"	Transfer bit specify	Start condition selection	Number of synchronous serial transfer bit			
0x03F22	SC1MD1	SC1IOM	SC1SBTS	SC1SBIS	SC1SBOS	SC1CKM	SC1MST	SC1DIV	SC1CMD	XVIII-28
		0	0	0	0	0	0	0	0	
		Serial data input pin selection	SBT function selection	Serial input control	SBO (TXD) function selection	Transfer clock dividing selection	Clock master/slave selection	Transfer clock dividing value selection	Clock synchronous/UART selection	
0x03F23	SC1MD2	SC1FM1	SC1FM0	SC1PM1	SC1PM0	SC1NPE	Reserved	SC1BRKF	SC1BRKE	XVIII-29
		0	0	0	0	0	0	0	0	
		Frame mode specify		Additional bit specify		Parity enable	Always set to "0"	Break status reception monitor	Break status transmission control	
0x03F24	SC1MD3	SC1FDC1	SC1FDC0	-	-	SC1PSCE	SC1PSC2	SC1PSC1	SC1PSC0	XVIII-30
		0	0	-	-	0	0	0	0	
		Output selection after SBO final data transmit		-	-	Prescaler count control	Clock selection			
0x03F25	SC1STR	SC1TBSY	SC1RBSY	SC1TEMP	SC1REMP	SC1FEF	SC1PEK	SC1ORE	SC1ERE	XVIII-31
		0	0	0	0	0	0	0	0	
		Serial bus transmission status	Serial bus reception status	Transmission buffer empty flag	Reception buffer empty flag	Frame error detection	Parity error detection	Overrun error detection	Error monitor flag	
0x03F26	RXBUF1	RXBUF17	RXBUF16	RXBUF15	RXBUF14	RXBUF13	RXBUF12	RXBUF11	RXBUF10	XVIII-25
		x	x	x	x	x	x	x	x	
0x03F27	TXBUF1	Serial interface 1 reception data buffer								XVIII-25
		TXBUF17	TXBUF16	TXBUF15	TXBUF14	TXBUF13	TXBUF12	TXBUF11	TXBUF10	
		Serial interface 1 transmission data buffer								
0x03F28	SCINTSEL	-	-	-	-	-	-	SC1IGC	SC0IGC	XVIII-33
		-	-	-	-	-	-	0	0	
		-	-	-	-	-	-	Serial 1 reception interrupt generation timing	Serial 0 reception interrupt generation timing	
0x03F2E	ACZCTR	P21IM	-	-	-	P20IM	-	-	-	VII-55
		0	-	-	-	0	-	-	-	
		ACZ1 input enable	-	-	-	ACZ0 input enable	-	-	-	
0x03F2F	LVIMD	LVIMOD	LVICTR2	LVICTR1	LVICTR0	-	-	LVIMON	LVION	VI-3
		0	0	0	0	-	-	0	0	
		Power supply voltage detection voltage setting				-	-	Power supply voltage monitor flag	Power supply detection enable	
0x03F30	SC2SEL	SBO2SEL	SC2BRP2	SC2BRP1	SC2BRP0	Reserved	SC2SEL2	SC2SEL1	SC2SEL0	XVIII-22
		0	0	0	0	0	0	0	0	
		UART reverse output selection	Timer clock output selection			Always set to "0"	Timer selection			
0x03F31	SC2MD0	SC2CE1	-	Reserved	SC2DIR	SC2STE	SC2LNG2	SC2LNG1	SC2LNG0	XVIII-27
		0	-	0	0	0	1	1	1	
		Communication data I/O edge	-	Always set to "0"	Transfer bit specify	Start condition selection	Number of synchronous serial transfer bit			
0x03F32	SC2MD1	SC2IOM	SC2SBTS	SC2SBIS	SC2SBOS	SC2CKM	SC2MST	SC2DIV	SC2CMD	XVIII-28
		0	0	0	0	0	0	0	0	
		Serial data input pin selection	SBT function selection	Serial input control	SBO (TXD) function selection	Transfer clock dividing selection	Clock master/slave selection	Transfer clock dividing value selection	Clock synchronous/UART selection	

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Address	Register	Bit symbol								Page
		bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0	
0x03F33	SC2MD2	SC2FM1	SC2FM0	SC2PM1	SC2PM0	SC2NPE	Reserved	SC2BRKF	SC2BRKE	XVIII-29
		0	0	0	0	0	0	0	0	
		Frame mode specification		Additional bit specification		Parity enable	Always set to "0"	Break status reception monitor	Break status transmission control	
0x03F34	SC2MD3	SC2FDC1	SC2FDC0	-	-	SC2PSCE	SC2PSC2	SC2PSC1	SC2PSC0	XVIII-30
		0	0	-	-	0	0	0	0	
		Output selection after SBO final data transmit		-	-	Prescaler count control				
0x03F35	SC2STR	SC2TBSY	SC2RBSY	SC2TEMP	SC2REMP	SC2FEF	SC2PEK	SC2ORE	SC2ERE	XVIII-31
		0	0	0	0	0	0	0	0	
		Serial bus transmission status	Serial bus reception status	Transmission buffer empty flag	Reception buffer empty flag	Frame error detection	Parity error detection	Overrun error detection	Error monitor flag	
0x03F36	RXBUF2	RXBUF27	RXBUF26	RXBUF25	RXBUF24	RXBUF23	RXBUF22	RXBUF21	RXBUF20	XVIII-25
		x	x	x	x	x	x	x	x	
		Serial interface 2 reception data buffer								
0x03F37	TXBUF2	TXBUF27	TXBUF26	TXBUF25	TXBUF24	TXBUF23	TXBUF22	TXBUF21	TXBUF20	XVIII-25
		x	x	x	x	x	x	x	x	
		Serial interface 2 transmission data buffer								
0x03F3E	KEYT3_1IMD	KEYT3SEL	-	-	-	KEYT3_1EN3	KEYT3_1EN2	KEYT3_1EN1	KEYT3_1EN0	VII-58
		0	-	-	-	0	0	0	0	
		Interrupt source selection	-	-	-	KEY interrupt selection				
0x03F3F	KEYT3_2IMD	-	-	-	-	KEYT3_2EN3	KEYT3_2EN2	KEYT3_2EN1	KEYT3_2EN0	VII-59
		-	-	-	-	0	0	0	0	
		KEY interrupt selection								
0x03F4C	WD2MDSEL	WD2MD7	WD2MD6	WD2MD5	WD2MD4	WD2MD3	WD2MD2	WD2MD1	WD2MD0	XVI-6
		1	0	1	0	1	0	1	0	
		WDT2 operation mode setup								
0x03F4D	WD2CLR	WD2CL7	WD2CL6	WD2CL5	WD2CL4	WD2CL3	WD2CL2	WD2CL1	WD2CL0	XVI-8
		0	0	0	0	0	0	0	0	
		Watchdog timer 2 counter clear								
0x03F4E	IRQEXPEN	Reserved	IRQEXPEN6	IRQEXPEN5	IRQEXPEN4	IRQEXPEN3	IRQEXPEN2	IRQEXPEN1	IRQEXPEN0	VII-38
		0	0	0	0	0	0	0	0	
		Always set to "0"	ATC1 interrupt enable	A/D conversion interrupt enable	24H timer alarm interrupt enable	PWM underflow interrupt enable	PWM overflow interrupt enable	TM8 compare 2 match interrupt enable	TM7 compare 2 match interrupt enable	
0x03F4F	IRQEXPDT	Reserved	IRQEXPDT6	IRQEXPDT5	IRQEXPDT4	IRQEXPDT3	IRQEXPDT2	IRQEXPDT1	IRQEXPDT0	VII-39
		0	0	0	0	0	0	0	0	
		Always set to "0"	ATC1 interrupt request	A/D conversion interrupt request	24H timer alarm interrupt request	PWM underflow interrupt request	PWM overflow interrupt request	TM8 compare 2 match interrupt request	TM7 compare 2 match interrupt request	
0x03F50	SC4MD0	SC4CE1	-	SC4DEM	SC4DIR	SC4STE	SC4LNG2	SC4LNG1	SC4LNG0	XVIII-35
		0	-	0	0	0	1	1	1	
		Communication data I/O edge	-	Operation selection after detecting instability of IIC slave communication	Transfer bit specify	Start condition selection	Number of transfer bit			
0x03F51	SC4MD1	SC4IOM	SC4SBTS	SC4SBIS	SC4SBOS	-	SC4MST	-	-	XVIII-36
		0	0	0	0	-	0	-	-	
		Serial data input pin selection	SBT function selection	Serial input control	SB0 (SDA) function selection	-	Clock master/slave selection	-	-	
0x03F52	SC4MD2	SC4FDC1	SC4FDC0	-	-	SC4PSCE	SC4PSC2	SC4PSC1	SC4PSC0	XVIII-37
		0	0	-	-	0	0	0	0	
		Output selection after SBO final data transmit		-	-	Prescaler count control	Clock selection			

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Address	Register	Bit symbol								Page
		bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0	
0x03F53	SC4MD3	Reserved	SC4ADM	SC4STPC	SC4TMD	SC4REX	SC4CMD	SC4ACKS	SC4ACK0	XVIII-38
		0	0	0	0	0	0	0	0	
		Always set to "0"	Address format selection	Stop condition generation flag in IIC	Communication mode selection in IIC	Transmission/reception selection in IIC master communication	Clock synchronous/IIC selection	ACK bit enable	ACK bit selection in transmission/reception mode	
0x03F54	SC4AD0	SC4ADR7	SC4ADR6	SC4ADR5	SC4ADR4	SC4ADR3	SC4ADR2	SC4ADR1	SC4ADR0	XVIII-39
		0	0	0	0	0	0	0	0	
		Serial interface 4 address setting register 0								
0x03F55	SC4AD1	-	-	-	-	-	-	SC4ADR9	SC4ADR8	XVIII-39
		-	-	-	-	-	-	0	0	
		Serial interface 4 address setting register 1								
0x03F56	SC4STR0	SC4BSY	-	SC4TEMP	SC4REMP	-	-	-	SC4ORE	XVIII-40
		0	-	0	0	-	-	-	0	
		Serial bus status in clock synchronous	-	Transmission buffer empty flag	Reception buffer empty flag	-	-	-	Overrun error detection	
0x03F57	SC4STR1	SC4WRS	SC4ABT_LST	SC4ADD_ACC	SC4STRT	SC4BUSBSY	SC4IICBSY	SC4GCALL	SC4DATA_ER R	XVIII-41
		0	0	0	0	0	0	0	0	
		Data transfer determination flag in slave mode	Arbitration lost detection flag	Slave address comparison flag	Start condition detection flag	Bus busy flag	Serial bus status in IIC	General-call detection flag	Communication abnormal detection flag	
0x03F58	RXBUF4	RXBUF47	RXBUF46	RXBUF45	RXBUF44	RXBUF43	RXBUF42	RXBUF41	RXBUF40	XVIII-34
		x	x	x	x	x	x	x	x	
		Serial interface 4 reception data buffer								
0x03F59	TXBUF4	TXBUF47	TXBUF46	TXBUF45	TXBUF44	TXBUF43	TXBUF42	TXBUF41	TXBUF40	XVIII-34
		x	x	x	x	x	x	x	x	
		Serial interface 4 transmission data buffer								
0x03F5A	SC4SEL	Reserved	SC4BRP2	SC4BRP1	SC4BRP0	Reserved	SC4SEL2	SC4SEL1	SC4SEL0	XVIII-24
		0	0	0	0	0	0	0	0	
		Always set to "0"	Timer clock output selection			Serial output port selection	Timer selection			
0x03F5E	STB_EDG	-	-	-	STB_EDG_IR Q4	STB_EDG_IR Q3	STB_EDG_IR Q2	STB_EDG_IR Q1	STB_EDGIRQ 0	VII-60
		-	-	-	0	0	0	0	0	
		Edge Interrupt control in STANDBY mode								
0x03F5F	STB_MSK	-	-	-	STB_MSK_IR Q4	STB_MSK_IR Q3	STB_MSK_IR Q2	STB_MSK_IR Q1	STB_MSK_IR Q0	VII-61
		-	-	-	0	0	0	0	0	
		Interrupt mask control at transition to STANDBY mode								
0x03F60	TM0BC	TM0BC7	TM0BC6	TM0BC5	TM0BC4	TM0BC3	TM0BC2	TM0BC1	TM0BC0	IX-22
		0	0	0	0	0	0	0	0	
		Timer 0 binary counter								
0x03F61	TM1BC	TM1BC7	TM1BC6	TM1BC5	TM1BC4	TM1BC3	TM1BC2	TM1BC1	TM1BC0	IX-22
		0	0	0	0	0	0	0	0	
		Timer 1 binary counter								
0x03F62	TM0OC	TM0OC7	TM0OC6	TM0OC5	TM0OC4	TM0OC3	TM0OC2	TM0OC1	TM0OC0	IX-20
		x	x	x	x	x	x	x	x	
		Timer 0 compare register								
0x03F63	TM1OC	TM1OC7	TM1OC6	TM1OC5	TM1OC4	TM1OC3	TM1OC2	TM1OC1	TM1OC0	IX-20
		x	x	x	x	x	x	x	x	
		Timer 1 compare register								
0x03F64	TM0MD	-	TM0POP	TM0MOD	TM0PWM	TM0EN	TM0CK2	TM0CK1	TM0CK0	IX-25
		-	0	0	0	0	0	0	0	
		-	Output signal start polarity selection	Pulse width measurement control	Operation mode selection	Count control	Clock source selection			

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Address	Register	Bit symbol								Page
		bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0	
0x03F65	TM1MD	Reserved	Reserved	TM1CAS1	TM1CAS0	TM1EN	TM1CK2	TM1CK1	TM1CK0	IX-27
		0	0	0	0	0	0	0	0	
		Always set to "0"	Always set to "0"	Operation mode		Count control	Clock source selection			
0x03F66	CK0MD	-	TM0ADD1	TM0ADD0	TM0ADDEN	TM0PSC2	TM0PSC1	TM0PSC0	TM0BAS	IX-17
		-	0	0	0	0	0	0	0	
		-	Additional pulse location		Additional pulse method PWM control	Clock source selection				
0x03F67	CK1MD	-	-	-	-	TM1PSC2	TM1PSC1	TM1PSC0	TM1BAS	IX-18
		-	-	-	-	0	0	0	0	
		-	-	-	-	Clock source selection				
0x03F68	TM2BC	TM2BC7	TM2BC6	TM2BC5	TM2BC4	TM2BC3	TM2BC2	TM2BC1	TM2BC0	IX-22
		0	0	0	0	0	0	0	0	
		Time 2 binary counter								
0x03F69	TM3BC	TM3BC7	TM3BC6	TM3BC5	TM3BC4	TM3BC3	TM3BC2	TM3BC1	TM3BC0	IX-22
		0	0	0	0	0	0	0	0	
		Time 3 binary counter								
0x03F6A	TM2OC	TM2OC7	TM2OC6	TM2OC5	TM2OC4	TM2OC3	TM2OC2	TM2OC1	TM2OC0	IX-20
		X	X	X	X	X	X	X	X	
		Time 2 compare register								
0x03F6B	TM3OC	TM3OC7	TM3OC6	TM3OC5	TM3OC4	TM3OC3	TM3OC2	TM3OC1	TM3OC0	IX-21
		X	X	X	X	X	X	X	X	
		Time 3 compare register								
0x03F6C	TM2MD	TM2BUFSEL	TM2POP	TM2MOD	TM2PWM	TM2EN	TM2CK2	TM2CK1	TM2CK0	IX-28
		0	0	0	0	0	0	0	0	
		Compare register start polarity selection	Output signal start polarity selection	Pulse width measurement control	Operation mode selection	Count control	Clock source selection			
0x03F6D	TM3MD	TM3BUFSEL	-	-	TM3CAS	TM3EN	TM3CK2	TM3CK1	TM3CK0	IX-30
		0	-	-	0	0	0	0	0	
		Compare register buffer selection	-	-	Operation mode selection	Count control	Clock source selection			
0x03F6E	CK2MD	-	TM2ADD1	TM2ADD0	TM2ADDEN	TM2PSC2	TM2PSC1	TM2PSC0	TM2BAS	IX-18
		-	0	0	0	0	0	0	0	
		-	Additional pulse location		Additional pulse method PWM control	Clock source selection				
0x03F6F	CK3MD	-	-	-	-	TM3PSC2	TM3PSC1	TM3PSC0	TM3BAS	IX-19
		-	-	-	-	0	0	0	0	
		-	-	-	-	Clock source selection				
0x03F70	TM4BC	TM4BC7	TM4BC6	TM4BC5	TM4BC4	TM4BC3	TM4BC2	TM4BC1	TM4BC0	IX-22
		0	0	0	0	0	0	0	0	
		Time 4 binary counter								
0x03F71	TMABC	TMABC7	TMABC6	TMABC5	TMABC4	TMABC3	TMABC2	TMABC1	TMABC0	X-5
		0	0	0	0	0	0	0	0	
		Time A binary counter								
0x03F72	TM4OC	TM4OC7	TM4OC6	TM4OC5	TM4OC4	TM4OC3	TM4OC2	TM4OC1	TM4OC0	IX-21
		X	X	X	X	X	X	X	X	
		Time 4 compare register								
0x03F73	TMAOC	TMAOC7	TMAOC6	TMAOC5	TMAOC4	TMAOC3	TMAOC2	TMAOC1	TMAOC0	X-5
		X	X	X	X	X	X	X	X	
		Time A compare register								
0x03F74	TM4MD	-	TM4POP	TM4MOD	TM4PWM	TM4EN	TM4CK2	TM4CK1	TM4CK0	IX-31
		-	0	0	0	0	0	0	0	
		-	Output signal start polarity selection	Pulse width measurement control	Operation mode selection	Count control	Clock source selection			

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Address	Register	Bit symbol								Page
		bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0	
0x03F75	TMAMD1	-	-	-	-	TMAEN	TMACK2	TMACK1	TMACK0	X-6
		-	-	-	-	0	0	0	0	
		-	-	-	-	Count control	Clock source selection			
0x03F76	CK4MD	-	TM4ADD1	TM4ADD0	TM4ADDEN	TM4PSC2	TM4PSC1	TM4PSC0	TM4BAS	IX-19
		-	0	0	0	0	0	0	0	
		-	Additional pulse location		Additional pulse method PWM control	Clock source selection				
0x03F77	TMAMD2	Reserved	PSCEN	-	-	-	-	-	-	X-6
		0	0	-	-	-	-	-	-	
		Always set to "0"	Prescaler operation control	-	-	-	-	-	-	
0x03F78	TM6BC	TM6BC7	TM6BC6	TM6BC5	TM6BC4	TM6BC3	TM6BC2	TM6BC1	TM6BC0	XIV-6
		0	0	0	0	0	0	0	0	
		Timer 6 binary counter								
0x03F79	TM6OC	TM6OC7	TM6OC6	TM6OC5	TM6OC4	TM6OC3	TM6OC2	TM6OC1	TM6OC0	XIV-6
		X	X	X	X	X	X	X	X	
		Timer 6 compare register								
0x03F7A	TM6MD	TM6CLRS	TM6IR2	TM6IR1	TM6IR0	TM6CK3	TM6CK2	TM6CK1	TM6CK0	XIV-8
		0	0	0	0	0	0	0	0	
		Binary counter clear selection	Time base timer interrupt cycle selection			Clock source selection			Time base timer clock source selection	
0x03F7B	TBCLR	TBCLR7	TBCLR6	TBCLR5	TBCLR4	TBCLR3	TBCLR2	TBCLR1	TBCLR0	XIV-6
		-	-	-	-	-	-	-	-	
		Time base timer clear control register								
0x03F7C	TM6BEN	-	-	-	-	-	Reserved	TBEN	TM6EN	XIV-7
		-	-	-	-	-	0	0	0	
		-	-	-	-	-	Always set to "0"	Time base timer operation control	Timer6 operation control	
0x03F7E	TM7MD4	Reserved	T7IGBTSFT	T7IGBTCNT	T7ONESHOT	T7NODED	-	T7ICT2	T7CAPCLR	XI-24
		0	0	0	0	0	-	0	0	
		Always set to "0"	IGBT software activation	BC operation at IGBT disable is invalid	One shot pulse selection	Dead time selection	-	Capture trigger selection	BC clear at capture clear	
0x03F7F	TM8MD4	Reserved	Reserved	Reserved	Reserved	Reserved	TM8SEL	T8ICT2	T8CAPCLR	XI-29
		0	0	0	0	0	0	0	0	
		Always set to "0"	Always set to "0"	Always set to "0"	Always set to "0"	Always set to "0"	Timer8/IGBT selection	Capture trigger selection	BC clear at capture clear	
0x03F80	TM7BCL	TM7BCL7	TM7BCL6	TM7BCL5	TM7BCL4	TM7BCL3	TM7BCL2	TM7BCL1	TM7BCL0	XI-14
		0	0	0	0	0	0	0	0	
		Timer 7 binary counter lower 8-bit								
0x03F81	TM7BCH	TM7BCH7	TM7BCH6	TM7BCH5	TM7BCH4	TM7BCH3	TM7BCH2	TM7BCH1	TM7BCH0	XI-14
		0	0	0	0	0	0	0	0	
		Timer 7 binary counter upper 8-bit								
0x03F82	TM7OC1L	TM7OC1L7	TM7OC1L6	TM7OC1L5	TM7OC1L4	TM7OC1L3	TM7OC1L2	TM7OC1L1	TM7OC1L0	XI-12
		X	X	X	X	X	X	X	X	
		Timer 7 compare register lower 8-bit								
0x03F83	TM7OC1H	TM7OC1H7	TM7OC1H6	TM7OC1H5	TM7OC1H4	TM7OC1H3	TM7OC1H2	TM7OC1H1	TM7OC1H0	XI-12
		X	X	X	X	X	X	X	X	
		Timer 7 compare register upper 8-bit								
0x03F84	TM7PR1L	TM7PR1L7	TM7PR1L6	TM7PR1L5	TM7PR1L4	TM7PR1L3	TM7PR1L2	TM7PR1L1	TM7PR1L0	XI-13
		X	X	X	X	X	X	X	X	
		Timer 7 preset register 1 lower 8-bit								
0x03F85	TM7PR1H	TM7PR1H7	TM7PR1H6	TM7PR1H5	TM7PR1H4	TM7PR1H3	TM7PR1H2	TM7PR1H1	TM7PR1H0	XI-13
		X	X	X	X	X	X	X	X	
		Timer 7 preset register 1 upper 8-bit								

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Address	Register	Bit symbol								Page
		bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0	
0x03F86	TM7ICL	TM7ICL7	TM7ICL6	TM7ICL5	TM7ICL4	TM7ICL3	TM7ICL2	TM7ICL1	TM7ICL0	XI-15
		X	X	X	X	X	X	X	X	
		Timer 7 input capture lower 8-bit								
0x03F87	TM7ICH	TM7ICH7	TM7ICH6	TM7ICH5	TM7ICH4	TM7ICH3	TM7ICH2	TM7ICH1	TM7ICH0	XI-15
		X	X	X	X	X	X	X	X	
		Timer 7 input capture upper 8-bit								
0x03F88	TM7MD1	Reserved	T7ICEDG1	TM7CL	TM7EN	TM7PS1	TM7PS0	TM7CK1	TM7CK0	XI-21
		0	0	1	0	0	0	0	0	
		Always set to "0"	Capture trigger edge selection	Timer output reset	Timer count control	Count clock selection		Clock source selection		
0x03F89	TM7MD2	TM7ICEDG0	T7PWMSL	TM7BCR	TM7PWM	TM7IRS1	T7ICEN	T7ICT1	T7ICT0	XI-22
		0	0	0	0	0	0	0	0	
		Capture trigger edge selection	PWM mode selection	Count clear factor selection	Timer output wave selection	Interrupt factor selection	Input capture enable	Capture trigger selection		
0x03F8A	TM7OC2L	TM7OC2L7	TM7OC2L6	TM7OC2L5	TM7OC2L4	TM7OC2L3	TM7OC2L2	TM7OC2L1	TM7OC2L0	XI-12
		X	X	X	X	X	X	X	X	
		Timer 7 compare register 2 lower 8-bit								
0x03F8B	TM7OC2H	TM7OC2H7	TM7OC2H6	TM7OC2H5	TM7OC2H4	TM7OC2H3	TM7OC2H2	TM7OC2H1	TM7OC2H0	XI-12
		X	X	X	X	X	X	X	X	
		Timer 7 compare register 2 upper 8-bit								
0x03F8C	TM7PR2L	TM7PR2L7	TM7PR2L6	TM7PR2L5	TM7PR2L4	TM7PR2L3	TM7PR2L2	TM7PR2L1	TM7PR2L0	XI-13
		X	X	X	X	X	X	X	X	
		Timer 7 preset register 2 lower 8-bit								
0x03F8D	TM7PR2H	TM7PR2H7	TM7PR2H6	TM7PR2H5	TM7PR2H4	TM7PR2H3	TM7PR2H2	TM7PR2H1	TM7PR2H0	XI-13
		X	X	X	X	X	X	X	X	
		Timer 7 preset register 2 upper 8-bit								
0x03F8E	TM7DPR1	TM7DPR17	TM7DPR16	TM7DPR15	TM7DPR14	TM7DPR13	TM7DPR12	TM7DPR11	TM7DPR10	XI-17
		0	0	0	0	0	0	0	0	
		Timer 7 dead-time preset register 1								
0x03F8F	TM7DPR2	TM7DPR27	TM7DPR26	TM7DPR25	TM7DPR24	TM7DPR23	TM7DPR22	TM7DPR21	TM7DPR20	XI-17
		0	0	0	0	0	0	0	0	
		Timer 7 dead-time preset register 2								
0x03F90	TM8BCL	TM8BCL7	TM8BCL6	TM8BCL5	TM8BCL4	TM8BCL3	TM8BCL2	TM8BCL1	TM8BCL0	XI-20
		0	0	0	0	0	0	0	0	
		Timer 8 binary counter lower 8-bit								
0x03F91	TM8BCH	TM8BCH7	TM8BCH6	TM8BCH5	TM8BCH4	TM8BCH3	TM8BCH2	TM8BCH1	TM8BCH0	XI-20
		0	0	0	0	0	0	0	0	
		Timer 8 binary counter upper 8-bit								
0x03F92	TM8OC1L	TM8OC1L7	TM8OC1L6	TM8OC1L5	TM8OC1L4	TM8OC1L3	TM8OC1L2	TM8OC1L1	TM8OC1L0	XI-18
		X	X	X	X	X	X	X	X	
		Timer 8 compare register 1 lower 8-bit								
0x03F93	TM8OC1H	TM8OC1H7	TM8OC1H6	TM8OC1H5	TM8OC1H4	TM8OC1H3	TM8OC1H2	TM8OC1H1	TM8OC1H0	XI-18
		X	X	X	X	X	X	X	X	
		Timer 8 compare register 1 upper 8-bit								
0x03F94	TM8PR1L	TM8PR1L7	TM8PR1L6	TM8PR1L5	TM8PR1L4	TM8PR1L3	TM8PR1L2	TM8PR1L1	TM8PR1L0	XI-19
		X	X	X	X	X	X	X	X	
		Timer 8 preset register 1 lower 8-bit								
0x03F95	TM8PR1H	TM8PR1H7	TM8PR1H6	TM8PR1H5	TM8PR1H4	TM8PR1H3	TM8PR1H2	TM8PR1H1	TM8PR1H0	XI-19
		X	X	X	X	X	X	X	X	
		Timer 8 preset register 1 upper 8-bit								
0x03F96	TM8ICL	TM8ICL7	TM8ICL6	TM8ICL5	TM8ICL4	TM8ICL3	TM8ICL2	TM8ICL1	TM8ICL0	XI-20
		X	X	X	X	X	X	X	X	
		Timer 8 input capture register lower 8-bit								
0x03F97	TM8ICH	TM8ICH7	TM8ICH6	TM8ICH5	TM8ICH4	TM8ICH3	TM8ICH2	TM8ICH1	TM8ICH0	XI-20
		X	X	X	X	X	X	X	X	
		Timer 8 input capture register upper 8-bit								

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Address	Register	Bit symbol								Page
		bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0	
0x03F98	TM8MD1	Reserved	T8ICEDG1	TM8CL	TM8EN	TM8PS1	TM8PS0	TM8CK1	TM8CK0	XI-26
		0	0	1	0	0	0	0	0	
		Always set to "0"	Capture trigger edge selection	Timer output reset	Timer count control	Count clock selection		Clock source selection		
0x03F99	TM8MD2	T8ICERDG0	TM8PWMSL	TM8BCR	TM8PWM	TM8IRS1	T8ICEN	T8ICT1	T8ICT0	XI-27
		0	0	0	0	0	0	0	0	
		Capture trigger edge selection	PWM mode selection	Count clear factor selection	Timer output wave selection	Interrupt factor selection	Input capture enable	Capture trigger selection		
0x03F9A	TM8OC2L	TM8OC2L7	TM8OC2L6	TM8OC2L5	TM8OC2L4	TM8OC2L3	TM8OC2L2	TM8OC2L1	TM8OC2L0	XI-18
		X	X	X	X	X	X	X	X	
		Timer 8 compare register 2 lower 8-bit								
0x03F9B	TM8OC2H	TM8OC2H7	TM8OC2H6	TM8OC2H5	TM8OC2H4	TM8OC2H3	TM8OC2H2	TM8OC2H1	TM8OC2H0	XI-18
		X	X	X	X	X	X	X	X	
		Timer 8 compare register 2 upper 8-bit								
0x03F9C	TM8PR2L	TM8PR2L7	TM8PR2L6	TM8PR2L5	TM8PR2L4	TM8PR2L3	TM8PR2L2	TM8PR2L1	TM8PR2L0	XI-19
		X	X	X	X	X	X	X	X	
		Timer 8 preset register 2 lower 8-bit								
0x03F9D	TM8PR2H	TM8PR2H7	TM8PR2H6	TM8PR2H5	TM8PR2H4	TM8PR2H3	TM8PR2H2	TM8PR2H1	TM8PR2H0	XI-19
		X	X	X	X	X	X	X	X	
		Timer 8 preset register 2 upper 8-bit								
0x03F9E	TM7MD3	TM7CKSMP	TM7BUFSEL	TM7CKEDG	TM7IGBTTR	T7IGBTDT	T7IGBTEN	T7IGBT1	T7IGBT0	XI-23
		0	0	0	0	0	0	0	0	
		Capture sampling selection	Buffer selection	TM7IO count edge selection	IGBT trigger level selection	IGBT dead-time insertion timing	IGBT operation enable	IGBT/timer activation factor selection		
0x03F9F	TM8MD3	TM8CKSMP	TM8BUFSEL	-	TM8CKEDG	-	TM8PWMF	TM8PWMO	TM8CAS	XI-28
		0	0	-	0	-	0	0	0	
		Capture sampling selection	Buffer selection	-	TM8IO count edge selection	-	PWM output value selection	PWM output polarity selection	Cascade selection	
0x03FA0	TM7ICL1	TM7ICL17	TM7ICL16	TM7ICL15	TM7ICL14	TM7ICL13	TM7ICL12	TM7ICL11	TM7ICL10	XI-15
		X	X	X	X	X	X	X	X	
		Timer 7 input capture 1 lower 8-bit								
0x03FA1	TM7ICH1	TM7ICH17	TM7ICH16	TM7ICH15	TM7ICH14	TM7ICH13	TM7ICH12	TM7ICH11	TM7ICH10	XI-15
		X	X	X	X	X	X	X	X	
		Timer 7 input capture 1 upper 8-bit								
0x03FA2	TM7ICLD	TM7ICLD7	TM7ICLD6	TM7ICLD5	TM7ICLD4	TM7ICLD3	TM7ICLD2	TM7ICLD1	TM7ICLD0	XI-16
		X	X	X	X	X	X	X	X	
		Timer 7 input capture D lower 8-bit								
0x03FA3	TM7ICHD	TM7ICHD7	TM7ICHD6	TM7ICHD5	TM7ICHD4	TM7ICHD3	TM7ICHD2	TM7ICHD1	TM7ICHD0	XI-16
		X	X	X	X	X	X	X	X	
		Timer 7 input capture D upper 8-bit								
0x03FA4	TM7ICLD1	TM7ICLD17	TM7ICLD16	TM7ICLD15	TM7ICLD14	TM7ICLD13	TM7ICLD12	TM7ICLD11	TM7ICLD10	XI-16
		X	X	X	X	X	X	X	X	
		Timer 7 input capture D1 lower 8-bit								
0x03FA5	TM7ICHD1	TM7ICHD17	TM7ICHD16	TM7ICHD15	TM7ICHD14	TM7ICHD13	TM7ICHD12	TM7ICHD11	TM7ICHD10	XI-16
		X	X	X	X	X	X	X	X	
		Timer 7 input capture D1 upper 8-bit								
0x03FA6	TM7MD5	-	-	-	-	-	-	T7DBUF	T7UD	XI-25
		-	-	-	-	-	-	0	0	
		-	-	-	-	-	-	IC register buffer selection	IC register 1 selection	
0x03FA8	TMRMD1	-	TMRIRSEL2	TMRIRSEL1	TMRIRSEL0	-	-	TMRALMEN	TMREN	XIII-5
		-	0	0	0	-	-	0	0	
		-	Interrupt generation cycle selection			-	-	Alarm function operation enable	24H timer operation enable	
0x03FA9	TMRMD2	Reserved	Reserved	Reserved	Reserved	TMRCK3	TMRCK2	TMRCK1	TMRCK0	XIII-6
		0	0	0	0	0	0	0	0	
		Always set to "0"				24H timer frequency selection		clock specification		

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Address	Register	Bit symbol								Page
		bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0	
0x03FAA	TMRBCL	-	TMRBC6	TMRBC5	TMRBC4	TMRBC3	TMRBC2	TMRBC1	TMRBC0	XIII-7
		-	X	X	X	X	X	X	X	
		24 H timer binary counter L								
0x03FAB	TMRBCM	-	-	TMRBC12	TMRBC11	TMRBC10	TMRBC9	TMRBC8	TMRBC7	XIII-7
		-	-	X	X	X	X	X	X	
		24 H timer binary counter M								
0x03FAC	TMRBCH	-	-	-	TMRBC17	TMRBC16	TMRBC15	TMRBC14	TMRBC13	XIII-7
		-	-	-	X	X	X	X	X	
		24 H timer binary counter H								
0x03FAD	TMRALL	-	TMRAL6	TMRAL5	TMRAL4	TMRAL3	TMRAL2	TMRAL1	TMRAL0	XIII-9
		-	X	X	X	X	X	X	X	
		24 H timer alarm setting register L								
0x03FAE	TMRALM	-	-	TMRAL12	TMRAL11	TMRAL10	TMRAL9	TMRAL8	TMRAL7	XIII-9
		-	-	X	X	X	X	X	X	
		24 H timer alarm setting register M								
0x03FAF	TMRALH	-	-	-	TMRAL17	TMRAL16	TMRAL15	TMRAL14	TMRAL13	XIII-9
		-	-	-	X	X	X	X	X	
		24 H timer alarm setting register H								
0x03FB0	TMCKSEL1	TM6CKSEL	-	-	TM4IOSEL	TM3IOSEL	TM2IOSEL	TM1IOSEL	TM0IOSEL	IX-32 XIV-9
		0	-	-	0	0	0	0	0	
		Selection of Time Base Timer clock dividing ratio	-	-	Timer 4 input selection	Timer 3 input selection	Timer 2 input selection	Timer 1 input selection	Timer 0 input selection	
0x03FB1	TMCKSEL2	-	TM8IOSEL2	TM7IOSEL3	TM7IOSEL2	-	TM9IOSEL	TM8IOSEL	TM7IOSEL	XI-30
		-	0	0	0	-	0	0	0	
		-	Timer 8 input selection 2	Timer 7 input selection 3	Timer 7 input selection 2	-	Timer 9 input selection	Timer 8 input selection	Timer 7 input selection	
0x03FB2	TMINSEL1	TMINSEL17	TMINSEL16	TMINSEL15	TMINSEL14	TMINSEL13	TMINSEL12	TMINSEL11	TMINSEL10	IX-34
		0	0	0	0	0	0	0	0	
		Timer 3 input selection		Timer 2 input selection		Timer 1 input selection		Timer 0 input selection		
0x03EB3	TMINSEL2	TMINSEL27	TMINSEL26	TMINSEL25	TMINSEL24	TMINSEL23	TMINSEL22	TMINSEL21	TMINSEL20	IX-36
		0	0	0	0	0	0	0	0	
		Timer 9 input selection		Timer 8 input selection		Timer 7 input selection		Timer 4 input selection		
0x03FBD	FBEWER	BEW7	BEW6	BEW5	BEW4	BEW3	BEW2	BEW1	BEW0	XXIV-25
		0	0	0	0	0	0	0	0	
		On-board rewriting control								
0x03FBF	FEWSPD	Reserved	-	-	-	NSTOP	-	-	Reserved	III-26
		0	-	-	-	0	-	-	1	
		Always set to "0"	-	-	-	Internal flash control	-	-	-	
0x03FC5	ANCTR0	ANSH1	ANSH0	ANCK1	ANCK0	ANLADE	ANCK2	-	-	XX-7
		0	0	0	0	0	0	-	-	
		Sample and hold time		A/D conversion clock		A/D resistor ladder control	A/D conversion clock	-	-	
0x03FC6	ANCTR1	-	-	-	Reserved	ANCHS3	ANCHS2	ANCHS1	ANCHS0	XX-9
		-	-	-	0	0	0	0	0	
		-	-	-	Always set to "0"	Analog input channel selection				
0x03FC7	ANCTR2	ANST	ANSTSEL1	ANSTSEL0	-	-	-	-	-	XX-10
		0	0	0	-	-	-	-	-	
		A/D conversion status	A/D conversion starting factor selection		-	-	-	-	-	
0x03FC8	ANBUF0	ANBUF07	ANBUF06	-	-	-	-	-	-	XX-11
		X	X	-	-	-	-	-	-	
		Conversion data storage buffer 0		-	-	-	-	-	-	
0x03FC9	ANBUF1	ANBUF17	ANBUF16	ANBUF15	ANBUF14	ANBUF13	ANBUF12	ANBUF11	ANBUF10	XX-11
		X	X	X	X	X	X	X	X	
		Conversion data storage buffer 1								

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Address	Register	Bit symbol								Page
		bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0	
0x03FCB	DACTR	-	DACHS0	DACKS1	DACKS0	DACMD2	DACMD1	DACMD0	DABUSY	XXI-6
		-	X	X	X	X	X	X	0	
		-	Monitor flag of DA conversion	DA scan clock		DA conversion mode selection			D/A conversion enable	
0x03FCC	DADR0	DA0BUF7	DA0BUF6	DA0BUF5	DA0BUF4	DA0BUF3	DA0BUF2	DA0BUF1	DA0BUF0	XXI-7
		X	X	X	X	X	X	X	X	
		D/A Converter Input Data Register 0								
0x03FCD	DADR1	DA1BUF7	DA1BUF6	DA1BUF5	DA1BUF4	DA1BUF3	DA1BUF2	DA1BUF1	DA1BUF0	XXI-7
		X	X	X	X	X	X	X	X	
		D/A Converter Input Data Register 1								
0x03FD0	IRQCNT	-	-	-	P24EN	P23EN	P22EN	P21EN	P20EN	VII-51
		-	-	-	0	0	0	0	0	
		-	-	-	Pin setting for External Interrupt					
0x03FD1	NF0CTR	NF0SCK2	NF0SCK1	NF0SCK0	NF0EN1	-	-	-	-	VII-54
		0	0	0	0	-	-	-	-	
		IRQ0 noise sampling frequency		Noise filter ON/OFF control	-	-	-	-		
0x03FD2	NF1CTR	NF1SCK2	NF1SCK1	NF1SCK0	NF1EN1	-	-	-	-	VII-54
		0	0	0	0	-	-	-	-	
		IRQ1 noise sampling frequency		Noise filter ON/OFF control	-	-	-	-		
0x03FD3	NF2CTR	NF2SCK2	NF2SCK1	NF2SCK0	NF2EN1	-	-	-	-	VII-54
		0	0	0	0	-	-	-	-	
		IRQ2 noise sampling frequency		Noise filter ON/OFF control	-	-	-	-		
0x03FD4	NF3CTR	NF3SCK2	NF3SCK1	NF3SCK0	NF3EN1	-	-	-	-	VII-54
		0	0	0	0	-	-	-	-	
		IRQ3 noise sampling frequency		Noise filter ON/OFF control	-	-	-	-		
0x03FD5	NF4CTR	NF4SCK2	NF4SCK1	NF4SCK0	NF4EN1	-	-	-	-	VII-54
		0	0	0	0	-	-	-	-	
		IRQ4 noise sampling frequency		Noise filter ON/OFF control	-	-	-	-		
0x03FD6	NF5CTR	NF5SCK2	NF5SCK1	NF5SCK0	NF5EN1	-	-	-	-	XIX-7
		0	0	0	0	-	-	-	-	
		TXD0 noise sampling frequency		Noise filter ON/OFF control	-	-	-	-		
0x03FD7	LVLMD	-	-	EXLVL4	LVLEN4	EXLVL3	LVLEN3	EXLVL2	LVLEN2	VII-57
		-	-	0	0	0	0	0	0	
		-	-	IRQ4 valid input level setting	IRQ4 valid input setting	IRQ3 valid input level setting	IRQ3 valid input setting	IRQ2 valid input level setting	IRQ2 valid input setting	
0x03FE1	NMICR	-	-	-	-	-	IRQNPG	IRQNWDG	IRQNPRI	VII-23
		-	-	-	-	-	0	0	0	
		-	-	-	-	-	Program interrupt request flag	Watchdog interrupt request flag	Peripheral functions NMI request flag	
0x03FE2	IRQ0ICR	IRQ0LV1	IRQ0LV0	REDG0	-	Reserved	-	IRQ0IE	IRQ0IR	VII-24
		0	0	0	-	0	-	0	0	
		Interrupt level specification flag		Interrupt valid edge flag	-	Always set to "0"	-	Interrupt enable flag	Interrupt request flag	
0x03FE3	IRQ1ICR	IRQ1LV1	IRQ1LV0	REDG1	-	Reserved	-	IRQ1IE	IRQ1IR	VII-24
		0	0	0	-	0	-	0	0	
		Interrupt level specification flag		Interrupt valid edge flag	-	Always set to "0"	-	Interrupt enable flag	Interrupt request flag	
0x03FE4	IRQ2ICR	IRQ2LV1	IRQ2LV0	REDG2	-	Reserved	-	IRQ2IE	IRQ2IR	VII-24
		0	0	0	-	0	-	0	0	
		Interrupt level specification flag		Interrupt valid edge flag	-	Always set to "0"	-	Interrupt enable flag	Interrupt request flag	
0x03FE5	IRQ3ICR	IRQ3LV1	IRQ3LV0	REDG3	-	Reserved	-	IRQ3IE	IRQ3IR	VII-24
		0	0	0	-	0	-	0	0	
		Interrupt level specification flag		Interrupt valid edge flag	-	Always set to "0"	-	Interrupt enable flag	Interrupt request flag	

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Address	Register	Bit symbol								Page
		bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0	
0x03FE6	IRQ4ICR	IRQ4LV1	IRQ4LV0	REDG4	-	Reserved	-	IRQ4IE	IRQ4IR	VII-24
		0	0	0	-	0	-	0	0	
		Interrupt level specification flag		Interrupt valid edge flag	-	Always set to "0"	-	Interrupt enable flag	Interrupt request flag	
0x03FE7	LINICR	LINLV1	LINLV0	LINREDG	-	Reserved	-	LINIE	LINIR	VII-25
		0	0	0	-	0	-	0	0	
		Interrupt level specification flag		Interrupt valid edge flag	-	Always set to "0"	-	Interrupt enable flag	Interrupt request flag	
0x03FE8	LVIICR	LVILV1	LVILV0	-	-	Reserved	-	LVIE	LVIR	VII-26
		0	0	-	-	0	-	0	0	
		Interrupt level specification flag		-	-	Always set to "0"	-	Interrupt enable flag	Interrupt request flag	
0x03FE9	TM0ICR	TM0LV1	TM0LV0	-	-	Reserved	-	TM0IE	TM0IR	VII-27
		0	0	-	-	0	-	0	0	
		Interrupt level specification flag		-	-	Always set to "0"	-	Interrupt enable flag	Interrupt request flag	
0x03FEA	TM1ICR	TM1LV1	TM1LV0	-	-	Reserved	-	TM1IE	TM1IR	VII-27
		0	0	-	-	0	-	0	0	
		Interrupt level specification flag		-	-	Always set to "0"	-	Interrupt enable flag	Interrupt request flag	
0x03FEB	TM2ICR	TM2LV1	TM2LV0	-	-	Reserved	-	TM2IE	TM2IR	VII-27
		0	0	-	-	0	-	0	0	
		Interrupt level specification flag		-	-	Always set to "0"	-	Interrupt enable flag	Interrupt request flag	
0x03FEC	TM3ICR	TM3LV1	TM3LV0	-	-	Reserved	-	TM3IE	TM3IR	VII-27
		0	0	-	-	0	-	0	0	
		Interrupt level specification flag		-	-	Always set to "0"	-	Interrupt enable flag	Interrupt request flag	
0x03FED	TM4ICR	TM4LV1	TM4LV0	-	-	Reserved	-	TM4IE	TM4IR	VII-27
		0	0	-	-	0	-	0	0	
		Interrupt level specification flag		-	-	Always set to "0"	-	Interrupt enable flag	Interrupt request flag	
0x03FEE	TMRICR	TMRLV1	TMRLV0	-	-	Reserved	-	TMRIE	TMRIR	VII-28
		0	0	-	-	0	-	0	0	
		Interrupt level specification flag		-	-	Always set to "0"	-	Interrupt enable flag	Interrupt request flag	
0x03FEF	TM6ICR	TM6LV1	TM6LV0	-	-	Reserved	-	TM6IE	TM6IR	VII-27
		0	0	-	-	0	-	0	0	
		Interrupt level specification flag		-	-	Always set to "0"	-	Interrupt enable flag	Interrupt request flag	
0x03FF0	TBICR	TBLV1	TBLV0	-	-	Reserved	-	TBIE	TBIR	VII-29
		0	0	-	-	0	-	0	0	
		Interrupt level specification flag		-	-	Always set to "0"	-	Interrupt enable flag	Interrupt request flag	
0x03FF1	TM7ICR	TM7LV1	TM7LV0	-	-	Reserved	-	TM7IE	TM7IR	VII-30
		0	0	-	-	0	-	0	0	
		Interrupt level specification flag		-	-	Always set to "0"	-	Interrupt enable flag	Interrupt request flag	
0x03FF2	TM8ICR	TM8LV1	TM8LV0	-	-	Reserved	-	TM8IE	TM8IR	VII-31
		0	0	-	-	0	-	0	0	
		Interrupt level specification flag		-	-	Always set to "0"	-	Interrupt enable flag	Interrupt request flag	
0x03FF3	TM9OC2ICR	TM9OC2LV1	TM9OC2LV0	-	-	Reserved	-	TM9OC2IE	TM9OC2IR	VII-32
		0	0	-	-	0	-	0	0	
		Interrupt level specification flag		-	-	Always set to "0"	-	Interrupt enable flag	Interrupt request flag	
0x03FF4	SC0TICR	SC0TLV1	SC0TLV0	-	-	Reserved	-	SC0TIE	SC0TIR	VII-33
		0	0	-	-	0	-	0	0	
		Interrupt level specification flag		-	-	Always set to "0"	-	Interrupt enable flag	Interrupt request flag	

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Address	Register	Bit symbol								Page
		bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0	
0x03FF5	SC0RICR	SC0RLV1	SC0RLV0	-	-	Reserved	-	SC0RIE	SC0RIR	VII-34
		0	0	-	-	0	-	0	0	
		Interrupt level specification flag		-	-	Always set to "0"	-	Interrupt enable flag	Interrupt request flag	
0x03FF6	SC1TICR	SC1TLV1	SC1TLV0	-	-	Reserved	-	SC1TIE	SC1TIR	VII-33
		0	0	-	-	0	-	0	0	
		Interrupt level specification flag		-	-	Always set to "0"	-	Interrupt enable flag	Interrupt request flag	
0x03FF7	SC1RICR	SC1RLV1	SC1RLV0	-	-	Reserved	-	SC1RIE	SC1RIR	VII-34
		0	0	-	-	0	-	0	0	
		Interrupt level specification flag		-	-	Always set to "0"	-	Interrupt enable flag	Interrupt request flag	
0x03FF8	SC2TICR	SC2TLV1	SC2TLV0	-	-	Reserved	-	SC2TIE	SC2TIR	VII-33
		0	0	-	-	0	-	0	0	
		Interrupt level specification flag		-	-	Always set to "0"	-	Interrupt enable flag	Interrupt request flag	
0x03FF9	SC2RICR	SC2RLV1	SC2RLV0	-	-	Reserved	-	SC2RIE	SC2RIR	VII-34
		0	0	-	-	0	-	0	0	
		Interrupt level specification flag		-	-	Always set to "0"	-	Interrupt enable flag	Interrupt request flag	
0x03FFC	SC4ICR	SC4LV1	SC4LV0	-	-	Reserved	-	SC4IE	SC4IR	VII-35
		0	0	-	-	0	-	0	0	
		Interrupt level specification flag		-	-	Always set to "0"	-	Interrupt enable flag	Interrupt request flag	
0x03FFD	SC4STPCICR	SC4STPCLV1	SC4STPCLV0	-	-	Reserved	-	SC4STPCIE	SC4STPCIR	VII-36
		0	0	-	-	0	-	0	0	
		Interrupt level specification flag		-	-	Always set to "0"	-	Interrupt enable flag	Interrupt request flag	
0x03FFE	PERILR	PERILV1	PERILV0	-	-	Reserved	-	-	-	VII-37
		0	0	-	-	0	-	-	-	
		Interrupt level specification flag		-	-	Always set to "0"	-	-	-	

25.2 Instruction Set

MN101E SERIES INSTRUCTION SET

Group	Mnemonic	Operation	Flag				Code Size	Cycle Repeat	Machine Code											Notes			
			VF	NF	CF	ZF			Ext.	1	2	3	4	5	6	7	8	9	10		11		
Data Move Instructions																							
MOV	MOV Dn,Dm	Dn→Dm	--	--	--	--	2	1	1010 DnDm														
	MOV imm8,Dm	imm8→Dm	--	--	--	--	4	2	1010DmDm <#8. ...>														
	MOV Dn,PSW	Dn→PSW	●	●	●	●	3	3	0010 1001 01Dn														
	MOV PSW,Dm	PSW→Dm	--	--	--	--	3	2	0010 0001 01Dm														
	MOV (An),Dm	mem8(An)→Dm	--	--	--	--	2	2	0100 1ADm														
	MOV (d8,An),Dm	mem8(d8+An)→Dm	--	--	--	--	4	2	0110 1ADm <d8. ...>														*1
	MOV (d16,An),Dm	mem8(d16+An)→Dm	--	--	--	--	7	4	0010 0110 1ADm <d16>														
	MOV (d4,SP),Dm	mem8(d4+SP)→Dm	--	--	--	--	3	2	0110 01Dm <d4>														*2
	MOV (d8,SP),Dm	mem8(d8+SP)→Dm	--	--	--	--	5	3	0010 0110 01Dm <d8. ...>														*3
	MOV (d16,SP),Dm	mem8(d16+SP)→Dm	--	--	--	--	7	4	0010 0110 00Dm <d16>														
	MOV (io8),Dm	mem8(IOTOP+io8)→Dm	--	--	--	--	4	2	0110 00Dm <io8 ...>														
	MOV (abs8),Dm	mem8(abs8)→Dm	--	--	--	--	4	2	0100 01Dm <abs 8..>														
	MOV (abs12),Dm	mem8(abs12)→Dm	--	--	--	--	5	2	0100 00Dm <abs 12.. ...>														
	MOV (abs16),Dm	mem8(abs16)→Dm	--	--	--	--	7	4	0010 1100 00Dm <abs 16..>														
	MOV Dn,(Am)	Dn→mem8(Am)	--	--	--	--	2	2	0101 1aDn														
	MOV Dn,(d8,Am)	Dn→mem8(d8+Am)	--	--	--	--	4	2	0111 1aDn <d8. ...>														*1
	MOV Dn,(d16,Am)	Dn→mem8(d16+Am)	--	--	--	--	7	4	0010 0111 1aDn <d16>														
	MOV Dn,(d4,SP)	Dn→mem8(d4+SP)	--	--	--	--	3	2	0111 01Dn <d4>														*2
	MOV Dn,(d8,SP)	Dn→mem8(d8+SP)	--	--	--	--	5	3	0010 0111 01Dn <d8. ...>														*3
	MOV Dn,(d16,SP)	Dn→mem8(d16+SP)	--	--	--	--	7	4	0010 0111 00Dn <d16>														
	MOV Dn,(io8)	Dn→mem8(IOTOP+io8)	--	--	--	--	4	2	0111 00Dn <io8 ...>														
	MOV Dn,(abs8)	Dn→mem8(abs8)	--	--	--	--	4	2	0101 01Dn <abs 8..>														
	MOV Dn,(abs12)	Dn→mem8(abs12)	--	--	--	--	5	2	0101 00Dn <abs 12.. ...>														
	MOV Dn,(abs16)	Dn→mem8(abs16)	--	--	--	--	7	4	0010 1101 00Dn <abs 16..>														
	MOV imm8,(io8)	imm8→mem8(IOTOP+io8)	--	--	--	--	6	3	0000 0010 <io8 ...> <#8. ...>														
	MOV imm8,(abs8)	imm8→mem8(abs8)	--	--	--	--	6	3	0001 0100 <abs 8..> <#8. ...>														
	MOV imm8,(abs12)	imm8→mem8(abs12)	--	--	--	--	7	3	0001 0101 <abs 12.. ...> <#8. ...>														
	MOV imm8,(abs16)	imm8→mem8(abs16)	--	--	--	--	9	5	0011 1101 1001 <abs 16..> <#8. ...>														
	MOV Dn,(HA)	Dn→mem8(HA)	--	--	--	--	2	2	1101 00Dn														
MOVW	MOVW (An),DWm	mem16(An)→DWm	--	--	--	--	2	3	1110 00Ad														
	MOVW (An),Am	mem16(An)→Am	--	--	--	--	3	4	0010 1110 10Aa														*4
	MOVW (d4,SP),DWm	mem16(d4+SP)→DWm	--	--	--	--	3	3	1110 011d <d4>														*2
	MOVW (d4,SP),Am	mem16(d4+SP)→Am	--	--	--	--	3	3	1110 010a <d4>														*2
	MOVW (d8,SP),DWm	mem16(d8+SP)→DWm	--	--	--	--	5	4	0010 1110 011d <d8. ...>														*3
	MOVW (d8,SP),Am	mem16(d8+SP)→Am	--	--	--	--	5	4	0010 1110 010a <d8. ...>														*3
	MOVW (d16,SP),DWm	mem16(d16+SP)→DWm	--	--	--	--	7	5	0010 1110 001d <d16>														
	MOVW (d16,SP),Am	mem16(d16+SP)→Am	--	--	--	--	7	5	0010 1110 000a <d16>														
	MOVW (abs8),DWm	mem16(abs8)→DWm	--	--	--	--	4	3	1100 011d <abs 8..>														
	MOVW (abs8),Am	mem16(abs8)→Am	--	--	--	--	4	3	1100 010a <abs 8..>														
	MOVW (abs16),DWm	mem16(abs16)→DWm	--	--	--	--	7	5	0010 1100 011d <abs 16..>														
	MOVW (abs16),Am	mem16(abs16)→Am	--	--	--	--	7	5	0010 1100 010a <abs 16..>														
	MOVW DWn,(Am)	DWn→mem16(Am)	--	--	--	--	2	3	1111 00aD														
	MOVW An,(Am)	An→mem16(Am)	--	--	--	--	3	4	0010 1111 10aA														*4
	MOVW DWn,(d4,SP)	DWn→mem16(d4+SP)	--	--	--	--	3	3	1111 011D <d4>														*2
	MOVW An,(d4,SP)	An→mem16(d4+SP)	--	--	--	--	3	3	1111 010A <d4>														*2
	MOVW DWn,(d8,SP)	DWn→mem16(d8+SP)	--	--	--	--	5	4	0010 1111 011D <d8. ...>														*3
	MOVW An,(d8,SP)	An→mem16(d8+SP)	--	--	--	--	5	4	0010 1111 010A <d8. ...>														*3
	MOVW DWn,(d16,SP)	DWn→mem16(d16+SP)	--	--	--	--	7	5	0010 1111 001D <d16>														
	MOVW An,(d16,SP)	An→mem16(d16+SP)	--	--	--	--	7	5	0010 1111 000A <d16>														
	MOVW DWn,(abs8)	DWn→mem16(abs8)	--	--	--	--	4	3	1101 011D <abs 8..>														
	MOVW An,(abs8)	An→mem16(abs8)	--	--	--	--	4	3	1101 010A <abs 8..>														
	MOVW DWn,(abs16)	DWn→mem16(abs16)	--	--	--	--	7	5	0010 1101 011D <abs 16..>														
	MOVW An,(abs16)	An→mem16(abs16)	--	--	--	--	7	5	0010 1101 010A <abs 16..>														
	MOVW DWn,(HA)	DWn→mem16(HA)	--	--	--	--	2	3	1001 010D														
	MOVW An,(HA)	An→mem16(HA)	--	--	--	--	2	3	1001 011A														
	MOVW imm8,DWm	sign(imm8)→DWm	--	--	--	--	4	2	0000 110d <#8. ...>														*5
	MOVW imm8,Am	zero(imm8)→Am	--	--	--	--	4	2	0000 111a <#8. ...>														*6
	MOVW imm16,DWm	imm16→DWm	--	--	--	--	6	3	1100 111d <#16>														

*1 d8 sign-extension *4 A=An, a=Am
*2 d4 zero-extension *5 #8 sign-extension
*3 d8 zero-extension *6 #8 zero-extension

MN101E SERIES INSTRUCTION SET

Group	Mnemonic	Operation	Flag				Code Size	Cycle	Repeat	Extension	Machine Code											Notes
			VF	NF	CF	ZF					1	2	3	4	5	6	7	8	9	10	11	
	MOVW imm16,Am	imm16→Am	--	--	--	--	6	3			1101	111a	<#16	...	...	...						
	MOVW SP,Am	SP→Am	--	--	--	--	3	3			0010	0000	100a									
	MOVW An,SP	An→SP	--	--	--	--	3	3			0010	0000	101A									
	MOVW DWn,DWm	DWn→DWm	--	--	--	--	3	3			0010	1000	00Dd									*1
	MOVW DWn,Am	DWn→Am	--	--	--	--	3	3			0010	0100	11Da									
	MOVW An,DWm	An→DWm	--	--	--	--	3	3			0010	1100	11Ad									
	MOVW An,Am	An→Am	--	--	--	--	3	3			0010	0000	00Aa									*2
PUSH	PUSH Dn	SP-1→SP, Dn→mem8(SP)	--	--	--	--	2	3			1111	10Dn										
	PUSH An	SP-2→SP, An→mem16(SP)	--	--	--	--	2	5			0001	011A										
POP	POP Dn	mem8(SP)→Dn, SP+1→SP	--	--	--	--	2	3			1110	10Dn										
	POP An	mem16(SP)→An, SP+2→SP	--	--	--	--	2	4			0000	011A										
EXT	EXT Dn,DWm	sign(Dn)→DWm	--	--	--	--	3	3			0010	1001	000d									*3

Arithmetic manipulation instructions

ADD	ADD Dn,Dm	Dm+Dn→Dm	●	●	●	●	3	2			0011	0011	DnDm									
	ADD imm4,Dm	Dm+sign(imm4)→Dm	●	●	●	●	3	2			1000	00Dm	<#4>									*6
	ADD imm8,Dm	Dm+imm8→Dm	●	●	●	●	4	2			0000	10Dm	<#8, ...>									
ADDC	ADDC Dn,Dm	Dm+Dn+CF→Dm	●	●	●	●	3	2	○		0011	1011	DnDm									
ADDW	ADDW DWn,DWm	DWm+DWn→DWm	●	●	●	●	3	3	○		0010	0101	00Dd									*1
	ADDW DWn,Am	Am+DWn→Am	●	●	●	●	3	3	○		0010	0101	10Da									
	ADDW imm4,Am	Am+sign(imm4)→Am	●	●	●	●	3	2			1110	110a	<#4>									*6
	ADDW imm8,Am	Am+sign(imm8)→Am	●	●	●	●	5	3			0010	1110	110a	<#8, ...>								*7
	ADDW imm16,Am	Am+imm16→Am	●	●	●	●	7	4			0010	0101	011a	<#16>								
	ADDW imm4,SP	SP+sign(imm4)→SP	--	--	--	--	3	2			1111	1101	<#4>									*6
	ADDW imm8,SP	SP+sign(imm8)→SP	--	--	--	--	4	2			1111	1100	<#8, ...>									*7
	ADDW imm16,SP	SP+imm16→SP	--	--	--	--	7	4			0010	1111	1100	<#16>								
	ADDW imm16,DWm	DWm+imm16→DWm	●	●	●	●	7	4			0010	0101	010d	<#16>								
ADDUW	ADDUW Dn,Am	Am+zero(Dn)→Am	●	●	●	●	3	3	○		0010	1000	1aDn									*8
ADDSW	ADDSW Dn,Am	Am+sign(Dn)→Am	●	●	●	●	3	3	○		0010	1001	1aDn									
SUB	SUB Dn,Dm(when Dn=Dm)	Dm-Dn→Dm	●	●	●	●	3	2	○		0010	1010	DnDm									
	SUB Dn,Dn	Dn-Dn→Dn	0	0	0	1	2	1			1000	01Dn										
	SUB imm8,Dm	Dm-imm8→Dm	●	●	●	●	5	3			0010	1010	DmDm	<#8, ...>								
SUBC	SUBC Dn,Dm	Dm-Dn-CF→Dm	●	●	●	●	3	2	○		0010	1011	DnDm									
SUBW	SUBW DWn,DWm	DWm-DWn→DWm	●	●	●	●	3	3			0010	0100	00Dd									*1
	SUBW DWn,Am	Am-DWn→Am	●	●	●	●	3	3			0010	0100	10Da									
	SUBW imm16,DWm	DWm-imm16→DWm	●	●	●	●	7	4			0010	0100	010d	<#16>								
	SUBW imm16,Am	Am-imm16→Am	●	●	●	●	7	4			0010	0100	011a	<#16>								
MULU	MULU Dn,Dm	Dm*Dn→DWk	0	●	●	●	3	8			0010	1111	111D									*4
DIVU	DIVU Dn,DWm	DWm/Dn→DWm-l...DWm-h	●	●	●	●	3	9			0010	1110	111d									*5
CMP	CMP Dn,Dm	Dm-Dn...PSW	●	●	●	●	3	2			0011	0010	DnDm									
	CMP imm8,Dm	Dm-imm8...PSW	●	●	●	●	4	2			1100	00Dm	<#8, ...>									
	CMP imm8,(abs8)	mem8(abs8)-imm8...PSW	●	●	●	●	6	3			0000	0100	<abs 8...> <#8, ...>									
	CMP imm8,(abs12)	mem8(abs12)-imm8...PSW	●	●	●	●	7	3			0000	0101	<abs 12...> <#8, ...>									
	CMP imm8,(abs16)	mem8(abs16)-imm8...PSW	●	●	●	●	9	5			0011	1101	1000	<abs 16...> <#8, ...>								
CMPW	CMPW DWn,DWm	DWm-DWn...PSW	●	●	●	●	3	3			0010	1000	01Dd									*1
	CMPW DWn,Am	Am-DWn...PSW	●	●	●	●	3	3			0010	0101	11Da									
	CMPW An,Am	Am-An...PSW	●	●	●	●	3	3			0010	0000	01Aa									*2
	CMPW imm16,DWm	DWm-imm16...PSW	●	●	●	●	6	3			1100	110d	<#16>									
	CMPW imm16,Am	Am-imm16...PSW	●	●	●	●	6	3			1101	110a	<#16>									

Logical manipulation instructions

AND	AND Dn,Dm	Dm&Dn→Dm	0	●	0	●	3	2			0011	0111	DnDm									
	AND imm8,Dm	Dm&imm8→Dm	0	●	0	●	4	2			0001	11Dm	<#8, ...>									
	AND imm8,PSW	PSW&imm8→PSW	●	●	●	●	5	3			0010	1001	0010	<#8, ...>								
OR	OR Dn,Dm	Dm Dn→Dm	0	●	0	●	3	2			0011	0110	DnDm									
	OR imm8,Dm	Dm imm8→Dm	0	●	0	●	4	2			0001	10Dm	<#8, ...>									
	OR imm8,PSW	PSW imm8→PSW	●	●	●	●	5	3			0010	1001	0011	<#8, ...>								
XOR	XOR Dn,Dm	Dm^Dn→Dm	0	●	0	●	3	2			0011	1010	DnDm									*9
	XOR imm8,Dm	Dm^imm8→Dm	0	●	0	●	5	3			0011	1010	DmDm	<#8, ...>								

*1 D=DWn, d=DWm

*2 A=An, a=Am

*3 d=DWm

*4 D=DWk

*5 D=DWm

*6 #4 sign-extension

*7 #8 sign-extension

*8 Dn zero extension

*9 m≠n

MN101E SERIES INSTRUCTION SET

Group	Mnemonic	Operation	Flag				Code Size	Cycle Repeat	Extension	Machine Code											Notes
			VF	NF	CF	ZF				1	2	3	4	5	6	7	8	9	10	11	
NOT	NOT Dn	$\neg Dn \rightarrow Dn$	0	●	0	●	3	2		0010	0010	10Dn									
ASR	ASR Dn	Dn.msb $\rightarrow$ temp, Dn.lsb $\rightarrow$ CF Dn $\gg 1 \rightarrow$ Dn, temp $\rightarrow$ Dn.msb	0	--	●	●	3	2	○	0010	0011	10Dn									
LSR	LSR Dn	Dn.lsb $\rightarrow$ CF, Dn $\gg 1 \rightarrow$ Dn 0 $\rightarrow$ Dn.msb	0	0	●	●	3	2	○	0010	0011	11Dn									
ROR	ROR Dn	Dn.lsb $\rightarrow$ temp, Dn $\gg 1 \rightarrow$ Dn CF $\rightarrow$ Dn.msb, temp $\rightarrow$ CF	0	●	●	●	3	2	○	0010	0010	11Dn									

Bit manipulation instructions

BSET	BSET (io8)bp	mem8(IOTOP+io8)&bpdata...PSW 1 $\rightarrow$ mem8(IOTOP+io8)bp	0	●	0	●	5	5		0011	1000	0bp.	<io8	...							
	BSET (abs8)bp	mem8(abs8)&bpdata...PSW 1 $\rightarrow$ mem8(abs8)bp	0	●	0	●	4	4			1011	0bp.	<abs	8.							
	BSET (abs16)bp	mem8(abs16)&bpdata...PSW 1 $\rightarrow$ mem8(abs16)bp	0	●	0	●	7	6		0011	1100	0bp.	<abs	16.		...					
BCLR	BCLR (io8)bp	mem8(IOTOP+io8)&bpdata...PSW 0 $\rightarrow$ mem8(IOTOP+io8)bp	0	●	0	●	5	5		0011	1000	1bp.	<io8	...							
	BCLR (abs8)bp	mem8(abs8)&bpdata...PSW 0 $\rightarrow$ mem8(abs8)bp	0	●	0	●	4	4			1011	1bp.	<abs	8.							
	BCLR (abs16)bp	mem8(abs16)&bpdata...PSW 0 $\rightarrow$ mem8(abs16)bp	0	●	0	●	7	6		0011	1100	1bp.	<abs	16.		...					
BTST	BTST imm8,Dm	Dm&imm8...PSW	0	●	0	●	5	3		0010	0000	11Dm	<#8.	...							
	BTST (abs16)bp	mem8(abs16)&bpdata...PSW	0	●	0	●	7	5		0011	1101	0bp.	<abs	16.		...					

Branch instructions

Bcc	BEQ label	if(ZF=1), PC+3+d4(label)+H $\rightarrow$ PC if(ZF=0), PC+3 $\rightarrow$ PC	--	--	--	--	3	2/3		1001	000H	<d4>									*1
	BEQ label	if(ZF=1), PC+4+d7(label)+H $\rightarrow$ PC if(ZF=0), PC+4 $\rightarrow$ PC	--	--	--	--	4	2/3		1000	1010	<d7.	...H								*2
	BEQ label	if(ZF=1), PC+5+d11(label)+H $\rightarrow$ PC if(ZF=0), PC+5 $\rightarrow$ PC	--	--	--	--	5	2/3		1001	1010	<d11		...H							*3
	BNE label	if(ZF=0), PC+3+d4(label)+H $\rightarrow$ PC if(ZF=1), PC+3 $\rightarrow$ PC	--	--	--	--	3	2/3		1001	001H	<d4>									1
	BNE label	if(ZF=0), PC+4+d7(label)+H $\rightarrow$ PC if(ZF=1), PC+4 $\rightarrow$ PC	--	--	--	--	4	2/3		1000	1011	<d7.	...H								*2
	BNE label	if(ZF=0), PC+5+d11(label)+H $\rightarrow$ PC if(ZF=1), PC+5 $\rightarrow$ PC	--	--	--	--	5	2/3		1001	1011	<d11		...H							*3
	BGE label	if((VF^NF)=0), PC+4+d7(label)+H $\rightarrow$ PC if((VF^NF)=1), PC+4 $\rightarrow$ PC	--	--	--	--	4	2/3		1000	1000	<d7.	...H								*2
	BGE label	if((VF^NF)=0), PC+5+d11(label)+H $\rightarrow$ PC if((VF^NF)=1), PC+5 $\rightarrow$ PC	--	--	--	--	5	2/3		1001	1000	<d11		...H							*3
	BCC label	if(CF=0), PC+4+d7(label)+H $\rightarrow$ PC if(CF=1), PC+4 $\rightarrow$ PC	--	--	--	--	4	2/3		1000	1100	<d7.	...H								*2
	BCC label	if(CF=0), PC+5+d11(label)+H $\rightarrow$ PC if(CF=1), PC+5 $\rightarrow$ PC	--	--	--	--	5	2/3		1001	1100	<d11		...H							*3
	BCS label	if(CF=1), PC+4+d7(label)+H $\rightarrow$ PC if(CF=0), PC+4 $\rightarrow$ PC	--	--	--	--	4	2/3		1000	1101	<d7.	...H								*2
	BCS label	if(CF=1), PC+5+d11(label)+H $\rightarrow$ PC if(CF=0), PC+5 $\rightarrow$ PC	--	--	--	--	5	2/3		1001	1101	<d11		...H							*3
	BLT label	if((VF^NF)=1), PC+4+d7(label)+H $\rightarrow$ PC if((VF^NF)=0), PC+4 $\rightarrow$ PC	--	--	--	--	4	2/3		1000	1110	<d7.	...H								*2
	BLT label	if((VF^NF)=1), PC+5+d11(label)+H $\rightarrow$ PC if((VF^NF)=0), PC+5 $\rightarrow$ PC	--	--	--	--	5	2/3		1001	1110	<d11		...H							*3
	BLE label	if((VF^NF) ZF=1), PC+4+d7(label)+H $\rightarrow$ PC if((VF^NF) ZF=0), PC+4 $\rightarrow$ PC	--	--	--	--	4	2/3		1000	1111	<d7.	...H								*2
	BLE label	if((VF^NF) ZF=1), PC+5+d11(label)+H $\rightarrow$ PC if((VF^NF) ZF=0), PC+5 $\rightarrow$ PC	--	--	--	--	5	2/3		1001	1111	<d11		...H							*3
	BGT label	if((VF^NF) ZF=0), PC+4+d7(label)+H $\rightarrow$ PC if((VF^NF) ZF=1), PC+4 $\rightarrow$ PC	--	--	--	--	4	3/4		0010	0010	0001	<d7.	...H							*2
	BGT label	if((VF^NF) ZF=0), PC+5+d11(label)+H $\rightarrow$ PC if((VF^NF) ZF=1), PC+5 $\rightarrow$ PC	--	--	--	--	5	3/4		0010	0010	0001	<d7.	...H							*2

*1 d4 sign-extension
*2 d7 sign-extension
*3 d11 sign-extension

MN101E SERIES INSTRUCTION SET

Group	Mnemonic	Operation	Flag				Code Size	Cycle Repeat	Exten- sion	Machine Code											Notes			
			VF	NF	CF	ZF				1	2	3	4	5	6	7	8	9	10	11				
Bcc	BGT label	if((VF^NF) ZF=0),PC+6+d11(label)+H→PC if((VF^NF) ZF=1),PC+6→PC	--	--	--	--	6	3/4		0010	0011	0001	<d11		...H								*3	
	BHI label	if(CF ZF=0),PC+5+d7(label)+H→PC if(CF ZF=1),PC+5→PC	--	--	--	--	5	3/4		0010	0010	0010	<d7.	...H									*2	
	BHI label	if(CF ZF=0),PC+6+d11(label)+H→PC if(CF ZF=1),PC+6→PC	--	--	--	--	6	3/4		0010	0011	0010	<d11		...H								*3	
	BLS label	if(CF ZF=1),PC+5+d7(label)+H→PC if(CF ZF=0),PC+5→PC	--	--	--	--	5	3/4		0010	0010	0011	<d7.	...H									*2	
	BLS label	if(CF ZF=1),PC+6+d11(label)+H→PC if(CF ZF=0),PC+6→PC	--	--	--	--	6	3/4		0010	0011	0011	<d11		...H								*3	
	BNC label	if(NF=0),PC+5+d7(label)+H→PC if(NF=1),PC+5→PC	--	--	--	--	5	3/4		0010	0010	0100	<d7.	...H									*2	
	BNC label	if(NF=0),PC+6+d11(label)+H→PC if(NF=1),PC+6→PC	--	--	--	--	6	3/4		0010	0011	0100	<d11		...H								*3	
	BNS label	if(NF=1),PC+5+d7(label)+H→PC if(NF=0),PC+5→PC	--	--	--	--	5	3/4		0010	0010	0101	<d7.	...H									*2	
	BNS label	if(NF=1),PC+6+d11(label)+H→PC if(NF=0),PC+6→PC	--	--	--	--	6	3/4		0010	0011	0101	<d11		...H								*3	
	BVC label	if(VF=0),PC+5+d7(label)+H→PC if(VF=1),PC+5→PC	--	--	--	--	5	3/4		0010	0010	0110	<d7.	...H									*2	
	BVC label	if(VF=0),PC+6+d11(label)+H→PC if(VF=1),PC+6→PC	--	--	--	--	6	3/4		0010	0011	0110	<d11		...H								*3	
	BVS label	if(VF=1),PC+5+d7(label)+H→PC if(VF=0),PC+5→PC	--	--	--	--	5	3/4		0010	0010	0111	<d7.	...H									*2	
	BVS label	if(VF=1),PC+6+d11(label)+H→PC if(VF=0),PC+6→PC	--	--	--	--	6	3/4		0010	0011	0111	<d11		...H								*3	
	BRA label	PC+3+d4(label)+H→PC	--	--	--	--	3	3		1110	111H	<d4>												*1
	BRA label	PC+4+d7(label)+H→PC	--	--	--	--	4	3		1000	1001	<d7.	...H											*2
	BRA label	PC+5+d11(label)+H→PC	--	--	--	--	5	3		1001	1001	<d11		...H										*3
CBEQ	CBEQ imm8,Dm,label	if(Dm=imm8),PC+6+d7(label)+H→PC if(Dm≠imm8),PC+6→PC	●	●	●	●	6	3/4		1100	10Dm	<#8.	...>	<d7.	...H								*2	
	CBEQ imm8,Dm,label	if(Dm=imm8),PC+8+d11(label)+H→PC if(Dm≠imm8),PC+8→PC	●	●	●	●	8	4/5		0010	1100	10Dm	<#8.	...>	<d11		...H						*3	
	CBEQ imm8,(abs8),label	if(mem8(abs8)=imm8),PC+9+d7(label)+H→PC if(mem8(abs8)≠imm8),PC+9→PC	●	●	●	●	9	6/7		0010	1101	1100	<abs	8.>	<#8.	...>	<d7.	...H					*2	
	CBEQ imm8,(abs8),label	if(mem8(abs8)=imm8),PC+10+d11(label)+H→PC if(mem8(abs8)≠imm8),PC+10→PC	●	●	●	●	10	6/7		0010	1101	1101	<abs	8.>	<#8.	...>	<d11		...H				*3	
	CBEQ imm8,(abs16),label	if(mem8(abs16)=imm8),PC+11+d7(label)+H→PC if(mem8(abs16)≠imm8),PC+11→PC	●	●	●	●	11	7/8		0011	1101	1100	<abs	16..		...>	<#8.	...>	<d7.	...H			*2	
	CBEQ imm8,(abs16),label	if(mem8(abs16)=imm8),PC+12+d11(label)+H→PC if(mem8(abs16)≠imm8),PC+12→PC	●	●	●	●	12	7/8		0011	1101	1101	<abs	16..		...>	<#8.	...>	<d11		...H			*3
CBNE	CBNE imm8,Dm,label	if(Dm≠imm8),PC+6+d7(label)+H→PC if(Dm=imm8),PC+6→PC	●	●	●	●	6	3/4		1101	10Dm	<#8.	...>	<d7.	..H>								*2	
	CBNE imm8,Dm,label	if(Dm≠imm8),PC+8+d11(label)+H→PC if(Dm=imm8),PC+8→PC	●	●	●	●	8	4/5		0010	1101	10Dm	<#8.	...>	<d11		...H						*3	
	CBNE imm8,(abs8),label	if(mem8(abs8)≠imm8),PC+9+d7(label)+H→PC if(mem8(abs8)=imm8),PC+9→PC	●	●	●	●	9	6/7		0010	1101	1110	<abs	8.>	<#8.	...>	<d7.	...H					*2	
	CBNE imm8,(abs8),label	if(mem8(abs8)≠imm8),PC+10+d11(label)+H→PC if(mem8(abs8)=imm8),PC+10→PC	●	●	●	●	10	6/7		0010	1101	1111	<abs	8.>	<#8.	...>	<d11		...H				*3	
	CBNE imm8,(abs16),label	if(mem8(abs16)≠imm8),PC+11+d7(label)+H→PC if(mem8(abs16)=imm8),PC+11→PC	●	●	●	●	11	7/8		0011	1101	1110	<abs	16..		...>	<#8.	...>	<d7.	...H			*2	
	CBNE imm8,(abs16),label	if(mem8(abs16)≠imm8),PC+12+d11(label)+H→PC if(mem8(abs16)=imm8),PC+12→PC	●	●	●	●	12	7/8		0011	1101	1111	<abs	16..		...>	<#8.	...>	<d11		...H			*3
TBZ	TBZ (abs8)bp,label	if(mem8(abs8)bp=0),PC+7+d7(label)+H→PC if(mem8(abs8)bp=1),PC+7→PC	0	●	0	●	7	6/7		0011	0000	0bp.	<abs	8.>	<d7.	...H							*2	
	TBZ (abs8)bp,label	if(mem8(abs8)bp=0),PC+8+d11(label)+H→PC if(mem8(abs8)bp=1),PC+8→PC	0	●	0	●	8	6/7		0011	0000	1bp.	<abs	8.>	<d11		...H						*3	

*1 d4 sign-extension
 *2 d7 sign-extension
 *3 d11 sign-extension

MN101E SERIES INSTRUCTION SET

Group	Mnemonic	Operation	Flag				Code Size	Cycle Re- peat	Exten- sion	Machine Code											Notes			
			VF	NF	CF	ZF				1	2	3	4	5	6	7	8	9	10	11				
TBZ	TBZ (io8)bp,label	if(mem8(IOTOP+io8)bp=0),PC+7+d7(label)+H→PC if(mem8(IOTOP+io8)bp=1),PC+7→PC	0	●	0	●	7	6/7		0011 0100 0bp. <io8 ...> <d7. ...H													*1	
	TBZ (io8)bp,label	if(mem8(IOTOP+io8)bp=0),PC+8+d11(label)+H→PC if(mem8(IOTOP+io8)bp=1),PC+8→PC	0	●	0	●	8	6/7		0011 0100 1bp. <io8 ...> <d11H													*2	
	TBZ (abs16)bp,label	if(mem8(abs16)bp=0),PC+9+d7(label)+H→PC if(mem8(abs16)bp=1),PC+9→PC	0	●	0	●	9	7/8		0011 1110 0bp. <abs 16..> <d7. ...H													*1	
	TBZ (abs16)bp,label	if(mem8(abs16)bp=0),PC+10+d11(label)+H→PC if(mem8(abs16)bp=1),PC+10→PC	0	●	0	●	10	7/8		0011 1110 1bp. <abs 16..> <d11H													*2	
TBNZ	TBNZ (abs8)bp,label	if(mem8(abs8)bp=1),PC+7+d7(label)+H→PC if(mem8(abs8)bp=0),PC+7→PC	0	●	0	●	7	6/7		0011 0001 0bp. <abs 8.> <d7. ...H													*1	
	TBNZ (abs8)bp,label	if(mem8(abs8)bp=1),PC+8+d11(label)+H→PC if(mem8(abs8)bp=0),PC+8→PC	0	●	0	●	8	6/7		0011 0001 1bp. <abs 8.> <d11H													*2	
	TBNZ (io8)bp,label	if(mem8(io)bp=1),PC+7+d7(label)+H→PC if(mem8(io)bp=0),PC+7→PC	0	●	0	●	7	6/7		0011 0101 0bp. <io8 ...> <d7. ...H													*1	
	TBNZ (io8)bp,label	if(mem8(io)bp=1),PC+8+d11(label)+H→PC if(mem8(io)bp=0),PC+8→PC	0	●	0	●	8	6/7		0011 0101 1bp. <io8 ...> <d11H													*2	
	TBNZ (abs16)bp,label	if(mem8(abs16)bp=1),PC+9+d7(label)+H→PC if(mem8(abs16)bp=0),PC+9→PC	0	●	0	●	9	7/8		0011 1111 0bp. <abs 16..> <d7. ...H													*1	
	TBNZ (abs16)bp,label	if(mem8(abs16)bp=1),PC+10+d11(label)+H→PC if(mem8(abs16)bp=0),PC+10→PC	0	●	0	●	10	7/8		0011 1111 1bp. <abs 16..> <d11H													*2	
JMP	JMP (An)	0→PC.17-16,An→PC.15-0,0→PC.H	---	---	---	---	3	4		0010 0001 00A0														
	JMP label	abs18(label)+H→PC	---	---	---	---	7	5		0011 1001 0aaH <abs 18.b p15~ 0.>													*5	
	JMP label	abs20(label)+H→PC	---	---	---	---	9	6		0011 1101 1010 000B bbbH <abs 20.b p15~ 0.>													*6*7	
JSR	JSR (An)	SP-3→SP,(PC+3).bp7-0→mem8(SP) (PC+3).bp15-8→mem8(SP+1) (PC+3).H→mem8(SP+2).bp7, 0→mem8(SP+2).bp6-4, (PC+3).bp19-16→mem8(SP+2).bp3-0 0→PC.bp19-16 An→PC.bp15-0,0→PC.H	---	---	---	---	3	7		0010 0001 00A1														
	JSR label	SP-3→SP,(PC+5).bp7-0→mem8(SP) (PC+5).bp15-8→mem8(SP+1) (PC+5).H→mem8(SP+2).bp7, 0→mem8(SP+2).bp6-4, (PC+5).bp19-16→mem8(SP+2).bp3-0 PC+5+d12(label)+H→PC	---	---	---	---	5	6		0001 000H <d12>													*3	
	JSR label	SP-3→SP,(PC+6).bp7-0→mem8(SP) (PC+6).bp15-8→mem8(SP+1) (PC+6).H→mem8(SP+2).bp7, 0→mem8(SP+2).bp6-4, (PC+6).bp19-16→mem8(SP+2).bp3-0 PC+6+d16(label)+H→PC	---	---	---	---	6	7		0001 001H <d16>													*4	
	JSR label	SP-3→SP,(PC+7).bp7-0→mem8(SP) (PC+7).bp15-8→mem8(SP+1) (PC+7).H→mem8(SP+2).bp7, 0→mem8(SP+2).bp6-4, (PC+7).bp19-16→mem8(SP+2).bp3-0 abs18(label)+H→PC	---	---	---	---	7	8		0011 1001 1aaH <abs 18.b p15~ 0.>													*5	
	JSR label	SP-3→SP,(PC+7).bp7-0→mem8(SP) (PC+7).bp15-8→mem8(SP+1) (PC+7).H→mem8(SP+2).bp7, 0→mem8(SP+2).bp6-4, (PC+7).bp19-16→mem8(SP+2).bp3-0 abs20(label)+H→PC	---	---	---	---	9	9		0011 1101 1011 000B bbbH <abs 20.b p15~ 0.>													*6 *7	
	JSRV (tbl4)	SP-3→SP,(PC+3).bp7-0→mem8(SP) (PC+3).bp15-8→mem8(SP+1) (PC+3).H→mem8(SP+2).bp7 0→mem8(SP+2).bp6-4, (PC+3).bp19-16→mem8(SP+2).bp3-0 mem8(x'004080+tbl4<<2>)→PC.bp7-0 mem8(x'004080+tbl4<<2+1>)→PC.bp15-8 mem8(x'004080+tbl4<<2+2>).bp7→PC.H mem8(x'004080+tbl4<<2+2>).bp3-0→ PC.bp19-16	---	---	---	---	3	9		1111 1110 <4>														
	NOP	NOP	PC+2→PC	---	---	---	---	2	1	○	0000 0000													

- *1 d7 sign-extension
- *2 d11 sign-extension
- *3 d12 sign-extension
- *4 d16 sign-extension
- *5 aa=abs18.17 - 16
- *6 B=abs20.19
- *7 bbb=abs20.18 - 16

MN101E SERIES INSTRUCTION SET

Group	Mnemonic	Operation	Flag				Code Size	Cycle Repeat	Exten- sion	Machine Code											Notes
			VF	NF	CF	ZF				1	2	3	4	5	6	7	8	9	10	11	
RTS	RTS	mem8(SP)→(PC).bp7-0 mem8(SP+1)→(PC).bp15-8 mem8(SP+2).bp7→(PC).H mem8(SP+2).bp3-0→(PC).bp19-16 SP+3→SP	---	---	---	---	2	7		0000	0001										
RTI	RTI	mem8(SP)→PSW mem8(SP+1)→(PC).bp7-0 mem8(SP+2)→(PC).bp15-8 mem8(SP+3).bp7→(PC).H mem8(SP+3).bp3-0→(PC).bp19-16 mem8(SP+4)→HA-l mem8(SP+5)→HA-h SP+6→SP	●	●	●	●	2	11		0000	0011										

Contorl instructions

REP	REP imm3	imm3-1→RPC	---	---	---	---	3	2		0010	0001	1rep									*1
BE	BE	PSW & x'3F'→PSW	---	---	---	---	3	3		0010	0010	0000									
BD	BD	PSW x'c0'→PSW	---	---	---	---	3	3		0010	0011	0000									

*1 no repeat whn imm3=0, (rep: imm3-1)



Other than the instruction of MN101E Series, the assembler of this Series has the following instructions as macro instructions.

The assembler will interpret the macro instructions below as the assembler instructions.

macro instructions		replaced instructions		remarks
INC	Dn	ADD	1,Dn	
DEC	Dn	ADD	-1,Dn	
INC	An	ADDW	1,An	
DEC	An	ADDW	-1,An	
INC2	An	ADDW	2,An	
DEC2	An	ADDW	-2,An	
CLR	Dn	SUB	Dn,Dm	n=m
ASL	Dn	ADD	Dn,Dm	n=m
LSL	Dn	ADD	Dn,Dm	n=m
ROL	Dn	ADDC	Dn,Dm	n=m
NEG	Dn	NOT ADD	Dn 1,Dn	
NOPL		MOVW	DWn,DWm	n=m
MOV	(SP),Dn	MOV	(0,SP),Dn	
MOV	Dn,(SP)	MOV	Dn,(0,SP)	
MOVW	(SP),DWn	MOVW	(0,SP),DWn	
MOVW	DWn,(SP)	MOVW	DWn,(0,SP)	
MOVW	(SP),An	MOVW	(0,SP),An	
MOVW	An,(SP)	MOVW	An,(0,SP)	

Ver3.3(2002.01.31)

25.3 Instruction Map

MN101E SERIES INSTRUCTION MAP

1st nibble2nd nibble

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F	
0	NOP	RTS	MOV #8,(io8)	RTI	CMP #8,(abs8)/(abs12)		POP An		ADD #8,Dm				MOVW #8,DWm		MOVW #8,Am		
1	JSR d12(label)		JSR d16(label)		MOV #8,(abs8)/(abs12)		PUSH An		OR #8,Dm				AND #8,Dm				
2	When the extension code is b'0010'																
3	When the extension code is b'0011'																
4	MOV (abs12),Dm				MOV (abs8),Dm				MOV (An),Dm								
5	MOV Dn,(abs12)				MOV Dn,(abs8)				MOV Dn,(Am)								
6	MOV (io8),Dm				MOV (d4,SP),Dm				MOV (d8,An),Dm								
7	MOV Dn,(io8)				MOV Dn,(d4,SP)				MOV Dn,(d8,Am)								
8	ADD #4,Dm				SUB Dn,Dn				BGE d7	BRA d7	BEQ d7	BNE d7	BCC d7	BCS d7	BLT d7	BLE d7	
9	BEQ d4		BNE d4		MOVW DWn,(HA)		MOVW An,(HA)		BGE d11	BRA d11	BEQ d11	BNE d11	BCC d11	BCS d11	BLT d11	BLE d11	
A	MOV Dn,Dm / MOV #8,Dm																
B	BSET (abs8)bp								BCLR (abs8)bp								
C	CMP #8,Dm				MOVW (abs8),Am		MOVW (abs8),DWm		CBEQ #8,Dm,d7				CMPW #16,DWm		MOVW #16,DWm		
D	MOV Dn,(HA)				MOVW An,(abs8)		MOVW DWn,(abs8)		CBNE #8,Dm,d7				CMPW #16,Am		MOVW #16,Am		
E	MOVW (An),DWm				MOVW (d4,SP),Am		MOVW (d4,SP),DWm		POP Dn				ADDW #4,Am		BRA d4		
F	MOVW DWn,(Am)				MOVW An,(d4,SP)		MOVW DWn,(d4,SP)		PUSH Dn				ADDW #8,SP		ADDW #4,SP	JSRV (tbl4)	

Extension code: b'0010'

2nd nibble 3rd nibble

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0	MOVW An,Am				CMPW An,Am				MOVW SP,Am		MOVW An,SP		BTST #8,Dm			
1	JMP (A0)	JSR (A0)	JMP (A1)	JSR (A1)	MOV PSW,Dm				REP #3							
2	BE	BGT d7	BHI d7	BLS d7	BNC d7	BNS d7	BVC d7	BVS d7	NOT Dn				ROR Dn			
3	BD	BGT d11	BHI d11	BLS d11	BNC d11	BNS d11	BVC d11	BVS d11	ASR Dn				LSR Dn			
4	SUBW DWn,DWm				SUBW #16,DWm		SUBW #16,Am		SUBW DWn,Am				MOVW DWn,Am			
5	ADDW DWn,DWm				ADDW #16,DWm		ADDW #16,Am		ADDW DWn,Am				CMPW DWn,Am			
6	MOV (d16,SP),Dm				MOV (d8,SP),Dm				MOV (d16,An),Dm							
7	MOV Dn,(d16,SP)				MOV Dn,(d8,SP)				MOV Dn,(d16,Am)							
8	MOVW DWn,DWm (NOPL @n=m)				CMPW DWn,DWm				ADDUW Dn,Am							
9	EXT Dn,DWm		AND #8,PSW	OR #8,PSW	MOV Dn,PSW				ADDSW Dn,Am							
A	SUB Dn,Dm / SUB #8,Dm															
B	SUBC Dn,Dm															
C	MOV (abs16),Dm				MOVW (abs16),Am		MOVW (abs16),DWm		CBEQ #8,Dm,d12				MOVW An,DWm			
D	MOV Dn,(abs16)				MOVW An,(abs16)		MOVW DWn,(abs16)		CBNE #8,Dm,d12				CBEQ #8,(abs8),d7/d11		CBNE #8,(abs8),d7/d11	
E	MOVW (d16,SP),Am		MOVW (d16,SP),DWm		MOVW (d8,SP),Am		MOVW (d8,SP),DWm		MOVW (An),Am				ADDW #8,Am		DIVU	
F	MOVW An,(d16,SP)		MOVW DWn,(d16,SP)		MOVW An,(d8,SP)		MOVW DWn,(d8,SP)		MOVW An,(Am)				ADDW #16,SP			MULU

Extension code: b'0011'

2nd nibble\ 3rd nibble

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0	TBZ (abs8)bp,d7								TBZ (abs8)bp,d11							
1	TBNZ (abs8)bp,d7								TBNZ (abs8)bp,d11							
2	CMP Dn,Dm															
3	ADD Dn,Dm															
4	TBZ (io8)bp,d7								TBZ (io8)bp,d11							
5	TBNZ (io8)bp,d7								TBNZ (io8)bp,d11							
6	OR Dn,Dm															
7	AND Dn,Dm															
8	BSET (io8)bp								BCLR (io8)bp							
9	JMP abs18(label)								JSR abs18(label)							
A	XOR Dn,Dm / XOR #8,Dm															
B	ADDC Dn,Dm															
C	BSET (abs16)bp								BCLR (abs16)bp							
D	BTST (abs16)bp								cmp #8,(abs16)	mov #8,(abs16)	JMP abs20(label)	JSR abs20(label)	CBEQ #8,(abs16),d7/11	CBNE #8,(abs16),d7/11		
E	TBZ (abs16)bp,d7								TBZ (abs16)bp,d11							
F	TBNZ (abs16)bp,d7								TBNZ (abs16)bp,d11							

Ver2.1(2001.03.26)

Record of Changes

The following shows the changes in the publication of "MN101EF56K/57G/76K LSI User's Manual" (From 3rd Edition dated in January, 2012 to the 3rd Edition 1st Printing dated in February, 2014)

Page	Line	Definition	Former Edition (3.0)	New Edition (3.1)
I-3	Package	Specification addition	-	Package(TQFP080-P-1212F) is added in MN101EF57G.
I-13	Package	Specification addition	80LQFP Top View	80LQFP/ <u>TQFP</u> Top View
I-73	Package	Specification addition	-	Package code: TQFP080-P-1212F (MN101EF57G) is added.

Remark: Definition in the above table is classified according to the content of changes as follows.

Error correction, Description change, Description addition, Description deletion: For the description of LSI manual.

Specification change, Specification addition, Specification deletion: For microcontroller's specification.

The following shows the changes in the publication of "MN101EF56K/57G/76K LSI User's Manual" (From 2nd Edition dated in September, 2011 to the 3rd Edition dated in January, 2012.)

Page	Line	Definition	Former Edition (2.0)	New Edition (3.0)
-	All Chapters	Description addition	-	Added the specification for MN101EF56K.
I-2 I-47	Note	Description change	...When using <u>ICE version</u> , it is...	...When using <u>In-circuit Emulator</u> , it is...
I-10	- Ports	Error correction	MN101EF76K Timer I/O <u>28</u> External interrupt <u>5</u>	MN101EF76K Timer I/O <u>34</u> External interrupt <u>10</u>
I-26	Table:1.3.4	Description addition	-	Added OCD_DATA and OSC_CLK in MN101EF76K.
I-35	Table:1.3.5	Description addition	-	Added OCD_DATA and OSC_CLK in MN101EF57G.
I-59	*17	Description change	...When using <u>ICE version</u> , it is...	...When using <u>In-circuit Emulator</u> , it is...
VII-3	Table:7.1.1	Error correction	Table address IVBM=0 0x04008 to <u>0x04074</u> IVBM=1 0x00108 to <u>0x00174</u>	Table address IVBM=0 0x04008 to <u>0x04078</u> IVBM=1 0x00108 to <u>0x00178</u>
VII-52	Note 2	Description addition	-	<u>When the external interrupt is not used, set IROnIE flag to "0"...</u>
	Note 3	Description addition	-	<u>The interrupt request flag may have been set when the interrupt edge is switched...</u>
VIII-2	8.1.1	Description addition	MN101EF76K has...which are allocated to the I/O ports (Port 0, Port 2...	MN101EF76K has...which are allocated to the I/O ports (Port 0, <u>Port 1</u> , Port 2...
	Table:8.1.1	Description addition	-	<u>Port 1 Input mod...</u>
VIII-4	Table:8.2.1	Description addition	-	MN101EF76K: <u>P1OUT, P1IN, P1DIR...</u>
		Description deletion	MN101EF76K: <u>P3IMD...</u>	MN101EF76K: <u>-</u>
IX-3	Note 1	Description addition	...Pin <u>A and B</u> , " <u>A</u> " and " <u>B</u> " of the pin names are omitted.	...Pin <u>A, B and C</u> , " <u>A</u> ", " <u>B</u> ", and " <u>C</u> " of the pin names are omitted.
IX-4	Table:9.1.2	Description addition	-	Event count <u>TMnIOC input</u> Timer pulse output <u>TMnIOC output</u> PWM output <u>TMnIOC output</u>
IX-21	Note 2	Description addition	-	<u>Under the following conditions, be sure to select a system clock (fs) as a clock source of timers to set compare register of Timer n...</u>
XI-14	Note	Description addition	-	<u>Under the following conditions, be sure to select a system clock (fs) as a clock source of timer to set preset register 1 and 2 of Timer 7 and Timer 8....</u>
XI-39	Table:11.4.1	Description addition	-	Timer 7 Timer 8 Event input <u>TM7IOC input</u> <u>TM8IOC input</u>
XI-46	Table:11.5.1	Description addition	-	Timer 7 Timer 8 Pulse output pin <u>TM7IOC output</u> <u>TM8IOC output</u>
XII-18	Note	Description addition	-	<u>Be sure to set the IROSEL2 to IROSEL0 flags before setting the other flags in PWMOFFL register.</u>
XII-26 XII-28	Note 3	Description addition	-	<u>Under the following conditions, be sure to set PWMCKSL1 to 0 flags of PWM timer operation control register to "00".</u>
XVIII-3	Table:18.1.3	Description deletion	Serial Interface 2 <u>SBO2B(P30), SBI2B(P31), SBT2B(P32), TXD2B(P30), RXD2B(P31)</u>	-

Page	Line	Definition	Former Edition (2.0)	New Edition (3.0)
XVIII-4	Table:18.1.4	Description addition	Serial Interface 0, 1, 2 and 3 Selection of transfer clock division ratio Divide by 8 Divide by16	Serial Interface 0, 1, 2 and 3 Selection of transfer clock division ratio <u>No dividing</u> Divide by 8 Divide by16
XVIII-12	Figure:18.1.6	Description deletion	<u>SBO2B/TXD2B/P30, SBI2B/RXD2B/P31, SBT2B/P32</u>	-
XVIII-35	SC4MD0	Description change	bp: 5 Flag: SC4DEM Description: Operation selection after detecting <u>instability of IIC slave communication data</u> 0: <u>Disable communication</u> 1: <u>Enable communciation</u>	bp: 5 Flag: SC4DEM Description: Operation selection after detecting <u>communication data instability in IIC slave communication mode</u> 0: <u>Start condition is not detected</u> 1: <u>Start condition is detected</u>
XVIII-38	SC4MD3	Description deletion	bp: 5 Flag: SC4STPC Description: Stop condition generation request flag in IIC communication <u>*1</u>	bp: 5 Flag: SC4STPC Description: Stop condition generation request flag in IIC communication
		Description change	bp: 0 Flag: SC4ACKO Description: At transmission mode (<u>SC4REX=0</u>) <u>ACK</u> bit detection flag At reception mode (<u>SC4REX=1</u>) <u>ACK</u> bit level specification <u>*2</u>	bp: 0 Flag: SC4ACKO Description: At transmission <u>ACK/NACK</u> bit detection flag At reception <u>ACK/NACK</u> bit transmission flag
	Note 1 (Note 1 and 2: Former Edition)	Description change	Note 1: <u>*1: Only "1" can be written. In addition, write is disabled when a forced reset...</u> Note 2: <u>Set data to SC4MD3 register with a MOV instruction according to each bit setting...</u>	Note 1: <u>SC4STPC and SC4ACKO flags are updated by hardware. Thus, when values of these flags are read out, the values might be...</u>
XVIII-40	Note	Description addition	-	Regarding SC4ORE flag: when "1" is written, the value of this flag... when "0" is written, the value of this flag...
XVIII-41	SC4STR1	Description deletion	bp: 6 Flag: SC4ABT_LST Description: Arbitration lost detection flag <u>*1</u> bp: 0 Flag: SC4DATA_ERR Description: Communication error detection flag <u>*1</u>	bp: 6 Flag: SC4ABT_LST Description: Arbitration lost detection flag bp: 0 Flag: SC4DATA_ERR Description: Communication error detection flag
			<u>*1 Only "0" can be written.</u>	<u>Regarding SC4ABT_LST flag and SC4DATA_ERR flag:</u> <u>when "1" is written, the value of this flag...</u> <u>when "0" is written, the value of this flag...</u>
XVIII-48	■ Overrun Error and...	Description addition	■ Overrun Error and Error Monitor Flag (<u>Serial Interface 0, 1 and 2</u>)	■ Overrun Error and Error Monitor Flag (<u>Serial Interface 0, 1, 2 and 3</u>)
XVIII-73	Table:18.4.5	Description addition	<u>Serial Interface 0, 1 and 2</u> Setup: <u>Serial Interface 0, 1 and 2</u> clock source... Register: SC0MD3, SC1MD3	<u>Serial Interface 0, 1, 2 and 3</u> Setup: <u>Serial Interface 0, 1, 2 and 3</u> clock source... Register: SC0MD3, SC1MD3, <u>SC2MD3, SC3MD3</u>
			-	When the SCnCMD flag of SCnMD1 register is set to "1"...
XVIII-79	Figure:18.4.4 Figure:18.4.5	Description deletion	-	The timing from a data setup to a startbit output is changed.
XVIII-87	Hint 3	Description addition	-	When the SCnCMD flag of SCnMD1 register is set to "1"...

Page	Line	Definition	Former Edition (2.0)		New Edition (3.0)	
XVIII-93	■ Communication Data Instability...	Description change	<u>When it detects a change in the data bus (SDA) while...the subsequent start condition and communication data instability cannot be detected</u>		<u>When a serial circuit detects a changes in the data bus (SDA4 pin) while...the circuit can detect start conditions and starts communication even if the SC4DATA_ERR flag is "1"</u>	
XXI-2	Table:21.1.2	Description deletion	<u>Input Range</u>		-	
XXII-4	22.1.4	Error correction	LCD output control registers <u>1 to 6 (LCCTR1 to LCCTR6)</u>		LCD output control register <u>1 to 7 (LCCTR1 to LCCTR7)</u>	
XXIV-30	Table:24.7.3	Error correction	Command library Program_1ms_Lib Erase_1ms_Lib	Address <u>0x6F889</u> <u>0x6F893</u>	Command library Program_1ms_Lib Erase_1ms_Lib	Address <u>0x6F899</u> <u>0x6F8A3</u>

Remark: Definition in the above table is classified according to the content of changes as follows.

Error correction, Description change, Description addition, Description deletion: For the description of LSI manual.

Specification change, Specification addition, Specification deletion: For microcontroller's specification.

The following shows the changes in the publication of "MN101EF57G/76K LSI User's Manual" (From 1st Edition dated in April, 2009 to the 2nd Edition dated in September, 2011.)

Page	Line	Definition	Former Edition (1.0)	New Edition (2.0)
-	All Chapters	Description addition	-	Added the specification for MN101EF76K.
-	All Chapters	Error correction	PAPLUD Port A pull-up/ <u>pull-down</u> resistor control register	PAPLU Port A pull-up resistor control register
-	All Chapters	Description deletion	-	Deleted SELUD2 (pull-up/pull-down resistor selection register 2).
-	-	Description deletion	-	Deleted reset pin of PnOUT registers.
I-2	22	Description deletion	The package is 80-pin, LQFP.	-
I-5	22	Description deletion	16-bit cascade connection (connected with timer 0), <u>timer synchronous output</u>	16-bit cascade connection (connected with timer 0)
I-28	Table: 1.3.4 <NRST>	Error correction	...it is recommended that a discharge diode be placed between NRST and <u>VDD</u> .	...it is recommended that a discharge diode be placed between NRST and <u>VDD5</u> .
I-30	Table: 1.3.4 <PA0 to PA7>	Description deletion	...A pull-up/ <u>pull-down</u> resistor for each bit can be selected individually by the PAPLUD register. <u>A pull-up/down resistor connection for each port can be selected individually by the SELUD2 register.(A pull-up/pull-down can not be mixed.)</u> At reset,...	...A pull-up resistor for each bit can be selected individually by the PAPLU register. At reset,...
I-32	Table: 1.3.4 <VREF+>	Error correction	...This pin is generally used as VREF+ = <u>VDD</u> .	...This pin is generally used as VREF+ = <u>VDD5</u> .
I-34	Table: 1.3.4 <MMOD>	Description change	Fix this pin at VSS level.	Set to VDD5-level or VSS-level.
I-35	Note	Error correction	...it is necessary to connect the pull-up resistor on the <u>substrate</u> .	...it is necessary to connect the pull-up resistor on the <u>circuit board</u> .
I-38	1, 2	Description addition	-	This LSI manual describes standard specifications. When....
I-41	B16, B18 Figure: 1.5.2, 1.5.3	Description change	<u>f_{hosc-1}</u> , <u>f_{sosc-1}</u>	<u>f_{hosc1}</u> , <u>f_{sosc1}</u>
I-42	B20, B25	Description change	<u>f_{hosc-2}</u> , <u>f_{sosc-2}</u>	<u>f_{hosc2}</u> , <u>f_{sosc2}</u>
I-44	C9, C10	Error correction	frc=20 MHz [Double-speed mode: fs= <u>fosc</u>]	frc=20 MHz [Double-speed mode: fs= <u>frc</u>]
I-46	*17	Error correction	...,it is necessary to connect the pull-up resistor on the <u>substrate</u> .	...,it is necessary to connect the pull-up resistor on the <u>circuit board</u> .
I-49	C68 to C72	Description change	Condition: <u>VDD5 = 5.0 V (Figure: 1.5.6)</u>	Condition: <u>Figure: 1.5.6</u>
I-49	C74	Error correction	Display output pin 2 SEG0 to <u>SEG31</u>	Display output pin 2 MN101EF57G: SEG0 to <u>SEG40</u>
I-49	*20	Error correction	<u>However</u> , COM0 to COM3 are also used as <u>P84 to P87</u> .	COM0 to COM3 are also used as <u>P81 to P84</u> .
I-49	*22	Description change	<u>Summation of 3 resistors among</u> V _{LC1} and...V _{SS} .	<u>Total of 3 resistor values between</u> V _{LC1} and...V _{SS} , <u>respectively</u> .
I-51	1.5.5	Description change	E13 Reference voltage pin input leakage current Condition: <u>Vref+ = OFF</u>	E13 Reference voltage pin input leakage current Condition: <u>Ladder resistance</u> OFF
I-51	1.5.5	Error correction	E14 Ladder resistance MIN:15 TYP: <u>35</u> MAX: 80	E14 Ladder resistance MIN:15 TYP: <u>40</u> MAX: 80

Page	Line	Definition	Former Edition (1.0)	New Edition (2.0)
I-54	1.5.8	Error correction	H21 Minimum pulse width Unit: <u>ms</u>	H21 Minimum pulse width Unit: <u>μs</u>
			H23 Consumption current in power supply voltage detection circuit Conditions: $V_{DD} = 5.0\text{ V}$	H23 Consumption current in power supply voltage detection circuit Conditions: $V_{DD5} = 5.0\text{ V}$
I-55	1.5.9	Description change	I. Internal High-speed Oscillation Circuit <u>I1, I2</u>	I. Internal High-speed Oscillation Circuit <u>I1 to I4</u> <u>*24 The specification values described in I3 and I4 are....</u>
I-56	1.5.10	Description addition	K. Flash EEPROM Program Conditions -	K. Flash EEPROM Program Conditions <u>*25 The specification value described in K1 is....</u>
I-58	Figure: 1.6.2	Description change	-	Replaced the figure for 80-pin LQFP package dimension.
I-60	Table: 1.7.1	Description change	-	Replaced the table for recommended oscillators.
I-63	2 to 5	Description change	Unused I/O pins should be set according to <u>pins'</u> condition at reset. If the output....	Unused I/O pins should be set according to <u>port</u> direction control. In the input mode....
	figure: 1.7.5	Description deletion	Figure: 1.7.5 <u>Unused I/O Pins (high impedance output at reset)</u>	Figure: 1.7.5 <u>Unused I/O Pins</u> <Deleted the figure for setting the unused Nch open-drain pins.>
	Table: 1.7.2	Description addition	-	Added the table: 1.7.2
I-65 I-66 I-67	Note	Description addition	-	<u>The circuit connected with the reset pin should be evaluated....</u>
I-65	5	Error correction	[The time of V_{DD18} is over 1.65 V ($t_{V_{DD18}}$)] <u>≤</u> [The time of Reset pin voltage should be low (t_{NRSTL})]	[The time of V_{DD18} is over 1.65 V ($t_{V_{DD18}}$)] <u>× 2 ≤</u> [The time of Reset pin voltage should be low (t_{NRSTL})]
I-66	Figure: 1.7.9	Description change	-	Changed the term of tATRSTH.
II-15	Note 2	Description deletion	<u>The MMOD pin should be fixed at "L" level.</u> Do not change the settings of the MMOD pin even after reset release.	Do not change the settings of MMOD pin even after reset is released.
	Hint	Description addition	-	For MMOD setup to rewrite the flash memory, refer to ...
II-23	3	Description addition	-	Address 0x03C00 to 0x03DEF are reserved.
II-24	Table: 2.2.5	Error correction	03EBF: <u>SELUD2</u>	03EBF: <u>Reserved</u>
II-25	Note 1	Description addition	-	<u>For details of the P4_VIL to P0_VIL flags....</u>
	Note 2	Description change	-	<Transferred from page II-26> <u>For details of the WD2OP2 to WD2OP0 flags....</u>
	Note 3	Description change	-	<Transferred from page II-26> <u>The flash option function is determined before internal reset release....</u>
II-26	Note	Description addition	-	<u>For details of the PB_VIL to P5_VIL flags....</u>
II-27 to II-35	Chapter 2 2.3	Description addition	-	Added Chapter 2 2.3 ROM Correction.
II-36	Figure: 2.4.1	Description deletion	-	Deleted ATC block.

Page	Line	Definition	Former Edition (1.0)	New Edition (2.0)
II-47	■Initiating a Reset	Error correction	<u>100</u> μs	<u>1</u> μs
	Note 2	Description deletion	<u>When NRST pin connected to low power voltage detection circuit that gives plus for...</u>	-
	Note 3	Description addition	-	<u>Refer to [Chapter 1 1.7.4 Power Supply] for reset control at power supply on.</u>
II-39 (Former Edition)	■Reset Sequence at Power-On	Description deletion	<Deleted the page II-39 in former edition> ■ <u>Reset Sequence at Power-On</u> <u>This LSI is equipped....</u> <u>Note....</u> <u>Figure: 2.6.2</u>	-
II-50	Note 1	Error correction	...which is set by <u>DLY1-0</u> flags...	...which is set by <u>DLYS1 and DLYS0</u> flags...
III-2	4 to 19	Description addition	-	1) ... <u>At reset releasing, oscillation circuit is operated at 20 MHz.</u> 2) ... <u>At reset releasing, oscillation circuit is operated.</u> 3) ... <u>At reset releasing, oscillation circuit is stopped.</u> 4) ... <u>At reset releasing, oscillation circuit is stopped.</u> 5) ... <u>At reset releasing, PLL circuit is stopped.</u>
III-7	Figure: 3.1.2	Error correction	RCCNT bp2: <u>RCFSEL</u> OSCCNT bp4 to bp 7: <u>HOSCIV...</u>	RCCNT bp2: - OSCCNT bp4 to bp 7: <u>HOSCIV...</u>
III-10	OSCCNT	Error correction	bp 7 to 4 3 2 1 to 0 At reset - <u>1</u> <u>0</u> - Access - <u>R/W</u> <u>R/W</u> -	bp 7 to 4 3 2 1 to 0 At reset <u>0</u> - - <u>0</u> Access <u>R/W</u> - - <u>R/W</u>
III-11	Note 1	Error correction	..., wait <u>more than 30 μs</u> after the setting the RCON flag of the RCCNT register. Then <u>clear</u> the HOSCSEL flag <u>and change the operation clock.</u>	..., wait for <u>30 μs or longer</u> after setting the RCON flag of the RCCNT register <u>to "0"</u> . Then <u>set</u> the HOSCSEL flag <u>of the OSCCNT register to "0"</u> .
III-12	Note 4	Error correction	..., <u>set the RCON flag and wait more than 30 μs.</u> Then <u>clear</u> the HOSCSEL flag of the OSCCNT register.	..., <u>wait for 30 μs or longer after setting the RCON flag of RCCNT register to "0"</u> . Then <u>set</u> the HOSCSEL flag of OSCCNT register <u>to "0"</u> .
III-14	Note 4	Error correction	...,wait more than 200 μs after <u>the setting</u> the RCSON flag of the RCSCNT register. Then <u>clear</u> the SOSCSEL flag <u>and change the operation clock.</u>	...,wait more than 200 μs after <u>setting</u> the RCSON flag of RCSCNT register <u>to "0"</u> . Then <u>set</u> the SOSCSEL flag <u>of OSCSCNT register to "0"</u> .
III-16	RCSCNT	Description change	bp 7 Flag <u>Reserved</u> At reset <u>1</u> Access <u>R/W</u> bp Flag Description 7 <u>Reserved</u> <u>Always set to "1"</u> .	bp 7 Flag - At reset - Access - bp Flag Description 7 - -
III-16	*1	Description change	Rewriting is <u>enable</u> when <u>SOSCSEL flag of the OSCCNT register is "1"</u> and <u>WD2MD7-0 flags of the WD2MDSEL register are "00"</u> .	Rewriting is <u>enabled</u> when <u>WD2MD7 to WD2MD0 flags of WD2MDESEL register are "0x00"</u> .
III-16	Note 5	Description change	...,wait more than 200 μs after <u>clearing</u> the SOSCSEL flag of the OSCSCNT register <u>and switch the operation clock.</u>	...,wait more than 200 μs after <u>setting the RCSON flag of RCSCNT register to "0"</u> . Then <u>set</u> the SOSCSEL flag of OSCSCNT register <u>to "0"</u> .

Page	Line	Definition	Former Edition (1.0)	New Edition (2.0)
III-20	PLL CNT	Description change	PLLCK3-0 4'h0: 2 <u>multiplication</u> ... 4'h1: 2 <u>multiplication</u> ... 4'h2: 3 <u>multiplication</u> ... 4'h3: 4 <u>multiplication</u> ... 4'h4: 4 <u>multiplication</u> ... 4'h5: 5 <u>multiplication</u> ... 4'h6: 5 <u>multiplication</u> ... 4'h7: 6 <u>multiplication</u> ... 4'h8: 8 <u>multiplication</u> ... 4'h9: 10 <u>multiplication</u> ...	PLLCK3, PLLCK2, PLLCK1, PLLCK0 0000: Multiply by 2.... 0001: Multiply by 2.... 0010: Multiply by 3.... 0011: Multiply by 4.... 0100: Multiply by 4.... 0101: Multiply by 5.... 0110: Multiply by 5.... 0111: Multiply by 6.... 1000: Multiply by 8.... 1001: Multiply by 10.... 1010: <u>Setting prohibited</u> 1011: <u>Setting prohibited</u> 1100: <u>Setting prohibited</u> 1101: <u>Setting prohibited</u> 1110: <u>Setting prohibited</u> 1111: <u>Setting prohibited</u>
III-21	Note 3	Error correction	changing the <u>OSCSEL</u> flag of the OSCCNT register....	changing the <u>HOSCSEL</u> flag of OSCCNT register.
III-22	CPUM	Description addition	 1011: STOP1	 1011: STOP1 <u>Others: Setting prohibited</u>
III-25	Hint 3	Description addition	-	<u>On the condition of $f_s \leq 4$ MHz ($1.8\text{ V} \leq V_{DD5} < 2.7\text{ V}$), $f_s \leq 10$ MHz...</u>
III-27 III-29 to III-31	Hint	Description addition	Refer to [Chapter 6 Power Supply Voltage Detection Function] for...	<u>The frequency used as a system clock varies depending on...</u> Refer to [Chapter 6 Power Supply Voltage Detection] for...
III-29	■Transition from OSC mode to RC mode	Error correction	Transition from <u>RC</u> mode to <u>OSC</u> mode requires...	Transition from <u>OSC</u> mode to <u>RC</u> mode requires...
	Note 1	Description change	..., <u>set the RCON flag and wait 30 μs and over</u> . Then <u>clear</u> the HOSCSEL flag of the OSCCNT register.	..., <u>wait more than 30 μs after setting the RCON flag of RCCNT register to "0"</u> . Then <u>set</u> the HOSCSEL flag of OSCCNT register <u>to "0"</u> .
III-30	Program 3	Description change	 BSET (RCCNT)1 Set the internal.... <u>MOV X'8C', D0</u> <u>LOOP ADD -1, D0</u> <u>Loop to wait....</u> <u>BNE LOOP</u> BCLR (CPUM)0 Set NORMAL....	 BSET (RCCNT)1 Set the internal.... <u>NOP</u> BCLR (CPUM)0 Set NORMAL....
III-33	Note 1	Description change	After setting the RCSON flag of the RCSCNT register to set the internal low-speed oscillation <u>circuit</u> , transition to SLOW mode <u>is</u> executed after waiting more than 200 μ s.	After setting the RCSON flag of RCSCNT register <u>to "0"</u> to set the internal low-speed oscillation <u>operation</u> , transition to SLOW mode <u>should be</u> executed after waiting more than 200 μ s.
	Note 2	Error correction	After the reset release, do not transit <u>300 ms</u> to SLOW, HALT1 and STOP modes.	After the reset is released, wait for <u>30 ms or longer</u> to change the operation mode to SLOW, HALT1 or STOP.
III-34	Hint	Description deletion	When setting the SOSSEL flag of the OSCSCNT register to change....	-
	Note 2	Description addition	-	After the reset is released, wait for 30 ms or longer to change the operation mode to SLOW, HALT1 or STOP.
III-35	Program 8	Description deletion	 BSET (OSCSCNT)1 Set the low-speed... <u>BSET (OSCSCNT)1</u> <u>Set XI mode</u> MOV X'88', (PRTKEY) Release....	 BSET (OSCSCNT)1 Set the low-speed... MOV X'88', (PRTKEY) Release....
	Hint	Description deletion	When setting the SOSSEL flag of the OSCSCNT register to change....	-

Page	Line	Definition	Former Edition (1.0)	New Edition (2.0)
III-36	Note 1	Description change	..., <u>set the RCSON flag of the RCSCNT register and</u> wait more than 200 μ s. Then <u>clear</u> the SOSSEL flag.	..., wait more than 200 μ s <u>after setting the RCSON flag of RCSCNT register to "0"</u> . Then <u>set</u> the SOSSEL flag of OSCSCNT register to "0".
	Note 2	Error correction	When the SOSCCNT flag is cleared to set <u>P90/P91</u> to...	When the SOSCCNT flag is cleared to set <u>P25/P26</u> to...
III-39	Note 2	Error correction	After the reset releasing, do not transfer <u>300</u> ms to SLOW,....	After the reset is released, wait for <u>30 ms or longer</u> to change...
III-41	■Transition to STOP mode	Error correction	Program <u>Z</u>	Program <u>14</u>
IV-15	■Setup Example: Clock Transition Mode is Not Used	Error correction	Upper limit frequency: 27 <u>MHz</u> Lower limit frequency: <u>15 MHz</u> Setup Procedure (6).... bp7-0: MNLSET[7:0] = <u>0x0f</u> Description (5)....the upper limit frequency to 27 <u>MHz</u> . (6)....the lower limit frequency to <u>15 MHz</u> .	Upper limit frequency: 27 <u>kHz</u> Lower limit frequency: <u>1 kHz</u> Setup Procedure (6).... bp7 to 0: MNLSET[7:0] = <u>0x01</u> Description (5)....the upper limit frequency to 27 <u>kHz</u> . (6)....the lower limit frequency to <u>1 kHz</u> .
IV-19, IV-20	■Setup Example: Clock Transition Mode is Used	Error correction	Upper limit frequency: 27 <u>MHz</u> Lower limit frequency: 15 <u>MHz</u> Description (5)....the upper limit frequency to 27 <u>MHz</u> . (6)....the lower limit frequency to 15 <u>MHz</u> .	Upper limit frequency: 27 <u>kHz</u> Lower limit frequency: 15 <u>kHz</u> Description (5)....the upper limit frequency to 27 <u>kHz</u> . (6)....the lower limit frequency to 15 <u>kHz</u> .
V-2	1	Description change	This LSI has a register protect function to prevent writing errors <u>of</u> the control registers <u>of</u> fail-safe function....	This LSI has a register protect function to prevent writing errors <u>to</u> the control registers <u>as</u> fail-safe functions....
V-5	Figure: 5.3.1	Error correction	<u>NM</u> interrupt	<u>NMI</u> interrupt
VI-3	LVIMD	Error correction	bp Description 7 to 4 : 1 1: $V_{DD5} \geq$ detection voltage	bp Description 7 to 4 : <u>1000: 3.0 V</u> 1 1: $V_{DD5} \geq$ detection voltage
VI-4	Hint	Error correction	The detection voltage which can be monitored by the <u>LVIMON</u> flag is the voltage detected....	The detection voltage which can be monitored by the <u>LVION</u> flag is the voltage detected....
	Note 2	Error correction	The <u>LVIMON</u> flag and the <u>LVICR0</u> to 2 flags must be set when the LVION flag is "0".	The <u>LVIMOD</u> flag and the <u>LVICTR0</u> to <u>LVICTR2</u> flags must be set when the LVION flag is "0".
VI-5, 6	6.3.1	Error correction	<u>LVI1ICR</u>	<u>LVIICR</u>
VII-4	Figure: 7.1.1	Description change	<u>Vector 29</u>	<u>Vector 30</u>
VII-7	Figure: 7.1.3	Description change	<u>Vector 8</u>	<u>Vector 7</u>
VII-10	Figure: 7.1.3	Description change	PC7 to <u>1</u>	PC7 to <u>0</u>
VII-11	Note 2	Description change	when program counter (PC [<u>bit20 to bit17, bit0</u>]) are saved to the stack....	when program counter (PC [<u>bit19 to bit16, bitH</u>]) are saved to the stack....
VII-32	Note	Description deletion	Do not word access to the register denoted by *.	-
VII-43 to VII-52	Figure: 7.3.1 to Figure: 7.3.10	Description change	-	Deleted IRQCNT register. Added AC zero-cross function (IRQ0, 1). Corrected EDGDT register (IRQ2, 3, 4). bp2: EDGSEL2 bp3: EDGSEL3 bp5: EDGSEL5
VII-51	3	Description addition	-	The IRQCNT register should be operated when...
	Hint	Description addition	-	Set P2nEN flag to "0" to fix the input of external interrupts...
	Note	Description addition	-	To enable maskable interrupts after setting P2nEN flag...

Page	Line	Definition	Former Edition (1.0)	New Edition (2.0)
VII-53	5	Description addition	-	The NFnEN1 flags should be operated when...
	Note	Description addition	-	To enable maskable interrupts after setting NFnEN1 flag...
VII-54	3	Description addition	-	The ACZCTR register should be operated when...
	Note	Description addition	-	To enable maskable interrupts after setting P2nIM flag...
VII-57	5	Description addition	-	The KEYT3_1ENn flags should be operated when...
	Note 2	Description addition	-	To enable maskable interrupts after setting KEYT3_1ENn flag...
VII-58	4	Description addition	-	The KEYT3_2IMD register should be operated when...
	Note	Description addition	-	To enable maskable interrupts after setting KEYT3_2ENn flag...
VIII-67, 68	7.3.8	Description addition	-	Added [Chapter 7 7.3.8 AC Zero-Cross Detection]
VII-80	4	Description deletion	...the previous level is sent. <u>It means that only...</u>	...the previous level is sent.
VIII-2	Table:8.1.1	Error correction	Port A: No pull-up/ <u>pull-down</u> resistor	Port A: No pull-up resistor
VIII-5	Table:8.2.1	Error correction	<u>PAPLUD</u> : Port A pull-up/ <u>pull-down</u> resistor control register	<u>PAPLU</u> : Port A pull-up resistor control register
		Description deletion	SEUD2 register	-
VIII-6	Table:8.3.1	Description addition	-	Added Table:8.3.1
VIII-7	15 to 28	Description addition	-	P05 is also used as the I/O pin for Serial Interface 3... For P01 to P04 each bit...by Port0 output mode register 2 (P0OMD2).
VIII-51	Figure:8.5.7 Figure:8.5.8	Description addition	-	Added AC-zero-cross detection circuit and selector.
VIII-68	Figure:8.6.8	Description addition	-	Added Port input mode control
VIII-34	Figure:8.6.9	Description addition	"SC4BTS"	"SC4 <u>S</u> BTS"
VIII-111	20 to 22	Description addition	-	For P62 to P63 each bit can be set individually...
VIII-134	P7ODC	Error correction	P7ODC: <u>0x03EF5</u>	P7ODC: <u>0x03EF7</u>
VIII-172 to VIII-174	Figure:8.11.14 to Figure:8.11.16	Description addition	-	Added switch for VLC3 to VLC1
VIII-175	■Special Function Pin Setup	Error correction	P92 to <u>P94</u> are also used... P92 to <u>P94</u> are also used... <u>the value of the port 9 input register is read to be "0"</u> .	P92 to <u>P95</u> are also used... P92 to <u>P95</u> are also used... <u>undefined value is read from Port 9 input register (P9IN)</u> .
		Description addition	-	P94 and P95 are also used as the analog output pins for D/A.
VIII-185	Figure:8.12.8	Description addition	-	Added [Figure:8.12.8 P91 Block Diagram (MN101EF57G)]
VIII-189	10	Description deletion	... <u>port A can be selected to add pull-up or pull-down resistor by bp2 of the pull-up/pull-down resistor selection register 2(SELUD2).</u> <u>For port A</u> , each bit can...	... Each bit can...

Page	Line	Definition	Former Edition (1.0)	New Edition (2.0)
VIII-190	■Special Function Pin Setup	Error correction	PA0 to PA7 are also used... <u>the value of the port A input register is read to be "0"</u> .	PA0 to PA7 are also used... <u>undefined value is read from Port A input register (PAIN)</u> .
		Description change	PA0 is also used as the I/O pin of timer 0. ... PA7 is also used as the I/O pin of timer 9....	PA0 to PA7 are also used as the I/O pins for Timer 0 to Timer 6 and Timer 9...
VIII-191	Table:8.13.2	Description deletion	SELUD2 register	-
VIII-103 (Former Edition)	-	Error correction	■ Pull-UP/pull-down Resistor Selection Register 2 (SELUD2: 0x03EBF)	Deleted the page
VIII-203 to VIII-209	Figure:8.13.8 to Figure:8.13.14	Description deletion	-	Deleted Pull-up/pull down resistor selection
VIII-210	■Special Function Pin Setup	Error correction	PB0 to <u>PB3</u> are also used... PB0 to <u>PB3</u> are also used... <u>the value of the port B input register is read to be "0"</u> .	PB0 to <u>PB7</u> are also used... PB0 to <u>PB7</u> are also used... <u>undefined value is read from Port B input register (PBIN)</u> .
VIII-244	8.19.4 Input Voltage Level	Description change	■ Input Voltage Level P0 to <u>P9</u> ,PB	■ Input Voltage Level P0 to PB
		Error correction	Parameter MAX Input voltage level B: Input low voltage <u>0.5</u> V _{DD5}	Parameter MAX Input voltage level B: Input low voltage <u>0.45</u> V _{DD5}
IX-5	Table:9.1.3	Error correction	Timer 0 to Timer 4 Clock source <u>Simple timer A input</u>	Timer 0 to Timer 4 Clock source <u>Timer A output</u>
			Timer 1 Synchronous <u>TM1IO input</u> Timer 3 Synchronous <u>TM3IO input</u>	Timer 1 Synchronous <u>TM1IO input</u> Timer 3 Synchronous <u>TM3IO input</u>
IX-9 IX-12 IX-14	Figure:9.1.3 Figure:9.1.6 Figure:9.1.8	Error correction	CKnMD (<u>bp5-3</u>)	CKnMD (<u>bp6 to 4</u>)
IX-21	Note	Description addition	-	In double buffer configuration, when the clear timing of binary counter occurs...
IX-25 IX-27	Note	Description addition	-	To load data from a compare register to an internal register...
IX-42	Note 1	Description addition	-	MN101EF57G does not have double-buffer function...
IX-47	Note 3	Description change	In double buffer configuration, <u>the event input (TM2IO input, TM3IO input) is selected as...</u>	In double-buffer configuration, <u>the clock other than fs is selected as...</u>
IX-50	Note	Description addition	-	When the synchronous TMnIO input is selected as the count clock source...
IX-57	Figure:9.7.5	Error correction	CKnMD <u>bit 4, 5</u>	CKnMD <u>bit 6, 5</u>
IX-60	Table:9.9.1	Description change	· Timer 0: Serial interface 0, 2, 4 · Timer 1: Serial interface 0, 2, 4 · Timer 2: Serial interface 0, 2, 4	Replaced to Table:9.9.1
IX-64 IX-68 IX-70	Table:9.11.1 Table:9.11.2 Table:9.11.3	Description addition	-	Added "Timer pulse output"
IX-71	Note 2	Error correction	At 32-bit cascade connection, always set TM3CAS flag of TM3MD register to " <u>0</u> ".	At 32-bit cascade connection, always set TM3CAS flag of TM3MD register to " <u>1</u> ".
IX-72	Note	Description addition	-	In cascade connection (16-bit, 24-bit, 32-bit) with double-buffered compare register...
	Figure:9.11.1	Description addition	-	Added Figure:9.11.1

Page	Line	Definition	Former Edition (1.0)	New Edition (2.0)																		
XI-8	Figure:11.1.5	Error correction																				
XI-22	Note	Description change	When T7ICT1 to 0 flags <u>are set to timer interrupt</u> , the <u>timer</u> set by T7ICT2 flag of the TM7MD4 register is <u>set</u> as a capture trigger.	When T7ICT1 to 0 flags <u>select "timer interrupt"</u> , the <u>timer interrupt</u> set by T7ICT2 flag of TM7MD4 register is <u>used</u> as a capture trigger.																		
XI-27	Note	Description change	When T8ICT1 to 0 flags <u>are set to timer interrupt</u> , the <u>timer</u> set by T8ICT2 flag of the TM8MD4 register is <u>set</u> as a capture trigger.	When T8ICT1 to 0 flags <u>select "timer interrupt"</u> , the <u>timer interrupt</u> set by T8ICT2 flag of TM8MD4 register is <u>used</u> as a capture trigger.																		
XI-29	TM8MD4	Error correction	<table><tr><th>bp</th><th>Flag</th><th>Description</th></tr><tr><td>3</td><td><u>TM8SEL_B</u></td><td><u>B port selection</u></td></tr><tr><td>2</td><td><u>TM8SEL_A</u></td><td><u>A port selection</u></td></tr></table>	bp	Flag	Description	3	<u>TM8SEL_B</u>	<u>B port selection</u>	2	<u>TM8SEL_A</u>	<u>A port selection</u>	<table><tr><th>bp</th><th>Flag</th><th>Description</th></tr><tr><td>3</td><td><u>Reserved</u></td><td><u>Always set to "0"</u></td></tr><tr><td>2</td><td><u>TM8SEL_A</u></td><td><u>A port selection</u></td></tr></table>	bp	Flag	Description	3	<u>Reserved</u>	<u>Always set to "0"</u>	2	<u>TM8SEL_A</u>	<u>A port selection</u>
bp	Flag	Description																				
3	<u>TM8SEL_B</u>	<u>B port selection</u>																				
2	<u>TM8SEL_A</u>	<u>A port selection</u>																				
bp	Flag	Description																				
3	<u>Reserved</u>	<u>Always set to "0"</u>																				
2	<u>TM8SEL_A</u>	<u>A port selection</u>																				
XI-37	Note 3	Description change	When the binary counter is read during the timer operation, <u>it is treated as the data by every 8 bits...</u>	When binary counter is read during the timer operation, <u>the data is processed as 8-bit unit data...</u>																		
XI-40	Hint	Description change	<u>If the binary counter is read out during operation, incorrect data at counting up may be read...</u>	<u>When event input (TMnIO input) is selected as a count clock source, do not read out values...</u>																		
	Note 4 (Former Edition)	Description deletion	<u>If the TM7IO input is selected as the count clock source, procedure such as pull-up is...</u>	-																		
XI-57	Figure:11.7.4	Error correction	Figure:11.7.4 <u>Pulse Width Measurement of External Interrupt 0</u>	Figure:11.7.4 <u>Output Waveform of TM7IO Output Pin</u>																		
XI-66	3	Description change	...the capture trigger is synchronized with the <u>capture sampling clock</u> which is set by...	...the capture trigger synchronized with the <u>clock</u> which is set by...																		
XI-67	Note	Error correction	...When setting the <u>capture clock as the count clock</u> to execute the event count operation...	...When setting the <u>TMnCKSMP flag to "Count clock"</u> to execute the event count operation...																		
XI-96	Table:11.13.1	Description addition	-	Added "Capture function" and "Pulse width measurement"																		
XII-2	Table:12.1.1	Description addition	-	Added Table:12.1.1 Timer 9 Pin Functions																		
XII-9	PWMMDH	Error correction	<table><tr><th>bp</th><th>Flag</th><th>Access</th></tr><tr><td>2</td><td><u>DTSEL</u></td><td><u>R/W</u></td></tr></table>	bp	Flag	Access	2	<u>DTSEL</u>	<u>R/W</u>	<table><tr><th>bp</th><th>Flag</th><th>Access</th></tr><tr><td>2</td><td><u>DTSEL</u></td><td><u>W</u></td></tr></table>	bp	Flag	Access	2	<u>DTSEL</u>	<u>W</u>						
bp	Flag	Access																				
2	<u>DTSEL</u>	<u>R/W</u>																				
bp	Flag	Access																				
2	<u>DTSEL</u>	<u>W</u>																				
XII-9	Note 2	Description addition	-	DTSEL flag can only write it...																		
XII-25	12.3	Description change	<u>12.3 Operation</u> <u>12.3.1 16-bit Timer Operation</u> <u>12.3.1.1 16-bit Timer Operation</u>	12.3 16-bit Timer Operation 12.3.1 Operation -																		
XII-29	12.3.2	Description change	<u>12.3.1.2 Setup Example</u>	12.3.2 Setup Example																		
XII-30	12.4	Description change	<u>12.3.2 16-bit Event Count</u> <u>12.3.2.1 Operation</u>	12.4 16-bit Event Count 12.4.1 Operation																		
XII-32	12.4.2	Description change	<u>12.3.2.2 Setup Example</u>	12.4.2 Setup Example																		
XII-32	■ Event Count Setup Example	Error correction	Setup Procedure (3) Set the special function pin to input PADIR(0x03E9A) bp7: <u>PADIR0</u> = 0	Setup Procedure (3) Set the special function pin to input PADIR(0x03E9A) bp7: <u>PADIR7</u> = 0																		
XII-33	12.5	Description change	<u>12.3.3 16-bit Timer Pulse Output</u> <u>12.3.3.1 Operation</u>	12.5 16-bit Timer Pulse Output 12.5.1 Operation																		
XII-34	12.5.2	Description change	<u>12.3.3.2 Setup Example</u>	12.5.2 Setup Example																		

Page	Line	Definition	Former Edition (1.0)		New Edition (2.0)	
XVIII-28	SCnMD1 (SCnCKM)	Description addition	Description Transfer clock division selection 0: Not divided 1: Divided		Description <u>Clock synchronous (SCnCMD=0)</u> Transfer clock division selection 0: Not divided 1: Divided <u>UART (SCnCMD=1)</u> <u>Transfer clock is divided regardless of the setting value.</u> <u>The data read out is the setting value.</u>	
	Note 2	Description addition	-		Writing to watchdog timer 2 operation mode register is possible only once...	
XVIII-34	18.2.4	Error correction	This serial interface consists of 2 buffers and <u>7</u> registers.		This serial interface consists of 2 buffers and <u>8</u> registers.	
XVIII-39	SC4AD0 SC4AD1	Error correction	<u>SC4AD9</u> to <u>SC4AD0</u>		<u>SC4ADR9</u> to <u>SC4ADR0</u>	
XVIII-57	Note 3	Description change	<u>When the SCnIGC flag of clock synchronous serial reception interrupt control register (SCINTSEL) is...</u>		<u>SCnIGC flag of clock synchronous serial reception interrupt control register (SCINTSEL) is set to...</u>	
XVIII-63	Table:18.3.8	Error correction	Setup item Set to Serial data input selection <u>SBI1A</u>		Setup item Set to Serial data input selection <u>SBI1</u>	
XVIII-63 XVIII-66	Setup Procedure	Description change	Description (1) Set the SC1PSCE flag of the SC1MD3 register to "1" to select <u>"prescaler operation"</u> .		Description (1) Set the SC1PSCE flag of SC1MD3 register to "1" to select <u>"Enable count" for prescaler count control.</u>	
XVIII-86	Setup Procedure	Description addition	Setup Procedure (6)... Select the clock dividing SC1MD1 (0x03F22) bp3: SC1CKM =1 ...	Description (6)... Set the SC1CKM flag... The SC1MST flag...	Setup Procedure (6)... Select the clock dividing SC1MD1 (0x03F22) bp3: SC1CKM =1 <u>bp1: SC1DIV =0</u> ...	Description (6)... Set the SC1CKM flag... <u>Set the SC1DIV flag...</u> <u>clock division value...</u> The SC1MST flag...
	Setup Procedure	Description change	Setup Procedure (<u>1</u>) Set the baud rate timer	Description (<u>1</u>) Set the baud rate by the TMAMD register and...	Setup Procedure (<u>7</u>) Set the baud rate timer	Description (<u>7</u>) Set the baud rate by TMAMD register and...
XVIII-91	2nd line under Table:18.5.4	Error correction	<u>SC4ADm</u> flag of SC4ADn register		<u>SC4ADRm</u> flag of SC4ADn register	
XVIII-106	18.5.2	Error correction	Pin setup (flag setup) SC4MD1 register ... <u>SBT2</u> pin function selection	... <u>SBO2 (RXD2)</u> pin function selection	Pin setup (flag setup) SC4MD1 register ... <u>SBT4</u> pin function selection	... <u>SBO4 (RXD4)</u> pin function selection
XVIII-107 XVIII-110	Table:18.5.6 Table:18.5.7	Error correction	Setup item Set to Serial data input selection <u>SDA4A</u>		Setup item Set to Serial data input selection <u>SDA4</u>	
XVIII-107 XVIII-110	Setup Procedure	Description change	Description (1) Set the SC4PSCE flag of the SC4MD2 register to "1" to select <u>prescaler operation.</u>		Description (1) Set the SC4PSCE flag of SC4MD2 register to "1" to select <u>"Enable count" for prescaler count control.</u>	
XVIII-112		Error correction	Description (13) Communication completion interrupt (<u>SC4TIRO</u>) is generated...		Description (1) Serial 4 communication completion interrupt (<u>SC4IRQ</u>) is generated...	
XX-2	Table:20.1.1	Description addition	-		Added Table:20.1.1 A/D Converter Pin Functions	
XX-7	ANCTR0 bp2, 5 to 4	Error correction	Flag ANCK <u>5</u> ANCK <u>4</u> ANCK <u>2</u>	Description -	Flag ANCK <u>2</u> ANCK <u>1</u> ANCK <u>0</u>	Description <u>111: Setting prohibited</u>
XX-13	Note 4	Description change	<u>...Change the flags when A/D conversion is not executed or after forced stop.</u>		<u>...Set the ANLADE flag of A/D converter control register 0 (ANCTR0) to "0" to disable the operation before changing the flags.</u>	

Page	Line	Definition	Former Edition (1.0)	New Edition (2.0)
XX-14	Table:20.3.1	Error correction	A/D conversion cycle (T_{AD}) at low speed oscillation fslow=32.768 <u>MHz</u> 244.14 μ s 488.28 μ s 976.56 μ s 15.26 μ s 1953.12 μ s	A/D conversion cycle (T_{AD}) Low-speed operation fslow=32.768 <u>kHz</u> 244.14 μ s (<u>setting prohibited</u>) 488.28 μ s (<u>setting prohibited</u>) 976.56 μ s (<u>setting prohibited</u>) 15.26 μ s 1953.12 μ s (<u>setting prohibited</u>)
XX-21	Hint	Description addition	-	This LSI contains a sample hold capacitor (C) (approximately. 10 pF)...
	Table:20.3.4	Description addition	-	Added Table:20.3.4 Impedance of analog signal output circuit
	Figure:20.3.5	Description addition	-	Added Figure:20.3.5 Circuit Example
XXI-2	Table:21.1.1	Description addition	-	Added Table:21.1.1 D/A Converter Pin Functions
XXI-2	Table:21.1.2	Description addition	-	Input Range: V_{SS} to V_{REF+}
XXII-2	Table:22.1.1	Description deletion	Clock Source ... (LCDCLK) <u>Timer 0 to 4 or simple timer A selection</u>	Clock Source ... (LCDCLK) <u>Timer 0 to Timer 4</u>
XXII-5	Figure:22.1.1	Description deletion	<u>Timer A</u>	$\pm$ (Deleted Timer A input)
XXII-9	Note	Description addition	-	To use Timer n (n= 0 to 4) as the LCD clock source, set the LCDEN flag...
XXIV-2	Figure:24.1.1	Description addition	-	Added Figure:24.1.1 Sector Structure
XXIV-4	■ Parallel... ■ Serial... ■ User...	Description deletion	Sector 0 to 6 can be rewritten.	-
	■ BOOT...		Sector 0 to 5 can be rewritten.	-
XXIV-4	Table:24.1.2	Description change	Rewritable area: <u>Sector 0 to 6</u> Features: <u>-Activating and rewriting areas...</u>	Rewritable area: <u>All sectors</u> Features: <u>-The last sector is used as activating...</u>
XXIV-5	Table:24.1.4	Description change	Sector ... SA6 (also used as BOOT area) BOOT area is used MAIN area ... (<u>Rewriting Prohibited</u>)	Sector ... SA6 (also used as BOOT area) When BOOT area is in used MAIN area ... <u>0x23000 to 0x23FFF</u>
	1st line under Table:24.1.4	Description deletion	*1 Unwritable. Reading or instruction execution...	-
XXIV-6	24.1.3	Description addition	...MAIN area is to store user's program, and the area is rewritable. Reserved area is to store...	...MAIN area stores user program, and it is rewritable. <u>BOOT area activates BOOT mode microcontroller rewriting method, and it is rewritable.</u> Reserved area stores...

Page	Line	Definition	Former Edition (1.0)	New Edition (2.0)
XXIV-7	24.2	Description deletion	Parallel programmer rewriting is used to rewrite the...<u>For parallel programmer writing, sector 0 to 6 can be rewritten. The following rewriting methods are provided depending on the usage pattern of users.</u> ■ <u>General Usage</u> <u>(1) Entire area rewrite</u> ... ■ <u>When Using BOOT Mode Microcontroller...</u> <u>For BOOT mode microcontroller rewriting...</u> <u>(1) BOOT area rewriting in BOOT mode...</u> ... <u>(2) MAIN area rewrite in BOOT mode...</u> ...	Parallel programmer rewriting is used to rewrite the ...the usage pattern of users.
XXIV-7	Hint	Description change	If using entire area rewrite, batch rewriting...<u>However, when using entire area rewrite in...</u> The following description shows how to create a data file using debugger or <u>programmer</u> for the batch rewriting of MAIN area and BOOT area. (a) Using PanaXSeries... ... (b) Using <u>programmer</u> 1. Prepare data... 2. Write the data file for BOOT area <u>using BOOT area rewriting in BOOT mode microcontroller rewriting method.</u> 3. Write the data file for MAIN area <u>using MAIN area rewriting in BOOT mode microcontroller rewriting method.</u> 4. Read the data of the entire area <u>using entire area rewrite to write into file.</u>	To rewrite entire area, batch rewriting... The following description shows how to create a data file using debugger or <u>PX-FW2 programmer</u> for the batch rewriting of MAIN area and BOOT area. <u>For using the other programmer, it is...manufacturer.</u> (a) Using PanaXSeries... ... (b) Using <u>PX-FW2 programmer</u> 1. Prepare data... 2. Write the data file for BOOT area <u>in BOOT area.</u> 3. Write the data file for MAIN area <u>in BOOT area.</u> 4. Read the data of the entire area, <u>and write it into the file.</u>
XXIV-8	Figure:24.2.1	Description change	-	Replaced Figure:24.2.1 Parallel Programmer Rewriting Memory Map
XXIV-9	Table:24.2.1	Description change	Product ... Model MN101EF57G ... <u>AF8723</u>	Product ... Model MN101EF57G ... <u>AF9723</u>
XXIV-10	24.3	Description deletion	Serial programmer rewriting is used to rewrite...in the flash memory in advance. <u>For serial programmer writing, sector 0 to 6 can be rewritten. The following rewriting methods are provided depending on the usage pattern of users.</u> ■ <u>General Usage</u> <u>(1) Entire area rewrite</u> ... ■ <u>When Using BOOT Mode Microcontroller...</u> <u>For BOOT mode microcontroller rewriting...</u> <u>(1) BOOT area rewriting in BOOT mode...</u> ... <u>(2) MAIN area rewrite in BOOT mode...</u> ...	Serial programmer rewriting is used to rewrite...in the flash memory in advance.

Page	Line	Definition	Former Edition (1.0)	New Edition (2.0)
XXIV-10	Hint	Description change	If using entire area rewrite, batch rewriting... <u>However, when using entire area rewrite in...</u> The following description shows how to... (a) Using DebugFactory Builder ... (b) Using <u>programmer</u> 1. Prepare data... 2. Write the data file for BOOT area <u>using BOOT area rewriting in BOOT mode microcontroller rewriting method.</u> 3. Write the data file for MAIN area <u>using MAIN area rewriting in BOOT mode microcontroller rewriting method.</u> 4. Read the data of the entire area <u>using entire area rewrite to write into file.</u>	To rewrite entire area, batch rewriting... The following description shows how to... (a) Using DebugFactory Builder ... (b) Using <u>PX-FW2 programmer</u> 1. Prepare data... 2. Write the data file for BOOT area <u>in BOOT area.</u> 3. Write the data file for MAIN area <u>in MAIN area.</u> 4. Read the data of the entire area, <u>and write it into the file.</u>
XXIV-11	Figure:24.3.1	Description change	-	Replaced Figure:24.3.1 Serial Programmer Rewriting Memory Map
XXIV-14	24.4.2	Description change	<u>Writing to MN101EF57G, it uses the oscillator...</u> $Crst = \frac{\text{The oscillator stabilization time}}{(3 * \text{Pull-up resistor } R1)} \dots \text{Equation (1)}$	<u>After applying a reset to the target microcomputer...</u> $Crst = (250 * 10^{-3} - T_{wait}) / R1 \dots \text{Formula (1)}$
XXIV-15	24.4.3	Description change	The maximum output current from...<u>R_{upMin}</u> can be determined <u>from equation (2)</u> $R_{upMin} = \text{Operating supply voltage (V}_{DD5}) / 12 \text{ mA} \dots \text{Equation (2)}$	The maximum output current from...<u>R_{upRst}</u> can be determined <u>according to Formula (2)</u> $R_{upRst} = \text{Operating supply voltage (V}_{DD5}) / 12 \text{ mA} \dots \text{Formula (2)}$
XXIV-15	24.4.4	Description change	<u>If you want to insert a resistor with a large value...</u>	<u>If a resistor with a large value is inserted in series...</u>
XXIV-15	24.4.5	Description change	Find the maximum output current, <u>I_{OH}</u>, for the pins... $R_{upMin} = \text{Operating supply voltage (V}_{DD5}) / \text{Pin maximum output current (I}_{OH}) \dots \text{Equation (4)}$	Find the maximum output current, <u>I_{OL}</u>, for the pins... $R_{upMin} = \text{Operating supply voltage (V}_{DD5}) / \text{Pin maximum output current (I}_{OL}) \dots \text{Formula (4)}$
XXIV-18 XXIV-23	Figure:24.5.2 Figure:24.6.2	Error correction	[Internal RAM] Address: 0x00000 to <u>0x07FFE</u>: 6 Kbyte Address: 0xF0000 to <u>0xF7FFE</u>: 6 Kbyte(Mirror RAM)	[Internal RAM] Address: 0x00000 to <u>0x017FE</u>: 6 Kbyte Address: 0xF0000 to <u>0xF17FE</u>: 6 Kbyte(Mirror RAM)
XXIV-22	24.6 5 to 6	Description addition	...the user's program is destroyed <u>by mistake</u>. The LSI activates by releasing reset...	...the user program is destroyed <u>accidentally</u>. <u>In this rewriting method, the entire area can be...this method cannot be performed.</u> The LSI activates by releasing reset...
XXIV-23	5 to 6	Description change	...the command library becomes available to enable the rewriting of MAIN area (sector 0 to 5). ...The setting routine of the rewriting enable register <u>and the rewriting program</u> must be allocated in the internal RAM. <u>For BOOT area (sector 6) after changing...</u> After rewriting, when the activation mode...	...the command library becomes available to enable the rewriting of MAIN area (Sector 0 to 12[Sector 0 to 5]) <u>and BOOT area (Sector 13[Sector 6]).</u> ...The setting routine of the rewriting enable register must be allocated in the internal RAM. After rewriting, when the activation mode...
XXIV-24	Figure:24.6.3	Description addition	-	"BOOT area" is added in <Transfer the writing data> and <Execute the rewriting>.
	*1	Description deletion	...The setting routine of rewriting enable register <u>and the rewriting program</u> must be allocated in the internal RAM...	...The setting routine of rewriting enable register must be allocated in the internal RAM...
XXIV-24	24.6.2	Description deletion	<u>BOOT area (sector 6) is mapped in MAIN area and can be rewritten in user mode microcontroller rewriting.</u> If the sector 6 is rewritten <u>by mistake</u> , rewriting with...	If <u>BOOT area (Sector 13[Sector 6])</u> is rewritten <u>accidentally</u> , rewriting with...
XXIV-25	Table:24.7.2	Description addition	Activation Area Rewriting Area BOOT area <u>MAIN area</u>	Activation Area Rewriting Area BOOT area <u>MAIN area + BOOT area</u>
XXIV-28	24.7.4	Error correction	- The command library needs up to <u>613</u> byte as...	- The command library needs up to <u>623</u> byte as...

Page	Line	Definition	Former Edition (1.0)		New Edition (2.0)	
XXIV-28	24.7.4	Description addition	-		- To execute rewriting flash memory using... - When calling command library, it is required...	
XXIV-31	24.8	Description addition	-		Added 24.8 ROM Programming Service	
XXV-33	0x03F05 WD2CTR	Error correction	bit7	bit6	bit7	bit6
			-	-	<u>Reserved</u>	<u>Reserved</u>
			-	-	0	0
			-	-	<u>Always set to "0"</u>	<u>Always set to "0"</u>
XXV-39	0x03F7F TM8MD4	Error correction	bit3	bit2	bit3	bit2
			<u>TM8SEL_B</u>	<u>TM8SEL_A</u>	<u>Reserved</u>	<u>TM8SEL</u>
			0	0	0	0
			<u>B port timer8/ IGBT selection</u>	<u>A port timer8/ IGBT selection</u>	<u>Always set to "0"</u>	<u>Timer8/IGBT selection</u>

Remark: Definition in the above table is classified according to the content of changes as follows.

Error correction, Description change, Description addition, Description deletion: For the description of LSI manual.

Specification change, Specification addition, Specification deletion: For microcontroller's specification.

Inquiries

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